

BOMBARDIER CHALLENGER 605

FCOM1

REV 67, Nov 22/22

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Flight Crew Operating Manual

MODEL CL-600-2B16
(Serial No. 5701 to 6049)

21,863 kg / 48,200 lb MTOW

FIRST ISSUED: Oct 05/06

REVISION 67: Nov 22/22

B O M B A R D I E R
CHALLENGER 605

CL-605 FLIGHT CREW OPERATING MANUAL – VOLUME 1

This manual applies to the Challenger 605. The Challenger 605 is a marketing designation that is applicable to the Model CL-600-2B16, Aircraft serial No. 5701 to 6049.

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BOMBARDIER CHALLENGER 605	TRANSMITTAL LETTER	Vol. 1	Transmittal-1
		REV 67, Nov 22/22	

This is revision **67** to the CL-605 Flight Crew Operating Manual, PSP 605-6.
To bring this manual up to date remove old pages and insert revised pages as detailed below:

	REMOVE	INSERT
Title Page	Revision 66	Revision 67
Record of Revisions	00-01-17	00-01-17
Record of Temporary Revisions	00-02-4	00-02-4
List of Effective Pages	00-03-1 to 00-03-8	00-03-1 to 00-03-8
Supplementary Procedures	06-12-26	06-12-26

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NOTE

Record the date you insert this revision in the Record of Revisions at the front of the manual.

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BOMBARDIER CHALLENGER 605	Record of Revisions	Vol. 1	00-01-1
		REV 6, Jun 08/07	

Record the date you insert each revision in your manual.

Revision No.	Subject	Incorporated By	Date
REV 1 Dec 21/06	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 1: Limitations. Revision to Volume 1 introducing additional information and miscellaneous changes to Limitations and Emergency, Abnormal and Supplementary Procedures and Techniques and In-Flight Checks. Revision to Volume 2 introducing editorial changes.		
REV 2 Jan 11/07	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 2: Limitations. Revision to Volume 1 introducing miscellaneous changes to Supplementary Procedures and Techniques and In-Flight Checks.		
REV 3 Mar 19/07	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 3: Limitations and Emergency, Normal and Abnormal Procedures. Revision to Volume 1 introducing miscellaneous changes to Introduction and Normal Procedures. Revision to Volume 2 updating the oxygen system information. Revision to Volume 2 introducing editorial changes.		
REV 4 Apr 16/07	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 4: Introduction, Limitations and Normal and Supplementary Procedures. Revision to Volume 2 updating the communication system information.		
REV 5 Jun 04/07	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 5: Emergency, Abnormal and Supplementary Procedures.		
REV 6 Jun 08/07	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 6: Limitations and Supplementary Procedures. Revision to Volume 1 to clarify hard landings.		

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BOMBARDIER CHALLENGER 605	Record of Revisions	Vol. 1	00-01-2
		REV 9, May 14/08	

Revision No.	Subject	Incorporated By	Date
REV 7 Oct 31/07	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 7: Introduction, Limitations and Emergency, Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to correct the requirements for anti-skid checks before landing. Revision to Volume 1 to delete instructions regarding FMS selection for overwater operations. Revision to Volume 1 to update the table for type II anti-icing fluids. Revision to Volume 1 to clarify airspeed requirement for ADG in-flight check.		
REV 8 Jan 22/08	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 8: Limitations.		
REV 9 May 14/08	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 9: Limitations. Revision to Volume 1 incorporating minor correction to Mode S limitations. Revision to Volume 1 incorporating note regarding momentary display of APU BATTERY OFF message. Revision to Volume 1 incorporating minor corrections to normal power-up check. Revision to Volume 1 incorporating data regarding fuel burn with ADG deployed. Revision to Volume 1 incorporating correction of APU start cycles in supplementary procedures.		

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Revision No.	Subject	Incorporated By	Date
REV 10 Nov 07/08	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 10: Limitations and Emergency, Normal and Abnormal Procedures, and Performance. Revision to Volume 1 incorporating temporary revision 605/1-4. Revision to Volume 1 to revise the deicing fluid holdover charts. Revision to Volume 1 to add information regarding infrared deicing. Revision to Volume 1 to clarify the in-flight check of the ADG. Revision to Volumes 1 and 2 to clarify information regarding the aircraft clock. Revision to Volume 2 introducing new FCC with revised FD take-off pitch. Revision to Volume 2 introducing changes to reflect FD software update. Revision to Volume 2 to correct the maximum usable fuel capacities. Revision to Volume 2 to clarify logic of HYD SOV caution message. Revision to Volume 2 introducing additional information and miscellaneous editorial changes.		
REV 11 Feb 17/09	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 11: Limitations, Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to add information regarding landing. Revision to Volume 1 to clarify bus inoperative list. Revision to Volume 1 to clarify runway contaminants. Revision to Volume 1 to clarify hydraulic pressure during ADG check. Revision to Volume 2 to introduce hard landing indication system. Revision to Volume 2 to introduce new CB panel for aircraft 5775 and subs. Revision to Volume 2 to clarify oxygen duration tables. Revision to Volume 2 to remove shock strut limits.		

BOMBARDIER CHALLENGER 605	Record of Revisions	Vol. 1	00-01-4
		REV 13, Oct 20/09	

Revision No.	Subject	Incorporated By	Date
REV 12 Jul 17/09	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 12: Limitations and Emergency and Abnormal Procedures. Revision to Volume 1 to clarify tire check during walkaround. Revision to Volume 1 to add information regarding APR failure. Revision to Volume 1 to remove instructions regarding flap setting during windshear. Revision to Volume 2 to revise HLIS operation information. Revision to Volume 2 to correct elevator position indicator description. Revision to Volume 2 to clarify TAWS mode 4B. Revision to Volume 2 to revise TCAS description and operation. Revision to Volume 2 to incorporate Temporary Revision 605/4. Revision to Volume 2 for editorial corrections.		
REV 13 Oct 20/09	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 13: Limitations, Normal and Abnormal Procedures. Revision to Volume 1 to correct minimum fuel requirement for go-around. Revision to Volume 1 to revise hard landing definition. Revision to Volume 1 to remove ATC tuning restriction.		

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Revision No.	Subject	Incorporated By	Date
REV 14 Feb 01/10	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 14: Limitations, Emergency, Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to add bounced landing procedure. Revision to Volume 1 to revise procedure for flap reset. Revision to Volume 1 to revise the deicing fluid holdover charts. Revision to Volume 1 to clarify ADG in-flight operational check. Revision to Volume 2 to clarify take-off pitch up command bar. Revision to Volume 2 to remove ATS go-around. Revision to Volume 2 to revise LDS range. Revision to Volume 2 to make minor corrections to numerous illustrations.		
REV 15 Jun 28/10	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 15: Limitations, Emergency, Normal, and Abnormal Procedures. Revision to Volume 1 to incorporate temporary revision 605/9. Revision to Volume 1 to clarify ADG in-flight check. Revision to Volume 2 to add circuit breaker panel 6. Revision to Volume 2 to correct autopilot EICAS messages. Revision to Volume 2 to make minor editorial and illustration corrections.		
REV 16 Sep 09/10	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 16: Emergency, Normal, and Abnormal Procedures. Revision to Volume 1 to incorporate temporary revision 605/8.		

Revision No.	Subject	Incorporated By	Date
REV 17 Dec 10/10	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 17: Emergency and Supplementary Procedures. Revision to Volume 1 to add a caution avoid false captures during precision approach. Revision to Volume 1 to clarify hard landing definition. Revision to Volume 1 to revise the deicing fluid holdover charts. Revision to Volume 1 to make minor editorial corrections. Revision to Volume 2 to correct the identification of the LDS antenna. Revision to Volume 2 to update CDU TUNE page description. Revision to Volume 2 to add datalink EICAS messages. Revision to Volume 2 to correct CB panel listing. Revision to Volume 2 to add alignment counter to ISI description.		
REV 18 Mar 16/11	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 18: Limitations and Emergency, Normal, Abnormal and Supplementary Procedures. Revision to Volume 2 to revise ISI description.		
REV 19 Jun 30/11	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 19: Emergency and Abnormal Procedures. Revision to Volume 1 to clarify anti-icing during descent. Revision to Volume 2 to clarify aft equipment bay door description. Revision to Volume 2 to include 406 MHz emergency frequency in ELT description.		
REV 20 Sep 20/11	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 20: Abnormal Procedures. Revision to Volume 1 to make miscellaneous editorial corrections. Revision to Volume 1 to revise the deicing fluid holdover charts. Revision to Volume 2 to revise bleed air leak detection illustration.		

Revision No.	Subject	Incorporated By	Date
REV 21 Jan 18/12	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 21: Emergency, Abnormal and Supplementary Procedures. Revision to Volume 1 to add power plant notes to after engine start check. Revision to Volumes 1 and 2 to introduce additional power supply for CVR. Revision to Volume 2 to introduce TCAS change 7.1 Revision to Volume 2 to clarify bleed air leak detection illustration.		
REV 22 Mar 16/12	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 22: Emergency, Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to add information regarding accumulator checks during external safety inspection.		
REV 23 Jun 26/12	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 23: Limitations, Emergency and Abnormal Procedures. Revision to Volume 2 to revise the oxygen system description. Revision to Volume 2 to add information regarding refuel shut-off valve operation.		
REV 24 Jan 11/13	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 24: Emergency, Normal and Abnormal Procedures. Revision to Volume 1 to clarify APU starting and shutdown procedures. Revision to Volume 1 to update the cold weather operations. Revision to Volume 1 to incorporate temporary revision 605/13. Revision to Volume 1 to update the hot weather operation. Revision to Volume 1 to incorporate temporary revision 605/15. Revision to Volume 2 to clarify APU shutdown procedure. Revision to Volume 2 to clarify the typical flight pressurization profile.		

BOMBARDIER CHALLENGER 605	Record of Revisions	Vol. 1	00-01-8
		REV 26, Sep 25/13	

Revision No.	Subject	Incorporated By	Date
REV 25 Jul 08/13	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 25: Introduction, Limitations, Emergency, Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to add a caution concerning interior materials, finishing and products. Revision to Volume 1 and 2 to add information regarding oil consumption. Revision to Volume 1 to add information regarding water system purging. Revision to Volume 2 to add optional LDS antenna location. Revision to Volume 2 to add landing eye-to-wheel height. Revision to Volume 2 to make minor editorial corrections to the air conditioning and pressurization descriptions. Revision to Volume 2 to add RDC FAIL status message. Revision to Volume 2 to correct ATS operation during AFCS transfer.		
REV 26 Sep 25/13	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 26: Emergency Procedures. Revision to Volume 2 to clarify waste container fire extinguisher option. Revision to Volume 2 to update power plant EICAS messages.		

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Revision No.	Subject	Incorporated By	Date
REV 27 Jan 06/14	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 27: Emergency, Normal and Abnormal Procedures and In-Flight Checks. Revision to Volume 1 to remove chip detector check during external safety inspection. Revision to Volume 1 to add information regarding push-back during the start check. Revision to Volume 1 and 2 to add information regarding fuel vent relief valves. Revision to Volume 1 to update the cold weather operations. Revision to Volume 2 to clarify door location and dimensions. Revision to Volume 2 to remove STC installed toggle switch on air conditioning panel. Revision to Volume 2 to introduce a 5 channel SELCAL decoder.		
REV 28 Mar 31/14	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 28: Limitations and Abnormal Procedures. Revision to Volume 1 to add information regarding transponder selection during after landing check. Revision to Volume 1 to clarify bus inoperative list. Revision to Volume 1 to clarify corrective actions for operation in volcanic ash/dust. Revision to Volume 1 to clarify AC essential bus verification during ADG in-flight check. Revision to Volume 2 to clarify APR operation.		

Revision No.	Subject	Incorporated By	Date
REV 29 Jul 11/14	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 29: Normal Procedures. Revision to Volume 1 to update external safety inspection and external walkaround. Revision to Volume 1 to add information to maintain N₂ speed during descent in icing conditions. Revision to Volume 1 to add information to APU battery failure procedure. Revision to Volume 1 to delete duplicate battery/external air start procedure in supplementary procedures. Revision to Volume 2 to remove reference to AFIS. Revision to Volume 2 to reinstate STC installed toggle switch on air conditioning panel. Revision to Volume 2 to provide additional information regarding 14th stage isolation valve.		
REV 30 Oct 01/14	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 30: Limitations and Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to delete requirement to check engine oil filter bypass indicators during external safety inspection. Revision to Volume 1 to clarify cowl anti-ice check after engine start. Revision to Volume 1 to update the cold weather operations. Revision to Volume 2 to add CVR FAIL status message.		
REV 31 Jan 21/15	Revision to Volume 1 to update the cold weather operations. Revision to Volume 1 to clarify speed requirement for ADG in-flight check. Revision to Volumes 1 and 2 to make minor editorial and illustration corrections.		

Revision No.	Subject	Incorporated By	Date
REV 32 Mar 18/15	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 32: Abnormal Procedures. Revision to Volume 1 to remove nose door safety pin from external safety inspection. Revision to Volume 1 to clarify drain checks during external walkaround. Revision to Volume 1 to revise note regarding nuisance windshear fail message during take-off check. Revision to Volume 1 to permit landing gear retraction after ADG in-flight check. Revision to Volume 2 to clarify note regarding APU bleed air load. Revision to Volume 2 to amend circuit breaker summaries for communications and navigation systems.		
REV 33 Jun 11/15	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 33: Abnormal Procedures. Revision to Volume 2 to make minor editorial corrections. Revision to Volume 2 to clarify autopilot engagement criteria. Revision to Volume 2 to clarify yaw damper disengagement. Revision to Volume 2 to add information regarding failed or intermittent radio altimeter. Revision to Volume 2 to clarify automatic transition between IAS and Mach. Revision to Volume 2 to correct AFCS and FD EICAS messages. Revision to Volume 2 to update datalink messages. Revision to Volume 2 to revise radio altimeter circuit breaker summary.		
REV 34 Sep 03/15	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 34: Emergency, and Abnormal Procedures. Revision to Volume 1 to clarify AC essential bus verification during ADG in-flight check. Revision to Volume 2 to make minor editorial corrections.		

BOMBARDIER CHALLENGER 605	Record of Revisions	Vol. 1	00-01-12
		REV 41, Nov 28/16	

Revision No.	Subject	Incorporated By	Date
REV 35 Oct 19/15	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 35: Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to update the cold weather operations.		
REV 36 Nov 27/15	Revision to Volumes 1 and 2 introducing changes to reflect Airplane Flight Manual revision 36: Introduction, Limitations, Emergency, Normal, and Abnormal, and Supplementary Procedures.		
REV 37 Dec 21/15	Revision to Volume 2 introducing changes to reflect release of Service Bulletins for avionic update.		
REV 38 Mar 11/16	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 38: Emergency, Abnormal and Supplementary Procedures. Revision to Volumes 1 and 2 to make minor editorial corrections. Revision to Volume 2 to introduce supplemental wing anti-icing system.		
REV 39 Jun 09/16	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 39: Limitations, Normal, Abnormal and Supplementary Procedures. Revision to Volume 2 to amend the description of the audio EMER selection. Revision to Volume 2 to introduce relocation of hydraulic No. 3 accumulator.		
REV 40 Aug 30/16	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 40: Limitations and Emergency, Normal, Abnormal and Supplementary Procedures. Revision to Volume 1 to clarify windshear information. Revision to Volume 2 to clarify starter engagement limits.		
REV 41 Nov 28/16	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 41: Limitations, Emergency, Normal and Abnormal Procedures. Revision to Volume 1 to update the cold weather operations. Revision to Volume 2 to correct wing anti-ice advisory message.		

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Revision No.	Subject	Incorporated By	Date
REV 42 Feb 23/17	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 42: Normal, Abnormal and Supplementary Procedures and Special Operations. Revision to Volume 2 to revise uninhibited caution message list. Revision to Volume 2 to clarify TCAS aural alerts.		
REV 43 Mar 13/17	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 43: Supplementary Procedures.		
REV 44 Jun 16/17	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 44: Limitations, Emergency, Normal, Abnormal and Supplementary Procedures. Revision to Volume 2 to revise 3 crew single bottle oxygen table at 10,000 feet.		
REV 45 Dec 04/17	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 45: Limitations, Emergency, Normal and Supplementary Procedures. Revision to Volume 1 to correct hyperlinks.		
REV 46 Mar 01/18	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 46: Limitations, Emergency, Abnormal and Supplementary Procedures. Revision to Volume 1 to update the cold weather operations. Revision to Volume 2 to clarify oxygen system description.		
REV 47 May 29/18	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 47: Limitations. Revision to Volume 1 to reverse anti-icing steps to prevent ATS disconnect.		
REV 48 Aug 23/18	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 48: Limitations, Abnormal and Supplementary Procedures. Revision to Volume 1 to clarify information regarding generator switching.		

Revision No.	Subject	Incorporated By	Date
REV 49 Nov 19/18	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 49: Supplementary Procedures and Special Operations. Revision to Volume 1 to provide additional information regarding temperature when selecting an alternate destination. Revision to Volume 1 to introduce a procedure for suspected external damage. Revision to Volume 1 to introduce information regarding single engine taxi operations. Revision to Volume 1 to update the cold weather operations. Revision to Volume 1 to make minor editorial corrections to Supplementary Procedures and Special Operations.		
REV 50 Mar 25/19	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 50: Limitations and Special Operations. Revision to Volume 1 to revise ADG in-flight check to reflect dry ADG and to address possible dormant failure of stabilizer trim channel 2. Revision to Volume 2 to update auto pressurization schedule. Revision to Volume 2 to introduce metric ISI. Revision to Volume 2 to provide description of transfer ejector float valve. Revision to Volume 2 to update operation of FSCU automatic transfer.		
REV 51 Apr 05/19	Updates the title page to harmonize the revision number.		
REV 52 Jun 20/19	Revision to Volume 2 to provide aerodrome categories for rescue and fire fighting. Revision to Volume 2 to clarify the refueling/defueling operation.		
REV 53 Sep 17/19	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 53: Limitations. Revision to Volume 1 to update the cold weather operations.		

BOMBARDIER CHALLENGER 605	Record of Revisions	Vol. 1	00-01-15
		REV 62, Nov 24/21	

Revision No.	Subject	Incorporated By	Date
REV 54 Dec 18/19	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 54: Limitations, Emergency, Abnormal and Supplementary Procedures.		
REV 55 Mar 18/20	Revision to Volume 1 to update the cold weather operations.		
REV 56 Jun 16/20	Updates the title page to harmonize the revision number.		
REV 57 Sep 11/20	Revision to Volume 1 introducing Operational Landing Distance (OLD) to the Performance chapter. Revision to Volume 2 to harmonize the revision number.		
REV 58 Dec 8/20	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 58: Limitations, and Abnormal Procedures. Revision to Volume 1 to include editorial corrections to Operational Landing Distance (OLD) in the Performance chapter. Revision to Volume 1 to incorporate TR 605/34 (Cold Weather Operations). Revision to Volume 2 to incorporate UC20-05-12 to improve details of windshield and side window cracking.		
REV 59 Jun 2/20	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 59: Emergency Procedures. Revision to Volume 1 to add Nosewheel Steering Inoperative guidance in the new Chapter 10 — ROPAT. Revision to Volume 2 to harmonize the revision number.		
REV 60 Jul 8/21	Revision to Volume 1 to harmonize the revision number. Revision to Volume 2 to harmonize the revision number.		
REV 61 Aug 27/21	Revision to Volume 1 to incorporate TR 605/35. Revision to Volume 2 to harmonize the revision number.		
REV 62 Nov 24/21	Revision to Volume 1 to update the cold weather operations. Revision to Volume 2 to harmonize the revision number.		

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BOMBARDIER CHALLENGER 605	Record of Revisions	Vol. 1	00-01-16
		REV 63, Mar 14/22	

Revision No.	Subject	Incorporated By	Date
REV 63 Mar 14/22	Revision to Volume 1 introducing changes to reflect Airplane Flight Manual revision 63: Abnormal Procedures. Revision to Volume 1 to add chapter 11. Revision to Volume 1 to add battery disconnect procedure in chapter 11. Revision to Volume 1 to add a sub-step and a note under step 7 at page 04-04-30 (UC20-11-01). Revision to Volume 1 to modify the illustration (Figure 04-04-1) for the Hyd accumulator gauge location at page 04-04-2 (UC19-08-07). Revision to Volume 1 to add steps and effectivity boxes related to the Hyd accumulator gauge check at page 04-04-4 and 04-04-6 (UC18-03-30). Revision to Volume 1 to add substeps for LEFT UPPER FUSELAGE (FORWARD) steps (4), (7) and RIGHT UPPER FUSELAGE (Forward) step (4) (UC18-02-27). Revision to Volume 1 to modify the illustration (Figure 04-04-1) at page 04-04-2 and to add check items (engine oil filter and engine chip detector) in the AFT EQUIPMENT BAY area at page 04-04-6 (UC20-12-07). Revision to Volume 1 to add a note at the cargo bay section of page 04-04-18 (UC19-11-09). Revision to Volume 1 to the cargo bay section, to change to word “as required” to “closed and latched” at page 04-04-18 (UC19-11-09). Revision to Volume 2 to change the word lock to latch at page 01-10-11 (UC19-11-09). Revision to Volume 2 to change the word RPA to RNP at page 17-10-179 (UC20-03-18). Revision to Volume 2 to change word HIGH to HI in the EICAS MESSAGES at page 02-10-26 (UC21-01-02). Revision to Volume 2 to update figure 16-10-12 at page 16-10-14 (UC17-08-21). Revision to Volume 2 to add instruction on how to use the oxygen quantity tables at page 08-10-8 and 08-10-13 (UC18-03-01).		

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Revision No.	Subject	Incorporated By	Date
REV 64 Jun 02/22	Volume 1 revised to: - Add the No.3 hyd accumulator gauge at Left rear fuselage section in Airplane preparation section (UC22-10-01). Volume 2 revised to: - Update figure 17-10-31 and figure 17-10-32 (UC21-09-26) - Update speed bug information related to the width of the speed bug indicator in the FLIGHTS INSTRUMENT chapter under Air Data System Indication section (UC21-05-17). - Update software attributes in Chapter 18 Pneumatic System Summary, the content, however, remains the same.		
REV 65 Jun 24/22	Volume 1, not issued Volume 2, not issued		
REV 66 Aug 29/22	Volume 1 revised to: - issued to keep in sync with Airplane Flight Manual Revision 66 - correct typographical errors in the procedure and add a note under section 3 of Ch.11 – SERVICING – FCOM1 11-02. Volume 2 revised to: - issued to keep in sync with FCOM 1 Revision 66.		
REV 67 Nov 22/22	Volume 1 revised to: - Revision to Volume 1 to update ROTR. Volume 2 revised to: - issued to update section 4 – AUTOMATIC FLIGHT CONTROL SYSTEM – AUTOTHROTTLE SYSTEM Chapter – Descent subsection		

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		REV 64, Jun 02/22	

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Record the date you insert each revision in your manual.

TR No.	Subject	Status	Date
605/1	Temporary revision to advise the flight crew of a clarification of the take-off procedures.	Superseded by TR 605/1-1	
605/1-1	Temporary revision to advise the flight crew of a clarification of the take-off procedures.	Superseded by TR 605/1-2	
605/1-2	Temporary revision to advise the flight crew of a clarification of the take-off procedures.	Superseded by TR 605/1-3	
605/1-3	Temporary revision to advise the flight crew of a clarification of the take-off procedures.	Superseded by TR 605/1-4	
605/1-4	Temporary revision to advise the flight crew of a clarification of the take-off procedures.	Incorporated by REV 10	
605/2	Temporary revision to advise the flight crew of additional operating limitations related to operation safety enhancement.	Superseded by TR 605/2-1	
605/2-1	Temporary revision to advise the flight crew of additional operating limitations related to operation safety enhancement.	Superseded by TR 605/2-2	
605/2-2	Temporary revision to advise the flight crew of additional operating limitations related to operation safety enhancement.	Incorporated by REV 13	
605/3	Temporary revision to advise the flight crew of mandatory deicing at 5°C (41°F) or below on AR certified airplanes.	Incorporated by REV 13	
605/4	Temporary revision to advise the flight crew of a clarification to the FMS power up mode.	Incorporated by REV 12	
605/5	Temporary revision to advise the flight crew of the introduction of a WAAS capable GPS and related changes.	Incorporated by REV 14	
605/6	Temporary revision to advise the flight crew of the introduction of procedures for enhanced vision system (EVS).	Incorporated by REV 14	
605/7	Temporary revision to advise the flight crew of the introduction of ADS-B capable Air Traffic Control transponder.	Superseded by TR 605/7-1	

TR No.	Subject	Status	Date
605/7-1	Temporary revision to advise the flight crew of the introduction of ADS-B capable Air Traffic Control transponder.	Incorporated by REV 18	
605/8	Temporary revision to advise the flight crew of an additional check to ensure that the aircraft is on the correct runway.	Incorporated by REV 16	
605/9	Temporary revision to advise the flight crew of a revision to the external walkaround to include a check of the wing leading edge sealant.	Incorporated by REV 15	
605/10	Temporary revision to advise the flight crew of the incorporation of WAAS capable GPS into production.	Superseded by TR 605/10-1	
605/10-1	Temporary revision to advise the flight crew of the incorporation of WAAS capable GPS into production.	Incorporated by REV 21	
605/11	Temporary revision to advise the flight crew of additional information regarding nose wheel steering failure.	Incorporated by REV 18	
605/12	Temporary revision to advise the flight crew of an additional procedure for dual GPS failure during oceanic and remote operations.	Incorporated by REV 22	
605/13	Temporary revision to advise the flight crew of additional information regarding engine starting during cold weather operations.	Incorporated by REV 24	
605/14	Temporary revision to advise the flight crew of the addition of an altimeter cross-check during the take-off check.	Incorporated by REV 24	
605/15	Temporary revision to advise the flight crew of additional information regarding hot weather operation.	Incorporated by REV 24	
605/16	Temporary revision to advise the flight crew of the introduction of Data Link 2000.	Incorporated by REV 25	
605/17	Temporary revision to advise the flight crew of the addition of a procedure for uncommanded yaw motion.	Incorporated by REV 25	
605/18	Temporary revision to advise the flight crew of the removal of the requirement for the daily check of the oil chip detector.	Incorporated by REV 25	
605/19	Temporary revision to advise the flight crew of the addition of instructions to address nuisance AUTO TAIL XFER caution message when AC power is applied during aircraft power up.	Incorporated by REV 27	
605/20	Temporary revision to advise the flight crew of a revision to the stabilizer trim system check.	Incorporated by REV 25	

TR No.	Subject	Status	Date
605/21	Temporary revision to advise the flight crew of the addition of a procedure for total IRS failure and the revision of instructions regarding IRS alignment.	Incorporated by REV 30	
605/22	Temporary revision to advise the flight crew of a revision of the limitations for taxi lights and the addition of limitations and a caution regarding landing lights.	Incorporated by REV 30	
605/23	Temporary revision to advise the flight crew of additional information regarding cold weather operations.	Incorporated by REV 31	
605/24	Temporary revision to advise the flight crew of the introduction of supplemental ground wing anti-icing for airplanes on the Canadian Registry.	Superseded by TR 605/24-1	
605/24-1	Temporary revision to advise the flight crew of the introduction of supplemental ground wing anti-icing.	Superseded by TR 605/24-2	
605/24-2	Temporary revision to advise the flight crew of the introduction of supplemental ground wing anti-icing.	Superseded by TR 605/24-3	
605/24-3	Temporary revision to advise the flight crew of the introduction of supplemental ground wing anti-icing.	Incorporated by REV 41	
605/25	Temporary revision to advise the flight crew of revised information regarding cold weather operations.	Incorporated by REV 41	
605/26	Temporary revision to advise the flight crew of revised information regarding cold weather operations.	Incorporated by REV 46	
605/27	Temporary revision to advise the flight crew of the addition of a note to include reference to semi-synthetic jet fuels.	Incorporated by REV 45	
605/28	Temporary revision to advise the flight crew of a revision of the procedure for loss of all normal electrical power, to address landing with ADG power only.	Incorporated by REV 46	
605/29	Temporary revision to advise the flight crew of a clarification to the tail tank to aux tank automatic transfer failure procedure.	Incorporated by REV 46	
605/30	Temporary revision to advise the flight crew of the addition of FAA datalink operational capabilities.	Incorporated by REV 48	
605/31	Temporary revision to advise the flight crew of a clarification to the ADS-B operational capabilities.	Incorporated by REV 48	
605/32	Temporary revision to advise the flight crew of revised information regarding cold weather operations.	Incorporated by REV 49	

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		REV 67, Nov 22/22	

TR No.	Subject	Status	Date
605/33	Temporary revision to advise the flight crew of addition of FMS and IRS limitations due to outdated magnetic variation data.	Incorporated by REV 54	
605/34	Temporary revision to advise the flight crew of revised information regarding cold weather operations.	Incorporated by REV 58	
605/35	Temporary revision to update the In-Flight Checks chapter.	Incorporated by REV 61	
605/36	Temporary revision to advise the flight crew of revised information regarding cold weather operations.	Active	

TRAINING PURPOSES ONLY

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1. FOREWORD

The CL-600 Model 2B16 Challenger airplane (Serial No. 5701 to 6049) Flight Crew Operating Manual is designed to provide the flight crew with readily accessible operational information. For optimum utilization of the manual, this introduction should be read carefully.

The purpose of the Flight Crew Operating Manual is:

- to provide information regarding operational procedures, performance and limitations
- to standardize terminology and behavioral patterns
- to provide rapid access to reference procedures
- to provide reference material for self-teaching
- to provide information on airplane systems and operations that are controlled and revised.

To accomplish this, the Flight Crew Operating Manual has been divided into two volumes as follows:

Volume 1 – PROCEDURES AND PERFORMANCE

Volume 2 – SYSTEMS DESCRIPTION

Throughout this manual, the experience of the typical Challenger crew has been recognized, and for this reason, basic system principles have been omitted. For example, the text is not intended to teach the crew how to fly an airplane, but to enable an experienced crew to operate the airplane safely and proficiently.

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2. VOLUME 1 – PROCEDURES AND PERFORMANCE

The Procedures and Performance volume is the complete operating procedures document.

It is the responsibility of pilots who are qualified to operate the airplane to be entirely familiar with the information contained in this publication, to ensure that the airplane is operated at all times within the approved flight envelope.

For clarity and simplicity, the manual is written in the imperative, in order that the information and operating instructions may be presented in a positive sense and require no interpretation by the user.

In some procedures, the designations “PF” and “PNF” have been used:

- “PF” means that the associated action is always performed by the pilot presently flying the airplane, whether it is the pilot or the copilot.
- “PNF” means the pilot not flying the airplane.

Specific items requiring emphasis are expanded upon and ranked in increasing order of importance, in the form of a NOTE, CAUTION or WARNING.

NOTE

Expands on information which is considered essential to emphasize.
Information contained in notes may also be safety related.



Provides information that may result in damage to equipment if not followed.



Emphasizes information that may result in personal injury or loss of life if not followed.

The contents and general format of the procedures and performance volume are as follows:

Chapter 1 – Introduction

Chapter 2 – Limitations

Chapter 2 contains the reproduction of the limitations governing operation of the airplane, found within PSP 605-1, the Airplane Flight Manual.

Chapter 3 – Emergency Procedures

Chapter 3 contains all procedures which can be related to foreseeable emergency situations. The procedures are presented in expanded form, describing in detail how and why the emergency procedure steps are accomplished.

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2. VOLUME 1 – PROCEDURES AND PERFORMANCE (CONT'D)

Chapter 4 – Normal Procedures

Chapter 4 contains detailed procedures for conducting a normal flight with all airplane systems operational. Procedures are listed sequentially by phase of flight, starting with exterior safety inspection and extending through post-flight duties at destination.

Line items define the steps to be accomplished during each phase of flight, and are expanded to define the action required to perform the steps.

Chapter 5 – Abnormal Procedures

Chapter 5 contains all procedures which can be related to foreseeable abnormal situations. The procedures are presented in expanded form, describing in detail how and why the abnormal procedure steps are accomplished.

Chapter 6 – Supplementary Procedures

Chapter 6 contains normal procedures which are either not related to a specific phase of flight, or are not performed as part of routine daily procedures.

Chapter 7 – Performance

Chapter 7 contains the instructions to flight crew that certified performance data on the airplane can be found within PSP 605-1, the Airplane Flight Manual.

Chapter 8 – Special Operations

Chapter 8 contains special operating procedures which are not routinely performed on each flight, and are accomplished only on an “as required” basis to cope with minor system irregularities or abnormal flight conditions.

Chapter 9 – In-Flight Checks

Chapter 9 contains operating procedures not routinely performed every flight, and are accomplished only when certain airplane systems are to be tested in flight.

3. VOLUME 2 – SYSTEMS DESCRIPTION

Volume 2 contains descriptive airplane systems information. This volume is divided into 19 chapters, listed alphabetically by systems. Refer to Volume 2 – Foreword.

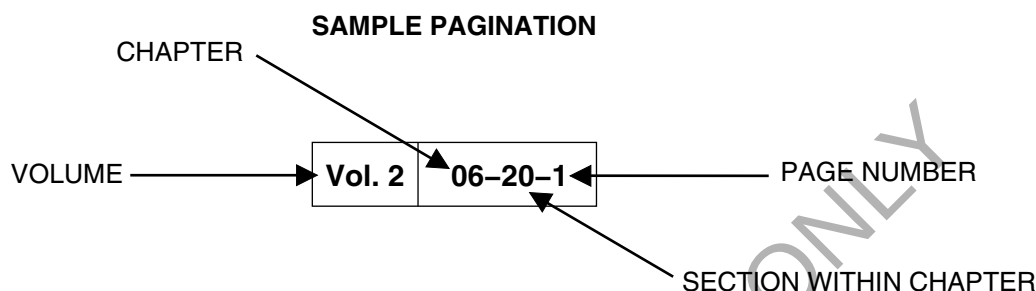
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4. PAGINATION

Each chapter of Volumes 1 and 2 of the Flight Crew Operating Manual is subdivided into sections categorized by the subject or type of material presented.

This permits issuance of small blocks of revision pages without re-numbering and reprinting complete sections of the manual.

Volumes 1 and 2 are paginated with a Volume Chapter/Section/Page numbering system as follows:



Additional identification data carried in the margin of the page are the date of issue or revision date, printed below the page number, and the chapter, title and subject carried in the masthead.

5. REVISION SYSTEM

Revisions to this manual will be issued when necessary, and are numbered consecutively. Each revision should be inserted immediately and entered in the Record of Revisions sheet.

Alterations to former pages will be identified by a strong vertical line, except when a whole chapter or section is revised. The reason for revision is then given under revision highlights only.

Beside the normal white pages, yellow pages are used to identify the Temporary Revisions. The Temporary Revisions are numbered consecutively. Each Temporary Revision should be inserted immediately and entered in the Record of Temporary Revisions sheet.

6. REGISTRY CODES

The following registry codes are utilized throughout the Flight Crew Operating Manual to identify data that is only applicable to a particular certification authority:

AR	Aviation Registry (Russian)
CAAI	Civil Aviation Authority of Israel
EASA	European Aviation Safety Agency
FAA	Federal Aviation Administration (US)
SAAU	State Aviation Administration of Ukraine

7. QUICK REFERENCE HANDBOOK

The Quick Reference Handbook (QRH) presents the Normal Checklist, Quick Reference Performance Data and combined Emergency/Abnormal Procedures. The QRH is a small booklet, bound so that it will lie flat when opened to any page.

The QRH contains Emergency and Abnormal Procedures in a system and EICAS message format that is readily accessible in the flight compartment. Line items and layout are identical to those used in the expanded version, however, procedures are abbreviated by deleting explanatory material from the expanded procedures.

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8. CIRCUIT BREAKER RESET

The following paragraphs summarize Bombardier Aerospace philosophy and policies with respect to resetting of tripped circuit breakers.

A. General

There is a latent danger in resetting a circuit breaker tripped by an unknown cause because the tripped condition is a signal that something may be wrong in the related electrical circuit.

Until it is positively determined what has caused a circuit breaker trip to occur, flight crews, maintenance personnel, or airplane ground servicing personnel usually have no way of knowing the consequences of resetting a tripped circuit breaker.

Resetting a circuit breaker tripped by an unknown cause should normally be a maintenance function conducted on the ground.

B. In-Flight

A circuit breaker must not be reset or cycled (i.e., opened or closed) unless doing so is consistent with explicit procedures specified in the Flight Crew Operating Manual or the Quick Reference Handbook or unless in the judgement of the Pilot-in-Command, the resetting or cycling of the circuit breaker is necessary for the safe completion of the flight.

A Logbook write-up is required for an in-flight circuit breaker reset/cycle.

C. On Ground

Bombardier Aerospace finds it acceptable for operators to attempt to clear the faults by performing system resets (power cycling on the ground) on an occasional basis.

If unable to clear the fault, maintenance action should be carried out. The MMEL must be used (if applicable) and/or the affected system condition must be satisfactory for airplane dispatch.

9. OPERATIONAL CAUTION TO CUSTOMERS



Please be advised that your interior is hand crafted using the highest quality materials, finishing and products available in the industry. They are fragile and exotic. In order to protect them and ensure their continued aesthetic qualities, they must not be subjected to extreme or rapid changes in temperature or humidity. If the Owner/Operator does not comply with this caution, the warranty coverage for a given component may be affected.

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1. GENERAL

The limitations included in this chapter contain items peculiar to the CL600, model 2B16 Challenger airplane (Serial No. 5701 to 6049).

Observance of these limitations is mandatory.

2. KINDS OF AIRPLANE OPERATION

The airplane is certified for operations in the following conditions, when the equipment and instruments required by the airworthiness and operating regulations are approved, installed and in an operable condition:

- Day and night
- VFR and IFR
- Flight in icing conditions

The airplane is certified for ditching when the safety equipment, specified by the applicable regulations, is installed.

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1. STRUCTURAL WEIGHT LIMITATION

Maximum taxi and ramp weight:	21,909 kg (48,300 lb)
Maximum take-off weight:	21,863 kg (48,200 lb)
Maximum landing weight:	17,237 kg (38,000 lb)
Maximum zero fuel weight:	14,515 kg (32,000 lb)
Minimum flight weight:	11,794 kg (26,000 lb)
Minimum operating empty weight:	10,047 kg (22,150 lb)

NOTE

1. The maximum take-off weight (MTOW) and/or maximum landing weight (MLW) may be further limited due to performance considerations.
2. The maximum landing weight (MLW) is further limited when landing at airport elevations above 10,000 feet (Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – OPERATIONS AT HIGH AIRPORT ELEVATIONS) .

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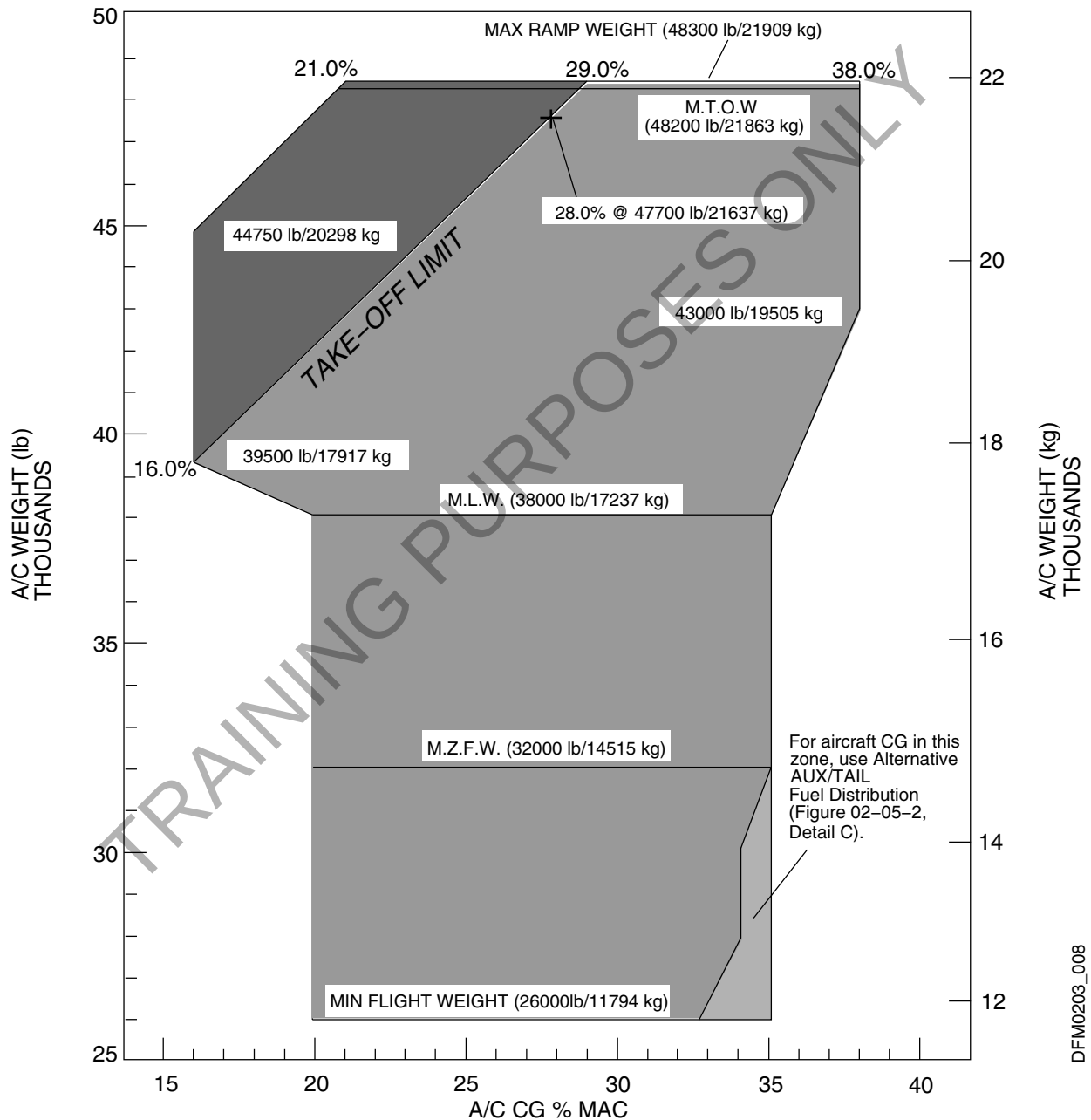
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1. CENTRE OF GRAVITY LIMITS (MTOW 48,200 LB)

The maximum permissible centre of gravity (CG) limits are as shown in Figure 02-03-1.

The airplane including interior payload, passengers and fuel (refer to Figure 02-05-2) must be loaded such that the airplane weight and centre of gravity are maintained within the specified limits (including variations due to fuel consumption, passenger movement, retraction of landing gear, etc.).

The stabilizer trim setting must be in accordance with Chapter 4; NORMAL PROCEDURES – TAXIING AND TAKE-OFF – Taxi Check.



Centre of Gravity Limits (MTOW 48,200 lb)
Figure 02-03-1

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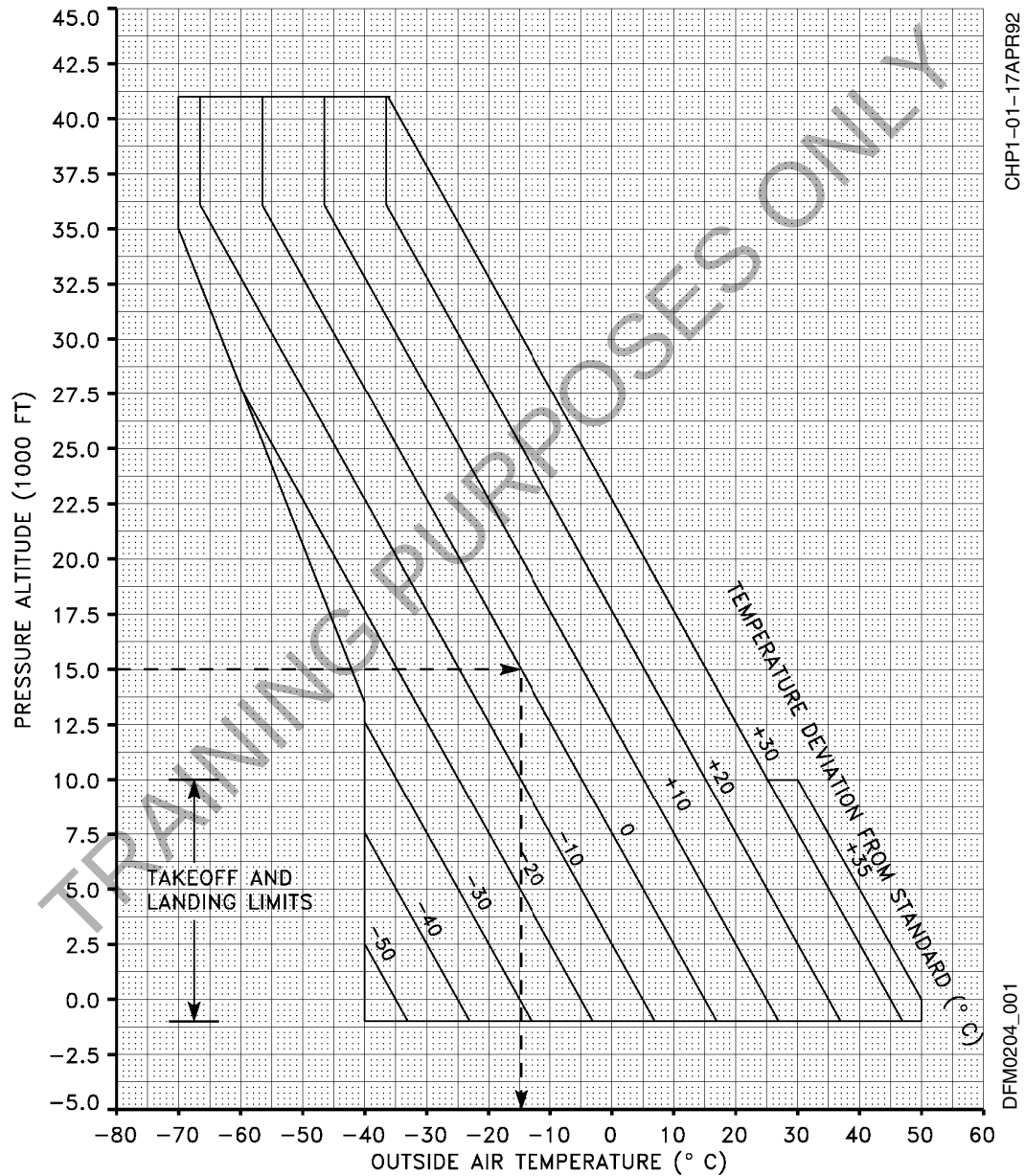
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1. ALTITUDE AND TEMPERATURE OPERATING LIMIT

The altitude and temperature operating limit is as shown in Figure 02-04-1.

- Maximum airport pressure altitude for take-off and landing is 10,000 feet.
- Maximum operating altitude is 41,000 feet.
- Maximum ambient air temperature approved for take-off and landing is ISA + 35°C.
- Minimum ambient temperature approved for take-off is -40°C (-40°F).
- At altitudes above 25,000 feet, a safety harness must be worn by at least one pilot.



Altitude and Temperature Operating Limits
Figure 02-04-1

2. TAKE-OFF

A. Rotation Rate and Pitch Attitude

WARNING

Excessive rotation rates (exceeding 3 degrees per second) or over-rotations may lead to high pitch attitudes and angles of attack being attained while the aircraft is near the ground. This can reduce stall margins significantly resulting in stick shaker/pusher activation and potentially loss of control. Pilots must rotate smoothly towards the target pitch attitude then transition to speed control.

B. Pitch Trim

WARNING

Failure to set the pitch trim appropriate to the computed centre of gravity may result in excessive rotation rate at take-off.

Pitch trim must be set according to the airplane's computed centre of gravity.

C. Take-Off Pitch Target

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

The initial target for rotation is 10 degrees.

If Flight Director is used for take-off, set pitch target to 10 degrees. Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – AUTOMATIC FLIGHT CONTROL SYSTEM – TAKE-OFF.

Take-off performance data contained in Chapter 6 – Performance of the Airplane Flight Manual remains applicable.

To obtain pitch mode, select TO as normal, then rotate the Flight Control Panel (FCP) VS pitch wheel to reduce the FD command to 10 degrees.

NOTE

The VS pitch wheel will adjust the FD command in 0.50 degree steps.

3. COLD WEATHER OPERATIONS

WARNING

Even small amounts of frost, ice, snow or slush on the wing leading edges and forward upper wing surface may adversely change the stall speeds, stall characteristics and the protection provided by the stall protection system, which may result in loss of control on take-off.

Take-off is prohibited with frost, ice, snow or slush adhering to any critical surface (wings, horizontal stabilizer, vertical stabilizer, control surfaces and engine inlets) except that take-off is permitted with frost adhering to:

- the upper surface of the fuselage; and/or
- the underside of the wing, that is caused by cold soaked fuel,

in accordance with the instructions provided in Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION – PRE-FLIGHT PREPARATION – External Safety Inspection.

Although the upper fuselage is not defined as a critical surface, it must nonetheless be deiced to remove contamination, other than allowable frost, anytime the wing and tail surfaces require deicing.

NOTE

Comprehensive procedures for operating in cold weather are provided in Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION.

In addition to a visual check, a tactile check of the wing leading edge, wing forward upper surface and wing rear upper surface is required during the External Walkaround inspection to determine that the wing is free from frost, ice, snow or slush when:

- the outside air temperature (OAT) is 5°C (41°F) or less, or
- the wing fuel temperature is 0°C (32°F) or less; or
- the atmospheric conditions have been conducive to frost formation.

NOTE

Ice and frost may continue to adhere to wing surfaces for some time even at outside air temperatures above 5°C (41°F).

4. OPERATION IN ICING CONDITIONS

During cold weather operations, the flight crew must ensure that the airplane fuselage, wings and tail surfaces are free from ice, snow or frost.

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WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics or the warning margin provided by the stall protection computer.

A. Cowl Anti-ice System

(1) Ground Operations

The engine cowl anti-ice system must be on when the OAT is 10°C (50°F) or below and visible moisture in any form is present (such as fog with visibility of one mile or less, rain, snow, sleet and ice crystals).

The engine cowl anti-ice system must also be on when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water or slush is present.

Single engine taxi operations are prohibited if the OAT is 10°C (50°F) or below.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

The engine cowl anti-ice system must be on:

At or above 22,000 feet:

- when ice is indicated by the ice detection system, or
- when in icing conditions, if an ice detector has failed.

Below 22,000 feet:

- when in icing conditions, or
- when ice is indicated by the ice detection system.

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4. OPERATION IN ICING CONDITIONS (CONT'D)

B. Wing Anti-ice System

(1) Ground Operations

NOTE

Icing conditions exist on the ground when the OAT is 5°C (41°F) or below and:

1. visible moisture in any form (such as clouds, fog or mist) is present below 400 feet AGL, or
2. the runway is wet or contaminated, or
3. in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

The wing anti-ice system must be selected ON, for final taxi prior to take-off, if the OAT is 5°C (41°F) or below, unless Type II, III or IV anti-icing fluids have been applied.

NOTE

1. If wing anti-ice is selected ON during taxi, **L (R) WING A/ICE** caution messages may be posted or L HEAT or R HEAT lights may go out, but caution messages must be verified out and **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message and L HEAT and R HEAT lights must be verified on, prior to take-off. If wing anti-ice is not required for take-off it should be selected OFF just prior to take-off.
2. To prevent wing contamination from reverse jet blast, operating the thrust reversers during taxi operations on wet and contaminated surfaces should be avoided.

The wing anti-ice system must be selected and confirmed ON for take-off, when the OAT is 5°C (41°F) or below and:

- visible moisture in any form (such as clouds, fog or mist) is present below 400 feet AGL, or
- the runway is wet or contaminated, or
- in the presence of any precipitation (such as rain, snow, sleet or ice crystals).

NOTE

If wing anti-ice is selected ON for take-off, the cowl anti-ice must also be selected ON.

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4. OPERATION IN ICING CONDITIONS (CONT'D)

B. Wing Anti-ice System (Cont'd)

When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off. The wing anti-ice system may be selected OFF only if it can be positively verified that all of the following conditions are met:

- the runway to be used is not wet or contaminated,
- there is no precipitation, and
- there is no visible moisture in any form (such as clouds, fog, mist, rain, snow, sleet or ice crystals) below 400 feet AGL.

Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION – PHASE OF FLIGHT PROCEDURES.

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WARNING

When the OAT is 5°C (41°F) or below use deicing/anti-icing treatment before take-off.

When the OAT is 5°C (41°F) or below the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off.

When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected ON, if required, just prior to thrust increase for take-off.

(2) Flight Operations

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

The wing anti-ice system must be on:

At or above 22,000 feet:

- when ice is indicated by the ice detection system, or
- when in icing conditions, if an ice detector has failed.

Below 22,000 feet:

- when in icing conditions, or
- when ice is indicated by the ice detection system, or

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4. OPERATION IN ICING CONDITIONS (CONT'D)

B. Wing Anti-ice System (Cont'd)

- prior to extending flaps for approach when TAT is 10°C (50°F) or below.

NOTE

The wing anti-ice system may be selected OFF when the WING L HEAT and R HEAT indicators turn green on the anti-ice panel and:

- ICE is not annunciated by the ice detection system, and
- not in icing conditions.

C. Super-Cooled Large Droplet Icing

Continued operation in areas where super-cooled large droplet (SLD) icing conditions exist is prohibited.

SLD icing conditions are indicated by ice accretion on the flight compartment side windows.

- The wing anti-icing system must be ON in SLD icing conditions.
- The cowl anti-icing system must be ON in SLD icing conditions.
- Leave icing conditions when side window icing occurs.

5. ENHANCED TAKE-OFF PROCEDURES AND WINTER OPERATIONS TRAINING

Take-offs conducted where the OAT is 5°C (41°F) or below, are subject to the following training limitations:

- the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off, if the OAT is 5°C (41°F) or below, or
- the Aircraft Commander has successfully completed specific training, within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

Additionally, no take-off shall be conducted by the First Officer where the OAT is 5°C (41°F) or below, unless that pilot has successfully completed the specific training within the preceding 12 calendar months, for take-off procedures, ground icing conditions and cold weather operations.

6. RUNWAY SLOPES

The maximum runway slopes approved for take-off and landing are:

- +2% (uphill)
- 2% (downhill)

7. TAILWIND CONDITIONS

The maximum tailwind component approved for take-off and landing is 10 knots.

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8. APPROACHES

Operations and limitations with a glidepath angle that exceeds 3.5 degrees are described in the following supplementary procedures:

- APPROACHES BELOW 4.5 DEGREES
- STEEP APPROACHES UP TO 5.5 DEGREES
- FAA STEEP APPROACH INTO LONDON CITY AIRPORT

9. MINIMUM FLIGHT CREW

The minimum flight crew is one pilot and one copilot.

10. MAXIMUM OCCUPANTS

The total number of occupants, including no more than nineteen passengers, must not exceed the lesser of the following:

- Twenty-two or,
- The number for which seating accommodation approved for take-off and landing is provided.

11. MAXIMUM CROSSWIND COMPONENT

AR Certified Airplanes

The maximum crosswind component for take-off and landing is 24 knots (12 meters/sec).

SAAU Certified Airplanes

The maximum crosswind component for take-off and landing on a dry runway is 24 knots (12 meters/sec).

The maximum crosswind component for take-off and landing on a wet runway with water depth no more than 3.00 mm (0.125 inch) is defined in the following table for different values of the coefficient of friction:

REPORTED COEFFICIENT OF FRICTION	MAXIMUM CROSSWIND COMPONENT
0.3 (poor braking)	10 knots (5 meters/sec)
0.4 (average braking)	15 knots (8 meters/sec)
0.5 (good braking, equivalent to dry)	24 knots (12 meters/sec)

The maximum crosswind component for take-off and landing on a contaminated runway is 10 knots (5 meters/sec).

Operation on runways with a coefficient of friction less than 0.3 is prohibited.

SAAU Certified Airplanes

12. GROUND OPERATIONS IN HIGH WIND CONDITIONS

In the event that the airplane is parked and sustains winds or gust loads in excess of 27 meters/second, an inspection and functional check of the aileron, elevator and rudder power control units is required.

SAAU Certified Airplanes**13. MINIMUM ENROUTE CLEARANCE**

It is recommended that all terrain and/or obstacles be cleared by at least 1,300 feet along the enroute flight path.

SAAU Certified Airplanes**14. MINIMUM GO-AROUND ALTITUDE**

Demonstrated minimum altitude for all engines go-around without touching the ground is 50 feet and for single engine go-around is 100 feet.

TRAINING PURPOSES ONLY

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TRAINING PURPOSES ONLY

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1. ENGINES

Type: General Electric CF34-3B, quantity two.

2. ENGINE INDICATIONS

The engine limit display markings on EICAS must be used to determine compliance with the maximum/minimum Limits and precautionary ranges. If EICAS markings show more conservative Limits than those specified below, the limit markings on the EICAS must be used.

INDICATION	RED (MAX/MIN LIMITS)	AMBER (CAUTION RANGE)	GREEN (NORMAL RANGE)
N₁ % RPM	98.6	—	0 to 98.5
ITT °C: — APR not operating — APR operating — Maximum Continuous Thrust (MCT)	900 (for first 2 min.)	—	0 to 900
	884 (for next 3 min.)	—	0 to 884
	928 (for first 2 min.)	—	0 to 928
	900 (for next 3 min.)	—	0 to 900
	900	—	0 to 900
N₂ % RPM: — wing anti-ice on — wing anti-ice off			
	99.3	0 to 77.9	78 to 99.2
	99.3	—	0 to 99.2
OIL TEMP °C	163	155 to 162	-40 to 154
OIL PRESS psi	0 to 25	116 to 156	26 to 115

3. ENGINE OPERATING LIMITS

A. Engine Operating Limits Table

CONDITION	CORE RPM N ₂ %	FAN RPM N ₁ %	ITT °C
Start	20		903 (50 seconds)
Acceleration			930 (5 seconds)
Max Continuous	99.2	98.6	899
Normal Take-Off	98.3	96.2	884 (5 minutes)* 900 (2 minutes out of 5 total transient)*
Maximum Take-Off (APR operating)	99.4	98.6	899 (5 minutes)* 928 (2 minutes out of 5 total transient)*
* Transient limits			

NOTE

- The take-off, go-around and maximum continuous thrust N₁ values for the CF34-3B engine are presented on the appropriate engine thrust setting charts contained in Chapter 6; PERFORMANCE – THRUST SETTINGS of the Airplane Flight Manual .
- If above 40,000 feet, one air-conditioning unit or cowl anti-ice must be selected on for each engine.
- Engine to engine N₂ split at ground idle power should not be greater than 2% N₂ .

B. Airplane Cold Soak

Before the first flight of a day, when the airplane is cold-soaked at an ambient temperature of –30°C (–22°F) or below for more than 8 hours, the engines must be motored for 60 seconds, and fan rotation must be verified, before the engine start is initiated.

Thrust reversers must be actuated until the deploy and stow cycles are less than 5 seconds.

C. Oil Temperature

Minimum for starting	–40°C
Maximum Continuous	+155°C
Maximum Permissible	+163°C (15 minutes maximum)

D. Oil Pressure

Steady state idle	25 psi minimum
Take-off power	45 psi minimum
Maximum continuous	115 psi maximum
Maximum transient	
After cold start	156 psi (130 psi at idle, 10 minutes maximum)

NOTE: Engine must remain at idle until oil pressure returns to normal range.

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4. CONTINUOUS ENGINE IGNITION

Continuous engine ignition must be used during the following:

- Take-offs and landings on contaminated runways;
- Take-offs with high crosswind components;
- Flight through moderate or heavier intensity rain;
- Flight through moderate or heavier intensity turbulence;
- Flight in the vicinity of thunderstorms.

5. AUTOTHROTTLE (ATS)

- ATS operation is restricted to two-engine operation only.
- ATS operation is prohibited during category II approach operations.
- Crew must confirm engine power is set to appropriate take-off N_1 limit prior to reaching 80 KIAS during take-off.
- At first indication of stall (stall buffet, stick shaker, or stick pusher), the crew must disengage the ATS and set thrust levers as required.
- ATS operation with APR selected on FMS performance THRUST LIMIT page is prohibited.
- If a **DISENG'D** or **FAIL** message occurs on the ATS MSD, the crew must position the thrust levers as required. Press either thrust lever ATS DISC switch to cancel ATS message.
- Selection of AFCS FLC mode following go-around mode is prohibited, unless the altitude preselector is set higher than current altitude when ATS is engaged.
- The AFM engine limits must not be exceeded when manually entering FMS thrust target (TGT) limit. The ATS will set engine thrust to this target without regard to engine operating limits.
- ATS operation during take-off with thrust target (TGT) selected on FMS performance THRUST LIMIT page is prohibited.
- ATS operation with maximum continuous thrust (MCT) selected as the FMS performance thrust limit is prohibited.
- If both mode status displays (MSDs) fail, the crew must disengage the ATS and consider the system inoperative.
- N_1 thrust limit data is required and must be available from the FMS for ATS operation.
- The radio altimeter must be valid for the ATS retard mode operation. If radio altitude is unavailable or invalid, the ATS must be disengaged prior to reaching 100 feet AGL.
- Use of the ATS during touch and go landings is prohibited. ATS must be disengaged if system engages during touch and go landings.
- Use of the ATS during go-around is prohibited. Pilot must manually set thrust to override ATS. ATS may be used to trim final N_1 , once manually set.
- If flaps are selected to less than 45° for landing, ATS must be disengaged prior to reaching 100 feet AGL.

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6. AUTOMATIC PERFORMANCE RESERVE (APR)

If take-off performance is predicated upon the use of APR, the APR system must be verified operative prior to take-off.

The APR system must be selected off, if an **APR INOP** caution message is displayed on EICAS.

APR OFF Performance must be used if an **APR INOP** caution message is displayed.

7. STARTER CRANKING LIMITS (GROUND AND AIR)

The starter must not be used if indicated N₂ rpm exceeds 55%.

A. Engine Start

Normal engine start – Three consecutive engine start cycles with 5 minutes cooling between additional cycles.

B. Associated Conditions

At initiation of thrust lever movement from SHUT OFF to IDLE:

(1) ITT must be 120°C or less for all ground starts.

(2) ITT must be 90°C or less for all air starts.

C. Dry Motoring Cycle

Dry motoring is performed with ignition OFF and thrust levers at SHUT OFF.

Dry motoring may be used for engine ground starts and engine airstarts.

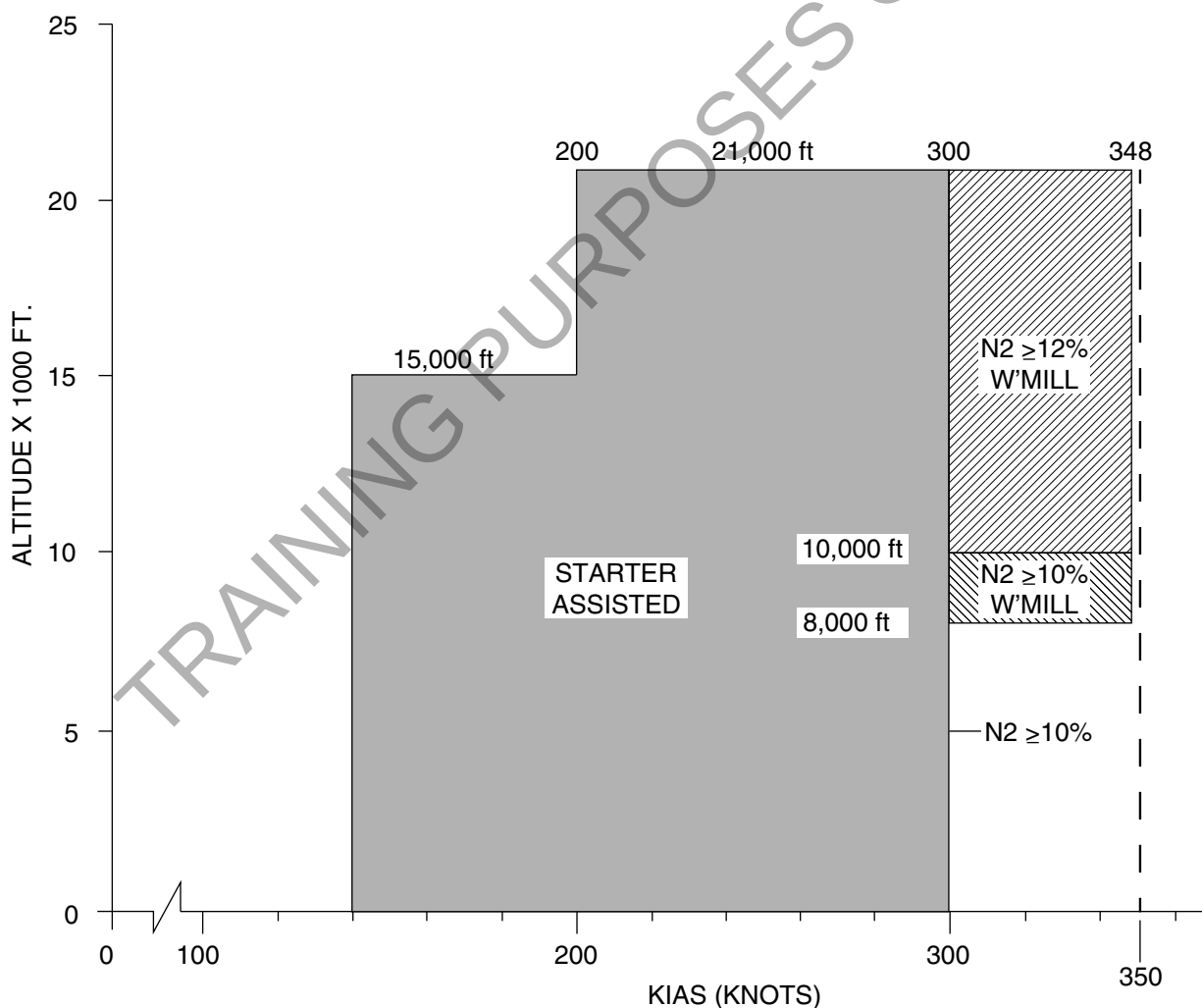
START	MAXIMUM TIME ON	FOLLOWED BY
1	90 seconds	5 minutes off
2 and subsequent	30 seconds	5 minutes off

8. ENGINE RELIGHT

Engine starting in-flight is only permitted within the envelope defined in Figure 02-05-1:

8. ENGINE RELIGHT (CONT'D)

RELIGHT TYPE	ENVELOPE (FIGURE 02-05-1)
Windmilling	Altitude from 21,000 to 10,000 feet: Airspeed from 300 to 348 KIAS, and from 12 to 55% N₂. Altitude from 10,000 to 8,000 feet: Airspeed from 300 to 348 KIAS, and from 10 to 55% N₂. Altitude from sea level to 8,000 feet: Airspeed 300 KIAS, and from 10 to 55% N₂. NOTE N₂ should be stable or increasing.
Starter-assisted Cross Bleed	Altitude from 21,000 feet to 15,000 feet: Airspeed from 200 KIAS up to 300 KIAS, and from 0 to 55% N₂. Altitude from 15,000 feet to sea level: Airspeed from 140 KIAS up to 300 KIAS, and from 0 to 55% N₂.

Engine Air Start Envelope (In Flight)
Figure 02-05-1

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9. FUEL

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 182 kg (400 lb) for take-off and taxi.

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 363 kg (800 lb) in flight.

The maximum permissible fuel imbalance between the contents of the main left tank and the main right tank is 182 kg (400 lb) for landing.

Fuel remaining in a tank when the appropriate fuel quantity indicator reads zero is not usable.

Based upon a fuel density of 6.75 lb/US gallon, the maximum usable fuel load achieved by pressure refueling for each tank is given below:

Left main tank	2,205 kg (4,860 lb)
Right main tank	2,205 kg (4,860 lb)
Auxiliary tank	3,251 kg (7,168 lb)
Tail tank	1,411 kg (3,112 lb)

Total	9,072 kg (20,000 lb)
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To determine approximate maximum usable fuel load achieved by gravity refueling, reduce weight by 7%.

Normal AUX/TAIL Fuel Distribution for Take-Off:

- Auxiliary tank quantity must be full, or at least 2.65 times the tail tank quantity, for take-off (as shown in Figure 02-05-2, Detail A, Normal Fuel Distribution for Take-Off).

OR

Provisional AUX/TAIL Fuel Distribution for Forward CG Limit Take-Off:

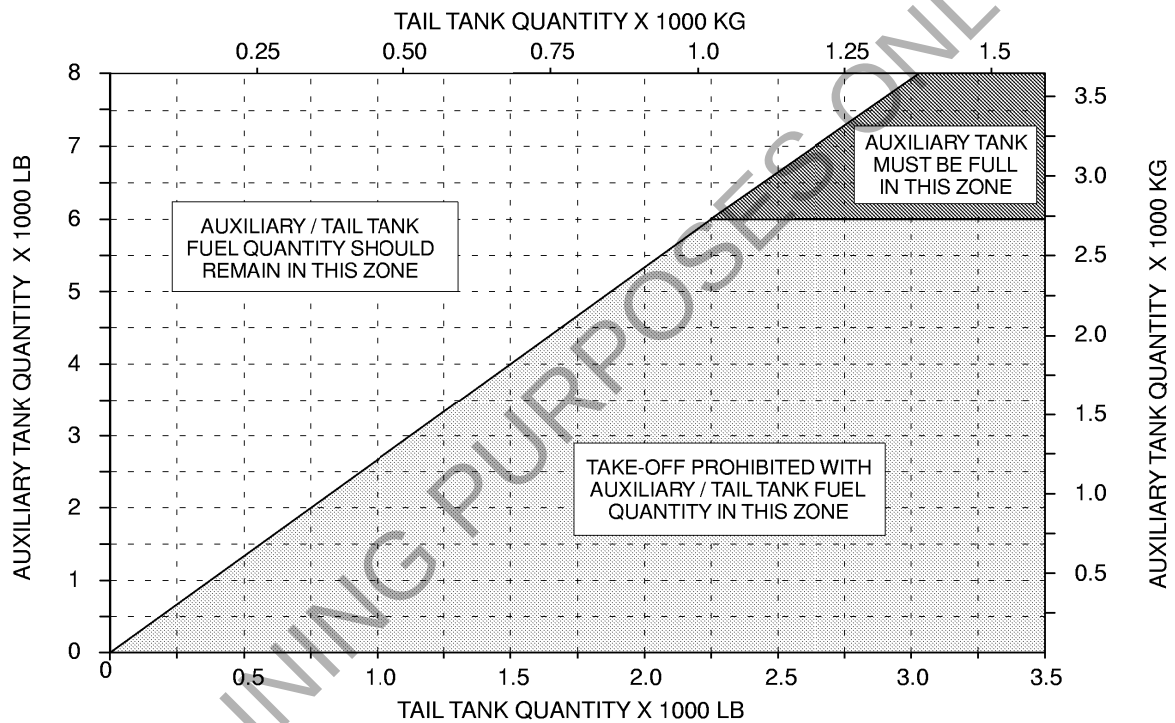
- Where the normal fuel distribution would cause the airplane centre of gravity to fall ahead of the forward limit for take-off, the provisional fuel distribution may be applied to bring the airplane centre of gravity within the permissible forward CG limit (as shown in Figure 02-05-2, Detail B, Provisional Fuel Distribution for Take-Off). This fuel distribution is permitted only if the resulting CG remains ahead of 34% MAC.

OR

Alternative AUX/TAIL Fuel Distribution for Aft CG Limit at ZFW:

- If the aircraft CG at ZFW is within the envelope aft lower zone (CG between 32,000 lb/35%, 30,000 lb/34%, 28,000 lb/34%, 26,000 lb/32.5% and 26,000 lb/35%) an alternative manual fuel distribution may be applied. Auxiliary tank quantity must be full, or at least 2.95 times the tail tank quantity (as shown in Figure 02-05-2, Detail C, Alternative Fuel Distribution for Take-Off).

9. FUEL (CONT'D)

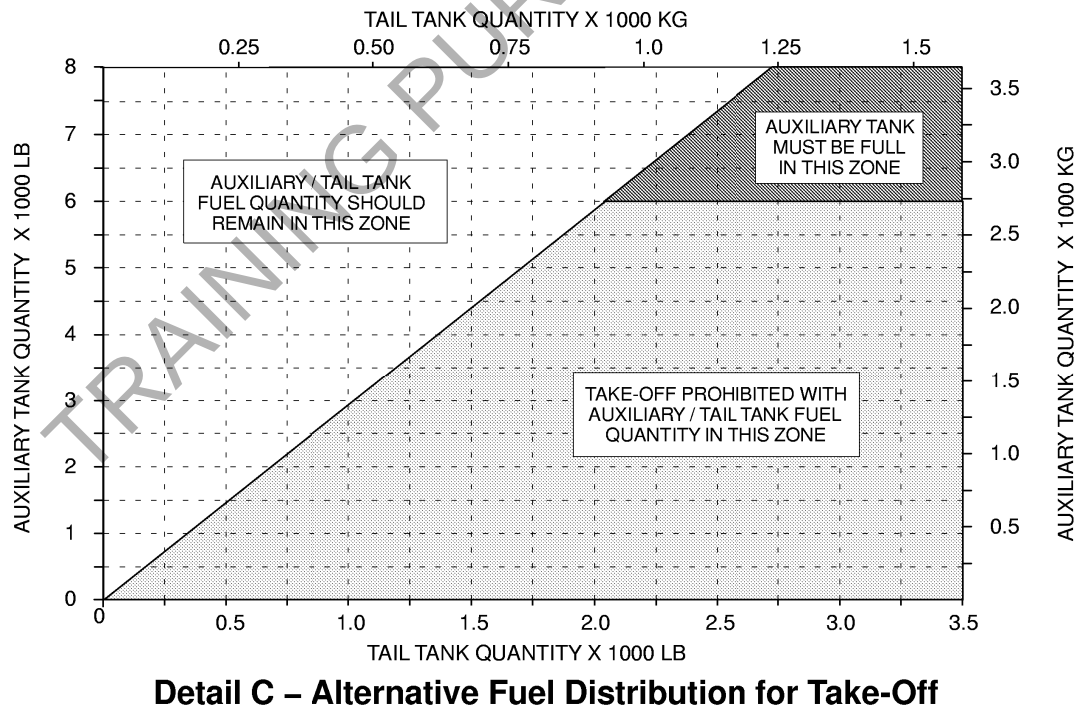
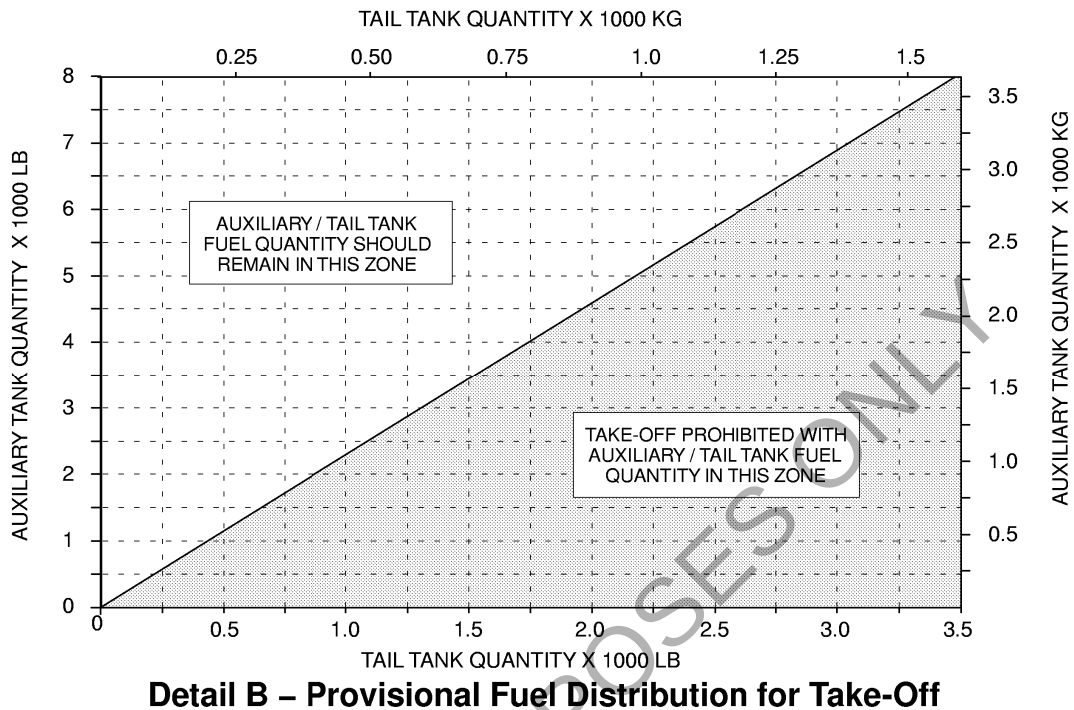
**Detail A – Normal Fuel Distribution for Take-Off**

NOTE: Based on a fuel density of 6.75 lb/US gallon the auxiliary tank must be full for tail tank quantities greater than 1227 kg (2705lb)

DFM0205_005

Auxiliary Tank Quantity/Tail Tank Quantity – Relation Verses Centre of Gravity (Page 1 of 2)
Figure 02-05-2

9. FUEL (CONT'D)



NOTE: Based on a fuel density of 6.75 lb/US gallon the auxiliary tank must be full for tail tank quantities greater than 1102 kg (2430 lb)

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Auxiliary Tank Quantity/Tail Tank Quantity – Relation Verses Centre of Gravity (Page 2 of 2)
Figure 02-05-2

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9. FUEL (CONT'D)

Take-off with up to 230 kg (500 lb) of fuel in the auxiliary tank is permitted, provided that there is at least 690 kg (1,500 lb) of fuel in each wing tank and no fuel in the tail tank.

Take-off with more than 230 kg (500 lb) of fuel in the auxiliary tank is permitted, provided that the minimum quantity of fuel in each wing tank is as per Figure 02-05-3.

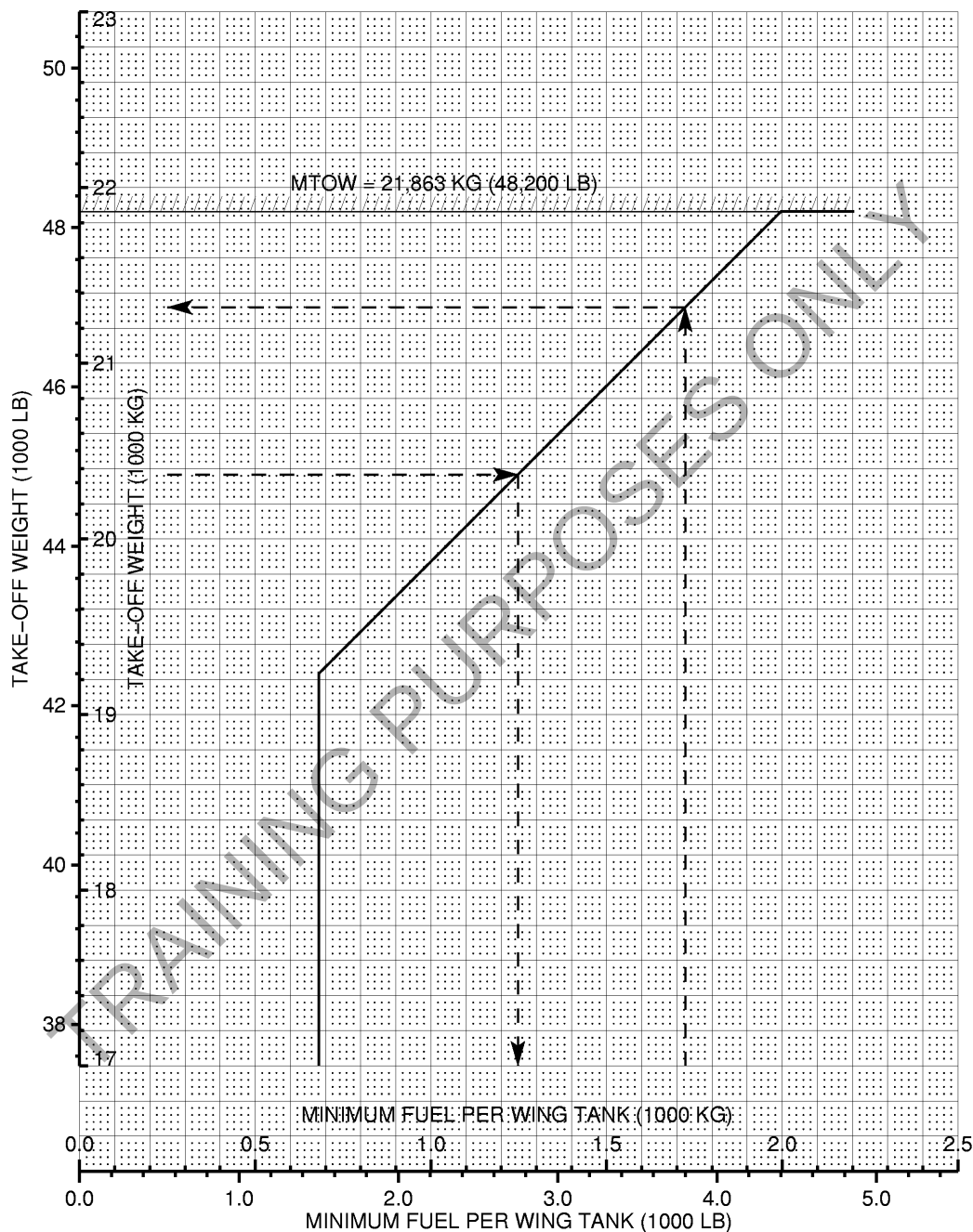
- For example, if there is at least 1,724 kg (3,800 lb) of fuel in each wing tank, then according to Figure 02-05-3, the take-off weight cannot exceed 21,319 kg (47,000 lb).
- Conversely, if the take-off weight is for example 20,366 kg (44,900 lb), then according to Figure 02-05-3, there must be at least 1,247 kg (2,750 lb) of fuel in each wing tank.

The minimum fuel quantity for go-around is 230 kg (500 lb) per wing (with the airplane level) and assuming a maximum airplane climb attitude of 10° nose up.

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9. FUEL (CONT'D)



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Minimum Wing Tank Fuel Quantity for Take-Off
Figure 02-05-3

9. FUEL (CONT'D)

A. Fuel Temperature

Take-off with engine fuel temperature indications below 5°C (41°F) is prohibited.

Take-off with bulk fuel temperature indications below the limits stated is prohibited.

During flight, bulk fuel temperature must remain above the applicable bulk fuel freezing point.

FUEL	BULK FUEL TAKE-OFF LIMIT	BULK FUEL FREEZING POINT
ASTM D 1655 JET A	−30°C	−40°C
ASTM D 1655 JET A1	−37°C	−47°C
ASTM D 6615 JET B	−40°C	−50°C
MIL-DTL-5624 JP 4	−48°C	−58°C
MIL-DTL-5624 JP 5	−36°C	−46°C
MIL-DTL-83133 JP 8	−37°C	−47°C
GB 6537-2006 No. 3 JET	−37°C	−47°C
Russian TS-1	−43°C	[†]
Russian RT	−40°C	[†]
Ukrainian TS-1	−42°C	−52°C
Ukrainian RT	−42°C	−52°C
[†] Russian TS-1 (TC-1 in Russian script) and RT fuels with a freeze point of not higher than −50°C are approved for use when the ground level OAT is not below −30°C during the 24 hours before departure. TS-1 fuel with a freeze point of not higher than −60°C and RT fuel with a freeze point of not higher than −55°C, for use in low temperature regions, are available at the operator's request.		

9. FUEL (CONT'D)

B. Fuel Grades

Fuels conforming to any of the following specifications are approved for use.

Mixing of fuels is permitted.

CANADIAN	AMERICAN	BRITISH	CHINESE	RUSSIAN/ UKRAINIAN
KEROSENE TYPE				
CAN 2 - 3.23	ASTM D1655 JET A	—	—	—
CAN 2 - 3.23	ASTM D1655 JET A1	DEF STAN 91-91	GB 6537-2006 No. 3 JET	TS-1 [††] or RT
—	MIL-DTL-83133 JP 8	DEF STAN 91-87	—	—
—	MIL-DTL-5624 JP 5	DEF STAN 91-86	—	—
WIDE CUT TYPE				
CAN 2 - 3.22	ASTM D6615 JET B	D. ENG. RD. 2486	—	—
CAN 2 - 3.22	MIL-DTL-5624 JP 4	D. ENG. RD. 2454	—	—
[††] When using TS-1 fuel (TC-1 in Russian script), engine fuel system components must be inspected in compliance with SB 605-73-001.				

NOTE

ASTM D1655, DEF STAN 91-91 and CAN 2 - 3.23 jet fuels whose origin is ASTM D7566 (Aviation Turbine Fuel Containing Synthesized Hydrocarbons) and are re-identified as Jet A or Jet A1 fuels are considered equivalent to fossil based fuels.

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9. FUEL (CONT'D)

C. Fuel Additives

The following additives, used individually or in combination, are approved:

(1) ANTI-ICING

Anti-icing additives to the latest revision of specifications MIL-I-27686E or MIL-DTL-85470B or any direct equivalent at a concentration of 0.10 to 0.15% by volume.



Do not add unblended PRIST additive directly into the fuel tank, as this may damage fuel tank components.

Alcohol-based anti-icing additives of Fuel System Icing Inhibitors (FSII) type Fluid I and Fluid I-M should not be used continuously. Russian and Ukrainian fuel containing (or pre-mixed with) alcohol-based additives of anti-icing Fuel System Icing Inhibitor (FSII) type Fluid I and Fluid I-M should not be used continuously. Contact engine manufacturer for details of impact on a reduced component inspection program if continuous use has been determined.

(2) BIOCIDES

SOHIO Biobor JF biocide additive at a concentration not in excess of 270 parts per million (20 parts per million elemental boron) for the initial dose to prevent the growth of micro-organisms. A maintenance dose of 135 parts per million should be used thereafter.

(3) ANTI-STATIC

Stadis 450 anti-static additive at a concentration of 3 g/m³ for the initial dose. Maximum dosage should not exceed 5 g/m³.

For Russian fuels, Sigbol static dissipater additive may be used in concentrations of less than 0.0005% by weight.

(4) CORROSION INHIBITOR

Corrosion inhibitors listed below are approved, by the concentrations indicated, for hydro-treated fuels only. It is recommended that corrosion inhibitors, conforming to MIL-I-25017, be blended with the fuel to provide lubricity. The corrosion inhibitor must be added after water removal and downstream of any clay filters (these processes remove the inhibitor).

- Appollo PRI-19, at a maximum concentration of 23 g/m³;
- Octel DCI-4A, at a maximum concentration of 23 g/m³;
- Octel DCI-6A, at a maximum concentration of 9 g/m³;
- Hitec E-515, at a maximum concentration of 21 g/m³;
- Hitec E-580, at a maximum concentration of 23 g/m³;
- Nalco 5403, at a maximum concentration of 23 g/m³;
- Nalco 5405, at a maximum concentration of 11 g/m³;
- Tolad 245, at a maximum concentration of 34 g/m³;

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9. FUEL (CONT'D)

D. Fuel Jettison

Fuel jettison must only be carried out with flaps set to 0.

Jettisoning of fuel in known lightning conditions is prohibited.

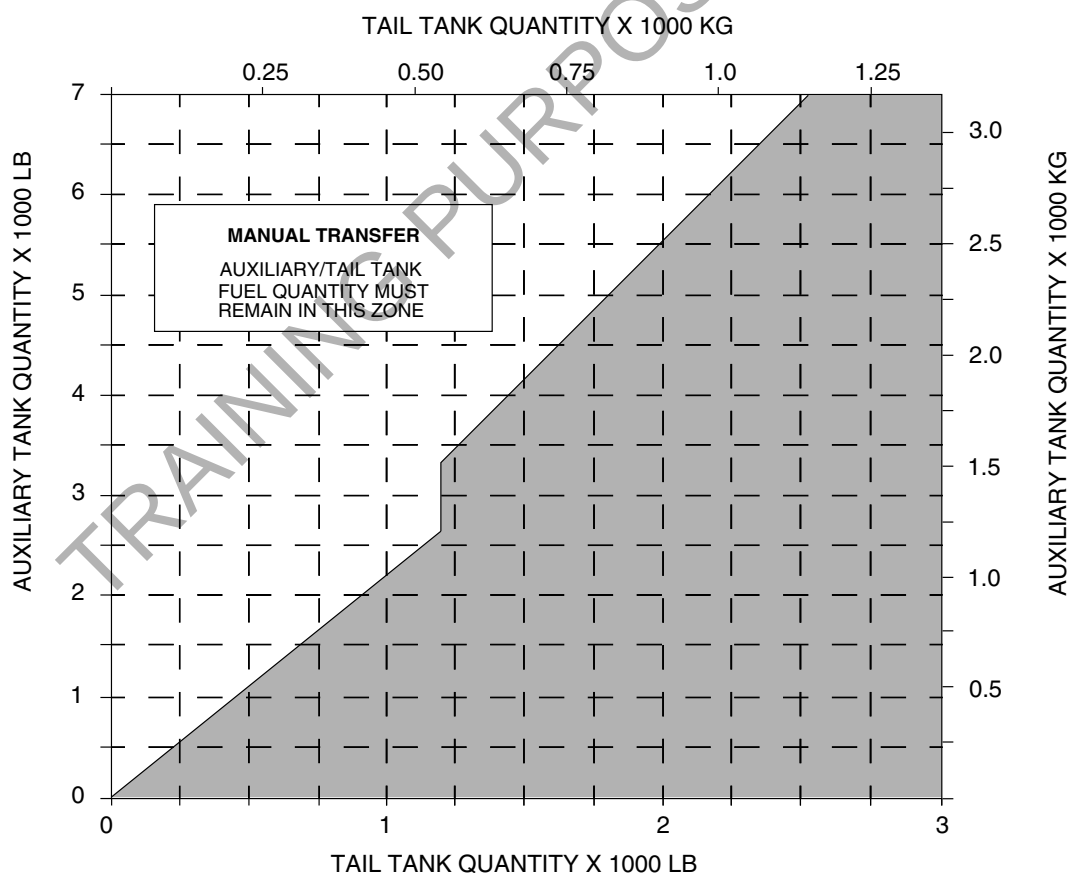
E. Fuel Transfer

Fuel transfer (left or right tank to auxiliary tank/gravity) must be off for take-off.

During normal operation, the tail tank transfer system must not be used as a centre of gravity control device; it must remain selected to automatic mode.

During manual transfer operation – Refer to Figure 02-05-4:

- Auxiliary tank quantity must be at least 2.75 times the tail tank quantity, if tail tank quantity is equal to or greater than 544 kg (1,200 lb).
- Auxiliary tank quantity must be at least 2.2 times the tail tank quantity, if tail tank quantity is less than 544 kg (1,200 lb).
- Manual tail tank fuel transfer is not permitted with more than 3,175 kg (7,000 lb) of fuel in auxiliary tank.
- Auxiliary tank quantity must be continuously monitored for overfill during manual tail tank fuel transfer.



DFM0205_003

Manual Fuel Transfer – Auxiliary Tank Quantity vs Tail Tank Quantity
Figure 02-05-4

10. OIL GRADES

Refer to appropriate maintenance or servicing manual for approved oil grades.

11. OIL CONSUMPTION

Maximum oil consumption on each engine is 189 milliliters per hour (6.4 ounces per hour or 0.05 US gallons per hour).

NOTE

Oil consumption calculation should be carried out over a minimum of 10 engine operating hours.

12. OIL LEVEL

The maximum duration without engine oil servicing is 16 operating hours.

13. OIL REPLENISHMENT SYSTEM

The engine oil level should be checked between 15 minutes to 2 hours after engine shutdown. The engines must be motored if the replenishment period is exceeded. Maximum refill allowable is 1.9 liters (2 US quarts) without dry motoring the engine.

14. AUXILIARY POWER UNIT**A. Type**

36–150 (CL)

B. Maximum RPM: 110%

The APU overspeed control will automatically shutdown the APU at 107% rpm.

C. Maximum EGT: 731°C

14. AUXILIARY POWER UNIT (CONT'D)

D. Starting:

- (1) Minimum ambient temperature for starting a cold soaked APU on the ground is -40°C.
- (2) The following APU start cycles are permitted:
 - (a) Using airplane batteries on the ground, or for normal in-flight start:
 - Three start attempts, each of 30 seconds continuous cranking
 - Followed by a 20-minute off-time
 - Followed by two further attempts each of 30 seconds continuous cranking.
 - (b) Using ground power:
 - Two start attempts, each of 15 seconds continuous cranking
 - Followed by a 20-minute off-time
 - Followed by two further attempts each of 15 seconds continuous cranking.
 - (c) If, in either case (2)(a) or (2)(b), a successful start is not obtained, a further start must not be attempted for a period of at least 35 minutes.
- (3) Maximum EGT:
974°C (not to be exceeded under any operating conditions).
- (4) Hung start:
If not greater than 30% rpm, within 60 seconds.
- (5) APU starting and operation is permitted within the following operating envelope:
 - Temperature – Refer to Figure 02-04-1.
 - Altitude and Airspeed – Refer to Figure 02-05-5.
- (6) In-flight start:
In-flight starting is guaranteed at altitudes up to 20,000 feet.

14. AUXILIARY POWER UNIT (CONT'D)**E. APU Bleed Air**

(1) Engine-start during ground operations up to 15,000 feet:

- (a) No bleed air extraction limitation. Each engine may be started using the APU as a bleed air source.
- (b) If both engines are to be started using the APU bleed air, then the operating engine's thrust must not exceed 70% N_2 .

(2) Engine-start during flight:

- (a) During single engine operations, APU bleed air extraction for an engine start is not permitted.
- (b) During double engine failure conditions, APU bleed air extraction for engine starts is permitted.

Inflight APU main engine starts have been demonstrated to 15,000 feet.

(3) Air Conditioning:

- (a) Bleed air extraction from the APU is not permitted above 15,000 feet.

F. APU Generator

The maximum permissible load on the APU generator in flight is 30kVA.

G. APU Indications

The APU limit display markings on the EICAS must be used to determine compliance with the maximum/minimum limits and precautionary ranges. If EICAS markings show more conservative limits than those specified below, the limit markings on the EICAS should be used.

INDICATION		RED (MAX LIMITATIONS)	AMBER (CAUTION RANGE)	GREEN (NORMAL RANGE)
APU EGT °C	start scale	974 and above	—	0 to 973
	normal scale	731 and above	680 to 730	0 to 679
APU RPM %		110 and above	105 to 109	0 to 104

14. AUXILIARY POWER UNIT (CONT'D)

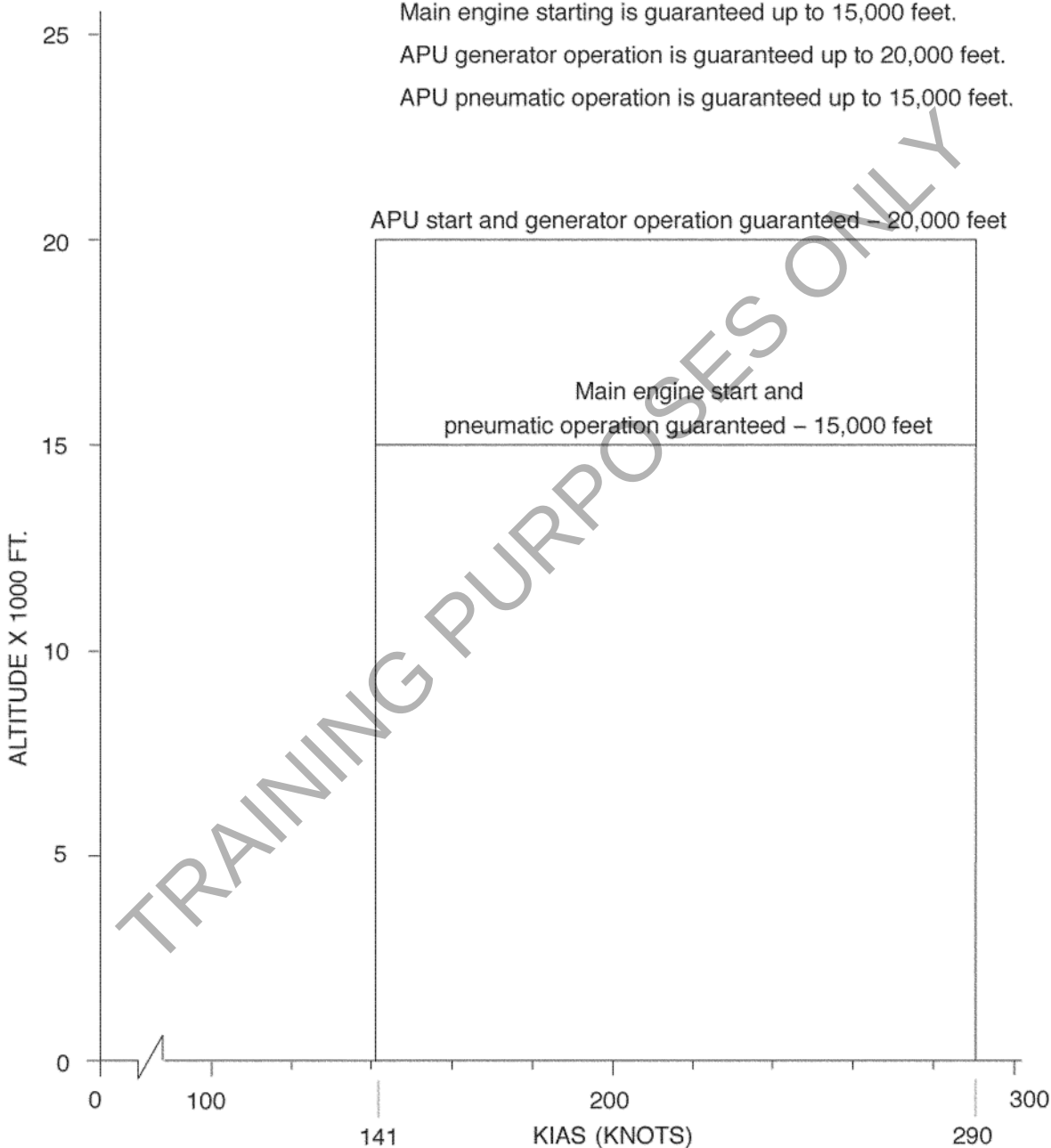
APU Operation is permitted up to 20,000 feet.

APU in-flight starting is guaranteed up to 20,000 feet.

Main engine starting is guaranteed up to 15,000 feet.

APU generator operation is guaranteed up to 20,000 feet.

APU pneumatic operation is guaranteed up to 15,000 feet.



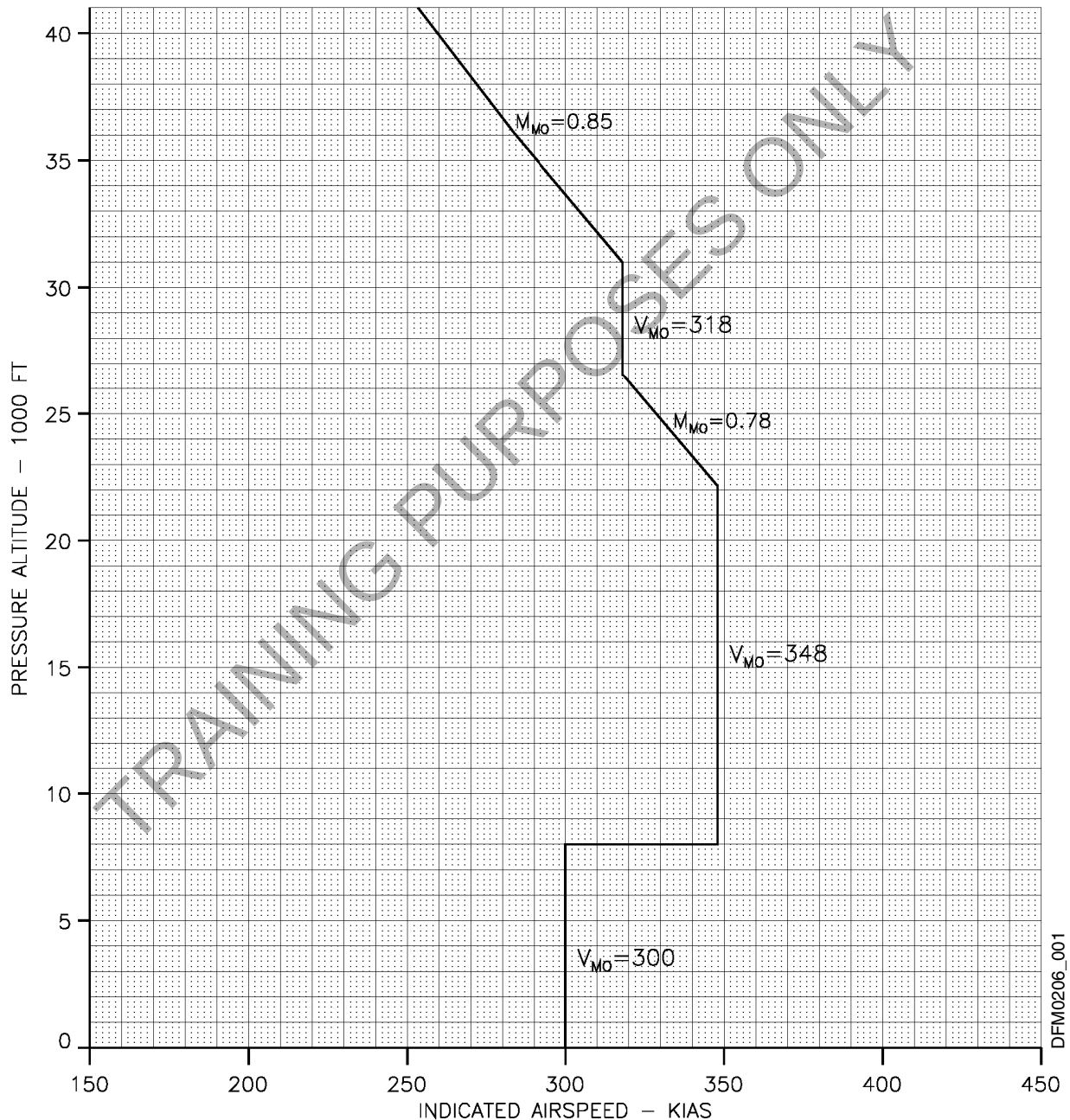
APU In-Flight Envelope
Figure 02-05-5

1. MAXIMUM OPERATING SPEED AND MACH NUMBER

Maximum operating limit speeds as given in Figure 02-06-1, must not be deliberately exceeded in any regime of flight (climb, cruise or descent), unless a higher speed is specifically authorized for flight test or training operations.

NOTE

Overspeed alerting in the flight deck is based on baro-corrected altitude. This is applicable to both the primary flight displays and to the integrated standby instrument.



Maximum Operating Speed and Mach Number
Figure 02-06-1

2. RVSM MAXIMUM OPERATING SPEED

The maximum cruise mach number during flight in RVSM airspace is 0.83.

3. FLAPS EXTENDED SPEED

The maximum speeds at which the flaps may be extended are:

Flaps to 20 degrees: 231 KIAS

Flaps to 30 degrees: 197 KIAS

Flaps to 45 degrees: 189 KIAS

4. MAXIMUM LANDING GEAR OPERATING SPEED

The maximum airspeed at which it is safe to extend the landing gear is 197 KIAS / 0.6 MI.

The maximum airspeed at which it is safe to retract the landing gear is 197 KIAS / 0.6 MI.

5. MAXIMUM LANDING GEAR EXTENDED SPEED

The maximum airspeed at which the airplane may be flown with the landing gear extended and locked is 250 KIAS / 0.7 MI.

Flight at altitudes above 20,000 feet with the landing gear extended is prohibited.

6. TIRE LIMIT SPEED

The tire limit speed is 182 knots ground speed.

7. TURBULENCE PENETRATION SPEED

Maximum air speed for turbulence penetration is 280 KIAS or 0.75 Mach, whichever is lower.

8. MINIMUM OPERATING LIMIT SPEED

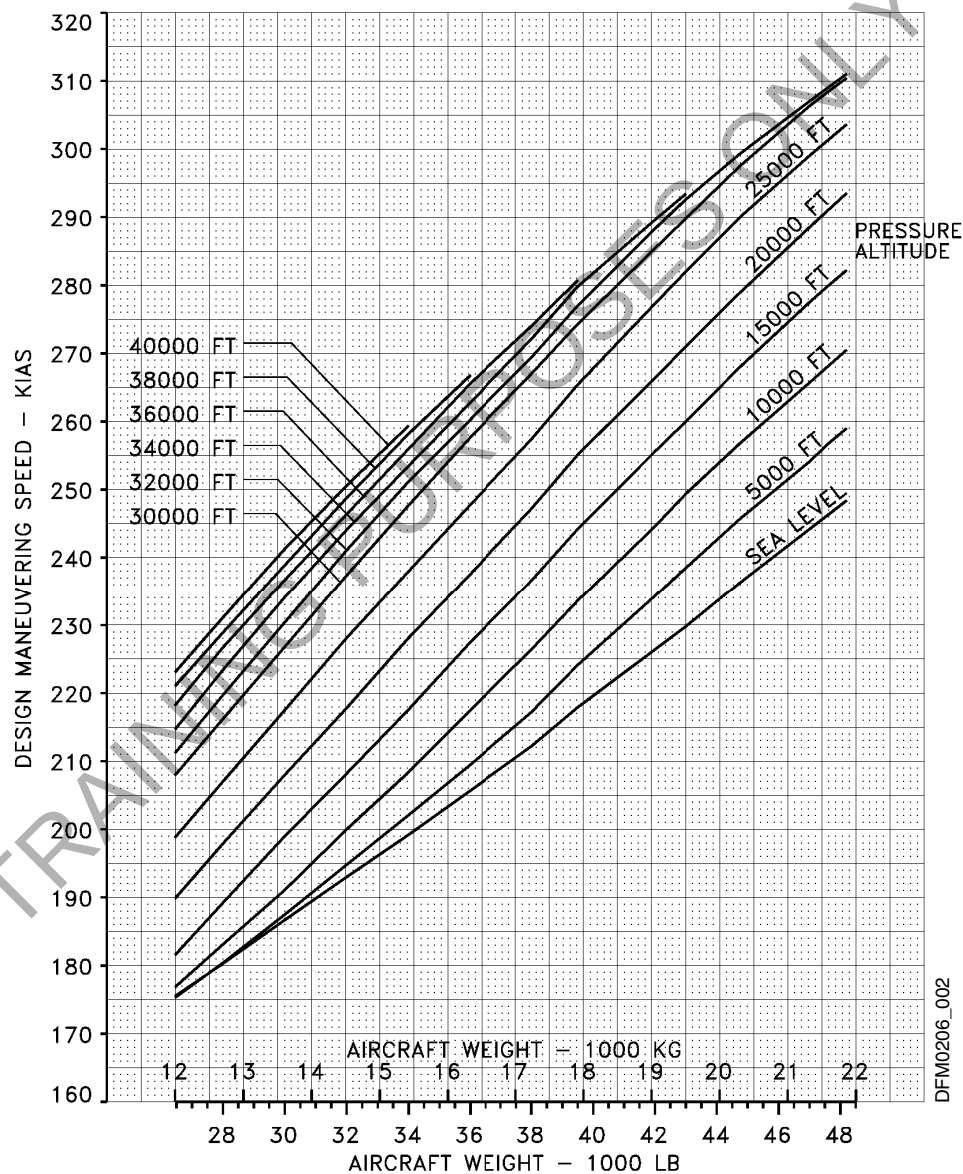
Intentional speed reduction below the onset of stall warning, as defined by stick shaker operation, is prohibited unless a lower speed is specifically authorized for flight test or training operations.

9. DESIGN MANEUVERING SPEED

Full application of rudder and aileron controls as well as maneuvers that involve angles of attack near the stall, must be confined to speeds below V_A . Values of V_A are given in Figure 02-06-2, for varying pressure altitudes and airplane weights.

CAUTION

Avoid rapid and large alternating control inputs, especially in combination with large changes in pitch, roll, or yaw (e.g. large side slip angles), as they may cause structural failure at any speed, including below V_A .



DFM0206_002

Design Maneuvering Speeds
Figure 02-06-2

BOMBARDIER CHALLENGER 605	LIMITATIONS Operating Speeds	Vol. 1	02-06-4
		Oct 05/06	

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TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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BOMBARDIER CHALLENGER 605	LIMITATIONS Maneuvering Loads	Vol. 1	02-07-1
		Oct 05/06	

1. MANEUVERING LIMIT LOAD FACTORS

These load factors limit the permissible angles of bank in turns and the severity of pull-up and push-over maneuvers:

- Flaps up: -1.0 G to 2.5 G.
- Flaps down: 0.0 G to 2.0 G.

2. SIDE SLIP MANEUVERS

Avoid unnecessary and large side-slip maneuvers during low speed high altitude cruise.

TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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BOMBARDIER CHALLENGER 605	LIMITATIONS Maneuvering Loads	Vol. 1	02-07-2
		Oct 05/06	

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	CL-605 Flight Crew Operating Manual PSP 605-6	
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BOMBARDIER CHALLENGER 605	LIMITATIONS Systems Limitations	Vol. 1	02-08-1
		REV 58, Dec 08/20	

1. AIR-CONDITIONING AND PRESSURIZATION

The maximum relief differential pressure is 9.2 psi.

The maximum negative differential pressure is -0.5 psi.

During taxi, take-off and landing, the pressure differential must not exceed 0.2 psi.

The airplane must be completely depressurized prior to opening any of the airplane doors.

2. AUTOMATIC FLIGHT CONTROL SYSTEM

WARNING

During the (V) ALTS CAP or (V) ALTV CAP modes the flight guidance/autopilot may not maintain the selected airspeed. If an engine failure occurs while in or just before (V) ALTS CAP or (V) ALTV CAP, the airspeed may rapidly decay below the safe operating speed. Crew intervention may be required to maintain safe operating speed and to avoid loss of control.

The minimum autopilot engage height after take-off is 320 feet AGL.

The minimum autopilot use height for visual and non-precision approaches is 320 feet AGL.

The minimum autopilot use height for precision approaches (ILS) with a glidepath angle of 3.5 degrees or less is 80 feet AGL.

On airplanes incorporating Service Bulletin 605-34-019:

The minimum autopilot use height for precision approaches (ILS) and LPV approaches, with a glidepath angle of 3.5 degrees or less, is 80 feet AGL.

It is prohibited to select the XFR button on the flight control panel while simultaneously making inputs on a FMS CDU.

3. BLEED AIR SYSTEMS

The bleed air 10th stage valves must be closed for take-off and landing if the engine cowl and/or wing anti-ice systems have been selected on.

If above 40,000 feet, one air-conditioning unit or cowl anti-ice system must be selected on for each engine.

	CL-605 Flight Crew Operating Manual PSP 605-6	
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4. ELECTRICAL SYSTEMS

A. Permissible Loads on AC System

Individual AC generator loading must not exceed the following values:

ALTITUDE (FEET)	LOAD LIMITATION (KVA)	
	MAIN GENERATOR (EACH)	APU GENERATOR
0 to 20,000	30	30
20,001 to 35,000	30	0
35,001 and above	25	0

B. Permissible Loads on DC Systems

The maximum permissible continuous load on each TRU is 100 amps.

C. Circuit Breaker Reset (In Flight)

A circuit breaker must not be reset or cycled (i.e., opened or closed) unless doing so is consistent with explicit procedures specified in the Airplane Flight Manual or unless, in the judgement of the Pilot-in-Command, that resetting or cycling of the circuit breaker is necessary for the safe completion of the flight.

5. FLIGHT CONTROLS – LIFT/DRAG DEVICES

A. Flaps

Enroute use of flaps is prohibited.

Flight with flaps extended at altitudes above 15,500 feet is prohibited.

B. Flight Spoilers

Flight below an altitude of 300 feet AGL with flight spoilers extended is prohibited.

To ensure adequate maneuver margins, flight spoilers must not be extended in flight at airspeeds below the recommended approach speed plus 10 KIAS (refer to Chapter 6; PERFORMANCE – LANDING PERFORMANCE of the Airplane Flight Manual).

6. STALL PROTECTION SYSTEM

Both stall protection system pusher switches must remain on for all phases of flight.

The stall protection test indicator must only be used for stall protection system functional test purposes.

The PFD normalized AOA indicator is advisory only, it does not replace the airspeed indicator as primary speed instrument, and it does not replace the stick shaker as primary stall warning.

The PFD normalized AOA indicator shall not be used during take-off.

7. THRUST REVERSERS

Thrust reversers are approved for ground use only.

The thrust reversers are intended for use during full stop landings. Do not attempt a go-around maneuver after deployment of the thrust reversers.

Backing the airplane with the use of reverse thrust is prohibited.

Take-off with any of the following thrust reverser icons or EICAS messages displayed is prohibited:

- REV icon on N_1 gauge,
- **L (R) REV UNLOCKED** caution message, and
- **L (R) REV UNSAFE** caution message.

During landing, application of maximum reverse thrust is not permitted at airspeeds below 60 KIAS. Below 60 KIAS, reverse thrust must be reduced to 60% N_1 or less.

The maximum demonstrated crosswind component approved for use of reverse thrust is 24 knots [at 33 feet (10 meters) tower height]. This value was demonstrated on a dry runway, and is considered limiting.

8. TAXI AND LANDING LIGHTS

The taxi lights must be selected off whenever the airplane is stationary.

NOTE

During prolonged taxiing, the nose landing lights may be used in place of the taxi lights, if necessary, at the pilot's discretion.

The left and right landing lights must be selected off during taxi or when the airplane is stationary.

The left and right landing lights must not be used on the ground, except for the take-off roll and landing.

9. WHEEL BRAKE COOLING LIMITATIONS

Brake cooling times (established in accordance with the procedures in Chapter 6; PERFORMANCE – TURN-AROUND TIME of the Airplane Flight Manual) must be observed between a landing or a low-energy rejected take-off (RTO) and a subsequent take-off, to ensure that sufficient brake energy is available to bring the airplane to a complete stop, if the subsequent take-off is rejected.

If a fusible plug releases, the wheels, brakes and tires and the anti-skid speed sensors must be inspected in accordance with the procedure in the time limits/maintenance checks, provided in the CL-605 Airplane Maintenance Manual, and any damage rectified before the next take-off.

10. CONFIGURATION DEVIATION LIST

If the aircraft is to be operated with certain secondary airframe and/or any nacelle parts missing, operation must be in accordance with the limitations specified in the basic Airplane Flight Manual, and as amended by the Configuration Deviation List (Appendix 1).

11. ELECTRICAL/AVIONICS EQUIPMENT

During ground operation at ambient temperatures above 40°C (104°F), operation of electrical/avionics equipment must be limited to 30 minutes, unless at least one air conditioning pack is operating and the passenger door is closed.

AR Certified Airplanes**12. AIRSPACE OPERATIONAL LIMITATIONS**

The airplane can fly in the former USSR airspace only on routes covered by ATC ground facilities using RBS mode. If the airplane is to fly in areas that are not completely covered by VHF stations and if the interruptions between VHF covered zones exceeds one flight hour, two HF radios must be installed on the airplane.

AR Certified Airplanes**13. LONG RANGE NAVIGATION ACCURACY**

When airplanes, not equipped with GPS, operate on routes, having a width of ± 5 kilometers, not covered by VOR/DME, it is recommended that the airplane position be confirmed by ATC after 1 hour 30 minutes.

When airplanes, not equipped with GPS, operate on routes, having a width of ± 10 kilometers, not covered by VOR/DME, it is recommended that the airplane position be confirmed by ATC after 3 hours.

AR Certified Airplanes**14. GROUND OPERATIONS IN HIGH WIND CONDITIONS**

In the event that the airplane is parked and sustains winds or gust loads in excess of 27 meters/second, an inspection and functional check of the aileron, elevator and rudder power control units is required.

AR Certified Airplanes**15. OPERATIONS FROM GRAVEL RUNWAYS**

Operations from gravel runways are prohibited.

BOMBARDIER CHALLENGER 605	LIMITATIONS Navigation Systems Limitations	Vol. 1	02-09-1
		REV 48, Aug 23/18	

1. FLIGHT MANAGEMENT SYSTEM - FMS-6000

The flight management system (FMS) must be operated in accordance with the latest edition of the following;

- Airplane Flight Manual, and
- Flight Management System FMS-6000 Pilot's Guide, part number 523-0807938-001117 or later applicable revision.

The FMS is approved for use only with the SCID 832-4117-094 (no SAR) or SCID 832-4117-095 (with SAR) software program versions or later applicable revisions.

FMS database information must be verified as being current. Waypoints must be checked for accuracy prior to use.

Use of pilot defined FMS approaches, in instrument meteorological conditions, is prohibited.

The FMS must not be used for navigation unless it is receiving suitable navigation information from the following sources;

- One VOR/DME, or
- Two DMEs, or
- One inertial reference system (IRS), or
- GPS.

Airplane operation must not be predicated upon performance data; ETE/ETA and fuel remaining.

FMS flight plans must be identical when conducting dual FMS approach operations.

Use of FMS instrument approaches past the Final Approach Fix is prohibited, unless APPR is annunciated on the PFD.

Use of FMS instrument approaches is prohibited, unless APPR is annunciated on the PFD.

Manual changes to altitudes in departure and missed approach procedures are prohibited.

Use of FMS temperature compensation (TEMP COMP) is prohibited.

The flight crew must verify any altitude/speed constraints on the last waypoint of any entered STAR procedure and/or the first waypoint of any entered approach procedure. The flight crew may correct any such altitude/speed constraint(s) as required to agree with current chart procedures.

	CL-605 Flight Crew Operating Manual PSP 605-6	
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1. FLIGHT MANAGEMENT SYSTEM - FMS-6000 (CONT'D)



Errors to VNAV defined paths may occur because of coding errors in the navigation data base and because of altimetry errors. The actual VNAV path may deviate significantly below the intended VNAV path in very cold temperatures, unless an approved temperature compensation function (if equipped) is used to correct for non-standard temperatures. Flight crew must coordinate use of temperature compensation with ATC prior to employing this function.

Database coding of VNAV altitudes for approach waypoints may result in VNAV paths continuing below Minimum Descent Altitude (MDA), Decision Altitude (DA), or Decision Height (DH), or ending at an altitude too high to continue a safe descent to landing. Baro VNAV paths (often called pseudo-glide paths) are not equivalent to an ILS glideslope.

Position along an approach must be verified prior to commencement of VNAV descent, as displayed by the FMS. The required visual reference must be obtained prior to commencing descent below published MDAs, DAs, or DHs.

Use of VFLC (FLC with VNAV) is prohibited during climb when the airspeed reference is displayed as Mach.

Use of VNAV vertical guidance is prohibited during QFE operation, when the altimeter is adjusted to show height above ground rather than sea level.

Use of VNAV vertical guidance for a V-MDA type approach between the final approach fix and the missed approach fix is prohibited.

The FMS, with inputs from GPS, may only be used for approach guidance if the reference coordinate data system for the instrument approach is WGS-84 or NAD-83.

It is prohibited to initiate a NAV RADIO SELF TEST from either the NAV radio tuning page on the FMS CDU or the MFD nav radio menu during flight.

The V_{ERC} data shown on the THRUST LIMIT page is advisory information only and must be cross checked with the AFM.

The FMS thrust setting data is the primary means of information, provided two air data computers and two flight management computers are available.

The FMS V speed data is the primary means of information, provided two active flight management computers are available.

AR Certified Airplanes

The FMS V speed data is advisory only.

The following performance database, part number 815-9079-001, must be verified to be current and valid.

1. FLIGHT MANAGEMENT SYSTEM - FMS-6000 (CONT'D)

The following V speed database, part number 815-9174-001, must be verified to be current and valid.

On FAA Certified Airplanes incorporating SB 605-34-007:

The following V speed database, part number 815-9082-001, must be verified to be current and valid.

The FMS does not reduce available runway lengths for runways with displaced thresholds. When using FMS approach performance data for a runway with a displaced threshold, the pilot must manually enter the RWY LENGTH value with the actual available landing distance from a published chart.

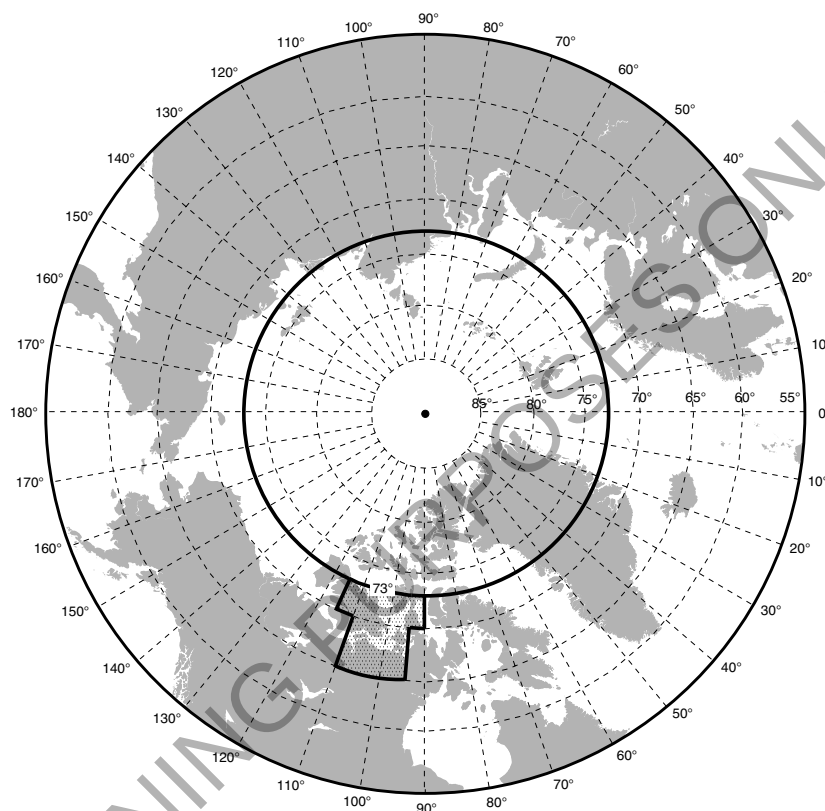
When RWY LENGTH is manually entered, the FMS does not compute headwind/crosswind; these must be manually entered on the CDU.

TRAINING PURPOSES ONLY

1. FLIGHT MANAGEMENT SYSTEM - FMS-6000 (CONT'D)

Due to magnetic variation errors, the indicated FMS course may differ from the published course by more than 3 degrees in the following regions.

LATITUDE	LONGITUDE
N70 – N73	W115 – W90
N65 – N70	W110 – W95



Area of Magnetic Unreliability
Figure 02-09-1

In areas of magnetic unreliability, the following operations shall only be performed using true north bearings:

- PLACE BRG/DIST to pilot-defined waypoints
- PLACE BRG/PLACE BRG to pilot-defined waypoints
- Holding patterns at waypoint except at present position (PPOS)
- Intercept course (INTC CRS)
- Radial crossing fix

NOTE

At latitudes greater than 73°N and greater than 60°S, the FMS always uses true course values, and local magnetic variation has no effect.

2. FLIGHT MANAGEMENT SYSTEM - FMS-6200***On airplanes incorporating Service Bulletin 605-34-019:***

The flight management system (FMS) must be operated in accordance with the latest revision of the following:

- Airplane Flight Manual, and
- Pro Line 21 Advanced Flight Management System Operators Guide (part number 523-0822377 at the latest applicable revision).

If a difference exists between the AFM and the Operators Guide, the AFM shall take precedence. The Pro Line 21 Advanced Flight Management System Operators Guide must be immediately available to the flight crew whenever navigation is predicated on the use of the FMS.

The FMS is approved for use with software version FMC HSCD 946-2720-110, or later applicable revision.

On airplanes NOT incorporating Service Bulletin 605-34-033:

The FMS is approved for use with software version FMC HSCD 946-2720-102, or later applicable revision.

The following performance database, part number 096-5752-101, must be verified to be current and valid.

On airplanes NOT incorporating Service Bulletin 605-34-033:

The following performance database, part number 096-5752-001, must be verified to be current and valid.

The following thrust and VERC database, part number 096-5753-101, must be verified to be current and valid.

On airplanes NOT incorporating Service Bulletin 605-34-033:

The following thrust database, part number 096-5753-001, must be verified to be current and valid.

The following V speed database must be verified to be current and valid:

	ALL REGISTRIES EXCEPT FAA	FAA REGISTRY
Airplanes incorporating Service Bulletin 605-34-033	part number 096-5750-101	part number 096-5751-101
Airplanes NOT incorporating Service Bulletin 605-34-033	part number 096-5750-002	part number 096-5751-002

AR Certified Airplanes:

The FMS V speed data is advisory only.

The FMS V speed data does not need to be verified with the AFM, provided both flight management computers are available.

2. FLIGHT MANAGEMENT SYSTEM - FMS-6200 (CONT'D)

The FMS V speed data is the primary means of information, provided two active flight management computers are available.

When FMS 3 is selected to 1 or 2 for take-off, manual V speed data (cyan) must be used for take-off.

On airplanes NOT incorporating Service Bulletin 605-34-033:

The AFM must be used for the V_{ERC} data. The use of the V_{ERC} data on the THRUST LIMIT page is prohibited.

On airplanes incorporating Service Bulletin 605-34-033:

The V_{ERC} data shown on the THRUST LIMIT page is advisory information only and must be cross checked with the AFM.

The FMS thrust setting data is the primary means of information, provided two air data computers and two flight management computers are available.

FMS database information must be verified as being current. Waypoints must be checked for accuracy prior to use.

It is prohibited to load pilot routes from the PILOT ROUTE LIST during flight.

It is prohibited to upload flight plans from ROUTE MENU (FPLN RECALL) during flight.

It is prohibited to select FPLN WIND UPDATE from the ROUTE MENU.

It is prohibited to tune a localizer on the TUNE page using the identifier only.

It is prohibited to initiate a NAV RADIO SELF TEST from either the NAV radio tuning page on the FMS CDU or the MFD nav radio menu during flight.

It is prohibited to use the FMS as a navigation source when it is not receiving suitable navigation information from the following:

- One GNSS; or
- Two DMEs; or
- One VOR / DME; or
- One IRS.

FMS navigation (FMS selected as NAV Source) and display of magnetic heading is prohibited in Canadian Northern Domestic Airspace and at the following locations:

- Latitudes greater than 82°N
- Latitudes greater than 73°N with a longitude between 80°W and 170°W
- Latitudes greater than 82°S
- Latitudes greater than 60°S with a longitude between 120°E and 160°E.

Refer to Inertial Reference System (IRS) for additional limitation regarding the usage of magnetic heading.

It is prohibited to load FMS procedures that are referenced to True Heading into the flightplan.

During periods of dead reckoning, indicated by the FMS DR annunciation, the FMS shall not be utilized as the primary source of navigation.

2. FLIGHT MANAGEMENT SYSTEM - FMS-6200 (CONT'D)

BRNAV and PRNAV operations with the following message are prohibited:

- IRS ONLY (on CDU)

When conducting RNP operations the pilot must verify the PFD CDI scale annunciation is appropriate for the RNP leg as follows:

- RNP APCH CDI Scale 0.3 nm
- BASIC RNP-1 CDI Scale 1.0 nm
- RNP-2 CDI Scale 2.0 nm
- RNP-4 CDI Scale 4.0 nm (selected by pilot)
- RNP-10 CDI Scale 4.0 nm (selected by pilot).

When conducting BASIC RNP-1 operations, the length of each individual flight plan leg or between the aircraft position and the TO waypoint must not exceed 200 NM.

When conducting RNP-2 operations, the length of each individual flight plan leg or between the aircraft position and the TO waypoint must not exceed 400 NM.

When conducting RNP-4 or RNP-10 operations, the length of each individual flight plan leg or between the aircraft position and the TO waypoint must not exceed 500 NM.

Pilot defined waypoints (e.g. along track offset) must not be used to create individual legs shorter than the limiting length.

When conducting non-AR RNP operations, pilots should maintain CDI lateral deviation less than ± 1 dot deviation except brief deviation in excess of ± 1 dot deviation during waypoint sequencing is permitted.

RNP operations with any of the following FMS messages annunciated, are prohibited:

- NO APPR (on PFD)
- UNABLE RNP (on PFD)

VERTICAL NAVIGATION

Vertical guidance deviation information is an aid to assist pilots to comply with altitude restrictions and plan descent profiles. When using vertical guidance the pilot must use the primary barometric altimeter to ensure compliance with altitude restrictions, particularly during instrument approach operations.

Use of VNAV vertical guidance is prohibited when the barometric altitude is corrected to the landing field elevation (QFE).

Use of VNAV vertical guidance is prohibited with a single ADC source when the 3rd FMS is reverted in.

It is prohibited to select MCRZ from the FMS VNAV CRUISE page.

Use of VFLC (FLC with VNAV) is prohibited during climb when the airspeed reference is displayed as Mach.

It is prohibited to select VNAV following a go-around during the missed approach procedure.

2. FLIGHT MANAGEMENT SYSTEM - FMS-6200 (CONT'D)

Errors to VNAV defined paths may occur because of coding errors in the navigation data base and because of altimetry errors. The actual VNAV path may deviate significantly below the intended VNAV path in very cold temperatures; the temperature compensation function is used to correct for nonstandard temperatures. Flight crew must coordinate use of temperature compensation with ATC prior to employing this function.

Database coding of VNAV altitudes for approach waypoints may result in VNAV paths continuing below Minimum Descent Altitude (MDA), Decision Altitude (DA), or Decision Height (DH), or ending at an altitude too high to continue a safe descent to landing. VNAV paths (often called pseudo glide paths) are not equivalent to an ILS glideslope.

Position along an approach must be verified prior to commencement of VNAV descent as displayed by the FMS. The required visual reference must be obtained prior to commencing descent below published MDAs, DAs, or DHs.

PERFORMANCE

The FMS will automatically display the displaced thresholds of runways provided that they are identified in the loaded active cycle FMS navigation database. If RUNWAY LENGTH is manually entered then the FMS will not compute headwind/crosswind; these must be manually entered.

Airplane operation must not be predicated upon performance data; ETE/ETA and fuel remaining.

FMS range, fuel management information and ETE/ETA is advisory only.

TEMPERATURE COMPENSATION

When changing the destination to ALTN airport by selecting direct-to a waypoint in the route to the ALTN airport, if Temp Comp is ON and an OAT value has been entered for the original DEST, the same OAT value will be retained for the new DEST (ALTN). Crew must verify the correctness of OAT after changing destination.

APPROACH

Instrument approaches must be accomplished only in accordance with instrument approach procedures that are contained in the FMS database.

NOTE

Vectored ILS approaches are exempt from this limitation as they do not require nor do they rely on FMS guidance. ILS back course approaches also are exempt if they are vectored.

The flight crew must verify any altitude/speed constraints on the last waypoint of any entered STAR procedure and/or the first waypoint of any entered approach procedure. The flight crew may correct any such altitude/speed constraint(s) as required to agree with current chart procedures.

2. FLIGHT MANAGEMENT SYSTEM - FMS-6200 (CONT'D)

It is prohibited to conduct LPV approach operations with FMS2 as pilot navigation source and FMS1 as copilot navigation source.

Use of pilot defined FMS approaches, in instrument meteorological conditions, is prohibited.

Use of FMS instrument approaches is prohibited unless RNP APPR or LPV APPR is annunciated on the PFD.

RNP operations down to LPV minimum when any of the following messages are present, are prohibited:

- Red boxed "VNV" flag (vertical deviation scale on PFD)
- NO APPR (on PFD)
- EFIS MISCOMPARE (CAS message)
- EFIS COMP INOP (CAS message)
- LPV NOT AVAIL (on CDU)

RNP operations down to LPV minimum must not be commenced or continued unless all required airborne equipment are operating satisfactorily.

The required airborne equipment are:

- Dual PFD
- Two GNSS sensors must be available to commence an approach and one GNSS sensor is required to continue an approach.
- Dual FMC

Use of VNAV vertical guidance for V-MDA type approach between the final approach fix and the missed approach fix is prohibited.

FMS as a primary navigation source beyond the FAF is prohibited for ILS, LOC and LOCBC approaches.

FMS as a supplementary navigation source beyond the FAF is permitted when APPR or REF APPR is annunciated on the PFD if the navigation aid referenced in the approach procedure is displayed and monitored by the flight crew during the approach.

NOTE

For the FMS approaches for which GNSS is not authorized as a primary navigation source (i.e. GNSS does not appear in the approach title), APPR will be annunciated on the PFD if the GNSS sensors are deselected; otherwise, REF APPR will be annunciated.

The FMS with inputs from GNSS may only be used for approach guidance if the reference coordinate data system for the instrument approach is WGS84 or NAD83.

GNSS sensors must be deselected when on approach in airspace not referenced to WGS84 or NAD83.

APPROVAL STATUS

The approvals listed in the OPERATIONAL CAPABILITIES do not constitute operational approval. Some operations require additional crew qualification, communication and surveillance equipment in order to obtain operational approval from regulatory authorities.

3. NAVIGATION RECEIVER

It is prohibited to conduct ILS approach operations with LOC2 as pilot navigation source and LOC1 as copilot navigation source.

On airplanes incorporating Service Bulletin 605-34-019:

It is prohibited to navigate using VOR or LOC as Nav source when displaying true heading mode.

4. TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

Navigation must not be predicated upon the use of the terrain awareness display.

TAWS TERRAIN OFF must be selected when using QFE operations, and GPS is not available.

5. RUNWAY AWARENESS AND ADVISORY SYSTEM (RAAS)

RAAS advisories are not intended for navigation purposes.

Pilots should not deviate from ATC clearances or otherwise correct aircraft position based solely on RAAS advisories.

Pilots may not use RAAS advisories as a replacement for NOTAM or ATIS information.

RAAS Short and Insufficient Runway functions do not take into account aircraft performance factors such as aircraft weight, wind, runway conditions, slope, air temperature and altitude of airport. The runway length settings for these functions were determined from a review of takeoff and landing performance and are considered to be acceptable guidelines for RAAS purposes. This information is intended to enhance the flight crew awareness by identifying that the runway length available may be marginal for the current conditions.

Data on newly constructed runways, changes to length of existing runways, or the rare change of runway identifier may not necessarily be included in the RAAS runway database.

RAAS does not have knowledge of where painted hold lines are located at airports. In some cases, the RAAS approaching runway advisory may be issued after an aircraft has crossed a hold line.

The RAAS system is a situational awareness tool to the flight crew and should not be used as the primary source of determining field position, runway length, field performance, etc. The absence of RAAS aural and/or visual messages such as "short runway" or "long landing" does not guarantee that the aircraft can be safely operated from or onto the runway.

Altimeter Monitor-Below Transition Altitude function, is not available at airports defined as QFE in the EGPWS database.

6. INTEGRATED FLIGHT INFORMATION SYSTEM (IF INSTALLED)

Adequate back-up documentation for IFIS must be immediately available to the flight crew.

IFIS functions must not be used unless the related databases (i.e., charts, airspace, airways, geographic, political, graphical weather) incorporate the current update cycle.

The IFIS does not include enroute charts.

The aircraft symbol on the electronic charts is advisory and is presented for increased situational awareness. It is not to be used for navigation.

The aircraft symbol displayed on airport diagram charts during ground operations is advisory and is presented for increased situational awareness. It is not to be used for precision maneuvering on the ground.

The display of geo-political boundaries, airspace and airways on the MFD (enhanced map overlays) is for enhanced situational awareness only, and use for navigation is prohibited.

All weather products satellite linked to the aircraft are advisory.

On airplanes incorporating Service Bulletin 605-34-019:

It is not possible to display landscape formatted approach charts in split view.

7. SYNTHETIC VISION SYSTEM (SVS)

TAWS alerts must take priority over the image depicted on SVS.

Use of the SVS for airplane navigation is prohibited.

Use of the SVS is prohibited when the barometric altitude is corrected to the landing field elevation (QFE).

When conducting an approach in airspace not referenced to WGS-84 or NAD-83, the SVS shall be deselected prior to passing the FAF.

When conducting an approach to a runway not entered in the FMS flight plan, the SVS shall be deselected, prior to passing the FAF.

The SVS Airport/Runway database must incorporate the current update cycle. If not, the SVS shall be deselected prior to take-off and prior to passing the FAF.

The SVS does not render a video image above latitude 85N and below latitude 85S. The SVS shall be deselected prior reaching these latitudes.

The SVS is intended as a situational awareness enhancement feature and shall not be the sole source of reference by the pilot for any of the following:

- Continuation of an approach, landing or takeoff below the minima published for that approach.
- Execution of an approach, landing or takeoff in low visibility conditions.
- As a replacement for natural vision where there are specific visual requirements to be achieved.
- As a replacement for the basic attitude, flight guidance or navigation displays and information.
- The performance of any specific flight related task.
- The flight crew making any judgment, decision or action that is solely based on the synthetic vision display.
- The application of any operational credit.

8. DATALINK

The datalink is approved for the transmission and receipt of messages that will not create an unsafe condition if the message is improperly received. An unsafe condition may exist if:

- The message, or part of the message, is delayed or not received,
- The message is delivered to the wrong recipient, or
- The message content is corrupted.

Crew action-based messages, such as pre-departure clearance, oceanic clearance, digital automatic terminal information service, weight and balance, take-off data (speeds, trim settings, runway distances) are prohibited, unless the approved operational procedures are used to verify that the message is received by the intended recipient, that the message is valid, and that the content is not corrupted.

All weather products data linked to the aircraft are advisory.

On airplanes incorporating Service Bulletin 605-23-007:

ATN B1 CPDLC installation is limited to providing a supplementary means of communication; voice shall remain the primary means of communication. Use of ATN B1 CPDLC should be avoided during periods of high pilot workload and critical phases of flight where pilot-ATC communications are time critical. This AFM entry does not, by itself, constitute an operational approval where such an approval is required.

On EASA registered airplanes:

ATN B1 CPDLC shall only be used for routine datalink exchanges during en-route operations in upper airspace.

On airplanes incorporating Service Bulletin 605-23-008:

FANS 1/A+ datalink system must be operated in accordance with ATN and FANS Interlocked CPDLC Applications Operators Guide (part number 523-0822435 at the latest applicable revision).

The crew must ensure that the FMS is updated with the proposed flight plan and its associated constraints and procedures when receiving a route clearance via CPDLC.

Before executing any flight plan modification received via CPDLC, the crew must ensure that the entire content of the flight plan has been inserted.

Use of FANS INSERT MOD is prohibited within 5NM of sequencing a waypoint. This is due to the fact that the waypoint sequence may be delayed for up to 15 seconds if a large route is loaded in the FMS.

It is prohibited to use the WX DEV TO from the FANS ROUTE REQ 1/2 page.

It is prohibited to use the DIVERT TO from the EMERGENCY MSG 1/2 page using the FANS REQUEST page.

Inmarsat L band satellite communication systems (SwiftBroadBand, Swift64 and Classical) interfere with Iridium systems. If both Inmarsat L band and Iridium systems are installed and the Iridium is capable of providing cockpit datalink services (including FANS 1/A+, and CDU datalink), then the Inmarsat L band system must be depowered and considered inoperative for the entire flight where cockpit datalink services are intended to be used.

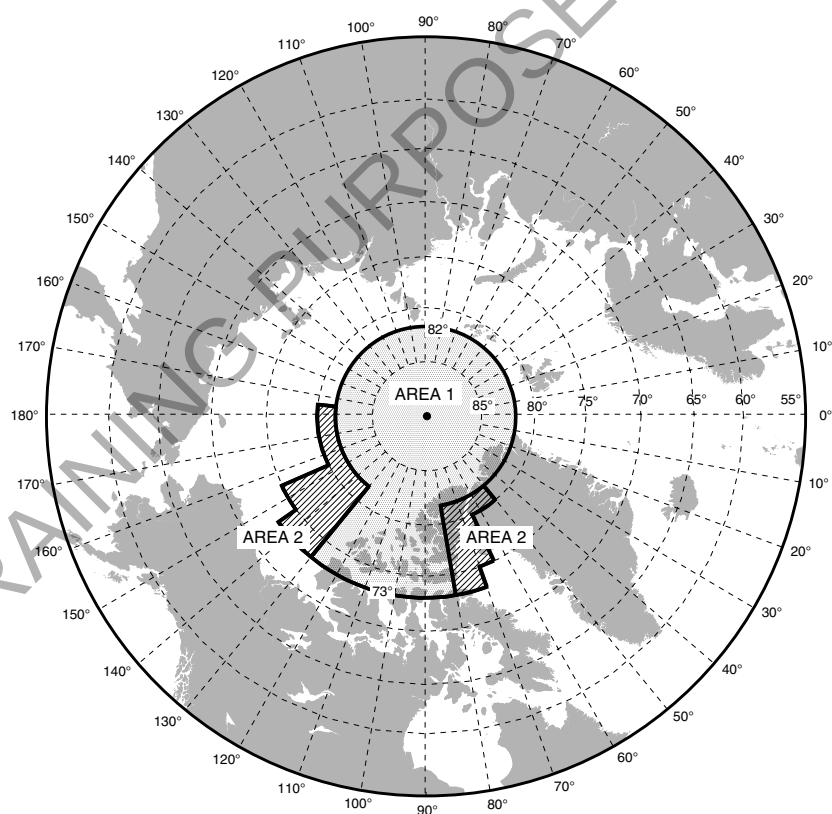
9. INERTIAL REFERENCE SYSTEM (IRS)***On airplanes 5701 to 5978:***

Magnetic north operation is prohibited in the following regions:

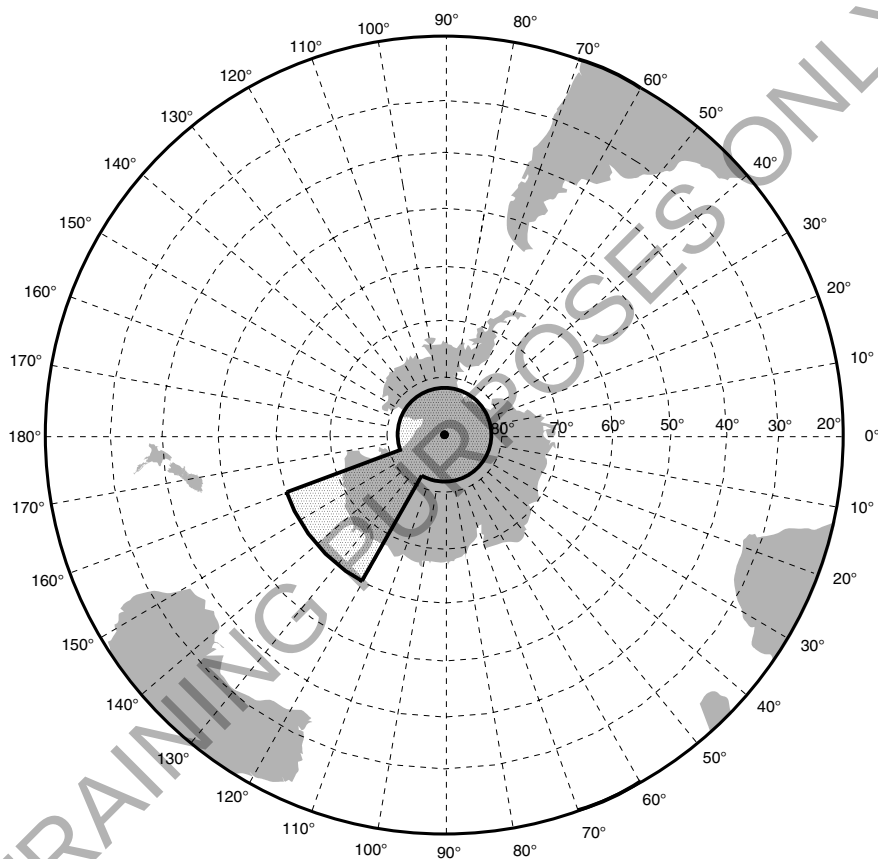
- Latitudes greater than 82°N and latitudes greater than 73.125°N for longitudes between 80°W and 130°W (refer to AREA 1 of Figure 02-09-2).
- Latitudes greater than 82°S and latitudes greater than 60°S for longitudes between 120°E and 160°E (refer to Figure 02-09-3).

The use of magnetic heading is prohibited when flying in the following regions as errors may exceed 10 degrees (refer to AREA 2 of Figure 02-09-2).

LATITUDE	LONGITUDE
N80 – N82	W180 – W130 W80 – W50 E175 – E180
N75 – N80	W155 – W130 W80 – W65
N73 – N75	W145 – W130 W80 – W70



Prohibited Areas of IRS Use – Northern Hemisphere
Figure 02-09-2

9. INERTIAL REFERENCE SYSTEM (IRS) (CONT'D)

Prohibited Areas of IRS Use – Southern Hemisphere
Figure 02-09-3

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1. GENERAL

Procedures given in this chapter of the Flight Crew Operating Manual are concerned with foreseeable but unusual situations in which immediate and precise crew action, as indicated by the items within the “boxed” area at the beginning of each procedure, will substantially reduce the risk of disaster.

In addition to the failure indications referenced in the following procedures, the master warning system and appropriate switch/lights operate where applicable.

The emergency procedures contained within this chapter have been developed and recommended by Bombardier and approved by Transport Canada for use in the operation of the CL-600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049). These procedures are provided as guidance, and should not be construed as prohibiting the development of equivalent Transport Canada-approved procedures.

2. LANDING DISTANCE FACTORS

Landing distance factors are provided herein with and without the use of thrust reversers. No operation should be predicated on the use of thrust reversers.

TRAINING PURPOSES ONLY

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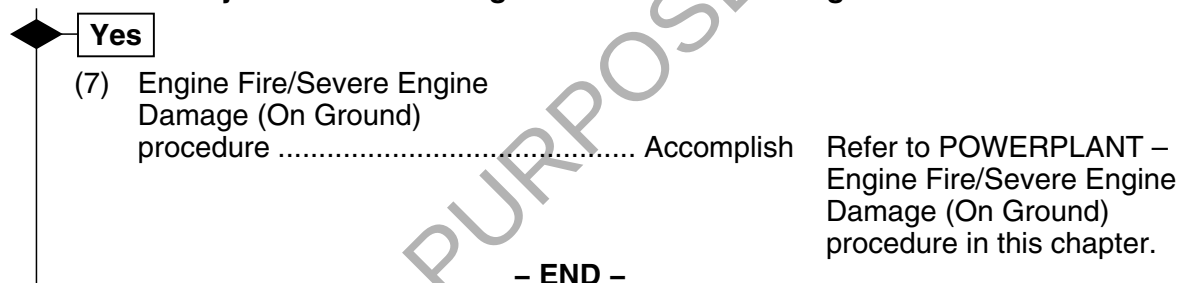
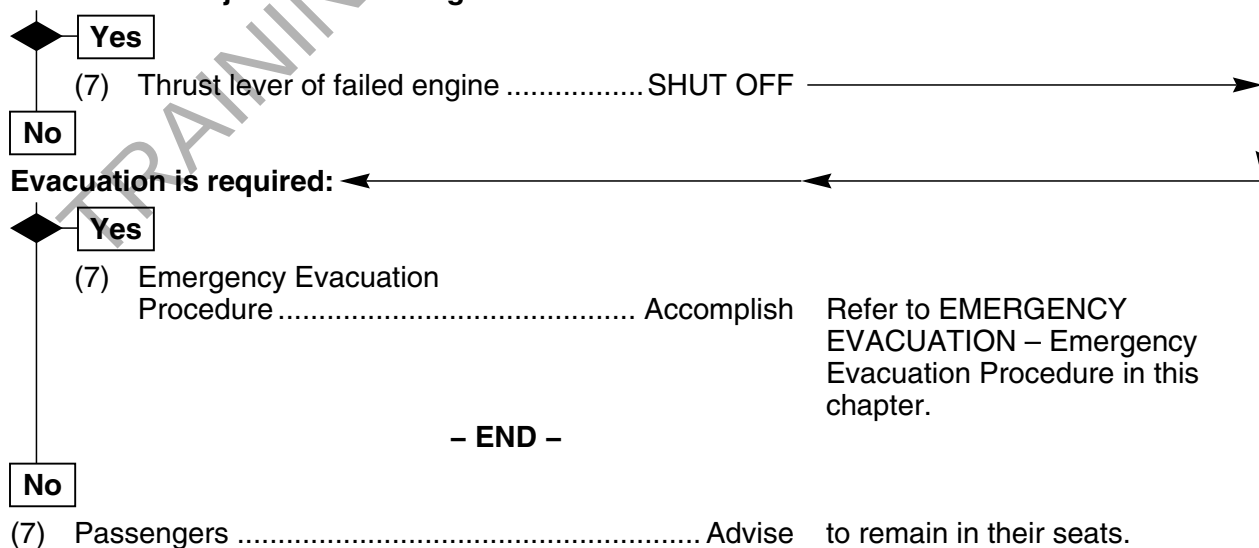
TRAINING PURPOSES ONLY

1. REJECTED TAKE-OFF**A. Rejected Take-Off Before Achieving V_1**

Rejection of the take-off must be initiated no later than V_1 .

Simultaneously:

- | | | |
|---|-------------|--|
| (1) Take-off | Discontinue | immediately. |
| (2) Thrust levers..... | IDLE | |
| (3) ATS DISC switch | Depress | |
| (4) Wheel brakes | Apply | maximum braking, until a safe stop on runway is assured. |
| (5) FLIGHT SPOILER lever | Select | to MAX. |
| (6) Thrust reverser(s)
[operating engine(s)] | Apply | maximum; consistent with directional control. |

Take-off was rejected due to an engine fire or severe damage:**Take-off was rejected due to engine failure:**

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1. REJECTED TAKE-OFF (CONT'D)

A. Rejected Take-Off Before Achieving V_1 (Cont'd)

After a rejected take-off, procedures associated with cooling of wheels, brakes and tires must be observed (Refer to Chapter 6; PERFORMANCE – TURN-AROUND TIME – Cooling of Wheels, Brakes and Tires, in the Airplane Flight Manual).

————— **END** —————

TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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1. POWER PLANT

NOTE

In the event of a lightning strike, simultaneous fire/overheat indications could occur. Abnormal values of secondary symptoms may be used to determine if an actual fire exists.

A. Engine Fire/Severe Engine Damage (In Flight)

Indication: L (R) ENG FIRE PUSH light on, **L (R) ENGINE FIRE** warning message and fire bell on.

During take-off after achieving V_1 :

Yes

(1) Take-off Continue

(2) Airplane Rotate at V_R to an initial take-off attitude of 10 degrees, and adjust pitch, as required, to achieve an airspeed of V_2 at 35 feet.

When positive rate of climb is achieved:

(3) Landing gear Retract

(4) Airspeed Maintain at V_2 to engine-out level-off height (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – OBSTACLE CLEARANCE).

NOTE

If engine failure occurs above V_2 , maintain airspeed at current value (not more than $V_2 + 10$ KIAS).

No

After take-off and at a safe altitude:

Affected engine:

(1) ATS DISC switch Depress

(2) Thrust lever Confirm and IDLE

(3) Thrust lever Confirm and SHUT OFF

(4) ENG FIRE PUSH switch/light Press in

Check the following:

- BOTTLE ARMED PUSH TO DISCH lights on,
- **L (R) ENG SOV CLSD** advisory message on and
- **HYD SOV 1 (2) CLSD** advisory message on.

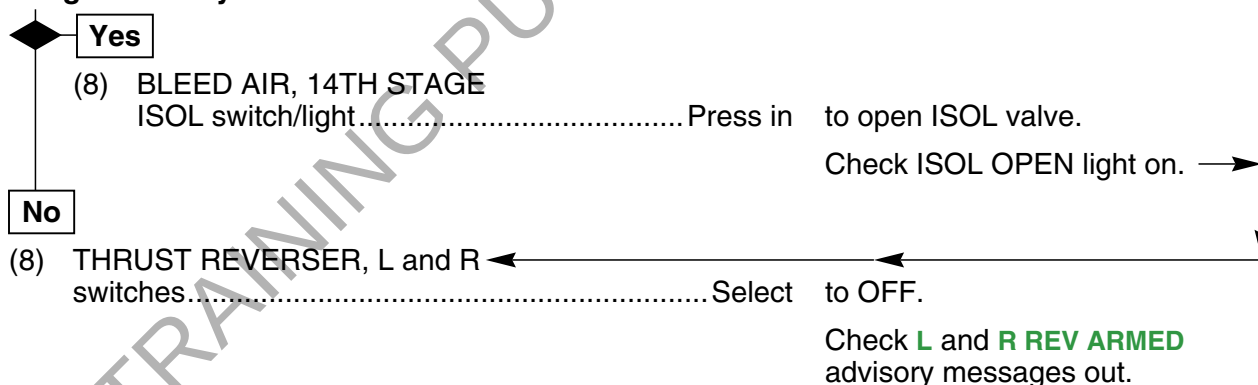
1. POWER PLANT (CONT'D)

A. Engine Fire/Severe Engine Damage (In Flight) (Cont'd)

- (5) FUEL, L (R) BOOST PUMP switch/light.....Press out to disarm affected boost pump.
Check **L (R) FUEL PUMP** caution message on.
- If warning persists (after 10 sec):**
- (6) BOTTLE ARMED PUSH TO DISCH switch/light.....Press in and hold until light goes out, to discharge firex bottle.
Check **ENG BOTTLE 1 (2) LO** caution message on.
- If warning persists (after an additional 30 sec):**
- (7) Other BOTTLE ARMED PUSH TO DISCH switch/light.....Press in and hold until light goes out, to discharge other firex bottle.
Check **ENG BOTTLE 1 (2) LO** caution message on.

After take-off (if applicable):

Wing anti-ice system selected on:



At not less than 400 feet AGL or the engine-out level off height:

- (9) Thrust Set thrust to maximum continuous thrust (MCT) (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS).
- (10) Airspeed Accelerate in level flight to V_{FTO} .
- (11) FLAPS Select to UP at a speed of not less than $V_{FTO} - 5$ KIAS.
- (12) Climb Initiate at V_{FTO} .



BOMBARDIER CHALLENGER 605	EMERGENCY PROCEDURES Power Plant	Vol. 1	03-03-3 REV 22, Mar 16/12
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1. POWER PLANT (CONT'D)

A. Engine Fire/Severe Engine Damage (In Flight) (Cont'd)

(13) CAS messages..... Review

At a safe altitude:

(14) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
– POWER PLANT – In-Flight
Engine Failure/Shutdown.

NOTE

Do not operate the APU in flight if engine rotor burst damage is suspected.

————— END —————

TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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1. POWER PLANT (CONT'D)

B. Engine Fire/Severe Engine Damage (On Ground)

Indication: L (R) ENG FIRE PUSH light on, L (R) ENGINE FIRE warning message and fire bell on.	
(1) Parking brake Set	
Affected engine:	
(2) Thrust lever SHUT OFF	
(3) ENG FIRE PUSH switch/light Press in	Check the following: • BOTTLE ARMED PUSH TO DISCH lights on, • L (R) ENG SOV CLSD advisory message on and • HYD SOV 1 (2) CLSD advisory message on.
(4) FUEL, L and R BOOST PUMP switch/lights Press out	to disarm both boost pumps. Check L and R FUEL PUMP caution messages on.
If warning persists (after 10 sec):	
(5) BOTTLE 1 and 2 ARMED PUSH TO DISCH switch/lights Press in	and hold until lights go out, to discharge firex bottles. Check ENG BOTTLE 1 and 2 LO caution messages on.
(6) Emergency Evacuation Procedure Accomplish	Refer to EMERGENCY EVACUATION – Emergency Evacuation Procedure in this chapter.

————— **END** —————

1. POWER PLANT (CONT'D)

C. Double Engine Failure

Indication: Engine failure indicated on EICAS; N_1 , N_2 , ITT and fuel indications.

- | | | |
|--------------------------------------|------------------|--|
| (1) IGNITION, CONT switch/light..... | Press in | Check IGNITION A/B advisory message on. |
| (2) Airspeed | 240 KIAS minimum | |

- (3) Engine instruments..... Verify N_1 , N_2 , ITT

If engines continue to run-down:

- (4) Thrust levers (both) SHUT OFF
- (5) ADG manual deploy handle Pull and leave extended.
- (6) AC ELECTRICAL page Check ESS BUS is powered.

When ADG power is established:

- (7) STAB TRIM, CH 2 switch Press in to engage STAB TRIM CH 2.
- (8) Oxygen masks (if required) Don Set to 100%.
- (9) Crew communication Establish Set MASK/BOOM switch to MASK.
- (10) PASS SIGNS, NO SMKG and SEAT BLTS switches Select both to ON.
- (11) APU (if available) Start when at or below 20,000 feet.
- (12) AC POWER, APU GEN switch Select to ON.
- (13) AIR-CONDITIONING, L and R PACK switch/lights Press out to select both packs off.
Check L and R PACK OFF lights on.
- (14) ANTI-ICE, WING switch Select to OFF.
- (15) ANTI-ICE, L and R COWL switch/lights Press out to select cowl anti-ice off.

WARNING

To avoid thermal seizure, increase airspeed as required to maintain at least 4% N_2 indicated. If the N_2 is allowed to drop to 0%, the engine may thermally seize and may not rotate even at higher airspeeds or with starter engagement.



1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

NOTE

- Between 21,000 feet and 10,000 feet, a minimum of 12% N_2 is necessary for a windmill relight.
- At 10,000 feet and below, a minimum of 10% N_2 is necessary for a windmill relight.
- Acceleration to V_{MO} is recommended to attain the necessary N_2 for a windmill relight.
- The altitude loss when accelerating from 240 KIAS to V_{MO} , could be more than 5,000 feet.
- A push-over to as steep as 15° nose down may be required.

Windmill relight is feasible:

Yes

(16) Glide airspeed Increase to V_{MO}



The resulting high rate of descent (up to 7800 fpm) will cause a significant reduction of available gliding distance.

(17) Descent Initiate to 21,000 feet or below, at V_{MO} .

(18) FUEL, L and R BOOST PUMP switch/lights Confirm ON

At 21,000 feet and below, attempt to start both engines at the same time.

When ITT is $\leq 90^\circ\text{C}$ and N_2 is $\geq 12\%$ (between 21,000 feet and 10,000 feet) or N_2 is $\geq 10\%$ (10,000 feet and below):

(19) IGNITION, CONT switch/light Confirm ON

(20) Thrust levers (both) IDLE

(21) Engine indications Monitor engine acceleration, ITT and oil pressure.

At least one engine relights within 25 seconds (stable IDLE within 2 minutes):

Yes

(22) Thrust lever(s) As required

Operating engine(s):

(23) Applicable AC POWER, GEN switch(es) Select to OFF/RESET then ON.

Check applicable **GEN 1 OFF** and/or **GEN 2 OFF** caution message(s) out.

1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

- (24) Applicable
AIR-CONDITIONING,
PACK switch/light(s) Press in to restart pack(s).

Check applicable PACK OFF
light(s) out.

NOTE

Use only one air-conditioning pack during single
engine operations, unless the APU is the bleed
source.

- (25) IGNITION, CONT
switch/light Press out to select continuous ignition
off.
Check **IGNITION A/B** advisory
message out.

Re-establish normal power:

- (26) ADG manual deploy
handle Stow
- (27) ADG, PWR TXFR
OVERRIDE switch Press to override.
- (28) CAS messages Review
- (29) Affected systems Restore
- (30) EFIS altimeters (2) Cross-check

Only one engine is operating:

◆ **Yes**

- (31) In-Flight Engine
Failure/Shutdown Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
– POWER PLANT – In-Flight
Engine Failure/Shutdown.

– END –

No

- (31) No further action required.

– END –

No

- (22) Thrust levers (both) SHUT OFF

1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

Another windmill relight attempt is feasible:

Yes

(23) Airspeed.....Maintain V_{MO}

(24) Wait 30 seconds, then repeat windmill relight procedure.

No

(23) Proceed to APU bleed air relight.

No

APU bleed air relight is feasible:

Yes

(16) Airspeed Maintain 240 KIAS minimum.

(17) FUEL, L and R BOOST
PUMP switch/lights..... Confirm ON(18) BLEED AIR, 10TH STAGE, L
and R switch/lights..... Press out to close bleed air valves.Check L and R 10TH STAGE
CLOSED lights on.(19) BLEED AIR, APU LCV
switch/light Press in to open LCV.Check APU LCV OPEN light
on.

Attempt to start one engine at a time.

At 15,000 feet and below:

(20) IGNITION, CONT switch/light..... Confirm ON

(21) ENG, L (R) START
switch/light Push Check L (R) ENG START light
on.When ITT is 90°C or less and N_2 reaches 28%:

(22) Thrust lever..... IDLE

(23) Engine indications Monitor engine acceleration, ITT and
oil pressure.

Engine relights within 25 seconds (stable IDLE within 2 minutes):

Yes

(24) Thrust lever.....As required

1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

Operating engine:

- | | | | |
|------|--|-----------|---|
| (25) | Applicable AC POWER,
GEN switch | Select | to OFF/RESET then ON.

Check applicable GEN 1 OFF
or GEN 2 OFF caution
message out. |
| (26) | BLEED AIR, APU LCV
switch/light | Press out | to close LCV.

Check APU LCV OPEN light
out. |
| (27) | Applicable BLEED AIR,
10TH STAGE switch/light | Press in | to open bleed air valve.

Check applicable 10TH
STAGE CLOSED light out. |
| (28) | Applicable
AIR-CONDITIONING,
PACK switch/light | Press in | to restart pack.

Check applicable PACK OFF
light out. |

NOTE

Use only one air-conditioning pack during single engine operations, unless the APU is the bleed source.

- | | | | |
|------|--------------------------------------|-----------|---|
| (29) | IGNITION, CONT
switch/light | Press out | to select continuous ignition
off.

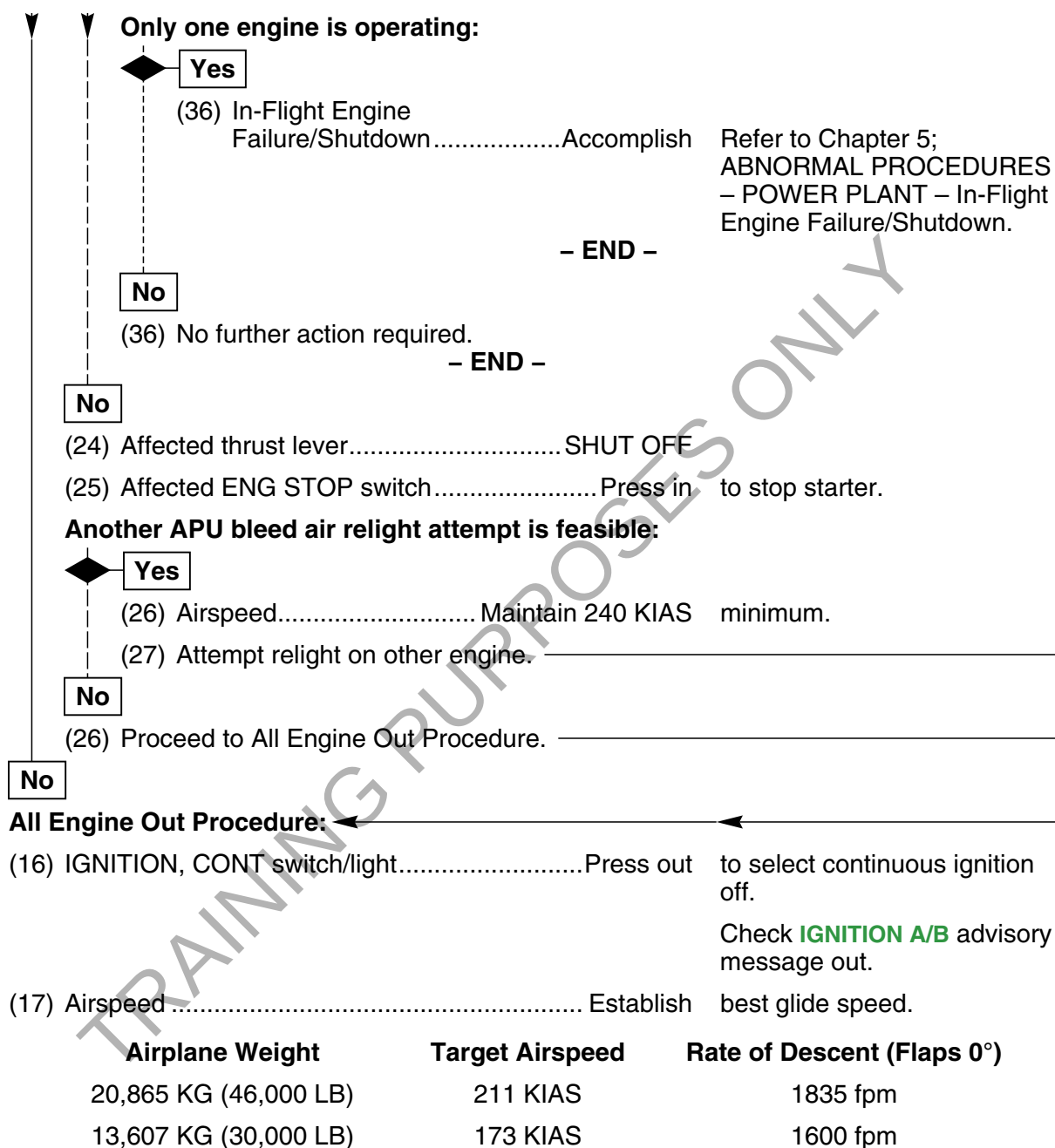
Check IGNITION A/B advisory
message out. |
|------|--------------------------------------|-----------|---|

Re-establish normal power:

- | | | | |
|------|--|-------------|--|
| (30) | ADG manual deploy
handle | Stow | |
| (31) | ADG, PWR TXFR
OVERRIDE switch | Press | to override. |
| (32) | CAS messages | Review | |
| (33) | Affected systems | Restore | |
| (34) | EFIS altimeters (2) | Cross-check | |
| (35) | Other engine | Start | Refer to Chapter 5;
ABNORMAL PROCEDURES
– AIR START. |

1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

**NOTE**

1. Target glide speed provides a range of approximately 2.5 NM for each 1,000 feet altitude.
2. Refer to Flight Planning and Cruise Control Manual, Chapter 6: Special Operational Data – Descent with Two Engines Inoperative, for detailed descent (time/distance) performance data.

1. POWER PLANT (CONT'D)

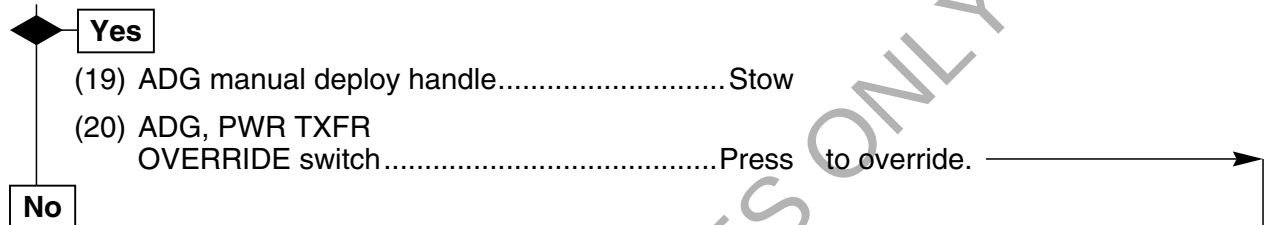
C. Double Engine Failure (Cont'd)

(18) Descent and approachPlan

NOTE

Recommended approach is a descending 360° turn starting at 5,000 feet above the landing area.

APU GEN available:



The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2
- Ground spoilers
- Flaps channel 2
- Stabilizer trim channel 1
- Mach trim, aileron and rudder trim
- Hydraulic pumps 1B, 2B & 3A
- DME, FMS 2
- Copilot's PFD, MFD, navigation & communication systems
- R Pack
- Right probe heaters, right AOA vane heater and right ice detector
- Nose wheel steering system, anti-skid system
- Both windshield heaters, right window heater
- Copilot's instrument lights, NAV lights, and L & R landing lights
- SPS altitude compensation
- Auto throttle system

At 10,000 feet MSL or lowest safe altitude:

- (19) AIR-CONDITIONING,
RAM-AIR switch/light.....Press in to open ram air valve.
Check RAM AIR OPEN light on.
- (20) TAWS WARNING, FLAPS OFF
switch/light.....Press in to mute flap aural warning.
- (21) TAWS WARNING, RAAS OFF
switch/light (if installed)Press in Check OFF light on.



1. POWER PLANT (CONT'D)

C. Double Engine Failure (Cont'd)

- (22) Approach start point

Establish

at 5,000 feet AGL over landing area.
- (23) Turn (20° to 30° bank angle)

Start

over landing area.
- (24) Target airspeed

Maintain
- (25) Downwind leg

Establish

at 2,500 feet AGL abeam landing area and 180° from final approach heading.
- (26) FLAPS

Select

to 20° for landing.
- (27) LDG GEAR lever

DN

prior to turning final.
- (28) Final approach

Establish

at 1,500 feet AGL.
- (29) Final approach speed

V_{REF} (Flaps 45°)
+ 14 KIAS

minimum.
- (30) Actual landing distance

Increase

as applicable by factor given below:

APU generator available	Emergency power only
1.55 (55%)	3.20 (220%)

When time and conditions permit:

- (31) Ditching and Forced Landing procedures

Review

Refer to DITCHING AND FORCED LANDING procedures in this chapter.

————— END —————

1. POWER PLANT (CONT'D)

D. Jet Pipe Overheat

Indication: **L (R) JETPIPE OVHT** warning message and “JETPIPE OVERHEAT” aural on.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and retard slowly until warning message goes out.

L (R) JETPIPE OVHT warning message persists:

Yes

(3) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

– END –

No

(3) No further action required.

END

1. POWER PLANT (CONT'D)

E. Engine Oil Pressure Low

Indication: **L (R) ENG OIL PRESS** warning message, "ENGINE OIL" aural on, or left or right engine oil pressure is below 25 psi.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and IDLE

(3) Oil pressure Check

L (R) ENG OIL PRESS warning message is on
and affected engine oil pressure is below 25 psi:

Yes

(4) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

– END –

No

Affected engine:

(4) Thrust lever Adjust as required.

(5) Engine indications Monitor

— END —

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment

Indication: **L (R) REV UNLOCKED** caution message.

On ground:

Yes



If the **L** or **R REV UNLOCKED** caution message is on during ground operations, ensure that the affected thrust lever is at IDLE. Do not cycle the thrust reverser(s). Maintenance inspection is required.

– END –

No

In flight:

Affected engine:

- (1) Thrust lever Confirm and IDLE
- (2) ATS DISC switch Depress
- (3) THRUST REVERSER, EMER
STOW switch/light Press in

Note that:

- **L (R) REV UNLOCKED** caution message will remain, and
- Stow indicated by improved airplane handling and normal operation of affected thrust lever.

- (4) Airspeed Not more than 200 KIAS.

Anti-ice is on:

Yes

- (5) ANTI-ICE, WING switch Select to OFF.

- (6) Affected ANTI-ICE, COWL
switch/light Press out to select cowl anti-ice off.

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

No

Thrust reverser stows:

Yes

CAUTION

Limit airspeed to not more than 200 KIAS.

(5) Thrust levers..... As required

(6) THRUST REVERSER, L and
R switches Select to OFF.Check **L** and **R REV ARMED**
advisory messages out.

CAUTION

Do not arm thrust reverser.

(7) Land at the nearest suitable airport.

Anti-ice is required:

Yes

(8) ANTI-ICE, WING switch..... Select to NORM.

(9) Affected ANTI-ICE,
COWL switch/light Press in to select cowl anti-ice on.

No

Prior to landing:

(8) TAWS WARNING, FLAPS
OFF switch/light..... Press in to mute flap aural warning.(9) TAWS WARNING,
RAAS OFF switch/light
(if installed) Press in Check OFF light on.

(10) FLAPS Select to 20° for landing.

(11) Final approach speed Not less than V_{REF} (Flaps 45°) + 22 KIAS.

(12) Actual landing distance..... Increase by a factor of 1.70 (70%).

- END -

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

No**Thrust lever locked at idle, and continued airplane buffet and yaw:****Affected engine:**

- (5) Thrust lever Confirm and
SHUT OFF



Do not relight engine.

- (6) Land at the nearest suitable airport.
- (7) FUEL, TAIL TANK TRANSFER,
MANUAL OVRD switch/light Press out if MANUAL light on.
- (8) Affected FUEL, L (R) BOOST
PUMP switch/light Press out to disarm affected boost
pump.
Check **L (R) FUEL PUMP**
caution message on.
- (9) Fuel system Check and monitor the following:
- Fuel quantity, and
 - Fuel balance.

Anti-ice is required:**Yes**

- (10) BLEED AIR, 14TH STAGE
ISOL switch/light Press in to open ISOL valve.
Check 14TH ISOL OPEN light
on.
- (11) ANTI-ICE, WING switch Select to NORM.
- (12) Leave icing conditions (if possible).

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

No

- (10) HYDRAULIC pump switch.....Select applicable pump to ON. ←
- If left engine shut down, turn 1B ON. Check that **HYD 1 LO PRESS** caution message out.
 - If right engine shut down, turn 2B ON. Check that **HYD 2 LO PRESS** caution message out.
- (11) AIR-CONDITIONING,
L (R) PACK switch/lightPress out to shutdown affected pack.
Check L (R) PACK OFF light on.
- (12) BLEED AIR, 10TH STAGE
L (R) switch/lightPress out to close affected shut-off valve.
Check L (R) CLOSED light on.

NOTE

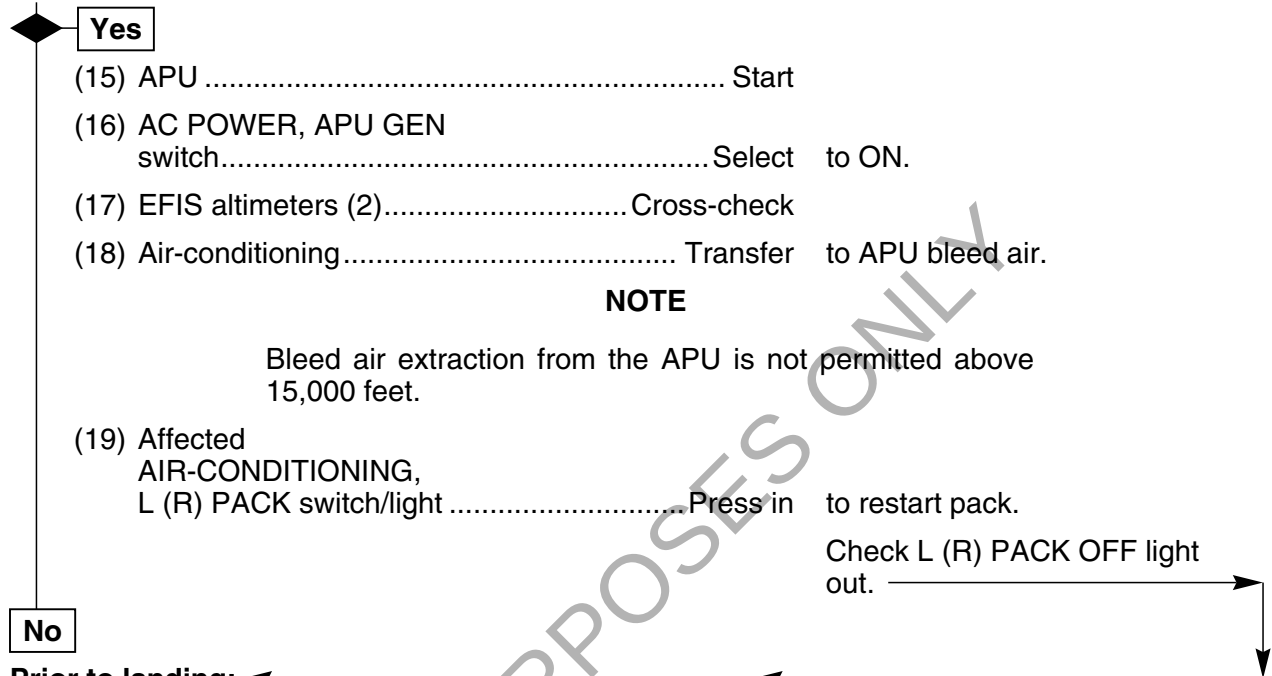
Use only one air-conditioning pack during single engine operations, when the operating engine is the only 10th-stage bleed source.

- (13) AC POWER, GEN 1 (2) switch.....Select affected generator to OFF.
Check **GEN 1 (2) OFF** caution message on.
- (14) EFIS altimeters (2)Cross-check

1. POWER PLANT (CONT'D)

F. Uncommanded Thrust Reverser Deployment (Cont'd)

Below 20,000 feet and APU is available:



Prior to landing: ←

- (15) TAWS WARNING, FLAPS OFF switch/light Press in to mute flap aural warning.
- (16) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (17) FLAPS Select to 20° for landing.
- (18) Final approach speed Not less than V_{REF} (Flaps 45°) + 22 KIAS.
- (19) Actual landing distance Increase by a factor of 1.70 (70%).

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

1. POWER PLANT (CONT'D)

G. Post-Shutdown Engine Tail Pipe Fire

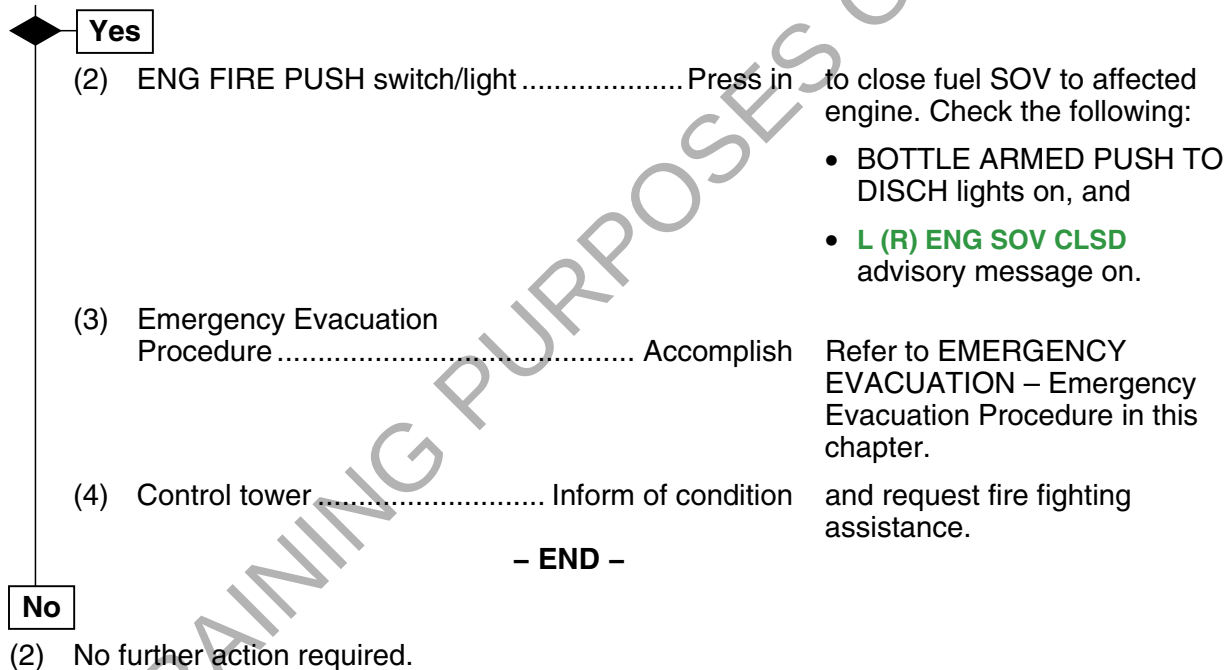
Indication: Engine ITT exceeds 350°C and ITT is rising rapidly, or if flame is reported at engine tailpipe.

Affected engine:

- (1) Dry motor..... Accomplish until ITT is reduced below 150°C or starter limit.

NOTE

The APU or ground air source (if available) may be used to dry motor the engine.

Tail pipe fire indications persist:

1. POWER PLANT (CONT'D)

H. N₁/N₂ Overspeed

Indication: **ENGINE** warning message. Engine N₁ or N₂ readout indicates engine limits being exceeded.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and retard until engine responds.

Engine limits continue to be exceeded:

Yes

(3) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

– END –

No

(3) Both thrust levers Retard to below 79% N₂.

(4) Both ENGINE CONTROL ENG SPEED switches OFF to prevent asymmetric thrust.

NOTE

The APR system is inoperative when the ENG SPEED switches are selected OFF.

(5) Both thrust levers Advance

NOTE

Thrust lever controls are more sensitive to changes.



1. POWER PLANT (CONT'D)

H. N₁/N₂ Overspeed (Cont'd)

Engine accelerates to desired power:

Yes

(6) Engine indications Monitor

NOTE

Exercise caution to prevent overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.

- END -

No

(6) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
- POWER PLANT - In-Flight
Engine Failure/Shutdown.

END

1. POWER PLANT (CONT'D)

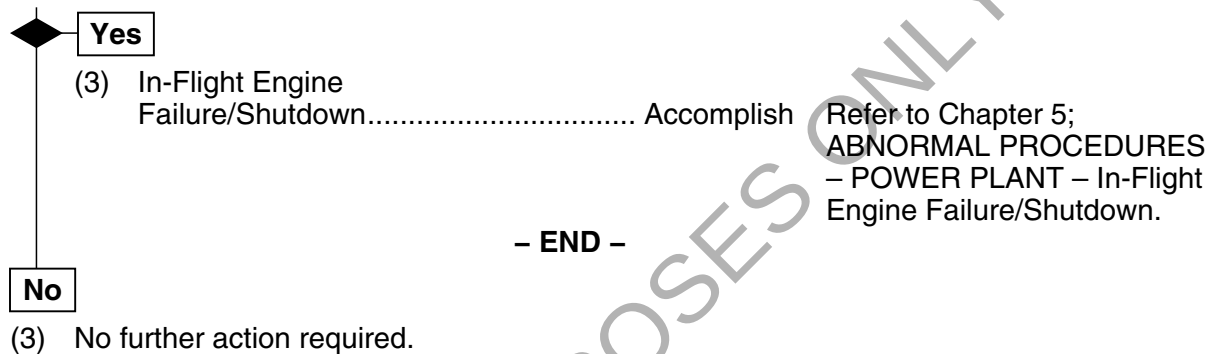
I. Engine Overtemperature

Indication: **ENGINE** warning message. Engine ITT readout indicates engine limits being exceeded.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Confirm and retard until ITT below limit range.

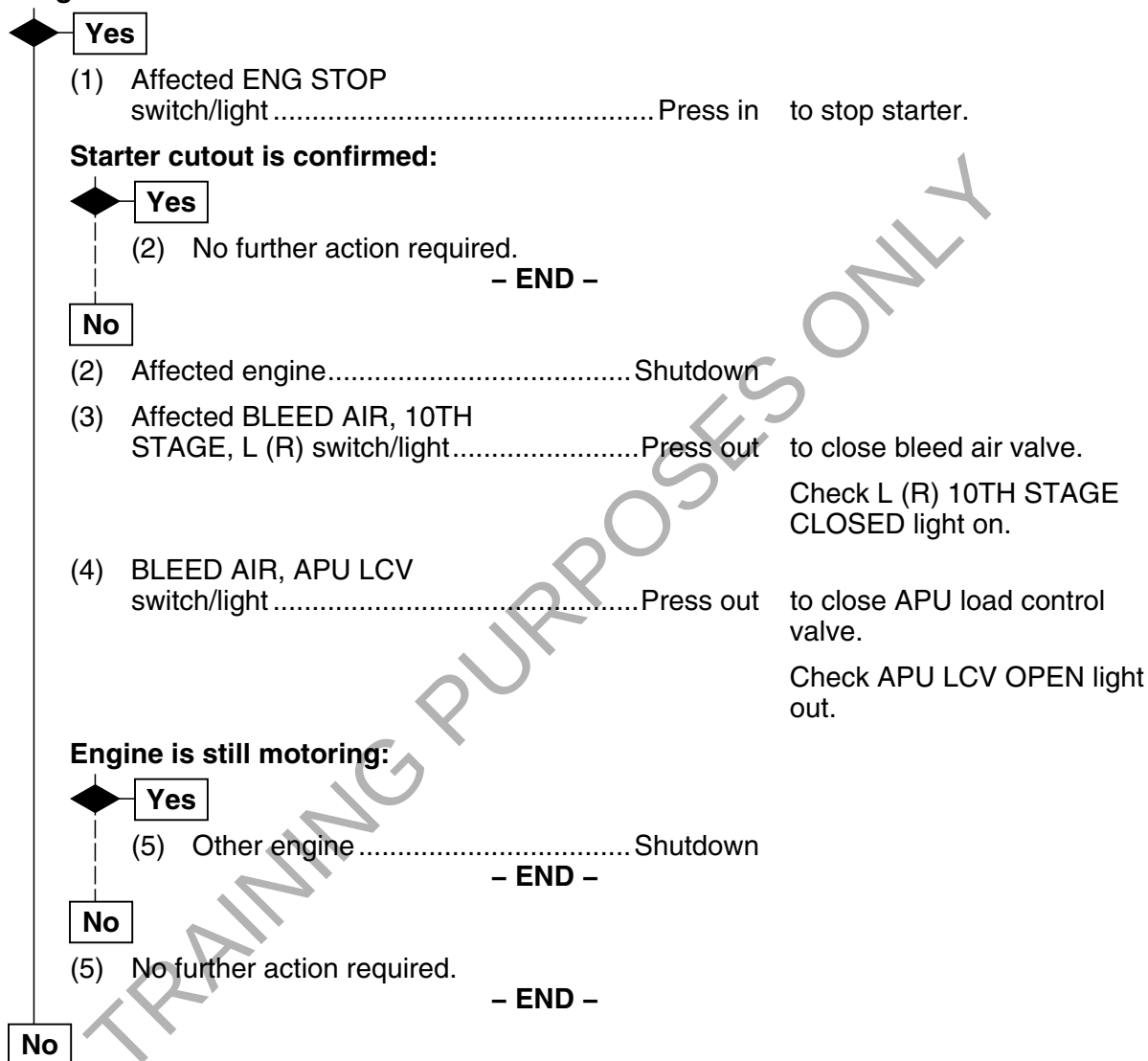
ITT limits continue to be exceeded:

1. POWER PLANT (CONT'D)

J. Engine Starter Fails to Cutout

Indication: **L (R) ENGINE START SOV** warning message. L (R) ENGINE START light and **IGNITION A (B)** advisory message on after 55% N₂.

On ground:



In-flight:

- (1) Affected ENG STOP switch/light Press in to stop starter.
- (2) IGNITION, A (B) switch/light Press out to disarm applicable ignition system.
Check ARM light out at applicable switch/light.
- (3) Affected ENG START circuit breaker (L-1L10) (R-1L9) Open



1. POWER PLANT (CONT'D)

J. Engine Starter Fails to Cutout (Cont'd)

Engine is still motoring:

 Yes

(4) Affected 10 ST B/AIR SOV
circuit breaker (L-4C11)
(R-2F11) Open

(5) Land at the nearest suitable airport.
- END -

No

(4) No further action required.

END

TRAINING PURPOSES ONLY

1. POWER PLANT (CONT'D)

K. Uncommanded Acceleration

Indication: Engine accelerates to thrust settings other than as selected with thrust lever.

(1) ATS DISC switch Depress

On ground, affected engine:

(2) Thrust lever Confirm and IDLE then SHUT OFF.

During flight, affected engine:

(2) Thrust lever Confirm and IDLE Check engine response.

CAUTION

The ATS **SPEED** mode N_1 synchronization will attempt inappropriate deceleration/acceleration of the remaining engine to maintain synchronization until disengaged.

Engine responds:

Yes

(3) Both thrust levers Advance to below 79% N_2 .

(4) Both ENGINE CONTROL
ENG SPEED switches OFF

NOTE

The APR system is inoperative when the ENG SPEED switches are selected OFF.

(5) Both thrust levers Advance

CAUTION

Do not change power settings abruptly as engine response is degraded.

(6) Engine indications Monitor

NOTE

Exercise caution to prevent overspeed during thrust reverser use, when operating with ENG SPEED switches OFF.

– END –

1. POWER PLANT (CONT'D)

K. Uncommanded Acceleration (Cont'd)

No

- (3) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5;
 ABNORMAL PROCEDURES
 – POWER PLANT – In-Flight
 Engine Failure/Shutdown.

————— **END** —————

TRAINING PURPOSES ONLY

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TRAINING PURPOSES ONLY

1. SMOKE/FIRE/FUMES

A. Smoke/Fire/Fumes Procedure

WARNING

Time is critical during smoke/fire/fumes emergencies. The flight crew should consider an immediate landing anytime the situation cannot be controlled.

CAUTION

Passenger oxygen masks should not be deployed when performing smoke or fire procedures.

- (1) Oxygen masks/smoke goggles
(if required) Don Set to 100% and EMERGENCY.
- (2) Crew communication Establish Set MASK/BOOM switch to MASK.
- (3) Land at the nearest suitable airport.

CAUTION

Continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished and the smoke or fumes have dissipated.

WARNING

Anytime smoke or fumes becomes the greatest threat, accomplish the Smoke or Fumes Removal Procedure. Refer to Smoke or Fumes Removal Procedure in this section.

Time and conditions permit and source of smoke/fire/fumes can be quickly identified and eliminated:

Yes

- (4) Appropriate procedure Accomplish
 - Air-Conditioning..See SOURCE IS IDENTIFIED AS AIR-CONDITIONING SYSTEM
 - Electrical.....See SOURCE IS IDENTIFIED AS ELECTRICAL SYSTEM
 - Cabin.....See SOURCE IS IDENTIFIED AS CABIN

- END -

1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

No

- (4) Smoke or Fumes Removal Procedure..... Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

- END -

SOURCE IS IDENTIFIED AS AIR-CONDITIONING SYSTEM

APU is supplying bleed air to the packs:

Yes

- (1) Air-conditioning..... Transfer to engine bleeds.

No

- (1) AIR-CONDITIONING, L PACK switch/light..... Press out to shutdown left pack.
Check L PACK OFF light on.

Monitor and determine if smoke/fire/fumes persists.

Smoke/fire/fumes persists:

Yes

- (2) AIR-CONDITIONING, L PACK switch/light Press in Check L PACK OFF light out.
- (3) AIR-CONDITIONING, R PACK switch/light..... Press out to shutdown right pack.
Check R PACK OFF light on.

Monitor and determine if smoke/fire/fumes persists.

Smoke/fire/fumes persists:

Yes

- (4) AIR-CONDITIONING, R PACK switch/light..... Press in Check R PACK OFF light out.

No

If required:

- (2) Smoke or Fumes Removal Procedure..... Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

- END -



1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

SOURCE IS IDENTIFIED AS ELECTRICAL SYSTEM

(1) AC and DC electrical loads Monitor

Source of electrical smoke/fire/fumes is positively identified:

Yes

WARNING

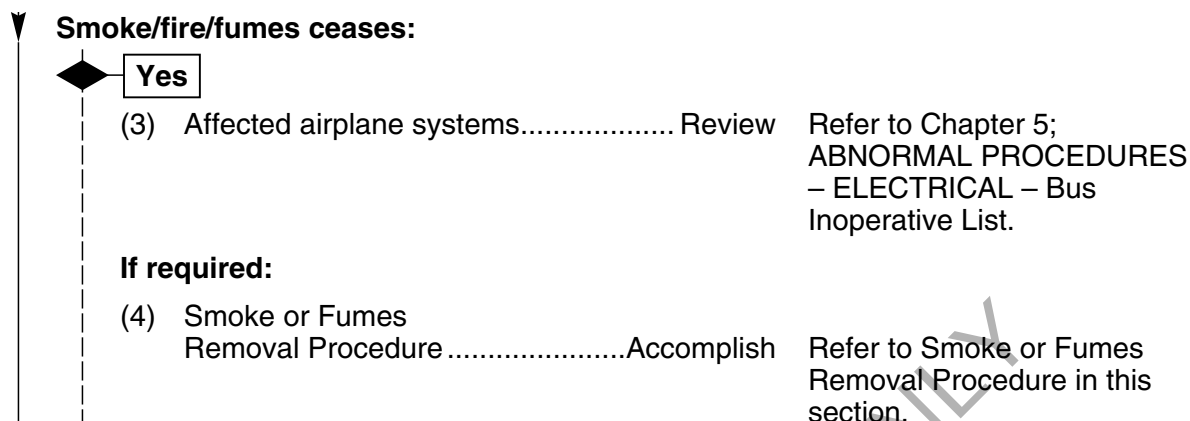
The table that follows provides guidance on how to isolate each electrical bus. Select only the steps required to isolate the positively identified bus. Do not use the table in an attempt to identify the source bus by trial and error.

(2) Affected electrical system..... Isolate See table below:

IDENTIFIED BUS	APPLICABLE CIRCUIT BREAKER/SWITCH
AC BUS 1 and DC BUS 1	AC ESS XFER..... PRESS IN LEFT AUTOXFER..... PRESS IN GEN 1 SWITCH OFF
AC BUS 2 and DC BUS 2	RIGHT AUTOXFER..... PRESS IN GEN 2 SWITCH OFF
DC BUS 1 CONTROL	FEED CB (1E8) OPEN
DC BUS 2 CONTROL	FEED CB (2E8) OPEN
AC ESS BUS	115 VAC ESS FEED CBs (1B5 and 2B8) OPEN
DC ESS BUS	RCCB CONT, FEED 1 and FEED 2 DC ESS CBs (2M6, 2M8 and 2M10) OPEN
DC BATT BUS	RCCB CONT MAIN and APU BATT CBs (2M4 and 2M5) and FEED 1 and FEED 2 BATT BUS CBs (2M7 and 2M9) OPEN
DC EMER BUS	APU BATT FEED CB (1S7) and EMERG BUS FEED CB (1M1) OPEN
AC UTLY 1 and 2 DC UTLY 1 and 2	AC/DC UTILITY PRESS IN

1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)



– END –

- No
- (2) ADG manual deploy handle Pull and leave extended.
- (3) AC ELECTRICAL page Check ESS BUS is powered.
- When ADG power is established:**
- (4) STAB TRIM, CH 2 switch Press in to engage STAB TRIM CH 2.
- (5) AIR-CONDITIONING, R PACK switch/light Press out to deselect.
- (6) AC POWER, GEN 1 and GEN 2 switches Select to OFF/RESET.
- (7) AC POWER, APU GEN switch Select to OFF/RESET.
- (8) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.
- (9) Leave icing conditions (if possible).

NOTE

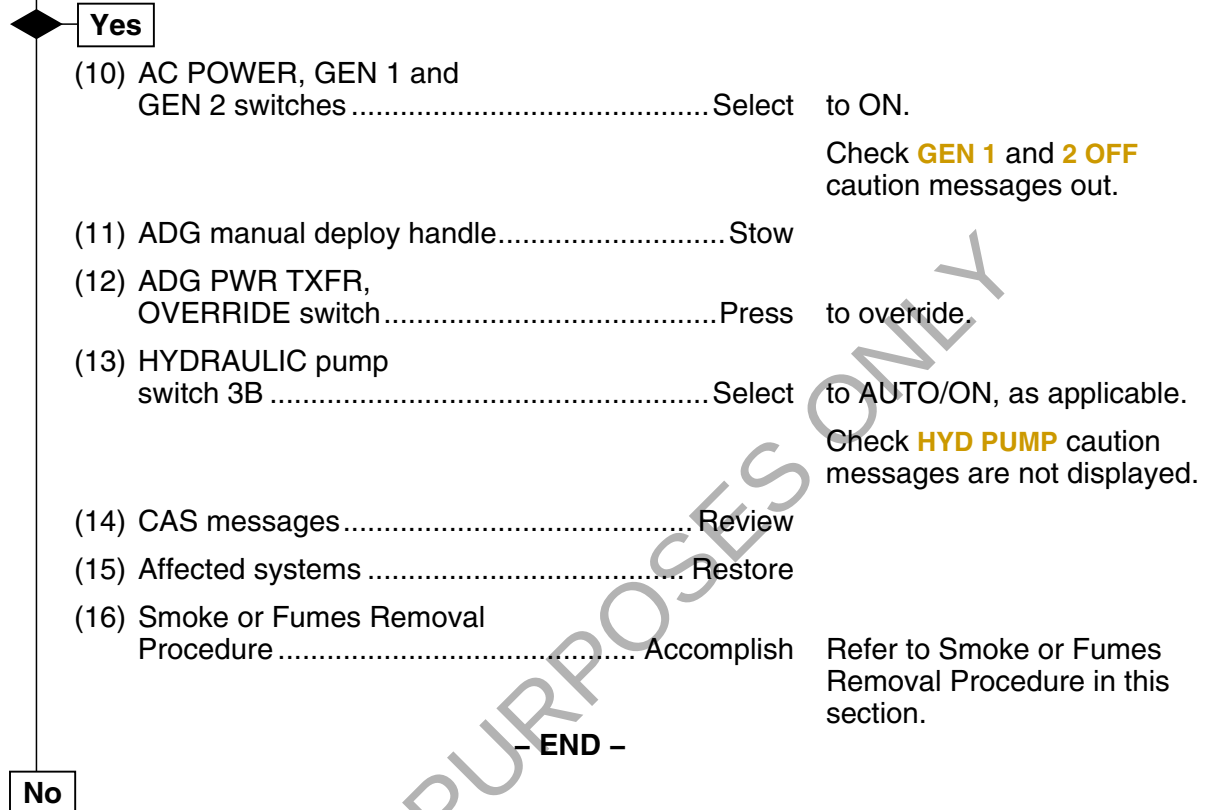
Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.



1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

Smoke/fire/fumes persists:



The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2
- Ground spoilers
- Flaps channel 2
- Stabilizer trim channel 1
- Mach trim, aileron and rudder trim
- Hydraulic pumps 1B, 2B & 3A
- DME, FMS 2
- Copilot's PFD, MFD, navigation & communication systems
- R Pack
- Right probe heaters, right AOA vane heater and right ice detector
- Nose wheel steering system, anti-skid system
- Both windshield heaters, right window heater
- Copilot's instrument lights, NAV lights, and L & R landing lights
- SPS altitude compensation
- Auto throttle system



1. SMOKE/FIRE/FUMES (CONT'D)

A. Smoke/Fire/Fumes Procedure (Cont'd)

NOTE

- With Mach trim not available, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.
- Without probe and AOA vane heaters, do not rely on stall protection system, ISI for altitude and airspeed, and cabin Δpressure on EICAS.
- Flaps will operate (extend or retract) at half speed.

(10) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.50 (150%)	1.85 (85%)



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

(11) Smoke or Fumes Removal Procedure Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

– END –

SOURCE IS IDENTIFIED AS CABIN

- Designated crew member Advise to isolate and extinguish source of fire or smoke, and to secure the area as necessary.
- AC/DC UTILITY switch/light Press in to isolate AC and DC utility busses.
Check OFF light on.
- Smoke or Fumes Removal Procedure Accomplish Refer to Smoke or Fumes Removal Procedure in this section.

————— END —————

1. SMOKE/FIRE/FUMES (CONT'D)

B. Smoke or Fumes Removal Procedure

- | | | |
|--|----------|---|
| (1) Descent | Initiate | to 10,000 feet or lowest safe altitude, whichever is higher. |
| (2) Passenger oxygen..... | Select | to CLOSED. |
| (3) CABIN PRESSURIZATION,
PRESS CONT switch/light | Press in | to select manual control.

Check PRESS CONT
MANUAL light on. |
| (4) CABIN PRESSURIZATION,
MAN ALT selector | Select | UP and hold. |
| (5) CABIN PRESSURIZATION,
MAN RATE control | Select | to maximum INCR. |
| (6) CABIN PRESSURIZATION,
EMER DEPRESS switch/light | Press in | Check the following: <ul style="list-style-type: none"> EMER DEPRESS ON light on, and EMER DEPRESS caution message on. |

NOTE

To assist in smoke clearance, the ram air vent may be used. Maintain at least 250 KIAS to clear smoke using the ram air vent.

Smoke/fire/fumes is uncontrolled:



- (7) Land immediately at the nearest suitable airport.



Dependant upon the severity of the situation, the flight crew should expedite the landing. The crew should also consider an overweight landing, tailwind landing, ditching or a forced off-airport landing.

- | | | |
|---|------------|--|
| (8) Emergency Evacuation
Procedure | Accomplish | Refer to EMERGENCY
EVACUATION – Emergency
Evacuation Procedure in this
chapter. |
|---|------------|--|

– END –

1. SMOKE/FIRE/FUMES (CONT'D)

B. Smoke or Fumes Removal Procedure (Cont'd)

No

If required:

(7) Smoke/Fire/Fumes Procedure Continue Refer to Smoke/Fire/Fumes Procedure in this section.

————— **END** —————

TRAINING PURPOSES ONLY

1. AIR-CONDITIONING AND PRESSURIZATION

A. Emergency Descent Procedure

(1) Oxygen masks.....	Don	set to 100%.
(2) Crew communication	Establish	set MASK/BOOM switch to MASK.
(3) PASS SIGNS, NO SMKG and SEAT BLTS switches	Select	both to ON.
(4) Passenger oxygen.....	Deploy	
(5) Descent	Initiate	to 10,000 feet or lowest safe altitude, whichever is higher.
(6) Thrust levers.....	IDLE	
(7) ATS DISC switch	Depress	
(8) FLIGHT SPOILER lever	Select	to MAX.
		Check EICAS indicates deployment.

- (9) Airspeed Do not exceed V_{MO}/M_{MO} .



If structural damage is suspected, do not exceed the airspeed at which the damage occurred, and minimize maneuvering loads.



Operation above 10,000 feet cabin altitude may result in hypoxia.

At a safe altitude:

- (10) Oxygen and masks..... As required

NOTE

- Closing the doors on the mask stowage compartments and pressing RESET will stop the flow of oxygen to the masks.
- If supplemental crew oxygen is still required, setting masks to normal (N) will reduce consumption.

- (11) Passenger oxygen..... Select to CLOSED then NORMAL to reset passenger oxygen.

————— **END** —————

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

B. Uncontrollable Loss of Cabin Pressure

Indication:

CABIN ALT warning message and “CABIN PRESSURE” aural on.

(1) Oxygen masks.....

Don

set to 100%.

(2) Crew communication.....

Establish

set MASK/BOOM switch to MASK.

(3) PASS SIGNS, NO SMKG and SEAT BLTS switches

Select

both to ON.

(4) Passenger oxygen.....

Deploy

(5) Descent

Initiate

to 10,000 feet or lowest safe altitude, whichever is higher.

(6) Thrust levers.....

IDLE

(7) ATS DISC switch

Depress

(8) FLIGHT SPOILER lever

Select

to MAX.

Check EICAS indicates deployment.

- (9) Airspeed

Do not exceed

V_{MO}/M_{MO} .



If structural damage is suspected, do not exceed the airspeed at which the damage occurred, and minimize maneuvering loads.



Operation above 10,000 feet cabin altitude may result in hypoxia.

At a safe altitude:

- (10) Crew oxygen masks

As required

NOTE

1.

Closing the doors on the crew mask stowage compartments and pressing RESET will stop the flow of oxygen to the crew masks.
2.

If supplemental crew oxygen is still required, setting masks to normal (N) will reduce consumption.

- (11) Passenger oxygen.....

Select

to CLOSED then NORMAL to reset passenger oxygen.

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

B. Uncontrollable Loss of Cabin Pressure (Cont'd)

Accompanied by any door/hatch message or unsafe condition:

Yes

- **PASSENGER DOOR** warning message and “DOOR” aural,
- **BAGGAGE DOOR** caution message, and
- **PAX DOOR OUT HNDL (STOW) (LATCH)** caution messages.

(12) Do not attempt to repressurize the airplane.

(13) Applicable door procedure..... Accomplish when at a safe cabin altitude (Refer to PASSENGER DOOR in this chapter or to Chapter 5; ABNORMAL PROCEDURES – DOORS).

- END -

No

Attempt to regain control of cabin pressurization:

(12) CABIN PRESSURIZATION, PRESS CONT switch/light Press in to select manual mode. Check PRESS CONT MANUAL light on.

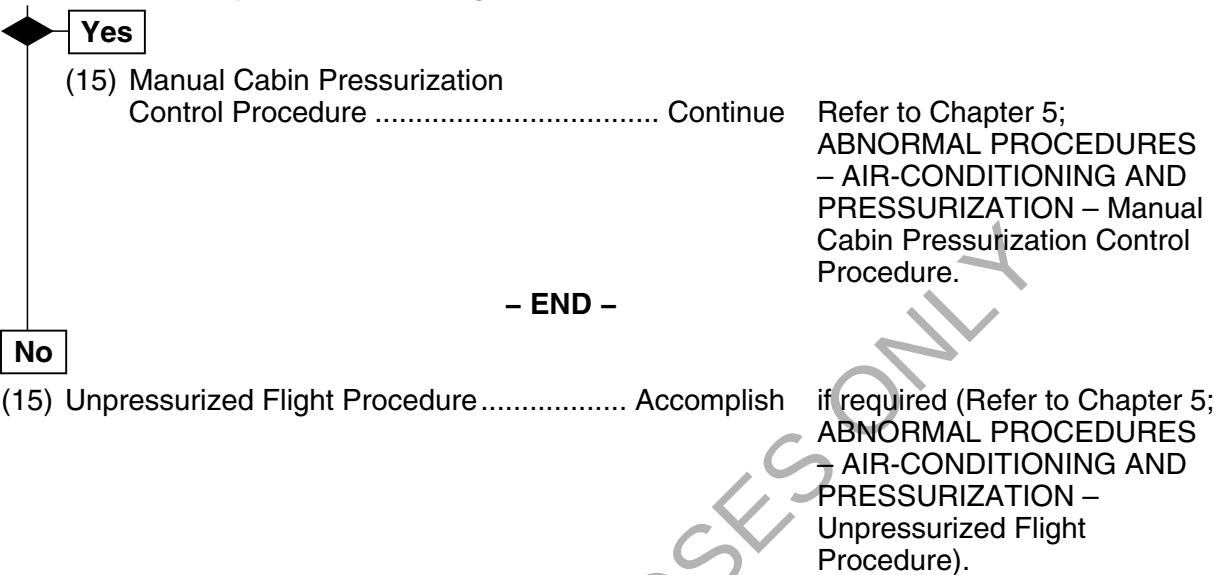
(13) CABIN PRESSURIZATION, MAN ALT selector Select DN and hold to decrease cabin altitude.

(14) CABIN PRESSURIZATION, MAN RATE switch Select to maximum INCR.

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

B. Uncontrollable Loss of Cabin Pressure (Cont'd)

Control of cabin pressurization regained:



NOTE

CABIN ALT warning message may re-appear during descent, if a static port sensing line is blocked, and should be considered a nuisance message.

— END —

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

C. High Rate of Cabin Pressurization or Over-Pressurization

Indication: **DIFF PRESS** warning message.
Differential pressure indicated as greater than 9.0 psi.

- (1) CABIN PRESSURIZATION, PRESS CONT switch/light Press in to select Manual mode.
Check PRESS CONT MANUAL light on.
- (2) CABIN PRESSURIZATION, MAN ALT selector Select UP and hold.
- (3) CABIN PRESSURIZATION, MAN RATE switch As required to increase cabin altitude and reduce differential pressure.

DIFF PRESS warning message persists:

- ◆ **Yes**
- (4) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.
 - (5) AIR-CONDITIONING, L (R) PACK switch/light Press out to shutdown one pack.
Check L (R) PACK OFF light on.

DIFF PRESS warning message persists:

- ◆ **Yes**
- (6) AIR-CONDITIONING, L and R PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.

DIFF PRESS warning message persists:

- ◆ **Yes**
- (7) CABIN PRESSURIZATION, EMER DEPRESS switch/light Press in to depressurize the airplane.
Check **EMER DEPRESS** caution message on.
 - (8) Descent Initiate to the lowest possible altitude.

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

C. High Rate of Cabin Pressurization or Over-Pressurization (Cont'd)

(9) Unpressurized Flight ProcedureAccomplish

Refer to Chapter 5; ABNORMAL PROCEDURES – AIR-CONDITIONING AND PRESSURIZATION – Unpressurized Flight Procedure.

No

- END -

- (4) Cabin altitude/differential pressureMonitor

(5) Manual Cabin Pressurization Control Procedure Continue

Refer to Chapter 5; ABNORMAL PROCEDURES – AIR-CONDITIONING AND PRESSURIZATION – Manual Cabin Pressurization Control Procedure.

NOTE

DIFF PRESS warning message may re-appear during descent, if a static port sensing line is blocked, and should be considered a nuisance message.

— END —

1. AUTOMATIC FLIGHT CONTROL SYSTEM

NOTE

After generator transfers, AFCS channel 2 will momentarily fail and the following must be manually re-engaged:

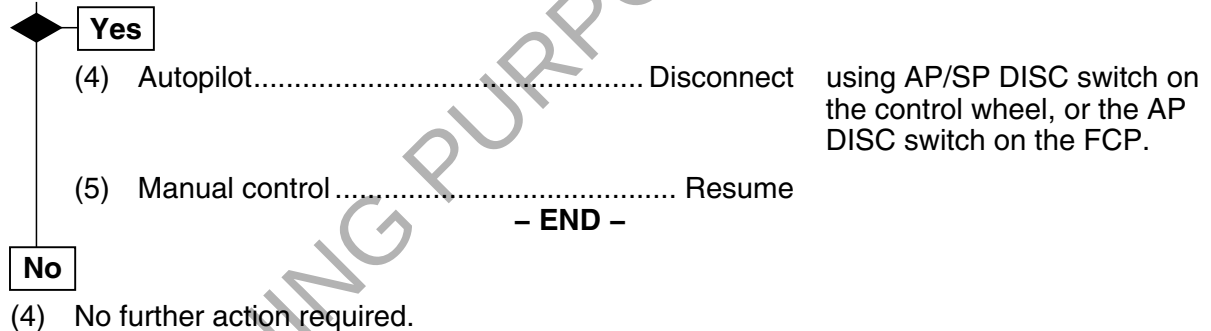
- Yaw damper 2 (YD 2), and
- Autopilot (if it was coupled before the power interruption, and operating on AFCS channel 2).

A. Autopilot Failure

Indication: Abnormal autopilot operation and/or "CAVALRY CHARGE" aural and/or flashing AP (red) message on the primary flight displays.

- (1) Autopilot Disconnect using AP/SP DISC switch on the control wheel, or the AP DISC switch on the FCP.
- (2) AFCS SEL 1 (2) switch (centre pedestal)..... Select to operative AFCS channel.
Check **AFCS 1 (2) INOP** status message on.
- (3) Autopilot Re-engage

Autopilot warning persists:



1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

B. AFCS Messages Failure

Indication: **AFCS MSGS FAIL** warning message and/or flashing AP (red) message on the primary flight displays.

- (1) Autopilot Disconnect
- using AP/SP DISC switch on the control wheel, or the AP DISC switch on the FCP.

NOTE

Do not use the autopilot when an **AFCS MSGS FAIL** warning message is displayed.

- I (2) Manual control Resume

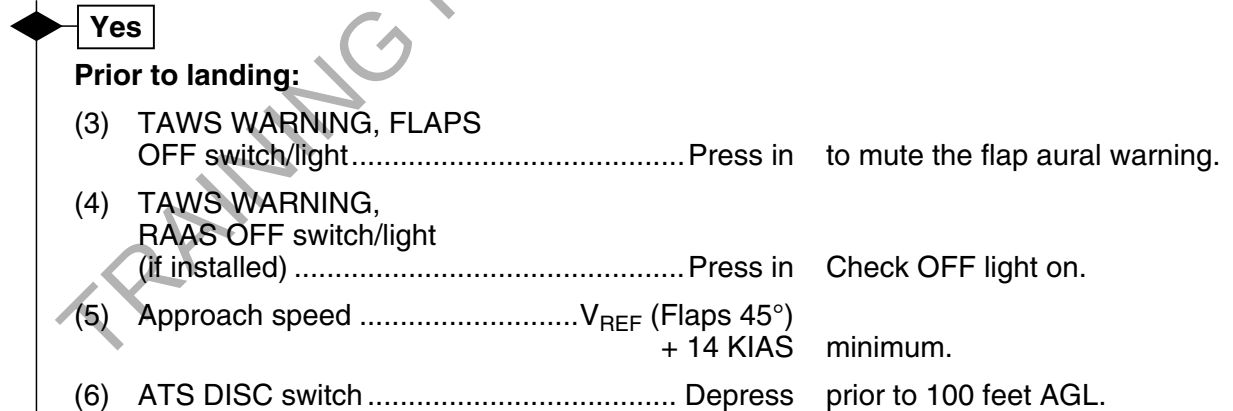
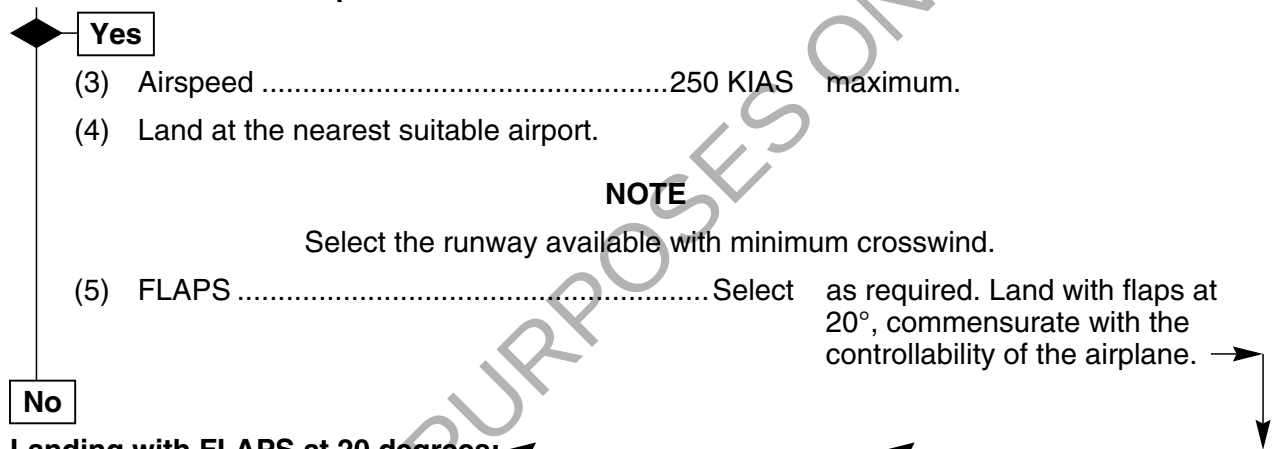
————— END —————

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

C. Uncommanded Yaw Motion

Indication:	Abnormal/uncommanded change in yaw attitude. Continuous or intermittent rudder "kicks". Sustained oscillation ("Dutch Roll").		
(1) Controls	Assume	manual control and counter aircraft motion using handwheel inputs.	
(2) YAW DAMPER, DISC button	Press in	to disconnect both yaw dampers.	

Uncommanded motion persists:



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

C. Uncommanded Yaw Motion (Cont'd)

- ▼ (7) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

- (3) No further action required.

— END —

1. AUXILIARY POWER PLANT

A. APU Fire

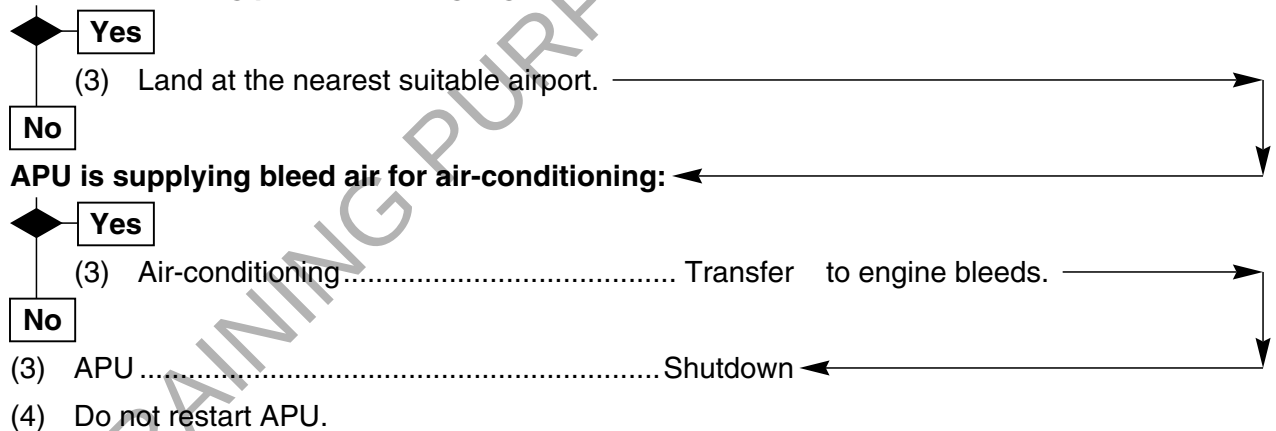
Indication: APU FIRE PUSH light on, **APU FIRE** warning message and fire bell on.

- (1) APU FIRE PUSH switch/light Press in Check the following:
- BOTTLE ARMED PUSH TO DISCH light on, and
 - **APU SOV CLSD** advisory message on.

After 5 seconds and APU fire warning persists:

- (2) BOTTLE ARMED PUSH TO DISCH switch/light..... Press in and hold until light out, to discharge firex bottle.
- Check the following:
- BOTTLE ARMED PUSH TO DISCH light out, and
 - **APU BOTTLE LO** caution message on.

APU fire warning persists during flight:



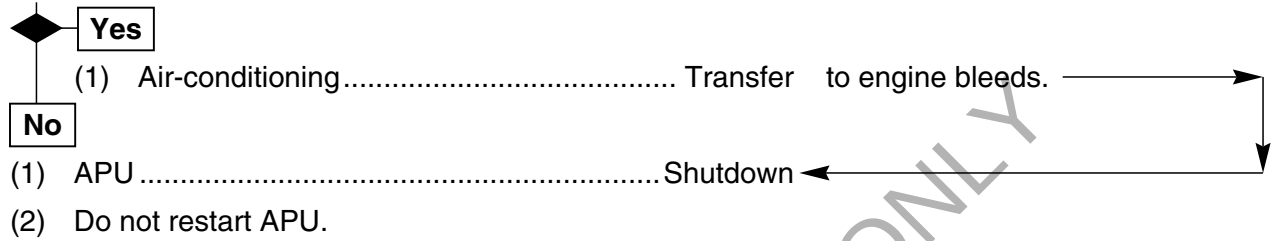
————— **END** —————

1. AUXILIARY POWER PLANT (CONT'D)

B. APU Overspeed

Indication: **APU OVERSPEED** warning message and “APU” aural on, APU RPM gauge and readout indicate speed equal to or greater than 110%, and APU automatically shuts down at 107%.

APU is supplying bleed air for air-conditioning:

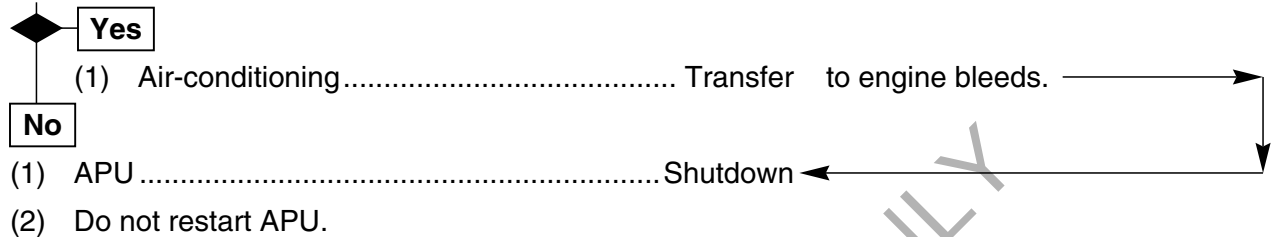


1. AUXILIARY POWER PLANT (CONT'D)

C. APU Overtemperature

Indication: **APU OVERTEMP** warning message and “APU” aural on, APU EGT gauge and readout indicate temperature greater than 732°C, and APU shuts down automatically.

APU is supplying bleed air for air-conditioning:



————— **END** —————

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TRAINING PURPOSES ONLY

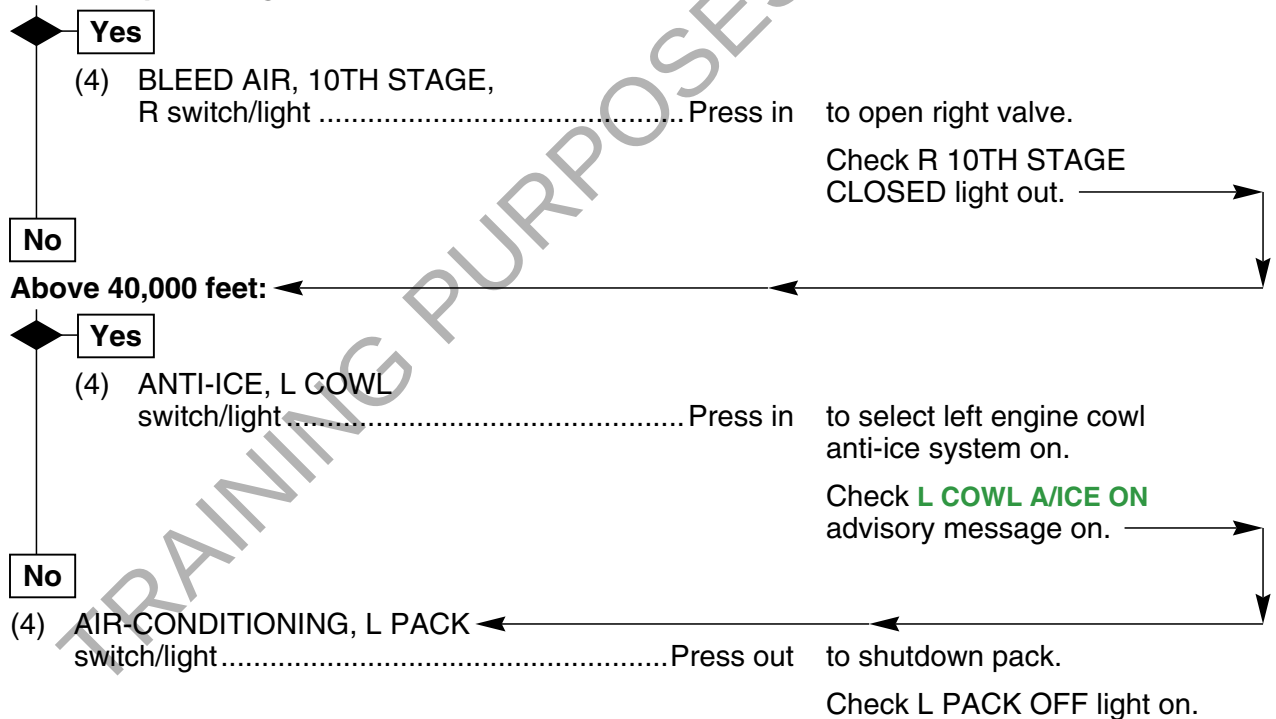
1. BLEED AIR LEAKS

A. 10TH Stage, Left Bleed Duct Failure

Indication: **L 10TH DUCT** warning message and “BLEED AIR DUCT” aural on:

- (1) BLEED AIR, 10TH STAGE, L switch/light Press out to close bleed air valve.
Check L 10TH STAGE CLOSED light on.
- (2) BLEED AIR, APU LCV switch/light Press out to close APU load control valve.
Check APU LCV OPEN light out.
- (3) BLEED AIR, 10TH STAGE ISOL switch/light Press out to close ISOL valve.
Check 10TH ISOL OPEN light out.

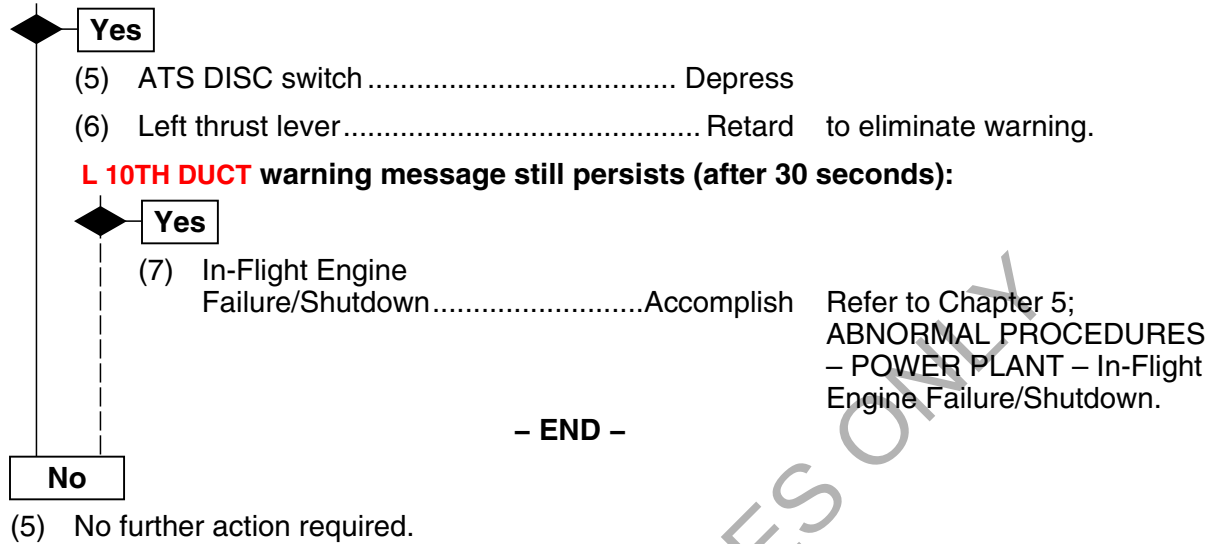
APU was providing bleed air:



1. BLEED AIR LEAKS (CONT'D)

A. 10TH Stage, Left Bleed Duct Failure (Cont'd)

Warning message persists:



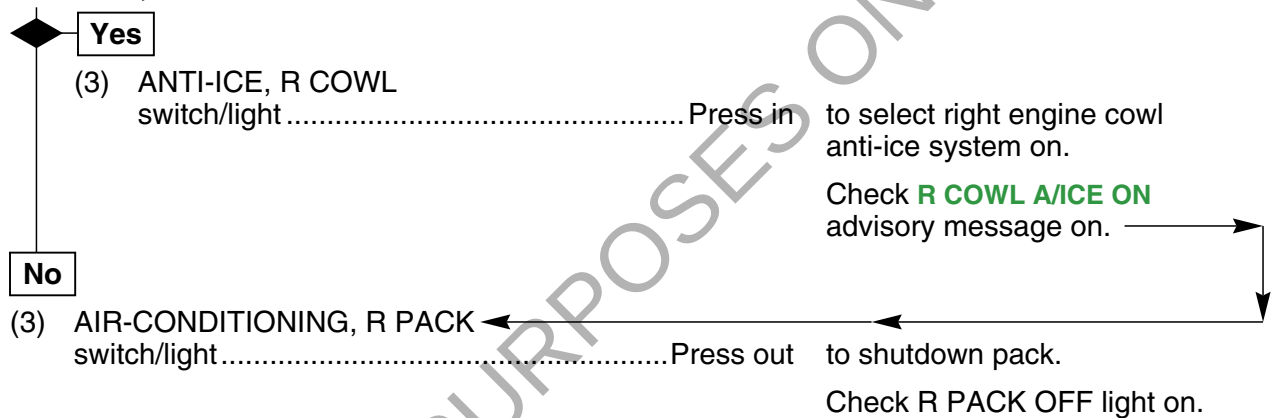
1. BLEED AIR LEAKS (CONT'D)

B. 10TH Stage, Right Bleed Duct Failure

Indication: **R 10TH DUCT** warning message and “BLEED AIR DUCT” aural on:

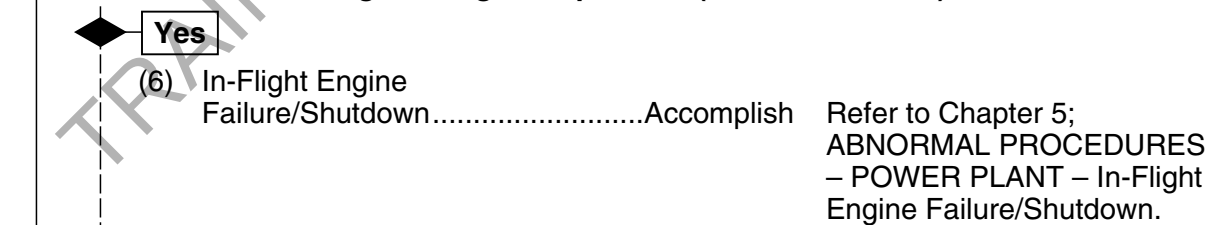
- (1) BLEED AIR, 10TH STAGE, R switch/light Press out to close bleed air valve.
Check R 10TH STAGE CLOSED light on.
- (2) BLEED AIR, 10TH STAGE ISOL switch/light Press out to close ISOL valve.
Check 10TH ISOL OPEN light out.

Above 40,000 feet:



Warning message persists:

- Yes**
- (4) ATS DISC switch Depress
- (5) Right thrust lever Retard to eliminate warning.

R 10TH DUCT warning message still persists (after 30 seconds):

END

1. BLEED AIR LEAKS (CONT'D)

C. 14TH Stage, Left/Right Bleed Duct Failure

Indication: **L (R) 14TH DUCT** warning message and “BLEED AIR DUCT” aural on.

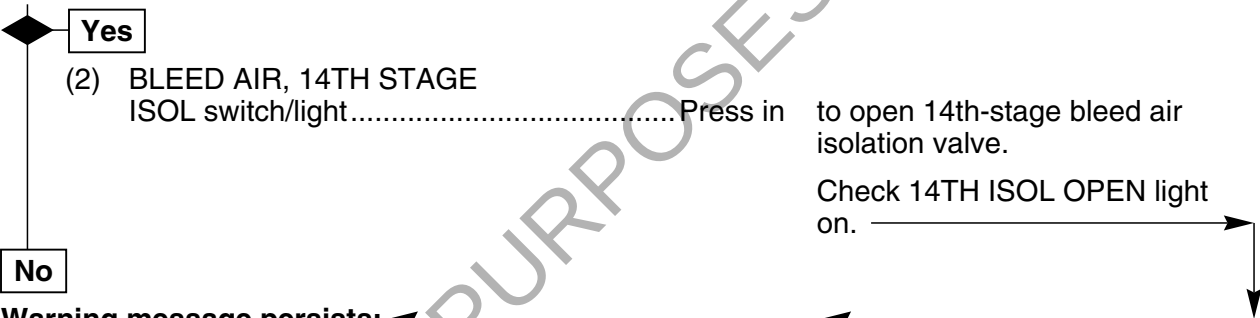
Affected engine:

- (1) BLEED AIR, 14TH STAGE L (R)
 switch/light.....Press out to close bleed air valve on affected engine.
 Check L (R) 14TH STAGE CLOSED light on.

NOTE

- Avoid icing conditions.
- Cowl anti-icing will not be available on the affected engine.
- The thrust reverser is inoperative on the affected engine.

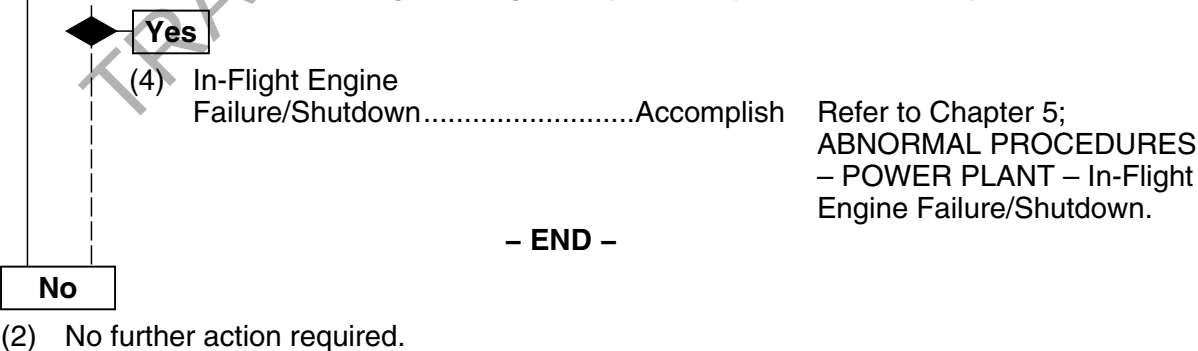
Wing anti-ice required:



Warning message persists:

- Yes
- (2) ATS DISC switch.....Depress
- Affected engine:
- (3) Thrust lever.....Retard to eliminate warning.

L (R) 14TH DUCT warning message still persists (after 30 seconds):



– END –

END

1. DITCHING AND FORCED LANDING

TO ENSURE A SAFE WATER LANDING:	TO ENSURE A SAFE FORCED LANDING:
• Approach speed..... V_{REF}	• Approach speed..... V_{REF}
• Descent rate (if thrust available)..... 200 – 300 fpm	• Descent rate (if thrust available)..... 200 – 300 fpm
• Landing gear Retracted	• Landing gear As required
• Flaps setting..... 45 degrees	• Flaps setting..... 45 degrees
• IF FLAPS 45 IS NOT POSSIBLE, USE THE MOST EXTENDED FLAP SETTING AVAILABLE. • FINAL APPROACH SPEED: • FLAPS 0: $V_{REF} + 30$ KTS • FLAPS 20: $V_{REF} + 14$ KTS • FLAPS 30: $V_{REF} + 7$ KTS	

A. Ditching or Forced Landing Imminent

This procedure is intended for use if an immediate airplane ditching is required.

PRELIMINARY

(1) AIR-CONDITIONING, L and R PACK switch/lights	Press out	to ensure minimum differential pressure within airplane.
(2) CABIN PRESSURIZATION, EMER DEPRESS switch/light	Press in	Prior to 200 feet AGL, to drive outflow valves open. Check EMER DEPRESS caution message on.

JUST BEFORE CONTACT

- | | | |
|--|----------|---|
| (1) Thrust levers..... | SHUT OFF | |
| (2) APU, LH and RH ENG FIRE
PUSH switch/lights (3)..... | Press in | Check all three BOTTLE ARMED – PUSH TO DISCH lights on.

Check the following advisory messages on: <ul style="list-style-type: none"> • L and R ENG SOV CLSD, • APU SOV CLSD, and • HYD SOV 1 and 2 CLSD. |
| (3) Water (or terrain) | Contact | with minimum forward speed, but not less than stick shaker speed, and at minimum sink rate. |



1. DITCHING AND FORCED LANDING (CONT'D)

A. Ditching or Forced Landing Imminent (Cont'd)

AFTER CONTACT

When the airplane has stopped:

- | | |
|---|---|
| <p>(1) BOTTLE ARMED – PUSH TO DISCH switch/lights (3)</p> <p>(2) Door and overwing exit.....</p> <p>(3) Passenger evacuation.....</p> <p>(4) BATT MASTER switch</p> <p>(5) Airplane</p> | <p>Press in and hold until lights go out, to discharge firex bottles.</p> <p>Open</p> <p>Initiate</p> <p>Select to OFF, to prevent CVR erasure.</p> <p>Abandon with survival equipment.</p> |
|---|---|

Check **ENG BOTTLE 1** and **2 LO** and **APU BOTTLE LO** caution messages on.

————— **END** —————

B. Planned Ditching

This procedure is intended for use where sufficient time is available.

PRELIMINARY

- | | |
|---|--|
| <p>(1) Descent</p> <p>(2) Crew</p> <p>(3) Air Traffic Control</p> <p>(4) MFD or CDU, ATC/TCAS tuning</p> <p>(5) PASS SIGNS, NO SMKG and SEAT BLTS switches</p> <p>(6) Loose Equipment</p> <p>(7) TAWS circuit breaker (1B13).....</p> | <p>Plan to reduce fuel to a minimum, if possible, while retaining sufficient fuel to make a controlled, powered approach.</p> <p>Alert and brief type of emergency, time available, airplane attitude after ditching and exits available for ditching.</p> <p>Notify</p> <p>Select to transponder code 7700.</p> <p>Select both to ON.</p> <p>Secure and stow all flight compartment loose equipment.</p> <p>Open to silence TAWS aural warnings.</p> <p>Check TAWS FAIL status message on.</p> |
|---|--|



1. DITCHING AND FORCED LANDING (CONT'D)

B. Planned Ditching (Cont'd)

- (8) AUDIO WARNING switchesDISABLE to silence EICAS (DCU) aural alerts.
Check **DCU 1** and **2 AURAL INOP** status messages on.
- (9) Survival equipmentCheck
- (10) Life vest, harness and belts On and tightened.

NOTE

Life vests should be donned, but not inflated, until outside of the airplane. Light plugs should be removed only if ditching at night.

- (11) Shoulder harness reel control Up to lock shoulder harness.
- (12) LDG GEAR lever Select UP for ditching.
- (13) FLAPS Select as required.

APPROACH

If possible, ditch in the vicinity of rescue vessels, near coastlines or islands.

At approximately 2,000 feet:

- (1) Sea conditions and wind direction Determine
- (2) Ditching heading Establish Ditching heading should be:
- Wind speed <15 kts; contact parallel to swells.
 - Wind speed between 15 and 45 kts; compromise between wind and swell.
 - Wind speed >45 kts; ditch into the wind.
- (3) Descent rate/approach speed Establish
- (4) AIR-CONDITIONING, L and R PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (5) CABIN PRESSURIZATION, PRESS CONT switch/light Press in to select manual control.
Check PRESS CONT MANUAL light on.
- (6) CABIN PRESSURIZATION, MAN ALT selector Select UP and hold.
- (7) CABIN PRESSURIZATION, MAN RATE control Select to MAX INCR.



1. DITCHING AND FORCED LANDING (CONT'D)

B. Planned Ditching (Cont'd)

When the airplane is completely depressurized:

- | | | | |
|------|---|-----------|--|
| (8) | BLEED AIR, 10TH STAGE L and R switch/lights | Press out | to close both valves.
Check L and R 10TH STAGE CLOSED lights on. |
| (9) | CABIN PRESSURIZATION, MAN ALT selector | Select | DN and hold to fully close outflow valves (approx 60 seconds). |
| (10) | FUEL, L and R BOOST PUMP switch/lights | Press out | to shutdown both boost pumps.
Check L and R FUEL PUMP ON advisory messages out. |
| (11) | APU (if not essential) | Shut down | |

At 500 feet (Radio Altitude):

- | | | | |
|------|--|----------|---|
| (12) | Crew | Alert | that ditching is imminent. |
| (13) | Radio | Transmit | Final Position. |
| (14) | BRACE FOR IMPACT | ORDER | over PA system : "BRACE – BRACE – BRACE". |
| (15) | EMER LTS switch | Select | to ON.
Check EMER LTS ON status message on. |
| (16) | LANDING LTS, LEFT and RIGHT switches | Select | to STDY. |
| (17) | LANDING LTS, NOSE switch | Select | to ON. |

JUST BEFORE CONTACT

- | | | | |
|-----|--|----------|--|
| (1) | Thrust levers | SHUT OFF | |
| (2) | APU, LH and RH ENG FIRE PUSH switch/lights (3) | Press in | Check all three BOTTLE ARMED – PUSH TO DISCH lights on.
Check the following advisory messages on:
<ul style="list-style-type: none"> • L and R ENG SOV CLSD, • APU SOV CLSD, and • HYD SOV 1 and 2 CLSD. |



1. DITCHING AND FORCED LANDING (CONT'D)

B. Planned Ditching (Cont'd)

- (3) Water Contact with minimum forward speed, but not less than stick shaker speed, and at minimum sink rate.

AFTER WATER CONTACT

When the airplane has stopped:

- (1) BOTTLE ARMED – PUSH TO DISCH switch/lights (3) Press in and hold until lights go out, to discharge firex bottles.
Check **ENG BOTTLE 1** and **2 LO** and **APU BOTTLE LO** caution messages on.
- (2) Door and overwing exit Open
- (3) Passenger evacuation Initiate
- (4) BATT MASTER switch Select to OFF, to prevent CVR erasure.
- (5) Airplane Abandon with survival equipment.

— **END** —

TRAINING PURPOSES ONLY

1. DITCHING AND FORCED LANDING (CONT'D)

C. Forced Landing

This procedure is intended for use where sufficient time is available.

PRELIMINARY

- | | | |
|--|-----------------|---|
| (1) Descent | Plan | to reduce fuel to a minimum, if possible, while retaining sufficient fuel to make a controlled, powered approach. |
| (2) Crew | Alert and brief | type of emergency, time available, airplane attitude after landing and exits available. |
| (3) Air Traffic Control | Notify | |
| (4) MFD or CDU, ATC/TCAS tuning | Select | to transponder code 7700. |
| (5) PASS SIGNS, NO SMKG and SEAT BLTS switches | Select | both to ON. |
| (6) Loose Equipment | Secure | and stow all flight compartment loose equipment. |
| (7) TAWS circuit breaker (1B13) | Open | to silence TAWS aural warnings.

Check TAWS FAIL status message on. |
| (8) AUDIO WARNING switches | DISABLE | to silence EICAS (DCU) aural alerts.

Check DCU 1 and 2 AURAL INOP status messages on. |
| (9) Survival equipment | Check | |
| (10) Crew harness and belts | On | and tightened. |
| (11) Shoulder harness reel control | Up | to lock shoulder harness. |
| (12) LDG GEAR lever | Select | as required. |
| (13) FLAPS | Select | as required. |

APPROACH

At approximately 2,000 feet:

- | | | |
|--|-----------|--|
| (1) Landing area conditions and wind direction | Determine | |
| (2) Descent rate/approach speed | Establish | |
| (3) AIR-CONDITIONING, L and R PACK switch/lights | Press out | to shutdown both packs.

Check L and R PACK OFF lights on. |



1. DITCHING AND FORCED LANDING (CONT'D)

C. Forced Landing (Cont'd)

- (4) CABIN PRESSURIZATION, PRESS CONT switch/light Press in to select manual control.

Check PRESS CONT MANUAL light on.

- (5) CABIN PRESSURIZATION, MAN ALT selector Select UP and hold.

- (6) CABIN PRESSURIZATION, MAN RATE control Select to MAX INCR.

When the airplane is completely depressurized:

- (7) BLEED AIR, 10TH STAGE L and R switch/lights Press out to close both valves.

Check L and R 10TH STAGE CLOSED lights on.

- (8) FUEL, L and R BOOST PUMP switch/lights Press out to shutdown both boost pumps.

Check **L** and **R FUEL PUMP ON** advisory messages out.

- (9) APU (if not essential) Shut down

At 500 feet (Radio altitude):

- (10) Crew Alert that forced landing is imminent.

- (11) Radio Transmit Final Position.

- (12) BRACE FOR IMPACT ORDER over PA system ; “BRACE – BRACE – BRACE”.

- (13) EMER LTS switch Select to ON.

Check **EMER LTS ON** status message on.

- (14) LANDING LTS, LEFT and RIGHT switches Select to STDY.

- (15) LANDING LTS, NOSE switch Select to ON.

JUST BEFORE CONTACT

- (1) Thrust levers SHUT OFF



1. DITCHING AND FORCED LANDING (CONT'D)

C. Forced Landing (Cont'd)

- (2) APU, LH and RH ENG FIRE

PUSH switch/lights (3)..... Press in

Check all three BOTTLE ARMED – PUSH TO DISCH lights on.

Check the following advisory messages on:

- **L** and **R ENG SOV CLSD**,
- **APU SOV CLSD**, and
- **HYD SOV 1** and **2 CLSD**.

- (3) Terrain Contact

with minimum forward speed, but not less than stick shaker speed, and at minimum sink rate.

AFTER CONTACT

When the airplane has stopped:

- (1) BOTTLE ARMED – PUSH TO

DISCH switch/lights (3) Press in

and hold until lights go out, to discharge firex bottles.

Check **ENG BOTTLE 1** and **2 LO** and **APU BOTTLE LO** caution messages on.

- (2) Door and overwing exit..... Open

- (3) Passenger evacuation..... Initiate

- (4) BATT MASTER switch..... Select

to OFF, to prevent CVR erasure.

- (5) Airplane Abandon

with survival equipment.

———— **END** ————

1. EMERGENCY EVACUATION

A. Emergency Evacuation Procedure

This procedure contains the steps required to prepare for passenger evacuation on land, and is initiated when a condition potentially endangering life or physical well-being of passengers and crew exists.

Pilot:

- | | | |
|--|----------|--|
| (1) Parking brake | Set | |
| (2) Evacuation..... | Command | |
| (3) GND SPOILERS switch | Select | to DISARM. |
| (4) FLIGHT SPOILER lever | Select | to RETRACT, to fair spoilers with wing surface.
Check EICAS indicates spoilers retracted. |
| (5) Thrust levers..... | SHUT OFF | |
| (6) APU, LH and RH ENG FIRE
PUSH switch/lights (3)..... | Press in | Check all three BOTTLE ARMED PUSH TO DISCH lights on.
Check the following advisory messages on:
<ul style="list-style-type: none"> • L and R ENG SOV CLSD, • APU SOV CLSD, and • HYD SOV 1 and 2 CLSD. |
| (7) BATT MASTER switch | OFF | to prevent CVR erasure. |

Copilot: On evacuation command

- | | | |
|---|-----------|--|
| (8) ATC | Notify | of emergency conditions and of intention to evacuate. |
| (9) AIR-CONDITIONING, L and R
PACK switch/lights | Press out | to ensure minimum differential pressure within airplane. |
| (10) CABIN PRESSURIZATION,
EMER DEPRESS switch/light | Press in | to drive outflow valves open.
Check EMER DEPRESS caution message on. |
| (11) EMER LTS switch | Select | to ON.
Check EMER LTS ON status message on. |

- (12) Exits..... Open



1. EMERGENCY EVACUATION (CONT'D)

I A. Emergency Evacuation Procedure (Cont'd)

- (13) Passenger evacuation Render assistance and direct passengers away from the airplane.
- (14) Airplane Abandon by any suitable exit.

————— END —————

TRAINING PURPOSES ONLY

1. ELECTRICAL

A. Loss of All Normal Electrical Power

Indication: ADG automatically deployed and **EMER PWR ONLY** warning message.

OR

AC ESS BUS caution message and no generator on-line.

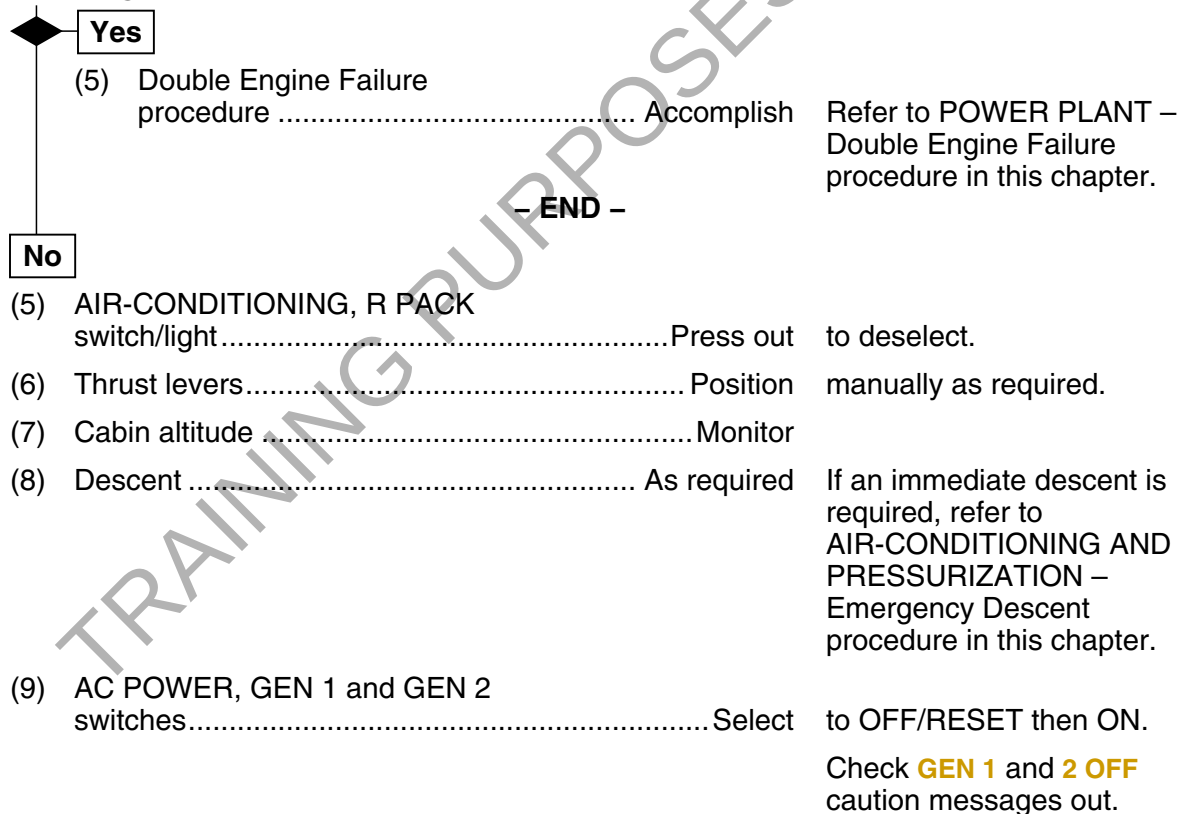
If ADG did not deploy:

- (1) ADG manual deploy handle Pull and leave extended.
- (2) AC ELECTRICAL page Check ESS BUS is powered.

When ADG power is established:

- (3) STAB TRIM, CH 2 switch Press in to engage STAB TRIM CH 2.
- (4) Engine instruments Verify and check N₁, N₂ and ITT indications.

Both engines confirmed failed:



1. ELECTRICAL (CONT'D)

A. Loss of All Normal Electrical Power (Cont'd)

Any generator comes on-line:

Yes

Re-establish normal power:

No

Below 20,000 feet and APU is available:

Yes

(10) APU Start

(11) AC POWER, APU GEN
switch..... Select to ON.
Check **APU GEN OFF** caution
message out.

APU generator comes on-line:

Yes

(12) ADG manual deploy
handle Stow

(13) ADG PWR TXFR,
OVERRIDE switch Press to override.

(14) HYDRAULIC pump
switch 3B Select to AUTO/ON, as applicable.
Check **HYD PUMP** caution
messages are not displayed.

(15) CAS messages Review

(16) Affected systems Restore

(17) EFIS altimeters (2) Cross-check

- END -

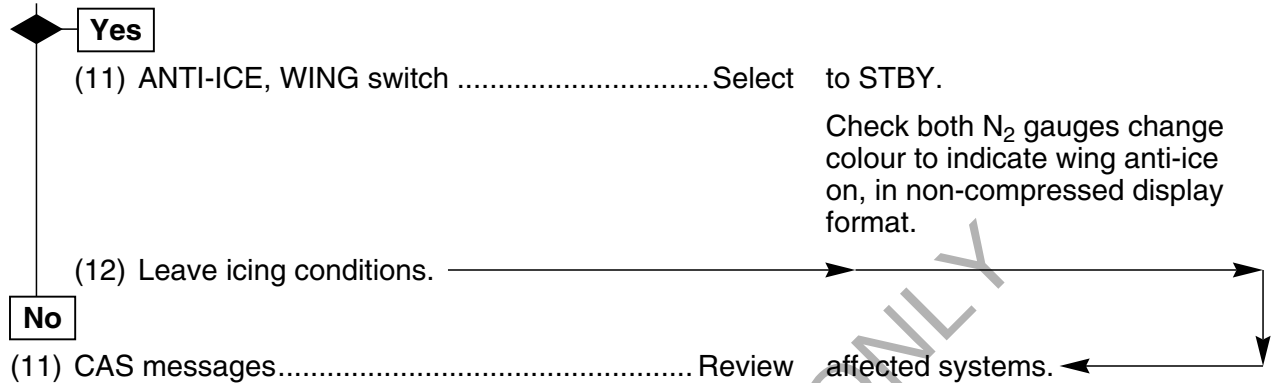
No

(10) Descent Continue to 10,000 feet or lowest safe
altitude (if required).

1. ELECTRICAL (CONT'D)

A. Loss of All Normal Electrical Power (Cont'd)

Wing anti-ice is required:



The following significant systems are not available when on emergency power only:

- Yaw damper 2 and AFCS 2
- Ground spoilers
- Flaps channel 2
- Stabilizer trim channel 1
- Mach trim, aileron and rudder trim
- Hydraulic pumps 1B, 2B & 3A
- DME, FMS 2
- Copilot's PFD, MFD, navigation & communication systems
- R Pack
- Right probe heaters, right AOA vane heater and right ice detector
- Nose wheel steering system, anti-skid system
- Both windshield heaters, right window heater
- Copilot's instrument lights, NAV lights, and L & R landing lights.
- SPS altitude compensation
- Auto throttle system

NOTE

1. With Mach trim not available, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.
2. Without probe and AOA vane heaters, do not rely on stall protection system, ISI for altitude and airspeed, and cabin Δpressure on EICAS.
3. Flaps will operate (extend or retract) at half speed.

(12) Land at the nearest suitable airport.



1. ELECTRICAL (CONT'D)

A. Loss of All Normal Electrical Power (Cont'd)

NOTE

For flight planning, a fuel burn correction of 5% should be used when the ADG is deployed.

NOTE

If the ADG is not operating properly, and the airplane is on battery power only, then all electrical power may be lost after 30 minutes.

Hydraulic pump 3B will be inoperative when on battery power only.

Prior to landing:

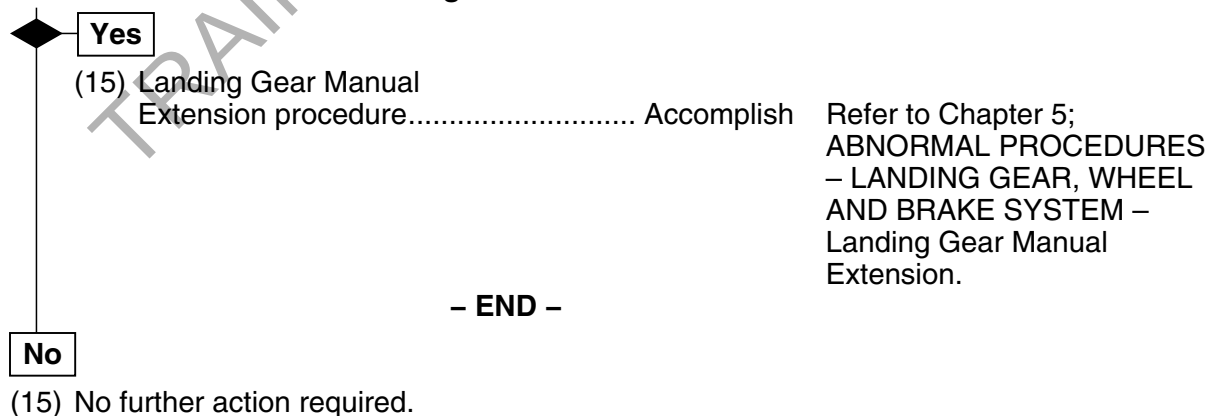
- (13) ADG manual deploy handle Pull
- (14) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.50 (150%)	1.85 (85%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

HYD 3 LO PRESS caution message is on:



END

1. FLIGHT CONTROLS

A. Aileron System Jammed

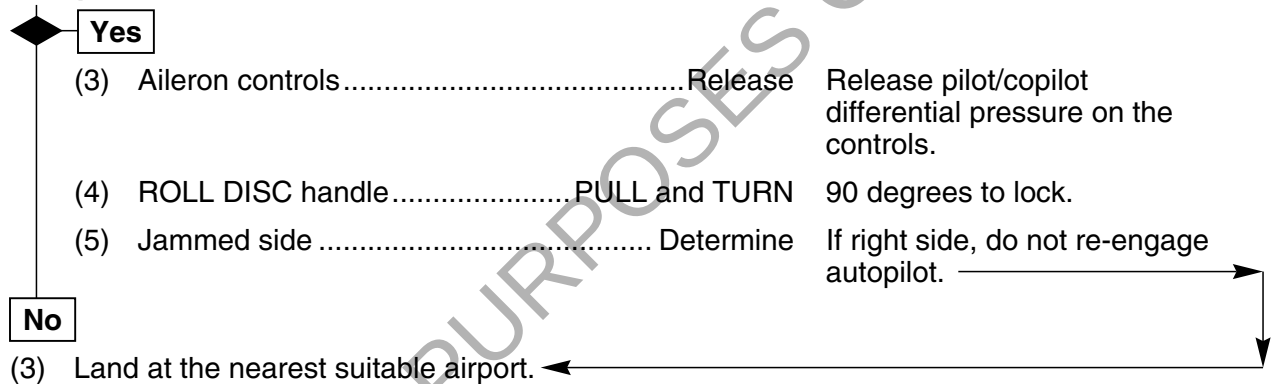
Indication: Higher than normal flight control forces.

NOTE

With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- | | | |
|---------------------------|------------|--|
| (1) Autopilot | Disconnect | if coupled. |
| (2) Aileron controls..... | Overpower | Attempt to overpower jammed condition. |
| | | Attempt to use aileron trim to alleviate forces on controls, if jam has occurred away from neutral position. |

Jam persists:



NOTE

Controllability is reduced. Select longest runway available with minimum turbulence and crosswind.

Prior to landing:

- | | | |
|--|---------------------------------|---------------------------------|
| (4) TAWS WARNING, FLAPS OFF switch/light | Press in | to mute the flap aural warning. |
| (5) TAWS WARNING, RAAS OFF switch/light (if installed) | Press in | Check OFF light on. |
| (6) FLAPS | Select | to 20° for landing. |
| (7) Approach speed | V_{REF} (Flaps 45°) + 14 KIAS | minimum. |
| (8) ATS DISC switch | Depress | prior to 100 feet AGL. |



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.



1. FLIGHT CONTROLS (CONT'D)

A. Aileron System Jammed (Cont'd)

(9) Actual landing distance

Increase

as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

END

1. FLIGHT CONTROLS (CONT'D)

B. Elevator System Jammed

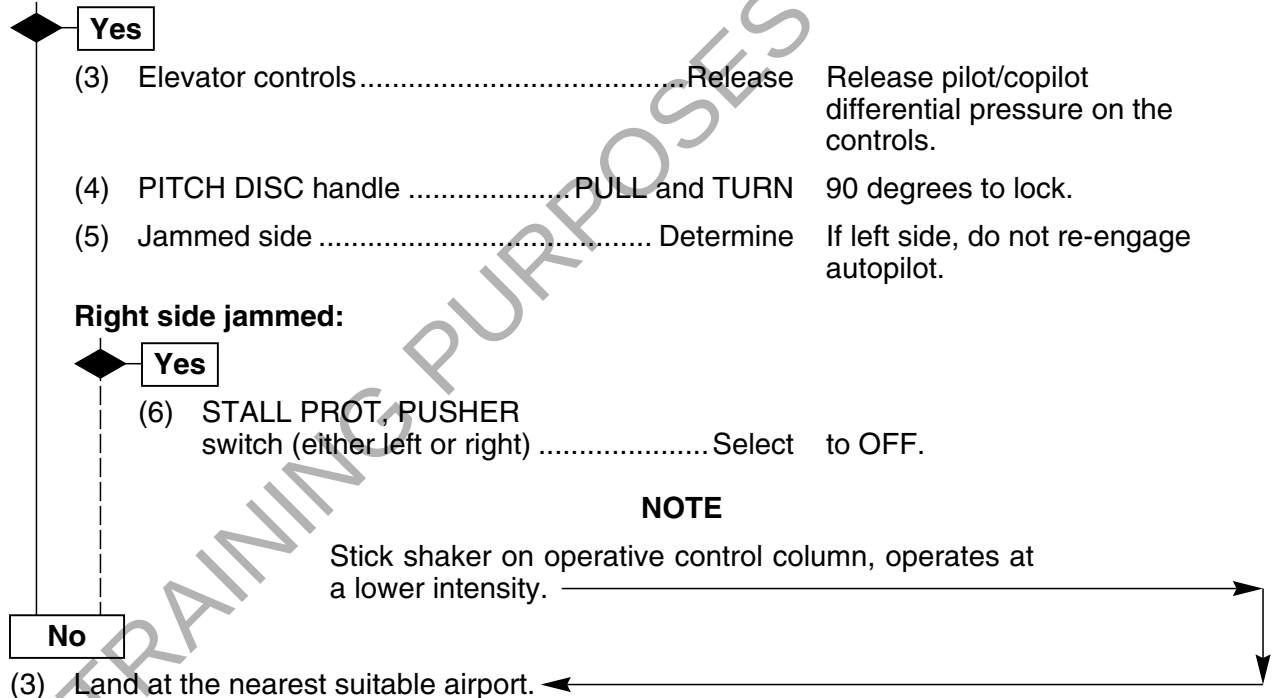
Indication: Higher than normal flight control forces.

NOTE

With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- | | | |
|----------------------------|------------|---|
| (1) Autopilot | Disconnect | if coupled. |
| (2) Elevator controls..... | Overpower | Attempt to overpower jammed condition. |
| | | Use stabilizer trim to compensate, if elevator jam has occurred away from the trimmed position. |

Jam persists:



NOTE

Stick shaker on operative control column, operates at a lower intensity.

NOTE

Controllability is reduced. Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

- | | | |
|--|----------|---------------------------------|
| (4) TAWS WARNING, FLAPS OFF switch/light..... | Press in | to mute the flap aural warning. |
| (5) TAWS WARNING, RAAS OFF switch/light (if installed) | Press in | Check OFF light on. |
| (6) FLAPS | Select | to 20° for landing. |



1. FLIGHT CONTROLS (CONT'D)

B. Elevator System Jammed (Cont'd)

- (7) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (8) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (9) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

1. FLIGHT CONTROLS (CONT'D)

C. Rudder System Jammed

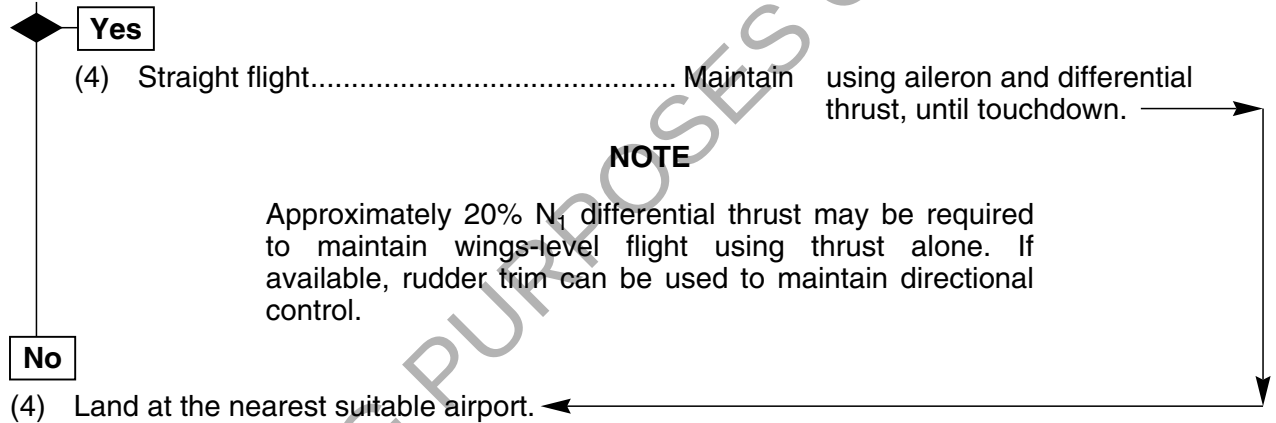
Indication: Higher than normal flight control forces.

NOTE

With a jammed primary flight control, it may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot Disconnect if coupled.
- (2) YD DISC pushbutton Press to disengage yaw damper.
Check **YAW DAMPER** caution message on.
- (3) Rudder pedals Overpower Attempt to overpower jammed condition.

Jam persists:



NOTE

Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (5) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (7) FLAPS Select to 20° for landing.
- (8) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.



1. FLIGHT CONTROLS (CONT'D)

C. Rudder System Jammed (Cont'd)

- (9) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (10) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

NOTE

After touchdown, use nose wheel steering and, if necessary, differential braking to maintain directional control, as the airplane will turn in the direction of the jammed rudder.

— END —

1. FLIGHT CONTROLS (CONT'D)

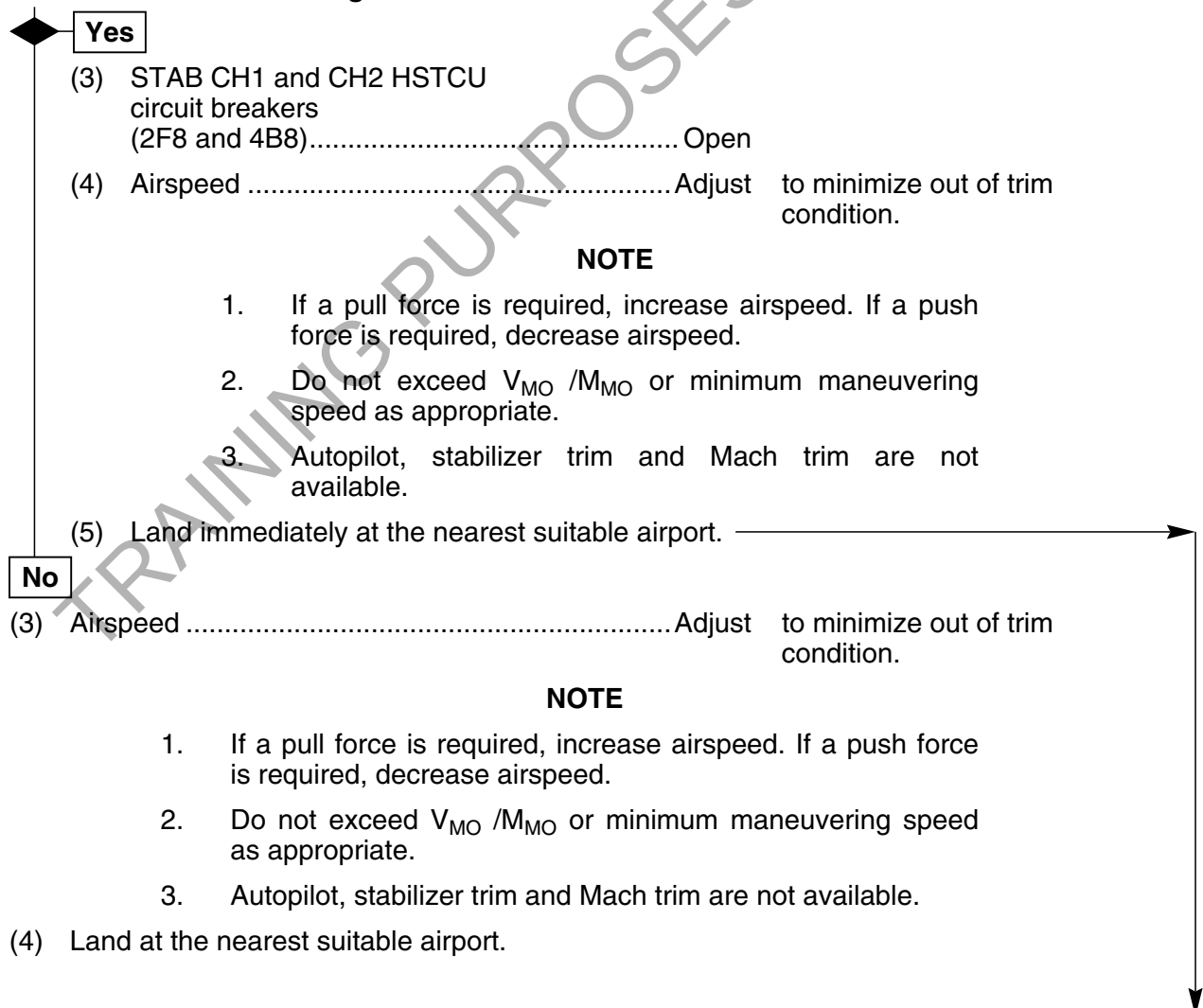
D. Stabilizer Trim Runaway

Indication:	Abnormal/uncommanded change in pitch attitude. Stabilizer trim changes without pilot input. Higher or lower than normal and increasing or decreasing pitch control forces.		
(1) Control column	Assume	manual control and override runaway.	
(2) Pilot's or copilot's STAB TRIM DISC switch	Press	hold and release.	

NOTE

- Flight spoilers will reduce airspeed and provide nose up pitch.
- Reducing thrust will reduce airspeed and provide nose up pitch.

STAB TRIM caution message is out or uncommanded stabilizer motion continues:



1. FLIGHT CONTROLS (CONT'D)

D. Stabilizer Trim Runaway (Cont'd)

Prior to landing: ←

- (5) TAWS WARNING, FLAPS OFF
switch/light Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (7) Recommended FLAPS Select to 20° for landing.



Selection of Flaps 20 will increase nose down pitching moment slightly, and therefore should be selected just prior to approach in the event of a full nose down stabilizer position. Selection of Flaps 20 significantly reduces landing field length compared to Flaps 0.

If control forces are excessive, some relief may be gained by adjusting airspeed or flap position. Reduced flap settings generally reduce pull forces. Increased flap settings generally reduce push forces.

- (8) Final approach speed V_{REF} (Flaps 45°)
+ ΔV_{REF} for applicable flap setting.
- (9) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (10) Actual landing distance Increase by factor given below:

Flaps Setting (Degrees)	ΔV_{REF} (KTS)	ALD Factor	
		Without Thrust Reversers	With Thrust Reversers
0	30	1.80 (80%)	1.75 (75%)
20	14	1.55 (55%)	1.50 (50%)
30	7	1.40 (40%)	1.35 (35%)



1. FLIGHT CONTROLS (CONT'D)

D. Stabilizer Trim Runaway (Cont'd)

CAUTION

Do not jettison fuel if trim runaway is in nose down direction.

Fuel jettison will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.

Consider use of fuel jettison in the event of a nose-up trim runaway condition.

A landing may be made with fuel in the tail tank.

WARNING

In the event of a go-around, increasing thrust will increase nose down forces. Increase thrust slowly to avoid excessive pitch force.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

E. Stall Recovery

- | | | | |
|--------------------------------|--|----------------------------|--|
| Indication: | Aircraft buffet, uncommanded roll, stick shaker activated and/or stall warbler on. | | |
| (1) Autopilot | Disconnect | if coupled. | |
| (2) Pitch attitude..... | Lower nose | to reduce angle of attack. | |
| (3) ATS DISC switch | Depress | | |
| (4) Thrust levers..... | Advance | to MAX POWER. | |
| (5) Roll attitude | Wings level | | |
| (6) FLIGHT SPOILER lever | Select | to RETRACT. | |

After airspeed increases and stall warning is extinguished:

- | | | |
|--|--------|----------------------------|
| (7) Pitch attitude..... | Adjust | to minimize altitude loss. |
| (8) Thrust levers and airplane configuration | Adjust | as required. |

NOTE

1. It is essential that the AOA be immediately reduced, even if this means a loss of altitude.
2. Avoid abrupt or aggressive pitch control inputs during recovery. Inappropriate recovery inputs can result in a secondary stall.
3. Height loss resulting from high AOA recovery, especially at cruise altitude and/or low initial thrust conditions, can be significant and may be in the order of thousands of feet .

———— **END** ————

BOMBARDIER CHALLENGER 605	EMERGENCY PROCEDURES Flight Controls	Vol. 1	03-12-11 REV 46, Mar 01/18
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1. FLIGHT CONTROLS (CONT'D)

F. Overspeed Warning

Indication: Overspeed warning clacker on.

(1) Airspeed Reduce until less than V_{MO}/M_{MO} or V_{FE} .

————— **END** —————

TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	EMERGENCY PROCEDURES Fuel System	Vol. 1	03-13-1 Oct 05/06
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1. FUEL SYSTEM

A. Low Fuel

Indication: The EICAS fuel quantities turn amber when either main tank is less than 230 kg (500 lb), or the total fuel quantity is less than 460 kg (1,000 lb).

- (1) AirplaneNot more than 5°
nose up
- (2) Fuel balance.....Check
- (3) Land at the nearest suitable airport.

NOTE

Do not attempt go-around. The minimum fuel quantity for go-around is 230 kg (500 lb) per wing (with the airplane level), and assuming a maximum airplane climb attitude of 10° nose up.

————— **END** —————

TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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BOMBARDIER CHALLENGER 605	EMERGENCY PROCEDURES Fuel System	Vol. 1	03-13-2
		Oct 05/06	

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TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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1. ICE AND RAIN PROTECTION

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is –40°C (–40°F) or below.

A. Wing Leading Edge Overheat

Indication: **WING OVHT** warning message and “WING OVERHEAT” aural on.

(1) ANTI-ICE, WING switchSelect to OFF.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message out.

After 40 seconds, **WING OVHT** warning message persists:

Yes

(2) BLEED AIR, 14TH STAGE, L and R switch/lights.....Press out to close shutoff valves.

Check L and R 14TH STAGE CLOSED lights on.

NOTE

With both 14th stage bleed air shut-off valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(3) ANTI-ICE, COWL L and R switch/lightsPress out to select cowl anti-ice off.

(4) Leave icing conditions.

If required:

(5) Ice Dispersal Procedure Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – ICE AND RAIN PROTECTION – Ice Dispersal Procedure.

– END –

1. ICE AND RAIN PROTECTION (CONT'D)

A. Wing Leading Edge Overheat (Cont'd)

No

(2) ANTI-ICE, WING switch Select to STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message on.

NOTE

WING OVHT warning message may be displayed when standby mode is selected at high power settings (N₂ above 85%).

L HEAT or **R HEAT** light is out and/or **L (R) WING A/ICE** caution message is on:

Yes

(3) BLEED AIR, 14TH STAGE, ISOL switch/light Press in to open isolation valve.

Check 14TH STAGE ISOL OPEN light on. →

No

WING OVHT warning message re-occurs: ←

Yes

(3) ANTI-ICE, WING switch Select to OFF.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message out.

(4) BLEED AIR, 14TH STAGE, L and R switch/lights Press out to close shutoff valves.

Check L and R 14TH STAGE CLOSED lights on.

NOTE

With both 14th stage bleed air shut-off valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.



1. ICE AND RAIN PROTECTION (CONT'D)

A. Wing Leading Edge Overheat (Cont'd)

(5) ANTI-ICE, COWL L and R
switch/lights Press out to select cowl anti-ice off.

(6) Leave icing conditions.

If required:

(7) Ice Dispersal Procedure Accomplish Refer to Chapter 5;
ABNORMAL PROCEDURES
– ICE AND RAIN
PROTECTION – Ice Dispersal
Procedure.

– END –

No

(3) No further action required.

END

TRAINING PURPOSES ONLY

1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure

Indication: **ANTI-ICE DUCT** warning message and “ANTI-ICE DUCT” aural on.

(1) ANTI-ICE, WING switch Select to OFF.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message out.

(2) Leave icing conditions.

ANTI-ICE DUCT warning message persists (after 30 seconds):

☐ **Yes**
 (3) BLEED AIR, 14TH STAGE, L and R switch/lights Press out to close shutoff valves.
 Check L and R 14TH STAGE CLOSED lights on.

NOTE

With both 14th stage bleed air shut-off valves closed, cowl anti-icing, wing anti-icing and thrust reversers are not available.

(4) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.

After leaving icing conditions:

Ice accumulation is observed on the heated portion of the wing leading edge:

☐ **Yes**
 (5) Airspeed Increase to V_{MO}/M_{MO} to disperse ice, if possible.

☐ **No**
 (5) No further action required.

– END –

1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure (Cont'd)

No

After leaving icing conditions:

Ice accumulation is observed on the heated portion of the wing leading edge:

Yes

- (3) Airspeed Increase to V_{MO}/M_{MO} to disperse ice, if possible.

Ice accumulation persists, perform manual anti-ice operation:

Yes

- (4) ANTI-ICE, WING switch..... Select to STBY.

Check the following:

- N_2 gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message on.

When **ANTI-ICE DUCT** warning message comes on:

- (5) ANTI-ICE, WING switch..... Select to OFF.

Check the following:

- N_2 gauges change colour to indicate wing anti-ice off, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message out.

- (6) Repeat steps (4) and (5), as required.

1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure (Cont'd)

Impossible to remove ice from the wing leading edge: ←

Yes

Assure adequate stall margin:

- (7) Maneuvering speed Not less than
200 KIAS

WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics, or the warning margins provided by the stall protection system.

Prior to landing:

- (8) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (9) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (10) FLAPS Select to 20° for landing.
- (11) Approach speed V_{REF} (Flaps 45°) + 27 KIAS
- (12) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

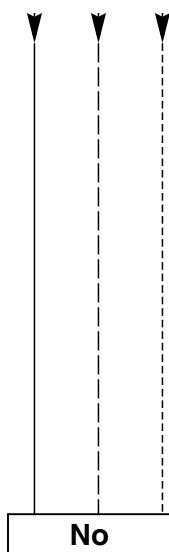
ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (13) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.80 (80%)	1.70 (70%)

1. ICE AND RAIN PROTECTION (CONT'D)

B. Fuselage/Wing Anti-Ice Duct Failure (Cont'd)



At 50 feet AGL:

(14) Thrust levers IDLE and do not prolong the flare.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- END -

(3) No further action required.

END

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TRAINING PURPOSES ONLY

1. INSTRUMENTS SYSTEM

A. Unreliable Airspeed

- Indication:
- Pitch attitude, thrust setting or external noise not consistent with indicated airspeed.
 - Loss of multiple airspeed indications.
 - Multiple airspeed indications remain constant regardless of pitch or thrust adjustments.
 - Airspeed difference between primary flight displays is more than 10 KIAS and/or more than 15 KIAS between any primary flight display and ISI and no reliable airspeed source can be determined.
 - **EFIS MISCAMP** with an amber IAS flag and/or **EFIS COMP INOP**

NOTE

1. If doubt exists about the reliability of aircraft speed indications, aircraft control must be established and maintained by the use of pitch and thrust settings.
2. Autopilot may not be available.
3. No crew action is required for the following messages as they are covered within:
 - Caution messages: **EFIS MISCAMP** with an amber IAS flag or **EFIS COMP INOP**.

- (1) ATS DISC switch Depress
- (2) Autopilot Disconnect
- (3) FDs Deselect
- (4) Airplane Altitude Maintain a safe altitude

If Climb or Go-Around is required:

- (5) Pitch/ N_1 15 degrees/TO from
SL to 15,000 ft
10 degrees / CLB
above 15,000 ft
- (6) Airplane Configuration Flaps 20/ Gear Up
for Go-Around
Flaps 0 / Gear Up
once Climbing



Respect stall warning/stick shaker and stick pusher.



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

At desired altitude:

- (7) Autopilot or Flight Director (if required and available) Engage basic PTCH/ROLL

NOTE

1. Using the FD/autopilot in PTCH, ALT, HDG and ROLL modes will help reduce workload.
2. If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.



1. Do not use any FD/autopilot SPEED mode.
 2. Do not use FLC mode or ATS.
- (8) Pitch/ N_1 Set as per Table A.

NOTE

Leave current environmental conditions if they are considered to be a reason for unreliable airspeed. Refer to Tables B or C as required for climb or descent.

- (9) ANTI-ICE, LH and RH COWL switches As required
- (10) ANTI-ICE, WING switch As required

NOTE

1. Autopilot/yaw damper should only be engaged once N_1 is set.
2. Cross check AOA after setting pitch attitude.
3. Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Adjust pitch to stop climb/descent.
4. Disregard low speed cue, flight path vector and airspeed trend vector as these may be erroneous.
5. If overspeed warning sounds, select the affected AUDIO WARNING switch to mute aural and disregard.
6. A nuisance **NOSE DOOR OPEN** warning message can be disregarded.
7. Altitude, Mode C and TCAS may be in error by up to 600 feet. Vertical speed indications may also be unreliable.
8. Groundspeed indication may be unreliable.



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

9. FMS cruise time and fuel predictions may be incorrect.
10. Mach trim may be inoperative or operate erratically. Disengage Mach trim if erratic stabilizer motion is observed.

Evaluate indicated airspeed sources in level flight as follows:



Airspeed should be considered reliable only if the following two conditions both exist:

1. Airspeed indication is consistent with pitch, thrust and airspeed range from Table A, and
 2. Airspeed indication changes and speed trend vector are consistent with pitch and thrust adjustments.
- (11) ADC source selector Select to NORM
- (12) Both PFDs and ISI Compare airspeed to expected speed range from Table A.
- (13) Determine which of the following conditions apply:
- If no indicated airspeed is considered reliable from Table A, proceed to step (14).
 - If one or more airspeed source is considered reliable from Table A, proceed to step (18).

NO RELIABLE AIRSPEED SOURCE

- (14) Airspeed Indications Disregard
- (15) Pitch/ N_1 Set using Table A, B, C, D 1, D 2, D 3 or E as appropriate for phase of flight.
- (16) Land at the nearest suitable airport.
- (17) When ready to start descent, proceed to step (23)

ONE OR MORE AIRSPEED SOURCE IS CONSIDERED RELIABLE

- (18) ADC source selector Confirm NORM
- (19) Reliable airspeed source MONITOR WITH CAUTION
- (20) Pitch/ N_1 Set using Table A, B, C, D 1, D 2, D 3 or E as appropriate for phase of flight and continue to monitor reliable airspeed source and airspeed range.



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)



1. Disregard airspeed indication if at any time it becomes unreliable.
 2. ADC source selector must remain in NORM position.
- (21) Land at the nearest suitable airport.
- (22) When ready to start descent, proceed to step (23)

Recovery to an airfield

- (23) DescentInitiate when ready



To avoid significant flap overspeed and flap damage:

1. Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight (Table D 1, D 2 or D 3).
2. Select FLAPS 45 at the start of final descent on the glideslope/glidepath (Table E).

NOTE

1. Plan to intercept a long final to allow time to re-configure for approach/landing and set pitch/ N_1 .
 2. Plan to recover to an airfield with an ILS, LNAV or VNAV under Visual Meteorological Conditions if possible.
- (24) Unreliable aircraft systems Review
- TAWS
 - Windshear detection
 - Landing gear warning horn
 - Flap overspeed
 - NOSE DOOR OPEN warning message
- (25) Actual landing distance Increase

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.25 (25%)	1.20 (20%)

NOTE

In the event of a go-around, refer to the immediate actions at the beginning of this procedure.



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
- Adjust pitch to stop climb/descent.
- If pitch is increased, REDUCE expected speed range by delta KIAS/delta Mach.
- If pitch is decreased, INCREASE expected speed range by delta KIAS/delta Mach.

Table A – Level Flight – FLAPS 0											
			WEIGHT kg (lb)					Expected SPD Range		Expected Speed Change for 0.5 deg change in pitch ΔKIAS/ ΔMach	
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)	KIAS		MACH
ALTITUDE (1,000 ft)	40	Pitch (°)	1.0	1.5	2.0	2.5			220 to 240	.71 to .82	5 / 0.01
		N ₁ (%)	83.0	84.6	87.2	89.9					
		AoA	0.4	0.4	0.5	0.5					
	35	Pitch (°)	1.5	2.0	2.5	2.5	3.0		240 to 260	.69 to .80	10 / 0.03
		N ₁ (%)	81.8	82.7	83.9	85.7	87.9				
		AoA	0.4	0.4	0.4	0.4	0.5				
	30	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	250 to 275		10
		N ₁ (%)	79.9	80.4	81.3	82.3	83.8	84.7			
		AoA	0.3	0.4	0.4	0.4	0.4	0.4			
	25	Pitch (°)	1.5	2.0	2.5	2.5	3.0	3.0	250 to 275		10
		N ₁ (%)	76.3	76.7	77.5	78.3	79.5	80.3			
		AoA	0.4	0.4	0.4	0.4	0.5	0.5			
If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.											



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

1. Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
2. Adjust pitch to stop climb/descent.
3. If pitch is increased, REDUCE expected speed range by delta KIAS/delta Mach.
4. If pitch is decreased, INCREASE expected speed range by delta KIAS/delta Mach.

Table A – Level Flight – FLAPS 0 (Cont'd)

Table A – Level Flight – FLAPS 0 (Cont'd)											
			WEIGHT kg (lb)					Expected SPD Range		Expected Speed Change for 0.5 deg change in pitch ΔKIAS/ ΔMach	
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)	KIAS		MACH
ALTITUDE (1,000 ft)	20	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	250 to 275		10
		N ₁ (%)	72.9	73.3	74.0	74.9	75.9	76.7			
		AoA	0.4	0.4	0.4	0.4	0.5	0.5			
	15	Pitch (°)	1.5	2.0	2.5	3.0	3.0	3.5	250 to 275		10
		N ₁ (%)	70.5	70.8	71.5	72.3	73.3	74.0			
		AoA	0.4	0.4	0.4	0.4	0.5	0.5			
	10	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	215 to 230		5
		N ₁ (%)	58.0	58.9	60.2	61.7	63.1	63.9			
		AoA	0.4	0.4	0.5	0.5	0.6	0.6			
	5	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225		5
		N ₁ (%)	54.8	55.7	57.0	58.3	59.7	60.5			
		AoA	0.4	0.4	0.4	0.5	0.5	0.5			
	0	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225		5
		N ₁ (%)	51.9	52.7	53.9	55.2	56.5	57.3			
		AoA	0.3	0.4	0.4	0.5	0.5	0.5			
If any indicated speed is outside the expected indicated airspeed range, it should be considered UNRELIABLE.											

1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

1. Use FD in PTCH, ALT, HDG or ROLL modes only.
2. If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.

CAUTION

1. Do not use any AP/FD SPEED mode.
2. Do not use FLC mode or ATS.

Table B – Climb – FLAPS 0

ALT (ft)	Pitch (°)	N ₁ (%)	AOA (°)	Expected SPD Range	
				KIAS	MACH
0	13.5	TO	0.3	225 to 300	
5000	13.0	TO	0.3	215 to 300	
10000	11.5	TO	0.4	215 to 315	
15000	10.0	CLB	0.4	225 to 310	
20000	8.0	CLB	0.4	230 to 295	
25000	6.0	CLB	0.4	255 to 305	
30000	4.0	CLB	0.4	265 to 290	.70 to .76
35000	3.5	CLB	0.4	245 to 265	.73 to .78



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

1. Use FD in PTCH, ALT, HDG or ROLL modes only.
2. If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.



1. Do not use any AP/FD SPEED mode.
2. Do not use FLC mode or ATS.

Table C – Descent – FLAPS 0					
ALT (ft)	Pitch (°)	N ₁ (%)	AOA (°)	Expected SPD Range	
				KIAS	MACH
40000	-1.5	IDLE	0.4	230 to 255	.75 to .83
35000	-1.5	IDLE	0.4	235 to 280	.69 to .82
30000	-2.0	IDLE	0.4	235 to 305	.62 to .79
25000	-2.0	IDLE	0.4	230 to 305	
20000	-2.0	IDLE	0.4	230 to 305	
15000	-1.0	IDLE	0.4	215 to 280	
10000	-1.0	IDLE	0.4	220 to 275	
5000	-1.0	IDLE	0.4	205 to 280	
0	-1.0	IDLE	0.4	205 to 280	

NOTE

With anti-ice ON, increase N₁ if required to obtain the minimum N₂ of 78%.

- For an N₁ increase of 5%, add 0.5° to the pitch from Table C.
- For an N₁ increase of 10%, add 1° to the pitch from Table C.
- For an N₁ increase of 15%, add 2° to the pitch from Table C.



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
- Adjust pitch to stop climb/descent.
- If pitch is increased, REDUCE expected speed range by delta KIAS.
- If pitch is decreased, INCREASE expected speed range by delta KIAS.
- Use FD in PTCH, ALT, HDG or ROLL modes only.
- If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.



- Do not use any AP/FD SPEED mode.
- Do not use FLC mode or ATS.

To avoid significant flap overspeed and flap damage:

Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight.

Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

Table D 1 – Approach – Level Flight – FLAPS 0 – GEAR UP										
			WEIGHT kg (lb)					Expected SPD Range KIAS	Expected ΔKIAS per 0.5° Δ pitch	
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)			21,772 (48,000)
ALT (1,000ft)	15	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	215 to 230	5
		N1 (%)	61.1	62.2	63.6	65.3	67.0	68.0		
		AoA (°)	0.4	0.4	0.5	0.6	0.6	0.6		
	10	Pitch (°)	2.0	2.5	3.5	4.0	4.5	5.0	215 to 230	5
		N1 (%)	58.0	58.9	60.2	61.7	63.1	63.9		
		AoA (°)	0.4	0.4	0.5	0.5	0.6	0.6		
	5	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225	5
		N1 (%)	54.8	55.7	57.0	58.3	59.7	60.5		
		AoA (°)	0.4	0.4	0.4	0.5	0.5	0.5		
	0	Pitch (°)	2.5	3.0	3.5	4.5	5.0	5.5	210 to 225	5
		N1 (%)	51.9	52.7	53.9	55.2	56.5	57.3		
		AoA (°)	0.3	0.4	0.4	0.5	0.5	0.5		



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
- Adjust pitch to stop climb/descent.
- If pitch is increased, REDUCE expected speed range by delta KIAS.
- If pitch is decreased, INCREASE expected speed range by delta KIAS.
- Use FD in PTCH, ALT, HDG or ROLL modes only.
- If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.



- Do not use any AP/FD SPEED mode.
- Do not use FLC mode or ATS.

To avoid significant flap overspeed and flap damage:

Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight.

Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

Table D 2– Approach – Level Flight – FLAPS 20 – GEAR UP										
			WEIGHT kg (lb)					Expected SPD Range KIAS	Expected ΔKIAS per 0.5° Δ pitch	
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)			21,772 (48,000)
ALT (1,000ft)	15	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N1 (%)	66.6	67.6	68.9	70.4	72.3	73.4		
		AoA (°)	0.3	0.4	0.4	0.5	0.5	0.5		
	10	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N1 (%)	63.1	64.0	65.2	66.6	68.4	69.4		
		AoA (°)	0.3	0.3	0.4	0.5	0.5	0.5		
	5	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N1 (%)	59.7	60.5	61.7	63.0	64.6	65.6		
		AoA (°)	0.3	0.3	0.4	0.5	0.5	0.5		
	0	Pitch (°)	0.5	1.0	2.0	3.0	4.0	4.5	165 to 185	5
		N1 (%)	56.4	57.2	58.4	59.6	61.2	62.1		
		AoA (°)	0.3	0.3	0.4	0.5	0.5	0.5		



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

- Pitch/Thrust settings will give approximate level flight, and may result in slight climb or descent. Interpolation is acceptable.
- Adjust pitch to stop climb/descent.
- If pitch is increased, REDUCE expected speed range by delta KIAS.
- If pitch is decreased, INCREASE expected speed range by delta KIAS.
- Use FD in PTCH, ALT, HDG or ROLL modes only.
- If FD/autopilot engaged in PTCH mode, one click of PTCH wheel results in 0.5 degrees of pitch change.



- Do not use any AP/FD SPEED mode.
- Do not use FLC mode or ATS.

To avoid significant flap overspeed and flap damage:

Transition from FLAPS 0 to FLAPS 30 must be accomplished in level flight.

Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

Table D 3– Approach – Level Flight – FLAPS 30 – GEAR DOWN										
			WEIGHT kg (lb)					Expected SPD Range KIAS	Expected ΔKIAS per 0.5° Δ pitch	
			11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)			21,772 (48,000)
ALT (1,000ft)	15	Pitch (°)	–2.0	–1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N1 (%)	77.1	78.0	79.0	80.1	81.3	82.2		
		AoA (°)	0.3	0.4	0.4	0.5	0.6	0.6		
	10	Pitch (°)	–2.0	–1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N1 (%)	73.1	73.9	74.8	75.9	77.0	77.8		
		AoA (°)	0.3	0.4	0.4	0.5	0.6	0.6		
	5	Pitch (°)	–2.0	–1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N1 (%)	69.1	69.9	70.9	72.0	73.0	73.7		
		AoA (°)	0.3	0.4	0.4	0.5	0.6	0.6		
	0	Pitch (°)	–2.0	–1.0	0.5	1.5	3.0	3.5	160 to 175	5
		N1 (%)	65.5	66.3	67.2	68.2	69.2	69.9		
		AoA (°)	0.3	0.4	0.4	0.5	0.6	0.6		



1. INSTRUMENTS SYSTEM (CONT'D)

A. Unreliable Airspeed (Cont'd)

NOTE

1. Set N_1 to maintain AOA.
2. Adjust pitch as necessary to maintain required glideslope guidance/glidepath.

CAUTION

Select FLAPS 45 at the start of final descent on the glideslope/glidepath.

TABLE E - APPROACH - FLAPS 45 - GEAR DN - 3 DEGREE G/S								
ALT (FT)		WEIGHT KG (LB)						EXPECTED SPD RANGE KIAS
		11,794 (26,000)	13,608 (30,000)	15,876 (35,000)	18,144 (40,000)	20,412 (45,000)	21,772 (48,000)	
15000	Pitch (°)	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	V _{REF} +/- 10
	N_1 (%)	65.5	67.7	71.2	74.9	77.3	76.7	
	AoA	0.5	0.5	0.5	0.5	0.5	0.5	
10000	Pitch (°)	-2.5	-2.5	-2.5	-2.5	-2.5	-2.5	
	N_1 (%)	60.8	62.5	66.3	69.8	72.8	74.6	
	AoA	0.5	0.5	0.5	0.5	0.5	0.5	
5000	Pitch (°)	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	
	N_1 (%)	55.9	57.9	61.4	65.0	67.9	69.4	
	AoA	0.6	0.6	0.6	0.6	0.6	0.6	
0	Pitch (°)	-2.0	-1.0	-1.0	-1.0	-1.0	-1.0	
	N_1 (%)	52.5	53.4	56.7	59.9	62.8	64.2	
	AoA	0.6	0.6	0.6	0.6	0.6	0.6	

— END —

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM

A. Gear Disagree

Indication: **GEAR DISAGREE** warning message and “GEAR DISAGREE” aural on and/or

One or more landing gear position indications displays red hash marks.

(1) Airspeed Reduce to not more than:

- Flaps 0° – 181 KIAS.
- Flaps 20° – 197 KIAS.
- Flaps 30° – 197 KIAS.
- Flaps 45° – 189 KIAS.



Leave hydraulic pump 3B OFF if pump was selected OFF due to a low or decreasing pressure, or a low quantity in system No. 3.

(2) HYDRAULIC pump switches
2B and 3B.....Select to ON.

GEAR DISAGREE warning message goes out and three gear position indications agree:

Yes

(3) No further action required.

– END –

No

Gear disagree occurred when LDG GEAR lever was selected UP:

Yes

(3) Accomplish manual gear extension.

– END –

No

Gear disagree occurred when LDG GEAR lever was selected DN:

(4) LDG GEAR leverSelect to UP.

NOTE

If necessary, press DN LCK REL button to release landing gear lever.

Wait 30 seconds, then:

(5) LDG GEAR leverSelect to DN.



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

A. Gear Disagree (Cont'd)

NOTE

- Steps (4) and (5) may be repeated as many times as desired.
- Quickly cycling the landing gear lever UP then DN may also provide a down and locked indication. For quick cycling, do not wait for the landing gear UP indications before re-selecting the landing gear lever down.

GEAR DISAGREE warning message goes out and three DN indications are displayed:

Yes

(6) No further action required.

- END -

No

(6) LDG GEAR lever Select to UP.

Gear disagree persists:

CAUTION

Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.

(7) LANDING GEAR MANUAL
RELEASE handle PULL to full extension.

NOTE

The **NOSE DOOR OPEN** warning message may be displayed when the LANDING GEAR MANUAL RELEASE handle is pulled. If the nose landing gear is in transition, this message will be inhibited.

(8) LDG GEAR lever Select to DN.

Gear disagree persists:

Yes

(9) HYDRAULIC pump switches
3A and 3B Select to OFF.

(10) Hydraulic pressure Verify system 3 pressure is decreasing.

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

A. Gear Disagree (Cont'd)

NOTE

Hydraulic system 3 pressure is decreased to 200 psi, in order to remove all hydraulic pressure from the landing gear system.

When hydraulic system 3 pressure is less than 200 psi:

(11) HYDRAULIC pump switches

3A and 3B.....Select to ON.

(12) Landing gear indicationCheck for three DN indications.

Gear disagree persists:

Yes

(13) Landing Gear Up/Unsafe

Landing ProcedureAccomplish Refer to Landing Gear Up/Unsafe Landing Procedure in this section.

- END -

No

CAUTION

Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering may not be available upon landing even if the **STEERING INOP** caution message is not displayed.

(9) No further action required.

END

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

B. Landing Gear Lever Jammed in the UP Position

- (1) Airspeed Reduce to not more than:
- Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.

- (2) HYDRAULIC pump switches
2B and 3B..... Select to ON.



Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.

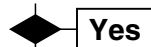
- (3) LANDING GEAR MANUAL
RELEASE handle PULL to full extension.

NOTE

1. The **NOSE DOOR OPEN** warning message may be displayed when the LANDING GEAR MANUAL RELEASE handle is pulled. If the nose landing gear is in transition, this message will be inhibited.
2. The **GEAR DISAGREE** warning message will appear with the LDG GEAR lever jammed in the UP position if any landing gear is not in its uplock.

- (4) Landing gear indication Check for three DN indications.

Any landing gear fails to lock in the down position:



- (5) HYDRAULIC pump switches
3A and 3B..... Select to OFF.
- (6) Hydraulic pressure..... Verify system 3 pressure is decreasing.

NOTE

Hydraulic system 3 pressure is decreased to 200 psi, in order to remove all hydraulic pressure from the landing gear system.



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

B. Landing Gear Lever Jammed in the UP Position (Cont'd)

When hydraulic system 3 pressure is less than 200 psi:

- (7) HYDRAULIC pump switches
3A and 3B.....Select to ON.
- (8) Landing gear indicationCheck for three DN indications.

Gear disagree persists:



Yes

NOTE

1. The LANDING GEAR MANUAL RELEASE handle can be cycled more than once during this procedure. Cycling of the LANDING GEAR MANUAL RELEASE handle should be performed if the flight crew experienced difficulty with the initial pull.
2. The Gear DN Disagree procedure can be repeated more than once.

- (9) Landing Gear Up/Unsafe
Landing ProcedureAccomplish Refer to Landing Gear
Up/Unsafe Landing Procedure
in this section.

— END —

No

Prior to landing:

- (5) NOSE STEER switchSelect to OFF.

CAUTION

Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering will not be available upon landing.

- (6) No further action required.

— END —

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

C. Landing Gear Up/Unsafe Landing Procedure

NOTE

1. IF ONE MAIN LANDING GEAR IS UP OR UNSAFE, hold applicable wing up for as long as possible. Maintain directional control with rudder and nosewheel steering (if considered safe). When wing touches the ground, apply asymmetrical braking for directional control.
2. IF NOSE LANDING GEAR IS UP OR UNSAFE, relocate passengers (if possible) to obtain aft CG. Trim stabilizer nose-up after touchdown. Gently lower the nose before elevator effectiveness is lost.
3. IF ALL WHEELS ARE UP OR UNSAFE, perform a nose high attitude touchdown, but do not reduce touchdown speed below stick shaker speed.
4. IF BOTH MAIN LANDING GEAR CANNOT BE LOCKED, consideration should be given to landing with all wheels up.

PRELIMINARY

- (1) Descent Plan to reduce fuel to the minimum, if possible, while retaining sufficient fuel to make a controlled, powered approach.

PREPARATION

- (1) Crew Alert and brief type of emergency, time available, airplane attitude after landing and exits available for use.
- (2) ATC Notify
- (3) PASS SIGNS, NO SMKG and SEAT BLTS switches Select both to ON.
- (4) Loose equipment Secure stow all loose equipment in the flight compartment.
- (5) TAWS circuit breaker (1B13) Open to silence TAWS aural warnings.
Check **TAWS FAIL** status message on.
- (6) AUDIO WARNING switches DISABLE to silence DCU aural alerts.
Check **DCU 1 and 2 AURAL INOP** status messages on.
- (7) Shoulder harness Tightened and locked.



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

C. Landing Gear Up/Unsafe Landing Procedure (Cont'd)

APPROACH

- (1) AIR-CONDITIONING, L and R
PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (2) CABIN PRESSURIZATION,
PRESS CONT switch/light Press in to select manual mode.
Check PRESS CONT MANUAL light on.
- (3) CABIN PRESSURIZATION, MAN
ALT selector Select UP and hold.
- (4) CABIN PRESSURIZATION, MAN
RATE switch As required

When depressurized:

- (5) BLEED AIR, L and R 10TH
STAGE BLEED switch/lights Press out to close both SOVs.
Check L and R 10TH STAGE CLOSED lights on.
- (6) FLAPS Select to 45°.
- (7) EMER LTS switch Select to ON.
Check **EMER LTS ON** status message on.
- (8) Brace for impact Order over the PA system,
"BRACE-BRACE-BRACE".

BEFORE IMPACT

- (1) Airplane attitude Maintain a nose high attitude.
- (2) Ground/landing field Contact with minimum forward speed,
but not less than stick shaker speed, and at a minimum sink rate.

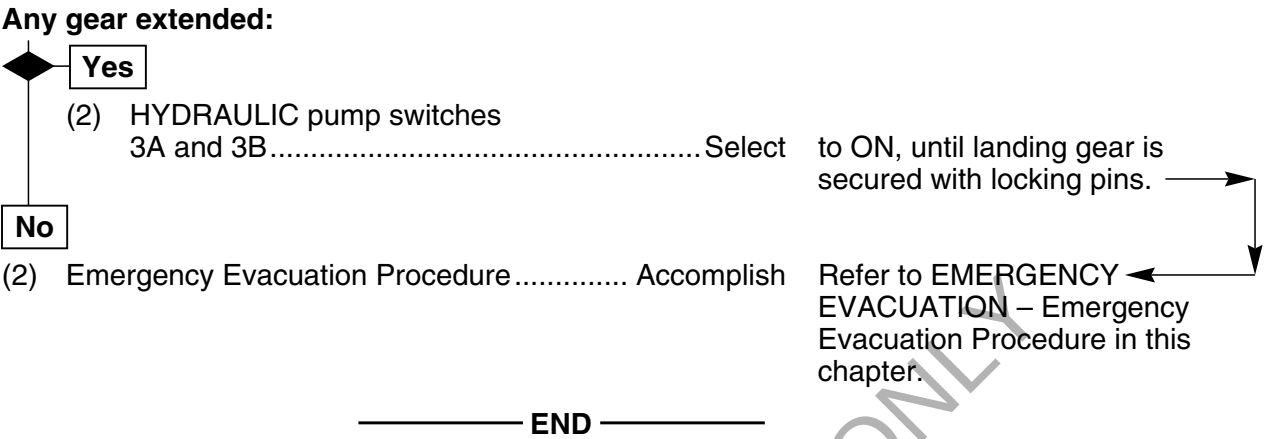
AFTER LANDING

- (1) Thrust levers SHUT OFF



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

C. Landing Gear Up/Unsafe Landing Procedure (Cont'd)



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

D. Main Landing Gear Bay Overheat Warning

Indication: **MLG BAY OVHT** warning message and "GEAR BAY OVERHEAT" aural on.

- (1) Airspeed Reduce to 197 KIAS.
(2) LDG GEAR lever Select to DN.

MLG BAY OVHT warning message persists:

◆ **Yes**

- (3) LDG GEAR lever Leave DN
(4) Airspeed Do not exceed 250 KIAS
(5) Land at the nearest suitable airport.

— END —

No

- (3) Flight Continue with gear down, for minimum of 5 minutes.
(4) LDG GEAR lever Select to UP.

— END —

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

E. Nose Door Fails Open

Indication: **NOSE DOOR OPEN** warning message and “NOSE DOOR” aural on.

- (1) Airspeed Reduce to prevent structural damage to nose gear doors, as follows:
- Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.
- (2) LDG GEAR lever Select to DN.
- (3) Land at the nearest suitable airport.

NOTE

Use of LANDING GEAR MANUAL RELEASE handle will cause **NOSE DOOR OPEN** warning message to be displayed.

————— END —————

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

F. Parking Brake On

Indication: **PARKING BRAKE** warning message, and "CONFIG BRAKES" aural on.

On ground (both engines N_1 greater than 70%):

- ◆ Yes
- (1) Thrust levers Retard
 - (2) Parking brake As required
- END -

No

In flight:

- (1) Parking brake handle Check released.

PARKING BRAKE warning message persists:

◆ Yes

CAUTION

Inboard and outboard anti-skid systems are inoperative. Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

- END -

No

- (2) No further action required.

- END -

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

G. Excessive Asymmetry or Loss of Braking

If excessive asymmetry or loss of braking is observed:

- (1) Wheel brakesRelease momentarily.
- (2) ANTI SKID switch.....Select to OFF.
- (3) Wheel brakes Re-apply as required.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

————— END —————

TRAINING PURPOSES ONLY

1. PASSENGER DOOR

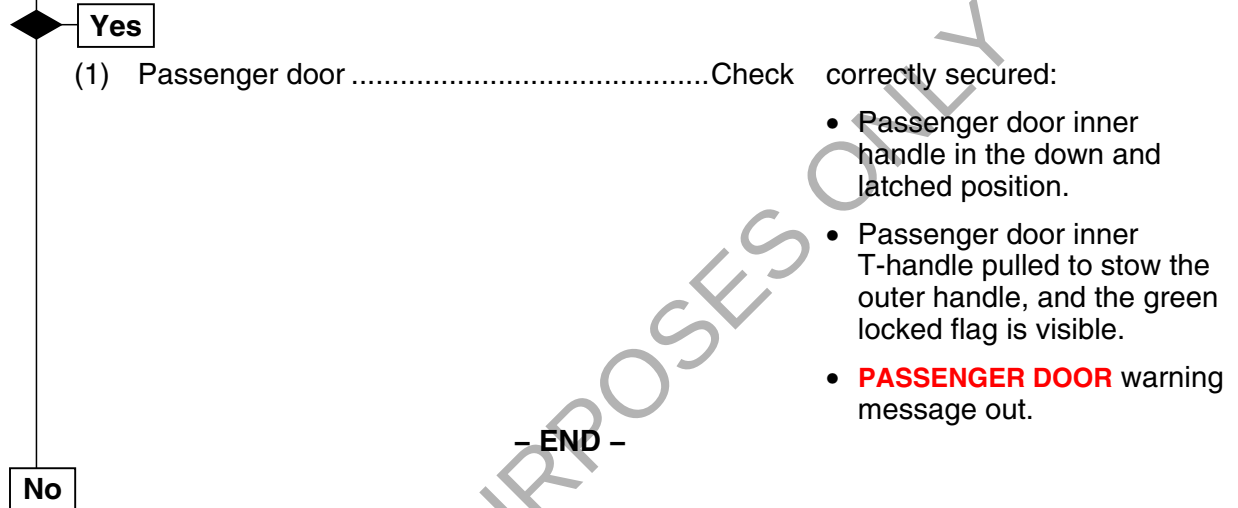
A. Passenger Door Failure

Indication: **PASSENGER DOOR** warning message and “DOOR” aural on (when engines are running).

NOTE

The passenger door warning comes on whenever two or more of the six door/handle sensors indicate that the door is open.

On ground:



In flight:

- (1) CABIN PRESSURIZATION, EMER DEPRESS switch/light Press in to depressurize the airplane. Check the following:
- **EMER DEPRESS** caution message on, and
 - **CABIN ALT** caution message on at 8,500 feet cabin altitude.
 - **CABIN ALT** warning message and “CABIN PRESSURE” aural on at 10,000 feet cabin altitude.



Do not attempt to pressurize the airplane.

- (2) Land at the nearest suitable airport.



BOMBARDIER CHALLENGER 605	EMERGENCY PROCEDURES Passenger Door	Vol. 1	03-17-2 REV 40, Aug 30/16
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1. PASSENGER DOOR (CONT'D)

A. Passenger Door Failure (Cont'd)

- (3) Emergency Descent Procedure Initiate

to safe altitude.

Refer to AIR-CONDITIONING
AND PRESSURIZATION –
Emergency Descent
Procedure in this chapter.

—————

END

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TRAINING PURPOSES ONLY

1. AURAL/VISUAL WARNING SYSTEM

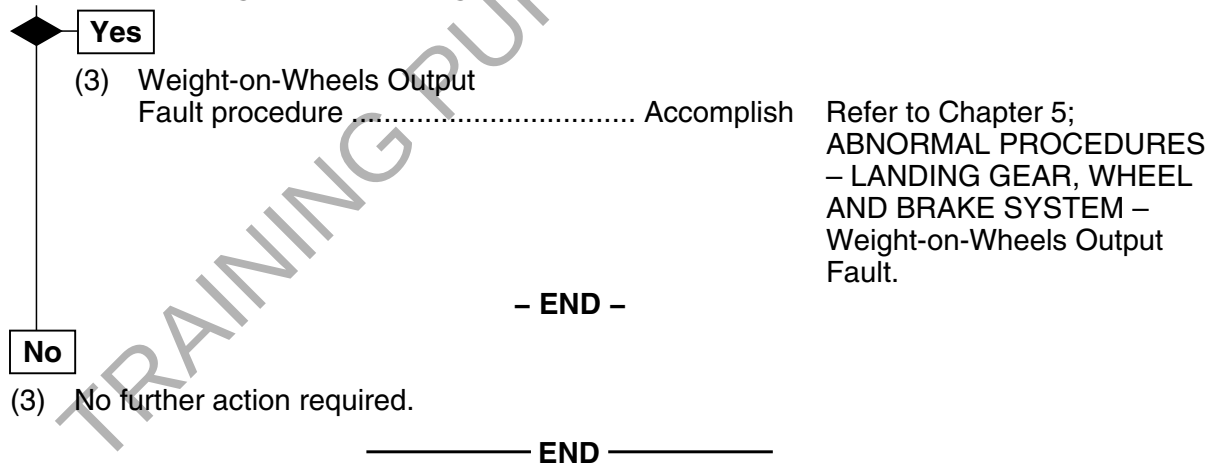
A. Configuration Warning

Indication: The following configuration warning messages and aural warnings come on when advancing thrust levers to take-off power (greater than 70% N_1):

WARNING MESSAGE	AURAL WARNING
CONFIG AP	“CONFIG AUTOPILOT”
CONFIG FLAPS	“CONFIG FLAPS”
CONFIG SPOILERS	“CONFIG SPOILERS”
CONFIG STAB	“CONFIG TRIM”
PARKING BRAKE	“CONFIG BRAKES”

- (1) Take-off Reject
- (2) Airplane configuration Confirm
- Check the position of the following items:
- Autopilot;
 - Flaps;
 - Spoilers;
 - Stabilizer; and
 - Parking brake.

Erroneous configuration messages and aural warnings occur:



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CHAPTER 4 - NORMAL PROCEDURES

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1. GENERAL

This chapter informs the crew about normal procedures which are necessary for the safe and orderly operation of the CL600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049). Normal procedures are presented in the form of a sequence of steps, with relevant expanded information.

Generally, all items of a given procedure are intended to prepare the airplane for the next phase of flight. The procedure items are listed for each phase of flight (BEFORE START – TAXI etc.), in a standardized scan of the flight compartment panels or equipment, except where required by the logic of action priority.

Besides standardization, this scan:

- Enables the flight crew to more readily memorize the required items,
- Ensures that all necessary actions have been performed in the most efficient manner, and
- Minimizes the time required.

2. AIRPLANE PREPARATION PROCEDURES

There are two possible ways to perform airplane preparation procedure:

- To perform the complete procedure including all actions prescribed in this manual, or
- To perform the reduced procedure following the same scan (as detailed in the complete procedure), omitting mainly system tests and functional checks to the extent indicated by brackets designated “Ø” in the expanded procedure.

The omit sign “Ø” in the procedure guides means either the associated step is completely omitted or some actions out of it are omitted.

The reduced procedure ensures correct positions of all controls and flight instruments for a safe and economic airplane operation.

The complete airplane preparation procedure is required after an airplane maintenance check, higher than a transit stop check, before every originating flight and before the first flight of the day.

It rests upon the judgement of the flight crew to perform the reduced procedure in all other cases, provided there is no doubt that all safety aspects are covered.

The “1” sign in the procedure guides means that the associated procedure or step is accomplished only on the first flight of the day.

The “2” sign in the procedure guides means that the associated procedure or step is accomplished only on the last flight of the day.

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3. NORMAL PROCEDURES

Normal procedures for each flight are performed by recall (memory). After completion of a given procedure, the related normal checklist is used to ascertain that the airplane is prepared and safe for the next phase of flight. This requires that all safety items and some selected determinant operational items be included in the checklist.

All crew members must report to the pilot-in-command all abnormalities discovered during airplane and systems inspections and checks.

NOTE

Throughout all normal procedures, even when not specifically written, it is assumed that crew members pay continuous attention to the airplane systems through periodic checks of the various instruments, displays and circuit breaker panels.

All checklists which have to be performed on the ground are initiated at the command of the pilot with every crew member at his/her station.

All checklists which have to be performed in flight are initiated at the command of the PF with every crew member at his/her station. However, when the copilot is the pilot-flying, the pilot-in-command will still retain the final authority for all actions directed or performed.

Some normal procedures, which are not routine at the same point of each flight, will be found in Chapter 6; SUPPLEMENTARY PROCEDURES.

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1. GENERAL

Normal Procedures describe actions which are necessary for routine airplane operation:

- The sequence of procedures follow the natural phases of flight.
- The sequence of actions in a procedure follow a standardized scan of panels or equipment, except when required by system logic or priority.
- For all crew actions, relevant expanded information is provided.

All actions (or steps) of a procedure must be performed, for the following reasons:

- They are required to operate the airplane safely in the related phase of flight.
- To prepare the airplane for the following phase of flight.

They should be performed by the assigned crew member, to ensure coordinated and safe operation.

2. NORMAL PROCEDURES

Normal procedures are usually performed by recall (memory), with the exception of less frequently used procedures, wherein reference to the FCOM is recommended.

During the airplane preparation, the complete pre-flight checks must always be performed:

- For every originating flight (flight crew's first flight of the day), or
- After any non-routine maintenance action, or
- After the airplane was left unattended by qualified personnel, or
- If there is any crew doubt about the airplane status.

3. NORMAL CHECKLISTS

Normal checklists are used to ensure that a standard flight compartment and airplane configuration, covering all safety aspects, is achieved.

Normal checklists are used after the appropriate procedures have been completed. Normal checklists are read by the copilot when the airplane is on the ground and by the pilot-not-flying (PNF) when the airplane is in flight.

After the complete pre-flight check or the reduced check procedure is completed, the pilot requests the checklist (pre-flight items) from the copilot. All on-ground checklists (except pre-flight and taxi) are initiated by the pilot by stating ".....items", triggering the copilot to do the necessary actions. After completion, the copilot offers the respective checklist by announcing its title. The pilot commands reading (or not) by saying: "go-ahead" or "standby".

Normal checklists are of two types; the challenge and response type, and the silent type. In the challenge and response checklist, the applicable crew member shall respond to the challenge after having verified the existing configuration. The other crew member shall then cross-check, whenever feasible, the validity of the response. The challenger shall wait for the response before proceeding with the checklist. If the actual configuration is not in accordance with the checklist standard, corrective action shall be initiated. If corrective action is not possible, the response must be modified to reflect the actual situation.

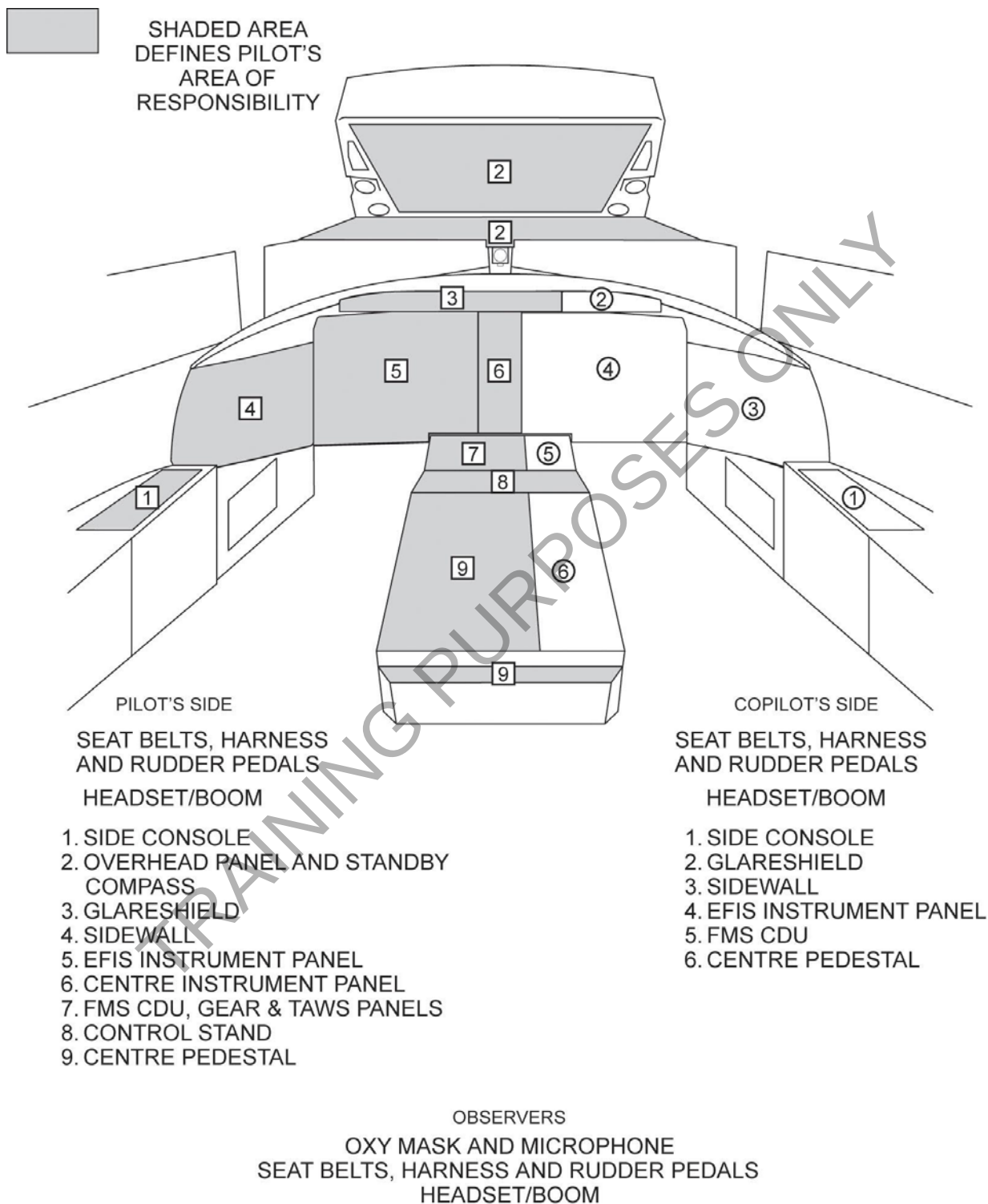
The silent checklist is accomplished by the designated crew member, and does not warrant any response or verification from the other crew member.

In both checklists, however, the applicable crew member shall announce "checklist completed", upon completion.

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4. PANEL SCAN SEQUENCE

Figure 04-02-1 illustrates the panel scan sequence recommended when accomplishing preflight checks.



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1. GENERAL

This section contains information pertaining to crew member responsibility during various phases of airplane operation.

Orders (callouts requesting a physical action) will be acknowledged by repeating the order prior to initiating the action, to prevent incorrect responses. This will be followed by a further acknowledgment when the system has reached the desired position. Other callouts will be acknowledged by the word "check".

2. CALLOUTS DURING TAKE-OFF

CONDITION	CALL	CREW MEMBER
When PF starts to advance thrust levers to take-off power	Set thrust – time	PF
When takeoff thrust is set	Thrust – set	PNF
When PFD shows: 80 knots At V_1 At V_R At V_2	80 knots V_1 Rotate V_2	PNF PNF PNF PNF
When take-off is to be rejected (for any reason)	Reject – max brakes	Pilot
When positive rate of climb is attained	Positive rate Gear up	PNF PF
LDG GEAR lever up and EICAS indicates 3 white UP lights on	Gear up – Indicated	PNF
At $V_2 + 10$ to 15 KIAS	Speed mode	PF
When speed mode is engaged on FCP	Speed mode – set	PNF
At 320 feet AGL, minimum	Autopilot – On	PF
When autopilot is engaged	Autopilot – set	PNF
Conditions for flaps retraction are fulfilled	Flaps up	PF
When EICAS indicates flaps at zero degrees	Flaps up – Indicated	PNF
Conditions for reduction to climb thrust are fulfilled	Climb thrust, After Take-off Check	PF
When climb thrust is set	Climb thrust - set	PNF
When AFTER TAKE-OFF CHECK is completed	After Take-off Check – Complete	PNF

3. CALLOUTS DURING CLIMB

CONDITION	CALL	CREW MEMBER
Passing transition altitude	Transition altitude	PNF
1,000 feet before reaching clearance altitude or flight level	1,000 feet to go	PNF

4. CALLOUTS DURING DESCENT

CONDITION	CALL	CREW MEMBER
1,000 feet before reaching clearance level or altitude	1,000 feet to go	PNF
Passing transition level	Transition level	PNF

5. CALLOUTS DURING APPROACH

CONDITION	CALL	CREW MEMBER
Conditions for initial flaps extension are fulfilled	Flaps 20	PF
When EICAS indicates flaps at 20 degrees	Flaps 20 – Indicated	PNF
When localizer course bar starts to move from full scale deflection	Localizer alive	PNF
When glideslope pointer starts to move downward	Glideslope alive	PNF

6. CALLOUTS DURING APPROACH AND LANDING

CONDITION	CALL	CREW MEMBER
When at the appropriate segment of the approach	Gear down, Before Landing Check	PF
When landing gear is down and EICAS indicates 3 green DN lights on	Gear down – 3 greens	PNF
When BEFORE LANDING CHECK is completed	Before Landing Check – Complete	PNF
Conditions for flaps extension to 30 degrees/45 degrees are fulfilled	Flaps 30/45	PF
When EICAS indicates flaps at 30 or 45	Flaps 30/45 – Indicated	PNF
At final approach fix (FAF)	Final approach fix (by name)	Both
If maximum localizer deviation is exceeded	Localizer	PNF
If maximum glideslope deviation is exceeded	Glide path	PNF
If maximum speed deviation is exceeded	Speed	PNF
If maximum descent rate is exceeded	Sink rate	PNF
If maximum altitude deviation is exceeded	Altitude	PNF
At 500 AGL (Precision approaches)	500 feet AGL	PNF
At 100 feet above decision height (DH) or decision altitude (DA)	100 above Decision	PNF
At decision height (DH) or decision altitude (DA)	Decision	PNF
At 100 feet above Minimum Descent Altitude (MDA)	100 above MDA	PNF
At Minimum Descent Altitude (MDA)	Minimums	PNF
At Missed Approach Point (MAP)	Missed approach point	PNF
At the MAP or DH/DA, when visual reference is established	Landing	PF
At pilot's decision to become PF	I have control	Pilot
At minimum height for autopilot use, if still engaged	Autopilot – disengage	PNF

7. CALLOUTS DURING GO-AROUND

CONDITION	CALL	CREW MEMBER
At the MAP or DH/DA and the runway is not in sight or at any instance when a go-around is called for	Go-Around	PF
While advancing the thrust levers to the vertical position	Go-Around thrust	PF
When Go-Around thrust is set	Go-Around thrust – set	PNF
While achieving G/A attitude and thrust	Flaps 20	PF
When EICAS indicates flaps at 20 degrees	Flaps 20 – Indicated	PNF
When positive rate of climb is attained	Positive rate Gear up	PNF PF
EICAS indicates 3 white UP lights on	Gear up – Indicated	PNF

8. CALLOUTS DURING LANDING ROLL

CONDITION	CALL	CREW MEMBER
EICAS page L and R REV ARMED advisory message is on at 70 knots	Two greens	PNF
When Pilot wants to take control	I have control	Pilot

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1. AIRPLANE PREPARATION

A. General

Before power (electrical, hydraulic or pneumatic) can be safely applied to the airplane, the external safety inspection must be made, either by qualified technical/maintenance ground staff or by the pilot. Once power has been applied to the airplane, the pilot performs the external walkaround and the systems check (if required).

The normal power-up check and the systems check are required:

- Before the airplane's first flight of the day, or
- After a crew change, or
- After any related non-routine maintenance action, or
- If the airplane was left unattended by qualified personnel, or
- If the pilot-in-command has any doubt about the airplane status.

B. External Safety Inspection (First Flight of the Day) Ø

This procedure must be performed before entering an airplane which has been left unattended, or which is taken over from maintenance responsibility. This procedure must be followed to ascertain that no obvious unsafe conditions exist prior to applying power (AC or DC, pneumatic or hydraulic) to the airplane. All panels, not required for the safety inspection, will be understood as closed and secured.

These procedures should be performed on through flights if there is any doubt that all safety aspects of the prevailing situation can be covered.

During cold weather operations, the flight crew must ensure that the airplane fuselage, wings and tail surfaces are free from ice, snow or frost (Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION).

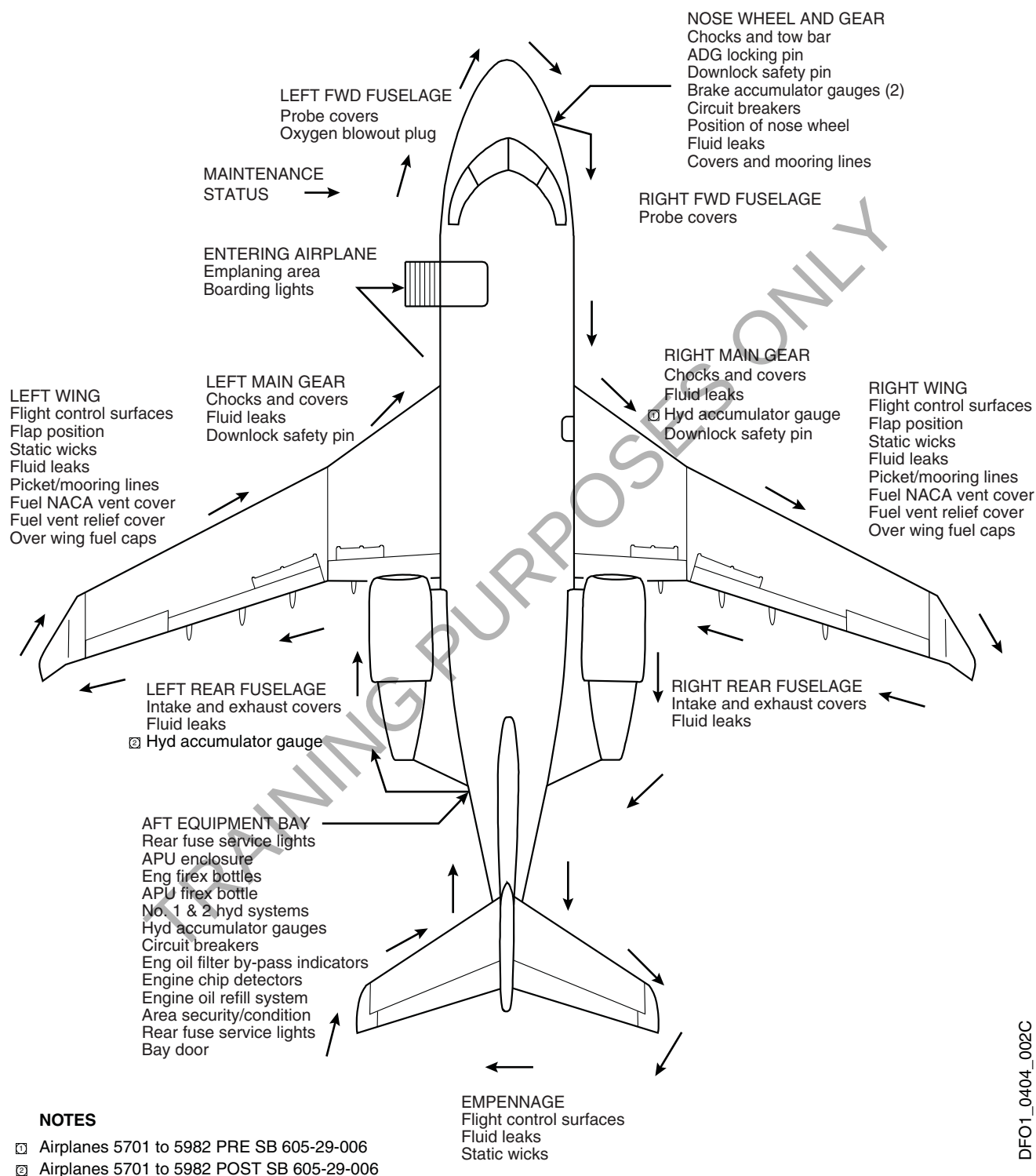
Start the inspection at the top of the boarding stairs, proceed in the direction indicated in Figure 04-04-1 and terminate at the bottom of the passenger stair unit.



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1. AIRPLANE PREPARATION (CONT'D)

B. External Safety Inspection (First Flight of the Day) Ø (Cont'd)

External Safety Inspection
Figure 04-04-1

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1. AIRPLANE PREPARATION (CONT'D)

B. External Safety Inspection (First Flight of the Day) Ø (Cont'd)

AIRPLANE DOCUMENTATION

- | | | |
|------------------------------|-------|---|
| (1) Maintenance status | Check | Ensure that airplane maintenance status is in accordance with Minimum Equipment List (MEL) and Configuration Deviation List (CDL) requirements. |
|------------------------------|-------|---|

LEFT FORWARD FUSELAGE

- | | |
|-------------------------------|---------|
| (1) Probe covers | Removed |
| (2) Oxygen blowout plug | Intact |

NOSEWHEEL AND GEAR

- | | | |
|--|-------------|---|
| (1) Chocks | Installed | |
| (2) Towbar | As required | |
| (3) ADG safety pin | Installed | |
| (4) Downlock safety pin | Installed | |
| (5) Brake accumulator gauges (2) | Check | Ensure pressure is between 700 and 800 psi. |
| (6) Circuit breakers (JB6) | Closed | |
| (7) Position of nosewheel | Check | Ensure steering angle is within nominal operating range of $\pm 55^\circ$. |
| (8) Fluid leaks | Check | |
| (9) Covers/mooring lines | Removed | |

RIGHT FORWARD FUSELAGE

- | | |
|------------------------|---------|
| (1) Probe covers | Removed |
|------------------------|---------|



1. AIRPLANE PREPARATION (CONT'D)

B. External Safety Inspection (First Flight of the Day) Ø (Cont'd)

RIGHT MAIN GEAR

- (1) Chocks As required
- (2) Wheel covers..... Removed
- (3) Fluid leaksCheck

5701 to 5982 NOT incorporating Service Bulletin 605-29-006

- (4) No. 3 hyd. accumulator gaugeCheck

Nominal pressure is 1,450 to 1,550 psi at 10°C (50°F). If pressure is less than 1,200 psi, verify that pressure drop is less than 250 psi per hour, and ensure that pressure prior to departure is 750 psi minimum.

- (5) Downlock safety pin Installed

RIGHT WING

- (1) Flight control area and surfaces Clear

Ensure proper clearance of all flight control surfaces from all ground equipment or other obstructions. Ensure surfaces clear of dirt, sand, ice, snow or frost.
- (2) Flap position Note
- (3) Static wicks..... Condition/secured
- (4) Fluid leaksCheck
- (5) Picket/mooring lines Removed
- (6) Fuel NACA vent cover..... Removed

Ensure vent is clear of any obstructions or contaminants. Check for evidence of fuel spillage.
- (7) Fuel vent relief cover Removed
- (8) Overwing fuel caps..... Secured

1. AIRPLANE PREPARATION (CONT'D)**B. External Safety Inspection (First Flight of the Day) Ø (Cont'd)****RIGHT REAR FUSELAGE**

- (1) Intake & exhaust covers Removed Ensure that the following areas are clear:
- RH engine (intake/exhaust)
 - APU (exhaust)
 - Ram air (dorsal intake)
 - RH pack (exhaust)

- (2) Fluid leaks Check

EMPENNAGE

- (1) Flight control area and surfaces Clear Ensure proper clearance of horizontal stabilizer and elevators from all ground equipment, buildings, trees or other obstructions. Ensure surfaces are clear of dirt, sand, ice, snow or frost.

- (2) Fluid leaks Check

- (3) Static wicks Check

AFT EQUIPMENT BAY

- (1) Rear fuse service lights On

- (2) APU enclosure Check

- (3) Engine fire bottles (2) Normal Pressure

- (4) APU fire bottle Normal Pressure

- (5) No. 1 & 2 hyd. systems Check

- (6) No. 1 & 2 hyd. accumulator gauges Check

Nominal pressure is 1,450 to 1,550 psi at 10°C (50°F). If pressure is less than 1,200 psi, verify that pressure drop is less than 250 psi per hour, and ensure that pressure prior to departure is 750 psi minimum.

- (7) Circuit breakers (JB5) Closed

- (8) Engine oil filter bypass indicators and engine chip detectors (JB5) Check

- (9) Engine oil refill system Check

- (10) Area security and condition Check

- (11) Rear fuse service lights Off



1. AIRPLANE PREPARATION (CONT'D)

B. External Safety Inspection (First Flight of the Day) Ø (Cont'd)

(12) Bay door Secure/closed Check proximate area for potential FOD items.

LEFT REAR FUSELAGE

- (1) Intake & exhaust covers Removed Ensure that the following areas are clear:
- LH engine (intake/exhaust)
 - APU (inlet)
 - LH pack (exhaust)
- (2) Fluid leaks Check

5701 to 5982 incorporating Service Bulletin 605-29-006

- (3) No. 3 hyd. accumulator gauge Check Nominal pressure is 1,450 to 1,550 psi at 10°C (50°F).
If pressure is less than 1,200 psi, verify that pressure drop is less than 250 psi per hour, and ensure that pressure prior to departure is 750 psi minimum.

LEFT WING

- (1) Flight control area and surfaces Clear Ensure proper clearance of all flight control surfaces from all ground equipment or other obstructions. Ensure surfaces clear of dirt, sand, ice, snow or frost.
- (2) Flap position Note
- (3) Static wicks Condition/secured
- (4) Fluid leaks Check
- (5) Picket/mooring lines Removed
- (6) Fuel NACA vent cover Removed Ensure vent is clear of any obstructions or contaminants. Check for evidence of fuel spillage.
- (7) Fuel vent relief cover Removed
- (8) Overwing fuel cap Secured

LEFT MAIN GEAR

- (1) Chocks As required
- (2) Wheel covers Removed
- (3) Fluid leaks Check



1. AIRPLANE PREPARATION (CONT'D)**B. External Safety Inspection (First Flight of the Day) Ø (Cont'd)**

(4) Downlock safety pin Installed

ENTERING AIRPLANE

(1) Enplaning area Clear, safe Tarmac and walkway to airplane clear of fluids, lubricants or ice. Area clear of proximate vehicles or ground equipment.

(2) Boarding lights.....Checked

TRAINING PURPOSES ONLY

1. AIRPLANE PREPARATION (CONT'D)

C. Normal Power-Up Check (First Flight of the Day) Ø

This is a check to determine if airplane systems are in a configuration to safely place AC electrical power on the airplane busses. These procedures should be performed on through flights (enroute stops) if there is any doubt that all safety aspects of the prevailing situation can be covered. Completion of this check ensures that there will be no danger to the airplane and/or personnel when powering the systems. This check should be executed prior to assuming normal crew positions.

- (1) Airplane and Flight Pubs On board Verify that airplane documentation, official documents, operating manual, quick reference handbook and navigation kits are aboard.

- (2) Circuit breakers (panels 1 – 4) Closed

NOTE

Refer to Chapter 1; INTRODUCTION – GENERAL, Circuit Breaker Reset for Bombardier Aerospace philosophy and policy with respect to the resetting of tripped circuit breakers.

- (3) Cabin and flight compartment emergency equipment Check Verify condition and security of the following:
- Crash axe
 - Halon fire extinguishers
 - Life vests (under seats)
 - Oxygen masks
 - Flashlights
 - Portable breathing equipment
- (4) Hydraulic pump switches Select to OFF.
- (5) NOSE STEER switch Select to OFF.
- (6) LDG GEAR lever Set to DN.
- (7) FLIGHT SPOILER lever Select to RETRACT.
- (8) FLAPS Check Lever and surface position should agree. If not, reposition FLAPS lever to the actual surface position.
- (9) ADG manual release Stowed



1. AIRPLANE PREPARATION (CONT'D)

C. Normal Power-Up Check (First Flight of the Day) Ø (Cont'd)

(10) BATT MASTER switch Select to ON.

NOTE

1. Flight compartment standby lights come on when the battery bus is powered and normal AC power is off.
2. When using the main battery as the only source of power, do not exceed 5 minutes of operation, as there will be insufficient cooling of the displays.

NOTE

On airplanes 5701 to 5978:

When starting the APU on ground, after selecting battery ON, wait a minimum of 3 minutes before selecting APU generator ON to avoid a nuisance **AUTO TAIL XFER** caution message.

(11) DC ELECTRICAL page Select and check that battery voltage is sufficient for APU start.
Check that:

- APU BATT above 22 VDC.
- MAIN BATT above 22 VDC.

(12) Firex monitor system Test 1

(a) FIREX MONITOR, ENG
BOTTLE 1 and 2 switches Select to TEST, and hold.

Check the following advisory messages come on:

- **L ENG SQUIB 1 OK.**
- **L ENG SQUIB 2 OK.**
- **R ENG SQUIB 1 OK.**
- **R ENG SQUIB 2 OK.**

(b) FIREX MONITOR, ENG
BOTTLE 1 and 2 switches Release

Check that the messages given above for both engine bottle squibs go out.

(c) FIREX MONITOR, APU
BOTTLE switch Select

to TEST, and hold for 5 seconds.

Check the following advisory messages on:

- **APU SQUIB 1 OK.**
- **APU SQUIB 2 OK.**



1. AIRPLANE PREPARATION (CONT'D)

C. Normal Power-Up Check (First Flight of the Day) Ø (Cont'd)

(d) FIREX MONITOR, APU

BOTTLE switch.....Release Check the following advisory messages stay on momentarily and then go out:

- **APU SQUIB 1 OK.**
- **APU SQUIB 2 OK.**

(13) Fire detection systems Test 1

(a) FIRE DET switch Select to WARN.

Hold switch in this position for 10 seconds and check:

- Fire bell on.
- "JETPIPE OVERHEAT" aural on.

Also check that:

The following warning messages are on:

- **L and R ENG FIRE.**
- **APU FIRE.**
- **L and R JETPIPE OVHT.**

The following switch/lights on:

- LH and RH ENG FIRE PUSH.
- APU FIRE PUSH.
- BOTTLE 1 and 2 ARMED PUSH TO DISCH.
- (APU) BOTTLE ARMED PUSH TO DISCH.

All indications go out when switch released.

(b) FIRE DET switch Select to FAIL.

Hold the switch in this position and check that the following caution messages come on:

- **L and R FIRE FAIL.**
- **APU FIRE FAIL.**
- **L and R JETPIPE OVHT FAIL.**

Messages go out when switch released.

(14) EXTERNAL LTS, NAV switch Select to ON.



1. AIRPLANE PREPARATION (CONT'D)

C. Normal Power-Up Check (First Flight of the Day) Ø (Cont'd)

- (15) APU (if required)..... Start Ensure that operating limitations/envelope is considered.

NOTE

The APU, PWR FUEL switch/light must be selected off between each APU start attempt. The APU, PWR FUEL switch/light should be firmly pressed once only. If pressed twice, an immediate shutdown may occur.

- (a) AC POWER, APU GEN switch..... Select to OFF.
- (b) BLEED AIR, APU LCV switch/light..... Check pressed out.
- (c) APU, PWR FUEL switch/light..... Press in Check APU readouts displayed on EICAS page.
- (d) APU, START/STOP switch/light..... Press in and check the following:
- START light on.
 - 35 to 45% rpm – START light out.
 - 95% rpm + 4 seconds (approx) – AVAIL light on.
 - 100% rpm + 15 seconds (approx) – **APU GEN OFF** caution message on.

NOTE

1. **APU BATTERY OFF** caution message may appear momentarily during APU start.
2. If time permits, allow APU rpm to stabilize at 100% for 2 minutes before applying bleed air load.

After APU RPM and EGT have stabilized:

- (e) AC POWER, APU GEN switch..... Select to ON.

If APU and AC external power are not available:

- (15) Start engine using battery and external air..... Complete then go to step (22).



1. AIRPLANE PREPARATION (CONT'D)

C. Normal Power-Up Check (First Flight of the Day) Ø (Cont'd)

- (16) Electrics (AC and DC) Established Check the following:
- Voltage, frequency and load indications are normal, and all AC busses powered and lit on AC electrical page.
 - TRUs are on-line, and voltage and load indications are normal on DC electrical page.
 - All DC busses powered and lit, and battery chargers operating on DC electrical page.
- (17) Lamps..... Test 1
- (a) LAMP TEST switch..... Select to 1 and hold.
- Check all switch/lights come on.
- Release switch and check all switch/lights go out.
- (b) LAMP TEST switch..... Select to 2 and hold.
- Check all switch/lights come on.
- Release switch and check all switch/lights go out.
- (18) Bleed leak detection Test 1
- (a) BLEED AIR, DUCT MON switch..... Set to TEST. Check that:
- L and R DUCT FAIL lights (10th and 14th) are on.
 - **L** and **R 10TH DUCT** warning messages on.
 - **L** and **R 14TH DUCT** warning messages on.
 - "BLEED AIR DUCT" aural on.
 - **ANTI-ICE DUCT** warning message on.
 - **DUCT TEST OK** advisory message on.



1. AIRPLANE PREPARATION (CONT'D)

C. Normal Power-Up Check (First Flight of the Day) Ø (Cont'd)

- (b) BLEED AIR, DUCT MON
switch.....Set to LOOP A. Check that:
- No warning messages come on.
 - **LOOP A TEST OK** advisory message on.
- (c) BLEED AIR, DUCT MON
switch.....Set to LOOP B. Check that:
- No warning messages come on.
 - **LOOP B TEST OK** advisory message on.
- (d) BLEED AIR, DUCT MON
switch.....Set to NORM.
Check that no warning messages come on.
- (19) BLEED AIR, APU LCV switch/light.....Press in to open load control valve.
Check APU LCV OPEN light on.
- (20) BLEED AIR, 10TH STAGE ISOL
switch/light.....Press in to open isolation valve.
Check ISOL OPEN light on.
- (21) AIR-CONDITIONING, L and R
PACK switch/lightsPress in to turn on packs.
Check L and R PACK OFF lights out.
- (22) FMS.....Set Position Select POS INIT on FMS status page.

1. AIRPLANE PREPARATION (CONT'D)

D. External Walkaround Ø

NOTE

A  symbol indicates the requirement to wear ear protection.

The external walkaround is primarily a visual check to ensure that the overall condition of the airplane and its visible components and equipment are safe for the following flight phases.

NOTE

The external walkaround may be omitted if qualified maintenance personnel have performed this inspection.

Prior to each flight, the flight crew or maintenance staff shall verify that the airplane condition is acceptable for flight.

Even though not noted individually, the airplane and its visible components must be checked for the following:

- Proximate area is free of potential FOD items;
- Passenger walkway and boarding stairs are safe and clear;
- Flight control surfaces are unobstructed and free from ice, snow or frost;
- All vents, ports, intakes and exhausts are unobstructed;
- Tire condition is acceptable;
- No evidence of wheel/brake overheat;
- Wheel brake wear pins (2 on each wheel) are visible;
- Condition of skin (visible damage) on radome section, fuselage, wings, nacelles, pylons, and empennage;
- All access panels, not actually involved in maintenance, are secured;
- Pitot-static probes for evidence of freezing, severe discoloration, condition and security.

LEFT UPPER FUSELAGE (FORWARD)

- | | | |
|-----|---------------------------------|-------------------------|
| (1) | Pass door emergency light | Condition |
| (2) | Oxygen blowout plug | Intact |
| (3) | Alternate static port | Clear |
| (4) | AOA vane | Free movement/condition |

- | | | | | |
|-------------|-----|---------------------------|-------|--|
| I
I
I | (5) | Standby pitot probe | Clear | Check the following: |
| | | | | <ul style="list-style-type: none"> • Clear of obstruction. • Ensure no distortion. |
| I
I
I | (6) | Ice detector probe | Clear | |
| | (7) | Pitot-static probe | Clear | Check the following: |
| | | | | <ul style="list-style-type: none"> • Clear of obstruction. • Ensure no distortion. |



1. AIRPLANE PREPARATION (CONT'D)

D. External Walkaround Ø (Cont'd)

- (8) Radome Check, secured
- (9) External lights Check
- (10) Windshields, windows Check
- (11) Nose landing lights Condition

NOSEWHEEL AND GEAR

WARNING

Hydraulically operated nose doors open and shut rapidly. Stay clear when doors are operated.

- (1) Area security Verified
- (2) Wheel chocks As required
- (3) Tires and wheels Condition
- (4) Oleo extension Check
- (5) Steering actuator Check
- (6) Rear door Check

NOTE

If abnormal resistance is felt during pin extraction, do not remove pin. Leave pin in place and request maintenance checkout.

- (7) Downlock safety pin Remove
- (8) ADG safety pin Remove

RIGHT UPPER FUSELAGE (Forward)

- (1) External AC power service door Secured
- (2) ADG door Secured
- (3) Oxygen service door Secured
- (4) Pitot-static probe Clear

Check the following:

- Clear of obstruction.
- Ensure no distortion.

- (5) Ice detector probe Clear
- (6) TAT probe Clear
- (7) AUX AOA probe Clear
- (8) AOA vane Free movement/condition
- (9) Alternate static port Clear
- (10) Refuel/defuel adapter door Secured



1. AIRPLANE PREPARATION (CONT'D)

D. External Walkaround Ø (Cont'd)

- (11) Wing inspection lights.....Condition
- (12) Refuel/defuel panel door Secured
- (13) Overwing emergency lights (3).....Condition
- (14) Overwing emergency exit..... Secured

RIGHT LOWER FUSELAGE (Forward)

- (1) AntennaeCheck ATC, ADF, TCAS (if installed),
VHF 2 and DME 1.

RIGHT MAIN GEAR (Front)

- (1) Area securityVerified
- (2) Wheel chocks As required
- (3) Tires and wheelsCondition
- (4) Brakes and wear pins.....Check
- (5) Gear door Secured

NOTE

If abnormal resistance is felt during pin extraction, do not remove pin.
Leave pin in place and request maintenance checkout.

- (6) Downlock safety pin Remove
- (7) Downlock dust cover Installed

RIGHT WING

- (1) Fuel filler caps (2) Secured
- (2) Landing and taxi lights.....Condition
- (3) Leading edge.....Condition
- (4) Leading edge sealantCondition
- (5) Engine intake..... Clear
- (6) Ram air inlet Clear
- (7) Upper wing surfaceCondition
- (8) Lower wing surfaceCondition
- (9) Wing anti-ice exhaust louvres Clear
- (10) Fuel and water drains.....Check
- (11) Fuel vent valveCheck
- (12) Navigation and strobe lightsCondition
- (13) WingletCondition
- (14) Static dischargers (7)Condition



1. AIRPLANE PREPARATION (CONT'D)

D. External Walkaround Ø (Cont'd)

(15) Flight control surfaces Clear Ensure proper clearance of all flight control surfaces from all ground equipment or other obstructions. Ensure surfaces clear of any contaminants.

(16) Trailing edge.....Condition

(17) AileronCondition

(18) Flaps.....Condition

(19) Flight spoilerCondition

(20) Ground spoilersCondition

(21) NACA vent..... Clear

RIGHT MAIN GEAR (Aft)

(1) Area security Verified

(2) Tires and wheels Condition

(3) Oleo extension Check

(4) Brakes and wear pins..... Check

RIGHT FUSELAGE (Aft)

(1) Engine anti-ice blowout plug Check/flush

(2) Engine cowls Condition/secured

(3) Thrust reverser Stowed

(4) Engine drains Clear

(5) Engine fuel vent ports..... Clear

(6) Engine exhaust/jetpipe Condition/clear

(7) Engine pylon..... Condition

WARNING

APU exhaust danger area (15 feet).

(8) APU exhaust Condition/clear

(9) External DC power door Secured

EMPENNAGE

(1) Flight control surfaces Clear Ensure proper clearance of all flight control surfaces from all ground equipment or other obstructions. Ensure surfaces clear of any contaminants.

(2) Horizontal stabilizer Condition



1. AIRPLANE PREPARATION (CONT'D)

D. External Walkaround Ø (Cont'd)

- (3) Vertical stabilizerCondition
- (4) ElevatorsCondition
- (5) Rudder.....Condition
- (6) Static dischargers (12)Condition
- (7) Navigation and strobe lightsCondition
- (8) Tail vent..... Clear
- (9) AntennaeCheck VOR/LOC (both sides of fin), and HF.
- (10) Dump mastCondition/clear

AFT EQUIPMENT BAY

- (1) Bay door Secured

LEFT LOWER FUSELAGE (Aft)

- (1) AntennaeCondition Marker beacon, Radio altimeter (2) and DME 2.
- (2) APU enclosure drains..... Clear

LEFT UPPER FUSELAGE (Aft)

- (1) External ground air door Secured
- (2) Engine pylon.....Condition
- (3) Engine exhaust/jetpipeCondition/clear
- (4) Thrust reverserStowed
- (5) Engine cowlsCondition/secured
- (6) Engine fuel vent ports..... Clear
- (7) Engine drains Clear
- (8) Engine anti-ice blowout plug Check/flush

CARGO BAY

- (1) Cargo bay door..... Closed and latched

NOTE

Make sure the cargo bay door is not key locked.

LEFT MAIN GEAR (Aft)

- (1) Area securityVerified
- (2) Tires and wheelsCondition
- (3) Oleo extensionCheck
- (4) Brakes and wear pins.....Check



1. AIRPLANE PREPARATION (CONT'D)

D. External Walkaround (Cont'd)

LEFT WING

- (1) NACA vent..... Clear
- (2) Flight control surfaces Clear
- (3) Ground spoilers Condition
- (4) Flaps..... Condition
- (5) Flight spoiler Condition
- (6) Aileron Condition
- (7) Trailing edge..... Condition
- (8) Static dischargers (7) Condition
- (9) Navigation and strobe lights Condition
- (10) Winglet Condition
- (11) Leading edge..... Condition
- (12) Leading edge sealant Condition
- (13) Fuel vent valve Check
- (14) Fuel and water drains Check
- (15) Lower wing surface Condition
- (16) Wing anti-ice exhaust louvres Clear
- (17) Upper wing surface Condition
- (18) Engine intake..... Clear
- (19) Fuel filler cap Secured
- (20) Landing and taxi lights..... Condition

Ensure proper clearance of all flight control surfaces from all ground equipment or other obstructions. Ensure surfaces clear of any contaminants.

LEFT MAIN GEAR (Front)

- (1) Area security Verified
- (2) Tires and wheels Condition
- (3) Brakes and wear pins Check
- (4) Gear door Secured

NOTE

If abnormal resistance is felt during pin extraction, do not remove pin.
Leave pin in place and request maintenance checkout.

- (5) Downlock safety pin Remove
- (6) Downlock dust cover Installed



1. AIRPLANE PREPARATION (CONT'D)

D. External Walkaround Ø (Cont'd)

LEFT UPPER FUSELAGE (Forward)

- (1) Wing inspection lights.....Condition
- (2) AntennaeCheck VHF 1, ATC 1 and TCAS.

BEFORE ENTERING AIRPLANE

- (1) NLG door safety switchCheck Set to FLT/CLOSE DOOR.

ENTERING AIRPLANE

- (1) Door area securityVerified
- (2) Pax door vent flapChecked 1

If door is closed and locked:

- (a) Passenger door outer handle trigger plate.....Push and as the outer handle comes out of the housing, check that the vent flap (flapper door) springs open.

If door is fully open:

- (a) Passenger door vent flap (flapper door)Check for spring tension by pulling door vent flap outboard, to verify that spring has not disconnected.
- (3) StairsChecked
- (4) Gear and safety pinsOn board

1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø

During the systems check, the airplane systems and components are checked and configured for the coming phases of flight.

The systems check will be performed by the flight crew, as applicable. Only the RH side console and RH instrument panel items are checked by the copilot.

The external and internal pre-flight checks will be accomplished by the pilot and the co-pilot prior to the start of this check.

COCKPIT

- (1) Seat and harnessAdjust Set seat to full aft position. Fasten lap belt and shoulder harness. Adjust height to obtain optimum eye reference position using indicator on center windscreen post. Check that seat position is not restrictive when moving the control column full forward and full aft.

NOTE

The correct eye reference position is obtained by adjusting the seat so that the white ball appears to cover the orange ball. The resulting eye level should be approximately in the centre of the windshield (Refer to Vol. 2, Chapter 1; AIRPLANE GENERAL – CREW SEATS).

- (2) Rudder pedalsAdjust to allow full travel when the knees are straightened.

When nosewheel area is confirmed clear:

- (3) Hydraulic pump 3ASelect to ON.
(4) Nosewheel doorClose



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

SIDE CONSOLES

- (1) Oxygen system.....Check
- (a) Crew oxygen system pressureCheck the following:
- Oxygen supply, pressure and quantity are sufficient.
 - **OXY LO PRESS** caution message out.
- (b) Crew oxygen masksCheck that the masks are fully operational. Check the following:
- Flow indicator indicates oxygen flow,
 - 100% is operational, and
 - Oxygen flow sound is picked up by mask microphone.
- (c) Passenger oxygen.....Select to NORMAL.
- (2) AUDIO WARNING switch/lights Guarded/Off
- (3) Cockpit voice recorder.....Press and hold TEST.

NOTE

When the Recording Independent Power Supply (RIPS) battery is being charged, testing of the cockpit voice recorder will not be possible. Wait 10 minutes for the battery to recharge and repeat the test.

OVERHEAD PANEL

- (1) ELECTRICAL POWER panel.....Check
- (a) BATT MASTER switchSelect to ON.
- (b) AC POWER, APU GEN switch.....As required
- (c) AC POWER, GEN 1 and 2 switches.....Select to OFF/RESET.
- (d) Switch/lights.....Check all lights out.
- (2) APU panel As required



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

- (3) HYDRAULIC panel.....Check



Ensure that the nose wheel area and flight control surface areas are clear before selecting hydraulic power.

- (a) Hydraulic synoptic page Select
- (b) Fluid quantities/temperatures Check
- (c) PUMP 3A pressure Check
- (d) PUMP 3A switch Select to OFF.
- (e) PUMP 3B switch Select to ON.
Check 3A pump outline is white.
- (f) PUMP 1B, and 2B switches Select to ON.
Check that **HYD 1** and **HYD 2 LO PRESS** caution messages out.
- (g) PUMP 1B, 3B and 2B switches Select to AUTO.
- (h) PUMP 3A switch Select to ON.
Check that no hydraulic system caution messages are on.
- (i) EICAS page Select
- (4) Miscellaneous/external lighting panel.....Check
- (a) BOARD switch As required
- (b) CABIN switch As required
- (c) OVHD selector As required
- (d) NAV switch Select to ON.
- (e) A/COLL switch Select to OFF.
- (f) WING-INSP switch As required
- (g) IND LTS switch As required



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

- (5) LANDING, RECOG TAXI LTS
panel.....Check
- (a) LEFT switch.....Select to OFF.
- (b) NOSE switchSelect to OFF.
- (c) RIGHT switchSelect to OFF.
- (d) RECOG TAXI LTS switch.....Select to OFF.
- (6) Start/IGNITION panelCheck
- (a) L and R ENG START
switch/lights Off
- (b) IGNITION, A, B and CONT
switch/lights Off Check that IGNITION A, B and
CONT ON lights are out.
- (7) FUEL panelCheck
- (a) BOOST PUMP switch/lightsCheck INOP lights on.
- (b) X/FLOW switch/lightsPress out Check ON and OPEN lights
out.
- (c) TAIL TANK TRANSFER and
DUMP switch/lights.....Press out Check lights out.
- (d) EICAS page.....Check tank quantities and balance.
- (8) AIR CONDITIONING panelCheck
- (a) CKPT mode selectorSet to NORM.
- (b) CABIN mode selectorSet to NORM.
- (c) RAM AIR switch/light.....Check pressed out.
- (d) CKPT TEMP control switchAs required
- (e) CABIN TEMP control switch.....As required
- (9) ELTSelect to ARM.
- (10) CABIN PRESSURIZATION panelCheck
- (a) EICAS page.....Check cabin differential pressure
zero, cabin altitude at
approximately field elevation
and cabin rate of climb zero.
- (b) MAN ALT switch.....Centered
- (c) MAN RATE selector Full decrease
- (d) PRESS CONT switch/light.....Press out Check MANUAL light out.
- (e) EMER DEPRESS switch/lightPress out Check ON light out.
- (f) Automatic pressure controller.....Set field elevation, barometer and
rate.



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

- (11) BLEED AIR panel Check
- (a) 14TH STAGE, L switch/light Pressed in Check CLOSED light out.
 - (b) 14TH STAGE ISOL switch/light Pressed out Check OPEN light out.
 - (c) 14TH STAGE, R switch/light Pressed in Check CLOSED light out.
 - (d) 10TH STAGE, L switch/light Pressed out Check CLOSED light on.
 - (e) 10TH STAGE, R switch/light Pressed out Check CLOSED light on.

- (12) ANTI-ICE panel Check
- (a) WING switch Select to OFF.
 - (b) L and R HEAT lights Check lights out.
 - (c) COWL switch/lights Check pressed out.
 - (d) WSHLD/WIND and PROBES switches Select to OFF.

On airplanes with Supplemental Ground Wing Anti-Icing:

- (13) Wing Anti-Ice Switch Test 1
- (a) ANTI-ICE, WING switch Check selected to OFF.
Check **WING A/ICE ON** advisory message out.
 - (b) SUPP GND WING ANTI-ICE, SYS switch/light Check pressed out.
 - (c) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice on.
Check **L** and **R COWL A/ICE** caution messages out.

NOTE

If **L** or **R COWL A/ICE** caution message comes on immediately, wing anti-ice switch has failed and maintenance is required.

- (d) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off.

- (14) Ice detector Test 1
- (a) ANTI-ICE, TEST switch Select to DET.
Check **ICE** caution message on.
- (15) PASS SIGNS panel Check
- (a) NO SMKG switch Select to ON.
 - (b) SEAT BLTS switch Select to ON.



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

- (16) EMER LTS panel..... Test 1
- (a) EMER LTS switch..... Select to OFF.
- Check **EMER LTS OFF** caution message on.
- (b) EMER LTS switch..... Select to ON.
- Check the following:
- Emergency lights are on.
 - **EMER LTS ON** status message on.
- (c) EMER LTS switch..... Select to ARM.
- Check the following:
- Emergency lights are off.
 - **EMER LTS ON** status message out.
- (17) Standby compass..... Check Compare headings with pilot's and copilot's HSI's on the PFD and MFD.

NOTE

Heading information may be affected by magnetic disturbances created by ground equipment.

GLARESHIELD

- (1) MASTER WARNING/CAUTION switch/lights Press out Check lights out.
- (2) BARO knobs..... Set barometric readout on the primary flight displays to station pressure.
- (3) SPEED knob Set speed bug.
- (4) ALT knob Set preselected altitude.

SIDE PANELS

- (1) Audio control panel..... As required
- (2) STALL PROT PUSHER switches..... Select to ON.



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

INSTRUMENT PANELS

NOTE

Ensure that display brightness is adjusted properly (using respective OFF/BRT switches) for the existing lighting conditions.

- (1) Primary flight displays Check Normal indications.
Cross-check ADIs and RMIs.
- (2) Multifunction displays Check Normal indications and messages.
- (3) Integrated Standby Instrument Cross-check
- (4) GPS clock Check/Set to correct local time.
- (5) SUPP GND WING ANTI-ICE panel (if installed) Check
- (a) TEST and SYS switch/lights Check pressed out.
Check TEST ON, TEST OK and OFF lights out.
- (b) TEMP lights Check lights out.
- UPPER PEDESTAL
- (1) HORN switch Guarded
- (2) ANTI-SKID switch ARMED Check that no **A/SKID INBD/OUTBD** caution messages are on.
- (3) MLG overheat detection system Check 1
- (a) MLG BAY OVHT TEST, OVHT switch Select to OVHT and hold.
Check the following:
- **MLG BAY OVHT** warning message on.
 - “GEAR BAY OVERHEAT” aural message on.
- Release switch and check that warning and aural messages go out.
- (b) MLG BAY OVHT TEST, FAIL switch Select to FAIL and hold.
Check **MLG OVHT FAIL** caution message on.
Release switch and check that caution message goes out.



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

- (4)

TAWS WARNING, TERRAIN OFF

switch/light

Select

as required.
- (5)

TAWS WARNING, FLAPS OFF

switch/light

Check

pressed out.
- (6)

TAWS WARNING, RAAS OFF

switch/light (if installed)

Check

pressed out.
- (7)

PITCH and ROLL DISC handles

In/stowed
- (8)

GND SPOILERS switch

Select

to AUTO.
- (9)

THRUST REVERSER panel

Check

(a)

THRUST REVERSER, L and R

switches

Select

to ARMED.

Check the following:

• L and R REV ARMED

advisory messages on.

• LH and RH UNLK lights out.

• L and R REV UNLOCKED

caution messages out.
- (b)

EMER STOW switch/lights

Check

pressed out.
- (10)

FLIGHT SPOILER lever

Retracted
- (11)

Thrust levers

Checked/SHUT OFF
- (12)

FLAPS

Check

setting agrees with flight control page.
- CENTRE PEDESTAL
- (1)

Engine vibration

Test

1

(a)

VIB switch

Select

to TEST. Check the following:

• VIB icon displayed on N₂ gauges or adjacent to N₂ readout.

• FAN VIB readout of 3.5 ±0.1 mils (amber) or VIB icon displayed on N₁ gauges in compressed display format.
- (2)

ENG SPEED, LH and RH

switches

Select

to ON.

1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

- (3) Aural warning system Check 1
- (a) AURAL WARN TEST switch Select to 1 and release.
- Check that all dedicated tones, chimes and voice messages within data concentrator unit 1 come on.
- (b) AURAL WARN TEST switch Select to 2 and release.
- Check that all dedicated tones, chimes and voice messages within data concentrator unit 2 come on.

NOTE

Re-selection of the TEST switch during the test will cancel the aural messages.

- (4) Stall protection system Check 1

NOTE

To ensure a successful test, all items within the following test must be verified to have occurred in the sequence indicated.

Both SPS TEST switches must be operated simultaneously.

- (a) Stabilizer trim switch Select to NOSE DN and hold until trim is 0 units.
- (b) FLIGHT CONTROLS page Select
- (c) STALL PROT, PUSHER switches Select to ON.
- (d) SPS TEST, L and R switches Select TEST and hold.
- Check **STALL FAIL** caution message comes on.



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

During test:

- (e) SPS TEST INDICATORS.....Verify that both test indicator needles sweep counter clockwise passed the red region and then sweep clockwise back to the red region and remain there.

Check the following:

- IGNITION A, B and CONT ON lights come on.
- **IGNITION A** and **B** advisory messages come on.

within the yellow region:

- Stick shakers are activated.

within the red region:

- Stick pusher is activated.
STALL lights flash.
Stall aural warning sounds.

While in stick pusher mode:

- (i) Pilot's AP/SP DISC switch.....Press Stick pusher is de-activated.
- (ii) Pilot's AP/SP DISC switch.....Release Stick pusher is re-activated.
- (iii) Copilot's AP/SP DISC switch.....Press Stick pusher is de-activated.

NOTE

During copilot's AP/SP DISC test, the control column will take several seconds to go back to neutral position. Copilot should pull (override) control column and note diminished feel force loads while pressing the AP/SP DISC switch.

- (iv) Copilot's AP/SP DISC switch.....Release Stick pusher is re-activated.
- (v) FLIGHT CONTROLS page.....Check right elevator surface pointer indicates full travel or copilot control column is against forward stop.
- (vi) G SWITCH.....Select TEST and hold.
Stick pusher is de-activated.
- (vii) G SWITCH.....Release Stick pusher is re-activated.



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

(f) SPS TEST, L and R switches.....Release Check the following:

- **STALL FAIL** caution message out.
- Stick pusher de-activated.
- Stick shakers de-activated.
- IGNITION A, B and CONT ON lights out.
- **IGNITION A** and **B** advisory messages out.
- STALL lights out.
- Stall aural warning stops.

(5) DCP As required

(6) CCP As required

(7) Radios/Avionics Set

(a) VHF 3 DATA or as required

NOTE

- U.S. Domestic CPDLC En Route and DCL operations require VHF 3 to be in DATA mode (SATCOM not supported).
- Europe ATN CPDLC network requires VHF 3 to be in DATA mode (SATCOM not supported).

(8) Flight control trim systems.....Check 1

(a) STAB TRIM, CH 1 and MACH TRIM switch/lights Press in

Check **STAB CH 2 INOP** status message on.

Check for independent CH 1 operation from both stabilizer trim switches in both directions, until trim clacker is activated.

(b) Copilot's STAB TRIM DISC switch.....Press

Check the following:

- **STAB TRIM** caution message on.
- **MACH TRIM** caution message on.
- Stabilizer trim is inoperative.



1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

- (c) STAB TRIM, CH 2 and MACH TRIM switch/lights Press in
- Check **STAB CH 1 INOP** status message on.
- Check for independent CH 2 operation from both stabilizer trim switches in both directions, until trim clacker is activated.
- (d) Pilot's STAB TRIM DISC switch Press
- Check the following:
- **STAB TRIM** caution message on.
 - **MACH TRIM** caution message on.
 - Stabilizer trim is inoperative.
- (e) STAB TRIM CH 1 and CH 2 switch/lights Press in
- to engage CH 1 and CH 2.
- Check the following:
- **STAB CH 1** and **CH 2 INOP** status messages out.
 - **STAB TRIM** caution message out.
- (f) MACH TRIM switch/light..... Press in
- to engage Mach trim.
- Check **MACH TRIM** caution message out.
- (g) Aileron and rudder trims Check
- for free movement in both directions.
- Set at neutral for take-off.
- (9) YD 1 and YD 2 switch/lights Press in
- to engage. Check that the **YAW DAMPER** caution message is out.
- (10) CDU and MFD TUNE INHIBIT switches..... Press out
- (11) Reversionary selectors Set to NORM.
- (12) AFCS SEL As required
- (13) IRS reversionary selectors As required
- (14) FMS 3 selector (if installed) As required
- (15) LANDING GEAR MANUAL RELEASE handle Stowed
- (16) ADG handle Stowed



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1. AIRPLANE PREPARATION (CONT'D)

E. Systems Check Ø (Cont'd)

(17) ADG AUTO DEPLOY CONTROL

switchSelect to LAMP. Check that TEST light illuminates.

TRAINING PURPOSES ONLY

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TRAINING PURPOSES ONLY

1. ENGINE STARTING

A. Before Start Check

- (1) Gravity crossflow Check
- (a) FUEL, GRAVITY XFLOW
switch/light Press in to open gravity crossflow valve.
Check GRAVITY XFLOW OPEN light on.
- (b) FUEL, GRAVITY XFLOW
switch/light Press out to close gravity crossflow valve.
Check GRAVITY XFLOW OPEN light out.
- 

CAUTION
- Do not leave the fuel system in manual or gravity crossflow while the airplane is on the ground. Uneven ground surface may create fuel imbalance.
- (2) Windshield/window heat Test
- (a) ANTI-ICE, WSHLD/WIND
L and R switches Select to HI.
- (b) ANTI-ICE, WSHLD/WIND,
TEST switch Press and hold. Check the following:
- **L** and **R WSHLD HEAT OK** advisory messages on.
 - **L** and **R WINDOW HEAT OK** advisory messages on.
- (c) ANTI-ICE, WSHLD/WIND
L and R switches Select as required.
- (3) Wing overheat detection system Test
- (a) ANTI-ICE, TEST switch Select to WING. Check the following:
- **WING OVHT** warning message on.
 - “WING OVERHEAT” aural on.
 - WING L HEAT and R HEAT lights on.
 - **WING A/ICE TEST OK** advisory message on.



1. ENGINE STARTING (CONT'D)

A. Before Start Check (Cont'd)

(4) CABIN PRESSURIZATION, LDG

ALT selector Set the following:

- Destination airfield altitude.
- Desired cabin pressure rate of change.
- Barometric pressure.

(5) EFIS altimeters (2) and Integrated

Standby Instrument altimeter (1) Set all three altimeters to the departure field elevation.



Reduced vertical separation with terrain may be encountered in mountainous terrain if altimeter is incorrectly set and LNAV is armed for departure.

(6) EFIS Set for departure

Review and disposition any boxed annunciations or reversion indications.

(7) CAS Checked

(8) Brakes and anti-skid system Check

(a) PARKING BRAKE Select to off.

Check **PARKING BRAKE ON** advisory message out.

(b) ANTI SKID, ARMED switch Select to ARMED.

(c) ANTI SKID, TEST switch Select to TEST momentarily.

Check **A/SKID IN TEST** advisory message on.

After approximately 6 seconds, check **A/SKID IN TEST** advisory message out.

NOTE

Anti-skid system is inoperative if the **A/SKID IN TEST** advisory message does not come on during test.

Anti-skid system is inoperative if **A/SKID INBD (OUTBD)** caution message comes on during test.

(d) PARKING BRAKE Select to on.

Check **PARKING BRAKE ON** advisory message on.



1. ENGINE STARTING (CONT'D)**A. Before Start Check (Cont'd)**

- (9) All FMS Set route, performance, and departure/arrival.
- (10) FMS Review and disposition messages.

NOTE

For any SBAS covered airspace (US RNAV routes, SIDs, and STARs), it is not required to run a pre-flight prediction of RAIM availability along the intended route if no satellites are NOTAM'd as out of service over the intended route.

For non-SBAS covered airspace, certain operations may require that an appropriate prediction program, such as Rockwell Collins part number 832-3443-008 or equivalent, be used prior to departure for RAIM availability along the intended route.

- (11) Radios/Nav aids Set for departure
- (12) Take-off data Set Ensure N_1 , V_1 , V_R , V_2 , V_T and stabilizer trim are set for take-off.
- (13) Take-off briefing Complete

An adequate crew briefing, covering the important aspects of a particular take-off, is a must, and cannot be overemphasized. It is the responsibility of the pilot to ensure that the flight crew is familiar with the pertinent take-off and departure procedures, and any emergency which may arise. The following items are recommended for review:

- Type of take-off required for the existing weather and runway conditions,
- Runway conditions,
- Anti-icing requirements,
- N_1 thrust setting, V speeds and bug settings,
- Airspeed calls,
- Rejected take-off considerations and procedures,
- SID or departure procedures as per clearance,
- Obstacle clearance requirements,
- Emergency plan and any other condition which may alter the normal take-off profile,
- Any questions, clarifications or other pertinent details (MEL/CDL).

1. ENGINE STARTING (CONT'D)

B. Start Check

NOTE

If push-back is required, release parking brake and turn off nosewheel steering. Safety pins can be removed and stowed for the push-back. Audio or visual communication with ground crew must be maintained at all times during the push-back. When push-back is complete, reset parking brake.

- | | | |
|--|---------------|--|
| (1) Passenger door | Closed | Check DOOR CLOSED advisory message on. |
| (2) Fuel quantity and balance | Check | the following: <ul style="list-style-type: none"> • Fuel quantity is within flight plan minimum. • Fuel contents are properly balanced. • Fuel temperature is within normal limits. |
| (3) BEACON switch (if installed) | Select | to ON. |
| (4) FUEL, L and R BOOST PUMP switch/lights | Press in | to turn on both boost pumps.
Check the following: <ul style="list-style-type: none"> • L and R BOOST PUMP, ON lights on. • L and R BOOST PUMP, INOP lights out. • L and R FUEL PUMP ON advisory messages on. |
| (5) BLEED AIR switch/lights | Set for start | Check the following: <ul style="list-style-type: none"> • L and R 10TH STAGE SOVs closed. • L and R 14TH STAGE SOVs open. |
| (6) IGNITION, A (B) switch/light | Press in | to arm applicable ignition system.

Check applicable ARM light on. |
| (7) Engines | START | |

1. ENGINE STARTING (CONT'D)

C. Engine Start Procedure (APU Assist)



If both engines are to be started using APU bleed air, the operating engine's thrust must not exceed 70% N_2 .

- | | |
|---|--|
| (1) Thrust levers.....Check
(2) BLEED AIR, APU LCV switch/light.....Check
(3) L (R) ENG START switch/lightPress | both selected to SHUT OFF.
pressed in.
to start applicable engine.
Check the following: <ul style="list-style-type: none"> • Applicable START light comes on. • IGNITION A (B) ON light comes on. • IGNITION A (B) advisory message comes on. • Check that there is positive bleed air pressure. • Check that N_1 increases. |
|---|--|

When N_2 reaches 20% RPM and ITT is below 120°C:

- | | |
|---|---------------------------|
| (4) Left or right thrust leverAdvance | applicable lever to IDLE. |
|---|---------------------------|

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.

- | | |
|------------------------------------|--|
| (5) Engine instruments.....Monitor | Check the following: <ul style="list-style-type: none"> • Starter disengages at 55% N_2. • IGNITION A (B) advisory message goes out. • ITT is 450 – 600°C (max. during start is 900°C). • N_1 stabilizes and does not drop below 22% RPM. • Check that oil pressure is within normal range [above 25 psi (174 kPa)]. |
|------------------------------------|--|



1. ENGINE STARTING (CONT'D)

C. Engine Start Procedure (APU Assist) (Cont'd)

(6) Fuel system check valve operationVerify 1

(a) FUEL, L and R BOOST PUMP switch/lightsPress out to turn both boost pumps off.

Check the following:

- L and R BOOST PUMP ON lights out.
- L and R BOOST PUMP INOP lights on.
- L and R FUEL PUMP ON advisory messages out.
- FUEL LO PRESS caution message is annunciated for the non-operating engine.

After 2 minutes, if FUEL LO PRESS caution message is still not displayed for the non-operating engine:

(b) Dry motor.....Accomplish for maximum of 30 seconds on the non-started engine.

Applicable FUEL LO PRESS caution message should be displayed during motoring as the residual fuel pressure is reduced.



Absence of the FUEL LO PRESS caution message during this check is an indication that the fuel feed cross flow check valve may have failed in the open position or that another component of the engine fuel feed sub-system may not be functioning properly. The fuel system needs further troubleshooting.



1. ENGINE STARTING (CONT'D)

C. Engine Start Procedure (APU Assist) (Cont'd)

Before starting other engine:

- (c) FUEL, L and R BOOST PUMP switch/lights Press in to turn both boost pumps on.
- Check the following:
- L and R BOOST PUMP ON lights on.
 - L and R BOOST PUMP INOP lights out.
 - **L** and **R FUEL PUMP ON** advisory messages on.
 - All **FUEL LO PRESS** caution message are out.

CAUTION

When operating on the ground with one engine operating, a fuel imbalance may occur with both boost pumps ON. This fuel imbalance may result in fuel spillage if fuel tanks are full. Select the inoperative engine boost pump to INOP. If a fuel imbalance occurs on the ground during single engine operation, open the respective L TO AUX or R TO AUX XFLOW valve to balance the fuel tanks.

If the other engine is to be started using APU bleed air:

- (7) Thrust lever of operating engine Check at IDLE.
- (8) 10th stage bleed air pressure Check that positive pressure is available.
- (9) L (R) ENG START switch/light Press to start applicable engine.
- Check the following:
- Applicable START light comes on.
 - IGNITION A (B) ON light comes on.
 - **IGNITION A (B)** advisory message comes on.
 - Check that N_1 increases.

When N_2 reaches 20% RPM and ITT is below 120°C:

- (10) Thrust lever of starting engine Advance to IDLE.

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.



1. ENGINE STARTING (CONT'D)

C. Engine Start Procedure (APU Assist) (Cont'd)

- (11) Engine instruments.....Monitor Check the following:
- Starter disengages at 55% N₂.
 - **IGNITION A (B)** advisory message goes out.
 - ITT is 450 – 600°C (max. during start is 900°C).
 - N₁ stabilizes and does not drop below 22% RPM.
 - Check that oil pressure is within normal range [above 25 psi (174 kPa)].
- (12) APU EGT indications.....Check within normal limits.
- (13) Idle speed.....Check N₂ variation between engines is within 2%. If greater, notify maintenance personnel.

TRAINING PURPOSES ONLY

1. ENGINE STARTING (CONT'D)

D. Engine Start Procedure (External Air with AC Power)

- (1) BLEED AIR, APU LCV switch/light.....Press out to close load control valve.
Check APU LCV OPEN light out.
- (2) External air Connected Verify 45 psi minimum bleed air pressure on EICAS.

NOTE

The external air source must be capable of supplying 45 psi manifold pressure at the initiation of start, however, lower pressure units, dependent upon altitude and flow, have been proven satisfactory.

- (3) L (R) ENG START switch/lightPress to start applicable engine.
Check the following:
- Applicable START light comes on.
 - IGNITION A (B) ON light comes on.
 - **IGNITION A (B)** advisory message comes on.
 - Check that there is positive bleed air pressure.
 - Check that N_1 increases.

When N_2 reaches 20% RPM and ITT is below 120°C:

- (4) Left or right thrust leverAdvance applicable lever to IDLE.

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.

- (5) Engine instruments.....Monitor Check the following:
- Starter disengages at 55% N_2 .
 - **IGNITION A (B)** advisory message goes out.
 - ITT is 450 – 600°C (max. during start is 900°C).
 - N_1 stabilizes and does not drop below 22% RPM.
 - Check that oil pressure is within normal range [above 25 psi (174 kPa)].



1. ENGINE STARTING (CONT'D)

D. Engine Start Procedure (External Air with AC Power) (Cont'd)

(6) Fuel system check valve operationVerify 1

(a) FUEL, L and R BOOST PUMP

switch/lightsPress out to turn both boost pumps off.

Check the following:

- L and R BOOST PUMP ON lights out.
- L and R BOOST PUMP INOP lights on.
- **L** and **R FUEL PUMP ON** advisory messages out.
- **FUEL LO PRESS** caution message is annunciated for the non-operating engine.

After 2 minutes, if **FUEL LO PRESS caution message is still not displayed for the non-operating engine:**

(b) Dry motor.....Accomplish for maximum of 30 seconds on the non-started engine.

Applicable **FUEL LO PRESS** caution message should be displayed during motoring as the residual fuel pressure is reduced.



Absence of **FUEL LO PRESS** caution message during this check is an indication that the fuel feed cross flow check valve may have failed in the open position or that another component of the engine fuel feed sub-system may not be functioning properly. The fuel system needs further troubleshooting.



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1. ENGINE STARTING (CONT'D)

D. Engine Start Procedure (External Air with AC Power) (Cont'd)

Before starting other engine:

- (c) FUEL, L and R BOOST PUMP

switch/lights Press in to turn both boost pumps on.

Check the following:

- L and R BOOST PUMP ON lights on.
- L and R BOOST PUMP INOP lights out.
- **L** and **R FUEL PUMP ON** advisory messages on.
- All **FUEL LO PRESS** caution message are out.



When operating on the ground with one engine operating, a fuel imbalance may occur with both boost pumps ON. This fuel imbalance may result in fuel spillage if fuel tanks are full. Select the inoperative engine boost pump to INOP. If a fuel imbalance occurs on the ground during single engine operation, open the respective L TO AUX or R TO AUX XFLOW valve to balance the fuel tanks.

- (7) Other engine Start

NOTE

When starting the other engine using cross-bleed, accomplish steps (8) to (10) first.

- (8) AC POWER, applicable GEN switch Select to ON.
- (9) External AC power (if connected) Disconnect
- (10) External air Disconnect

1. ENGINE STARTING (CONT'D)

E. Engine Start Procedure (Battery and External Air)

- | | | |
|--|----------------------|---|
| (1) Normal Power-Up Check..... | Carry out | as instructed. |
| (2) BLEED AIR, APU LCV switch/light..... | Press out | to close load control valve.
Check APU LCV OPEN light out. |
| (3) External air | Connected | |
| (4) External cart | Not less than 45 psi | Confirm with ground personnel. |

NOTE

The external air source must be capable of supplying 45 psi manifold pressure at the initiation of start, however, lower pressure units, dependent upon altitude and flow, have been proven satisfactory.

- | | | |
|--|----------|--|
| (5) FUEL, R BOOST PUMP switch/light..... | Press in | to arm right boost pump and to turn on left boost pump.
Check the following:
<ul style="list-style-type: none"> • L BOOST PUMP, ON light on. • L FUEL PUMP ON advisory message on. |
| (6) IGNITION, B switch/light | Press in | to arm ignition system B.
Check IGNITION B ARM light on. |
| (7) R ENG START switch/light..... | Press | to start applicable engine.
Check the following:
<ul style="list-style-type: none"> • Applicable START light comes on. • IGNITION B, ON light comes on. • IGNITION B advisory message comes on. • Check that N₁ increases. |

When N₂ reaches 20% RPM and ITT is below 120°C:

- | | | |
|------------------------------|---------|----------|
| (8) Right thrust lever | Advance | to IDLE. |
|------------------------------|---------|----------|

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.



1. ENGINE STARTING (CONT'D)

E. Engine Start Procedure (Battery and External Air) (Cont'd)

- (9) Engine instruments..... Monitor Check the following:
- Starter disengages at 55% N₂.
 - **IGNITION B** advisory message goes out.
 - ITT is 450 – 600°C (max. during start is 900°C).
 - N₁ stabilizes and does not drop below 22% RPM.

NOTE

Oil pressure indication is not available without AC power.

- (10) AC POWER, GEN 2 switch Select to ON.
Check that oil pressure is within normal range [above 25 psi (174 kPa)].

- (11) Electrics (AC and DC) Established Check the following:
- Voltage, frequency and load indications are normal and all AC busses powered and lit on AC electrical page.
 - TRUs are on-line and that voltage and load indications are normal on DC electrical page.
 - MAIN BUS TIE symbol indicates tie open.

- (12) Lamps..... Test 1

- (a) LAMP TEST switch..... Select to 1 and hold.

Check all switch/lights come on.

Release switch and check all switch/lights go out.

- (b) LAMP TEST switch..... Select to 2 and hold.

Check all switch/lights come on.

Release switch and check all switch/lights go out.



1. ENGINE STARTING (CONT'D)

E. Engine Start Procedure (Battery and External Air) (Cont'd)

(13) Fuel system check valve

operationVerify 1

(a) Boost pump status..... Confirm Check the following:

- L and R BOOST PUMP ON lights out.
- L BOOST PUMP INOP light on.
- **L** and **R FUEL PUMP ON** advisory messages out.
- **L FUEL LO PRESS** caution message on.

After 2 minutes, if **L FUEL LO PRESS caution message is still not displayed:**

(b) Dry motor left engineAccomplish for maximum of 30 seconds.

L FUEL LO PRESS caution message should be displayed during motoring as the residual fuel pressure is reduced.



Absence of the **FUEL LO PRESS** caution message during this check is an indication that the fuel feed cross flow check valve may have failed in the open position or that another component of the engine fuel feed sub-system may not be functioning properly. The fuel system needs further troubleshooting.



1. ENGINE STARTING (CONT'D)

E. Engine Start Procedure (Battery and External Air) (Cont'd)

Before starting other engine:

- (c) FUEL, L and R BOOST PUMP switch/lights Press in to turn both boost pumps on.
- Check the following:
- L and R BOOST PUMP ON lights on.
 - L and R BOOST PUMP INOP lights out.
 - **L** and **R FUEL PUMP ON** advisory messages on.
 - All **FUEL LO PRESS** caution messages are out.



When operating on the ground with one engine operating, a fuel imbalance may occur with both boost pumps ON. This fuel imbalance may result in fuel spillage if fuel tanks are full. Select the inoperative engine boost pump to INOP. If a fuel imbalance occurs on the ground during single engine operation, open the respective L TO AUX or R TO AUX XFLOW valve to balance the fuel tanks.

- (14) External air Disconnect
- (15) Bleed leak detection system Test 1
- (a) BLEED AIR, DUCT MON switch Set to TEST. Check that:
- L and R DUCT FAIL lights (10th and 14th) are on.
 - **L** and **R 10TH DUCT** warning messages on.
 - **L** and **R 14TH DUCT** warning messages on.
 - “BLEED AIR DUCT” aural on.
 - **ANTI-ICE DUCT** warning message on.
 - **DUCT TEST OK** advisory message on.
- (b) BLEED AIR, DUCT MON switch Set to LOOP A. Check that:
- No warning messages come on.
 - **LOOP A TEST OK** advisory message on.



1. ENGINE STARTING (CONT'D)

E. Engine Start Procedure (Battery and External Air) (Cont'd)

- (c) BLEED AIR, DUCT MON
switch.....Set to LOOP B. Check that:
- No warning messages come on.
 - **LOOP B TEST OK** advisory message on.
- (d) BLEED AIR, DUCT MON
switch.....Set to NORM.
Check that no warning messages come on.
- (16) BLEED AIR, 10TH STAGE, R
switch/light.....Press in Check R 10TH STAGE bleed CLOSED light out.
- (17) AIR-CONDITIONING, R PACK
switch/light.....Press in
- (18) Normal Power-Up Check..... Complete

TRAINING PURPOSES ONLY

1. ENGINE STARTING (CONT'D)

F. Engine Start Procedure (Cross-Bleed)

CAUTION

When starting engines in close quarters, consideration should be given to the effects of jet blast.

This procedure requires the use of higher than normal thrust settings on the ground. Ensure that intake and exhaust areas, of the operating engine, are secure.

- (1) BLEED AIR, APU LCV switch/light.....Press out to close load control valve.
Check APU LCV OPEN light out.
- (2) BLEED AIR, L (R) 10TH STAGE
switch/light.....Press in to open 10th stage SOV for operating engine.
Check L (R) 10TH STAGE CLOSED light out.
- (3) EICAS page.....Select and monitor bleed pressure.
- (4) Thrust lever of operating engine.....Advance to achieve a minimum of 60 psi bleed air pressure (approximately 85% N₂).
- (5) L (R) ENG START switch/lightPress to start applicable engine.
Check the following:
 - Applicable START light comes on.
 - IGNITION A (B) ON light comes on.
 - **IGNITION A (B)** advisory message comes on.
 - Check that N₁ increases.

When N₂ reaches 20% RPM and ITT is below 120°C:

- (6) Thrust lever of starting engine.....Advance to IDLE.

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored to lower ITT to 120°C before moving thrust lever to IDLE. The starter cranking limit of 90 seconds for first attempt must be adhered to.



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1. ENGINE STARTING (CONT'D)

F. Engine Start Procedure (Cross-Bleed) (Cont'd)

- (7) Engine instruments.....Monitor Check the following:
- Starter disengages at 55% N₂.
 - **IGNITION A (B)** advisory message goes out.
 - ITT is 450 – 600°C (max. during start is 900°C).
 - N₁ stabilizes and does not drop below 22% RPM.
 - Check that oil pressure is within normal range [above 25 psi (174 kPa)].
- (8) Thrust lever of first engineSelect to IDLE.
- (9) Idle speedCheck N₂ variation between engines is within 2%. If greater, notify maintenance personnel.
- (10) BLEED AIR, 10TH STAGE L and R switch/lightsPress in to open both 10th stage SOVs.
- Check L and R 10TH STAGE CLOSED lights out.

G. Failure to Start Procedure

At ambient temperatures of greater than 15°C (59°F), the start must be aborted if the engine fails to start (no light-off) **after 25 seconds** from thrust lever movement to IDLE.

After 60 seconds of start operation and the engine has not reached starter cutout speed, the start should be aborted by selecting STOP.

At temperatures below 15°C (59°F), the start sequence may exceed 60 seconds.

- (1) Thrust lever of starting engineSelect to SHUT OFF.
- (2) IGNITION A (B) switch/light.....Press out to disarm applicable ignition system.

If one engine is operating:

- (3) Inoperative engine FUEL, L (R) BOOST PUMP switch/light.....Press out to disarm boost pump.
- Check **L (R) FUEL PUMP** caution message on.
- (4) Dry motor until ITT is below 120°C (maximum 90 seconds) Accomplish Observe starter limits (Refer to Chapter 2; LIMITATIONS – POWER PLANT).

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1. ENGINE STARTING (CONT'D)

H. After Engine Start Check



If N₂ is 57% or below and OAT is above -20°C (-4°F), do not accelerate above idle.

NOTE

1. Do not accelerate engine until oil pressure has stabilized below 116 psi.
 2. A high vortex noise may occur during stationary engine run-up to high power in crosswind or tailwind conditions. This phenomenon is not harmful to the engines and may be ignored.
- (1) AC POWER, GEN 1 and 2 switches..... Select to ON.
- (2) Electrics (AC and DC) Check the following:
- voltage, frequency and load indications are normal, and all AC busses powered and lit on AC electrical page.
 - TRUs are on-line, and that voltage and load indications are normal on DC electrical page.
 - MAIN BUS TIE symbol indicates tie open.

On airplanes NOT incorporating Service Bulletin 605-34-019:

NOTE

FMS 2 may re-initialize and/or revert to independent operation after generator/APU switching.

- (3) AC POWER, APU GEN switch..... As required
- (4) APU As required
- (5) IGNITION, A (B) switch/light..... Press out to disarm applicable ignition system.
Check applicable ARM light out.
- (6) AIR-CONDITIONING, L and R PACK switch/lights Press in to turn packs on.
Check L and R PACK OFF lights out.



1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

- (7) 14TH stage valves.....Check
- (a) BLEED AIR, 14TH STAGE L
switch/lightPress out Check L CLOSED light on.
- (b) BLEED AIR, 14TH STAGE R
switch/lightPress out Check R CLOSED light on.
- (c) BLEED AIR, 14TH STAGE L
and R switch/lights..... Press in Check L and R CLOSED lights
out.
- (8) Wing anti-ice.....Check
- (a) ANTI-ICE, WING switchSelect to NORM, to open both wing
anti-icing valves.
- Left and right engines ITT increases.
 - N₂ gauges change colour to indicate wing anti-ice mode, in non-compressed display format.
 - **L** and **R WING A/ICE** caution messages come on briefly, then go out. If messages do not go out, advance thrust levers to not more than 75% N₂.
 - **WING A/ICE ON** advisory message on.
 - L HEAT and R HEAT lights on.
- (9) Wing anti-ice/isolation valve
operationCheck 1
- (a) BLEED AIR, 14TH STAGE
ISOL switch/light..... Press in to open isolation valve.
Check ISOL OPEN light on.
- (b) BLEED AIR, 14TH STAGE
ISOL switch/light.....Press out to close isolation valve.
Check ISOL OPEN light out.
- (c) ANTI-ICE, WING switchSelect to OFF.
Check the following:
- N₂ gauges change colour to indicate wing anti-ice off, in non-compressed display format.
 - **WING A/ICE ON** advisory message out.



1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

(10) Anti-ice cowl Check 1

(a) ANTI-ICE, COWL L and R
switch/lights

Press in

to open both cowl anti-icing valves.

Check the following:

- ITT gauges indicate rise in temperature.
- **L** and **R COWL A/ICE** caution messages out.
- **COWL A/ICE ON** advisory message on.

NOTE

It may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.

On airplanes with Supplemental Ground Wing Anti-Icing:

Forty-five seconds after selecting cowl anti-ice on:

(b) CAS messages..... Review

Check the following:

- **COWL A/ICE ON** advisory message remains on.
- **L** and **R COWL A/ICE** caution messages remain out.

NOTE

The COWL L and R switch/lights must be kept pressed in for a minimum of 45 seconds for a valid test. After 45 seconds, it may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.

(c) ANTI-ICE, COWL L and R
switch/lights

Press out

to select cowl anti-ice off.



1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

(11) Supplemental ground wing anti-ice (if installed) Check 1

(a) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice on. Check the following:

- **COWL A/ICE ON** advisory message on.
- **L** and **R COWL A/ICE** caution messages out.

Forty-five seconds after selecting cowl anti-ice on:

(b) CAS messages Review Check **COWL A/ICE ON** advisory message remains on.

NOTE

It may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.

(c) SUPP GND WING ANTI-ICE, SYS switch/light Check pressed out.

(d) SUPP GND WING ANTI-ICE, TEST switch/light Press in Check the following:

- TEST ON light comes on.
- Within 2 minutes, check TEST OK light comes on.

(e) SUPP GND WING ANTI-ICE, TEST switch/light Press out Check all lights out.

(f) ANTI-ICE, COWL L and R switch/lights Press out to select cowl anti-ice off. Check **COWL A/ICE ON** advisory message out.



1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

(12) ANTI-ICE, COWL L and R switch/lightsSelect as required (See Note).

If selected OFF, check the following:

- ITT gauges indicate drop in temperature.
- **COWL A/ICE ON** advisory message out.

NOTE

1. The engine cowl anti-ice system must be on when the OAT is 10°C (50°F) or below and visible moisture in any form is present (such as fog with visibility of one mile or less, rain, snow, sleet and ice crystals).
2. The engine cowl anti-ice system must also be on when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water or slush is present.
3. Single engine taxi operations are prohibited if the OAT is 10°C (50°F) or below.

On airplanes with Supplemental Ground Wing Anti-Icing:

4. If the cowl anti-ice is selected ON, wait a minimum of 45 seconds to ensure that **L** and **R COWL A/ICE** caution messages remain out. It may be necessary to advance thrust levers up to 70% N₂ to open the cowl anti-ice valves.



1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

(13) ANTI-ICE, WING switchSelect as required (See Note).

If selected OFF, check the following:

- N₂ gauges change colour to indicate wing anti-ice off when system is switched off, in non-compressed display format.
- **WING A/ICE ON** advisory message out.

AR Certified Airplanes

WARNING

When the OAT is 5°C (41°F) or below use deicing/anti-icing treatment before take-off.

When the OAT is 5°C (41°F) or below the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off.

NOTE

When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected ON, if required, just prior to thrust increase for take-off.



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1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

NOTE

1. The wing anti-ice system must be selected ON, for final taxi prior to take-off, if the OAT is 5°C (41°F) or below, unless Type II, III or IV anti-icing fluids have been applied.
2. If wing anti-ice is selected ON during taxi, **L (R) WING A/ICE** caution messages may be posted or L HEAT or R HEAT lights may go out, but caution messages must be verified out and **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message and L HEAT and R HEAT lights must be verified on, prior to take-off. If wing anti-ice is not required for take-off it should be selected OFF just prior to take-off.
3. To prevent wing contamination from reverse jet blast, operating the thrust reversers during taxi operations on wet and contaminated surfaces should be avoided.
4. The wing anti-ice system must be selected and confirmed ON for take-off, when the OAT is 5°C (41°F) or below and:
 - (a) visible moisture in any form (such as clouds, fog or mist) is present below 400 feet AGL, or
 - (b) the runway is wet or contaminated, or
 - (c) in the presence of any precipitation (such as rain, snow, sleet or ice crystals).
5. If wing anti-ice is selected ON for take-off, the cowl anti-ice must also be selected ON.
6. When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off.
7. The wing anti-ice system may be selected OFF only if it can be positively verified that all of the following conditions are met:
 - (a) the runway to be used is not wet or contaminated,
 - (b) there is no precipitation, and
 - (c) there is no visible moisture in any form (such as clouds, fog, mist, rain, snow, sleet or ice crystals) below 400 feet AGL.
8. Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION – PHASE OF FLIGHT PROCEDURES.



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1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

(14) ATSCheck

NOTE

1. Prior to all sensors becoming valid, and upon completion of the ATS self-test, the ATS Mode Status Display (MSD) will display a **FAIL** message. The **FAIL** message can be cancelled by depressing either ATS DISC switch. The ATS system will perform an automatic self-test after all sensors become valid following power-up.
2. Satisfactory self-test will result in illumination of the two green LED annunciators on the ATS Control Panel and scrolling **SELFTEST** message in each MSD. At the completion of the self-test, the green LED annunciators will extinguish and both MSDs will blank.
3. The LED displays in the MSDs and the ATS Control can be tested by holding the LAMP TEST switch in position 1 or 2. The MSD will display the scrolling **SELFTEST** message described above. The ATS system will perform a system self-test at this time also. If the ATS is engaged, the lamp test and self-test are inhibited.

(15) Rudder.....Operate over full travel range.

Check the following:

- Full free movement.
- Corresponding surface displacement on flight controls page.

(16) AileronsOperate over full travel range.

Check the following:

- Full free movement.
- Corresponding surface displacement on flight controls page.
- Ensure that no hydraulic messages come on.



1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

- (17) Elevators Operate over full travel range.
Check the following:
- Free full movement.
 - Corresponding surface displacement on flight control page.
 - Ensure that no hydraulic messages come on.
 - **ELEVATOR SPLIT** caution message out.
- (18) Ground spoilers check..... Test ☐
- (a) GND SPOILERS switch..... Select to DISARM.
Verify that both ground spoiler panels are retracted.

NOTE

If on the ground and both engines are above 70% N_1 , with the GND SPOILERS switch not armed the **GND SPOILERS NOT ARMED** caution message will come on.

- (b) GND SPOILERS switch..... Select to ARM.
Verify that both ground spoiler panels deploy to maximum travel.
- (c) GND SPOILERS switch..... Select to AUTO.
Verify that ground spoiler panels are retracted.
- (19) Flight spoilers check..... Test ☐
- (a) FLIGHT SPOILER lever Select to MAX.
Verify that flight spoiler panels deploy.

NOTE

When flight spoilers are fully deployed (on the ground with the throttles at IDLE), ground spoilers will deploy if selected to AUTO.

- (b) FLIGHT SPOILER lever Select to RETRACT.
Verify that flight spoiler panels retract.
- (20) NOSE STEER switch Select to ARMED.
Check **STEERING INOP** caution message out.



1. ENGINE STARTING (CONT'D)

H. After Engine Start Check (Cont'd)

- (21) APR.....Check
- (a) ENGINE CONTROL, ENG SPEED switches.....Select

both to ON.
- (b) ENGINE CONTROL, right APR switch.....Select

to TEST 1 and TEST 2 in turn.

Check the following:

• **APR TEST 1 OK** advisory message on.

• **APR TEST 2 OK** advisory message on.

NOTE

APR TEST 1 and **2 OK** advisory messages go out after 10 seconds.

- (c) ENGINE CONTROL, left APR switch.....Select
- to ARM.

NOTE

AT 79% N₁ during take-off, the **APR ARM** advisory message should come on.



If an **APR INOP** caution message is displayed, select the ENGINE CONTROL, APR switch to OFF and use APR-off performance. Refer to Chapter 5; ABNORMAL PROCEDURES – APR Failure.

- (22) ADG AUTO DEPLOY CONTROL switch
- Select to UNIT. Check that TEST light comes on for two (2) seconds, then goes out.

On airplanes NOT incorporating Service Bulletin 605-34-019:

- (23) CDU, FMS CONTROL page
- Select SYNC mode.

1. TAXIING AND TAKE-OFF

A. Taxi Check



Always make sure that the steering is within the nominal operating range of $\pm 55^\circ$ before re-cycling the arming switch.

NOTE

At airports where runway structural repair or debris is known to exist, use thrust reversers with extreme caution to preclude the possibility of foreign object damage (FOD) from occurring.

- (1) Transponder/TCAS Set if required.
- (2) Brakes Check
- (a) Airplane Roll forward and apply brakes.
- Check that no anti-skid or brake messages come on.

If conditions exist that could result in water saturated brakes:

- (b) Brakes Warm Perform the following number of firm brake applications from 25 knots to 10 knots. Perform the prescribed number of brake applications during the last mile/kilometer of taxi prior to and not including the final stop or snub before take-off. Do not drag the brakes.

AIRPLANE WEIGHT	NUMBER OF BRAKE APPLICATIONS
Greater than 42,000 lb	5
36,000 to 42,000 lb	6
Less than 36,000 lb	7

NOTE

- Use of symmetric braking is recommended, to ensure uniform brake heating.
 - Warming of the brakes will preclude the chance of water saturated brakes freezing at altitude and being locked for landing touchdown.
- (3) Flaps..... Set for take-off.
- Check corresponding surface displacement on flight controls page.



1. TAXIING AND TAKE-OFF (CONT'D)

A. Taxi Check (Cont'd)

- (4) Thrust reversersCheck 1
- (a) Left thrust reverse lever.....Deploy Check the following:
- REV icon appears on left N₁ gauge only.
 - **R REV UNLOCKED** caution message out.
 - Left and right UNLK lights out.
- (b) Left thrust reverse lever.....Stow Check **L REV UNLOCKED** caution message out.
- (c) Right thrust reverse leverDeploy Check the following:
- REV icon appears on right N₁ gauge only.
 - **L REV UNLOCKED** caution message out.
 - Left and right UNLK lights out.
- (d) Right thrust reverse leverStow Check **R REV UNLOCKED** caution message out.
- (5) Aileron and rudder trimsCheck Set at neutral for take-off.



1. TAXIING AND TAKE-OFF (CONT'D)

A. Taxi Check (Cont'd)

(6) Stabilizer trim..... Set for take-off.

Check the following:

- Both channels of the stabilizer trim are engaged.
- Trim is set to the correct take-off CG range.

EICAS STABILIZER TRIM SETTING FOR TAKE-OFF	
C.G. Range (%MAC)	Trim Setting
16.0 – 16.2	7.2
16.3 – 17.2	7.0
17.3 – 18.3	6.8
18.4 – 19.3	6.6
19.4 – 20.3	6.4
20.4 – 21.3	6.2
21.4 – 22.4	6.0
22.5 – 23.4	5.8
23.5 – 24.4	5.6
24.5 – 25.5	5.4
25.6 – 26.5	5.2
26.6 – 27.5	5.0
27.6 – 28.5	4.8
28.6 – 29.6	4.6
29.7 – 30.6	4.4
30.7 – 31.6	4.2
31.7 – 32.6	4.0
32.7 – 33.7	3.8
33.8 – 34.7	3.6
34.8 – 36.5	3.4
36.6 – 38.0	3.2

NOTE

Trim data from the FMS should be rounded down to the next lowest multiple of 0.2 trim units.



1. TAXIING AND TAKE-OFF (CONT'D)

A. Taxi Check (Cont'd)

- (7) Flight Instruments..... Set Check the following:
- No abnormal annunciations are displayed on the EFIS.
 - EFIS slip/skid indications are normal.
 - EFIS is set for departure.
- (8) Fuel balance.....Check
- (9) ANTI-ICE, WING switchSelect as required.

NOTE

1. The wing anti-ice system must be selected ON, for final taxi prior to take-off, if the OAT is 5°C (41°F) or below, unless Type II, III or IV anti-icing fluids have been applied.
2. If wing anti-ice is selected ON during taxi, **L (R) WING A/ICE** caution messages may be posted or L HEAT or R HEAT lights may go out, but caution messages must be verified out and **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message and L HEAT and R HEAT lights must be verified on, prior to take-off. If wing anti-ice is not required for take-off it should be selected OFF just prior to take-off.
3. To prevent wing contamination from reverse jet blast, operating the thrust reversers during taxi operations on wet and contaminated surfaces should be avoided.
4. When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off.

WARNING

When Type II, Type III or Type IV anti-icing fluids have been applied, use of wing anti-ice heat prior to take-off can cook the fluid onto the leading edges of the wing, creating contamination.



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1. TAXIING AND TAKE-OFF (CONT'D)

A. Taxi Check (Cont'd)

(10) ANTI-ICE, COWL

L and R switch/lightsSelect as required.

NOTE

1. The engine cowl anti-ice system must also be on when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water or slush is present.
2. Single engine taxi operations are prohibited if the OAT is 10°C (50°F) or below.
3. During taxi with the anti-ice selected ON, the **L** or **R COWL A/ICE** caution message may be posted and will go out if thrust is increased.

TRAINING PURPOSES ONLY

1. TAXIING AND TAKE-OFF (CONT'D)

B. Take-Off Check

- (1) Correct runway Confirm
- (2) IGNITION, CONT switch/light.....Press to turn ON, if required.
- (3) ANTI-ICE, WING switchSelect as required.

NOTE

1. Engage ATS only prior to take-off. A valid N_1 value is required to keep the ATS engaged. With wing anti-ice only selected, ATS is unavailable.
2. The wing anti-ice system must be selected ON, for final taxi prior to take-off, if the OAT is 5°C (41°F) or below, unless Type II, III or IV anti-icing fluids have been applied.
3. If wing anti-ice is selected ON during taxi, **L (R) WING A/ICE** caution messages may be posted or L HEAT or R HEAT lights may go out, but caution messages must be verified out and **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message and L HEAT and R HEAT lights must be verified on, prior to take-off. If wing anti-ice is not required for take-off it should be selected OFF just prior to take-off.
4. The wing anti-ice system must be selected and confirmed ON for take-off, when the OAT is 5°C (41°F) or below and:
 - (a) visible moisture in any form (such as clouds, fog or mist) is present below 400 feet AGL, or
 - (b) the runway is wet or contaminated, or
 - (c) in the presence of any precipitation (such as rain, snow, sleet or ice crystals).
5. If wing anti-ice is selected ON for take-off, the cowl anti-ice must also be selected ON.
6. When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected and confirmed ON just prior to thrust increase for take-off.

WARNING

When Type II, Type III or Type IV anti-icing fluids have been applied, use of wing anti-ice heat prior to take-off can cook the fluid onto the leading edges of the wing, creating contamination.



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1. TAXIING AND TAKE-OFF (CONT'D)

B. Take-Off Check (Cont'd)

NOTE

- The wing anti-ice system may be selected OFF only if it can be positively verified that all of the following conditions are met:
 - the runway to be used is not wet or contaminated,
 - there is no precipitation, and
 - there is no visible moisture in any form (such as clouds, fog, mist, rain, snow, sleet or ice crystals) below 400 feet AGL.
- Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION – PHASE OF FLIGHT PROCEDURES.

AR Certified Airplanes

WARNING

When the OAT is 5°C (41°F) or below use deicing/anti-icing treatment before take-off.

When the OAT is 5°C (41°F) or below the wing anti-ice system must be selected ON for take-off, just prior to thrust increase for take-off.

NOTE

When Type II, III or IV anti-icing fluids have been applied, the wing anti-ice system must only be selected on, if required, just prior to thrust increase for take-off.

- ANTI-ICE, COWL
L and R switch/lightsSelect as required.

NOTE

- The engine cowl anti-ice system must be on when the OAT is 10°C (50°F) or below and visible moisture in any form is present (such as fog with visibility of one mile or less, rain, snow, sleet and ice crystals).
- The engine cowl anti-ice system must also be on when the OAT is 10°C (50°F) or below when operating on runways, ramps, or taxiways where surface snow, ice, standing water or slush is present.
- Single engine taxi operations are prohibited if the OAT is 10°C (50°F) or below.
- With the anti-ice selected ON, the **L** or **R COWL A/ICE** caution message may be posted and will go out if thrust is increased.

- ANTI-ICE, WSHLD/WIND
L and R switchesSelect to HI or LOW, as required.



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1. TAXIING AND TAKE-OFF (CONT'D)

B. Take-Off Check (Cont'd)

- (6) ANTI-ICE, PROBES switchesSelect to ON.

NOTE

If departure is not imminent, delay selecting the PROBES to ON until just prior to take-off. A lack of cooling airflow with the PROBES selected ON may result in a potential nuisance **WINDSHEAR FAIL** status message.

On airplanes incorporating Service Bulletin 605-34-023:

If departure is not imminent, delay selecting the PROBES to ON until just prior to take-off. A lack of cooling airflow with the PROBES selected ON may result in a potential nuisance **WINDSHEAR FAIL** status message or a **TURB FAULT** message on the PFD.

- (7) Cabin Advise not less than one minute prior to take-off.
- (8) Cockpit lighting Set as required. Check EMER LTS switch selected to ARM.
- (9) EXTERNAL LTS switches Set as required.



The left and right landing lights must not be used on the ground, except for the take-off roll and landing.

- (10) Crew Alerting System (CAS)Check the following:
- **T/O CONFIG OK** advisory message is on.
 - No warning, caution or status messages are present.
- (11) Take-off data (V₁, V_R, V₂ and distance)Check displayed.
- (12) SUPP GND WING ANTI-ICE panel (if installed)Check
- (a) TEST and SYS switch/lightsCheck pressed out.
- Check TEST ON, TEST OK and OFF lights out.
- (b) TEMP lightsCheck lights out.
- (13) Weather radar selectors Set as required.
- (14) Transponder/TCAS Set
- (15) EFIS altimeters (2) Cross-check

1. TAXIING AND TAKE-OFF (CONT'D)

C. Normal Take-Off Procedure

AR Certified Airplanes**Take-Off Speeds**

Add 5 knots to the take-off speeds (V_1 , V_R and V_2) derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE.

Take-Off Distance

The take-off distances, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be increased by 10%.

Maximum Allowable Take-Off Weight

The take-off weight limited by brake energy, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be reduced by 500 kg (1,100 lb).

The take-off weight limited by tire speed, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be reduced by 1,200 kg (2,650 lb).

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:**Prior to entering the runway or when aligned on the runway centre line:**

- (1) Take-Off/Go-Around switch Press
- (2) VS pitch wheel Reduce the flight director command cue to 10 degrees.

NOTE

Auto throttle will not be available for take-off.

When cleared for take-off and aligned on the runway centre line:

- (3) Brakes Apply
- (4) Take-Off/Go-Around switch Press Ensure N1 TO message is posted on ATS Mode Status Display (MSD).

NOTE

If N1 TO message is not posted, depress ATS DISC switch and deselect flight director. Ensure N₁ is posted and re-select TOGA switch.



1. TAXIING AND TAKE-OFF (CONT'D)

C. Normal Take-Off Procedure (Cont'd)

- (5) Thrust levers.....Advance Set thrust to appropriate take-off N_1 setting.
- Check the following:
- Thrust indications are stable.
 - **APR ARM** advisory message comes on at above 79% N_1 .
 - Flight director cue bar at 12° on PFDs.

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

- (5) Thrust levers.....Advance Set thrust to appropriate take-off N_1 setting.
- Check the following:
- Thrust indications are stable.
 - **APR ARM** advisory message comes on at above 79% N_1 .

- (6) BrakesRelease
- (7) Directional control..... Maintain with rudder and aileron inputs, as required.
- (8) Flight and engine instrumentsMonitor Check that engine indications do not exceed limits.
- (9) N_1 setting.....Crosscheck
- (10) AirspeedMonitor



1. TAXIING AND TAKE-OFF (CONT'D)

C. Normal Take-Off Procedure (Cont'd)

WARNING

Excessive rotation rates (exceeding 3 degrees per second) or over-rotations may lead to high pitch attitudes and angles of attack being attained while the aircraft is near the ground. This can reduce stall margins significantly resulting in stick shaker/pusher activation and potentially loss of control. Pilots must rotate smoothly towards the target pitch attitude then transition to speed control.

NOTE

1. The flight director guidance represents an initial target for rotation only and does not guarantee that the recommended climb speed will be achieved/maintained under all conditions. Pilots must transition to speed control immediately after initial rotation.
2. Stall margins may be improved by reducing initial pitch attitude.

At V_R :

(11) Airplane Rotate smoothly towards an initial pitch target of 12 degrees in one continuous motion at a pitch rate not exceeding 3 degrees per second.

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

(11) Airplane Rotate smoothly towards an initial pitch target of 10 degrees in one continuous motion at a pitch rate not exceeding 3 degrees per second.

(12) Pitch attitude.....Adjust to achieve a minimum airspeed of $V_2 + 10$ KIAS.



1. TAXIING AND TAKE-OFF (CONT'D)

C. Normal Take-Off Procedure (Cont'd)

When positive rate of climb is indicated:

(13) LDG GEAR lever Set to UP.

Check three amber hash marks are displayed on the EICAS page, indicating that the landing gear is in transit, followed by three white UP indications, indicating that the landing gear is up and locked.

NOTE

If taxi and/or take-off were on snow or slush covered surfaces, unless weather conditions or performance requirements prohibit, delay retraction of the landing gear until excess water, snow or slush is thrown off by wheel rotation and/or slip stream force.

1. TAXIING AND TAKE-OFF (CONT'D)

D. High Crosswind Take-Off Procedure

When the crosswind component during take-off is greater than 10 knots, the following procedure should be used to avoid engine instabilities. Continuous engine ignition must be used during take-offs with high crosswind components.

AR Certified Airplanes**Take-Off Speeds**

Add 5 knots to the take-off speeds (V_1 , V_R and V_2) derived from the Airplane Flight Manual, Chapter 6: Performance – Take-Off Performance.

Take-Off Distance

The take-off distances, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be increased by 10%.

Maximum Allowable Take-Off Weight

The take-off weight limited by brake energy, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be reduced by 500 kg (1,100 lb).

The take-off weight limited by tire speed, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be reduced by 1,200 kg (2,650 lb).

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:**Prior to entering the runway or when aligned on the runway centre line:**

- (1) Take-Off/Go-Around switch.....Press
- (2) VS pitch wheel..... Reduce the flight director command cue to 10 degrees.

NOTE

Auto throttle will not be available for take-off.

When cleared for take-off and aligned on the runway centre line:

- (3) Brakes.....Apply
- (4) Thrust levers..... Set thrust to take-off power minus 5% N_1 .
Check **APR ARM** advisory message comes on at above 79% N_1 .
- (5) BrakesRelease



1. TAXIING AND TAKE-OFF (CONT'D)

D. High Crosswind Take-Off Procedure (Cont'd)

- (6) Thrust levers.....Advance to achieve take-off N_1 by the time 60 KIAS is reached, and press Take-Off/Go-Around (TOGA) switches.

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

- (6) Thrust levers.....Advance to achieve take-off N_1 by the time 60 KIAS is reached.

- (7) Flight and engine instrumentsMonitor Check that engine indications do not exceed limits.

- (8) N_1 setting.....Crosscheck

- (9) AirspeedMonitor

WARNING

Excessive rotation rates (exceeding 3 degrees per second) or over-rotations may lead to high pitch attitudes and angles of attack being attained while the aircraft is near the ground. This can reduce stall margins significantly resulting in stick shaker/pusher activation and potentially loss of control. Pilots must rotate smoothly towards the target pitch attitude then transition to speed control.

NOTE

- The flight director guidance represents an initial target for rotation only and does not guarantee that the recommended climb speed will be achieved/maintained under all conditions. Pilots must transition to speed control immediately after initial rotation.
- Stall margins may be improved by reducing initial pitch attitude.

At V_R :

- (10) Airplane.....Rotate smoothly towards an initial pitch target of 12 degrees in one continuous motion at a pitch rate not exceeding 3 degrees per second.

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

- (10) Airplane.....Rotate smoothly towards an initial pitch target of 10 degrees in one continuous motion at a pitch rate not exceeding 3 degrees per second.

- (11) Pitch attitude.....Adjust to achieve a minimum airspeed of $V_2 + 10$ KIAS.



1. TAXIING AND TAKE-OFF (CONT'D)

D. High Crosswind Take-Off Procedure (Cont'd)

When positive rate of climb is indicated:

(12) LDG GEAR lever Set to UP.

Check three amber hash marks are displayed on the EICAS primary page, indicating that the landing gear is in transit, followed by three white UP indications, indicating that the landing gear is up and locked.

NOTE

If taxi and/or take-off were on snow or slush covered surfaces, unless weather conditions or performance requirements prohibit, delay retraction of the landing gear until excess water, snow or slush is thrown off by wheel rotation and/or slip stream force.

1. TAXIING AND TAKE-OFF (CONT'D)

E. Rolling Take-Off Procedure

Rolling take-off procedures are described below. If used, the corresponding take-off distance penalties must be applied (refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE – ROLLING TAKE-OFF PROCEDURE).

AR Certified Airplanes**Take-Off Speeds**

Add 5 knots to the take-off speeds (V_1 , V_R and V_2) derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE.

Take-Off Distance

The take-off distances, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be increased by 10%.

Maximum Allowable Take-Off Weight

The take-off weight limited by brake energy, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be reduced by 500 kg (1,100 lb).

The take-off weight limited by tire speed, derived from the Airplane Flight Manual, Chapter 6; PERFORMANCE – TAKE-OFF PERFORMANCE, must be reduced by 1,200 kg (2,650 lb).

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:**Prior to entering the runway or when aligned on the runway centre line:**

- (1) Take-Off/Go-Around switch.....Press
- (2) VS pitch wheel..... Reduce the flight director command cue to 10 degrees.

NOTE

Auto throttle will not be available for take-off.

When cleared for take-off and aligned on the runway centre line:

- (3) BrakesRelease
- (4) Take-Off/Go-Around switch.....Press Ensure N1 TO message is posted on ATS Mode Status Display (MSD).

NOTE

If N1 TO message is not posted, depress ATS DISC switch and deselect flight director. Ensure N₁ is posted and re-select TOGA switch.



1. TAXIING AND TAKE-OFF (CONT'D)

E. Rolling Take-Off Procedure (Cont'd)

- (5) Thrust levers.....Advance promptly to pre-determined take-off N_1 by the time 60 KIAS is reached.
Check **APR ARM** advisory message comes on at above 79% N_1 .
- (6) Flight and engine instrumentsMonitor Check that engine indications do not exceed limits.
- (7) N_1 setting.....Crosscheck
- (8) AirspeedMonitor

WARNING

Excessive rotation rates (exceeding 3 degrees per second) or over-rotations may lead to high pitch attitudes and angles of attack being attained while the aircraft is near the ground. This can reduce stall margins significantly resulting in stick shaker/pusher activation and potentially loss of control. Pilots must rotate smoothly towards the target pitch attitude then transition to speed control.

NOTE

- The flight director guidance represents an initial target for rotation only and does not guarantee that the recommended climb speed will be achieved/maintained under all conditions. Pilots must transition to speed control immediately after initial rotation.
- Stall margins may be improved by reducing initial pitch attitude.

At V_R :

- (9) Airplane Rotate smoothly towards an initial pitch target of 12 degrees in one continuous motion at a pitch rate not exceeding 3 degrees per second.

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

- (9) Airplane Rotate smoothly towards an initial pitch target of 10 degrees in one continuous motion at a pitch rate not exceeding 3 degrees per second.

- (10) Pitch attitude.....Adjust to achieve a minimum airspeed of $V_2 + 10$ KIAS.



1. TAXIING AND TAKE-OFF (CONT'D)

E. Rolling Take-Off Procedure (Cont'd)

When positive rate of climb is indicated:

(11) LDG GEAR lever Set to UP.

Check three amber hash marks are displayed on the EICAS primary page, indicating that the landing gear is in transit, followed by three white UP indications, indicating that the landing gear is up and locked.

NOTE

If taxi and/or take-off were on snow or slush covered surfaces, unless weather conditions or performance requirements prohibit, delay retraction of the landing gear until excess water, snow or slush is thrown off by wheel rotation and/or slip stream force.

F. After Take-Off Procedure

(1) Airspeed $V_2 + 10$ to 15 KIAS

(2) Flight and engine instruments Monitor

Check that engine indications remain within limits, and flight instruments indicate the desired climb profile.

At a safe altitude, not below 320 ft AGL:

(3) AP switch/light Press in to engage, if required.

Check the following:

- Heading mode engaged.
- Speed bug set to 200 KIAS.

NOTE

The use of excessive pitch mode (VS) on the FCP could cause autopilot uncommanded disengagement.

Autopilot can be re-engaged after disengagement.

BOMBARDIER CHALLENGER 605	NORMAL PROCEDURES Taxiing and Take-Off	Vol. 1	04-06-19 REV 47, May 29/18
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1. TAXIING AND TAKE-OFF (CONT'D)

G. After Take-Off Check

- (1) Landing gearCheck retracted.

NOTE

During normal operations, landing gear and flap indications are normally occulted (removed from EICAS) when the landing gear are stowed and the flaps are up.

At not less than $V_{FTO} + 5$ KIAS:

- (2) Flaps..... Set to 0°.
- (3) THRUST REVERSER,
L and R switchesSelect to OFF.
Check **L** and **R REV ARMED** advisory messages out.
- (4) CASCheck/clear Verify and analyze any messages, and clear (if any).
Monitor during flight.
- (5) IGNITION, CONT switch/light..... As required

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

The engine cowl and wing anti-ice systems must be on when in icing conditions or when ice is indicated by the ice detection system.

- (6) ANTI-ICE, COWL
L and R switch/lightsSelect as required.
- (7) ANTI-ICE, WING switchSelect as required.

If wing anti-ice is selected ON and WING L HEAT or R HEAT light goes out or **WING A/ICE ON or **WING/COWL A/ICE ON** advisory message goes out:**

- (8) Thrust Increase until the following come on:
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message.
 - WING L HEAT and R HEAT lights.

If **WING A/ICE ON or **WING/COWL A/ICE ON** advisory message remains out or WING L HEAT or R HEAT light remains out:**

- (9) Leave icing conditions.

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TRAINING PURPOSES ONLY

1. CLIMB, CRUISE AND DESCENT

A. Climb Check

NOTE

When the auxiliary tank fuel quantity is low, the fuel quantity will fluctuate during the climb due to fuel migration.

- (1) Pressurization.....Check

If the APU is supplying bleed air:

- (2) Air-conditioning..... Transfer to engine bleeds.

NOTE

Bleed air extraction from the APU is not permitted above 15,000 feet.

- (a) BLEED AIR, 10TH STAGE R
switch/light..... Press in to open right valve.
Check the following:
- R 10TH CLOSED light out.
 - APU LCV OPEN light out.
- (b) BLEED AIR, 10TH STAGE
ISOL switch/light..... Press out to close isolation valve.
Check the following:
- 10TH ISOL OPEN light out.
 - APU LCV OPEN light on.
- (c) BLEED AIR, 10TH STAGE L
switch/light..... Press in to open left valve and close
APU load control valve.
Check the following:
- L 10TH CLOSED light out.
 - APU LCV OPEN light out.
- (d) BLEED AIR, APU LCV
switch/light..... Press out to match APU load control
valve closed position.
- (3) Climb thrust Set the computed climb thrust
setting.



1. CLIMB, CRUISE AND DESCENT (CONT'D)

A. Climb Check (Cont'd)

CAAI Certified Airplanes

When above 10,000 feet AGL:

(4) AC POWER, APU GEN switch.....Select to OFF/RESET.
Check **APU GEN OFF** caution message on.

(5) APUShutdown
(a) APU, START/STOP switch/lightPress out to shutdown the APU.
Check AVAIL light out.

When rpm decreases to below 50% and EGT to below 350°C (662°F):

(b) APU, PWR FUEL switch/lightPress out Check the following:

- **APU GEN OFF** caution message out.
- APU readouts disappear from EICAS page after approximately 60 seconds.

(6) TAWS WARNING, TERRAIN OFF switch/lightCheck Press out.
Check OFF light out.

(7) PASS SIGNS switches As required

(8) EXTERNAL LTS switches As required

At transition level:

(9) EFIS altimeters (2) and Integrated Standby Instrument altimeter Set to 29.92 inches Hg.

1. CLIMB, CRUISE AND DESCENT (CONT'D)

B. Cruise Check

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

At or above 22,000 feet, the engine cowl and wing anti-ice systems must be on when ice is indicated by the ice detection system.

Below 22,000 feet, the engine cowl and wing anti-ice systems must be on when in icing conditions or when ice is indicated by the ice detection system.

- (1) ANTI-ICE, WING switch Select as required.
- (2) ANTI-ICE, COWL L and R switch/lights Select as required.
- (3) Cruise thrust Set
- (4) Pressurization Check that pressure differential and cabin altitude are normal.
- (5) Fuel Monitor fuel balance.

NOTE

During uncoordinated flight, fuel indications are inaccurate. Accurate fuel indications will be obtained after 30 seconds when level and stabilized in coordinated flight.

During gravity crossflow operations, a steady heading sideslip must be used to ensure proper operation and to accelerate gravity crossflow.

To enhance gravity crossflow operation, fly one wing low slip (on the side with less fuel).



Continued or repetitive automatic crossflow operation could be indicative of a fuel tank leak.

1. CLIMB, CRUISE AND DESCENT (CONT'D)

C. Descent Check

A thorough and comprehensive planning is essential to ensure smooth transitions from one phase of flight to another. Prior to commencing descent, the following procedures should be performed to ensure that the airplane is properly configured, and the various airplane systems set-up for the subsequent phase of flight.

- (1) ATIS Copy
- (2) CAS Check/Clear Verify and analyze any messages, and clear (if any).

Within 15 NM of approach, if any of the following conditions apply:

- Runway is less than 3,500 feet in length, or
- Airport is not in the database, or
- Intended approach is not compatible with terrain awareness alerting, or
- QFE operation with no GPS available.

- (3) TAWS WARNING, TERRAIN OFF switch/light Press Check TAWS WARNING, TERRAIN OFF switch/light on.

- (4) Landing data Set the following:
- ATC clearance should have been acquired.
 - Landing speeds, N_1 for all engine go-around (take-off thrust setting), N_1 for engine-out go-around (single engine climb thrust setting, APR rating) and MCT calculated for the landing weight and bugs set accordingly.
 - Set the RADIO and NAV equipment for the approach.
 - Set the associated bugs accordingly.

FAA Certified Airplanes

- (4) Landing data Set the following:
- ATC clearance should have been acquired.
 - Set the RADIO and NAV equipment for the approach.
 - Set the associated bugs accordingly.

1. CLIMB, CRUISE AND DESCENT (CONT'D)

C. Descent Check (Cont'd)

- (5) Approach briefing Complete

The approach briefing should be accomplished well before entering the terminal control zone. The pilot-flying shall conduct the briefing and review and/or outline the operational aspects of the expected approach, which should include the following:

- (a) Type of approach
- (b) Runway in use
- (c) Landing minima
- (d) Review of V_{REF} and N_1 values
- (e) Altitudes (MSA, Field elevation, Threshold elevation, Descent crossing altitudes)
- (f) Outbound and procedure turn courses
- (g) Final inbound course
- (h) Decision Height (DH) or Minimum Descent Altitude (MDA)
- (i) Missed approach point (non-precision)
- (j) Missed approach procedure
- (k) NAV equipment set-up
- (l) Any questions or clarifications and other pertinent details.

- (6) Shoulder harnesses Secured

On airplanes incorporating Service Bulletin 604-34-023:

NOTE

The weather radar AUTO mode of operation may not accurately display the weather threat to the crew below the flight path of the airplane. MAN mode may be used to scan altitudes below the flight path.

- (7) CABIN PRESSURIZATION, LDG

ALT selector Check selected to the following:

- Destination airfield altitude.
- Destination airfield barometric pressure.
- Desired cabin pressure descent rate.

Check FAULT lamp out.

1. CLIMB, CRUISE AND DESCENT (CONT'D)

C. Descent Check (Cont'd)

NOTE

1. Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.
2. The use of anti-icing should always be anticipated during descent when operating in cold weather conditions.
3. Wing and engine cowl anti-icing should be activated when descending through areas of visible moisture (Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION).
4. If icing conditions are anticipated during descent, adjust descent profile to maintain a minimum of 78% N₂ to prevent ice accumulation. If ATS is engaged, the following actions can be taken to maintain a minimum of 78% N₂:
 - Increase speed and/or,
 - Decrease rate of descent and/or,
 - Use of flight spoiler if necessary or,
 - A combination of any of the three above, or
 - Disengage ATS and set thrust manually.

At or above 22,000 feet, the engine cowl and wing anti-ice systems must be on when ice is indicated by the ice detection system.

Below 22,000 feet, the engine cowl and wing anti-ice systems must be on when in icing conditions, or when ice is indicated by the ice detection system.

(8) ANTI-ICE, COWL L and R
switch/lights Select as required.

(9) ANTI-ICE, WING switch Select as required.

If wing anti-ice is selected ON and WING L HEAT or R HEAT light goes out or WING A/ICE ON or WING/COWL A/ICE ON advisory message goes out:

(10) Thrust Increase until the following come on:

- **WING A/ICE ON or WING/COWL A/ICE ON** advisory message.
- WING L HEAT and R HEAT lights.

If WING A/ICE ON or WING/COWL A/ICE ON advisory message remains out or WING L HEAT or R HEAT light remains out:

(11) Leave icing conditions.

1. CLIMB, CRUISE AND DESCENT (CONT'D)**C. Descent Check (Cont'd)**

(12) ANTI-ICE, WSHLD/WIND L and R

switches.....Select to HI or LOW, as required.

(13) Fuel quantity and balanceCheck the following:

- Cross-check all the readouts in the fuel quantity gauging system.
- Check fuel balance.

At transition level:

(14) EFIS altimeters (2) and Integrated

Standby Instrument altimeter (1) Set to destination airport
barometric setting.

1. CLIMB, CRUISE AND DESCENT (CONT'D)

D. In Range Check

- (1) APU (if required)..... Start Ensure that operating limitations/envelope is considered.

CAAI Certified Airplanes

When above 10,000 feet AGL:

- (1) APU (if required)..... Start Ensure that operating limitations/envelope is considered.

NOTE

The APU, PWR FUEL switch/light must be selected off between each APU start attempt. The APU, PWR FUEL switch/light should be firmly pressed once only. If pressed twice, an immediate shutdown may occur.

- (a) BLEED AIR, APU LCV switch/light..... Check pressed out.
- (b) APU, PWR FUEL switch/light..... Press in and check the following:
- PWR FUEL light on, and
 - APU readouts displayed on EICAS page.

NOTE

During in flight starts at high altitudes, the APU EGT may momentarily spike into the amber temperature zone and then immediately return to normal green values. This is normal operation and no crew action is required.

- (c) APU, START/STOP switch/light..... Press in and check the following:
- START light on,
 - 60% rpm - START light out.
 - 95% rpm + 4 seconds (approx) - AVAIL light on.
 - 100% rpm + 15 seconds (approx) - **APU GEN OFF** caution message on.

NOTE

1. **APU BATTERY OFF** caution message may appear momentarily during APU start.
2. If time permits, allow APU rpm to stabilize at 100% for 2 minutes before applying bleed air load.



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1. CLIMB, CRUISE AND DESCENT (CONT'D)

D. In Range Check (Cont'd)

After APU RPM and EGT have stabilized:

- (d) AC POWER, APU GEN switch.....Select as required.

If transferring air-conditioning to APU bleed air:

- (2) Air-conditioning.....Transfer to APU bleed.

NOTE

1. The ATS will disengage if the N₁ value is removed from the EICAS (no active selection present).
2. The 10th-stage engine bleeds must be closed for landing with the cowl and/or wing anti-ice system on. Refer to Chapter 2; LIMITATIONS – SYSTEM LIMITATIONS .
 - (a) BLEED AIR, APU LCV
switch/light Press in to preselect load control valve to the open position.
 - (b) BLEED AIR, 10TH STAGE L
switch/light Press out to select left valve closed.
Check the following:
 - APU LCV OPEN light on.
 - L 10TH CLOSED light on.
 - (c) BLEED AIR, 10TH STAGE
ISOL switch/light..... Press in to open isolation valve.
Check the following:
 - 10TH ISOL OPEN light on.
 - APU LCV OPEN light out.
 - (d) BLEED AIR, 10TH STAGE R
switch/light Press out to select right valve closed.
 - R 10TH CLOSED light on.
 - APU LCV OPEN light on.
- (3) EXTERNAL LTS switches As required Select RECOG TAXI light on.
- (4) PASS SIGNS switches As required

NOTE

Normally, the PASS SIGNS switches are set at AUTO, although the pilot may elect to have the Fasten Seat Belts sign turned ON earlier due to weather, turbulence, or any other consideration.

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TRAINING PURPOSES ONLY

1. APPROACH AND LANDING

A. Approach

It is assumed that all the necessary preparations for the approach have been accomplished prior to reaching the terminal control zone. The descent check should have been initiated prior to the start of the descent and completed before arriving at the terminal area. Plan the descent such that the airplane will be at the traffic pattern altitude at flaps-up maneuvering speed, at about 12 miles (19 km) out when proceeding straight-in, or at about 8 miles (13 km) out when entering the pattern abeam.

On airplanes incorporating Service Bulletin 605-34-019:

NOTE

When changing the destination to ALTN airport by selecting direct-to a waypoint in the route to the ALTN airport, if Temp Comp is ON and an OAT value has been entered for the original DEST, the same OAT value will be retained for the new DEST (ALTN). Crew must verify the correctness of OAT after changing destination.

As a rule-of-thumb, the descent should be planned so as to arrive at the following conditions:

- Altitude (AGL) 10,000 feet
- Distance from the airport 30 miles (48 km)
- Airspeed.....250 KIAS

NOTE

If FMS is in use, TOD will put the airplane at 15,000 feet AGL, 10 NM back.

B. Maneuvering and Approach Speeds

The minimum maneuvering speed is obtained by adding 10 knots to the approach speed as defined in the Airplane Flight Manual. The 10-knot additive provides adequate margin to stick shaker actuation for an inadvertent 15° overshoot beyond the normal 30° bank, when recommended procedures are followed.

1. APPROACH AND LANDING (CONT'D)

C. Holding

Prior to entering the hold, it is assumed that a holding clearance and all other pertinent data have been acquired. All other necessary items (entry procedures, timing, fuel, etc.) relevant to the hold and subsequent approach should also have been determined.

- (1) Airspeed Flaps 0° speed
+ 30 KIAS

- Reduce the airplane speed at least 3 minutes prior to arrival at the holding fix.
- Review the performance data for thrust setting if a prolonged holding is expected.

Procedures

Upon receipt of a holding clearance:

- (1) Request:
- Estimated approach time,
 - Type of approach in use.
- (2) Check:
- Minimum diversion fuel,
 - Weather at the destination and at the alternate,
 - Endurance speed.
- (3) Procedure:
- Determine entry procedure, and
 - Brief pilot not flying.

NOTE

If time does not permit determining entry procedure upon station passage, turn immediately outbound to parallel outbound track. After the required timing, turn inbound towards the holding track.

Holding patterns must be entered and flown at or below the airspeed set by regulatory agencies. Air Traffic Control (ATC) must be advised if it is impossible to comply with the appropriate maximum holding airspeed.

The initial outbound leg should be flown for 1 minute or 1 1/2 minutes, as required, for the altitude. Timing for subsequent outbound legs should be adjusted, as necessary, to achieve the proper inbound leg timing. ATC should be advised immediately if an increase in airspeed is necessary due to turbulence, or if it is impossible to accomplish any part of the holding procedure.

The maximum holding airspeeds and timing are shown in the table below:

ALTITUDE	AIRSPEED (KIAS)	TIMING (MINUTES)
Up to and including 14,000 feet	230	1.0
Above 14,000 feet	265	1.5

1. APPROACH AND LANDING (CONT'D)

D. Precision (ILS/PAR) Approach

Precision approaches utilizing the autopilot and/or the flight director are essentially the same.

Approaching the fix (or beacon), when performing a full approach (with procedure turn), or coming from a hold:

NOTE

When performing an ILS autopilot-coupled approach, ensure that both flight directors are displayed.

- (1) Airspeed Flaps 20° speed
+ 20 KIAS

Fix outbound:

- (2) Flaps 20 degrees

For straight-in approach (radar vectored):

- (1) Flaps 20 degrees
(2) Airspeed Flaps 20° speed
+ 20 KIAS

NOTE

When performing a straight-in approach, assume the initial approach configuration (as in 1 and 2 above), at a distance of 12 miles away from the airport.



To avoid possible false localizer captures, do not arm approach until airplane is close to, or established on, an inbound localizer course.

Cleared for the approach; Fix inbound (for a full approach), or when on the localizer intercept heading (for a straight-in approach):

- (3) FCP mode APPR
(4) Mode annunciations Verify
- LOC 1 (2) and GS armed – white indications on both PFDs.
 - LOC 1 (2) – green indications on both PFDs.

When 1 dot below the glideslope:

- (5) LDG GEAR lever Set to DN.
(6) Flaps 30 degrees

At glideslope capture (indicated by green GS annunciation on both PFDs):

- (7) Flaps 45 degrees
(8) Airspeed $V_{REF} + 10$ KIAS
(9) Before landing check Complete
(10) Preselect ALT Set to missed approach altitude.



1. APPROACH AND LANDING (CONT'D)

D. Precision (ILS/PAR) Approach (Cont'd)

At 1,500 feet AGL:

- | | | |
|---------------------|------------------|--|
| (11) Airspeed | $V_{REF} + WIND$ | Wind correction is half steady state crosswind plus all gust (regardless of direction). Maximum correction is + 20 KIAS. |
|---------------------|------------------|--|

At the final approach fix (FAF):

- | | | |
|---|---------|---|
| (12) Fix name, altitude and flags | Recheck | correct fix name, at correct altitude and no flags in view. |
|---|---------|---|

At decision height (DH) or decision altitude (DA) and landing is feasible:

- | | | |
|--------------------------------|------------|------------------------------|
| (13) Autopilot (if used) | Disconnect | at no less than 80 feet AGL. |
|--------------------------------|------------|------------------------------|



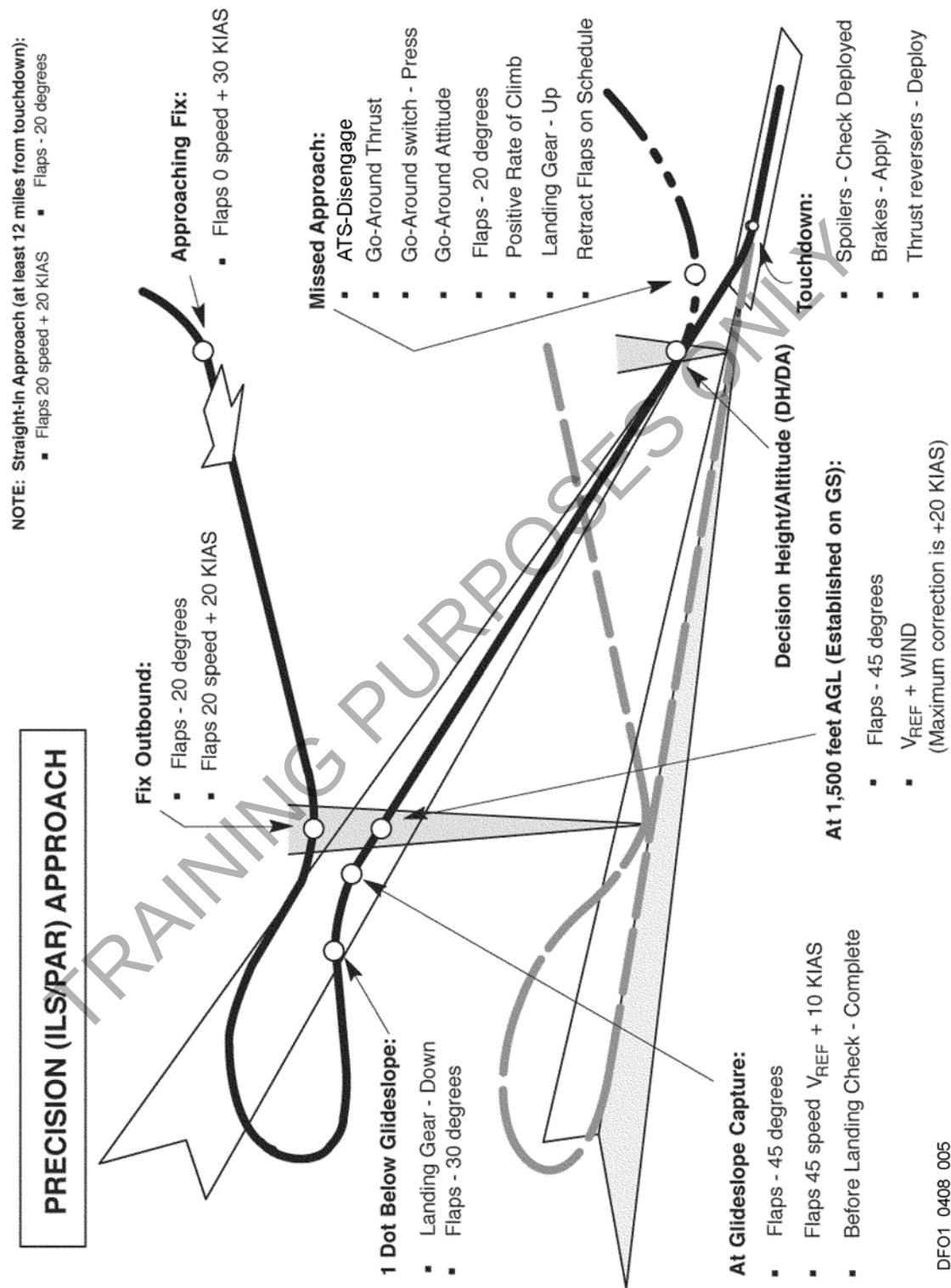
PFD negative barometric altitude is indicated by a yellow NEG flag within the barometric altitude window.

NOTE

- If the missed approach is not selected for display on the MFD, it will auto pop-up for display when the TOGA switch is depressed, but not until the missed approach point is sequenced by the FMS.
- The NAV preview needle will remain displayed after transfer to LOC. This indication should be disregarded.

1. APPROACH AND LANDING (CONT'D)

D. Precision (ILS/PAR) Approach (Cont'd)



Precision (ILS/PAR) Approach
Figure 04-08-1

1. APPROACH AND LANDING (CONT'D)

E. Non-Precision Approach

When approaching the fix (or beacon):

- (1) Airspeed Flaps 0° speed
+ 30 KIAS

When at fix outbound or when abeam the approach fix for straight-in approach (radar vectored):

- (2) Flaps 20 degrees
(3) Airspeed Flaps 20° speed
+ 20 KIAS

When tracking procedure turn altitude:

- (4) Preselect altitude Set to the final approach fix (FAF) altitude. Verify that ALTS white indication is on (armed).

When established inbound:

- (5) FCP mode FLC
(6) FLC Set as required
(7) Thrust levers Adjust to maintain the desired rate of descent.

At approximately 3 miles before FAF:

- (8) LDG GEAR lever Set to DN.
(9) Flaps 30 degrees
(10) Airspeed Flaps 30° speed
+ 10 KIAS
(11) Before landing check Accomplish

Approaching the FAF:

- (12) Flaps 45 degrees If a circling approach is planned, maintain flaps at 30°.
(13) Airspeed $V_{REF} + WIND$ Wind correction is half steady state crosswind plus all gust (regardless of direction). Maximum correction is + 20 KIAS. If a circling approach is planned, maintain flaps 30° speed + 10 KIAS.

Upon FAF altitude capture (also indicated by green ALT CAP and ALTS messages on both PFDs):

- (14) Preselect altitude Set to the minimum descent altitude (MDA).

At the FAF:

- (15) Fix name Call



1. APPROACH AND LANDING (CONT'D)**E. Non-Precision Approach (Cont'd)**

- (16) Chronometer Start
- (17) FCP mode FLC
- (18) FLC..... Set as required
- (19) Thrust levers..... Adjust to maintain the desired descent rate.
- (20) Autopilot (if used) Disconnect prior to commencing final turn.

When the MDA is captured (as indicated by yellow MIN alert flag and flashing yellow RA or BARO MIN reference on both PFDs):

- (21) Preselect altitude Set to missed approach altitude.

At the missed approach point (MAP) and runway is in sight:

- (22) Landing.....Continue visually Continue the approach on a 3° glide path.
- (23) Flight director..... Off

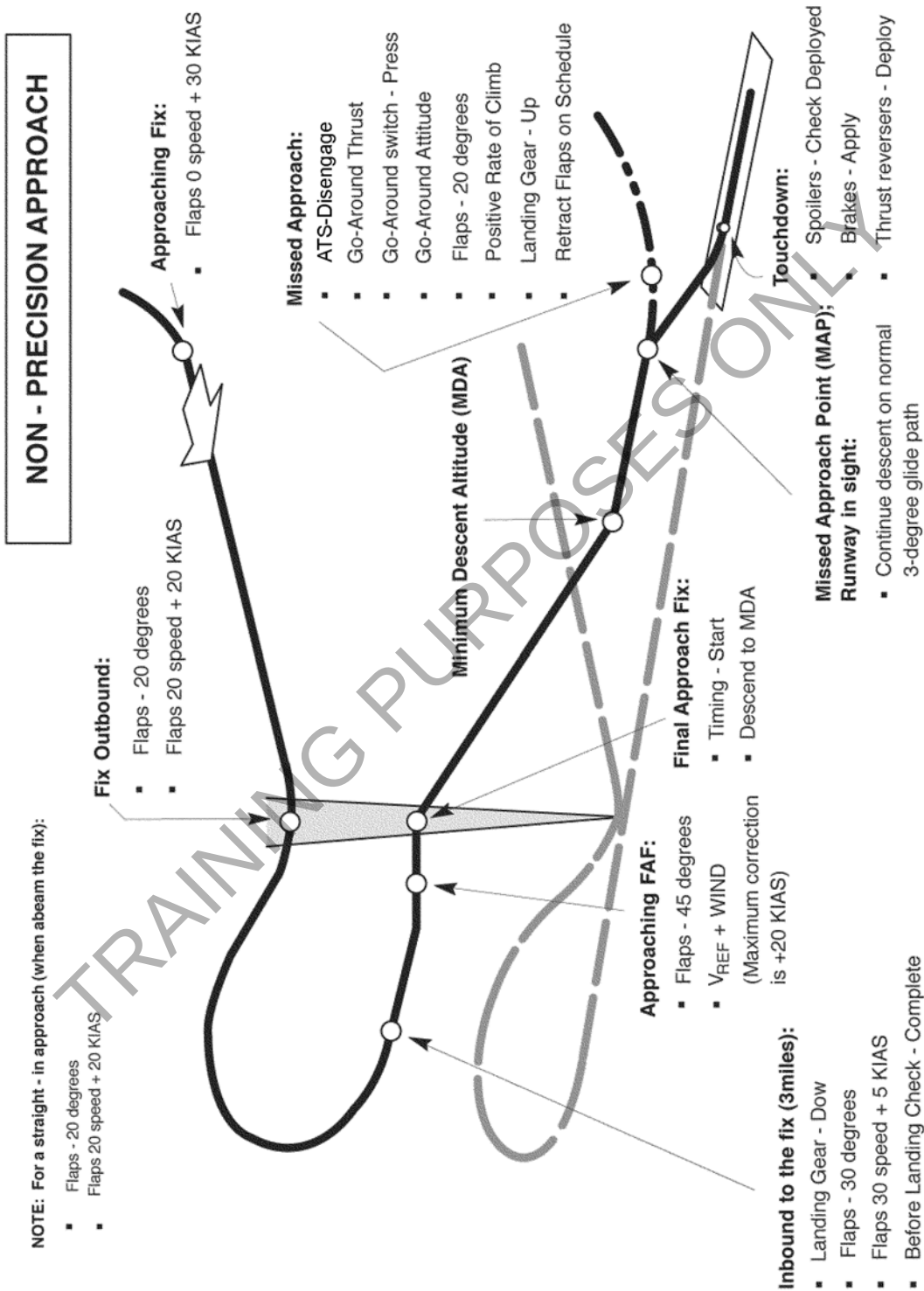


PFD negative barometric altitude is indicated by a yellow NEG flag within the barometric altitude window.



1. APPROACH AND LANDING (CONT'D)

E. Non-Precision Approach (Cont'd)



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Non-Precision Approach
Figure 04-08-2

1. APPROACH AND LANDING (CONT'D)

F. Circling Approach

When performing a circling approach, maintain the airplane configuration from the final approach fix (FAF) onwards (flaps 30° and landing gear down). At the circling MDA with the field in sight, maneuver to establish a downwind leg parallel to the runway at a distance of approximately 1 1/2 miles.

At the established downwind:

- (1) Circling MDA Maintain
- (2) Flaps 30 degrees
- (3) Airspeed Flaps 30° speed
+ 10 KIAS

When abeam the runway threshold:

- (4) Chronometer Start Time for 15 to 30 seconds,
plus or minus wind correction.

After the desired timing has elapsed, start the turn towards the base leg:

- (5) Descent Initiate

Approaching 400 feet, start turn towards final, and when landing is assured:

- (6) Flaps 45 degrees
- (7) Airspeed $V_{REF} + WIND$ Wind correction is half steady
state crosswind plus all gust
(regardless of direction).
Maximum correction is
+ 20 KIAS.

NOTE

The autopilot, if used, should be disconnected no later than 320 feet AGL.

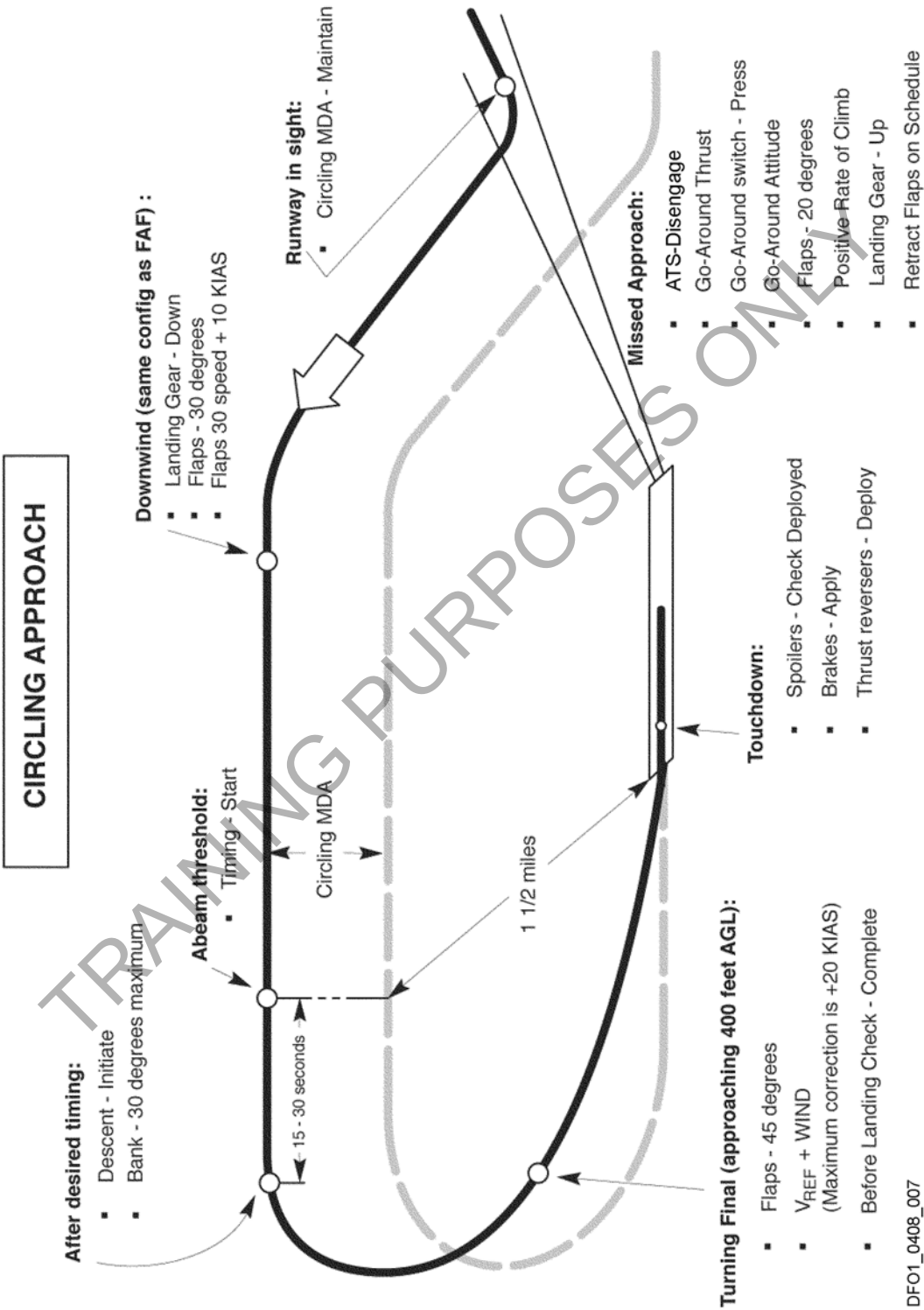


PFD negative barometric altitude is indicated by a yellow NEG flag within the barometric altitude window.



1. APPROACH AND LANDING (CONT'D)

F. Circling Approach (Cont'd)



Circling Approach
Figure 04-08-3

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1. APPROACH AND LANDING (CONT'D)

G. Standard Visual Approach

Enter the downwind leg according to the prescribed regulatory procedures.

- (1) Downwind altitude 1,500 feet
- (2) Flaps.....20 degrees
- (3) Airspeed Flaps 20° speed
+ 20 KIAS
- (4) Thrust As required to maintain the downwind
airspeed.
- (5) LDG GEAR lever Set to DN, prior to turning towards
the base leg.

Turning base:

- (6) Descent Initiate
- (7) Flaps.....30 degrees
- (8) Airspeed Flaps 30° speed
+ 10 KIAS
- (9) Rate of descent Approximately -700
fpm

At 500 to 700 feet altitude, initiate turn to final:

- (10) Flaps.....45 degrees

NOTE

The autopilot, if used, should be disconnected no later than 320 feet AGL.

- (11) Airspeed $V_{REF} + WIND$ Wind correction is half steady
state crosswind plus all gust
(regardless of direction).
Maximum correction is
+ 20 KIAS.

- (12) Before landing check..... Complete

At 50 feet AGL:

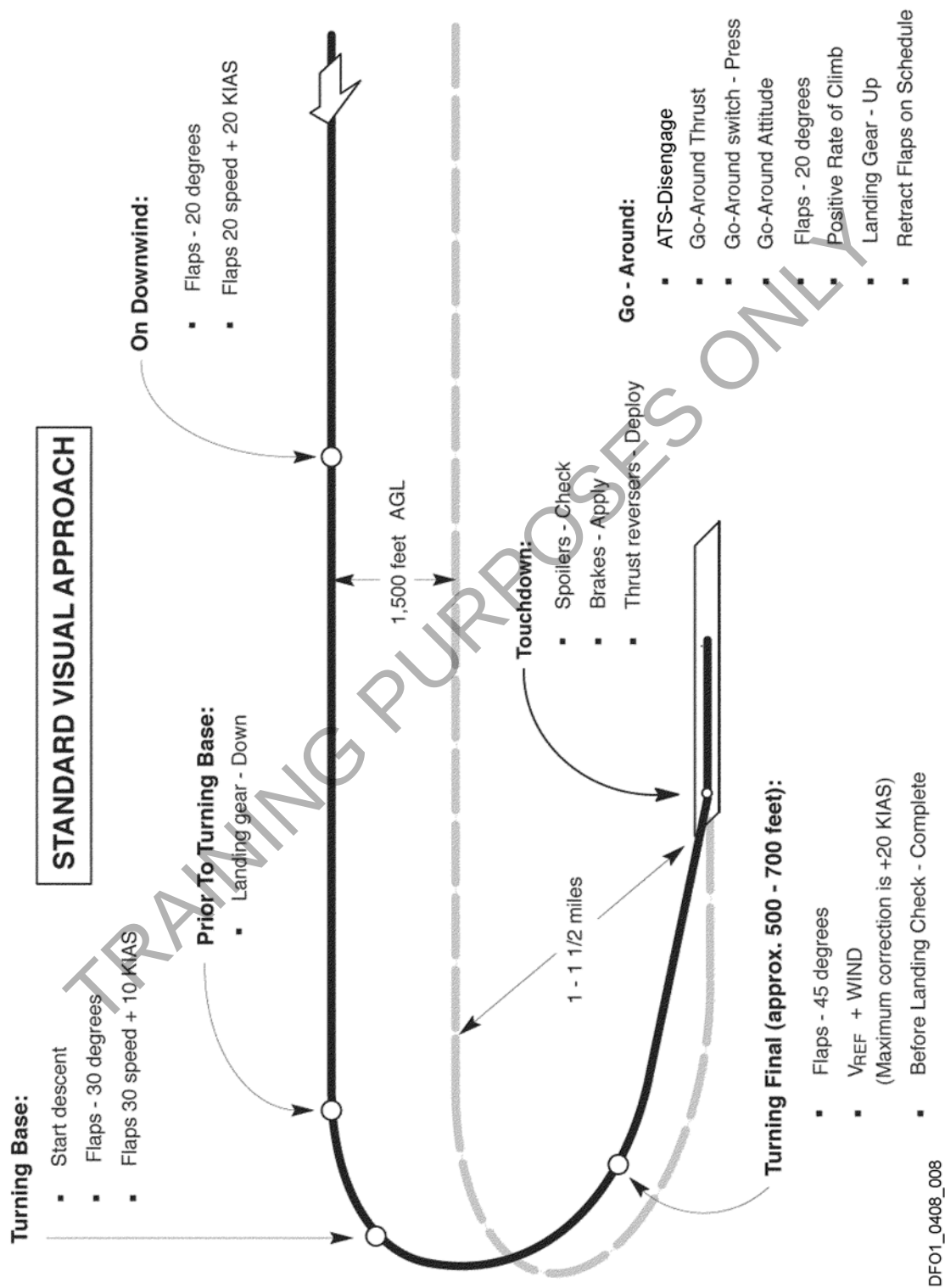
- (13) Thrust levers..... Select to IDLE.



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1. APPROACH AND LANDING (CONT'D)

G. Standard Visual Approach (Cont'd)



Standard Visual Approach
Figure 04-08-4

1. APPROACH AND LANDING (CONT'D)

H. Before Landing Check

- (1) IGNITION, CONT switch/light..... As required

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

The engine cowl and wing anti-ice systems must be on when in icing conditions, or when ice is indicated by the ice detection system:

- (2) ANTI-ICE, WING switchSelect as required.
- (3) ANTI-ICE, COWL L and R switch/lightsSelect as required.

If wing anti-ice is selected ON and WING L HEAT or R HEAT light goes out or WING A/ICE ON or WING/COWL A/ICE ON advisory message goes out:

- (4) Thrust Increase until the following come on:
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message.
 - WING L HEAT and R HEAT lights.

If WING A/ICE ON or WING/COWL A/ICE ON advisory message remains out or WING L HEAT or R HEAT light remains out:

- (5) Leave icing conditions.
- (6) Cabin Advise not less than five minutes prior to landing.
- (7) PASS SIGNS, NO SMKG switch.....Select to ON.
- (8) PASS SIGNS, SEAT BLTS switchSelect to ON.
- (9) LDG GEAR lever Set to DN. Check that three green DN indications are displayed, indicating that the landing gear is down and locked.

NOTE

To avoid nuisance **A/SKID INBD** and **A/SKID OUTBD** caution messages, wait 5 seconds after landing gear is down and locked before checking anti-skid system.

- (10) Brake and anti-skid systemCheck
- (a) ANTI SKID, ARMED switchCheck selected to armed.



1. APPROACH AND LANDING (CONT'D)

H. Before Landing Check (Cont'd)

- (b) ANTI SKID, TEST switch.....Select to TEST momentarily, and check the following:
- **A/SKID IN TEST** advisory message on.
- After approximately 3 seconds check:
- **A/SKID IN TEST** advisory message out.

NOTE

Anti-skid system is inoperative if the **A/SKID IN TEST** advisory message does not come on during test.

Anti-skid system is inoperative if **A/SKID INBD (OUTBD)** caution messages come on during test.

- (c) NOSE STEER switchCheck selected to ARMED.

NOTE

When the nose wheel steering system is armed in flight, the tiller should be centered.

- (11) THRUST REVERSER, L and R switches.....Select to ARMED.
- Check the following:
- **L** and **R REV ARMED** advisory messages on.
 - **L** and **R REV UNSAFE** caution messages out.
 - **L** and **R REV UNLOCKED** caution messages out.



Do not arm thrust reverser if **L (R) REV UNSAFE** caution message is displayed.

- (12) Flaps..... As required Check that the flap indications are displayed as selected.

At 100 feet AGL:

- (13) ATS MSDCheck **SPEED** message flashes. If message is not displayed, disengage ATS and manually control thrust.

1. APPROACH AND LANDING (CONT'D)

I. Go-Around Procedure

The following procedures are recommended in the event of a missed approach or any other situation which would necessitate making a go-around maneuver, with the airplane in the landing configuration. It is assumed that the flight instruments, radios and navigation aids have been previously set-up for the missed approach.

Use of the ATS during go-around is prohibited. Pilot must manually set thrust to override ATS. ATS may be used to trim final N_1 , once manually set.

An all engine go-around maneuver after touchdown during a normal landing is entirely the prerogative of the pilot to employ if conditions are not conducive for a full-stop landing.



A go-around maneuver should NOT be attempted after the thrust reversers have been deployed.

If unreliable airspeed is suspected, disregard all reference to TOGA and airspeed. Do not use TOGA switch.

NOTE

The minimum fuel quantity for go-around is 230 kg (500 lb) per wing tank (with the airplane level), with a maximum airplane climb attitude of 10° nose up.

From a gear-down, flaps 45 degrees approach; At the MAP, DH, DA or during circling, and a decision to go-around was made:

- (1) Thrust levers.....Advance to the pre-determined go-around N_1 setting, while simultaneously pressing the Take-Off/Go-Around (TOGA) switch and ATS DISC switch.



ATS must be disengaged as thrust levers are advanced to go-around. If ATS is not disengaged and the pilot overrides the ATS as the thrust levers are advanced, the ATS may advance the thrust levers to the forward stop, causing an engine exceedance.

- (2) FLIGHT SPOILER lever (if extended)Select to RETRACT.
- (3) AirplaneRotate smoothly, at a speed of not less than V_{REF} .
- (4) Pitch attitude.....Adjust to achieve a speed of not less than $V_2 + 10$ KIAS as the flaps are retracted to 20°.
- (5) FLAPS.....Select to 20°.



1. APPROACH AND LANDING (CONT'D)

I. Go-Around Procedure (Cont'd)

When a positive rate of climb is achieved:

- (6) LDG GEAR lever Select to UP.

NOTE

When a turn is required in the missed approach procedure, wait for the landing gear to retract fully before commencing the turn.

At a safe altitude (not below 400 feet AGL):

- (7) FLAPS Select to 0°.

- (8) THRUST REVERSER,
L and R switches Select to OFF.

Check **L** and **R REV ARMED**
advisory messages out.

- (9) Normal climb-out procedures Accomplish

J. Touch-and-Go Landing Procedure

Touch-and-go landing maneuvers are only intended for use during flight training, primarily for approach and landing practice, and should not be utilized as part of the normal take-off and landing procedures.

NOTE

- For touch-and-go landings, GROUND SPOILERS switch may be left at AUTO.
- The landing gear may be left extended throughout the whole exercise, if required for brake cooling.

When stabilized at the downwind leg:

- (1) V speeds Recompute and set the speed bugs accordingly.

1. APPROACH AND LANDING (CONT'D)

K. Full Stop Landing Procedure

The procedures outlined below are done simultaneously or in quick succession, as the situation requires.

Approach through 50 feet height point at V_{REF} (Refer to the Airplane Flight Manual; Chapter 6; PERFORMANCE – LANDING PERFORMANCE) on stabilized glide slope of 3 degrees, with landing gear down and flaps at 45 degrees.

Thrust reversers may be used after touchdown to supplement the use of wheel brakes. At airports where runway structural repair or debris is known to exist, use thrust reversers with extreme caution to preclude the possibility of foreign object damage (FOD) from occurring.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- (1) Thrust levers.....Select to IDLE at or below 50 feet AGL.

NOTE

If icing accumulation on the wheels and brakes is suspected, carry out a positive landing to ensure initial wheel spin-up and breakout of frozen brakes.

- (2) FLIGHT SPOILER leverSelect to MAX.
- (3) BrakesApply as appropriate for landing and runway conditions.
- (4) Thrust reverse levers.....Lift and pull back.

When interlocks release (REV icons inside N_1 gauges are green):

- (5) Thrust reverse levers.....Pull back to required reverse thrust setting.
- (6) Rudder and aileronsUse as required, to maintain directional control.
- (7) Engine instruments and airspeed.....Monitor Monitor any engine limitations about to be reached or any discrepancy.



1. APPROACH AND LANDING (CONT'D)

K. Full Stop Landing Procedure (Cont'd)

When speed falls to 60 KIAS:

- (8) Thrust reverse levers.....Select to not more than 60% N_1 reverse thrust or STOW.

NOTE

1. Reverse thrust idle is 25 to 30% N_1 .
 2. Thrust reverse (piggy-back) levers must be pushed fully down in order to do the following:
 - Achieve thrust reverser stow, and
 - Ensure that the main thrust lever mechanical lock is disengaged.
 3. When reducing reverse thrust, carefully move thrust levers to prevent an inadvertent stowing of thrust reversers.
- (9) Nose wheel steering As required

L. Bounced Landing

If the pilot believes that thrust must be added and maintained until touchdown to salvage a landing, then a rejected landing should be executed.

Should the aircraft bounce on landing, a rejected landing should be executed. Go-around thrust should be set and the normal landing attitude or slightly higher should be maintained. Aircraft configuration should not be changed at this time. Once the aircraft is accelerating above V_{REF} and climbing through a safe height, the go-around maneuver should be continued.

Improper landing technique (thrust levers not at IDLE) may result in a shallow bounce. Should the pilot decide not to execute a rejected landing, then the normal landing attitude should be maintained and the thrust levers reduced to IDLE. Be aware that following the bounce, the ground spoilers may deploy as soon as the thrust levers are set to IDLE, even if the aircraft is still in the air.

A poorly executed approach and touchdown with a high rate of descent can generate a high, hard bounce that can quickly develop into a hard landing accident. A rejected landing should always be executed following such a bounce.

1. APPROACH AND LANDING (CONT'D)

M. After Landing Check

The After Landing checklist will be accomplished when the airplane is clear of the runway, at a safe taxi speed, with the engines stabilized at forward thrust.

- (1) Flaps..... As required

NOTE

It may be necessary to leave the flaps at 20° if the approach was made in icing conditions, or if the runway is covered with slush or snow. Otherwise, retract the flaps.

- (2) FLIGHT SPOILER leverSelect to RETRACT.
(3) Transponder/TCAS Set to STBY, if required.

NOTE

Some airports require the transponder to remain active for surface movement management. The transponder/TCAS will automatically switch to STBY, 65 seconds after landing. If required, set the transponder/TCAS to other than STBY, until shutdown.

- (4) ANTI-ICE, PROBES switchesSelect to OFF.
(5) ANTI-ICE, WING switchSelect as required.

NOTE

Wing and cowl anti-icing may be required during cold weather operations. Otherwise, select to OFF.

- (6) ANTI-ICE, COWL L and R switch/lightsSelect as required.
(7) ANTI-ICE, WSHLD/WIND L and R switches.....Select as required.
(8) IGNITION, CONT switch/light.....Press out to turn off.
(9) EXTERNAL LTS switches As required Leave NAV light ON.
(10) LANDING LTS switchesSelect to OFF.
(11) Weather radar selectors Set to STANDBY.
(12) Trims Reset

1. APPROACH AND LANDING (CONT'D)

N. Hard Landing

An overweight landing is defined as a landing at more than the maximum landing weight. Overweight landings should be avoided. However, no inspection is required provided that neither a hard landing nor a hard de-rotation has occurred.

Following a hard landing or hard de-rotation at any weight, it is possible that damage may have occurred to the aircraft structure and systems. This damage may be visible or hidden. In this instance, it is strongly recommended that the aircraft be inspected for damage, prior to next flight, in accordance with the Hard Landing or Overweight Landing checks contained in Chapter 5 of the Aircraft Maintenance Manual.

A hard landing is described as:

- A landing at a vertical descent rate greater than 600 feet per minute or 10 feet per second when the aircraft gross weight is less than or equal to the maximum landing weight.
- A landing at a vertical descent rate greater than 360 feet per minute or 6 feet per second when the aircraft gross weight is greater than the maximum landing weight, but less than or equal to the maximum take-off weight.
- A landing where the aircraft lands with an uncontrolled sideways skid or continues to move from the prepared surface to a runway surface that is not prepared.
- A landing where the aircraft touches the ground with one or more landing gear on a surface other than the prepared surface.
- A landing where the aircraft nose landing gear touches the ground before both main landing gear have touched the ground.
- A landing where the aircraft bounces or skips after initial impact and the ground spoilers subsequently deploy while the aircraft is still in the air.
- A touchdown that results in a burst tire.

Factors contributing to a hard de-rotation are:

- Application of brakes before the nose landing gear touches the ground resulting in an unchecked, high de-rotation rate.
- Full or nearly full nose-down elevator applied before the nose landing gear touches the ground.
- Forward column pressure applied following main landing gear touchdown resulting in an unchecked, high de-rotation rate.

1. SHUTDOWN

A. Shutdown Check

The following procedures will be accomplished when the airplane has come to a complete stop.

- (1) PARKING BRAKE As required Ensure that chocks are in place before releasing parking brake.

NOTE

Inform ground crew of 'Hot Brakes' condition as soon as possible.

- (2) Thrust reverse levers.....Stow Check the following:
- REV icon goes out.
 - **REV UNLOCKED** caution message out.
- (3) ANTI-ICE, WING switchSelect to OFF.
- (4) ANTI-ICE, COWL
L and R switch/lightsPress out to select cowl anti-ice off.
- (5) ANTI-ICE, WSHLD/WIND
L and R switchesSelect to OFF.
- (6) APU Start Ensure that operating limitations/envelope is considered.

NOTE

The APU, PWR FUEL switch/light must be selected off between each APU start attempt. The APU, PWR FUEL switch/light should be firmly pressed once only. If pressed twice, an immediate shutdown may occur.

- (a) BLEED AIR, APU LCV
switch/lightCheck pressed out.
- (b) APU, PWR FUEL switch/lightPress in and check the following:
- PWR FUEL light on, and
 - APU readouts displayed on EICAS page.
- (c) APU, START/STOP switch/lightPress in and check the following:
- START light on.
 - 60% rpm – START light out.
 - 95% rpm + 4 seconds (approx) – AVAIL light on.
 - 100% rpm + 15 seconds (approx) – **APU GEN OFF** caution message on.



1. SHUTDOWN (CONT'D)

A. Shutdown Check (Cont'd)

After APU RPM and EGT have stabilized:

- (d) AC POWER, APU GEN switch.....Select to ON.
- (7) AC POWER, GEN 1 and 2 switches.....Select to OFF/RESET.

After the engine two minute cool-down period:

NOTE

The engine may be shutdown with the following indications present, however maintenance action is required after shutdown to stow the reverser prior to a subsequent start:

- REV icon in the N₁ gauge
- **L** or **R REV UNLOCKED** caution message displayed.

- (8) Engine that was started first Shutdown

NOTE

Prior to shutdown, operate the engine at or near IDLE for a minimum of two minutes, to dissipate heat and stabilize internal operating temperatures.

Taxi time at 70% N₂ or below may be credited in the two minute cool-down period.

If ITT rises above 350°C following engine shutdown, carry out an engine motoring cycle immediately to reduce ITT.

- (9) FUEL, L and R BOOST PUMP switch/lights Press out to turn both boost pumps off.

Check the following:

- L and R BOOST PUMP ON lights out.
- L and R BOOST PUMP INOP lights on.
- **L** and **R FUEL PUMP ON** advisory messages out.
- **FUEL LO PRESS** caution message is annunciated for the engine that now has been shutdown.



1. SHUTDOWN (CONT'D)

A. Shutdown Check (Cont'd)

- (10) Fuel system check valve operation Complete 1 the test initiated before engine start.

After 2 minutes, if FUEL LO PRESS caution message is still not displayed for the non-operating engine:

- (a) Dry motor Accomplish for maximum of 30 seconds on the non-operating engine.

Applicable **FUEL LO PRESS** caution message should be displayed during motoring as the residual fuel pressure is reduced.



Absence of the **FUEL LO PRESS** caution message during this check is an indication that the fuel feed cross flow check valve may have failed in the open position or that another component of the engine fuel feed sub-system may not be functioning properly. The fuel system needs further troubleshooting.

- (11) Other engine Shutdown

- (12) Fuel crossflow valve operation Verify 2

- (a) FUEL, L TO AUX XFLOW switch/light Press in to open left crossflow valve.
Check L TO AUX XFLOW ON light on.
- (b) FUEL, L TO AUX XFLOW switch/light Press out to close left crossflow valve.
Check L TO AUX XFLOW ON light out.
- (c) FUEL, R TO AUX XFLOW switch/light Press in to open right crossflow valve.
Check R TO AUX XFLOW ON light on.
- (d) FUEL, R TO AUX XFLOW switch/light Press out to close right crossflow valve.
Check R TO AUX XFLOW ON light out.



1. SHUTDOWN (CONT'D)

A. Shutdown Check (Cont'd)

- (13) PASS SIGNS, NO SMKG and SEAT BLTS switchesSelect to OFF.
- (14) BEACON switch (if installed)Select to OFF.
- (15) NOSE STEER switchSelect to OFF.

On airplanes incorporating Service Bulletin 605-23-008:

- (16) ATNLog OFF

- (17) Transponder/TCAS Set to STBY.

TRAINING PURPOSES ONLY

1. TERMINATING

A. Securing Check

The following procedures will be performed when the airplane is to be left unattended or handed over to maintenance personnel, after all the passengers have deplaned.

- (1) PARKING BRAKE As required Ensure that chocks are in place before releasing parking brake.
- (2) THRUST REVERSER, L and R switches Select to OFF.
- (3) EXTERNAL LTS switches Select all to OFF.
- (4) EMER LTS switch Select to OFF.
- (5) Hydraulic pump switches Select all to OFF.
- (6) AC POWER, APU GEN switch Select to OFF/RESET.
- (7) APU Shutdown
 - (a) APU, START/STOP switch/light Press out to shutdown the APU.
Check AVAIL light out.

When rpm decreases to below 50% and EGT to below 350°C (662°F):

 - (b) APU, PWR FUEL switch/light Press out Check the following:
 - **APU GEN OFF** caution message out.
 - APU readouts disappear from EICAS page after approximately 60 seconds.
- (8) AIR-CONDITIONING, L and R PACK switch/lights Press out to turn off packs.
Check L and R PACK OFF lights on.
- (9) BLEED AIR, APU LCV switch/light Press out to close load control valve.
Check APU LCV OPEN light out.
- (10) BLEED AIR, 10TH STAGE ISOL switch/light Press out to match isolation valve position.
Check ISOL OPEN light out.
- (11) BATT MASTER switch Select to OFF.
- (12) MISC LTS switches Select to OFF.

NOTE

When leaving the airplane with no maintenance personnel in attendance, the doors must be closed.

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TRAINING PURPOSES ONLY

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1. GENERAL

The abnormal operating procedures included in this chapter contain items peculiar to the CL 600, model 2B16 Challenger airplane (Serial No. 5701 to 6049). These checks address foreseeable situations, usually involving a failure condition, in which the use of the normal or alternate systems can be expected to maintain an acceptable level of airworthiness.

In addition to the failure indications referenced in the following abnormal procedures, the master caution system operates where applicable.

Unless otherwise specified in these abnormal procedures, the landing configuration of the airplane will be landing gear down and flaps 45.

These procedures have been developed and recommended by Bombardier and approved by Transport Canada for use in the operation of the Challenger airplane. These procedures are provided as guidance and should not be construed as prohibiting the development of equivalent Transport Canada-approved procedures.

2. LANDING DISTANCE FACTORS

Landing distance factors are provided herein with and without the use of thrust reversers. No operation should be predicated on the use of thrust reversers.

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TRAINING PURPOSES ONLY

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1. POWER PLANT**A. Reverser Unsafe to Arm**

Indication: **L (R) REV UNSAFE** caution message.



Do not arm affected thrust reverser during flight.

(1) Affected THRUST REVERSER

ARM switchOFF **L (R) REV ARMED** advisory message out.



Use remaining thrust reverser carefully upon landing.

— END —

1. POWER PLANT (CONT'D)

B. In-Flight Engine Failure/Shutdown

Indication: **L (R) ENGINE** caution message.

Accomplish an engine shutdown only when flight conditions permit.

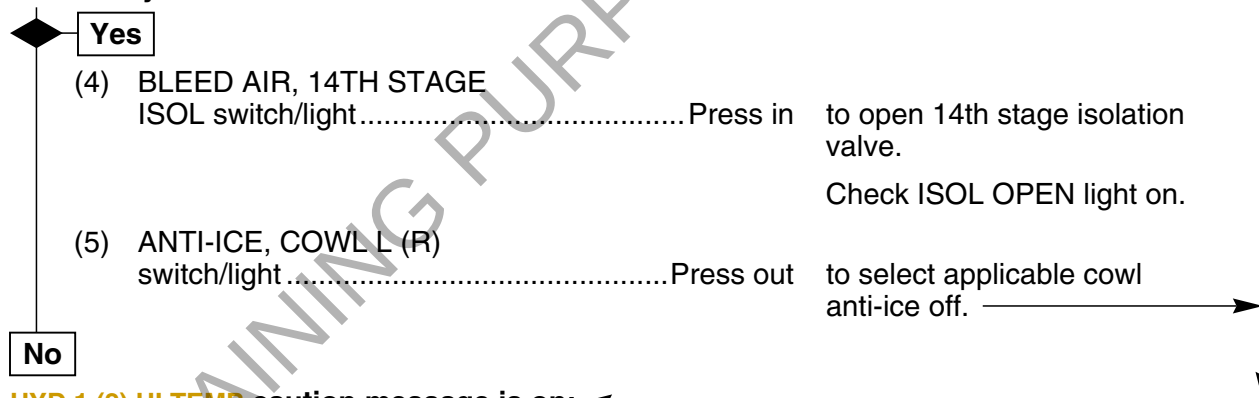
Shut down the engine if any of the following malfunction symptoms/indications occur:

- Excessive or uncontrollable power changes, or
- Indications outside the engine operating limitations, or
- Unsuccessful restart attempts after an unexplained flameout, or
- Any other condition that indicates advisability of engine shutdown.

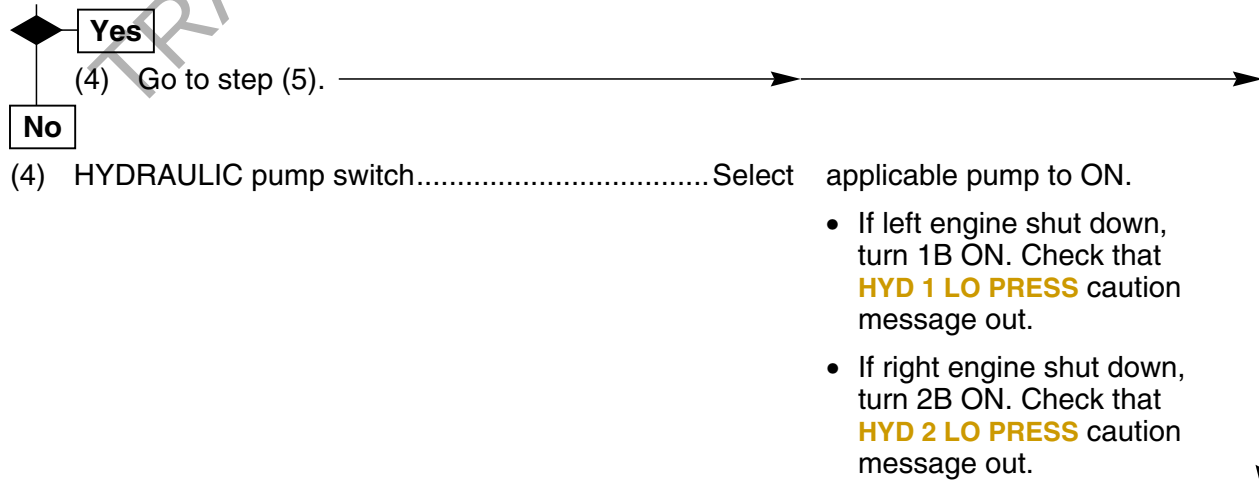
Affected engine:

- (1) Thrust lever Confirm and IDLE
- (2) Thrust lever Confirm and SHUT OFF
- (3) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press out if MANUAL light on, and if rotor burst is suspected.

Anti-ice system is on:



HYD 1 (2) HI TEMP caution message is on:



1. POWER PLANT (CONT'D)

B. In-Flight Engine Failure/Shutdown (Cont'd)

- (5) AIR-CONDITIONING, L (R) PACK switch/light Press out to shutdown affected pack.
Check L (R) PACK OFF light on.
- (6) BLEED AIR, 10TH STAGE L (R) switch/light Press out to close affected shut-off valve.
Check L (R) CLOSED light on.

NOTE

Use only one air-conditioning pack during single engine operations, when the operating engine is the only 10th-stage bleed source.

- (7) AC POWER, GEN 1 (2) switch Select affected generator to OFF.
Check **GEN 1 (2) OFF** caution message on.

Below 20,000 feet and APU is available:

- Yes
- (8) APU Start
- (9) AC POWER, APU GEN switch Select to ON.
- (10) Air-conditioning Transfer to APU bleed air.

NOTE

Bleed air extraction from the APU is not permitted above 15,000 feet.

- (11) Affected AIR-CONDITIONING, L (R) PACK switch/light Press in to restart pack.
Check L (R) PACK OFF light out.
- (12) EFIS altimeters (2) Cross-check

No

- (8) Fuel system Check and monitor the following:
- Fuel quantity, and
 - Fuel balance.

1. POWER PLANT (CONT'D)

B. In-Flight Engine Failure/Shutdown (Cont'd)

- (9) FUEL, L (R) BOOST PUMP switch/light.....Press out to disarm affected boost pump.

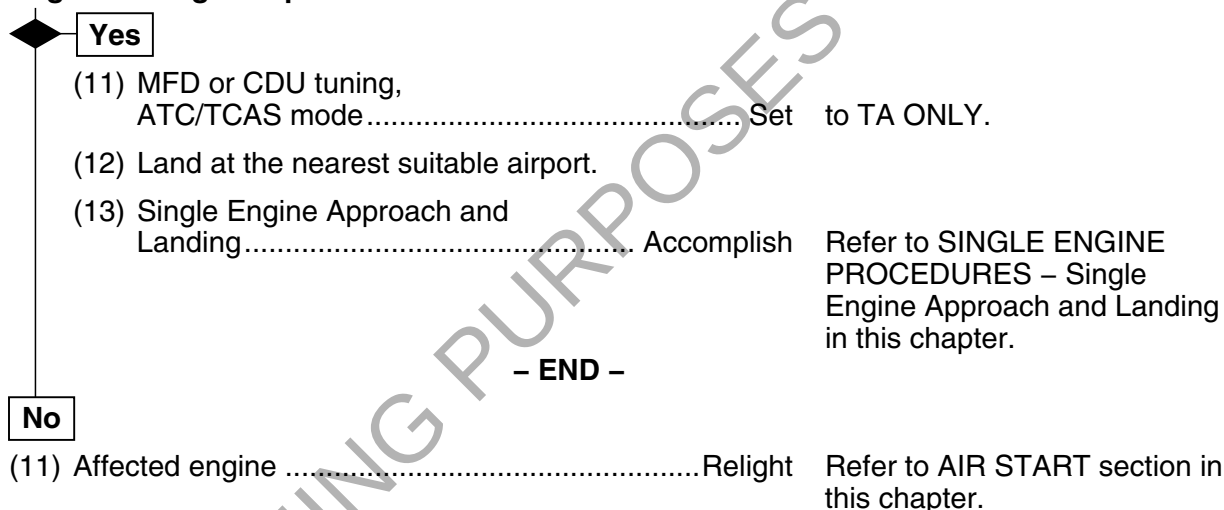
Check **L (R) FUEL PUMP** caution message on.

- (10) Leave icing conditions (if possible).

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

Engine damage suspected or intentional shutdown:



NOTE

- Do not attempt to relight an engine that is suspected to be not serviceable.
- Relight engine using starter-assisted cross bleed whenever possible.

END

1. POWER PLANT (CONT'D)

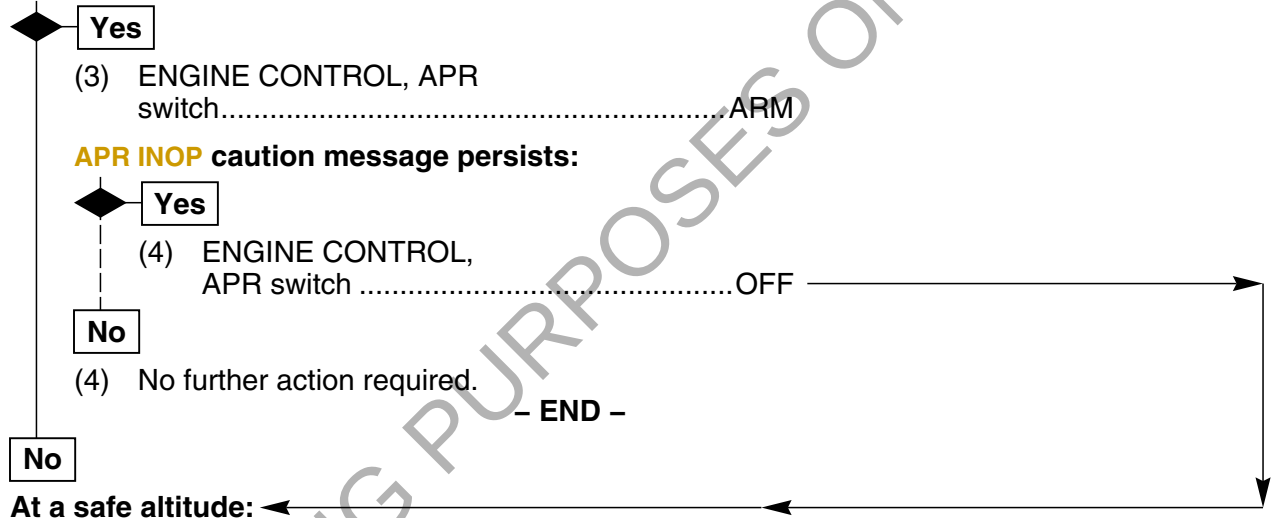
C. APR Failure

Indication: **APR INOP** caution message on.**NOTE**

After take-off, an **APR INOP** caution message may be displayed.
If the message goes out within 2 minutes, it can be disregarded
and be considered a nuisance message.

- (1) ENGINE CONTROL, APR switch.....OFF
- (2) ENGINE CONTROL, both ENG
SPEED switchesCheck ON and monitor engine
indications.

Engine indications are normal:



At a safe altitude:

- (3) ATTS DISC switch.....Depress
- (4) Both thrust leversRetard to below 79% N_2 .
- (5) ENGINE CONTROL, both ENG
SPEED switchesOFF to prevent asymmetric thrust.

NOTE

Use APR-off performance. Refer to the Airplane Flight Manual,
Chapter 6; PERFORMANCE.

- (6) Thrust levers.....Advance as required.

NOTE

Thrust lever controls are more sensitive to changes.

Due to the possibility of N_1 overspeed, monitor the N_1 if more
than idle reverse thrust is used after landing. Do not exceed
90% N_1 reverse thrust.

END

1. POWER PLANT (CONT'D)

D. Inadvertent APR Command

Indication: **APR CMD SET** caution message. (**APR CMD SET** caution message may be accompanied by APR operation on one or both engines.)

On the ground:

◆ **Yes**

(1) Do not take off.

— END —

□ **No**

In flight:

- (1) Thrust levers.....Adjust to within normal thrust range.
- (2) ENGINE CONTROL, APR switch.....OFF
- (3) Engine performanceMonitor during approach.

————— **END** —————

TRAINING PURPOSES ONLY

1. POWER PLANT (CONT'D)

E. Uncommanded Deceleration or Abnormal Engine Oscillation/Fluctuation

Indication: Engine thrust does not agree with thrust lever selection.

- (1) ATS DISC switch Depress



The ATS SPEED mode N_1 synchronization will attempt inappropriate deceleration/acceleration of the remaining engine, to maintain synchronization until disengaged.

NOTE

To prevent engine overspeed when selecting ENG SPEED switches OFF, N_2 indications should be referenced instead of N_1 .

Affected engine:

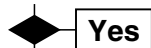
- (2) Thrust lever Adjust to below 79% N_2 or IDLE (whichever occurs first).
- (3) ENGINE CONTROL, ENG SPEED switch OFF
- (4) Thrust lever Advance



Do not change power settings abruptly as engine response is degraded.

- (5) Engine indications Monitor

Engine responds:



NOTE

- To minimize asymmetric thrust, both ENG SPEED switches should be selected OFF prior to approach and landing.
- Exercise caution to prevent overspeed during thrust reverse use, when operating with ENG SPEED switches OFF.
- The APR system is inoperative when the ENG SPEED switches are selected OFF.

— END —



- (6) In-Flight Engine Failure/Shutdown Accomplish Refer to In-Flight Engine Failure/Shutdown procedure in this section.

————— END —————

1. POWER PLANT (CONT'D)

F. N₁ Fan Vibration

Indication: FAN VIB readout greater than 2.7 Mils or
VIB icon on N₁ gauge in compressed display format.

CAUTION

It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality, i.e., high oil temperature, high oil pressure, abnormal increase in ITT, ITT exceedance, or abnormal engine vibration is felt through the airframe.

NOTE

1. If engine was started within 2 hours of shutdown, allow 10 minutes for vibration levels to return to normal.
2. An ITT increase of more than 10°C from stable, matched N₁ power is an abnormal increase.

(1) ATS DISC switch Depress

Icing conditions:**Yes****Affected engine:**

(2) Thrust lever Adjust N₂ to above 80%, and N₁ to above 70%.

Do not exceed maximum continuous thrust setting.

Maintain thrust settings until vibration is reduced.

NOTE

1. Vibration may increase during the ice shedding process.
2. Vibrations may momentarily indicate full scale during the ice shedding process.
3. Thrust settings up to and including maximum continuous may be required to shed ice.

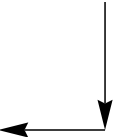
(3) Engine indications Monitor ITT.

No**Affected engine:**

(2) Thrust lever Adjust to a setting that maintains vibration level within normal range.

(3) Engine Monitor

1. POWER PLANT (CONT'D)

F. N₁ Fan Vibration (Cont'd)Vibration persists and the condition indicates advisability of an engine shutdown:  **Yes**

(4) Affected engine.....Shutdown

Refer to In-Flight Engine
Failure/Shutdown procedure in
this section.

- END -

No

(4) No further action required.

————— END —————

TRAINING PURPOSES ONLY

1. POWER PLANT (CONT'D)

G. N₂ Core Vibration

Indication: VIB icon on N₂ gauge or adjacent to N₂ readout.



It is not recommended that an engine be shut down unless there is another indication of a severe engine abnormality, i.e., high oil temperature, high oil pressure, high ITT, or abnormal engine vibration is felt through the airframe.

NOTE

N₂ VIB icon indicates core vibration is greater than 1.7 Mils.

(1) ATS DISC switch Depress

Affected engine:

(2) Thrust lever Adjust to a setting that maintains vibration level within normal range.

NOTE

If engine was started within 2 hours of shutdown, allow 10 minutes for vibration levels to return to normal.

(3) Engine indications Monitor

Vibration can be controlled or reduced:

Yes

(4) No further action required.

— END —

No

(4) Affected engine Shutdown Refer to In-Flight Engine Failure/Shutdown procedure in this section.

————— END —————

1. POWER PLANT (CONT'D)

H. Engine Hot Start

Indication: HOT icon displayed on ITT gauge or adjacent to ITT readout and/or readout indicates 900°C or greater.

Affected engine:

- (1) Thrust lever SHUT OFF
- (2) IGNITION switch/lights (all) Press out to turn off all ignition. ON lights go out. **IGNITION A** and **B** advisory messages and CONT IGNITION light out.

If one engine is operating:

- (3) Inoperative engine FUEL, L (R) BOOST PUMP switch/light Press out to disarm boost pump. Check **L (R) FUEL PUMP** caution message on.
- (4) Dry motor Accomplish until ITT is reduced to below 120°C or starter limit, whichever comes first.

————— **END** —————

I. Engine DCU Parameter Disagreement

Indication: **L (R) ENG MISCOMP** caution message.
Affected engine's (either left or right engine) DCU parameters disagree. (N₁ and N₂ by 5%, and ITT by 40°C).

- (1) Multifunction Displays (both) Select to EICAS page.
- (2) Affected engine's parameters (either left or right) Cross-check to find disagreement.
- (3) Affected engine Maintain within limits, using reliable EICAS page.

————— **END** —————

1. POWER PLANT (CONT'D)

J. Thrust Lever Inoperative

Indication: Affected thrust lever inoperative.

At a safe altitude, before commencing the approach:

Affected engine:

- (1) ENG FIRE PUSH switch/light Press in to shutdown affected engine.
 BOTTLE ARMED PUSH TO DISCH lights and **L (R) ENG SOV CLSD** advisory message on.

NOTE

Ground spoilers may not be available upon landing.

- (2) Actual landing distance Increase by a factor of 1.60 (60%), for a flaps 20° landing without the use of reverse thrust.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- (3) Affected engine Shutdown Refer to In-Flight Engine Failure/Shutdown procedure in this section.

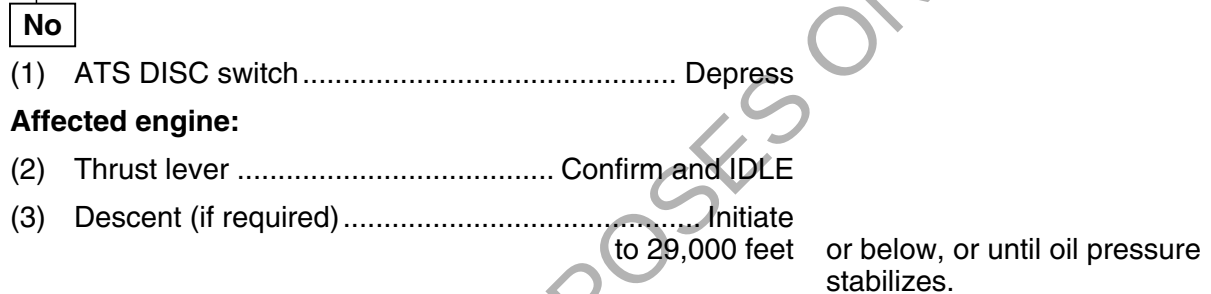
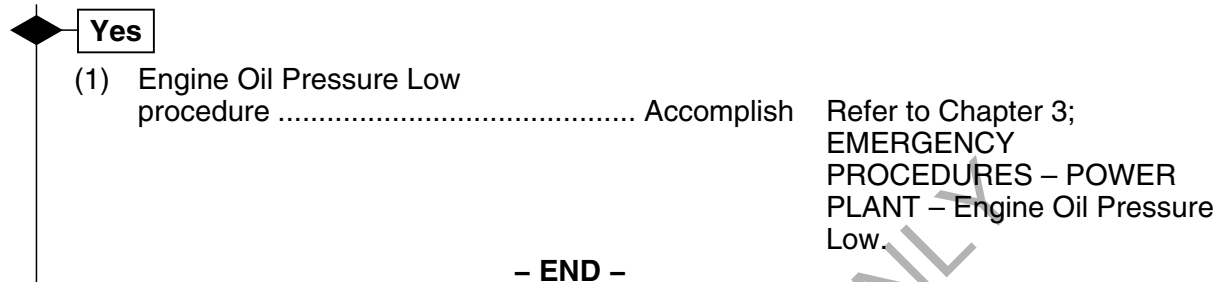
— END —

1. POWER PLANT (CONT'D)

K. Fluctuating Engine Oil Pressure

Indication: Left or right oil pressure readout fluctuates in excess of ± 5 psi and gradually decreases, possibly with increasing oil temperature.

Oil pressure is below 25 psi:



When affected engine oil pressure stabilizes:

- (4) Thrust lever Adjust as required.

NOTE

1. If necessary, keep descending to a lower altitude to maintain a stable oil pressure, as cruise power setting is resumed.
2. If oil pressure fluctuations in excess of ± 5 psi resume, select affected thrust lever to IDLE, and descend to a lower altitude.

————— **END** —————

1. POWER PLANT (CONT'D)

L. ATS Failure

Indication: **FAIL** message displayed in ATS MSD.

Flashing **FAIL** message:

**NOTE**

The ATS may be re-engaged by depressing the ATS push button switch on the ATS Control Panel. If the fault is still present, the ATS flashing **FAIL** message will reappear, and ATS will not re-engage.

— END —

Decision diamond: **No**

Steady **FAIL** message:

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

NOTE

The steady **FAIL** message is displayed if the MSD is not receiving data from the ATS computer. Depress the ATS DISC switch to confirm disengagement of the ATS. Do not re-engage the ATS.

— END —

M. Dual ATS MSD Failure

Indication: Loss of displayed data in both ATS Mode Status Displays.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

NOTE

In the event of a dual MSD failure causing the displays to go blank, the crew must depress an ATS DISC switch to confirm ATS disengagement, and consider the system inoperative.

— END —

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1. POWER PLANT (CONT'D)

N. ATS Uncommanded Disengage

Indication: Uncommanded flashing **DISENG'D** message in ATS MSD, and ATS disengages.

- (1) ATS DISC switch Depress to cancel message.
- (2) Thrust levers Position manually as required.

NOTE

The ATS will automatically disengage and illuminate the flashing **DISENG'D** message in the MSD, if an N_1 split of 13% or greater is detected, or an N_1 greater than 98.4% is detected, or if a Thrust Reverser unlocks. After fault correction, the ATS can be re-engaged by depressing the ATS push button on the ATS control panel.

The ATS will disengage if the N_1 value is removed from the EICAS (no active selection present). This could happen when the bleeds are changed from ENGINES to APU during the IN RANGE check.

————— **END** —————

O. ATS Overspeed

Indication: Flashing **VMO LIMIT** or **MMO LIMIT** message displayed in ATS MSD.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

————— **END** —————

P. ATS Thrust Limited

Indication: Flashing **SPEED** message displayed in ATS MSD.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually and re-adjust attitude as required.

————— **END** —————

Q. ATS Low Speed

Indication: Flashing **AOA LIMIT** message displayed in ATS MSD.

- (1) Airspeed Set AFCS airspeed reference (speed bug) higher than AOA limit.

————— **END** —————

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1. POWER PLANT (CONT'D)

R. ATS Radio Altimeter Input Failure

Indication: Lack of RETARD mode at 50 ft. or
 Early RETARD mode with **RETARD** or flashing **SPEED**
 message displayed in ATS MSD.

- (1) ATS DISC switch Depress
- (2) Thrust levers Position manually as required.

————— **END** —————

TRAINING PURPOSES ONLY

1. SINGLE ENGINE PROCEDURES

A. Engine Failure During Take-Off After Achieving V_1

- | | |
|--|--|
| (1) Take-off Continue
(2) Airplane Rotate | at V_R to an initial take-off attitude of 10 degrees, and adjust pitch, as required, to achieve an airspeed of V_2 at 35 feet. |
|--|--|

When positive rate of climb is achieved:

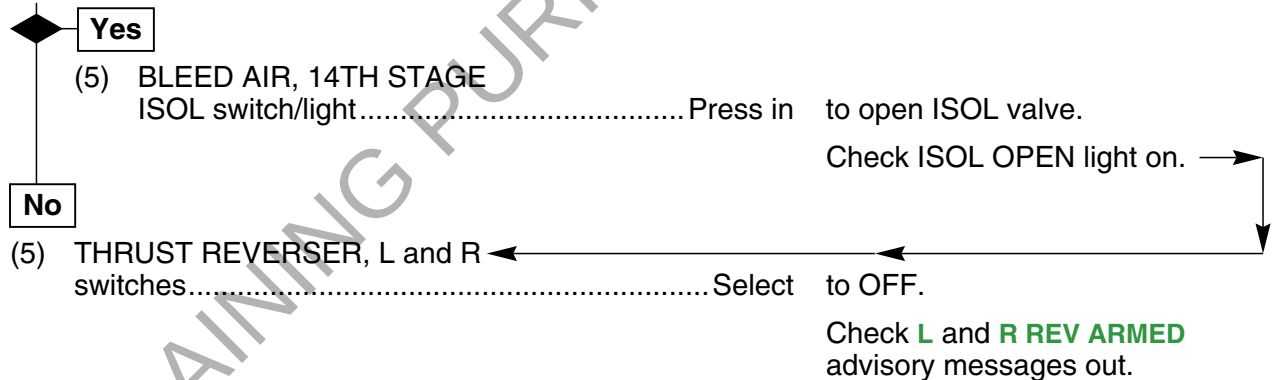
- | | |
|---|---|
| (3) Landing gear Retract
(4) Airspeed Maintain | at V_2 to engine-out level-off height (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – OBSTACLE CLEARANCE). |
|---|---|

NOTE

If engine failure occurs above V_2 , maintain airspeed at current value (not more than $V_2 + 10$ KIAS).

After take-off:

Wing anti-ice system selected on:



At not less than 400 feet AGL or the engine-out level off height:

- | | |
|--|---|
| (6) Thrust Set
(7) Airspeed Accelerate
(8) FLAPS Select
(9) Climb Initiate
(10) CAS messages..... Review | thrust to maximum continuous thrust (MCT) (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS).
in level flight to V_{FTO} .
to UP at a speed of not less than $V_{FTO} - 5$ KIAS.
at V_{FTO} . |
|--|---|



1. SINGLE ENGINE PROCEDURES (CONT'D)

A. Engine Failure During Take-Off After Achieving V₁ (Cont'd)

At a safe altitude:

- (11) In-Flight Engine Failure/Shutdown Accomplish Refer to POWER PLANT – In-Flight Engine Failure/Shutdown procedure in this chapter.

————— **END** —————

B. Engine Failure in Climb During (V) ALTS CAP or (V) ALTV CAP

- (1) Autopilot/FD Disconnect
- (2) Pitch attitude Adjust to maintain the required single engine operating airspeed.

NOTE

Vertical FD commands may be used and autopilot re-engaged after FCC mode changes to (V) ALTS or (V) ALTV at desired altitude.

- (3) In-Flight Engine Failure/Shutdown Accomplish Refer to POWER PLANT – In-Flight Engine Failure/Shutdown procedure in this chapter.

————— **END** —————

C. Engine Failure During Approach

- (1) Autopilot Disconnect if coupled approach.
- (2) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (3) Operating engine Increase thrust as required.
- (4) FLIGHT SPOILER lever Select to RETRACT.
Check EICAS indicates spoilers retracted.
- (5) FLAPS Select to 20° for landing.
- (6) Airspeed Increase to V_{REF} (Flaps 45°) + 14 KIAS.



1. SINGLE ENGINE PROCEDURES (CONT'D)

C. Engine Failure During Approach (Cont'd)

- (7) Airplane Retrim and continue approach, or go-around at pilot's discretion (Refer to Single Engine Go-Around, in this section).
If required, autopilot may be re-engaged, if above 800 feet AGL.

If continuing the approach:

- (8) Air-conditioning Transfer to APU bleed air.
(9) Go-Around thrust Determine applicable single engine climb thrust (APR rating) (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS).
(10) TAWS WARNING, FLAPS OFF switch/light Press in to mute flap aural warning.
(11) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
(12) Final approach speed Maintain V_{REF} (Flaps 45°) + 14 KIAS.
(13) Actual landing distance Increase by a factor of 1.55 (55%), for a flap 20° landing without the use of reverse thrust.

NOTE

Use remaining thrust reverser carefully upon landing.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

After landing:

- (14) Affected engine Shutdown

————— **END** —————

1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing

DURING DESCENT

- (1) ATIS Copy
- (2) CAS Check/Clear Verify and analyze any messages, and clear (if any).

Within 15 NM of approach, if any of the following conditions apply:

- Runway is less than 3,500 feet in length, or
 - Airport is not in the database, or
 - Intended approach is not compatible with terrain awareness alerting, or
 - QFE operation with no GPS available.
- (3) TAWS WARNING, TERRAIN OFF
switch/light Press in Check TAWS WARNING, TERRAIN OFF switch/light on.
- (4) TAWS WARNING, FLAPS OFF
switch/light Press in to mute flap aural warning.
- (5) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (6) Landing data Set the following:
- ATC clearance should have been acquired.
 - Landing speeds, N_1 for all engine go-around (take-off thrust setting), N_1 for engine-out go-around (single engine climb thrust setting, APR rating) and MCT calculated for the landing weight and bugs set accordingly.
 - Set the RADIO and NAV equipment for the approach.
 - Set the associated bugs accordingly.



1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing (Cont'd)

FAA Certified Airplanes

- (6) Landing data..... Set the following:
- ATC clearance should have been acquired.
 - Set the RADIO and NAV equipment for the approach.
 - Set the associated bugs accordingly.
- (7) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (8) Actual landing distance Increase by a factor of 1.55 (55%), for a flaps 20° landing without the use of reverse thrust.
- (9) Approach briefing Complete
- The approach briefing should be accomplished well before entering the terminal control zone. The pilot-flying shall conduct the briefing and review and/or outline the operational aspects of the expected approach, which should include the following:
- (a) Type of approach
 - (b) Runway in use
 - (c) Landing minima
 - (d) Review of V_{REF} and N_1 values
 - (e) Altitudes (MSA, Field elevation, Threshold elevation, Descent crossing altitudes)
 - (f) Outbound and procedure turn courses
 - (g) Final inbound course
 - (h) Decision Height (DH) or Minimum Descent Altitude (MDA)
 - (i) Missed approach point (non-precision)
 - (j) Missed approach procedure
 - (k) NAV equipment set-up
 - (l) Any questions or clarifications and other pertinent details.
- (10) Shoulder harnesses Secured



1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing (Cont'd)

(11) CABIN PRESSURIZATION, LDG

- ALT selector Check selected to the following:
- Destination airfield altitude.
 - Destination airfield barometric pressure.
 - Desired cabin pressure descent rate.
- Check FAULT lamp out.

NOTE

1. Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.
2. The use of anti-icing should always be anticipated during descent when operating in cold weather conditions.
3. Wing and engine cowl anti-icing should be activated when descending through areas of visible moisture (Refer to Chapter 6; SUPPLEMENTARY PROCEDURES – COLD WEATHER OPERATION).

At or above 22,000 feet, the engine cowl and wing anti-ice systems must be on when ice is indicated by the ice detection system.

Below 22,000 feet, the engine cowl and wing anti-ice systems must be on when in icing conditions, or when ice is indicated by the ice detection system.

- (12) ANTI-ICE, COWL L and R switch/lights Select as required.
- (13) ANTI-ICE, WING switch Select as required.
- (14) ANTI-ICE, WSHLD/WIND L and R switches Select to HI or LOW, as required.
- (15) Fuel quantity and balance Check the following:
- Cross-check all the readouts in the fuel quantity gauging system.
 - Check fuel balance.

At transition level:

- (16) EFIS altimeters (2) and Integrated Standby Instrument altimeter (1) Set to destination airport barometric setting.



1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing (Cont'd)

WHEN IN RANGE

- (1) APU Start Ensure that operating limitations/envelope is considered.
- (2) Air-conditioning Transfer to APU bleed air.

NOTE

Bleed air extraction from the APU is not permitted above 15,000 feet.

- (3) Go-Around thrust Determine applicable single engine climb thrust (APR rating) (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS).
- (4) Lights As required Select RECOG TAXI light on.
- (5) PASS SIGNS, NO SMKG switch Select to ON.
- (6) PASS SIGNS, SEAT BLTS switch Select to ON.

BEFORE LANDING

- (1) IGNITION, CONT switch/light As required

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below, and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

The engine cowl and wing anti-ice systems must be on when in icing conditions, or when ice is indicated by the ice detection system:

- (2) ANTI-ICE, WING switch Select as required.
- (3) ANTI-ICE, COWL L and R switch/lights Select as required.
- (4) Cabin Advise not less than five minutes prior to landing.
- (5) LDG GEAR lever Select to DN. Check that three green DN indications are displayed, indicating that the landing gear is down and locked.



1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing (Cont'd)

- (6) Brake and anti-skid systemCheck
- (a) ANTI SKID, ARMED switchCheck selected to ARMED.
- (b) ANTI SKID, TEST switchSelect to TEST momentarily, and check **A/SKID IN TEST** advisory message on. After approximately 3 seconds check **A/SKID IN TEST** advisory message out.

NOTE

Anti-skid system is inoperative if the **A/SKID IN TEST** advisory message does not come on during test.

Anti-skid system is inoperative if **A/SKID INBD (OUTBD)** caution messages come on during test.

- (c) NOSE STEER switchCheck selected to ARMED.

NOTE

When the nose wheel steering system is armed in flight, the tiller should be centered.

- (7) Operative THRUST REVERSER, L (R) switchSelect to ARMED.
- Check the following:
- **L (R) REV ARMED** advisory message on.
 - **L (R) REV UNSAFE** caution message out.
 - **L (R) REV UNLOCKED** caution message out.



Do not arm thrust reverser if **L (R) REV UNSAFE** caution message is displayed.

- (8) FLAPSSelect to 20° for landing. Verify proper indication.
- (9) Final approach speed Maintain V_{REF} (Flaps 45°) + 14 KIAS.



1. SINGLE ENGINE PROCEDURES (CONT'D)

D. Single Engine Approach and Landing (Cont'd)

- (10) Actual landing distance Increase by a factor of 1.55 (55%), for a flaps 20° landing without the use of reverse thrust.

NOTE

If required, use remaining thrust reverser carefully upon landing.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

————— END —————

1. SINGLE ENGINE PROCEDURES (CONT'D)

E. Single Engine Go-Around

NOTE

1. The minimum fuel for go-around is 230 kg (500 lb) per wing, with a maximum airplane climb attitude of 5° nose up.
2. Single engine go-around capability is not assured with flaps greater than 20°.

- (1) Single Engine Climb Thrust
(APR rating).....

Set immediately (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS).

FAA Certified Airplanes

- (1) Normal Take-off Thrust

Set immediately (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS).

- (2) TOGA push switch Depress

- (3) FLIGHT SPOILER lever
(If extended)

Select to RETRACT.

Check EICAS indicates spoilers retracted.

- (4) Airplane Rotate

smoothly to initial pitch attitude of 10°, and adjust pitch as required to achieve the approach climb speed. (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – APPROACH AND LANDING CLIMB).

- (5) FLAPS Select

to 20°.

When a positive rate of climb is achieved:

- (6) LDG GEAR lever Select

to UP.

At a safe altitude (not below 400 feet AGL):

- (7) FLAPS Select

to 0° at a speed of not less than $V_{FTO} - 5$ KIAS.

- (8) THRUST REVERSER, L and R
switches.....

Select to OFF.

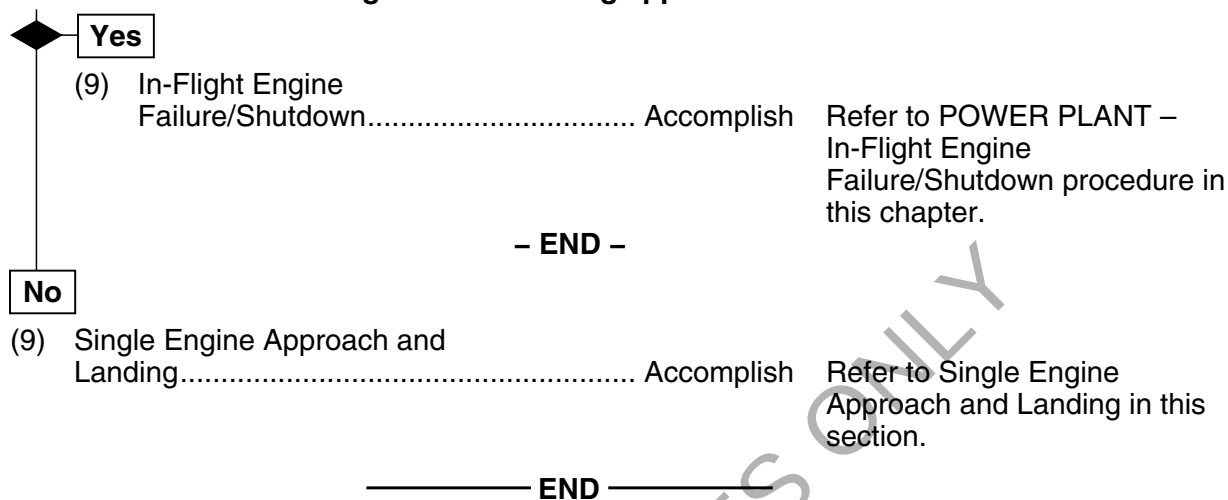
Check **L** and **R REV ARMED** advisory messages out.



1. SINGLE ENGINE PROCEDURES (CONT'D)

E. Single Engine Go-Around (Cont'd)

Go-around was due to engine failure during approach:



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TRAINING PURPOSES ONLY

	CL-605 Flight Crew Operating Manual PSP 605-6	
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1. AIR START

A. Engine Cross Bleed – Air Start Procedure

NOTE

Use cross bleed for in-flight air starts whenever possible.

Refer to Chapter 2; LIMITATIONS – POWER PLANT for starter cranking limits.

Affected engine:

- (1) Thrust leverCheck affected lever SHUT OFF.
- (2) FUEL, L (R) BOOST PUMP switch/light.....Check the following:
 - PUMP ON lights on.
 - **L and R FUEL PUMP ON** advisory messages on.
- (3) BLEED AIR, APU LCV switch/light.....Press out to close load control valve.
Check APU LCV OPEN light out.

Operating engine:

- (4) BLEED AIR, 10TH STAGE, L (R) switch/light.....Press in to open operative shut-off valve.
Check L (R) CLOSED light out.
- (5) Bleed air pressure 60 psi Adjust operating engine's thrust lever to attain 60 psi **minimum**.
- (6) Start envelope Establish

ENGINE CROSS BLEED AIR START ENVELOPE			
ALTITUDE	AIRSPEED	ITT	% N ₂
21,000 feet to 15,000 feet	200 KIAS up to 300 KIAS	90°C or less	0 to 55
15,000 feet to SL	140 KIAS up to 300 KIAS		

When ready to start:

- (7) IGNITION , CONT switch/light.....Press in for continuous ignition.
Check the following:
 - CONT ON, A and B lights on.
 - **IGNITION A/B** advisory message on.



1. AIR START (CONT'D)

A. Engine Cross Bleed – Air Start Procedure (Cont'd)

Affected engine:

- (8) ENG, L (R) START switch/lightPress to cross bleed-air start affected engine.
Check L (R) START light on.

When ITT is 90°C or less and N₂ is 28%:

- (9) Thrust leverSelect to IDLE when within start envelope.

NOTE

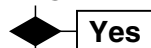
If ITT is above 90°C, dry motor to lower ITT to 90°C before moving throttle to IDLE.

- (10) Engine indicationsMonitor carefully.

NOTE

If engine does not ignite in 25 seconds, retard thrust lever to SHUT OFF, press associated engine STOP switch/light, wait 20 seconds and repeat procedure.

Engine start is successful:



When the engine is stabilized at Flight Idle:

- (11) Thrust levers Set as required.
- (12) IGNITION, CONT switch/lightPress out to select continuous ignition off.
Check **IGNITION A/B** advisory message out.
- (13) BLEED AIR, 10TH STAGE, L (R) switch/light Press in to open affected shut-off valve.
Check L (R) CLOSED light out.
- (14) BLEED AIR, APU LCV switch/light As required
- (15) AIR-CONDITIONING, L and R PACK switch/lights Press in to start both packs.
Check L and R PACK OFF lights out.
- (16) ANTI-ICE, COWL, L and R switches As required

1. AIR START (CONT'D)

A. Engine Cross Bleed – Air Start Procedure (Cont'd)

- ▼
- | | | |
|--|-------------|---|
| (17) BLEED AIR, 14TH STAGE, ISOL switch/light..... | Press out | to close isolation valve.
Check 14TH ISOL OPEN light out. |
| (18) AC POWER, GEN 1 (2) switch..... | Select | affected generator ON.
Check GEN 1 (2) OFF caution message out. |
| (19) Affected HYDRAULIC, pump switch..... | AUTO | Check the following:
<ul style="list-style-type: none"> • HYD EDP 1A and HYD PUMP 2B caution messages out, or • HYD EDP 2A and HYD PUMP 1B caution messages out. |
| (20) TAWS WARNING, FLAPS OFF switch/light..... | Press out | Check OFF light out. |
| (21) EFIS altimeters (2)..... | Cross-check | |
- END -

No

- | | | |
|---|-------------|---|
| (11) IGNITION, CONT switch/light..... | Press out | to select continuous ignition off.
Check IGNITION A/B advisory message out. |
| (12) BLEED AIR, 10TH STAGE L (R) switch/light..... | Press out | to select affected 10th-stage SOV closed.
Check L (R) 10TH CLOSED light on. |
| (13) BLEED AIR, APU LCV switch/light..... | As required | |
| (14) AIR-CONDITIONING, L (R) PACK switch/light..... | Press in | to turn on operative pack.
Check L (R) PACK OFF light out. |
| (15) MFD or CDU tuning, ATC/TCAS mode | Set | to TA ONLY. |
| (16) Land at the nearest suitable airport. | | |



1. AIR START (CONT'D)

A. Engine Cross Bleed – Air Start Procedure (Cont'd)

(17) Single Engine Approach and
Landing..... Accomplish

Refer to SINGLE ENGINE
PROCEDURES – Single
Engine Approach and Landing
in this chapter.

————— **END** —————

TRAINING PURPOSES ONLY

1. AIR START (CONT'D)

B. Windmilling Air Start Procedure

NOTE

1. At thrust lever advance, N_2 should be stable or increasing.
2. Windmill airstart efficiency is enhanced by attaining the highest practical airspeed and N_2 within the relight envelope.
3. 300 knots is a target airspeed only, and may be exceeded to obtain the minimum required N_2 .

Affected engine:

- (1) Thrust lever Check affected lever SHUT OFF.
- (2) FUEL, L (R) BOOST PUMP switch/light Check the following:
- PUMP ON lights on.
 - **L** and **R FUEL PUMP ON** advisory messages on.
- (3) BLEED AIR, 10TH STAGE, L (R) switch/light Press out to close affected shut-off valve.
Check 10TH CLOSED light on.
- (4) BLEED AIR, 10TH STAGE, ISOL switch/light Press out to close isolation valve.
Check 10TH ISOL OPEN light out.
- (5) BLEED AIR, APU LCV switch/light Press out to close load control valve.
Check APU LCV OPEN light out.
- (6) Start envelope Establish

WINDMILLING AIR START ENVELOPE			
ALTITUDE	AIRSPEED	ITT	% N_2
21,000 feet to 10,000 feet	300 KIAS up to 348 KIAS	90°C or less	12 to 55
10,000 feet to 8,000 feet			10 to 55
8,000 feet to SL	300 KIAS		



1. AIR START (CONT'D)

B. Windmilling Air Start Procedure (Cont'd)

When ready to start:

(7) IGNITION, CONT switch/light..... Press in for continuous ignition.

Check the following:

- CONT ON, A and B lights on.
- **IGNITION A/B** advisory message on.

When ITT is 90°C or less and N₂ is at least 12% (10% if below 10,000 feet):

Affected engine:

(8) Thrust lever Select to IDLE when within start envelope.

(9) Engine indications Monitor carefully.

NOTE

1. If engine does not ignite in 25 seconds, retard thrust lever to SHUT OFF, maintain at least 300 KIAS for 30 seconds, and repeat procedure. After ignition is achieved, monitor N₂ acceleration rate. N₂ acceleration should be uninterrupted and positive at all times. Stable idle must be achieved within 2 minutes of advancing thrust lever to IDLE.
2. During windmilling airstart, attempt to keep the flight control inputs to a minimum.

Engine start is successful:

◆ **Yes**

When the engine is stabilized at Flight Idle:

(10) Thrust levers..... Set as required.

(11) IGNITION, CONT switch/light..... Press out to select continuous ignition off.

Check **IGNITION A/B** advisory message out.

(12) BLEED AIR, 10TH STAGE, L (R) switch/light As required

(13) BLEED AIR, APU LCV switch/light As required

(14) AIR-CONDITIONING, L (R) PACK switch/light Press in to start affected pack.

Check L (R) PACK OFF light out.

(15) ANTI-ICE, COWL, L (R) switches As required

1. AIR START (CONT'D)

B. Windmilling Air Start Procedure (Cont'd)

- ▼
- | | | |
|---|-------------|---|
| (16) BLEED AIR, 14TH STAGE,
ISOL switch/light..... | Press out | to close isolation valve.

Check 14TH ISOL OPEN light out. |
| (17) AC POWER, GEN 1 (2)
switch..... | Select | affected generator ON.

Check GEN 1 (2) OFF caution message out. |
| (18) Affected HYDRAULIC, pump
switch..... | AUTO | Check the following:
<ul style="list-style-type: none"> • HYD EDP 1A and HYD PUMP 2B caution messages out, or • HYD EDP 2A and HYD PUMP 1B caution messages out. |
| (19) TAWS WARNING, FLAPS
OFF switch/light..... | Press out | Check OFF light out. |
| (20) EFIS altimeters (2)..... | Cross-check | |

- END -

No

- | | | |
|--|-------------|---|
| (10) IGNITION, CONT switch/light..... | Press out | to select continuous ignition off.

Check IGNITION A/B advisory message out. |
| (11) BLEED AIR, 10TH STAGE L (R)
switch/light..... | Press out | to select affected 10th-stage SOV closed.

Check L (R) 10TH CLOSED light on. |
| (12) BLEED AIR, APU LCV switch/light..... | As required | |
| (13) AIR-CONDITIONING, L (R) PACK
switch/light..... | Press in | to turn on operative pack.

Check L (R) PACK OFF light out. |
| (14) MFD or CDU tuning, ATC/TCAS
mode | Set | to TA ONLY. |
| (15) Land at the nearest suitable airport. | | |



1. AIR START (CONT'D)

B. Windmilling Air Start Procedure (Cont'd)

- (16) Single Engine Approach and Landing.....

Accomplish

Refer to SINGLE ENGINE PROCEDURES – Single Engine Approach and Landing in this chapter.

————— END —————

TRAINING PURPOSES ONLY

1. AIR-CONDITIONING AND PRESSURIZATION

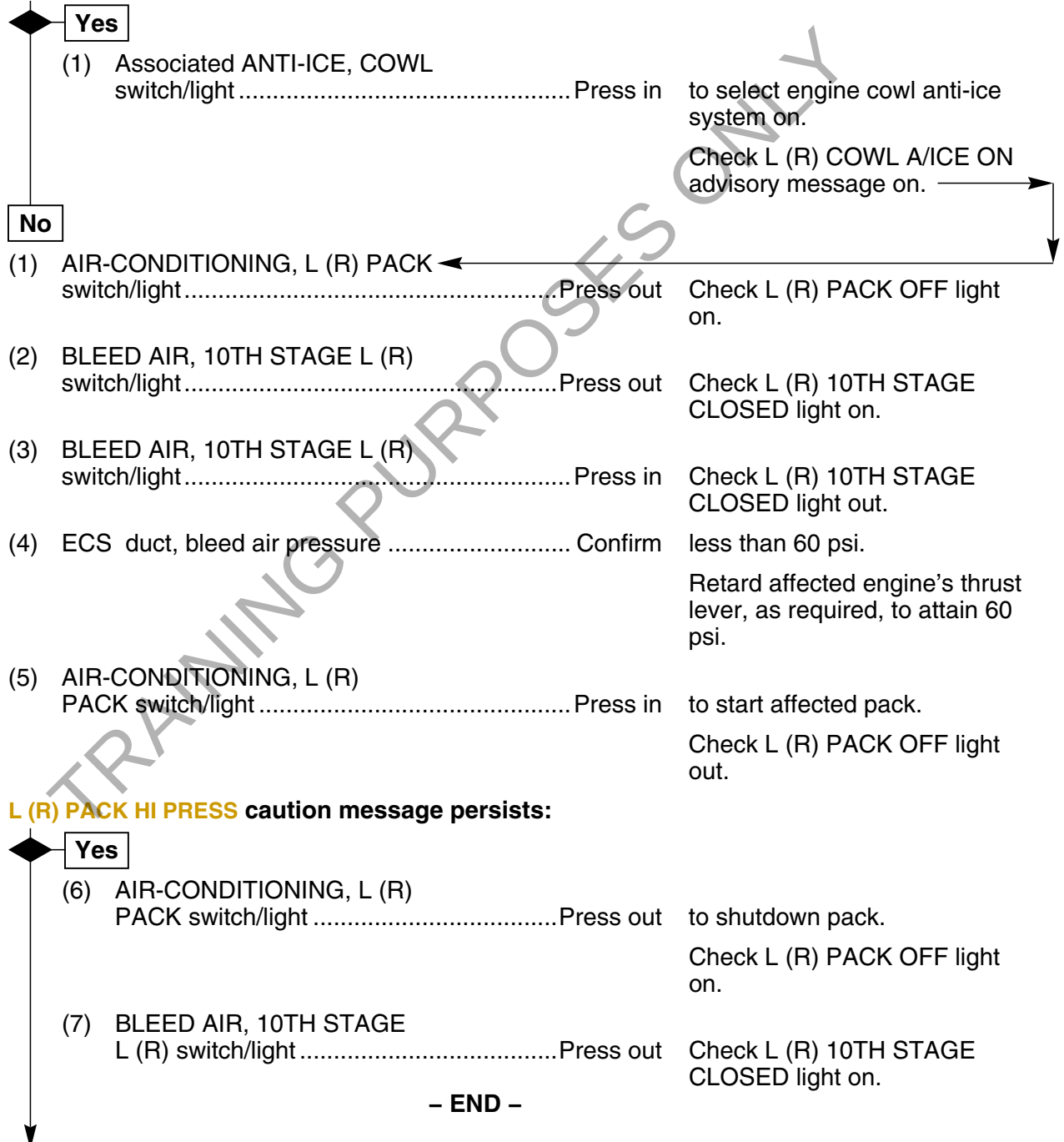
A. Air-Conditioning Pack High Pressure

Indication: **L (R) PACK HI PRESS** caution message and L (R) PACK FAIL light on.

NOTE

The isolation valve, pack valve and the 10th stage bleed valve will close automatically during an overpressure condition.

Above 40,000 feet:



1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

A. Air-Conditioning Pack High Pressure (Cont'd)

No

(6) No further action required.

END

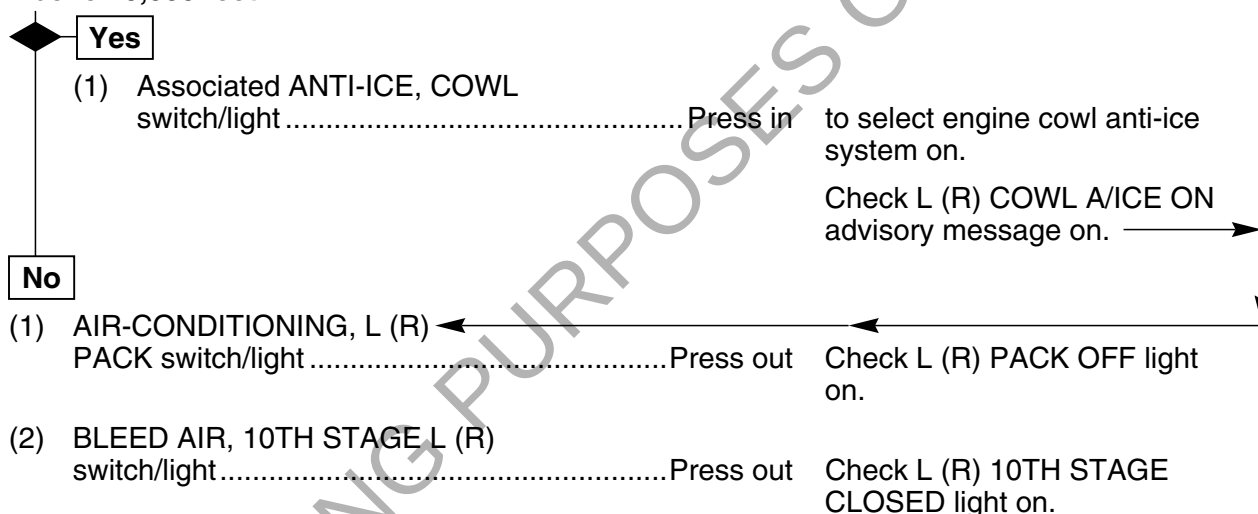
B. Air-Conditioning Pack High Temperature

Indication: **L (R) PACK HI TEMP** caution message and L (R) PACK FAIL light on.

NOTE

The pack valve will close automatically during an overtemperature condition.

Above 40,000 feet:



NOTE

If in manual mode with full HOT selected, decrease temperature before resetting pack.

When **L (R) PACK HI TEMP** caution message goes out:

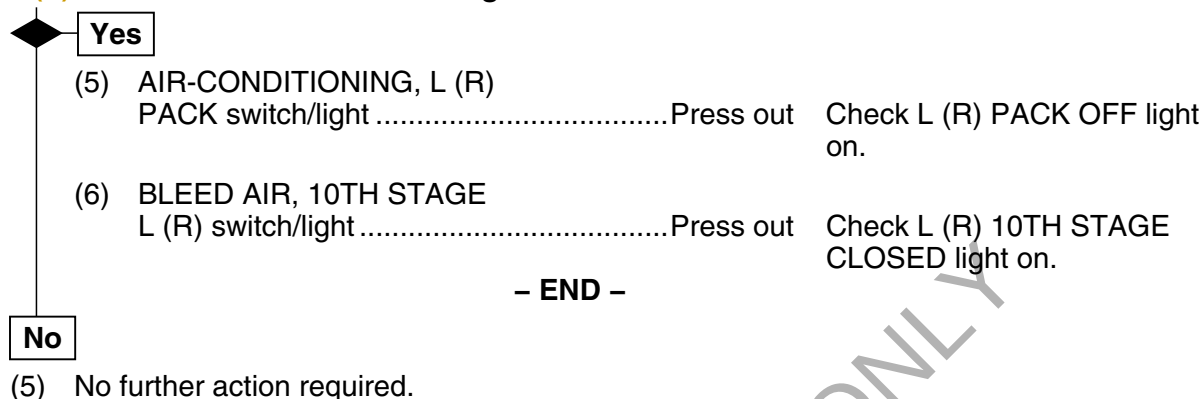
- (3) BLEED AIR, 10TH STAGE L (R) switch/light Press in Check L (R) 10TH STAGE CLOSED light out.
- (4) AIR-CONDITIONING, L (R) PACK switch/light Press in to start affected pack.
Check L (R) PACK OFF light out.



1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

B. Air-Conditioning Pack High Temperature (Cont'd)

L (R) PACK HI TEMP caution message on:



C. Automatic Pressurization System Malfunction

Indication: **AUTO PRESS** caution message and PRESS CONT FAIL light on.

- (1) Manual Cabin Pressurization
Control Procedure Accomplish Refer to Manual Cabin Pressurization Control Procedure in this section.

NOTE

Do not attempt to reselect auto mode.

- END -

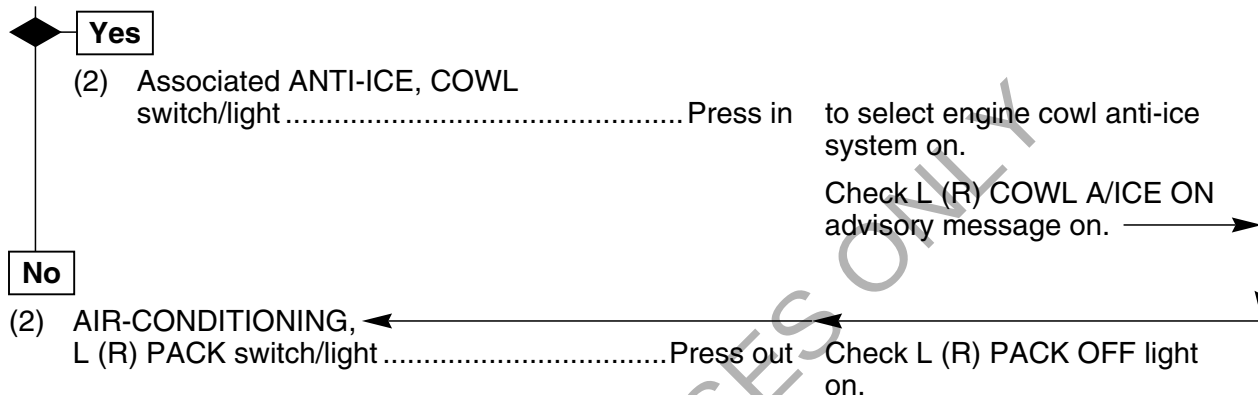
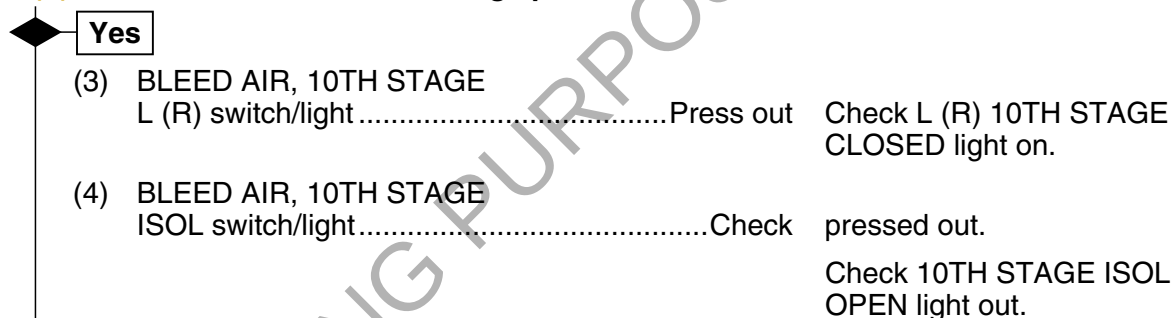
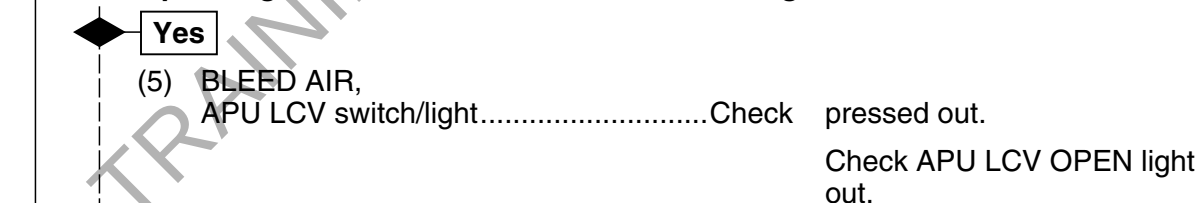
1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

D. Air-Conditioning Pack Pressure/Regulating Valve Failure

Indication: **L (R) PACK NOT OFF** caution message on and L (R) PACK FAIL light on.

- (1) AIR-CONDITIONING,
L (R) PACK switch/light Press in Check L (R) PACK OFF light out.

Above 40,000 feet:

**L (R) PACK NOT OFF** caution message persists:APU operating and **L PACK NOT OFF** caution message on:

- END -

- Decision: **No**
- (3) No further action required.

END

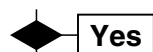
1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

E. Cabin Altitude Too High

Indication: **CABIN ALT** caution message on.

- (1) BLEED AIR, 10TH STAGE
L and R switch/lightsCheck pressed in.
Check L and R 10TH STAGE
CLOSED lights out.
- (2) AIR-CONDITIONING,
L and R Pack switch/lightsCheck pressed in.
Check L and R PACK OFF
lights out.
- (3) Engine thrust Increase
- (4) CABIN PRESSURIZATION,
EMER DEPRESS switch/lightCheck pressed out.
Check the following:
- EMER DEPRESS ON light out.
 - **EMER DEPRESS** caution message out.
- (5) CABIN PRESSURIZATION,
MAN ALT selectorCheck set to mid position/centre.

Automatic pressurization is restored:

Yes

(6) No further action required.

— END —

No

- (6) CABIN PRESSURIZATION,
PRESS CONTPress in to select manual mode.
Check PRESS CONT
MANUAL light on.
Operate MAN ALT and MAN
RATE switches as required.
(Refer to Manual Cabin
Pressurization Control
Procedure in this section).

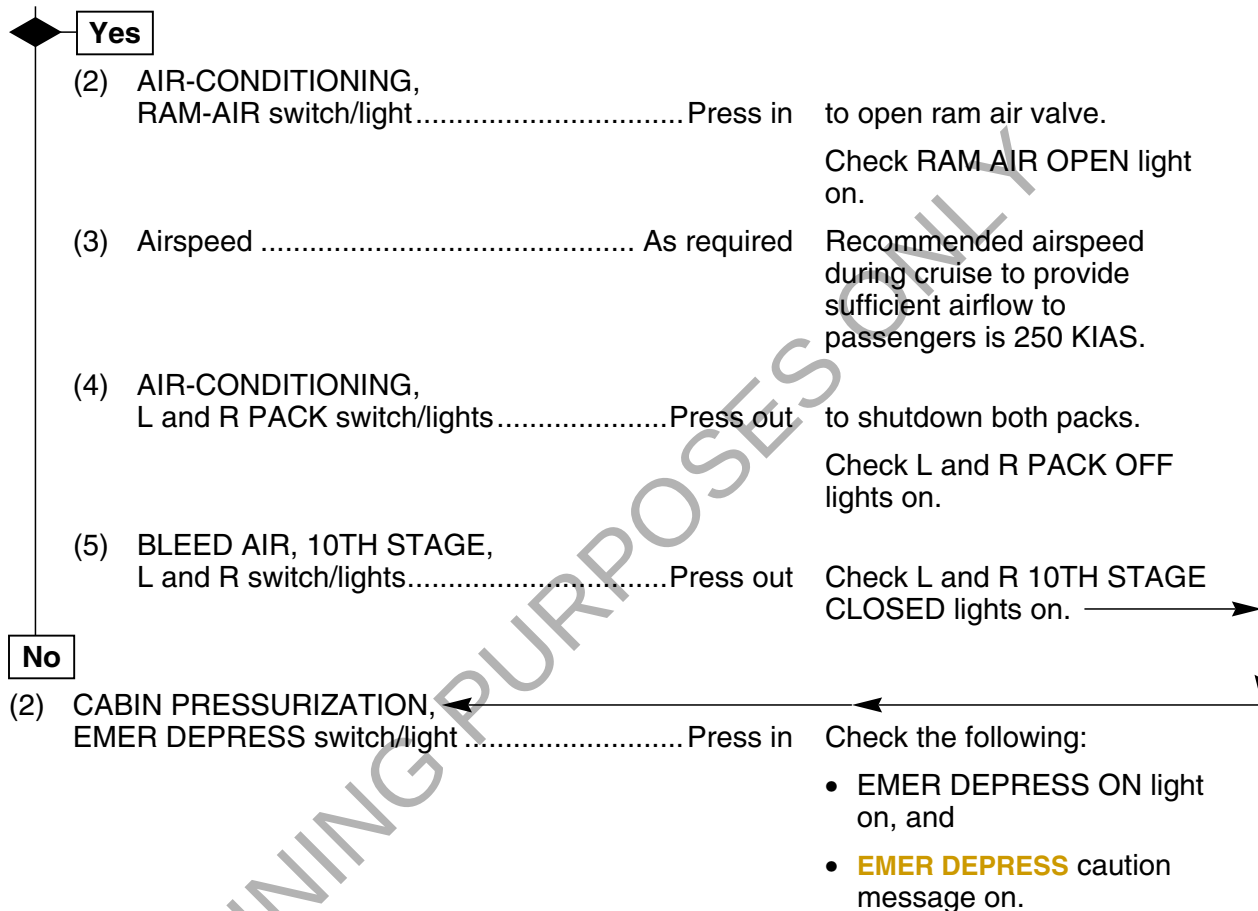
— END —

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

F. Unpressurized Flight Procedure

- (1) Airplane altitude..... Maintain 10,000 feet maximum or lowest safe altitude, whichever is higher.

PACKs OFF:



END

1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

G. Manual Cabin Pressurization Control Procedure

NOTE

The MAN ALT selector is a three (3) position switch.

The switch is spring-loaded to mid position/centre.

- (1) CABIN PRESSURIZATION,
MAN ALT selector Check set to mid position/centre.
- (2) CABIN PRESSURIZATION,
PRESS CONT switch/light Press in to select manual mode.
Check PRESS CONT
MANUAL light on.
Operate CABIN PRESS, MAN
ALT and MAN RATE switches
as required.



Do not exceed the maximum normal cabin differential pressure of 8.8 psid.

To increase cabin altitude:

- (3) CABIN PRESSURIZATION,
MAN ALT selector Select UP and hold.
- (4) CABIN PRESSURIZATION,
MAN RATE switch As required

To decrease cabin altitude:

- (3) CABIN PRESSURIZATION,
MAN ALT selector Select DN and hold.
- (4) CABIN PRESSURIZATION,
MAN RATE switch As required

To maintain cabin altitude:

When reaching target cabin altitude as per the chart below:

- (5) CABIN PRESSURIZATION,
MAN ALT selector Release Ensure selector returns to mid position/centre.

Cruise Flight Level	180	200	220	240	260	280	300	350	400	410
Target Cabin Altitude	2100	2400	2700	3000	3400	3800	4200	5200	6600	6900



1. AIR-CONDITIONING AND PRESSURIZATION (CONT'D)

G. Manual Cabin Pressurization Control Procedure (Cont'd)

Before landing:

- (6) Cabin altitude Adjust to landing field elevation.

NOTE

Do not set cabin altitude below landing field elevation.

If differential pressure is not zero upon landing:

- (7) CABIN PRESSURIZATION,
MAN ALT selector Select UP and hold.
- (8) CABIN PRESSURIZATION,
MAN RATE switch Set to maximum INCR.

————— END —————

H. Cabin Pressure Acquisition Module Failure

Indication: **CPAM FAIL** caution message on.

- (1) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.
- (2) Unpressurized Flight Procedure Accomplish if required (Refer to Unpressurized Flight Procedure in this section).

NOTE

EICAS pressurization system readouts (e.g., cabin altitude, rate of change, and differential pressure) are not available. EICAS cabin pressure warning and caution messages are also not available.

————— END —————

1. AUTOMATIC FLIGHT CONTROL SYSTEM

NOTE

Autopilot disconnect warnings are indicated by “Cavalry Charge” aural and flashing red AP message on primary flight display.

A. Autopilot Mistrim

Indication: **AP HOLDING LWD (RWD, NOSE DOWN, NOSE UP)** caution messages on.

(1) Autopilot Monitor operation.

Retrim as required.

WARNING

An abrupt change in control force, or an out of trim situation may be experienced when disconnecting the autopilot.

SEAT BLTS should be selected ON.

Caution message persists:

Yes

(2) Flight controls Hold firmly

(3) Autopilot..... Disconnect

(4) AFCS SEL 1(2) switch
(centre pedestal)..... Select

to operative AFCS channel.

Check **AFCS 1 (2) INOP** status message on.

(5) Autopilot..... Engage

and monitor operation.

Retrim as required.

Caution message persists:

Yes

(6) Flight controls Hold firmly

(7) Autopilot..... Disconnect

and assume manual control.

— END —

No

(2) No further action required.

— END —

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

B. Autopilot Pitch Trim Failure

Indication: **AP PITCH TRIM** caution message on.

NOTE

Failure is resultant of autopilot pitch trim not being able to keep up with the pitch change.

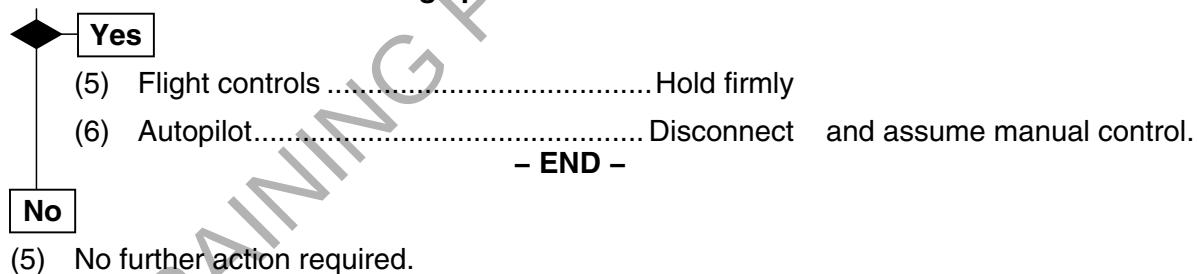
WARNING

An abrupt change in control force, or an out of trim situation may be experienced when disconnecting the autopilot.

SEAT BLTS should be selected ON.

- | | | |
|---|-------------|--|
| (1) Flight controls | Hold firmly | |
| (2) Autopilot | Disconnect | Retrim as required. |
| (3) AFCS SEL 1(2) switch
(centre pedestal) | Select | to operative AFCS channel.
Check AFCS 1 (2) INOP status
message on. |
| (4) FDs and autopilot | Engage | and monitor operation.
Retrim as required. |

AP PITCH TRIM caution message persists:



————— **END** —————

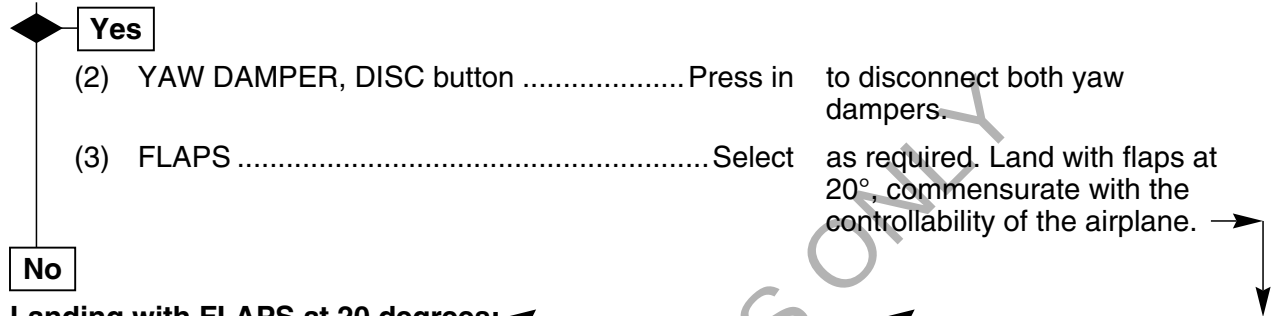
1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

C. Double Yaw Damper Failure

Indication: **YAW DAMPER** caution message on. Autopilot will disengage if it was coupled.

- (1) YAW DAMPER, YD1 and YD2 switches..... Press in to engage both yaw dampers.

YAW DAMPER caution message persists:



- Decision diamond: **Yes**
- Prior to landing:**
- (2) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (3) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (4) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (5) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

C. Double Yaw Damper Failure (Cont'd)

- (6) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

NOTE

Select the runway available with minimum crosswind.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

- (2) No further action required.

— END —

1. AUTOMATIC FLIGHT CONTROL SYSTEM (CONT'D)

D. Single Flight Director Failed

Indication: **FD 1 (2) FAIL** status message on, red FD flag displayed on affected PFD, loss of flight director guidance cue on affected PFD, and autopilot may disengage.

NOTE

- If a red FD flag is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to INSTRUMENTS SYSTEM – Air Data Computer Failure or Inertial Reference System Failure in this chapter.
- Anticipate elevator out of trim situation when disconnecting the autopilot.

- (1) Flight controls Hold firmly
- (2) Autopilot Disconnect
- (3) AFCS SEL 1(2) switch
(centre pedestal) Select to operative AFCS channel.
Check **AFCS 1 (2) INOP** status message on.
- (4) FDs and autopilot Engage and monitor operation.
Retrim as required.

Red FD flag displayed on both PFDs:

- Yes
- (5) Flight control panel XFR switch Press to transfer to operable side.
- END -

No

Sensor failure indicated (red line across affected flight director mode annunciator):

- Yes
- (5) Flight control panel affected mode switch Deselect
- END -

No

- (5) No further action required.

- END -

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TRAINING PURPOSES ONLY

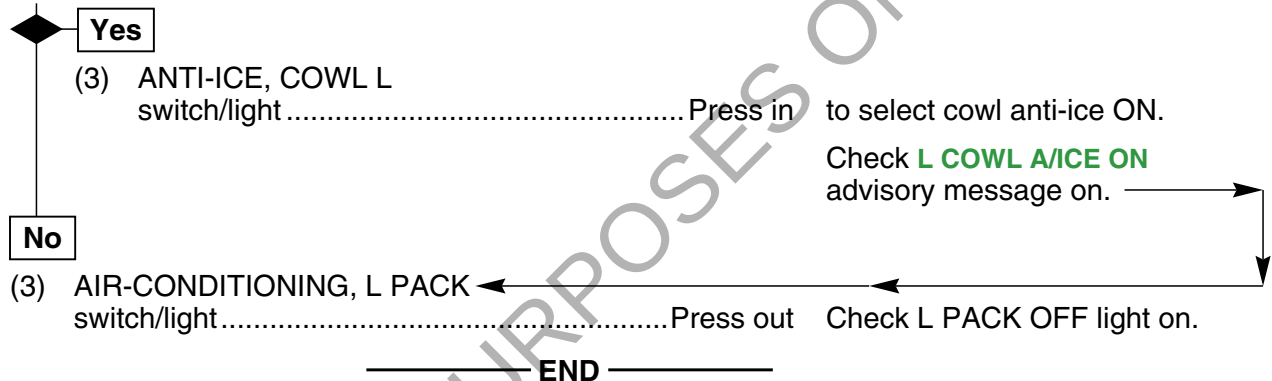
1. AUXILIARY POWER UNIT

A. APU Load Control Valve Fails to Close

Indication: **APU LCV FAIL** caution message and APU LCV FAIL light on.

- (1) BLEED AIR, 10TH STAGE, L
switch/light Press out to close left 10th stage bleed valve.
Check L CLOSED light on.
- (2) BLEED AIR, 10TH STAGE, ISOL
switch/light Press out to close 10th stage isolation valve.
Check 10TH ISOL OPEN light out.

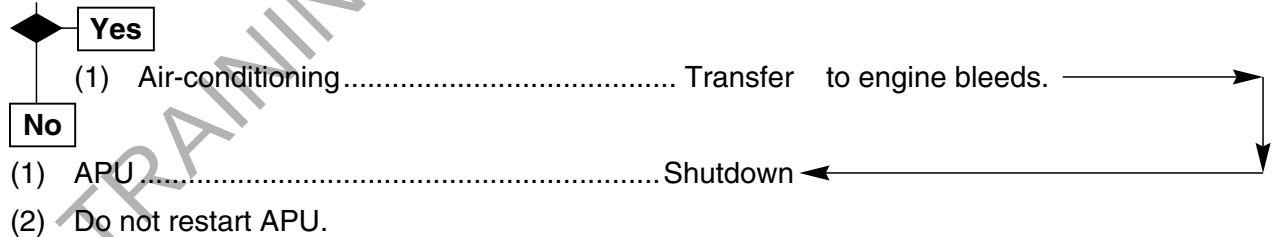
Above 40,000 feet:



B. APU Low Oil Pressure

Indication: **APU OIL PRESS** caution message on, and APU automatically shuts down.

APU is supplying bleed air for air-conditioning:

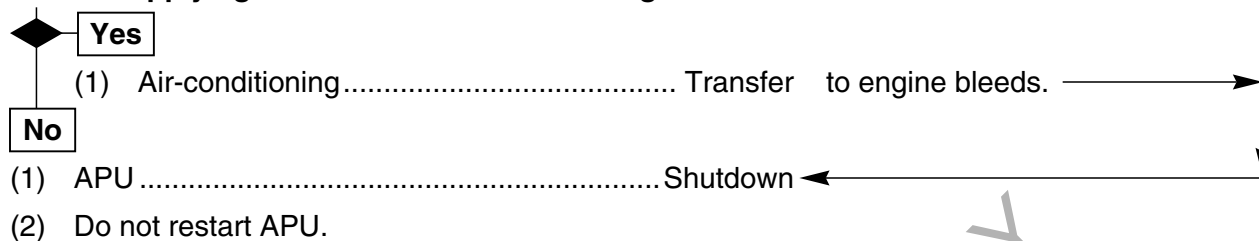


1. AUXILIARY POWER UNIT (CONT'D)

C. APU High Oil Temperature

Indication: **APU OIL TEMP** caution message, and APU automatically shuts down.

APU is supplying bleed air for air-conditioning:

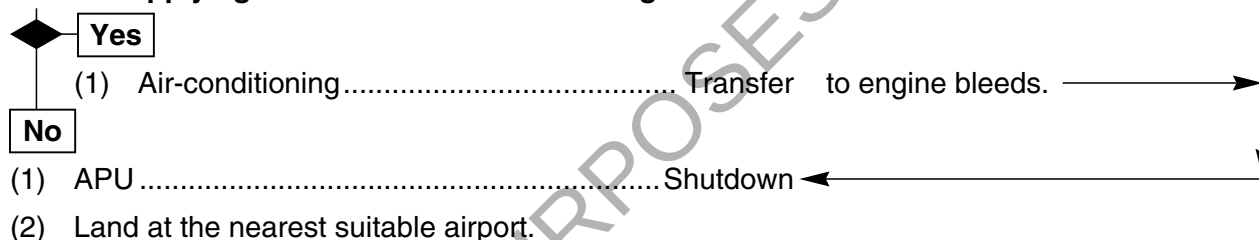


————— **END** —————

D. APU Shut-Off Valve Failed to Close, After APU Shutdown Due to Fire

Indication: **APU SOV** caution message displayed after APU shutdown due to fire.

APU is supplying bleed air for air-conditioning:



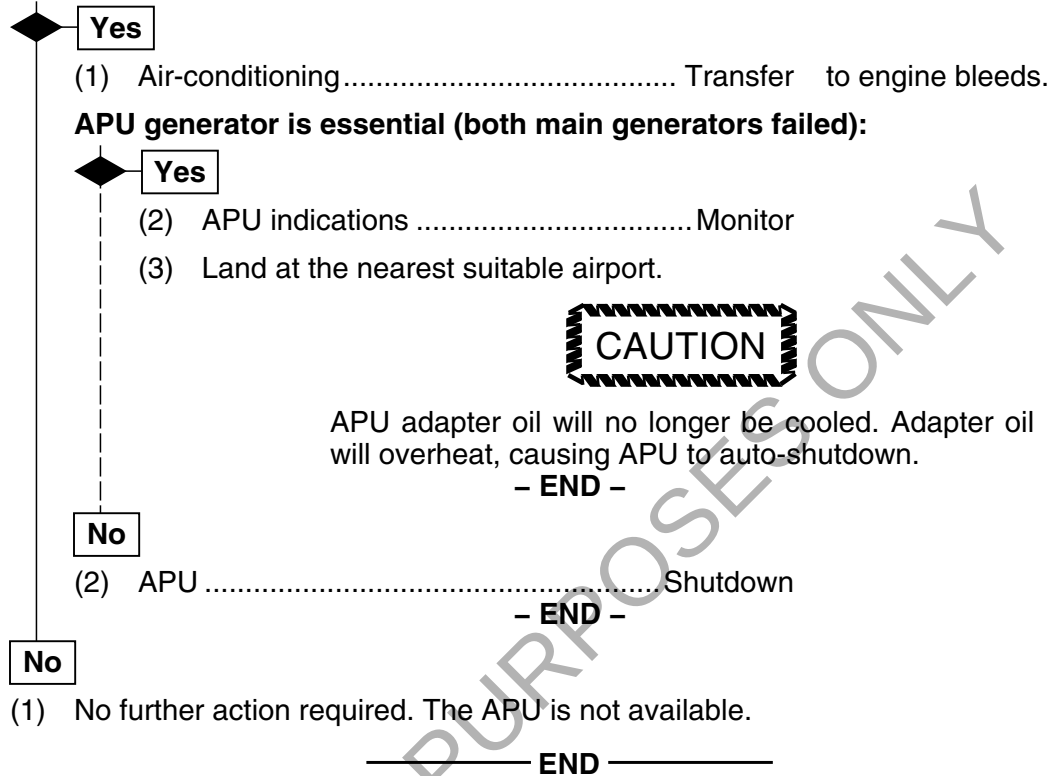
————— **END** —————

1. AUXILIARY POWER UNIT (CONT'D)

E. APU Fuel Shut-Off Valve not in the Selected Position

Indication: **APU SOV** caution message on.

APU is operating:



F. APU Negative-G Shut-Off Valve Failed to Close After APU Shutdown Due to Fire

Indication: **APU NEG-G SOV** caution message on.

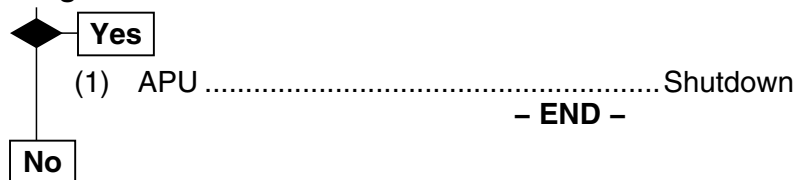
- (1) FUEL, L BOOST PUMP switch/light Press out to disarm affected boost pump.
- Check the following:
- **L FUEL PUMP ON** advisory message out, and
 - L FUEL PUMP INOP light on.
- (2) Left engine Shutdown Refer to POWER PLANT – In-Fight Engine Failure/Shutdown procedure in this chapter.
- END —

1. AUXILIARY POWER UNIT (CONT'D)

G. APU Pump Failure

Indication: **APU PUMP** caution message on and PUMP FAIL light on.

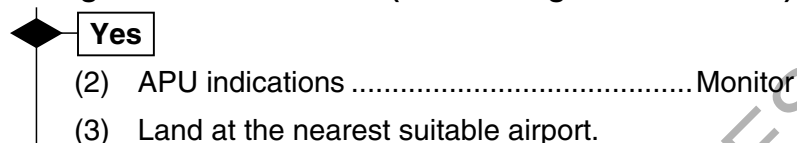
On ground:



In flight:

(1) Air-conditioning Transfer to engine bleeds.

APU generator is essential (both main generators failed):



APU adapter oil will no longer be cooled. Adapter oil will overheat, causing APU to auto-shutdown.

- END -



END

1. DOORS

A. Passenger Door – Outer Handle not Stowed

Indication: **PAX DOOR OUT HNDL** caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

◆ **Yes**

(3) Passenger door Check visually, and ensure that four roll-over catch green witness marks are correctly aligned.

All 4 witness marks are aligned:

◆ **Yes**

(4) Passenger door, inner T-handle..... Pull to stow outer handle. Green LOCKED flag-indicator visible.

When the handle is locked:

(5) Passenger door, inner T-handle..... Push in to stow inner T-handle, and then close inner T-handle access cover.

Check **PAX DOOR STOW** caution message out.

PAX DOOR OUT HNDL caution message persists:

◆ **Yes**

(6) Land at the nearest suitable airport.
 – END –

◆ **No**

(6) No further action required.
 – END –

◆ **No**

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.



1. DOORS (CONT'D)

A. Passenger Door – Outer Handle not Stowed (Cont'd)

- (4) CABIN PRESSURIZATION, MAN
ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

- (5) CABIN PRESSURIZATION, MAN
RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

- (6) CABIN PRESSURIZATION, MAN
ALT selector Release Ensure selector returns to mid position/centre.
- (7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.

- (8) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

- (9) CABIN PRESSURIZATION, MAN
ALT selector Select DN and hold to decrease cabin altitude to landing field elevation.

- (10) CABIN PRESSURIZATION, MAN
RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

- (11) CABIN PRESSURIZATION, MAN
ALT selector Release Ensure selector returns to mid position/centre.

————— **END** —————

1. DOORS (CONT'D)

B. Passenger Door – Inner Handle not Stowed

Indication: **PAX DOOR STOW** caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

Yes

(3) Passenger door, inner handle Stow Stow inner handle by pushing it down into the latched position.

PAX DOOR STOW caution message persists:

(4) No further action required.

– END –

No

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.

(4) CABIN PRESSURIZATION, MAN ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(5) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

(6) CABIN PRESSURIZATION, MAN ALT selector Release Ensure selector returns to mid position/centre.

(7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.

(8) Land at the nearest suitable airport.



1. DOORS (CONT'D)

B. Passenger Door – Inner Handle not Stowed (Cont'd)

At 10,000 feet or lowest safe altitude:

(9) CABIN PRESSURIZATION, MAN
 ALT selector Select DN and hold to decrease cabin altitude to landing field elevation.

(10) CABIN PRESSURIZATION, MAN
 RATE switch..... As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

(11) CABIN PRESSURIZATION, MAN
 ALT selector Release Ensure selector returns to mid position/centre.

————— **END** —————

TRAINING PURPOSES ONLY

1. DOORS (CONT'D)

C. Passenger Door – Not Latched

Indication: **PAX DOOR LATCH** caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

Yes

(3) No further action required.

– END –

No

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.

(4) CABIN PRESSURIZATION, MAN ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(5) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

(6) CABIN PRESSURIZATION, MAN ALT selector Release Ensure selector returns to mid position/centre.

(7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.

(8) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

(9) CABIN PRESSURIZATION, MAN ALT selector Select DN and hold to decrease cabin altitude to landing field elevation.

(10) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.



1. DOORS (CONT'D)

C. Passenger Door – Not Latched (Cont'd)

When reaching target cabin altitude:

(11) CABIN PRESSURIZATION, MAN

ALT selectorRelease Ensure selector returns to mid position/centre.

————— END —————

TRAINING PURPOSES ONLY

1. DOORS (CONT'D)

D. Baggage Door – Not Closed/Locked

Indication: **BAGGAGE DOOR** caution message on.

(1) PASS SIGNS, SEAT BLTS switch Select to ON.

(2) Cabin pressure Check

Cabin pressure is normal:

Yes

(3) No further action required.

– END –

No

Switch to manual pressurization control:

(3) CABIN PRESSURIZATION, PRESS CONT switch/light Select to manual mode.

Check PRESS CONT MANUAL light on.

(4) CABIN PRESSURIZATION, MAN ALT selector As required to maintain cabin altitude at 8,000 feet, to minimize the pressure differential across the affected door.

NOTE

Select and hold to “UP” to increase cabin altitude; “DN” to reduce cabin altitude.

(5) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.

When reaching target cabin altitude:

(6) CABIN PRESSURIZATION, MAN ALT selector Release Ensure selector returns to mid position/centre.

(7) Descent Initiate to 10,000 feet or lowest safe altitude, whichever is higher.

(8) Land at the nearest suitable airport.

At 10,000 feet or lowest safe altitude:

(9) CABIN PRESSURIZATION, MAN ALT switch Select DN and hold to decrease cabin altitude to landing field elevation.

(10) CABIN PRESSURIZATION, MAN RATE switch As required commensurate with crew and passenger comfort.



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1. DOORS (CONT'D)

D. Baggage Door – Not Closed/Locked (Cont'd)

When reaching target cabin altitude:

(11) CABIN PRESSURIZATION, MAN

ALT selectorRelease Ensure selector returns to mid position/centre.

————— END —————

TRAINING PURPOSES ONLY

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1. ELECTRICAL

A. AC Bus 1 Transfer Failure

Indication: **AC 1 AUTOXFER** caution message and associated AUTO XFER FAIL light on.

AC Bus 1 is powered:

Yes

(1) No further action required.

— END —

No

(1) Affected airplane systems Review Refer to Bus Inoperative List in this section.

— END —

B. AC Bus 2 Transfer Failure

Indication: **AC 2 AUTO XFER** caution message and associated AUTO XFER FAIL light on.

AC Bus 2 is powered:

Yes

(1) EFIS altimeters (2)..... Cross-check

— END —

No

(1) Affected airplane systems Review Refer to Bus Inoperative List in this section.

— END —

C. AC Bus 1 Failure

Indication: **AC BUS 1** caution message on.

(1) AC POWER, GEN 1 switch Select to OFF/RESET then ON.

AC BUS 1 caution message persists:

Yes

(2) Leave icing conditions.

(3) Affected airplane systems Review Refer to Bus Inoperative List in this section.

— END —

No

(2) No further action required.

— END —

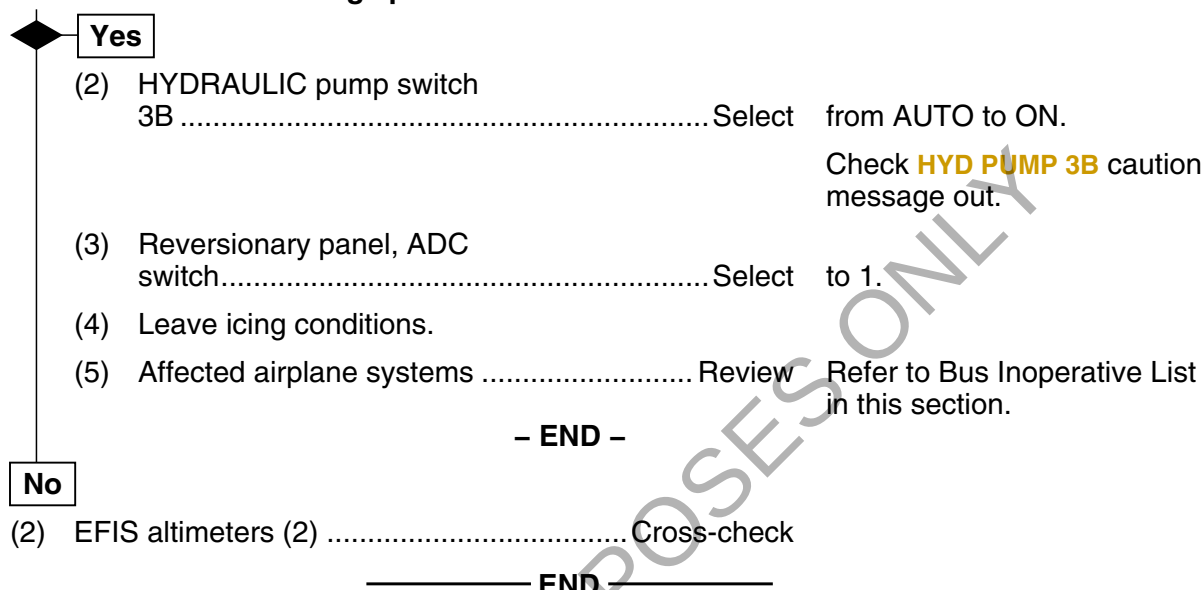
1. ELECTRICAL (CONT'D)

D. AC Bus 2 Failure

Indication: **AC BUS 2** caution message on.

(1) AC POWER, GEN 2 switch Select to OFF/RESET then ON.

AC BUS 2 caution message persists:



E. AC Essential Bus Failure

Indication: **AC ESS BUS** caution message on.

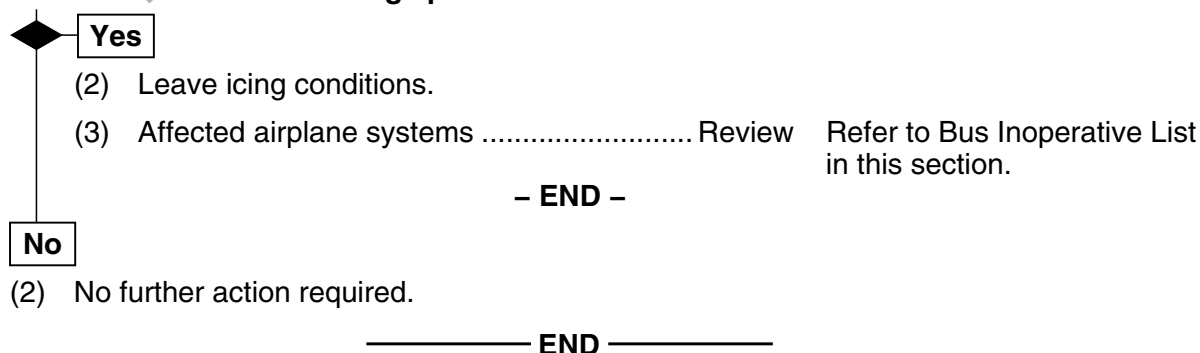
Manually transfer AC essential to alternate source:

(1) AC POWER, AC ESS XFER switch/light Press in to manually transfer the essential bus to AC BUS 2.

Check the following:

- ALTN light on, and
- **AC ESS ALTN** status message on.

AC ESS BUS caution message persists:



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1. ELECTRICAL (CONT'D)

F. ADG Inadvertent Deployment

Indication: ADG operating.

NOTE

The AC essential bus and hydraulic pump 3B can be transferred back to either main generator, or to the APU generator, if operating, by pressing the PWR TXFR OVERRIDE switch on the ADG control panel after the ADG manual deploy handle is re-stowed.

To re-establish normal operations:

- (1) ADG manual deploy handleSelect to STOW (if ADG was manually deployed).
- (2) ADG PWR TXFR OVERRIDE switchPush in to override.
- (3) HYDRAULIC pump switch 3BCheck selected to AUTO.
- (4) EFIS altimeters (2)Cross-check

NOTE

For flight planning, a fuel burn correction of 5% should be used when the ADG is deployed.

————— **END** —————

G. APU Battery Failure

Indication: **APU BATTERY OFF** caution message on. DC ELECTRICAL page – APU battery indicates less than 18 volts DC.

NOTE

APU BATTERY OFF caution message may appear momentarily during APU start.

- (1) BATT MASTER switchCheck ON.
- (2) DC ELECTRICAL pageMonitor APU battery voltage. Assume loss of APU battery. The APU is not available.
- (3) Affected airplane systemsReview Refer to Bus Inoperative List in this section.

————— **END** —————

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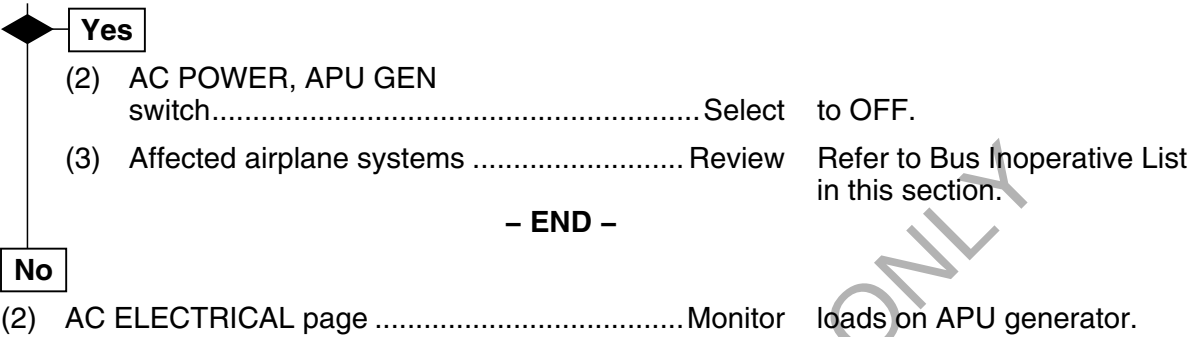
1. ELECTRICAL (CONT'D)

H. APU Generator Failure

Indication: **APU GEN OFF** caution message on.

- (1) AC POWER, APU GEN switch.....Select to OFF/RESET then ON.

APU GEN OFF caution message persists:



I. APU Generator Overload

Indication: **APU GEN OVLD** caution message on. AC ELECTRICAL page – APU generator load readout is greater than 35 KVA.

- (1) AC ELECTRICAL pageMonitor load on APU generator.
- (2) AC loads..... Reduce as necessary.

----- END -----

J. Battery Bus Failure

Indication: **BATTERY BUS** caution message on.

- (1) Land at the nearest suitable airport.
- (2) Affected airplane systems Review Refer to Bus Inoperative List in this section.

----- END -----

1. ELECTRICAL (CONT'D)

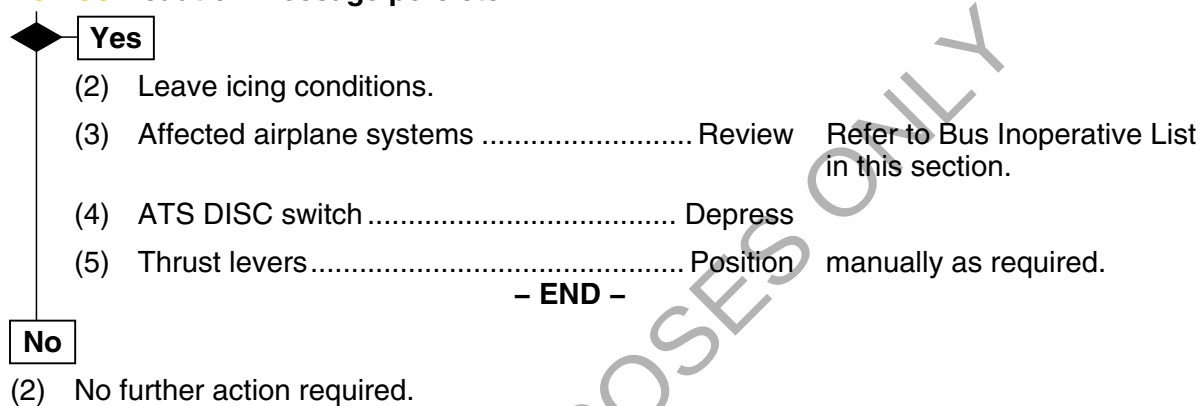
K. DC Bus 1 Failure

Indication: **DC BUS 1** caution message on. DC ELECTRICAL page – DC BUS 1 inoperative.

- (1) DC POWER, MAIN BUS TIE switch/light Press in to close bus tie manually.

Check MAIN BUS TIE
CLOSED light on.

DC BUS 1 caution message persists:



END

1. ELECTRICAL (CONT'D)

L. DC Bus 2 Failure

Indication: **DC BUS 2** caution message on. DC ELECTRICAL page – DC BUS 2 inoperative.

- (1) DC POWER, MAIN BUS TIE switch/light Press in to close bus tie manually.
Check MAIN BUS TIE CLOSED light on.

DC BUS 2 caution message persists:

- ◆ **Yes**
- (2) Affected airplane systems Review Refer to Bus Inoperative List in this section.

NOTE

MACH TRIM is inoperative.

- (3) Leave icing conditions as soon as possible.
- (4) Approach speed V_{REF} (Flaps 45°) + 10 KIAS minimum.
- (5) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

- (6) STAB TRIM channel Engage With at least one STAB TRIM channel engaged, manually trim airplane as required.

NOTE

With MACH TRIM inoperative, do not exceed 250 KIAS (0.7 M) unless the autopilot is engaged and functioning normally.

– END –

No

- (2) EFIS altimeters (2) Cross-check

————— END —————

1. ELECTRICAL (CONT'D)

M. DC Essential Bus Failure

Indication: **DC ESS BUS** caution message on. DC ELECTRICAL page – DC ESS BUS inoperative.

- (1) Leave icing conditions.
- (2) STAB TRIM channel..... Engage With at least one STAB TRIM channel engaged, manually trim airplane as required.

NOTE

MACH TRIM is inoperative. Do not exceed 250 KIAS (0.7 M) unless the autopilot is engaged and functioning normally.

- (3) L DISPLAYS switch..... Select to MFD REV.
- (4) AFCS SEL switch..... Select AFCS 2.
- (5) YAW DAMPER, YD 2 switch..... Press in to engage YD 2.

NOTE

YD 1 will be inoperative. If YD 2 does not re-engage, follow procedure for double yaw damper failure.

- (6) Affected airplane systems Review Refer to Bus Inoperative List in this section.
- (7) Land at the nearest suitable airport.

————— **END** —————

1. ELECTRICAL (CONT'D)

N. AC Generator 1 Failure

Indication: **GEN 1 OFF** caution message on.

(1) AC POWER, GEN 1 switch Select to OFF/RESET then ON.

GEN 1 OFF caution message persists:

◆ **Yes**

(2) AC POWER, GEN 1 switch Select to OFF.

NOTE

1. With generator 1 shutdown, HYD PUMP 2B is load shed.
2. During single generator operations, AC utility busses are shed, unless on ground with flaps at 0°.

(3) APU (if available) Start when at or below 20,000 feet.

(4) AC POWER, APU GEN switch Select to ON.

— END —

◆ **No**

(2) AC ELECTRICAL page Monitor voltage, load and frequency readouts on all generators.

— END —

1. ELECTRICAL (CONT'D)

O. AC Generator 2 Failure

Indication: **GEN 2 OFF** caution message on.

(1) AC POWER, GEN 2 switch Select to OFF/RESET then ON.

GEN 2 OFF caution message persists:

◆ **Yes**

(2) AC POWER, GEN 2 switch Select to OFF.

NOTE

1. With generator 2 shutdown, HYD PUMP 1B is load shed.
2. During single generator operations, AC utility busses are shed, unless on ground with flaps at 0°.

(3) APU (if available) Start when at or below 20,000 feet.

(4) AC POWER, APU GEN switch Select to ON.

(5) EFIS altimeters (2) Cross-check

— END —

◆ **No**

(2) AC ELECTRICAL page Monitor voltage, load and frequency readouts on all generators.

(3) EFIS altimeters (2) Cross-check

— END —

P. AC Generator Overload

Indication: **GEN 1 (2) OVLD** caution message on. AC ELECTRICAL page – generator load readout indicates greater than 35 KVA.

(1) AC ELECTRICAL page Monitor generator loads, and load shed as necessary.

(2) Affected airplane systems Review Refer to Bus Inoperative List in this section.

— END —

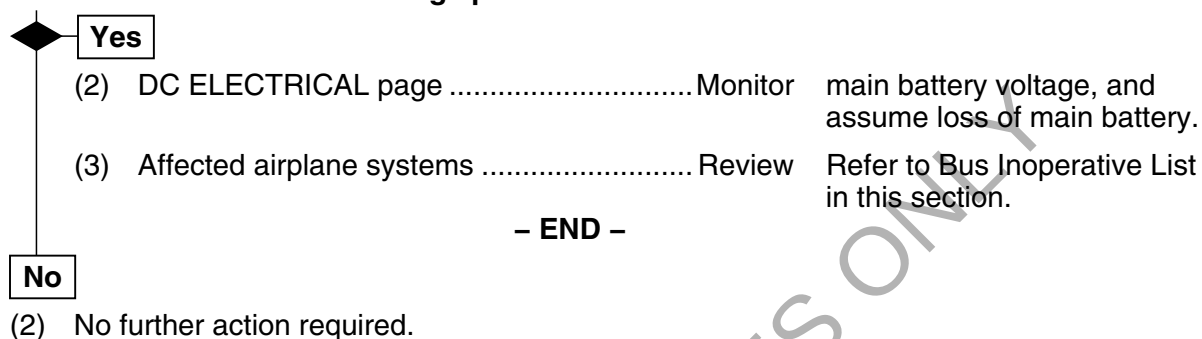
1. ELECTRICAL (CONT'D)

Q. Main Battery Failure

Indication: **MAIN BATT OFF** caution message on. DC ELECTRICAL page – Main battery indicates less than 18 volts DC, and main battery direct bus outline is amber.

- (1) BATT MASTER switchCheck ON.

MAIN BATT OFF caution message persists:



R. DC Emergency Bus Failure

Indication: **DC EMER BUS** caution message on. DC ELECTRICAL page – DC emergency bus outline is amber.

- (1) Affected airplane systemsReview Refer to Bus Inoperative List in this section.
- (2) Land at the nearest suitable airport.

————— **END** —————

1. ELECTRICAL (CONT'D)

S. Bus Inoperative List

NOTE

The bus inoperative list covers systems that are important to the approach and landing phases of flight. The list covers items that are NOT available. The list does not cover all electrically controlled and/or powered systems and is not a complete check of all circuit breaker panels to determine affected systems.

LEGEND:

♦ – Indicates that the applicable system is powered by a second bus. The failure of the two marked busses (♦) will render the applicable system inoperative.

* – If applicable (optional equipment).

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	APU BATT DIR BUS	MAIN BATT DIR BUS
ENGINE	IGN					A inop		B inop		
	START							L & R inop		
	OIL PRESS						R inop	L inop	BYPASS INDIC inop	
	THRUST REV						inop			
NAV/FT INST	PFD				2 inop		ISI, 1 inop			
	MFD				2 inop			1 inop		
	CCP/DCP				CCP 2 DCP 2 inop		DCP 1 inop	CCP 1 inop		
	ADC				2 inop		1 inop	STBY inop		
	IRS			♦ 3*	♦ 2		♦ 1	♦ 1,2,3*		
	CLK						inop			STBY PWR inop
	RAD ALT			1 inop	2 inop					
	NAV RCVR				2 inop		1 inop			
	DME			1 inop	2 inop					
	ATC XPDR				2 inop		1 inop			
	TCAS					inop				
	FMS				CDU 2 inop		CDU 3* inop	CDU 1 inop		
	GPS				2 inop			1 inop		
	TAWS	inop								
	WXR			inop	XM* inop					



1. ELECTRICAL (CONT'D)

S. Bus Inoperative List (Cont'd)

SYSTEM	SUB- SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	ADG BUS	DC EMER BUS
AIR COND. PRESS COOL	PRESS CONT				CPAM 2 inop		CPAM 1, EMER DEPRESS inop	AUTO inop		
	PACK				R inop		L inop			
	RAM AIR							SOV inop		
	10TH- STAGE BLEED				R inop		L inop, ISOL inop			
	LEAK DET					BOTH inop				
	AVIONIC COOLING	DISPLAY FAN inop								
FUEL	PUMPS				R inop			L & APU inop.		
	SOV's									L, R & APU inop
	XFLOW							inop		
	TAIL TANK XFER	PRIM PUMP inop	SEC PUMP inop	AUTO XFER, PRIM PUMP CONT inop	SEC PUMP CONT inop			inop		
	GRAV XFLOW							inop		
HYD	PUMPS	2B inop, ♦ 3B	1B, 3A inop	2B CONT inop	1B, 3A CONT inop			3B CONT inop	♦ 3B	
	GAUGE			2 inop	1 inop			3 inop		
	EDP SOV									L, R inop
ICE/RAIN	COWL A/ICE							L, R inop		
	WING A/ICE			AUTO 1 inop	AUTO 2 inop		MAN 1 inop	MAN 2 inop		
	14TH- STAGE BLEED			L & ISOL inop	R inop					
	PROBE HEAT	TAT, AUX AOA inop	R AOA, R PITOT R STATIC inop		ADS HTR R CONT inop	L AOA, L & STBY PITOT, L STATIC inop	ADS HTR L CONT inop			
	WSHLD	L inop	R inop	L CONT inop	R CONT inop					
	WINDOW		R inop		R CONT inop	L inop	L CONT inop			
	ICE DET		2 inop			1 inop				



1. ELECTRICAL (CONT'D)

S. Bus Inoperative List (Cont'd)

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	MAIN BATT DIR BUS
LDG GEAR	BRAKES							MLG OVHT DET inop	
	ANTI-SKID			♦	♦				
	NWS			♦	♦				
	PSEU			♦ LGC DOOR 1, DOOR 2 ♦ SPLR 1	♦ LGC ♦ SPLR 2		DOOR WARN inop	♦ LGC DOOR1, DOOR2 ♦ SPLR 1 ♦ SPLR 2 NGDR CONT inop	
	WOW			♦ 1	♦ 2			♦ 1 & 2	
LIGHT	LAND/TAXI	L inop	R inop	PULSE inop					
	EXT	NAV inop		REAR, A. COLL inop	BEACON, WING, A. COLL inop				
	INSTR			CHART HOLDER inop			PLT, CPLT CB PNL inop	PED, OVHD MAP LT 1, 2 inop	
EICAS	DCU 1						♦ A & B	♦ A & B	
	DCU 2							♦ A & B	♦ A & B
	LDU			L inop				R inop	
AUTO FLT	IAPS			♦ L	♦ R		♦ R	♦ L	
APU								♦ ECU	♦ ECU
COMM	VHF				2 inop		3* inop	1 inop	
	AUDIO CONT						♦ PLT, C/PLT	♦ PLT, C/PLT	
	HF			1 inop	2 inop				
	SELCAL			inop					



1. ELECTRICAL (CONT'D)

S. Bus Inoperative List (Cont'd)

SYSTEM	SUB-SYSTEM	AC BUS 1	AC BUS 2	DC BUS 1	DC BUS 2	AC ESS BUS	DC ESS BUS	DC BATT BUS	DC EMER BUS
ELEC	GCU							1, 2, 3 inop	
	ADG							DEPLOY inop	
OXY	INDIC							QTY inop	
FIREX	LOOPS							BOTH inop	
	ENG BTL								BOTH inop
	APU BTL								inop
FLT CONT	FLAPS		PDU 2 inop		CH 2 inop	PDU 1 inop	CH 1 inop		
	TRIMS			RUD inop	AIL inop				
	STAB TRIM		HSTA CH 1 inop		HSTCU CH 1 inop	HSTA CH 2 inop	HSTCU CH 2 inop		
	STALL PROT						R CH inop	L CH, STICK PUSH inop	
	GROUND SPOILERS SOLENOID VALVE			PWR 1 inop	PWR 2 inop				

END

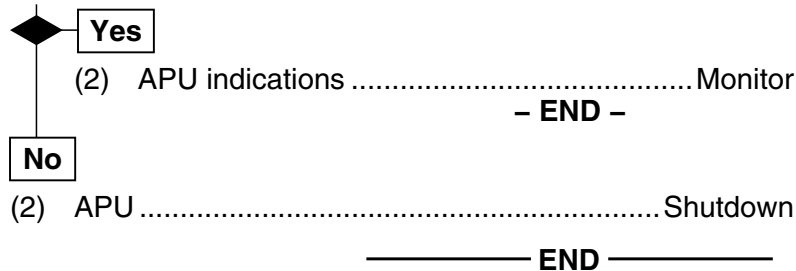
1. FIRE PROTECTION

A. APU Fire Detector Loop Failure

Indication: **APU FIRE FAIL** caution message on.

(1) Air-conditioning Transfer to engine bleeds.

APU generator is essential (both main generators failed):



B. Engine Fire Detector Loop Failure

Indication: **L (R) FIRE FAIL** caution message on.

(1) Affected engine Monitor
 — END —

C. Jetpipe Overheat Detector Failure

Indication: **L (R) JETPIPE OVHT FAIL** caution message on.

(1) Affected engine Monitor
 — END —

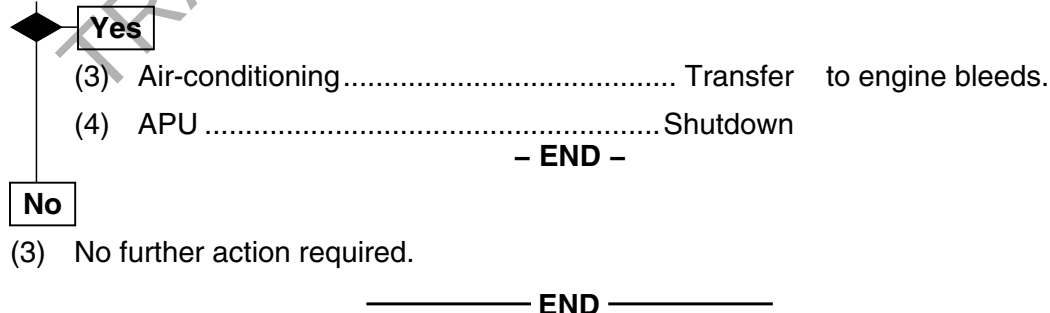
D. APU Firex Bottle Discharged

Indication: **APU BOTTLE LO** caution message on (without previous APU fire).

(1) FIRE DETECTION, TEST switch Select to WARN.

(2) Glareshield, APU BOTTLE switch/light Check on (confirming that bottle is charged).

APU BOTTLE light is out:



1. FIRE PROTECTION (CONT'D)

E. Engine Firex Bottle Discharged

Indication: **ENG BOTTLE 1 (2) LO** caution message on (without engine fire condition).

- (1) FIRE DETECTION, TEST switchSelect to WARN.
- (2) Glareshield, Engine BOTTLE 1
and 2 switch/lightsCheck on, confirming that engine
bottles are charged.
- (3) No further action required.

If any one engine bottle light fails to come on, assume that the affected bottle is discharged.

_____ **END** _____

TRAINING PURPOSES ONLY

1. FLIGHT CONTROLS

A. Lightning Strike

Following a lightning strike, failure of the stall protection system may occur without an associated indication to the crew. Failure of the stall protection system must be assumed.

- (1) STALL PROT, PUSHER switch
(either left or right) Select to OFF.

NOTE

Stick shakers remain operative.

Prior to landing:

- (2) Approach speed V_{REF} (Flaps 45°)
+ 10 KIAS minimum, to maintain margin above stall.
- (3) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

B. Stall Protection System Failure

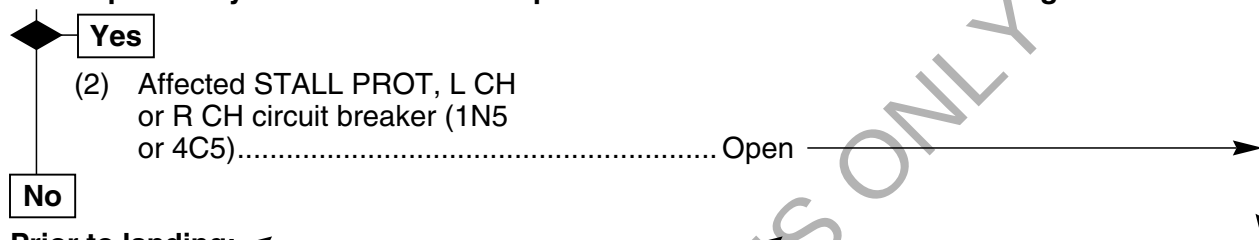
Indication: **STALL FAIL** caution message on.

- (1) STALL PROT, PUSHER switch
(either left or right) Select to OFF.

NOTE

Stick shakers remain operative.

Accompanied by a nuisance shaker operation and/or warbler aural warning:

**Prior to landing:**

- (2) Approach speed V_{REF} (Flaps 45°) + 10 KIAS minimum, to maintain margin above stall.
- (3) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

END

C. Stall Protection System Altitude Compensation Failure

Indication: **ALT COMP FAIL** caution message on.**NOTE**

The stick pusher may be inoperative. One or both channels of the stick shaker will remain operative.

Prior to landing:

- (1) Approach speed V_{REF} (Flaps 45°) + 10 KIAS minimum.
- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.40 (40%)	1.30 (30%)

END

1. FLIGHT CONTROLS (CONT'D)

D. Flight Spoilers Deployed During Flight

Indication: **FLT SPLRS DEPLOY** caution message on.

- (1) FLIGHT SPOILER lever Select to RETRACT.

————— **END** —————

E. Flight Spoilers Lever Jam

Indication: Flight spoilers cannot be retracted.

- (1) Approach speed V_{REF} (Flaps 45°) + 10 KIAS minimum.
- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.50 (50%)	1.40 (40%)

————— **END** —————

F. Flight Spoilers Failure to Deploy

Indication: Flight spoilers cannot be deployed.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

————— **END** —————

G. Flight Spoiler Asymmetric Extension

Indication: An uncommanded roll during flight spoiler extension.

- (1) FLIGHT SPOILER lever Retract Do not reselect in flight.
- (2) Runway Select to minimize crosswind.
- (3) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

H. Flight Spoiler Asymmetric Retraction

Indication: An uncommanded roll during flight spoiler retraction.

- | | | |
|-----------------------------------|------------------------------------|--|
| (1) Lateral control..... | Establish | using ailerons, and if necessary, rudder inputs. |
| (2) FLIGHT SPOILER lever | Retract/adjust | to eliminate roll asymmetry.
Make no further adjustments to spoilers in flight. |
| (3) Runway | Select | to minimize crosswind. |
| (4) Approach speed | V_{REF} (Flaps 45°)
+ 10 KIAS | minimum. |
| (5) Actual landing distance | Increase | as applicable by factor given below: |

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.50 (50%)	1.40 (40%)

————— **END** —————

I. Flight Spoilers Failure

Indication: **FLT SPLRS** caution message on.

On ground:

- (1) Do not take off.

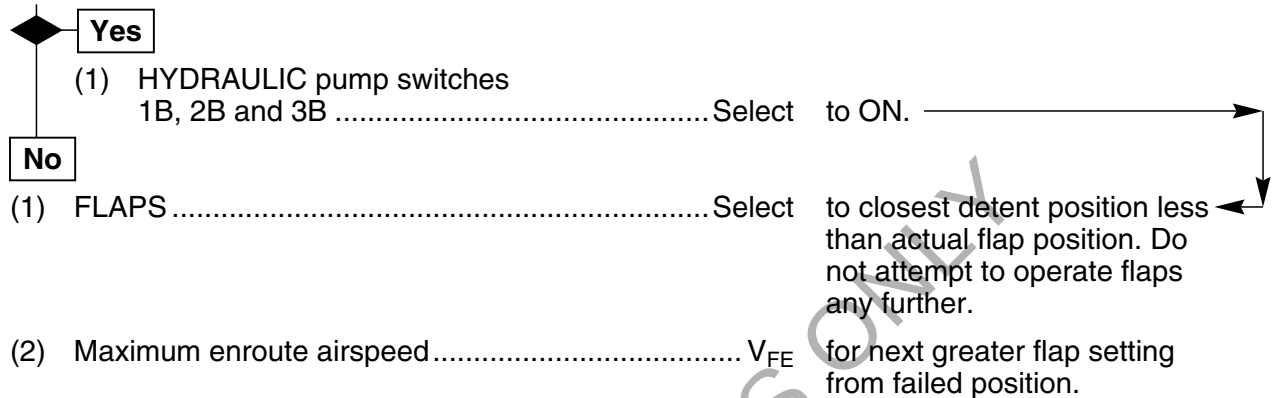
————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

J. Flaps Failure

Indication: **FLAPS FAIL** caution message on. FLIGHT CONTROLS page displays flap outline in amber.

Flaps failure occurred at less than 8 degrees:



NOTE

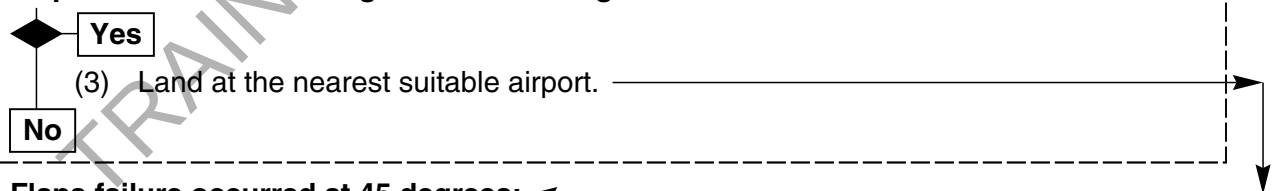
If flaps are confirmed retracted (0 degrees), reduction of cruise airspeed is not required.

CAUTION

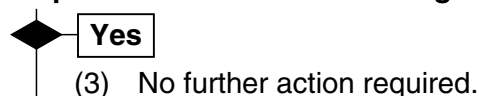
If possible, avoid prolonged flight in icing. If icing is encountered for an extended period of time, aerodynamic buffet may be experienced.

AR Certified Airplanes

Flaps failure occurred at greater than 5 degrees:



Flaps failure occurred at 45 degrees:



NOTE

If aerodynamic buffet is encountered after flying in icing conditions, reduce buffet by increasing the minimum approach speed to V_{REF} (Flaps 45°) + 5 KIAS.

- END -

1. FLIGHT CONTROLS (CONT'D)

J. Flaps Failure (Cont'd)

No

Prior to landing:

Flaps failure occurred at less than 20 degrees:

Yes

(3) Fuel balance Check

No

(3) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.

(4) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.

(5) Final approach speed V_{REF} (Flaps 45°) + ΔV_{REF} for next smaller flap setting from failed position.

(6) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

(7) Thrust levers Position manually as required.

(8) Actual landing distance Increase as applicable by factor given below:

Flaps Setting (Degrees)	ΔV_{REF} (KTS)	ALD Factor (Without Thrust Reversers)	ALD Factor (With Thrust Reversers)
0	30	1.80 (80%)	1.75 (75%)
20	14	1.55 (55%)	1.50 (50%)
30	7	1.40 (40%)	1.35 (35%)

NOTE

Accomplish a normal descent typically with a vertical speed of not more than 400 fpm at touchdown.



1. FLIGHT CONTROLS (CONT'D)

J. Flaps Failure (Cont'd)



Improper landing technique during a flaps failure can generate nose gear loads sufficient to cause structural damage.

To prevent damage:

- After main gear touchdown, gently lower the nose to the runway.
- Apply brakes only after nose wheel touchdown.

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

Example: If failure occurs at Flaps – 28 degrees:

Maximum Enroute Airspeed	= V_{FE} for Flaps 30 degrees
Approach & Landing Speed	= V_{REF} (Flaps 45°) + 14 KIAS
Factor Applied to Actual Landing Distance (without Thrust Reversers)	= 1.55 (55%)
Factor Applied to Actual Landing Distance (with Thrust Reversers)	= 1.50 (50%)

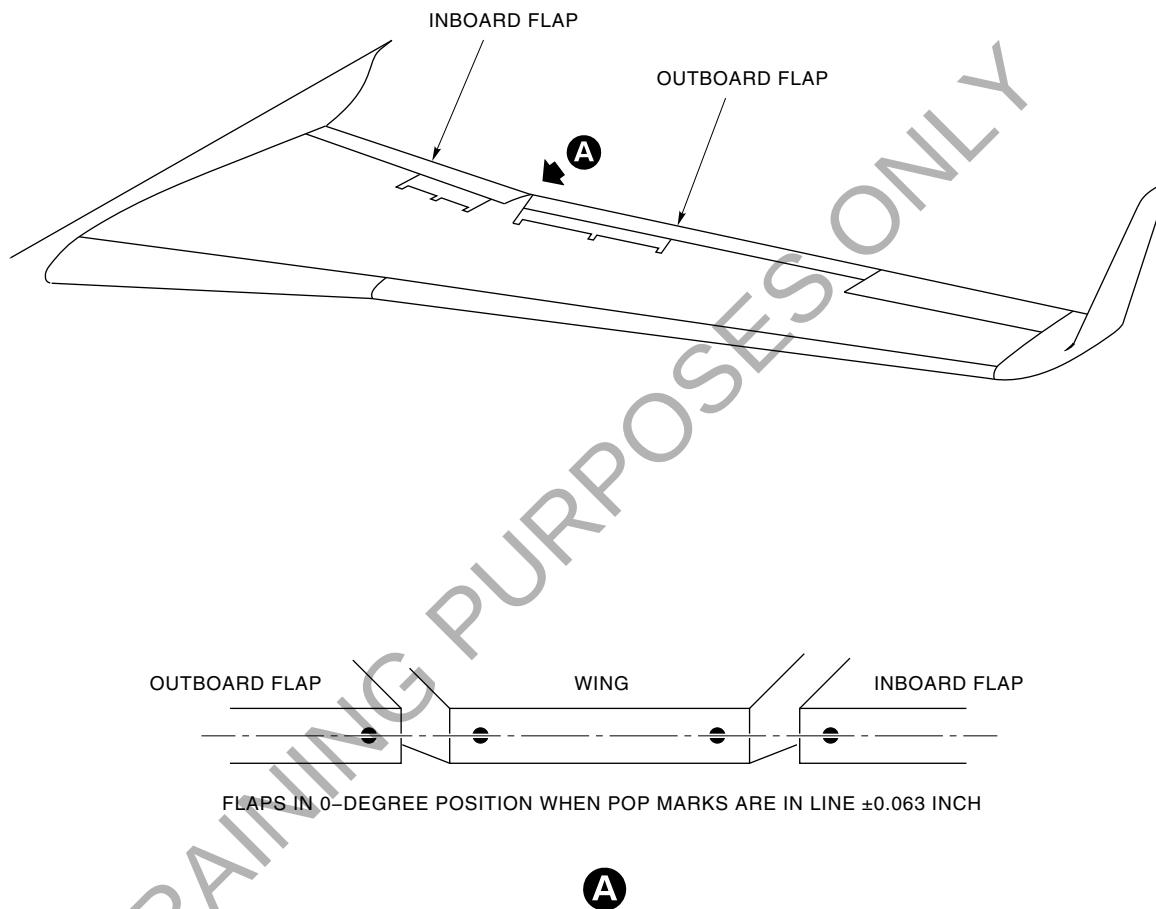
NOTE

If the failure occurs with the flaps at a detented flap position, use the V_{FE} , V_{REF} and landing distance factor relative to the flap position.

————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

K. Flap System Reset (On Ground) (Cont'd)



LF604_05_001

Flap System Reset
Figure 05-11-1

END

1. FLIGHT CONTROLS (CONT'D)

L. Ground Spoilers Failure

Indication: **GND SPLRS NOT ARMED** caution message on.

(1) GND SPOILERS switch Select to ARM.

GND SPLRS NOT ARMED caution message persists:



Yes

(2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

After touchdown:

(3) FLIGHT SPOILER lever Select to MAX.

– END –

No

(2) No further action required.

– END –

M. Ground Spoilers Unsafe

Indication: **GND SPLRS** caution message on.

On ground:



Yes

NOTE

GND SPLRS caution message may come on at taxi speeds above 16 knots.

(1) Airplane Stop

(2) Ground spoilers pre-flight check Accomplish

Refer to the Airplane Flight Manual, Chapter 4; NORMAL PROCEDURES – FLIGHT CONTROLS. Ground spoilers are unserviceable if message persists.

– END –

No

During flight:

(1) GND SPOILERS switch Select to DISARM.



1. FLIGHT CONTROLS (CONT'D)

M. Ground Spoilers Unsafe (Cont'd)

GND SPLRS caution message persists:

Yes

Prior to landing:

- (2) TAWS WARNING, FLAPS
OFF switch/light..... Press in to mute the flap aural warning.
- (3) TAWS WARNING,
RAAS OFF switch/light
(if installed) Press in Check OFF light on.
- (4) FLAPS Select to 20° for landing.
- (5) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (6) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (7) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

1. FLIGHT CONTROLS (CONT'D)

M. Ground Spoilers Unsafe (Cont'd)

No

Prior to landing:

- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

After touchdown:

- (3) GND SPOILERS switch Select to ARM.
 (4) FLIGHT SPOILER lever Select to MAX.

END

1. FLIGHT CONTROLS (CONT'D)

N. Ground Spoilers Deployed During Flight

Indication: **GND SPLRS DEPLOY** caution message on.

- (1) GND SPOILERS switch Select to DISARM.

Prior to landing:

- (2) TAWS WARNING, FLAPS OFF
switch/light Press in to mute the flap aural warning.
- (3) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (4) FLAPS Select to 20° for landing.
- (5) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (6) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (7) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

NOTE

Landing distance factors for failed ground spoilers are based upon serviceable flight spoilers.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

After touchdown:

- (8) GND SPOILERS switch Select to ARM, to deploy ground spoilers.
- (9) FLIGHT SPOILER lever Select to MAX.

END

1. FLIGHT CONTROLS (CONT'D)

O. Stabilizer Trim Failure

Indication: **STAB TRIM** caution message on and both stabilizer trim are disengaged.
MACH TRIM INOP light on and **MACH TRIM** caution message on.

- (1) Pilot's or copilot's STAB TRIM DISC switch Press hold and release.
- (2) STAB TRIM, CH1 and CH2 switch/lights Press in to engage both channels.
Check **STAB CH1** and **CH2 INOP** status messages out.

STAB TRIM caution message persists:

- Yes
- (3) STAB TRIM, CH2 switch/light Press in to engage.
Check **STAB CH1 INOP** status message on.

STAB TRIM caution message still persists:

- Yes
- NOTE**
1. Minimize changes to airspeed and configuration to minimize control forces or autopilot servo loads.
 2. Disregard **MACH TRIM** caution message.
- No
- (3) MACH TRIM switch/light Press in to engage. Check the following:
 - MACH TRIM INOP light out.
 - **MACH TRIM** caution message out.

– END –

1. FLIGHT CONTROLS (CONT'D)

O. Stabilizer Trim Failure (Cont'd)

Autopilot is engaged:

Yes

(4) Autopilot..... Consider leaving engaged as long as needed to reduce workload.

AP HOLDING NOSE DOWN (NOSE UP) caution message is displayed:

Yes

(5) Airspeed..... Adjust as required to keep autopilot engaged as long as possible.

NOTE

1. If **AP HOLDING NOSE DOWN** is displayed, decrease airspeed. If **AP HOLDING NOSE UP** is displayed, increase airspeed.
2. **AP HOLDING NOSE DOWN (NOSE UP)** caution message will go out if the airspeed is adjusted.

AP HOLDING NOSE DOWN (NOSE UP) caution message persists:

Yes

NOTE

Anticipate an out-of-trim condition with autopilot disengagement.

(6) Autopilot..... Disconnect

No

Prior to landing:

Failure occurred with flaps at 45°:

Yes

(7) FLAPS Leave at 45°.

CAUTION

Do not jettison fuel if trim runaway is in nose down direction.

Fuel jettison will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.

Consider use of fuel jettison in the event of a nose-up trim runaway condition.

A landing may be made with fuel in the tail tank.

- END -

1. FLIGHT CONTROLS (CONT'D)

O. Stabilizer Trim Failure (Cont'd)

No

- (7) TAWS WARNING, FLAPS OFF
switch/light Press in to mute the flap aural warning.
- (8) TAWS WARNING, RAAS OFF
switch/light (if installed) Press in Check OFF light on.
- (9) FLAPS Select to 20° for landing.
- (10) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (11) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (12) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

Do not jettison fuel if trim runaway is in nose down direction.

Fuel jettison will cause the CG to move forward requiring a larger pull force to counter and causing any push force to be reduced.

Consider use of fuel jettison in the event of a nose-up trim runaway condition.

A landing may be made with fuel in the tail tank.

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

1. FLIGHT CONTROLS (CONT'D)

P. Stabilizer Trim Malfunction

Indication: Stabilizer cannot be trimmed to the desired position. **STAB CH 1 INOP** status message is not displayed.

- (1) Pilot's or copilot's STAB TRIM DISC switch Press hold and release.

Check the following:

- **STAB TRIM** caution message on.
- **MACH TRIM** caution message on.

- (2) STAB TRIM, CH2 switch/light Press in to engage.

Check **STAB CH1 INOP** status message on.

NOTE

Do not engage STAB TRIM CH 1.

- (3) MACH TRIM switch/light Press in to engage.

Check the following:

- MACH TRIM INOP light out, and
- **MACH TRIM** caution message out.

————— **END** —————

Q. Mach Trim Failure

Indication: **MACH TRIM** caution message and MACH TRIM INOP light on.

With at least one STAB TRIM channel engaged:

- (1) MACH TRIM switch/light Press in to engage.

Check MACH TRIM INOP light out.

NOTE

With a Mach trim failure, do not exceed 250 KIAS (0.7 M), unless the autopilot is engaged and functioning normally.

————— **END** —————

1. FLIGHT CONTROLS (CONT'D)

R. Aileron PCU Malfunction

Indication: **AILERON PCU** caution message on.**NOTE**

It may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot Disconnect if coupled.

If necessary:

- (2) Aileron controls..... Overpower

Difficult to maintain desired flight path: **Yes****At a safe altitude:**

- (3) Aileron controls..... Release pilot/copilot differential pressure on the controls.
- (4) ROLL DISC handle..... PULL and TURN 90 degrees to lock.

NOTE

1. The pilot with the operative aileron will be required to maintain a roll control input to offset the effects of the aileron PCU malfunction.
2. If the right aileron control circuit is inoperative, the autopilot system is inoperative.

No

- (3) Land at the nearest suitable airport.

NOTE

Controllability is reduced. Select longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (4) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (5) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (6) FLAPS Select to 20° for landing.
- (7) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.



1. FLIGHT CONTROLS (CONT'D)

R. Aileron PCU Malfunction (Cont'd)

- (8) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (9) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

END

1. FLIGHT CONTROLS (CONT'D)

S. Elevator System Malfunction

Indication: **ELEVATOR SPLIT** caution message on.**NOTE**

It may be necessary to reduce airspeed to improve airplane handling.

- (1) Autopilot Disconnect if coupled.
- (2) Elevator controls..... Move slowly and monitor position on EICAS to identify malfunctioning elevator surface.

If necessary:

- (3) Elevator controls..... Overpower

Difficult to maintain desired flight path:

 **Yes**

At a safe altitude:

- (4) Elevator controls..... Release pilot/copilot differential pressure on the controls.
- (5) PITCH DISC handle PULL and TURN 90 degrees to lock.
- (6) Jammed side Determine If left side, do not re-engage autopilot.

Right side jammed:

 **Yes**

- (7) STALL PROT, PUSHER switch (either left or right) Select to OFF.

NOTE

Stick shaker on operative control column, operates at a lower intensity.

 **No**

- (4) Land at the nearest suitable airport.

NOTE

Controllability is reduced. Select the longest runway available with minimum turbulence and crosswind.

Prior to landing:

- (5) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.



1. FLIGHT CONTROLS (CONT'D)

S. Elevator System Malfunction (Cont'd)

- (7) FLAPS Select to 20° for landing.
- (8) Approach speed V_{REF} (Flaps 45°)
+ 14 KIAS minimum.
- (9) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (10) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

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TRAINING PURPOSES ONLY

1. FUEL

A. Fuel Imbalance

Indication: **FUEL IMBALANCE** caution message on. Left and right fuel quantity readouts indicate amber.



If a **FUEL IMBALANCE** caution message is displayed:

- Do not taxi until imbalance is rectified.
- Do not land until imbalance is rectified, unless a greater emergency exists.

NOTE

1. The maximum permissible fuel imbalance is 182 kg (400 lb) for take-off, taxi, and landing.
2. The maximum permissible fuel imbalance is 363 kg (800 lb) in flight.

(1) Fuel distribution Monitor to identify appropriate action required.

Affected high tank:

(2) FUEL, L (R) TO AUX XFLOW switch/light Press in to open crossflow valve.
Check L (R) TO AUX XFLOW ON light on.

(3) Fuel tank quantity readouts Monitor

When tank quantities have equalized:

(4) FUEL, affected L (R) TO AUX XFLOW switch/light Press out to close crossflow valve.
Check L (R) TO AUX XFLOW ON light out.



1. FUEL (CONT'D)

A. Fuel Imbalance (Cont'd)

Fuel imbalance persists:

◆ **Yes**

- (5) FUEL, GRAVITY/XFLOW switch/light Press in to open gravity crossflow valve.
 Check GRAVITY XFLOW OPEN light on.

NOTE

1. A steady heading sideslip will accelerate gravity crossflow operation. Fly one wing low slip (on the side with less fuel) to ensure proper operation and to accelerate gravity crossflow.
2. During uncoordinated flight, fuel indications are inaccurate. Accurate fuel indications will be obtained after 30 seconds when level and stabilized in coordinated flight.

On the low quantity side:

- (6) Affected engine Reduce thrust.
 (7) Fuel tank quantity readouts Monitor

When main tank quantities have equalized:

- (8) FUEL, GRAVITY/XFLOW switch/light Press out to close gravity crossflow valve.
 Check GRAVITY XFLOW OPEN light out.
 (9) Fuel tank quantity readouts Monitor

Imbalance persists (fuel leak from wing tank is suspected):

◆ **Yes**

- (10) Fuel Leak Procedure Accomplish Refer to Fuel Leak Procedure in this section.

— END —

No

- (5) No further action required.

— END —

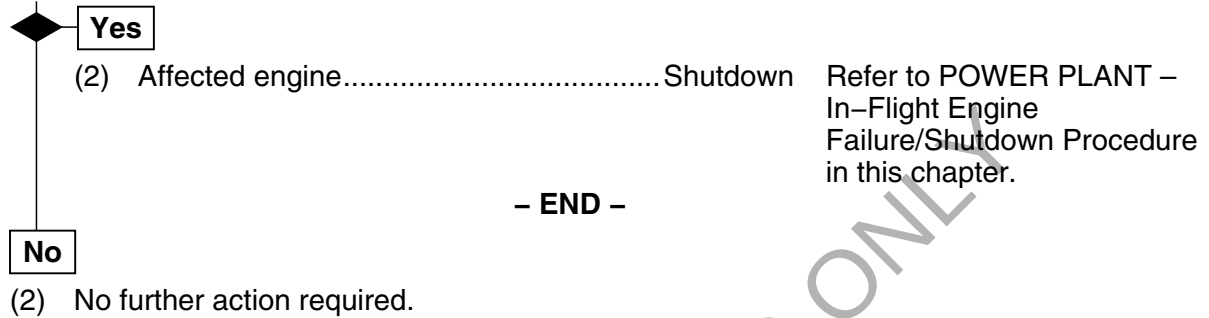
1. FUEL (CONT'D)

B. Engine Shut-Off Valve Failed Closed

Indication: **L (R) ENG SOV** caution message on.

- (1) Engine instruments..... Confirm engine failure using N_1 , fuel flow, ITT and N_2 .

Engine failure is confirmed:



C. Engine Shut-Off Valve Failed to Close After Engine Shutdown Due to Fire

Indication: **L (R) ENG SOV** caution message on.

- (1) Affected FUEL, BOOST PUMP switch/light..... Confirm pressed out.

Check the following:

- L (R) BOOST PUMP ON light out, and **L (R) FUEL PUMP ON** advisory message out,
- **L (R) FUEL PUMP** caution message on, and
- L (R) FUEL PUMP INOP light on.

- (2) Land at the nearest suitable airport.

————— **END** —————

D. Fuel Filter Impending Bypass

Indication: **L (R) FUEL FILTER** caution message on.

- (1) Affected engine Monitor engine indication and performance.

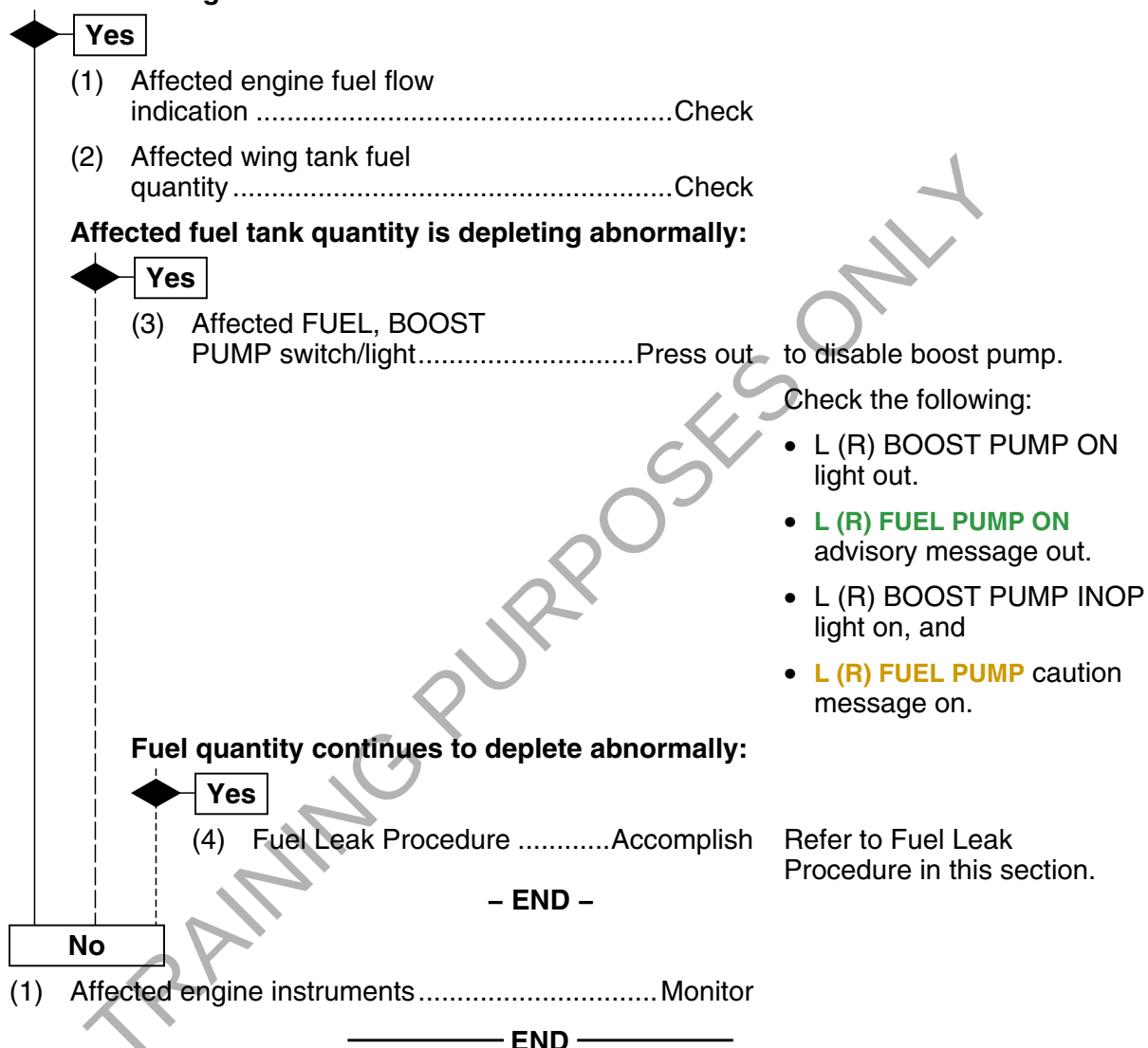
————— **END** —————

1. FUEL (CONT'D)

E. Low Fuel Pressure

Indication: **L (R) FUEL LO PRESS** caution message on.

Associated L (R) MAIN EJECTOR FAIL status message and/or L (R) SCAV EJECTOR caution message on:

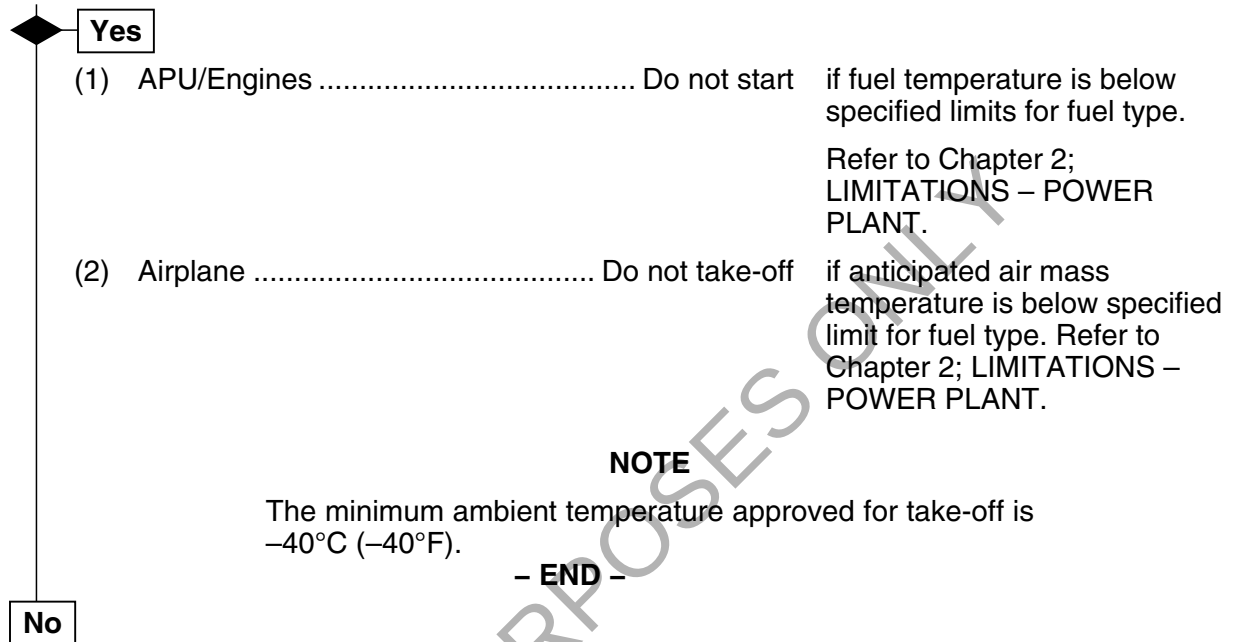


1. FUEL (CONT'D)

F. Bulk Fuel Temperature is Less than -37°C

Indication: **BULK FUEL TEMP** caution message on, and amber bulk fuel temperature readout on the SUMMARY page.

On ground:



During flight:

- (1) Airplane Operate as follows:
- Descend or deviate to warmer air mass, or
 - Increase airspeed.

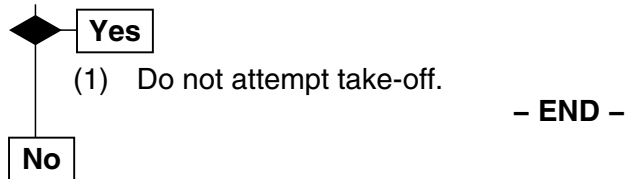
END

1. FUEL (CONT'D)

G. Fuel Temperature Less than 5°C

Indication: **L (R) FUEL LO TEMP** caution message on. Fuel temperature readout indicates caution range (amber).

On ground:



In flight:

(1) Affected engine instruments Monitor

NOTE

Under these conditions, the fuel feed to the engine is not preheated. **L (R) FUEL FILTER** caution message may come on, indicating an impending bypass.

————— **END** —————

H. Fuel Boost Pump Failure

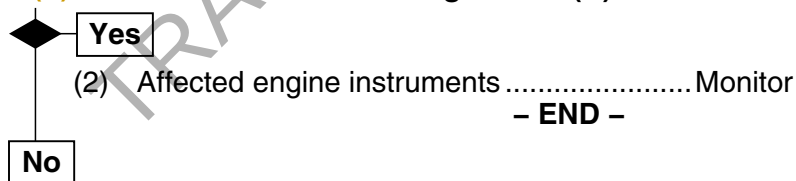
Indication: **L (R) FUEL PUMP** caution message and **L (R) BOOST PUMP INOP** light on.

(1) FUEL, L (R) BOOST PUMP switch/light Press out then press in.

Check the following:

- **L (R) BOOST PUMP INOP** light out.
- **L (R) FUEL PUMP** caution message out.

L (R) FUEL PUMP caution message and **L (R) BOOST PUMP INOP** light persists:



(2) No further action required.

————— **END** —————

1. FUEL (CONT'D)

I. Motive Flow Failure

Indication: **L (R) SCAV EJECTOR** caution message accompanied by **L (R) MAIN EJECTOR FAIL** status message.

(1) FUEL, L and R BOOST PUMP

switch/lights Check pressed in.

Check the following:

- L and R BOOST PUMP ON lights on,
- **L** and **R FUEL PUMP ON** advisory messages on, and
- **L** and **R FUEL PUMP** caution messages out.

Engine thrust is inadequate:

Yes

(2) ATS DISC switch Depress

(3) Unaffected engine Adjust thrust to maintain flight path. →

No

(2) Fuel quantity indicators Monitor for abnormal fuel depletion. ←

Total fuel quantity is depleting abnormally:

Yes

(3) Fuel Leak Procedure Accomplish Refer to Fuel Leak Procedure in this section.

— END —

No

(3) No further action required.

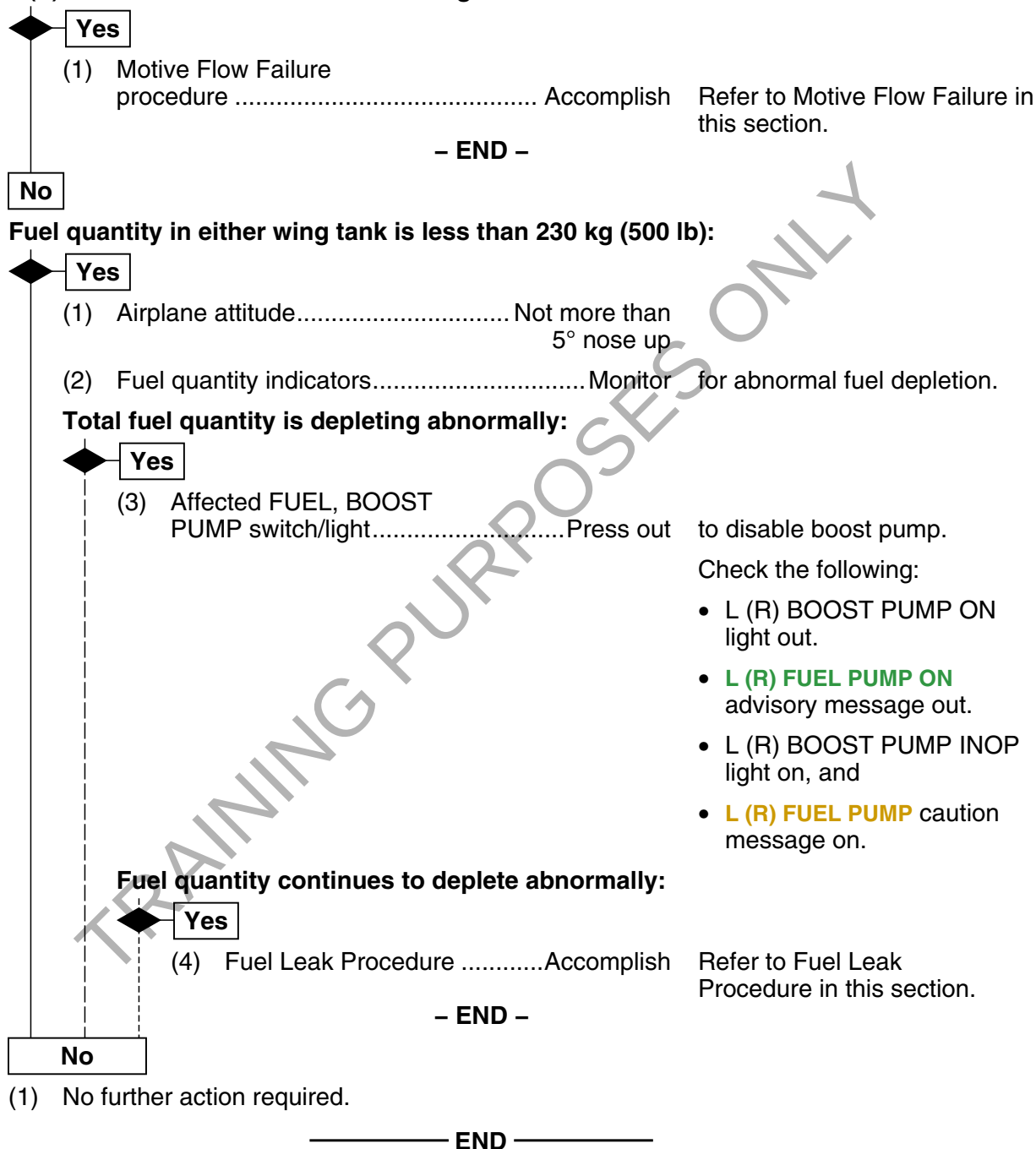
— END —

1. FUEL (CONT'D)

J. Scavenge Ejector Failure

Indication: **L (R) SCAV EJECTOR** caution message on.

L (R) MAIN EJECTOR FAIL status message on:



1. FUEL (CONT'D)

K. Primary Tail Tank Pump Failure (Manual Mode)

Indication: **PRI TAIL XFER PUMP** caution message on and PRI TAIL TANK TRANSFER FAIL light on.

- (1) FUEL, SEC TAIL TANK TRANSFER switch/light Press in to start secondary pump. Check the following:
 - SEC TAIL TANK TRANSFER ON light on, and
 - MANUAL OVRD, MANUAL light on.
- (2) FUEL, PRI TAIL TANK TRANSFER switch/light Press out
- (3) Fuel tank quantity readouts Monitor that secondary transfer pump maintains tail and auxiliary tank quantities normal (i.e., tail tank is depleting).

Transfer is inoperative:



- (4) Airplane attitude Level flight if practical.
- (5) FUEL, DUMP switch/light Press in to jettison fuel. Check DUMP OPEN light on.

NOTE

1. Fuel jettison must only be carried out with flaps set to 0.
2. Jettisoning of fuel in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

- (6) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting. Tail tank fuel is unusable.
 Continue to dump fuel until level flight is achieved.
 In level flight, continue to dump fuel until tail tank quantity is 635 kg (1,400 lb).
 If auxiliary tank quantity is below 1,361 kg (3,000 lb), dump fuel until tail tank is empty.

– END –



1. FUEL (CONT'D)

K. Primary Tail Tank Pump Failure (Manual Mode) (Cont'd)

No

When tail tank is depleted:

- (4) FUEL, SEC TAIL TANK TRANSFER switch/lightPress out

END

L. Secondary Tail Tank Pump Failure (Manual Mode)

Indication: SEC TAIL XFER PUMP caution message on and SEC TAIL TANK TRANSFER FAIL light on.

- (1) FUEL, PRI TAIL TANK TRANSFER switch/lightPress in to start primary pump. Check the following:
 - PRI TAIL TANK TRANSFER ON light on.
 - MANUAL OVRD, MANUAL light on.
- (2) FUEL, SEC TAIL TANK TRANSFER switch/lightPress out
- (3) Fuel tank quantity readoutsMonitor that primary transfer pump maintains tail and auxiliary tank quantities normal (i.e., tail tank is depleting).

Transfer is inoperative:

Yes

- (4) Airplane attitude.....Level flight if practical.
- (5) FUEL, DUMP switch/lightPress in to jettison fuel.
Check DUMP OPEN light on.

NOTE

- Fuel jettison must only be carried out with flaps set to 0.
- Jettisoning of fuel in known lightning conditions is prohibited.
- Tail tank contents will deplete from full to empty, in approximately 30 minutes.

1. FUEL (CONT'D)

L. Secondary Tail Tank Pump Failure (Manual Mode) (Cont'd)

- ▼ (6) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.
 Continue to dump fuel until level flight is achieved.
 In level flight, continue to dump fuel until tail tank quantity is 635 kg (1,400 lb).
 If auxiliary tank quantity is below 1,361 kg (3,000 lb), dump fuel until tail tank is empty.

- END -

No

When tail tank is depleted:

- (4) FUEL, PRI TAIL TANK TRANSFER switch/light Press out

- END -

M. Tail Tank to Aux Tank Automatic Transfer Failure

Indication: **AUTO TAIL XFER** caution message on.

NOTE

If a level sensor fails during fuel transfer, tail tank to auxiliary tank fuel transfer may continue after the **AUTO TAIL XFER** caution message is displayed.

On ground:

Yes

After AC power is applied:

- (1) FUEL FSCU CH 1 and FSCU CH 2 circuit breakers (1M11 and 4A10) Open

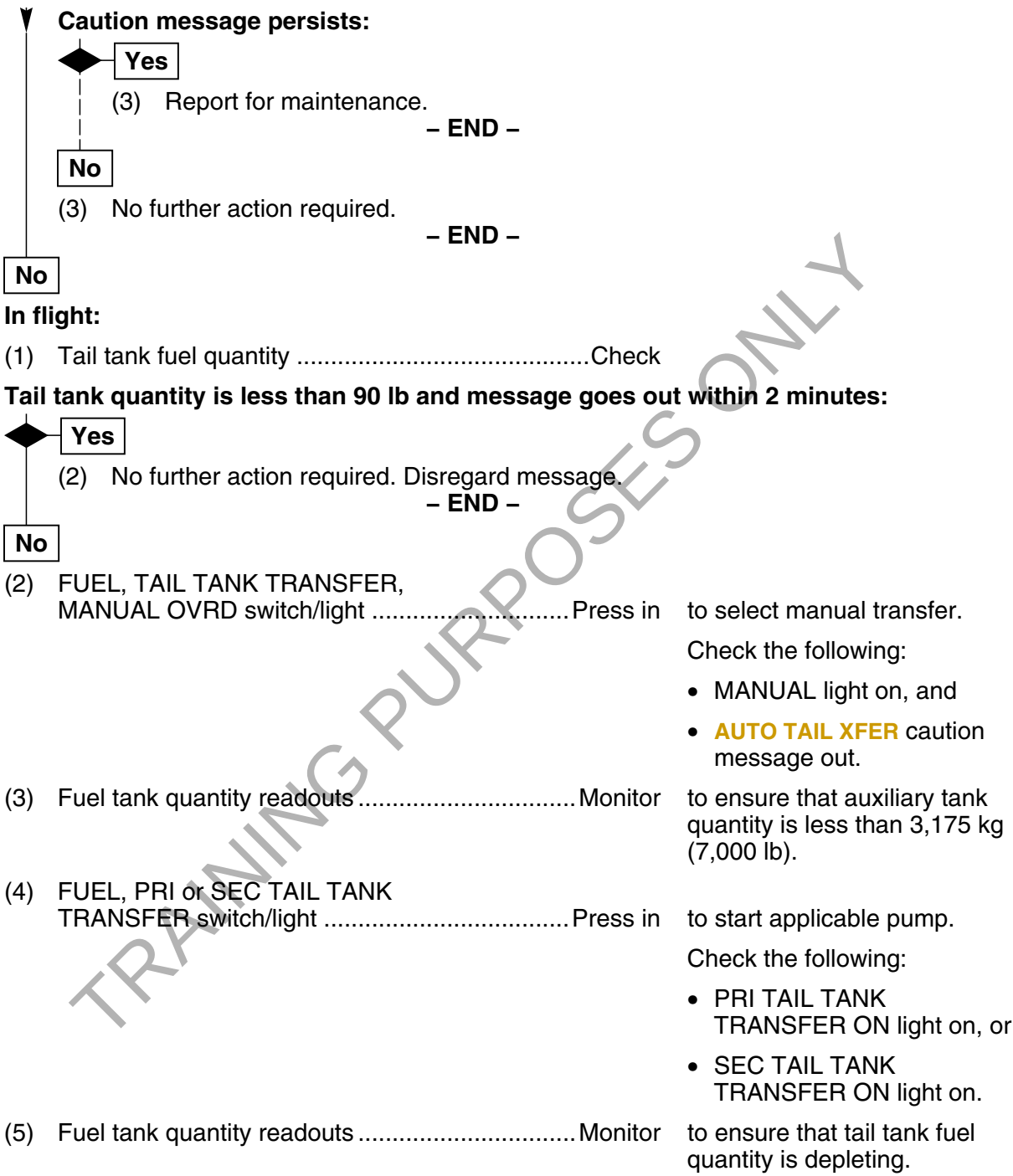
After 3 minutes:

- (2) FUEL FSCU CH 1 and FSCU CH 2 circuit breakers (1M11 and 4A10) Close



1. FUEL (CONT'D)

M. Tail Tank to Aux Tank Automatic Transfer Failure (Cont'd)



1. FUEL (CONT'D)

M. Tail Tank to Aux Tank Automatic Transfer Failure (Cont'd)

When tail tank is depleted:

- (6) FUEL, PRI or SEC TAIL TANK
TRANSFER switch/light Press out to stop applicable pump.

NOTE

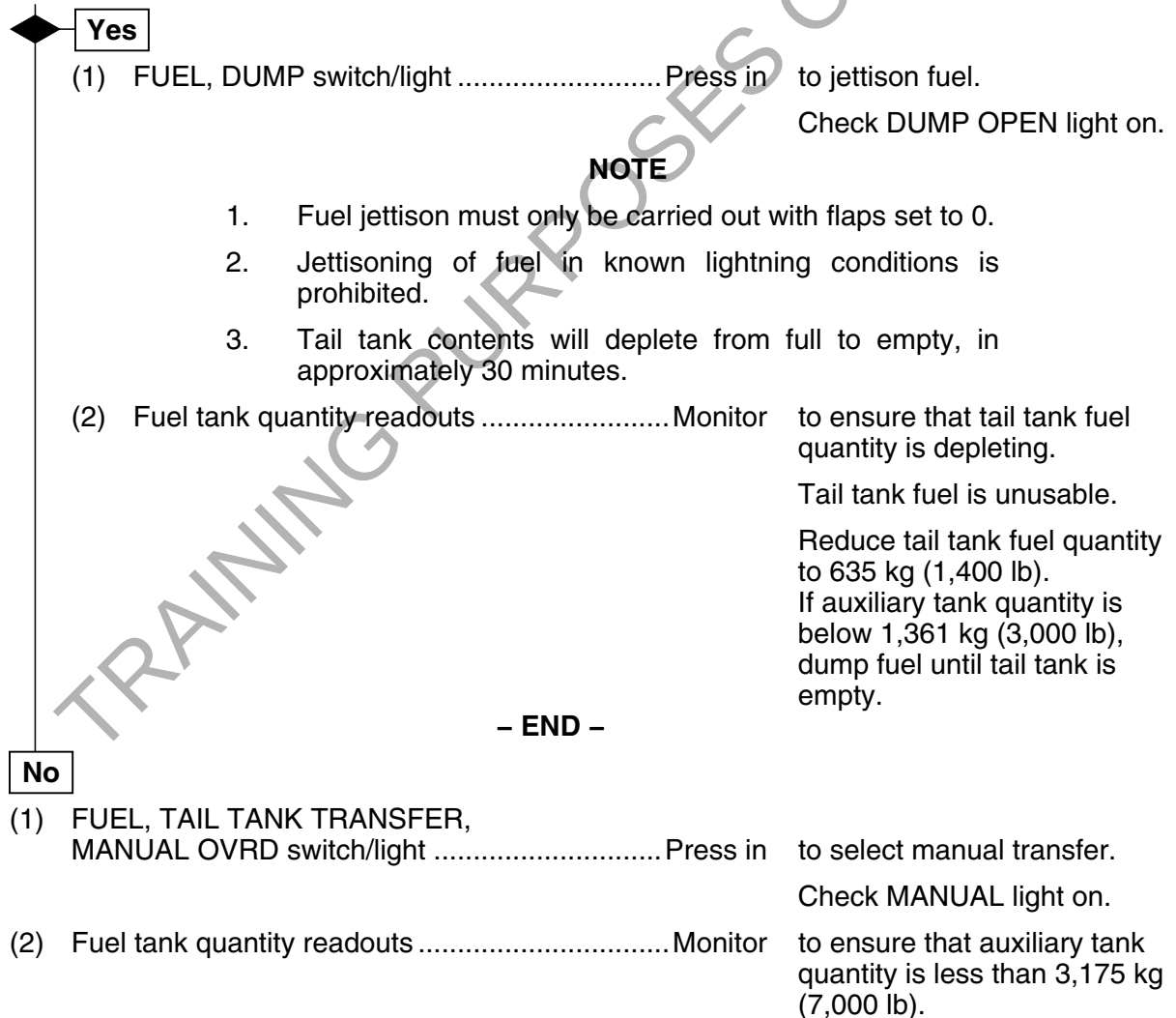
Do not deselect manual transfer until after landing.

— END —

N. Tail Tank Transfer is Inhibited

Indication: **AUTO TAIL XFER INHIB** caution message on.

Engine failure is suspected:



1. FUEL (CONT'D)

N. Tail Tank Transfer is Inhibited (Cont'd)

- (3) FUEL, PRI or SEC TAIL TANK
TRANSFER switch/light Press in to start applicable pump.
Check the following:
- PRI TAIL TANK
TRANSFER ON light on, or
 - SEC TAIL TANK
TRANSFER ON light on.
- (4) Fuel tank quantity readouts Monitor to ensure that fuel transfers from tail tank to auxiliary tank.

When tail tank is depleted:

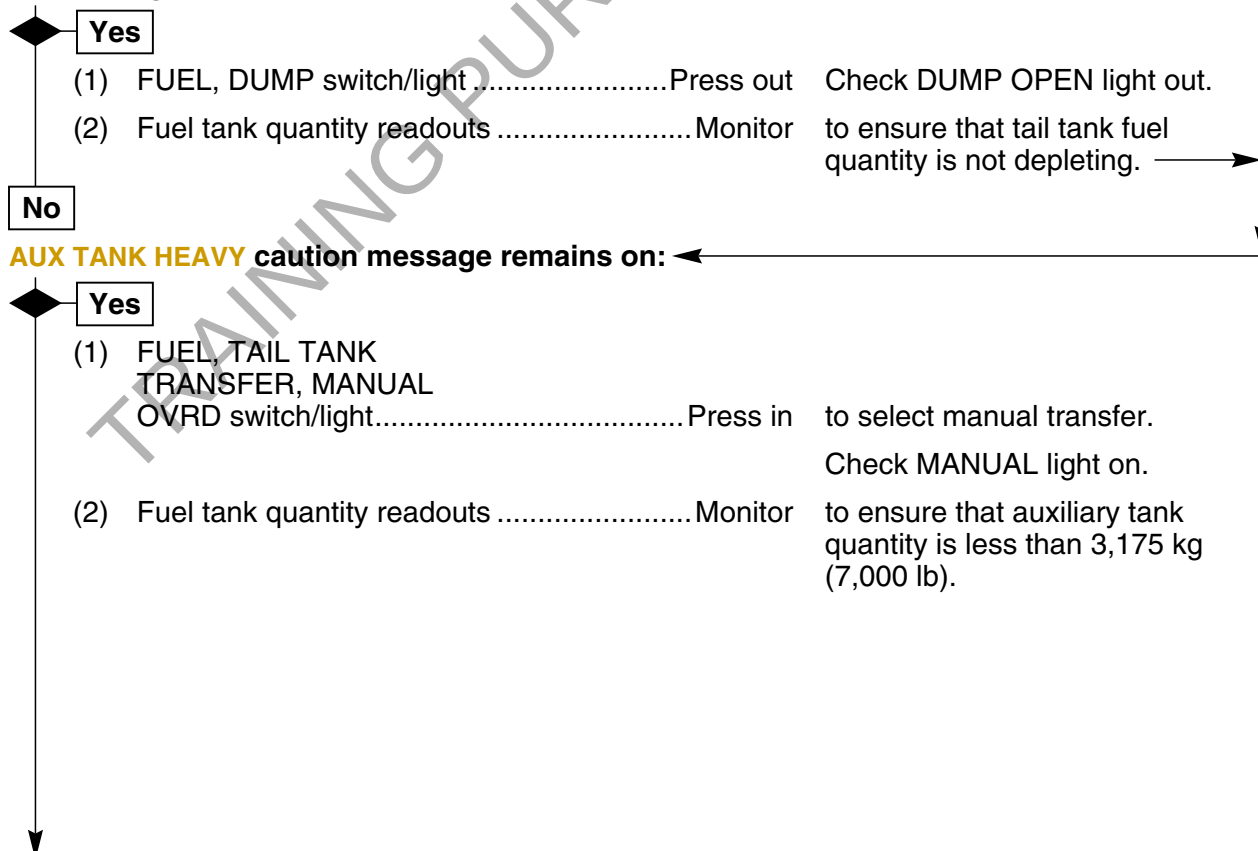
- (5) FUEL, PRI or SEC TAIL TANK
TRANSFER switch/light Press out to stop applicable pump.

END

O. Excessive Fuel Load – Aux Tank

Indication: **AUX TANK HEAVY** caution message on.

Fuel distribution between auxiliary and tail tank is abnormal.

Jettisoning fuel:

1. FUEL (CONT'D)

O. Excessive Fuel Load – Aux Tank (Cont'd)

- (3) FUEL, PRI and SEC TAIL
TANK TRANSFER

switch/lights Press out to stop both pumps.

Check the following:

- PRI TAIL TANK TRANSFER ON light out, and
- SEC TAIL TANK TRANSFER ON light out.

- (4) Fuel tank quantity readouts Monitor to ensure the following:

- Auxiliary tank fuel quantity is depleting, and
- Tail tank fuel quantity is not depleting.

NOTE

If auxiliary tank is not depleting, auxiliary and tail tank fuel is unusable.

AUX TANK HEAVY caution message persists:



Yes

Prior to landing:

- (5) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (6) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (7) FLAPS Select to 20° for landing.
- (8) Approach speed..... V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (9) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. FUEL (CONT'D)

O. Excessive Fuel Load – Aux Tank (Cont'd)

(10) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

– END –

No

(1) No further action required.

END

P. Excessive Fuel Load – Tail Tank

Indication: **TAIL TANK HEAVY** caution message on.

Fuel distribution between auxiliary and tail tank is abnormal.

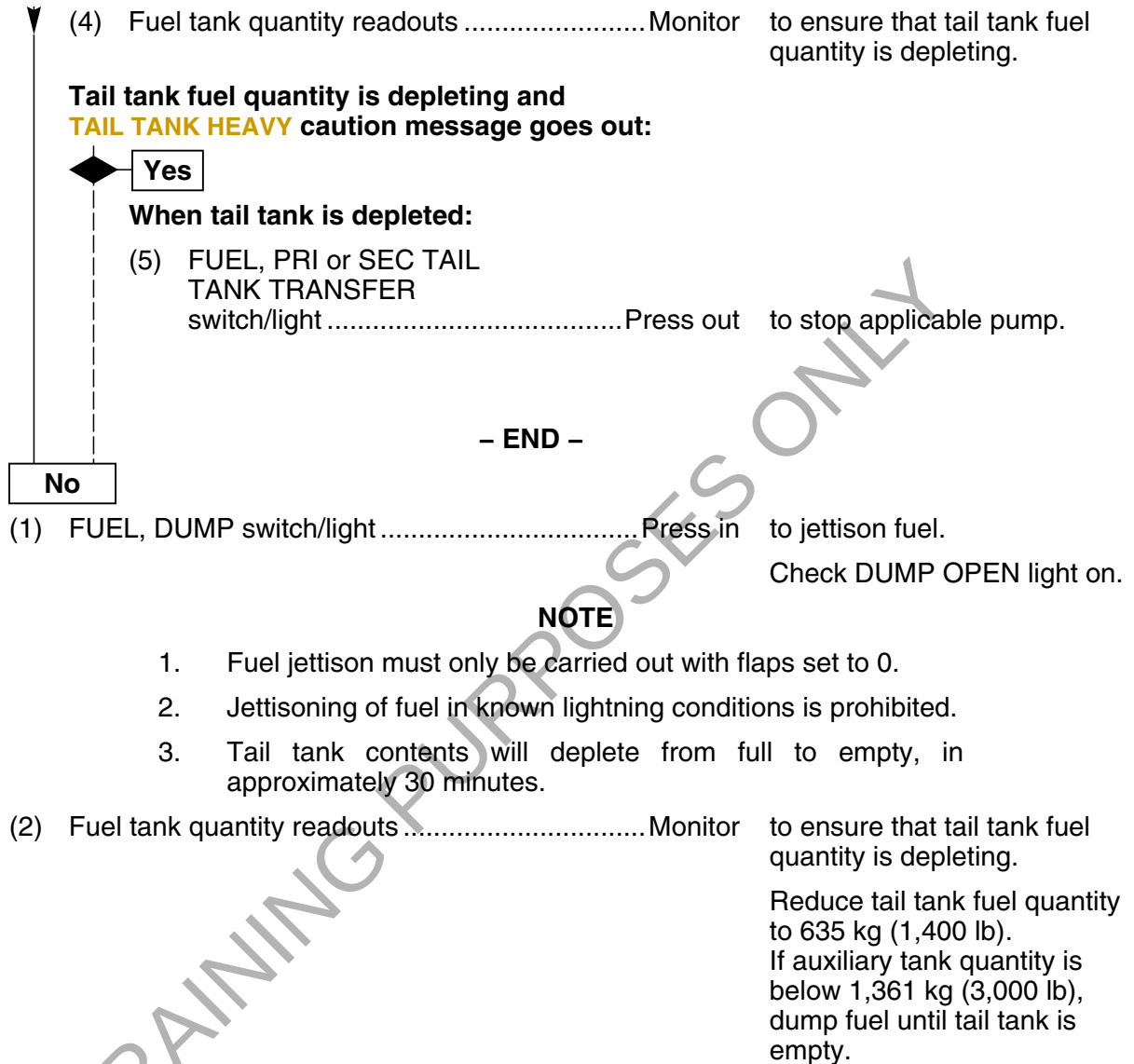
TAIL TANK TRANSFER AUTO INHIB light out:

Yes

- (1) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light..... Press in to select manual transfer.
Check MANUAL light on.
- (2) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).
- (3) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.

1. FUEL (CONT'D)

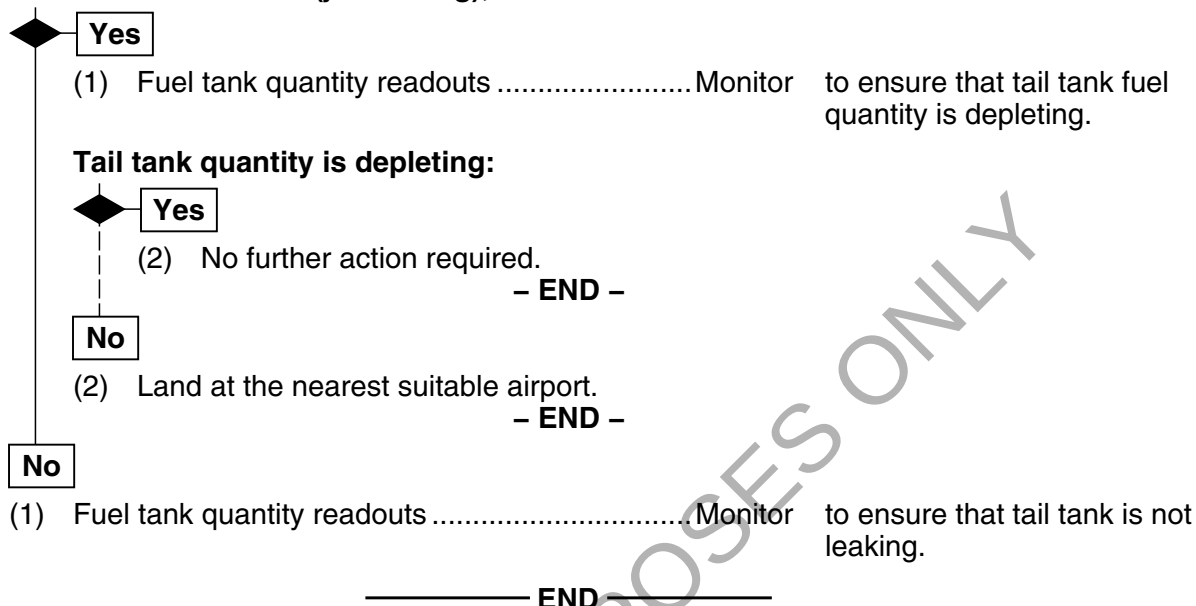
P. Excessive Fuel Load – Tail Tank (Cont'd)



END

1. FUEL (CONT'D)

Q. Tail Dump SOV Failed (SOV not in commanded position)

Indication: **TAIL DUMP SOV** caution message on.**FUEL DUMP** selected (jettisoning);

R. Tail Transfer SOV Failed Closed

Indication: **TAIL XFER SOV CLSD** caution message on.

- (1) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is not depleting.
- (2) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer. Check MANUAL light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).
- (4) FUEL, PRI TAIL TANK TRANSFER switch/light Press in to start primary pump. Check PRI TAIL TANK TRANSFER ON light on.

TAIL XFER SOV CLSD caution message persists:

1. FUEL (CONT'D)

R. Tail Transfer SOV Failed Closed (Cont'd)

(6) FUEL, PRI TAIL TANK
TRANSFER switch/light.....Press out

(7) Fuel tank quantity readoutsMonitor to ensure that tail tank fuel quantity is depleting.

Transfer is inoperative:



Yes

(8) Airplane attitude.....Level flight if practical.

(9) FUEL, DUMP switch/lightPress in to jettison fuel.
Check DUMP OPEN light on.

NOTE

1. Fuel jettison must only be carried out with flaps set to 0.
2. Jettisoning of fuel in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

(10) Fuel tank quantity
readouts.....Monitor to ensure that tail tank fuel quantity is depleting.
Tail tank fuel is unusable.
Continue to dump fuel until level flight is achieved.
In level flight, continue to dump fuel until tail tank quantity is 635 kg (1,400 lb).
If auxiliary tank quantity is below 1,361 kg (3,000 lb), dump fuel until tail tank is empty.

- END -

No

When tail tank is depleted:

(5) FUEL, PRI or SEC TAIL TANK
TRANSFER switch/lightPress out to stop applicable pump.

END

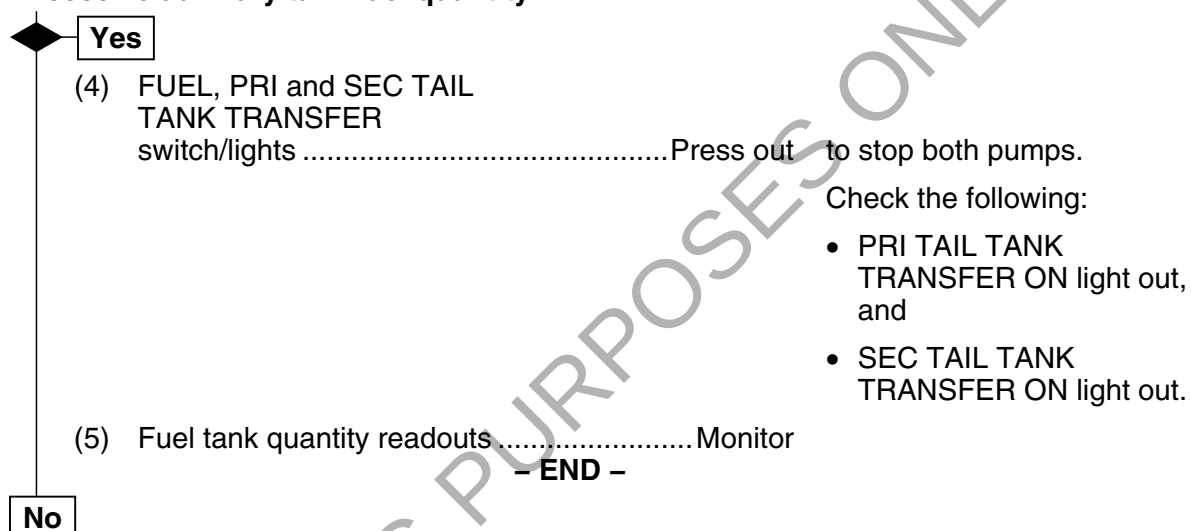
1. FUEL (CONT'D)

S. Tail Transfer SOV Failed Open

Indication: **TAIL XFER SOV OPEN** caution message on.

- (1) Fuel tank quantity readouts Monitor
- (2) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).

Excessive auxiliary tank fuel quantity:



Excessive tail tank fuel quantity:

- (4) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (5) Fuel tank quantity readouts Monitor

When tail tank is depleted:

- (6) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

————— **END** —————

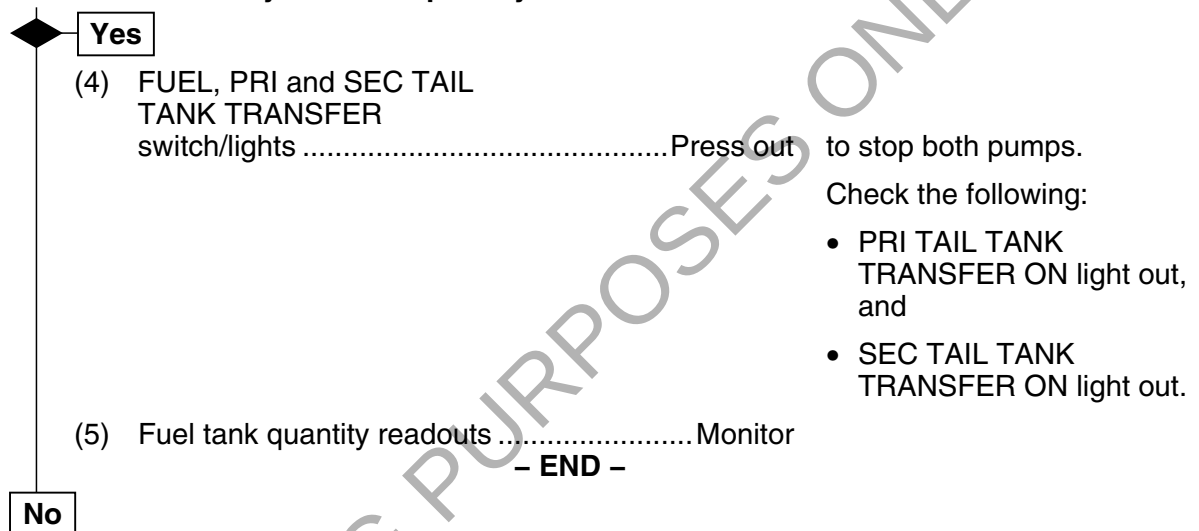
1. FUEL (CONT'D)

T. Tail Transfer SOV Failed in Middle Position

Indication: **TAIL XFER SOV** caution message on.

- (1) Fuel tank quantity readouts Monitor
- (2) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity is less than 3,175 kg (7,000 lb).

Excessive auxiliary tank fuel quantity:



Excessive tail tank fuel quantity:

- (4) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (5) Fuel tank quantity readouts Monitor

When tail tank is depleted:

- (6) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

————— **END** —————

1. FUEL (CONT'D)

U. L/R TO AUX Crossflow Valve Open

Indication: **FUEL XFLOW SOV OPEN** caution message on.

- (1) Fuel tank quantity readouts Monitor
- until contents of left and right main tanks are equalized.

When main tanks are equalized:

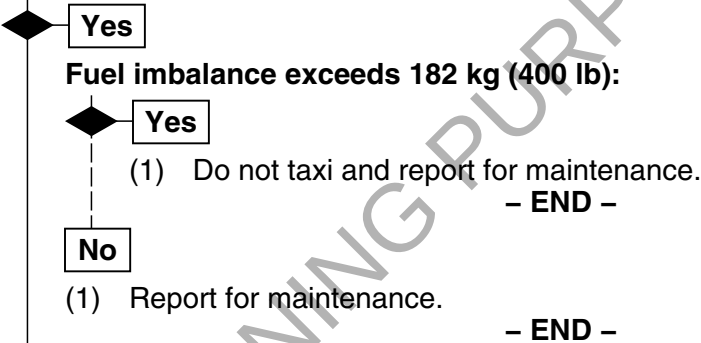
- (2) FUEL, L (R) TO AUX XFLOW switch/light.....Press out
- to close affected crossflow valve.
- Check the following:
- L (R) TO AUX XFLOW ON light out.
 - **FUEL XFLOW SOV OPEN** caution message out.

————— END —————

V. L/R TO AUX Crossflow Valve Fails to Close

Indication: Fuel crossflow continues after crossflow valve selected closed.

On ground:



Fuel imbalance exceeds 363 kg (800 lb):

- (1) Airplane attitude..... Level
- Steady heading, wings level.
- (2) FUEL, GRAVITY XFLOW switch/light.....Press in
- to open gravity crossflow valve.
- Check GRAVITY XFLOW OPEN light on.
- (3) Fuel tank quantity readouts Monitor
- until contents of left and right main tanks are equalized.

1. FUEL (CONT'D)

V. L/R TO AUX Crossflow Valve Fails to Close (Cont'd)

▼ When main tanks have equalized:

- (4) FUEL, GRAVITY XFLOW switch/light Press out to close gravity crossflow valve.
Check GRAVITY XFLOW OPEN light out.
- (5) Fuel tank quantity readouts Monitor
- END —

No

- (1) No further action required.

— END —

W. Auxiliary Tank Contents Increasing (abnormally) During L/R TO AUX Crossflow

Indication: Lateral imbalance.

- (1) Fuel tank quantity readouts Monitor and verify that lateral imbalance is corrected.
- (2) FUEL, L (R) TO AUX XFLOW switch/light Press out to close affected crossflow valve.
Check L (R) TO AUX XFLOW ON light out.

NOTE

Auxiliary tank contents may increase during L (R) TO AUX crossflow, depending upon fuel level.

— END —

X. Dump Valve Open for More than 5 Minutes

Indication: **DUMP VALVE OPEN** caution message on. Tail tank contents depleting rapidly.

- (1) Fuel tank quantity readouts Monitor
- (2) FUEL DUMP switch/light As required

NOTE

The tail tank contents will deplete from a full tank to empty, in approximately 30 minutes following selection.

— END —

1. FUEL (CONT'D)

Y. Loss of Auxiliary Tank Fuel Quantity Indication

Indication: Auxiliary tank fuel quantity indication dashed.

- (1) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (2) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press in to start applicable pump.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, or
 - SEC TAIL TANK TRANSFER ON light on.
- (3) Fuel tank quantity readouts Monitor to ensure that tail tank fuel quantity is depleting.

Tail tank quantity is not depleting:

- ☒ **Yes**
 (4) FUEL, DUMP switch/light Press in to jettison fuel.
 Check DUMP OPEN light on.

NOTE

1. Fuel jettison must only be carried out with flaps set to 0.
2. Jettisoning of fuel in known lightning conditions is prohibited.
3. Tail tank contents will deplete from full to empty, in approximately 30 minutes.

— END —

☐ **No**

When tail tank is depleted:

- (4) FUEL, PRI or SEC TAIL TANK TRANSFER switch/light Press out to stop applicable pump.

————— **END** —————

1. FUEL (CONT'D)

Z. Loss of Wing Tank Fuel Quantity Indication

Indication: Wing tank fuel quantity indication dashed.

- (1) ENGINE CONTROL, ENG SPEED switches Check both selected to ON.
- (2) Airplane attitude Level Steady heading, wings level.
- (3) Engine fuel flow indications Match left and right fuel flow rates.
- (4) Air-conditioning Transfer to engine bleeds.
- (5) APU Shutdown if not required.

NOTE

APU operation will increase fuel burn from right wing tank, by a rate of 45 to 136 kg/hr (100 to 300 lb/hr).

- (6) Airplane trim Adjust to achieve hands off flight.
- (7) Missing fuel tank quantity Calculate using FMS fuel used page and displayed quantity indications.
- (8) Fuel distribution Monitor for wing tank fuel imbalance.

Excessive imbalance exists (more than 25% of available trim required): **Yes****Affected high tank:**

- (9) FUEL, L (R) TO AUX XFLOW switch/light Press in to open crossflow valve.
Check L (R) TO AUX XFLOW ON light on.

NOTE

Cross flow transfers fuel at a rate of approximately 7 kg/min (15 lb/min) or higher.

When excessive imbalance has been corrected:

- (10) FUEL, affected L (R) TO AUX XFLOW switch/light Press out to close crossflow valve.
Check L (R) TO AUX XFLOW ON light out.

No

- (9) Fuel reserves Increase by 5%.

END

1. FUEL (CONT'D)

AA. Loss of Tail Tank Fuel Quantity Indication

Indication: Tail tank fuel quantity indication dashed.

- (1) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light Press in to select manual transfer.
Check MANUAL light on.
- (2) FUEL, PRI and SEC TAIL TANK TRANSFER switch/lights..... Press in to start both pumps.
Check the following:
 - PRI TAIL TANK TRANSFER ON light on, and
 - SEC TAIL TANK TRANSFER ON light on.
- (3) Fuel tank quantity readouts Monitor to ensure that auxiliary tank quantity does not exceed 3,175 kg (7,000 lb).

When tail tank is depleted:

- (4) FUEL, PRI and SEC TAIL TANK TRANSFER switch/lights..... Press out

END

AB. Loss of Total Fuel Quantity Indication

Indication: Total fuel quantity indication dashed.

- (1) Fuel tank quantity readouts Monitor sum of each tank vs fuel burn.

END

1. FUEL (CONT'D)

AC. Fuel Leak Procedure

Indication: Fuel leak is evident, suspected or confirmed.

- (1) Diversion may be required.

NOTE

If visibility permits, a visual check from the cabin may enable identification of the leak source.

Leak is confirmed or suspected:

- (2) Land immediately at the nearest suitable airport.

CAUTION

Do not delay landing while attempting to determine location of leak.

Expedite landing if tank quantity gauge indications are amber.

NOTE

1. The minimum fuel quantity for go-around is 230 kg (500 lb) per wing (with the airplane level), and assuming a maximum airplane climb attitude of 10° nose up.
 2. Do not action **FUEL IMBALANCE**, **L (R) FUEL LO PRESS** or **L (R) SCAV EJECTOR** caution messages or **L (R) MAIN EJECTOR FAIL** status messages. Disregard fuel imbalance limitations.
- (3) Autopilot Monitor operation.

NOTE

Anticipate out of trim situation when disconnecting autopilot.

Fuel leak from engine is suspected:

Yes

On low quantity side:

- (4) Thrust lever..... Confirm and IDLE
- (5) Thrust lever..... Confirm and SHUT OFF
- (6) Affected ENG FIRE PUSH switch/light Press in to close engine fuel feed SOV.
- (7) Affected engine..... Shutdown Refer to POWER PLANT – In-Flight Engine Failure/Shutdown Procedure in this chapter.

– END –

1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)

No

Fuel leak from wing tank is suspected:

- (4) FUEL, GRAVITY/XFLOW switch/light.....Press out to close gravity crossflow valve.
 Check GRAVITY XFLOW OPEN light out.

Range is a consideration:

Yes

NOTE

Consider use of asymmetric thrust to use as much fuel as possible from the leaking tank.

Leak in auxiliary tank is confirmed:

Yes

- (5) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light.....Press in to select manual transfer.
 Check MANUAL light on.
- (6) Auxiliary fuel tank quantity readout.....Monitor

When auxiliary tank quantity has stabilized:

- (7) FUEL, PRI TAIL TANK TRANSFER switch/light.....Press in to start primary pump.
 Check PRI TAIL TANK TRANSFER ON light on.
 Ensure auxiliary and tail tank quantities remain within limits.
 Refer to Figure 02-05-4.

– END –

No

Leak in tail tank is confirmed:

Yes

- (5) FUEL, TAIL TANK TRANSFER, MANUAL OVRD switch/light.....Press in to select manual transfer.
 Check MANUAL light on.

1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)

(6) FUEL, PRI and SEC TAIL
TANK TRANSFER

switch/lights Press in to start both pumps.

Check the following:

- PRI TAIL TANK TRANSFER ON light on, and
- SEC TAIL TANK TRANSFER ON light on.

Ensure auxiliary and tail tank quantities remain within limits. Refer to Figure 02-05-4.

When **AUX TANK HEAVY** caution message is displayed:

(7) FUEL, PRI and SEC TAIL
TANK TRANSFER

switch/lights Press out to stop both pumps.

Check the following:

- PRI TAIL TANK TRANSFER ON light out, and
- SEC TAIL TANK TRANSFER ON light out.

- END -

No

Leak in wing tank is confirmed:

Yes

(5) FUEL, L (R) TO AUX
XFLOW switch/light

Press in to open affected crossflow valve.

Check L (R) TO AUX XFLOW ON light on.

NOTE

Anticipate increasing lateral trim and control forces.

When trim or control forces becomes excessive:

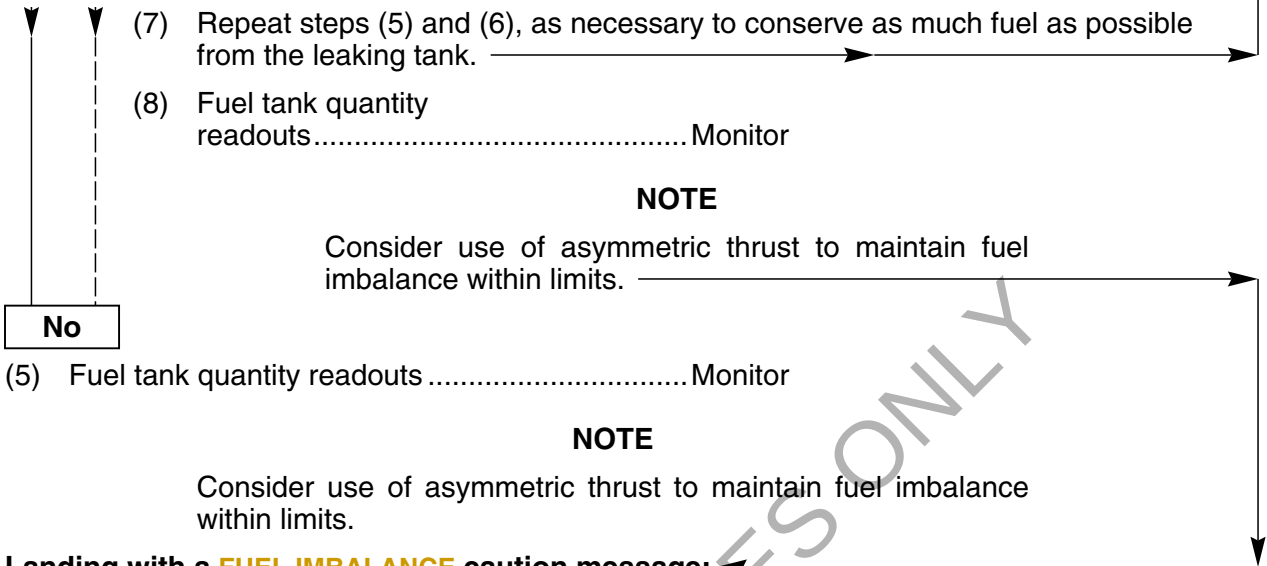
(6) FUEL, L (R) TO AUX
XFLOW switch/light

Press out to close affected crossflow valve.

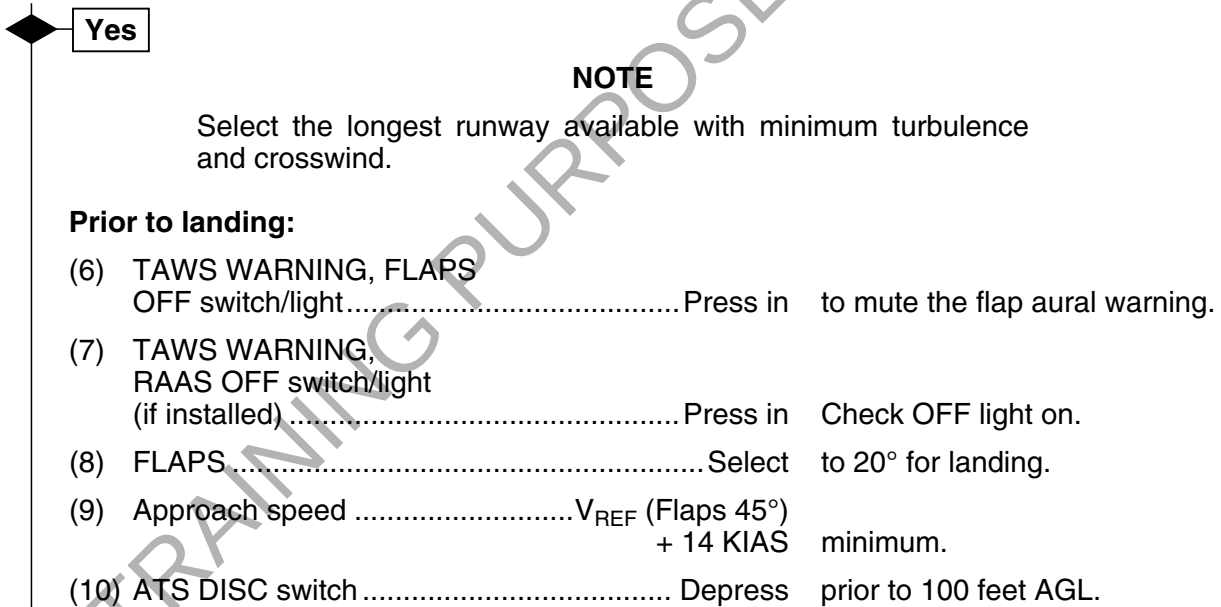
Check L (R) TO AUX XFLOW ON light out.

1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)



Landing with a **FUEL IMBALANCE** caution message:



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. FUEL (CONT'D)

AC. Fuel Leak Procedure (Cont'd)

- ▼ (11) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.55 (55%)	1.50 (50%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

- (6) No further action required.

— END —

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TRAINING PURPOSES ONLY

1. HYDRAULIC POWER

A. System No. 1 Failure

Indication: **HYD 1 LO PRESS** caution message on.

- (1) HYDRAULIC pump switch 1B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 1 quantity readout is less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing: **Yes**

- (3) HYDRAULIC pump switch 1B Select to OFF.
- (4) Land at the nearest suitable airport.
- (5) Inoperative systems Review Refer to HYDRAULIC page.

NOTE

The engine-driven pump (EDP) is operating without hydraulic fluid; log length of time that EDP has been running dry. Monitor fluid temperature readouts.

- (6) ANTI SKID switch Leave ARMED.
- (7) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (8) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (9) FLAPS Select to 20° for landing.
- (10) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (11) ATS DISC switch Depress prior to 100 feet AGL.

 CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. HYDRAULIC POWER (CONT'D)

A. System No. 1 Failure (Cont'd)

- ▼ (12) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.60 (60%)	1.55 (55%)

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

- (3) HYDRAULIC pump switch 1B Leave at ON.
(4) Hydraulic pressure and fluid quantity Monitor

— END —

1. HYDRAULIC POWER (CONT'D)

B. System No. 2 Failure

Indication: **HYD 2 LO PRESS** caution message on.

- (1) HYDRAULIC pump switch 2B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 2 quantity readout is less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:**Yes**

- (3) HYDRAULIC pump switch 2B Select to OFF.
- (4) Land at the nearest suitable airport.
- (5) Inoperative systems Review Refer to HYDRAULIC page.

NOTE

The engine-driven pump (EDP) is operating without hydraulic fluid; log length of time that EDP has been running dry. Monitor fluid temperature readouts.

- (6) ANTI SKID switch Leave ARMED.
- (7) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE

Maximize the use of reverse thrust.

— END —

No

- (3) HYDRAULIC pump switch 2B Leave at ON.
- (4) Hydraulic pressure and fluid quantity Monitor

— END —

1. HYDRAULIC POWER (CONT'D)

C. System No. 3 Failure

Indication: **HYD 3 LO PRESS** caution message on.

- (1) HYDRAULIC pump switch 3B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 3 quantity readout is less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:**Yes**

- (3) HYDRAULIC pump switches 3A and 3B Select to OFF.
- (4) Land at the nearest suitable airport.
- (5) Inoperative systems Review Refer to HYDRAULIC page.
- (6) ANTI SKID switch Leave ARMED.
- (7) Landing Gear Manual Extension procedure Accomplish Refer to LANDING GEAR, WHEEL AND BRAKE SYSTEM – Landing Gear Manual Extension procedure in this chapter.
- (8) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE

Maximize the use of reverse thrust.

– END –**No**

- (3) HYDRAULIC pump switch 3B Leave at ON.
- (4) Hydraulic pressure and fluid quantity Monitor

END

1. HYDRAULIC POWER (CONT'D)

D. System No. 1 High Temperature

Indication: **HYD 1 HI TEMP** caution message on and temperature readouts greater than 96°C (205°F).

NOTE

I Disregard **HYD 1 HI TEMP** caution message if one other hydraulic system has already failed.

During flight:

Temperature is less than 107°C (225°F):

- Yes**
- (1) HYDRAULIC pump switch 1B Select to OFF.
 - (2) System No. 1 temperature Monitor
- END -

No

Temperature is less than 135°C (275°F):

- Yes**
- (1) HYDRAULIC pump switch 1B Select to OFF.
 - (2) System No. 1 temperature Monitor
 - (3) Land at the nearest suitable airport.
- END -

No

- (1) HYDRAULIC pump switch 1B Select to OFF.
- (2) Left engine Shutdown Refer to POWER PLANT – In-Flight Engine Failure/Shutdown in this chapter.
- (3) Inoperative systems Review Refer to HYDRAULIC page.
- (4) Actual landing distance Increase by a factor of 1.60 (60%).

- END -

1. HYDRAULIC POWER (CONT'D)

E. System No. 2 High Temperature

Indication: **HYD 2 HI TEMP** caution message on and temperature readouts greater than 96°C (205°F).

NOTE

I Disregard **HYD 2 HI TEMP** caution message if one other hydraulic system has already failed.

During flight:

Temperature is less than 107°C (225°F):

- Yes**
- (1) HYDRAULIC pump switch 2B Select to OFF.
 - (2) System No. 2 temperature Monitor
- END -

No

Temperature is less than 135°C (275°F):

- Yes**
- (1) HYDRAULIC pump switch 2B Select to OFF.
 - (2) System No. 2 temperature Monitor
 - (3) Land at the nearest suitable airport.
- END -

No

- (1) HYDRAULIC pump switch 2B Select to OFF.
- (2) Right engine Shutdown Refer to POWER PLANT – In-Flight Engine Failure/Shutdown in this chapter.
- (3) Inoperative systems Review Refer to HYDRAULIC page.
- (4) Actual landing distance Increase by a factor of 2.25 (125%).

————— **END** —————

1. HYDRAULIC POWER (CONT'D)

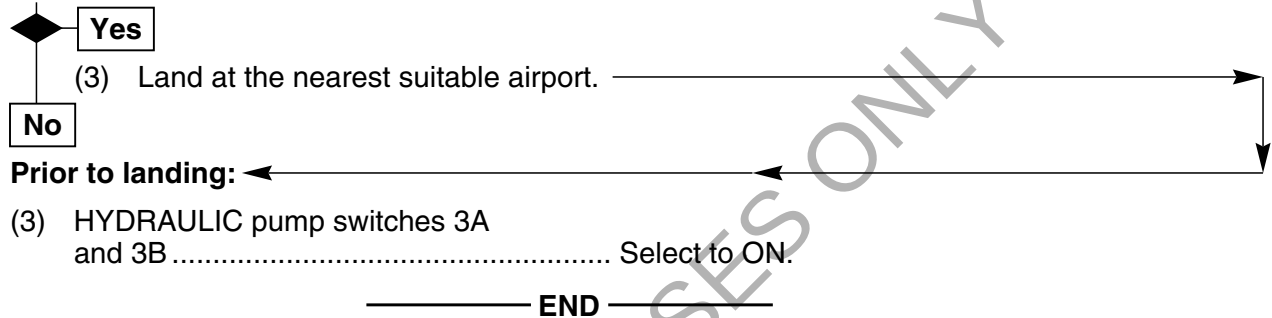
F. System No. 3 High Temperature

Indication: **HYD 3 HI TEMP** caution message on and temperature readouts greater than 96°C (205°F).

During flight:

- (1) HYDRAULIC pump switches 3A and 3B Select to OFF.
- (2) System No. 3 temperature Monitor

Temperature is more than 107°C (225°F):

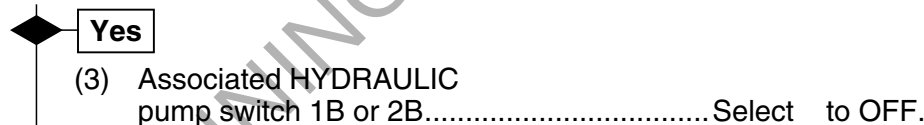


G. Engine-Driven Pump Failure

Indication: **HYD EDP 1A (2A)** caution message on.

- (1) Associated HYDRAULIC pump switch 1B or 2B Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

System quantity low:

**NOTE**

The engine-driven pump (EDP) is operating without hydraulic fluid; log length of time that EDP has been running dry. Monitor fluid temperature readouts.

– **END** –

- If **No**:
- (3) No further action required.

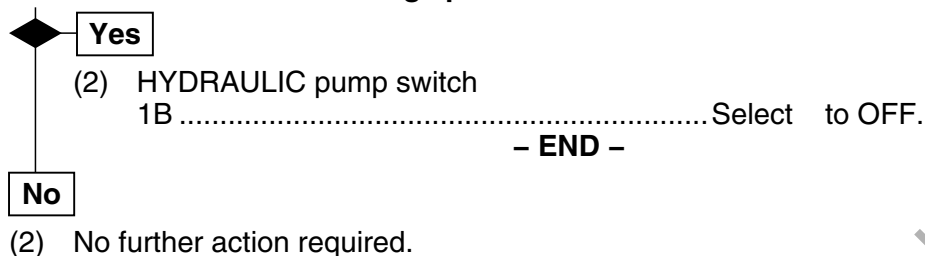
_____ **END** _____

1. HYDRAULIC POWER (CONT'D)

H. Electric Pump 1B Failure

Indication: **HYD PUMP 1B** caution message on.

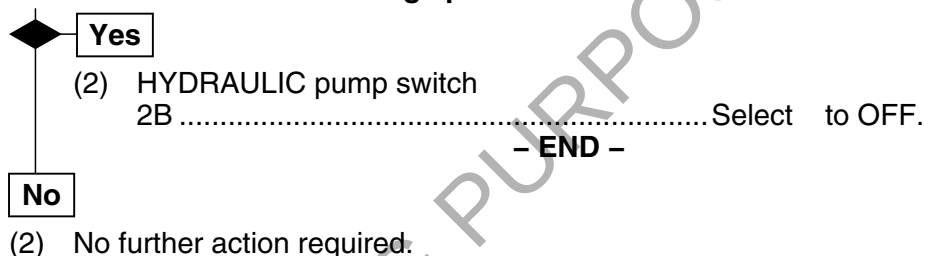
(1) HYDRAULIC pump switch 1B Select to ON.

HYD PUMP 1B caution message persists:

I. Electric Pump 2B Failure

Indication: **HYD PUMP 2B** caution message on.

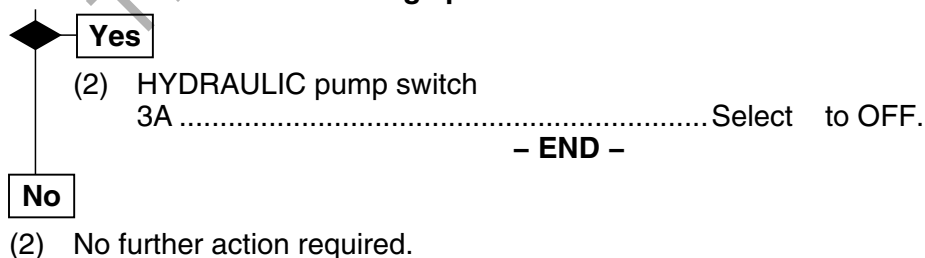
(1) HYDRAULIC pump switch 2B Select to ON.

HYD PUMP 2B caution message persists:

J. Electric Pump 3A Failure

Indication: **HYD PUMP 3A** caution message on.

(1) HYDRAULIC pump switches 3A and 3B Select both to ON.

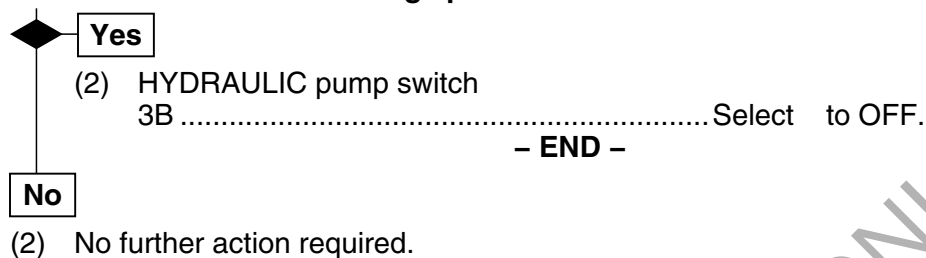
HYD PUMP 3A caution message persists:

1. HYDRAULIC POWER (CONT'D)

K. Electric Pump 3B Failure

Indication: **HYD PUMP 3B** caution message on.

- (1) HYDRAULIC pump switches 3A
and 3B Select both to ON.

HYD PUMP 3B caution message persists:

L. Hydraulic Shut-Off Valve Failure After Engine Shutdown Due to Fire

Indication: **HYD SOV 1 (2)** caution message on.

- (1) Land at the nearest suitable airport.
- _____ **END** _____

1. HYDRAULIC POWER (CONT'D)

M. System No. 1 and No. 2 Failure

Indication: **HYD 1 LO PRESS** and **HYD 2 LO PRESS** caution messages on.

- (1) HYDRAULIC pump switches (all) Select to ON.
- (2) Hydraulic pressure and fluid quantity Check

No. 1 and No. 2 quantity readouts are less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

◆ **Yes**

- (3) HYDRAULIC pump switches 1B and 2B Select to OFF, to prevent pump overheat.
 - (4) Inoperative systems Review Refer to HYDRAULIC page.
 - (5) ANTI SKID switch Leave ARMED.
 - (6) Land at the nearest suitable airport.
- Prior to landing:**
- (7) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
 - (8) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
 - (9) FLAPS Select to 20° for landing.
 - (10) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
 - (11) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

1. HYDRAULIC POWER (CONT'D)

M. System No. 1 and No. 2 Failure (Cont'd)

(12) Actual landing distance..... Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.55 (155%)	1.95 (95%)

NOTE

Maximize the use of reverse thrust.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

— END —

No

(3) No further action required.

— END —

1. HYDRAULIC POWER (CONT'D)

N. System No. 2 and No. 3 Failure

Indication: **HYD 2 LO PRESS** and **HYD 3 LO PRESS** caution messages on.

- (1) Airspeed200 KIAS maximum.
- (2) HYDRAULIC pump switches (all)Select to ON.
- (3) Hydraulic pressure and fluid quantityCheck

No. 2 and No. 3 quantity readouts are less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing: **Yes**

- (4) HYDRAULIC pump switches 2B, 3A and 3BSelect to OFF, to prevent pump overheat.
- (5) Inoperative systems Review Refer to HYDRAULIC page.
- (6) Land at the nearest suitable airport.

NOTE

1. Control is limited with No. 2 and No. 3 hydraulic systems failed.
2. Select the longest runway available with minimal crosswind and turbulence.
3. Rudder control is adequate for normal flight and should be used in coordination with aileron, if necessary, during turns.

Prior to landing:

- (7) NOSE STEER switchSelect to OFF.
- (8) ANTI SKID switchSelect to OFF.
- (9) Landing Gear Manual Extension procedure Accomplish Refer to LANDING GEAR, WHEEL AND BRAKE SYSTEM – Landing Gear Manual Extension procedure in this chapter.

NOTE

1. Unless it is possible to restore use of No. 2 hydraulic system, landing gear extension relies upon gear free fall after use of gear manual extension control.
2. A sideslip, for each side, must be performed to achieve MLG down-lock. It may be required to increase airspeed to ensure down-lock during sideslip.

- (10) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.

1. HYDRAULIC POWER (CONT'D)

N. System No. 2 and No. 3 Failure (Cont'd)

- (11) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (12) FLAPS Select to 20° for landing.
- (13) Approach speed V_{REF} (Flaps 45°) + 14 KIAS minimum.
- (14) ATS DISC switch Depress prior to 100 feet AGL.



ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

NOTE

The above double hydraulic system failure has an extremely remote probability. If it occurs, maximum use of reverse thrust must be used upon landing.

- (15) Actual landing distance Increase by a factor of 3.50 (250%).



Extreme caution is required during braking to avoid tire damage. Maximize use of reverse thrust.

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

Upon landing:

- (16) Both engines Apply maximum reverse thrust.
- (17) Brakes Apply light to moderate braking until safe taxi speed is maintained.
- (18) Brake pressure Monitor Approximately six (6) full brake applications are possible, using brake accumulator pressure.

NOTE

INBD and **OUTBD BRAKE PRESS** caution messages come on at less than 1,015 psi.

– END –

1. HYDRAULIC POWER (CONT'D)

N. System No. 2 and No. 3 Failure (Cont'd)

No

(4) No further action required.

————— **END** —————

TRAINING PURPOSES ONLY

1. HYDRAULIC POWER (CONT'D)

O. System No. 1 and No. 3 Failure

Indication: **HYD 1 LO PRESS** and **HYD 3 LO PRESS** caution messages on.

- (1) Airspeed200 KIAS maximum.
- (2) HYDRAULIC pump switches (all)Select to ON.
- (3) Hydraulic pressure and fluid quantityCheck

No. 1 and No. 3 quantity readouts are less than 5%, or pressure is less than 1,800 psi, or pressure is rapidly decreasing:

◆ **Yes**

- (4) HYDRAULIC pump switches 1B, 3A and 3BSelect to OFF.
- (5) Inoperative systems..... Review Refer to HYDRAULIC page.
- (6) NOSE STEER switchSelect to OFF.
- (7) ANTI SKID switchLeave ARMED.
- (8) Land at the nearest suitable airport.

NOTE

1. Flight path control is limited with No. 1 and No. 3 hydraulic systems failed.
2. Select the longest runway available with minimal crosswind and turbulence.
3. Rudder control is adequate for normal flight and should be used in coordination with aileron during turns.

- (9) Landing Gear Manual Extension procedure..... Accomplish Refer to LANDING GEAR, WHEEL AND BRAKE SYSTEM – Landing Gear Manual Extension procedure in this chapter.

Prior to landing:

- (10) TAWS WARNING, FLAPS OFF switch/light..... Press in to mute the flap aural warning.
- (11) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (12) FLAPSSelect to 20° for landing.
- (13) Approach speedV_{REF} (Flaps 45°) + 14 KIAS minimum.



1. HYDRAULIC POWER (CONT'D)

O. System No. 1 and No. 3 Failure (Cont'd)

- ▼ (14) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (15) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.40 (140%)	1.90 (90%)

NOTE

Maximize use of reverse thrust.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

– END –

No

- (4) No further action required.

————— **END** —————

1. ICE AND RAIN PROTECTION

A. Ice Dispersal Procedure

After leaving icing conditions, and if ice accumulation is observed on the heated portion of the wing leading edge:

- (1) Airspeed Increase to V_{MO}/M_{MO} , if possible, to disperse ice.

Impossible to remove ice from the wing leading edge:

◆ **Yes**

Assure adequate stall margin:

- (2) Maneuvering airspeed 200 KIAS minimum

WARNING

Even small accumulations of ice on the wing leading edge can change the stall speed, stall characteristics or the warning margins provided by the stall protection system.

Prior to landing:

- (3) TAWS WARNING, FLAPS OFF switch/light Press in to mute the flap aural warning.
- (4) TAWS WARNING, RAAS OFF switch/light (if installed) Press in Check OFF light on.
- (5) FLAPS Select to 20° for landing.
- (6) Approach speed V_{REF} (Flaps 45°) + 27 KIAS
- (7) ATS DISC switch Depress prior to 100 feet AGL.

CAUTION

ATS will not transition to landing mode and will not retard thrust levers at 50 feet AGL.

- (8) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.80 (80%)	1.70 (70%)

1. ICE AND RAIN PROTECTION (CONT'D)

A. Ice Dispersal Procedure (Cont'd)

▼ At 50 feet AGL:

- (9) Thrust levers IDLE and do not prolong the landing flare.

CAUTION

With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

- END -

No

- (2) No further action required.

— END —

1. ICE AND RAIN PROTECTION (CONT'D)

B. Ice Detected

Indication: **ICE** caution message on.

(1) Thrust Check N₂ RPM 78% minimum.

(2) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice ON.
Check **COWL A/ICE ON** advisory message on.

(3) ANTI-ICE, WING switch Select to NORM.

Check the following:

- **L** and **R WING A/ICE** caution messages may come on briefly, then out.
- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- **WING/COWL A/ICE ON** advisory message on.
- **ICE** advisory message on.
- WING L HEAT and R HEAT lights on.

If required:

(4) ANTI-ICE, WSHLD/WIND L and R switches Select to HI.

ICE caution message remains on, or
WING/COWL A/ICE ON advisory message is out or
WING L HEAT or R HEAT light is out:

Yes

(5) Engine thrust Increase until the following come on:

- **WING/COWL A/ICE ON** advisory message.
- **ICE** advisory message.
- WING L HEAT and R HEAT lights.

ICE caution message remains on, or
WING/COWL A/ICE ON advisory message remains out or
WING L HEAT or R HEAT light remains out:

Yes

(6) Leave icing conditions.

– END –

1. ICE AND RAIN PROTECTION (CONT'D)

B. Ice Detected (Cont'd)

No

(5) No further action required.

————— **END** —————

C. One Ice Detector Failed

Indication: **ICE DETECTOR 1 (2) FAIL** status message on.

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

When in icing conditions:

(1) ANTI-ICE, COWL L and R switch/lights Press in to select cowl anti-ice ON.

Check **COWL A/ICE ON** advisory message on.

(2) ANTI-ICE, WING switch Select to NORM.

Check the following:

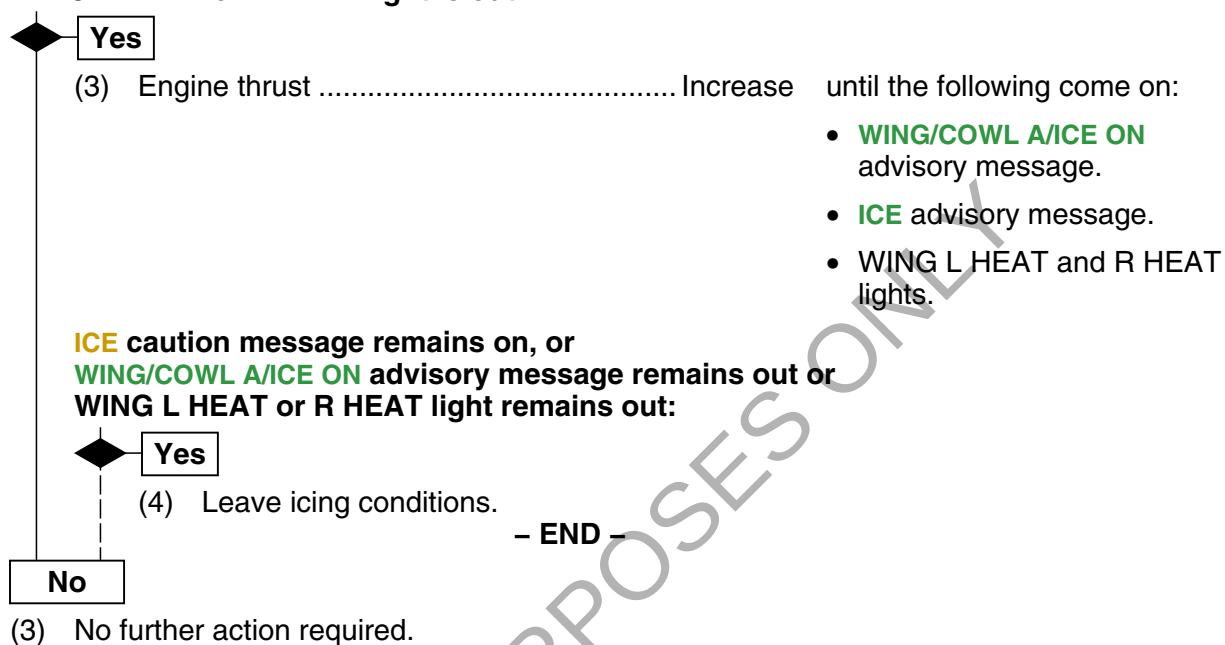
- **L** and **R WING A/ICE** caution messages may come on briefly, then out.
- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- **WING/COWL A/ICE ON** advisory message on.
- WING L HEAT and R HEAT lights on.



1. ICE AND RAIN PROTECTION (CONT'D)

C. One Ice Detector Failed (Cont'd)

ICE caution message remains on, or
WING/COWL A/ICE ON advisory message is out or
WING L HEAT or **R HEAT** light is out:

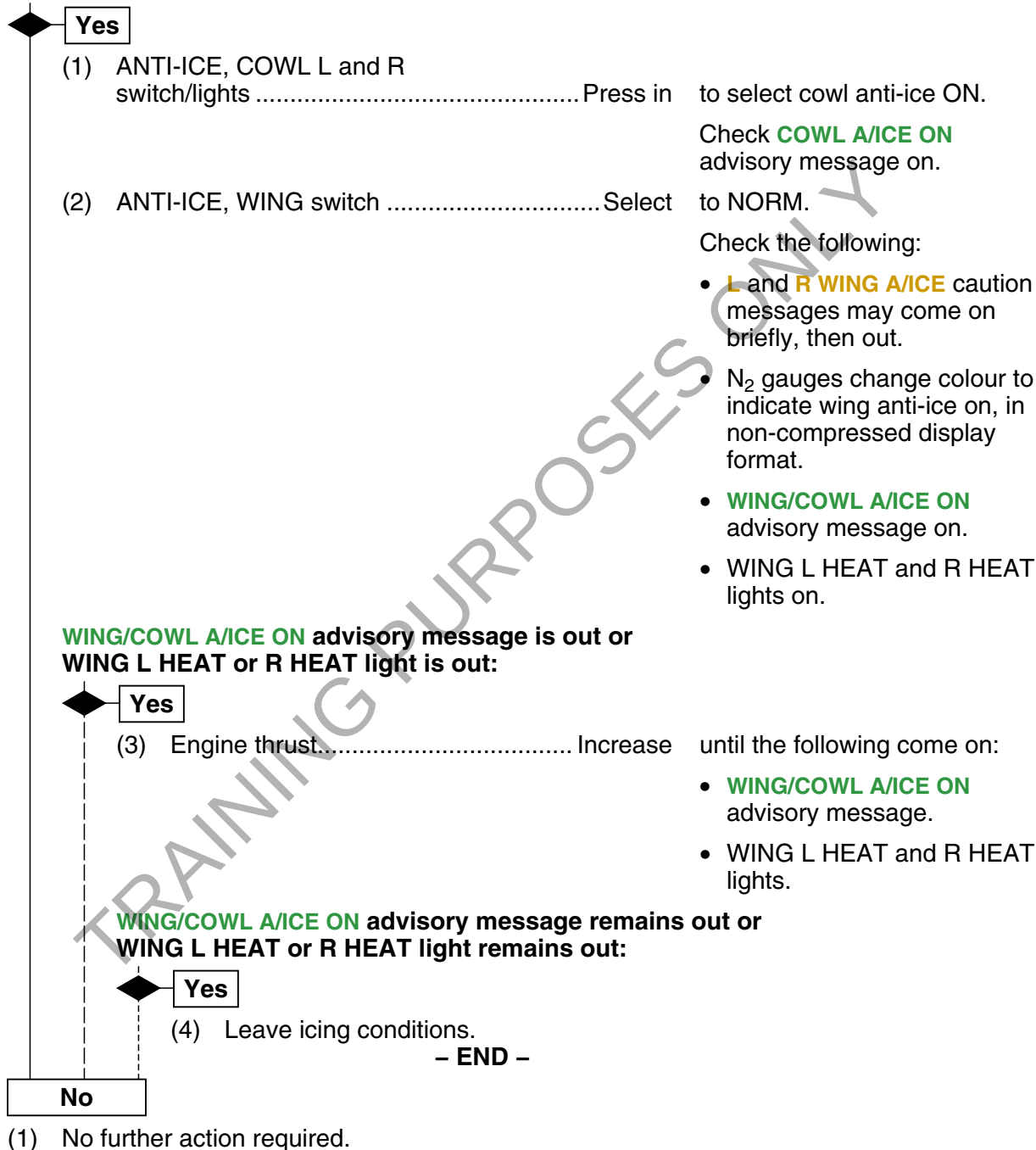


1. ICE AND RAIN PROTECTION (CONT'D)

D. Both Ice Detectors Failed

Indication: **ICE DETECTORS** caution message on.

Icing conditions are present or are anticipated:



1. ICE AND RAIN PROTECTION (CONT'D)

E. Wing Anti-Ice Sensor Failure

Indication: **WING A/ICE SNSR** caution message on.

(1) ANTI-ICE, WING switch Select to STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message on.

L (R) WING A/ICE caution message displayed:

Yes

(2) BLEED AIR, 14TH STAGE ISOL switch/light Press in to open isolation valve.

Check 14TH STAGE ISOL OPEN light on.

(3) ANTI-ICE, WING switch Check at STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message on.

L (R) WING A/ICE caution message remains displayed:

Yes

(4) Leave icing conditions.

If required:

(5) Ice Dispersal Procedure Accomplish Refer to Ice Dispersal Procedure in this section.

– END –

No

(2) No further action required.

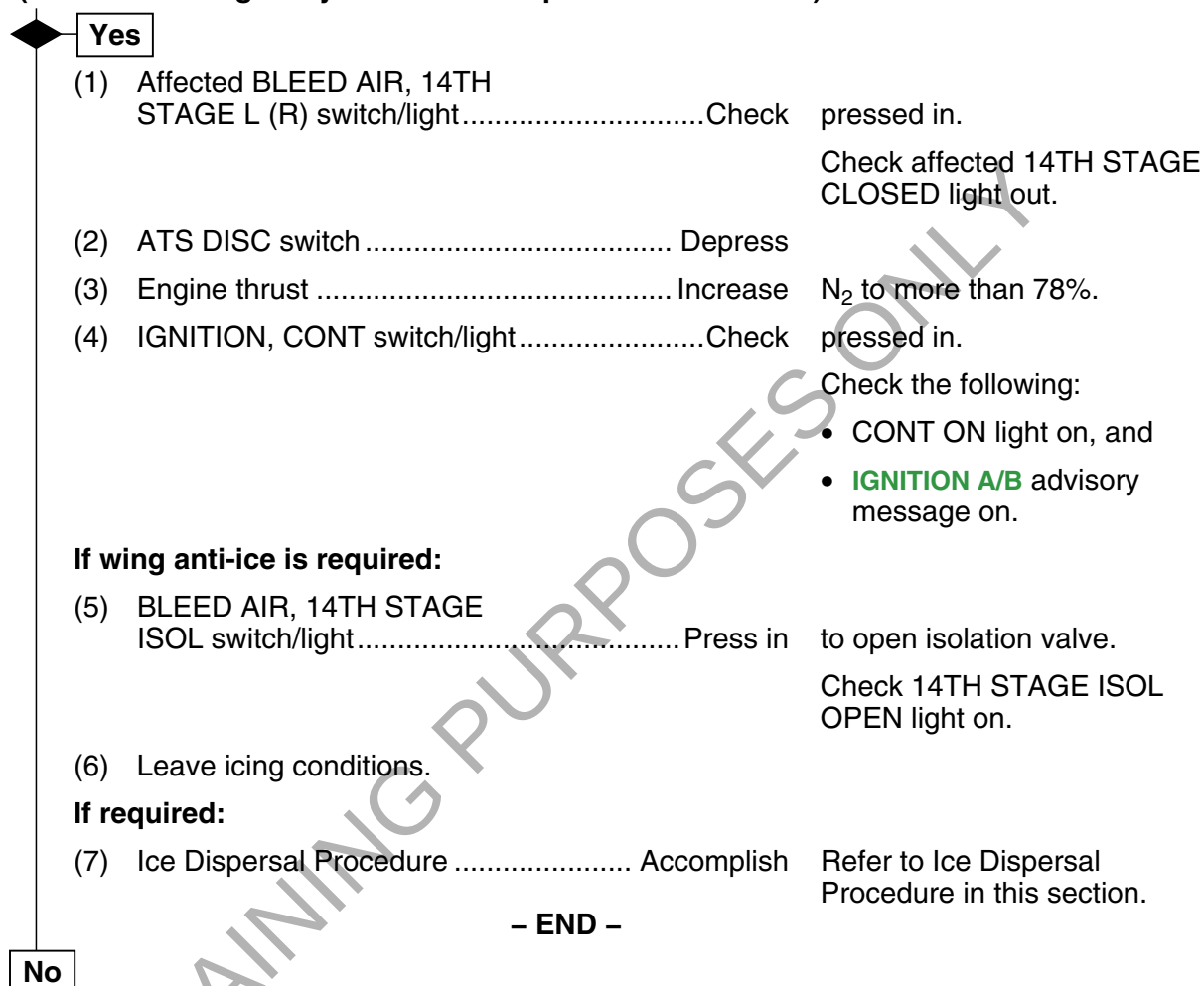
END

1. ICE AND RAIN PROTECTION (CONT'D)

F. Cowl Anti-Ice Failure

Indication: **L (R) COWL A/ICE** caution message on.

Cowl anti-ice is selected on:
(Caution message may indicate a low pressure condition.)



Cowl anti-ice is selected off:
(Caution message may indicate a high pressure condition.)

- (1) ATS DISC switch Depress
- (2) Affected engine thrust Reduce N₂ to less than 85%.
- (3) Affected BLEED AIR, 14TH STAGE L (R) switch/light.....Press out to close shutoff valve.
Check affected 14TH STAGE CLOSED light on.

————— END —————

1. ICE AND RAIN PROTECTION (CONT'D)

G. Wing Leading Edge Temperature and Pressure Low

Indication: **L (R) WING A/ICE** caution message on.

(1) Affected BLEED AIR, 14TH

STAGE L (R) switch/light.....Check pressed in.

Check affected 14TH STAGE
CLOSED light out.

(2) Engine thrust Increase

L (R) WING A/ICE caution message remains on:

◆ **Yes**

(3) BLEED AIR, 14TH STAGE

ISOL switch/light.....Press in to open isolation valve.

Check 14TH STAGE ISOL
OPEN light on.

After 30 seconds, and L (R) WING A/ICE caution message persists:

◆ **Yes**

(4) ANTI-ICE, WING switch.....Select to STBY.

Check the following:

- N₂ gauges change colour to indicate wing anti-ice on, in non-compressed display format.
- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message on.

(5) Leave icing conditions.

If required:

(6) Ice Dispersal ProcedureAccomplish Refer to Ice Dispersal Procedure in this section.

— END —

◆ **No**

(3) No further action required.

————— **END** —————

1. ICE AND RAIN PROTECTION (CONT'D)

H. Insufficient Heat for Wing Anti-Icing

Indication: Wing anti ice selected ON and WING L HEAT or R HEAT light goes out or
WING A/ICE ON or **WING/COWL A/ICE ON** advisory message goes out.

(1) Thrust Increase until the following come on:

- **WING A/ICE ON** or **WING/COWL A/ICE ON** advisory message.
- WING L HEAT and R HEAT lights.

WING A/ICE ON or **WING/COWL A/ICE ON** advisory message remains out or
WING L HEAT or R HEAT light remains out:

Yes

(2) Leave icing conditions.

– END –

No

(2) No further action required.

END

1. ICE AND RAIN PROTECTION (CONT'D)

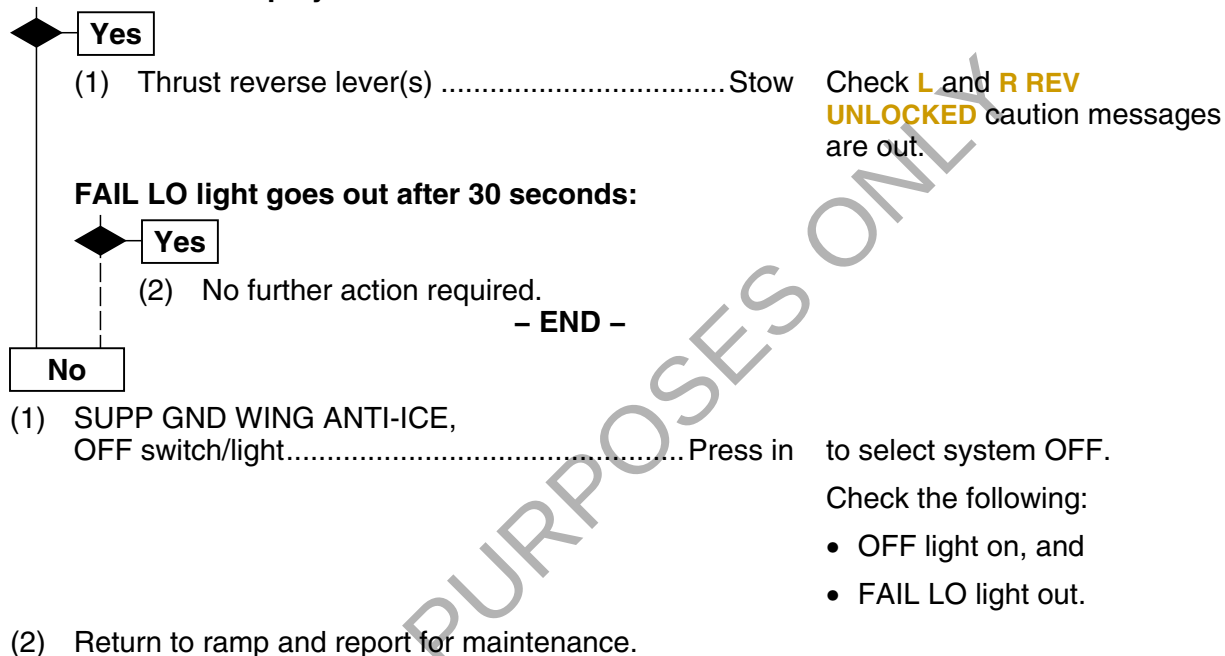
I. Insufficient Heat for Supplemental Ground Wing Anti-Icing

Indication: SUPP GND WING ANTI-ICE, FAIL LO light on.

NOTE

Extended taxi with a thrust reverser deployed may cause FAIL LO light to come on.

Thrust reverser deployed:

**END**

J. Supplemental Ground Wing Anti-Icing Overheat

Indication: SUPP GND WING ANTI-ICE, FAIL HI light on.

- (1) SUPP GND WING ANTI-ICE, OFF switch/light..... Press in to select system OFF.
Check OFF light on.
- (2) Return to ramp and report for maintenance.

END

1. ICE AND RAIN PROTECTION (CONT'D)

K. Probe Heater Failures

Indication:

- **L (R) AOA HEAT** caution message,
- **AUX AOA CASE (VANE) HEAT** caution message,
- **L (R) PITOT BASE (TUBE) HEAT** caution message,
- **L (R) STATIC HEAT** caution message,
- **STBY PITOT HEAT** caution message,
- **TAT PROBE HEAT** caution message,
- **L (R) PROBE HEAT OFF** caution message.

- (1) ANTI-ICE, PROBES L and R
switches.....Select OFF and then ON.

Applicable caution message still on:

 Yes

MESSAGE	ACTION
L (R) AOA HEAT	Do not rely on stall protection system or TAWS. (2) No further action required.
AUX AOA CASE HEAT AUX AOA VANE HEAT	Do not rely on low speed cue or Normalized AOA displayed on both PFDs. (2) No further action required.
L (R) PITOT TUBE HEAT L (R) PITOT BASE HEAT	(2) Select ADC source to operative side.
L (R) STATIC HEAT	Do not rely on associated instruments: • ISI for altitude and airspeed, and • Cabin ΔP on EICAS. (2) No further action required.
STBY PITOT HEAT	Do not rely on ISI for airspeed indication. (2) No further action required.
TAT PROBE HEAT	Do not rely on air temperature indications. (2) No further action required.
L (R) PROBE HEAT OFF	During flight – No further action required.

L and R AOA HEAT caution messages persist: Yes

(2) Avoid icing conditions.

(3) Stall Protection System

Failure ProcedureAccomplish Refer to FLIGHT CONTROLS
– Stall Protection System
Failure in this chapter.

– END –

1. ICE AND RAIN PROTECTION (CONT'D)

K. Probe Heater Failures (Cont'd)

☐ No

(2) No further action required.

————— END —————

L. Windshield Heat Failure

Indication: **L (R) WSHLD HEAT** caution message on.

- (1) Affected ANTI-ICE, WSHLD/WIND switch Select to RESET, then LO or HI as applicable.

L (R) WSHLD HEAT caution message remains on:☒ Yes

(2) Leave icing conditions as soon as possible.

— END —

☐ No

(2) No further action required.

NOTE

Moving the COCKPIT and CABIN COLD-HOT knobs to HOT may help reduce fogging.

————— END —————

M. Window Heat Failure

Indication: **L (R) WINDOW HEAT** caution message on.

- (1) Affected ANTI-ICE, WSHLD/WIND switch Select to RESET, then LO or HI as applicable.

NOTE

Moving the COCKPIT and CABIN COLD-HOT knobs to HOT may help reduce fogging.

————— END —————

1. ICE AND RAIN PROTECTION (CONT'D)

N. Windshield Outer Face Ply Failure

- (1) Affected ANTI-ICE, WSHLD/WIND switch Select to OFF.
- (2) Flight..... Continue minimizing flight into icing conditions, precipitation or other conditions that could cause fogging or misting, whenever possible.

NOTE

Icing conditions exist in flight at a TAT of 10°C (50°F) or below and visible moisture in any form is encountered (such as clouds, rain, snow, sleet or ice crystals), except when the SAT is -40°C (-40°F) or below.

————— **END** —————

TRAINING PURPOSES ONLY

1. INSTRUMENTS SYSTEM

A. Primary Flight Display Failure

Indication: PFD is erratic or goes blank.

- (1) Reversionary panel, applicable
L (R) DISPLAYS switch Select to MFD REV.
- (2) Reversionary panel, EICAS switch Select to BOTH PFD'S.

————— **END** —————

B. Display Control Panel Failure

Indication: Amber boxed DCP flag displayed on both PFD and MFD.
DCP is inoperative. Baro, course and HDG knobs may be inoperative.

- (1) Flight control panel, XFR switch Select side corresponding to operative display control panel.

NOTE

When the DCP flag is displayed on the PFD, the display features normally controlled by the DCP may be controlled using the Cursor Control Panel and the PFD submenu on the MFD.

————— **END** —————

C. EFIS Comparator Failure

Indication: **EFIS COMP INOP** caution message.

NOTE

If message is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to Air Data Computer Failure or Inertial Reference System Failure in this section.

- (1) Affected flight instruments Monitor
- (2) Integrated Standby Instrument Use to cross-check.

NOTE

If any of the following occurs, refer to Chapter 3; EMERGENCY PROCEDURES – INSTRUMENTS SYSTEM – Unreliable Airspeed:

- Pitch attitude, thrust setting or external noise not consistent with indicated airspeed
- Large airspeed differences between PFDs and/or ISI
- Loss of multiple airspeed indication.

————— **END** —————

1. INSTRUMENTS SYSTEM (CONT'D)

D. Multifunction Display Failure

Indication: MFD is erratic or displays stale information or goes blank.

- (1) Reversionary panel, applicable
L (R) DISPLAYS switch Select to PFD REV.
- (2) Reversionary panel, EICAS switch Select to BOTH PFD'S.

On airplanes incorporating Service Bulletin 605-34-019:

- (3) MFD TUNE INHIB switch/light Press in

————— **END** —————

E. Cursor Control Panel Failure

Indication: Cursor control panel is inoperative and CCP flag is displayed.

EICAS displayed on MFD corresponding to failed CCP:

◆ **Yes**

- (1) No further action required.

NOTE

If paging of CAS messages is required, the EICAS reversion switch must be selected to the operative CCP side.

— **END** —

◆ **No**

- (1) Reversionary panel, EICAS switch Select to BOTH PFD'S.

————— **END** —————

F. PFD Cross-Talk Failure

Indication: **PFD X-TALK FAIL** caution message on.

PFD is blank:

◆ **Yes**

- (1) Primary Flight Display Failure procedure Accomplish Refer to Primary Flight Display Failure procedure in this section.

— **END** —

◆ **No**

- (1) Instruments Cross-check
- (2) Verify that PFD reference values, speed and Vspeed bugs, and selected altitude are set on both DCPs or are correctly synchronized between both PFDs.

————— **END** —————

1. INSTRUMENTS SYSTEM (CONT'D)

G. EFIS Parameter Miscompare

Indication: **EFIS MISCOMP** caution message on, and one of the following comparator annunciators shown on the primary flight display:

- **ALT** • **ATT** • **GS** • **HDG** • **IAS**
- **LOC** • **PIT** • **RA** • **ROL**

PFD	ACTION
ALT and/or IAS	(1) Flight instruments and Integrated Standby Instrument.....Cross-check (2) Centre pedestal, AIR DATA source selector switch..... Select to reliable side. If any of the following occurs, refer to Chapter 3; EMERGENCY PROCEDURES – INSTRUMENTS SYSTEM – Unreliable Airspeed: • Pitch attitude, thrust setting or external noise not consistent with indicated airspeed • Large airspeed differences between PFDs and/or ISI • Loss of multiple airspeed indication.
ATT , ROL or PIT	(1) Flight instruments and Integrated Standby Instrument.....Cross-check (2) Reversionary panel, IRS switch..... Select to reliable alternate source.
HDG	IRS 1 (2) (3) IN ATT status message displayed: Yes (1) FMSEnter present airplane heading on the IRS CONTROL page. No (1) Flight instruments and Integrated Standby Instrument.....Cross-check (2) Reversionary panel, IRS switch..... Select to reliable alternate source.
RA	(1) Radio altimeter failure procedureAccomplish Refer to Radio Altimeter Failure procedure, in this section.



1. INSTRUMENTS SYSTEM (CONT'D)

G. EFIS Parameter Miscompare (Cont'd)

PFD	ACTION
LOC or GS	If non-visual, during approach: (1) Go-aroundInitiate

END

H. FMS V Speed Failure

Indication: Amber VSPD flag on pilot or copilot PFD.

- (1) V speeds.....

Set manually through the PFD REFS to clear the Flag. Use the AFM to determine V speeds.

NOTE

Do not use the FMS take-off and approach performance calculation function.

END

1. INSTRUMENTS SYSTEM (CONT'D)

I. Air Data Computer Failure

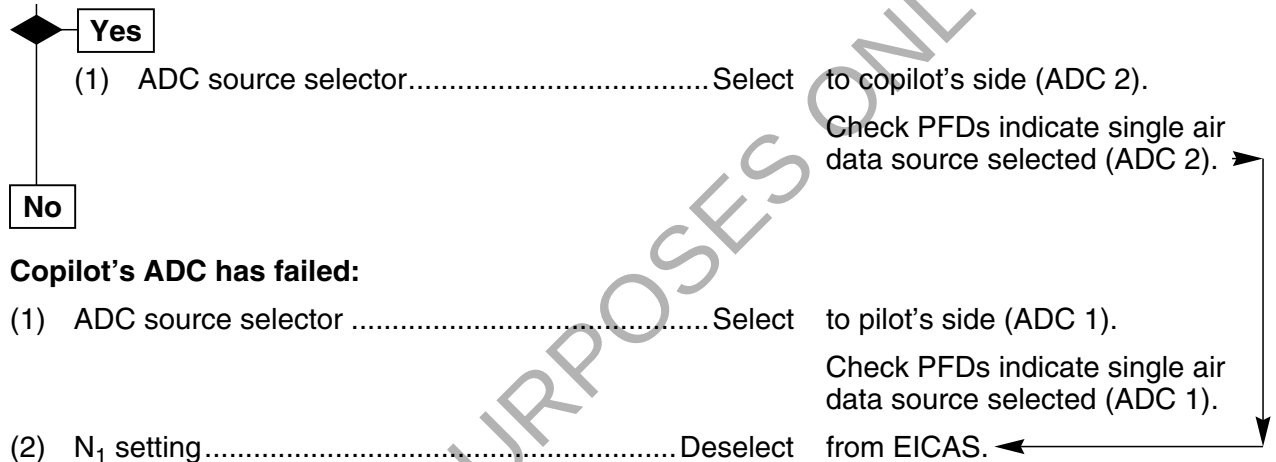
Indication: ALT, AOA, IAS, MACH and/or VS red flags;
erratic primary flight display data.

On airplanes incorporating Service Bulletin 605-34-019:

NOTE

If FMS 3 is reverted in, use of VNAV guidance is prohibited following reversion.

Pilot's ADC has failed:



NOTE

Data from the CDU can be used if crossed checked with the AFM. TGT thrust mode may be used to set the N_1 setting on EICAS after confirmation of the data.

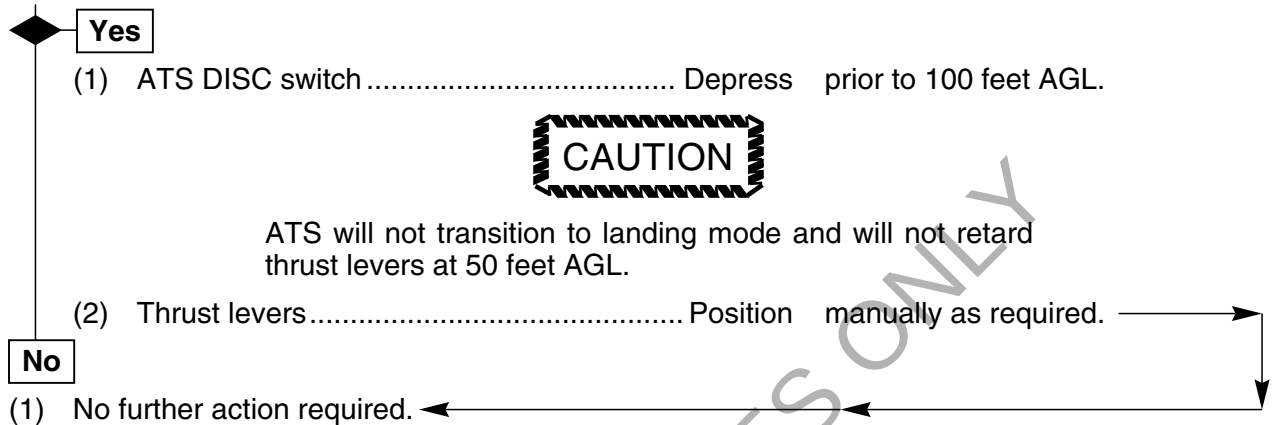
— END —

1. INSTRUMENTS SYSTEM (CONT'D)

J. Radio Altimeter Failure

Indication: False or intermittent radio altimeter height and/or
Amber RA miscompare flag and/or red RA fail flag on PFD.

Pilot's radio altimeter has failed:



NOTE

1. Do not rely on affected radio altitude displays.
2. TCAS, TAWS, RAAS (if installed) and windshear detection are inoperative. TCAS FAIL message on PFD or MFD and **TAWS BASIC FAIL**, **RAAS FAIL** and **TAWS WINDSHEAR FAIL** status messages on.

On airplanes incorporating Service Bulletin 605-34-005:

If only one radio altimeter has failed, TCAS, TAWS, RAAS and windshear detection remain operational.

3. For an intermittent or undetected radio altimeter error, the following may be incorrect or inactive:
 - Displayed radio altitudes
 - LOC/GS comparators
 - Excessive deviation alerts
 - Expanded lateral deviation scale
 - Approach minimums annunciation.
4. For an intermittent or undetected radio altimeter error, ATS RETARD mode may occur early or not occur. Refer to POWER PLANT – ATS Radio Altimeter Input Failure in this chapter.
5. For an intermittent or undetected radio altimeter error, localizer and/or glideslope AP/FD commands could be adversely affected.

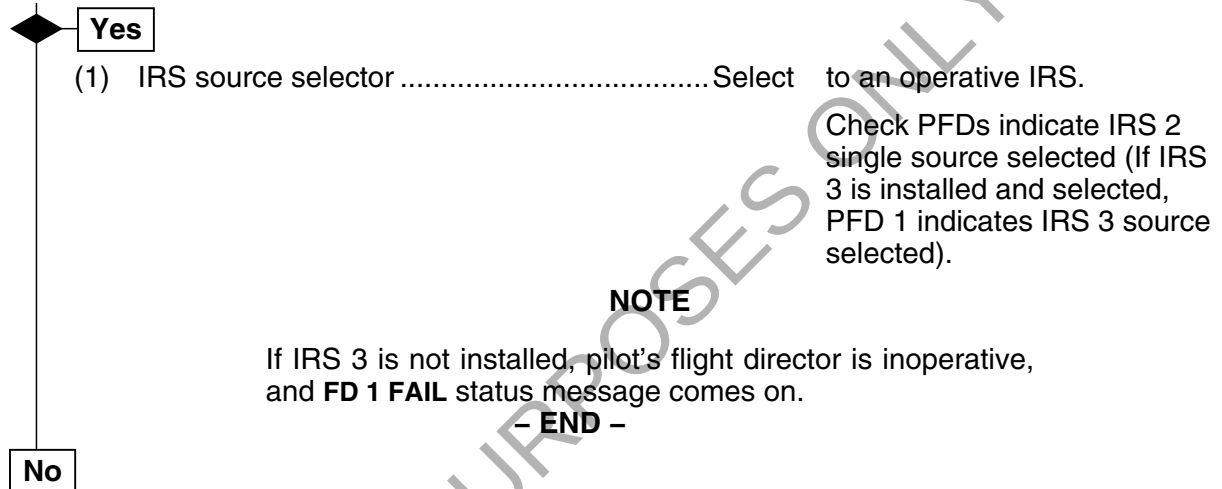
————— **END** —————

1. INSTRUMENTS SYSTEM (CONT'D)**K. Inertial Reference System Failure**

Indication: Erratic primary flight display data and/or red ATT and/or HDG failure flags.

NOTE

If an IRS is in attitude mode, a heading drift rate of up to 15 degrees per hour can occur. The FMS heading on the IRS CONTROL page should be updated periodically from the best available alternate navigation source.

Pilot's IRS has failed:**Copilot's IRS has failed:**

(1) IRS source selector Select to an operative IRS.

Check PFDs indicate IRS 1 single source selected (If IRS 3 is installed and selected, PFD 2 indicates IRS 3 source selected).

NOTE

If IRS 3 is not installed, copilot's flight director is inoperative, and **FD 2 FAIL** status message comes on.

END

L. Total Inertial Reference System Failure

Indication: Erratic primary flight display data and/or red ATT and/or HDG failure flags on both PFDs.

- (1) ISI and standby compass Monitor
- (2) Land at the nearest suitable airport.

END

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TRAINING PURPOSES ONLY

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM

A. Inboard Anti-Skid Failure

Indication: **A/SKID INBD** caution message on.

NOTE

A/SKID INBD caution message indicates anti-skid braking is lost on 1 or 2 inboard wheels, or parking brake shut-off valve has failed (**PARK BRAKE SOV** caution message on).



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

If excessive asymmetry or loss of braking is observed:

- (2) Wheel brakes Release momentarily.
- (3) ANTI-SKID switch Select to OFF.
- (4) Wheel brakes Re-apply as required.

END

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

B. Outboard Anti-Skid Failure

Indication: **A/SKID OUTBD** caution message on.

NOTE

A/SKID OUTBD caution message indicates anti-skid braking is lost on 1 or 2 outboard wheels.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

If excessive asymmetry or loss of braking is observed:

- (2) Wheel brakes Release momentarily.
 (3) ANTI-SKID switch Select to OFF.
 (4) Wheel brakes Re-apply as required.

END

C. Anti-Skid (Inboard and Outboard) Failure

Indication: **A/SKID INBD** caution message and **A/SKID OUTBD** caution message on.

NOTE

A/SKID INBD and **A/SKID OUTBD** caution messages indicate anti-skid braking is lost on 2 or more wheels.

- (1) ANTI SKID switch Select to OFF.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

END

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

D. Inboard/Outboard Brake Pressure Low

Indication: **INBD (OUTBD) BRAKE PRESS** caution message on.

- (1) Hydraulic pressure and fluid quantity Monitor for the affected system.
- (2) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.70 (70%)	1.50 (50%)

NOTE

Maximize the use of reverse thrust.

————— END —————

E. Parking Brake Shut-Off Valve Failure

Indication: **PARK BRAKE SOV** caution message on.

NOTE

PARK BRAKE SOV caution message indicates that the parking brake SOV is failed or is closed with the parking brake not set, possibly affecting inboard anti-skid system performance.

CAUTION

Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

- (1) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.05 (105%)	1.65 (65%)

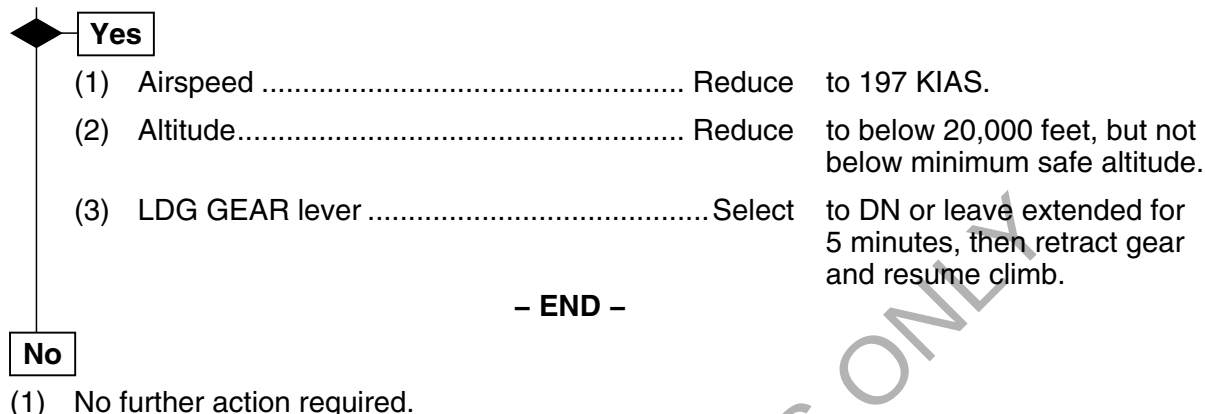
————— END —————

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

F. Main Landing Gear Overheat Detector Failure

Indication: **MLG OVHT FAIL** caution message on.

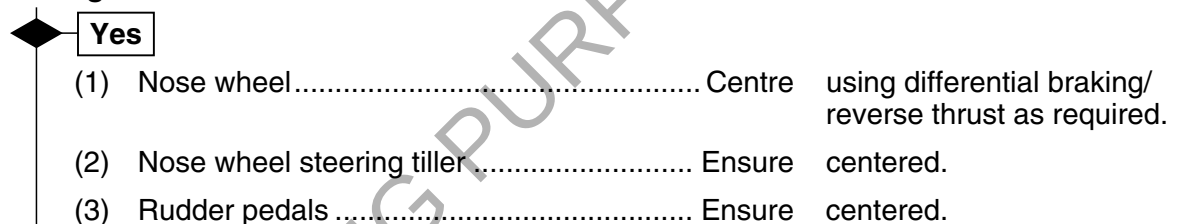
Failure occurs within 30 minutes after take-off:



G. Nose Wheel Steering System Failure

Indication: **STEERING INOP** caution message on.

During taxi:



NOTE

The NOSE STEER switch must be selected to ARMED to enable the nosewheel steering circuit, and centering of the nosewheel.

- (4) NOSE STEER switch Cycle OFF and ARMED.

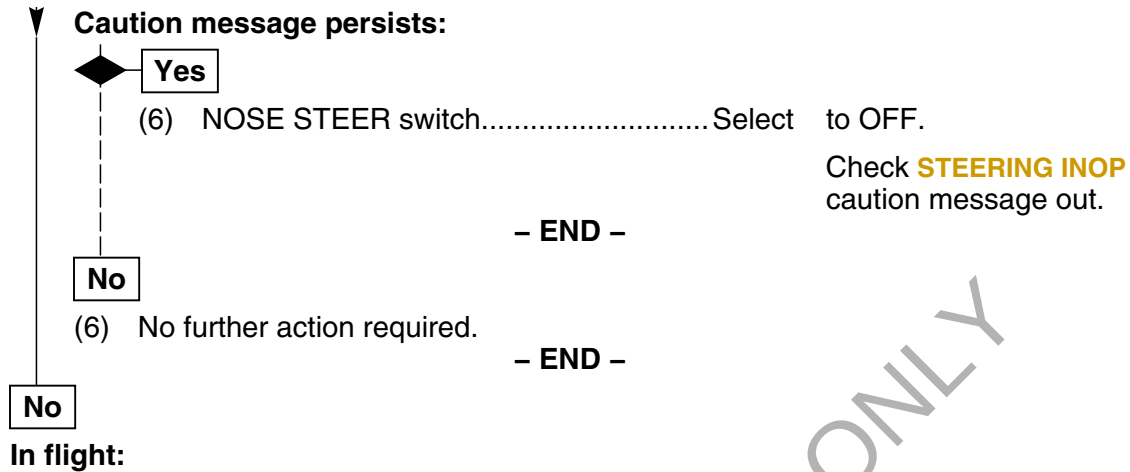
(5) Nose wheel steering Check responsiveness and ensure normal operation.

NOTE

Check nose wheel steering responsiveness by taxiing forward and checking proper aircraft response to left/right tiller inputs and rudder pedal inputs.

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

G. Nose Wheel Steering System Failure (Cont'd)



NOTE

Centering of the nosewheel steering tiller, rudder pedals, and/or cycling the switch from OFF to ARMED will not enable centering of the nosewheel.

- (1) NOSE STEER switchSelect to OFF.
Check **STEERING INOP** caution message out.

NOTE

- When landing with the nosewheel steering inoperative, select the longest runway available with minimum turbulence and crosswind.
- Use differential braking, rudder and engine thrust, as required, to assist directional control.

— END —

H. Weight-on-Wheels Input Fault

Indication: **WOW INPUT** caution message on.

- NOSE STEER switch Ensure selected to ARMED.
- GND SPOILERS switch Set to DISARM.



Do not override the landing gear downlock.

NOTE

If landing gear are down and locked, gear will not retract. If a go-around is required in this configuration, add 2.0 kts to the approach climb speed and reduce the approach climb gradient by 2.4%.



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

H. Weight-on-Wheels Input Fault (Cont'd)

Prior to landing:

- (3) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

After touchdown:

- (4) GND SPOILERS switch Select to ARM.
(5) FLIGHT SPOILER lever Select to MAX.

————— **END** —————

TRAINING PURPOSES ONLY

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

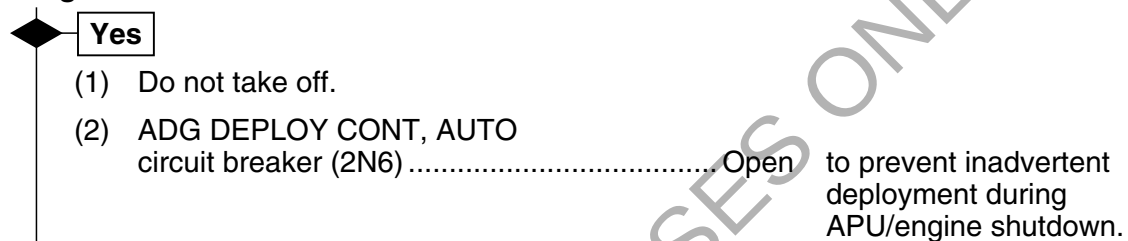
I. Weight-on-Wheels Output Fault

Indication: **WOW OUTPUT** caution message on.

The following may occur:

- PFD, speed trend vector inoperative.
- PFD, low speed cue comes on; HSI portion indicates TA ONLY,
- CDU/MFD indicates ATC/TCAS STANDBY,
- Inadvertent take-off configuration warnings, and
- Fuel quantity fluctuates during pitch and bank maneuvers.

On ground:

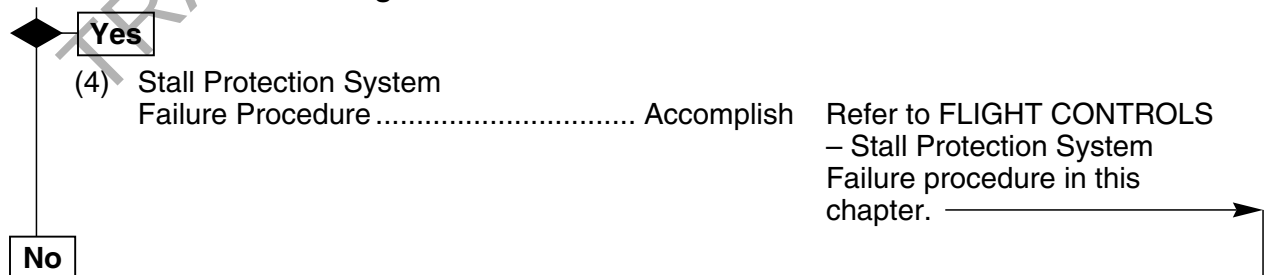


No

In flight:

- (1) Manual Cabin Pressurization Control Procedure Accomplish Refer to AIR-CONDITIONING AND PRESSURIZATION – Manual Cabin Pressurization Control Procedure in this chapter.
- (2) GND SPOILERS switch Set to DISARM.
- (3) NOSE STEER switch Select to OFF.

STALL FAIL caution message is on:



Prior to landing:

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

I. Weight-on-Wheels Output Fault (Cont'd)

- (4) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
1.30 (30%)	1.25 (25%)

After touchdown:

- (5) GND SPOILERS switch Select to ARM.
 (6) FLIGHT SPOILER lever Select to MAX.

If loss of braking occurs below 35 knots:

- (7) ANTI SKID switch Select to OFF.



Extreme caution is required during braking to avoid tire damage or blowout. Maximize use of reverse thrust.

If nose wheel steering is required:

- (8) NOSE STEER switch Select to ARMED.

After landing:

- (9) ADG DEPLOY CONT, AUTO circuit breaker (2N6) Open to prevent inadvertent ADG deployment during shutdown.

— END —

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

J. Proximity Sensing System Failure (Total System Shutdown)

Indication: **PROX SYS** caution message on, **GND SPLRS** caution message on, and landing gear symbols (nose, left and right main gear) display amber dash marks (position unknown) with landing gear lever UP or DN.

NOTE

1. EICAS landing gear position indications are inoperative.
 2. Normal landing gear extension/retraction is not available.
 3. Anti-skid, ground spoilers and nose wheel steering systems are inoperative.
 4. Nose door open warning system is not available.
- (1) Airspeed Reduce to not more than:
- Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.
- (2) HYDRAULIC pump switches 2B and 3B Select to ON.
- (3) LDG GEAR lever Select to DN.

CAUTION

Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.

- (4) LANDING GEAR MANUAL RELEASE handle Pull to release landing gear.
Confirm gear position with control tower.
- (5) GND SPOILERS switch Set to DISARM.
- (6) ANTI SKID switch Set to OFF.



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

J. Proximity Sensing System Failure (Total System Shutdown) (Cont'd)



Extreme caution is required during braking to avoid tire damage or blow out. Maximize use of reverse thrust.

- (7) NOSE STEER switchSelect to OFF.

NOTE

If landing with nosewheel steering inoperative, select the longest runway available with minimum turbulence and crosswind. Use differential braking, rudder, and engine thrust as required during taxi.

- (8) Actual landing distance Increase as applicable by factor given below:

WITHOUT THRUST REVERSERS	WITH THRUST REVERSERS
2.20 (120%)	1.75 (75%)

————— END —————

1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

K. Landing Gear Manual Extension

- (1) Airspeed Reduce to not more than:
- Flaps 0° – 181 KIAS.
 - Flaps 20° – 197 KIAS.
 - Flaps 30° – 197 KIAS.
 - Flaps 45° – 189 KIAS.



Leave hydraulic pump 3B OFF if pump was selected OFF due to a low or decreasing pressure, or a low quantity in system No. 3.

- (2) HYDRAULIC pump switches 2B and 3B Select to ON.
- (3) LDG GEAR lever Select to DN.



Failure to pull the LANDING GEAR MANUAL RELEASE handle to its full extension may prevent successful landing gear extension.

NOTE

The force required to operate the LANDING GEAR MANUAL RELEASE handle is greater than 40 pounds. The flight crew may choose to reposition the seat prior to handle deployment. In safe flight conditions, the flight crew may also choose to stand up to pull the handle.

- (4) LANDING GEAR MANUAL RELEASE handle PULL to full extension.

NOTE

The **NOSE DOOR OPEN** warning message may be displayed when the LANDING GEAR MANUAL RELEASE handle is pulled.

If the nose landing gear is in transition this message will be inhibited.

- (5) Landing gear indication Check for three DN indications.



1. LANDING GEAR, WHEEL AND BRAKE SYSTEM (CONT'D)

K. Landing Gear Manual Extension (Cont'd)

Three landing gear symbols indicate DN:

 **Yes**



Do not stow LANDING GEAR MANUAL RELEASE handle until gear locking pins are installed.

Nose wheel steering may not be available upon landing even if the **STEERING INOP** caution message is not displayed.

(6) No further action required.

– END –

No

Any landing gear fails to lock in the down position:

- (6) Gear Disagree Procedure Accomplish Refer to Chapter 3;
EMERGENCY
PROCEDURES – LANDING
GEAR, WHEEL AND BRAKE
SYSTEM – Gear Disagree.

— **END** —

1. MISCELLANEOUS SYSTEMS

A. Emergency Lighting System Selected Off

Indication: **EMER LTS OFF** caution message on.

- (1) EMER LTS switchCheck status of EMER LTS.

————— **END** —————

B. Crew Oxygen Low Pressure

Indication: **OXY LO PRESS** caution message on.

OXY pressure readout indicates less than 825 psi:

◆ **Yes**

- (1) Descent Initiate to a safe altitude.

— **END** —

No

- (1) Crew oxygen masks Test and check pressure readout.

- (2) Oxygen quantity Monitor

OXY pressure readout drops to 0 psi (bottle(s) closed):

◆ **Yes**

- (3) Descent Initiate to a safe altitude.

— **END** —

No

- (3) No further action required.

————— **END** —————

C. Passenger Oxygen Deployed and Activated (If Installed)

Indication: **PAX OXY ON** caution message.

- (1) Passenger oxygen.....Check status.

————— **END** —————

1. MISCELLANEOUS SYSTEMS (CONT'D)

D. Radio Tuning Failure

Indication: Amber radio frequencies displayed and/or inability to tune radios.

NOTE

1. Note all frequencies being used prior to starting procedure.
2. During procedure, continue tuning and attempting communication even if frequency is displayed in amber.

(1) CDU 1 TUNE INHIB switch/light..... Press in

If unable to restore radio tuning on CDU 2 or MFD:

(2) CDU 1 TUNE INHIB switch/light..... Press out

(3) CDU 2 TUNE INHIB switch/light..... Press in

If unable to restore radio tuning on CDU 1 or MFD:

(4) CDU 1 TUNE INHIB switch/light..... Press in

If unable to restore radio tuning on MFD:

(5) CDU 1 TUNE INHIB and CDU 2
TUNE INHIB switch/lights..... Press out

(6) MFD TUNE INHIB switch/light..... Press in

If unable to restore COM tuning on at least one side:

(7) CDU 1 TUNE INHIB and CDU 2
TUNE INHIB switch/lights..... Press in to force COM 1 to 121.5.

NOTE

1. To go straight to the emergency frequency (121.50 MHz), press in CDU 1 TUNE INHIB, CDU 2 TUNE INHIB and MFD TUNE INHIB.
2. If 8.33 KHz channel spacing was previously tuned on VHF 1, the emergency frequency (121.50) may be displayed in amber. The amber display does not prevent VHF from operating normally on this frequency.

————— END —————

1. MISCELLANEOUS SYSTEMS (CONT'D)

E. Dual HF Transmission Failure

Indication: No HF transmission. Dashes displayed for HF 1 and HF 2, on CDUs and MFDs.

- (1) HF COMM 2 circuit breakers
(2L8, 2L9 and 2L10) Open
- (2) HF 1 Attempt communication.

Transmission successful:

◆ **Yes**

- (3) No further action required.

— **END** —

□ **No**

- (3) HF COMM 2 circuit breakers
(2L8, 2L9 and 2L10) Close
- (4) HF COMM 1 circuit breakers
(1K5, 1K6 and 1K7) Open
- (5) HF 2 Attempt communication.

— **END** —

F. Transponder Failure

Indication: **XPDR 1 (2) FAIL** caution message on and/or
XPDR FAIL on CDU or MFD tuning.

- (1) MFD or CDU, ATC/TCAS
CONTROL menu Select alternate transponder.

NOTE

TCAS and ADS-B OUT will stop functioning until alternate transponder is activated.

— **END** —

G. Dual Transponder Failure

Indication: **XPDR 1 & 2 FAIL** caution message on.

- (1) Contact ATC on radio.

— **END** —

1. MISCELLANEOUS SYSTEMS (CONT'D)

H. ADS-B Out Fault

Indication: **ADS-B OUT FAULT** caution message on.

- (1) MFD or CDU, ATC/TCAS CONTROL menu Select alternate transponder.

————— **END** —————

I. ADS-B Out Failure

Indication: **ADS-B OUT FAIL** caution message on.

- (1) Contact ATC on radio.

————— **END** —————

J. Dual GPS Failure During Oceanic and Remote Operations

Indication: IRS ONLY message on PFD, or FMS DR message on PFD or MFD, or DR EXCEEDS 5 MIN message on MFD.

- (1) GPS 1 and GPS 2 circuit breakers (1N3 and 2H11) Open and close.

Message persists after approximately 60 seconds:

◆ **Yes**

- (2) Notify controlling agency.

— **END** —

□ **No**

- (2) No further action required.

————— **END** —————

K. FSU Failure

Indication: IFIS electronic charts, MFD graphical weather and enhanced map overlays are inoperative.

FSU INOP displayed on both MFDs:

◆ **Yes**

- (1) Use alternate source for electronic charts.

— **END** —

□ **No**

- (1) Reversionary panel, EICAS switch Select to BOTH PFD'S.

————— **END** —————

1. MISCELLANEOUS SYSTEMS (CONT'D)

L. Flight Management System CDU/FMC Failure

Indication: No data available to CDU or CDU is unresponsive. Red FMS flag on PFD.

NOTE

If a red FMS flag is displayed during an LPV approach with indications of an air data or attitude/heading failure, refer to INSTRUMENTS SYSTEM – Air Data Computer Failure or Inertial Reference System Failure in this chapter.

- (1) Reversionary panel, FMS 3 switch
(if installed) Set to failed FMS position.
- (2) CDU 1 (2) TUNE INHIB
switch/light for affected CDU Press in

One FMS is operative:

Yes

- (3) N₁ setting Deselect from EICAS.

NOTE

1. Data from the CDU can be used if crossed checked with the AFM. TGT thrust mode may be used to set the N₁ setting on EICAS after confirmation of the data.
2. The input parameters on the THRUST LIMIT, TAKEOFF REF and APPROACH REF CDU pages must be re-confirmed by the pilot following an FMS 3 reversion (if installed), an FMS SYNC command or an FMS cold start.

On airplanes incorporating Service Bulletin 605-34-019:

The input parameters on the THRUST LIMIT, TAKEOFF REF and APPROACH REF CDU pages must be re-confirmed by the pilot following an FMS 3 reversion (if installed) or an FMS cold start.

— END —

No

- (3) No further action required.

— END —

1. MISCELLANEOUS SYSTEMS (CONT'D)

M. RAAS Advisories In Air or During High Speed Operations

If advisory is in conflict with expectations:

- (1) Verify position and/or distance remaining or distance available.
- (2) Contact ATC for assistance if necessary.

NOTE

If aural alerts become distracting or interfere with communications between flight crew or ATC, select the TAWS WARNING RAAS OFF switch/light OFF.

————— END —————

N. RAAS Advisories On Ground During Low Speed Operations

If advisory is in conflict with expectations:

- (1) Stop.
- (2) Verify position.
- (3) Contact ATC for assistance if necessary.

NOTE

If aural alerts become distracting or interfere with communications between flight crew or ATC, select the TAWS WARNING RAAS OFF switch/light OFF.

————— END —————

O. RAAS Inoperative

Indication: **RAAS FAIL** status message on.

- (1) TAWS WARNING, RAAS OFF
switch/light..... Press in Check OFF light on.

————— END —————

P. RAAS Not Available

Indication: **RAAS NOT AVAILABLE** status message on.

NOTE

RAAS is enabled, but the system either has no position information, the accuracy of the position information is insufficient to allow RAAS to function, or the aircraft is at an airport that has not been validated for RAAS in the TAWS terrain database.

- (1) TAWS WARNING, RAAS OFF
switch/light..... Press in Check OFF light on.

————— END —————

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1. MISCELLANEOUS SYSTEMS (CONT'D)

Q. EVS Heater Failure

Indication: **EVS HEAT FAIL** caution message.

- (1) EVS image Monitor for degradation.

————— **END** —————

R. EVS Heater Overheat

Indication: **EVS HEAT OVHT** caution message.

- (1) EVS image Monitor for degradation.

————— **END** —————

S. Synthetic Vision System Runway Failure

Indication: **SVS RWY INOP** status message on PFD.

- (1) SVS Select to OFF prior to take-off and prior to passing the final approach fix (FAF).

————— **END** —————

T. Synthetic Vision System Failure

Indication: **SVS INOP** status message on PFD.

- (1) SVS Select to OFF.

————— **END** —————

TRAINING PURPOSES ONLY

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TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	ABNORMAL PROCEDURES Aural/Visual Warning System	Vol. 1	05–18–1 Oct 05/06
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1. AURAL/VISUAL WARNING SYSTEM

A. Loss of EICAS

Indication: MFD displaying EICAS goes blank or EICAS data is not available on selected MFD.

If display blanks, the system automatically transfers the EICAS primary page data to the operational MFD.

- (1) EICAS switch.....Select to BOTH PFD'S.

————— **END** —————

B. Data Concentrator Unit Failure

Indication: **DCU 1 (2) INOP** status message.

- (1) Affected AUDIO WARNING switch.....Select to **DISABLE**.
(2) Other AUDIO WARNING switch.....Select to **NORMAL**.

————— **END** —————

C. Data Concentrator Unit Aural Failure

Indication: **DCU 1 (2) AURAL INOP** status message.

- (1) Affected AUDIO WARNING switch.....Select to **DISABLE**.
(2) Other AUDIO WARNING switch.....Select to **NORMAL**.

————— **END** —————

D. EICAS Comparator Inoperative

Indication: **EICAS COMP INOP** caution message.

- (1) MFD displays (both)Select to EICAS page.
(2) Engine and instruments, parameters and messagesCross-check for any anomalies.
(3) MFD, EICAS pageUse reliable side.
(4) Affected engine and/or systemMaintain within limits.

————— **END** —————

E. TCAS System Failure

Indication: **TCAS FAIL** caution message on PFD and/or MFD.

- (1) MFD or CDU tuning, ATC/TCAS modeSet to **ALT ON**, to select TCAS **OFF**.
Check TCAS OFF indicated on MFDs/PFDs.

————— **END** —————

1. AURAL/VISUAL WARNING SYSTEM (CONT'D)

F. TCAS VSI Resolution Advisory Failure

Indication: NO VSI RA on PFD or MFD.

- (1) MFD or CDU tuning, ATC/TCAS
mode Set to TA ONLY.

————— **END** —————

TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	ABNORMAL PROCEDURES Other Procedures	Vol. 1	05–19–1 REV 49, Nov 19/18
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1. OTHER PROCEDURES

A. Suspected External Damage (Ground Only)

If the crew suspects that any external damage to the airplane has been caused by impact from a foreign object, (bird, animal, drone, ground equipment, maintenance vehicle, etc.), a visual inspection similar to a walkaround must be performed prior to the next dispatch.

- (1) External Walkaround Accomplish Refer to Chapter 4; NORMAL PROCEDURES – AIRPLANE PREPARATION – External Walkaround

Particular attention should be given to the following areas:

- Antennae
- Engine fan blades and cowl leading edges
- Flight control surfaces
- Landing gear areas (nose and main strut, hydraulic lines, wire harnesses, proximity sensors)
- Nose cone
- Probes (pitot/static, TAT, T2, etc.)
- Taxi and landing lights
- Windshield and window surfaces
- Wing and stab leading edges
- Winglet

Damage may take many forms such as bent or missing components, puncture marks or deformed skin, delaminated or cracked windows etc. An indication of bird or animal remains, paint transfer, scuff marks and scratches may also be evidence of possible underlying damage.

If evidence of damage is found or if the crew has a reason to believe that a component should be further examined before flight (i.e. due to engine ingestion, composite surface strike, etc.), dispatch is **not allowed**. Crew must contact Maintenance personnel, or call the Bombardier Customer Response Center for further guidance.

Non-Composite areas:

Dispatch may be possible, but **only** if no visual evidence of damage is found **and** the impact area is not made from composite material. Suspected impact areas must also be reevaluated before each subsequent leg of the flight. At the end of the flight the crew must report the occurrence to Maintenance, and provide all available information, (suspected impact areas, physical evidence, photos, etc.) to trigger an appropriate maintenance follow-up.

Composite areas:

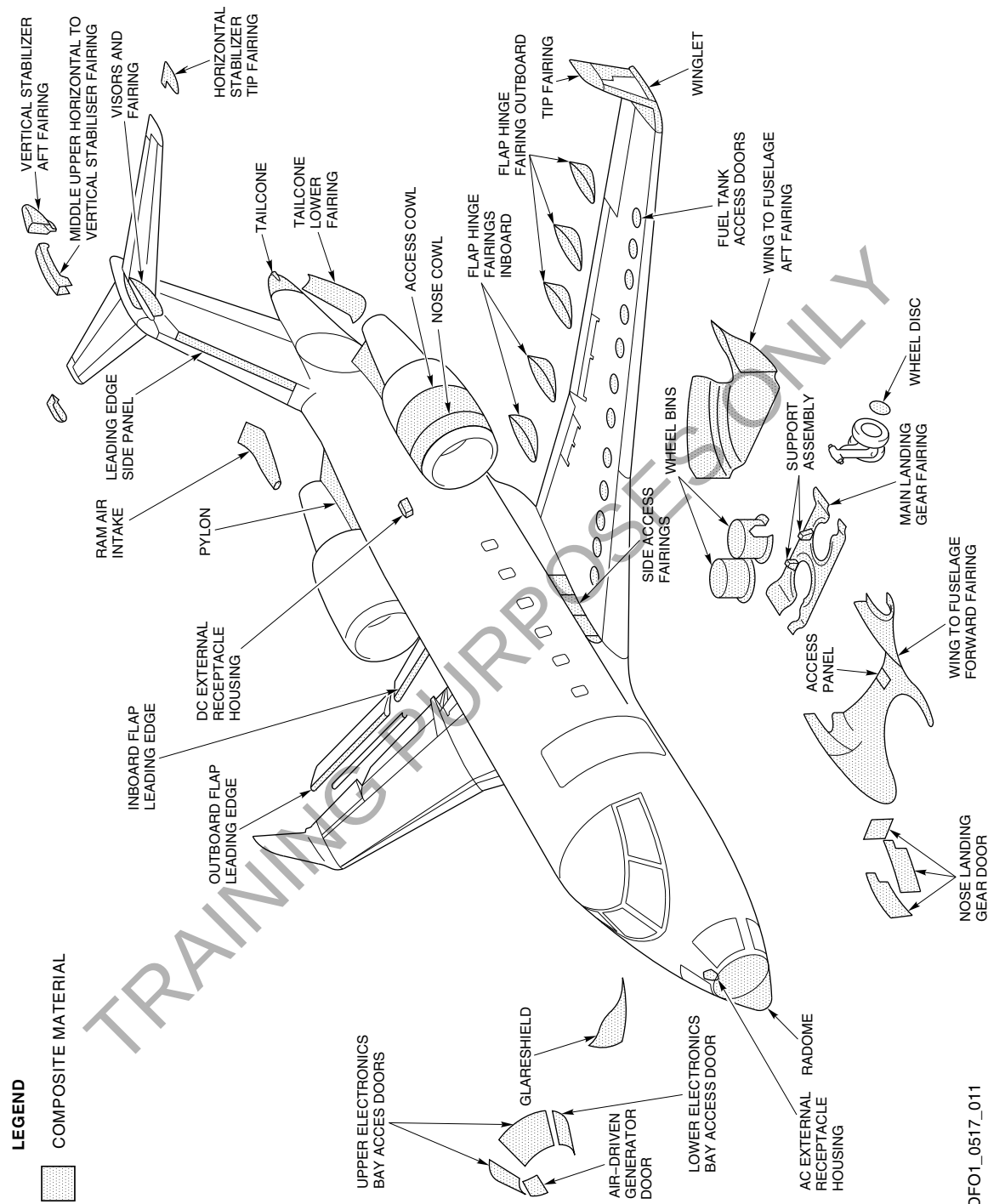
Damage to surfaces and structures made from composite materials may be particularly difficult to assess by means of a visual inspection alone. Therefore, if damage to these areas is suspected, maintenance action is required before attempting dispatch. Figure 05–19–1 shows the composite areas of the airplane.



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1. OTHER PROCEDURES (CONT'D)

A. Suspected External Damage (Ground Only) (Cont'd)



Composite Components
Figure 05-19-1

END

CHAPTER 6 - SUPPLEMENTARY PROCEDURES

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1. INTRODUCTION

Supplementary procedures are normal procedures not related to a specific phase of flight, and are accomplished “as required”, and not routinely performed on each flight. That may include procedures to conform with Air Traffic Control (ATC) instructions and other considerations, and for the operation of airplane systems which are used either as an option or as the situation warrants (e.g. Cat II operations, etc).

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1. AIR-CONDITIONING ON THE GROUND USING THE APU

The APU is used to provide air-conditioning on the ground, as follows:

- (1) APU Operating Check APU AVAIL light on.
- (2) BLEED AIR, 10TH STAGE,
L and R switch/lights Check pressed out.
Check L and R 10TH STAGE
CLOSED lights on.
- (3) BLEED AIR, APU LCV switch/light..... Press in to open load control valve.
Check APU LCV OPEN light
on.
- (4) BLEED AIR, 10TH STAGE, ISOL
switch/light Press in to open isolation valve.
Check 10TH ISOL OPEN light
on.
- (5) AIR-CONDITIONING, L and R
PACK switch/lights Press in to turn on packs.
Check L and R PACK OFF
lights out.
- (6) AIR-CONDITIONING, CKPT
TEMP and CABIN TEMP
selectors Set mode and temperature as
desired.

NOTE

Flight compartment and duct temperatures should not exceed 71°C (160°F) or be less than 3°C (37°F) during manual mode operations.

2. AIR-CONDITIONING FROM ENGINE BLEEDS DURING TAKE-OFF

If the engine bleed is desired to provide air-conditioning during take-off, transfer bleeds as follows:

- (1) BLEED AIR, 10TH STAGE R
switch/light Press in to open right valve.
Check R 10TH CLOSED light out.
- (2) BLEED AIR, 10TH STAGE ISOL
switch/light Check pressed out, to ensure isolation valve is closed.
Check the following:
 - 10TH ISOL OPEN light out.
 - APU LCV OPEN light on.
- (3) BLEED AIR, 10TH STAGE L
switch/light Press in to open left valve and close APU load control valve.
Check the following:
 - L 10TH CLOSED light out.
 - APU LCV OPEN light out.
- (4) BLEED AIR, APU LCV switch/light Press out to match APU load control valve closed position.

3. UNPRESSURIZED TAKE-OFF AND LANDING PROCEDURE

The procedures below are recommended when performing a take-off or landing without using engine and APU bleeds for pressurization.

A. Before Take-Off

- (1) AIR-CONDITIONING, RAM AIR switch/light Press in to open ram air valve.
Check RAM AIR OPEN light on.
- (2) CABIN PRESSURIZATION, EMER DEPRESS switch/light Press in Check the following:
 - EMER DEPRESS ON light on, and
 - **EMER DEPRESS** caution message on.
- (3) AIR-CONDITIONING, L and R PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (4) BLEED AIR, 10TH STAGE, L and R switch/lights Press out Check L and R 10TH STAGE CLOSED lights on.

B. After Take-Off

At a safe altitude (not below 1,500 feet AGL):

- (1) BLEED AIR, 10TH STAGE, L and R switch/lights Press in Check L and R 10TH STAGE CLOSED lights out.

NOTE

If an engine failure occurs, delay selecting the packs on until after obstacle clearance is assured.

- (2) AIR-CONDITIONING, R PACK switch/light Press in to start right air-conditioning pack.
Check R PACK OFF light out.

NOTE

A pressure bump may occur.

- (3) AIR-CONDITIONING, RAM AIR switch/light Press out to close ram air valve.
Check RAM AIR OPEN light off.



3. UNPRESSURIZED TAKE-OFF AND LANDING PROCEDURE (CONT'D)

B. After Take-Off (Cont'd)

- (4) AIR-CONDITIONING, L PACK
switch/light Press in to start left air-conditioning pack.
Check L PACK OFF light out.
- (5) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Press out Check the following:
- EMER DEPRESS ON light out, and
 - **EMER DEPRESS** caution message out.

C. Landing

- (1) Airplane altitude Descend to 10,000 feet maximum or lowest safe altitude, whichever is higher.
- (2) CABIN PRESSURIZATION,
PRESS CONT switch/light Check pressed out, thus enabling automatic operation.
Check MANUAL light out.

At the turn to final approach:

- (3) AIR-CONDITIONING, L and R
PACK switch/lights Press out to shutdown both packs.
Check L and R PACK OFF lights on.
- (4) BLEED AIR, 10TH STAGE,
L and R switch/lights Press out Check L and R 10TH STAGE CLOSED lights on.
- (5) AIR-CONDITIONING, RAM AIR
switch/light Press in to open ram air valve.
Check RAM AIR OPEN light on.
- (6) CABIN PRESSURIZATION,
EMER DEPRESS switch/light Press in Check the following:
- EMER DEPRESS ON light on, and
 - **EMER DEPRESS** caution message on.

NOTE

Avoid high rates of descent for passenger comfort.

1. GENERAL

The procedures outlined in this section address the operation of the Automatic Flight Control System (AFCS), and its interface with the other airplane systems peculiar to the CL-600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049). The presentation is by phase of flight, as applicable, and it is assumed that the AFCS and other correlated systems are fully operational, and are being utilized to their full capacity. The following procedures only serve to supplement the normal procedures, and should be employed as applicable.

The AFCS automatically moves and controls the airplane's flight control surfaces, or gives commands to the flight crew to follow guidance commands on the PFDs, depending on the AFCS selection made. Selection of any of the mode switches on the flight control panel (FCP) will illuminate the Mode/FCC Channel indicators, located adjacent to each mode switch.

For a more elaborate description of the procedures outlined herein, refer to Chapter 4; NORMAL PROCEDURES – APPROACH AND LANDING.

For a more detailed description of the AFCS and its controls and indications, refer to the Flight Crew Operating Manual (FCOM), Vol. 2.

2. BEFORE TAKE-OFF

- (1) Take-off data Computed/Set PLT
- (2) Navigation instruments Set Both
- (3) Course Set/Cross-check Both
 - For departure, using the course select knobs – CRS 1 & CRS 2 on the FCP.
 - Pilot and Copilot will cross-check each other's PFD for concurrence.
- (4) Heading Set PLT
 - To runway heading, using the heading (HDG) select knob.
- (5) Target altitude Set PLT
 - As required, using the altitude (ALT) preselect knob.

3. TAKE-OFF

Prior to entering the runway or when aligned on the runway centre line:

- (1) Take-Off/Go-Around switch Press PLT
- Flight director command cues will automatically appear on the PFDs.

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

- (2) VS pitch wheel Reduce PLT
- Flight director command cue to 10°.

NOTE

Auto throttle will not be available for take-off.

At V_R :

- (3) Airplane Rotate PF
- Smoothly towards an initial pitch target of 12° in one continuous motion at a pitch rate not exceeding 3° per second.

On airplanes 5701 to 5784 NOT incorporating Service Bulletin 605-22-001:

- (3) Airplane Rotate PF
- Smoothly towards an initial pitch target of 10° in one continuous motion at a pitch rate not exceeding 3° per second.

- (4) Pitch attitude Adjust PF
- To achieve a minimum airspeed of $V_2 + 10$ KIAS.

At a safe altitude:

- (5) FLC mode Select PNF
- Using the FLC switch.
- (6) Autopilot Engage PNF
- Using the autopilot engage (AP) switch.
- (7) Heading mode Select PF
- Using the heading select (HDG) switch.

At 1,000 feet above airport elevation (AAE):

- (8) Target speed Select PF
- Using the speed bug (SPEED) knob.

If a turn is required after take-off:

- (9) Heading Set as required PF

At 3,000 feet AAE:

- (10) Target speed Select PF

4. CLIMB

- (1) Preselect speed..... Set
 - To required climb speed.
- (2) Navigation mode Select
 - Using the navigation mode (NAV) switch.
 - System will operate in the current lateral mode with reference to the active (and valid) navigation signal of the selected NAV source, as displayed on the PFDs.
- (3) Target altitude Set
 - To cruise altitude or as instructed by ATC.

5. LEVEL-OFF

The AFCS will automatically capture and hold the pre-selected altitude. The flight crew can monitor the operation of the system by maintaining the flight directors on. Annunciations in the vertical mode indicator portion of the PFD will change as each condition is met (Refer to FCOM Vol. 2, Chapter 4; AUTOMATIC FLIGHT CONTROL SYSTEM).

If required to level-off at other than the desired cruise altitude:

- (1) Altitude mode Select
 - Using the altitude mode (ALT) switch, which will maintain the pressure altitude (altitude hold) at the time of selection.
- (2) Altitude Set
 - Select required altitude.
- (3) Thrust lever Adjust
 - As necessary.

6. CRUISE

Cruise procedures only involve monitoring of the system operation in the NAV mode. Depending on the NAV source selected, the flight control computer (FCC) generates lateral commands to fly the active (and valid) navigation signal.

If the active source is a Flight Management System (FMS), lateral commands are continuously sent by the flight management computer to the FCC after capture of the desired course, to follow the programmed flight plan. Deviations or changes to the pre-programmed flight plan can be accomplished by following the procedures outlined in the Pilot's Operating Manual, provided by the manufacturer.

If the active NAV source is other than a FMS (e.g. VOR, LOC, etc.), the FCC generates commands to maintain the selected course (or beam) once captured. Heading or course changes can be accomplished by going into the heading mode and selecting a new heading, or by selecting another course, and/or by tuning-in to another NAV source.

7. DESCENT

Climbs and descents can normally be accomplished by selecting the speed mode, with the system adjusting to the selected speed.

NOTE

Some excursions above V_{MO}/M_{MO} may occur when using the autopilot or flight director in IAS or vertical speed mode. Care should be taken not to exceed V_{MO}/M_{MO} .

If desired:

- (1) Vertical speed mode..... Select
 - Using the vertical speed mode (VS) switch.
- (2) Vertical reference value..... Set
 - Using the speed/pitch wheel.
 - Push the wheel forward to decrease the vertical speed value (also a pitch down command), and pull backward to increase the vertical speed value (also a pitch up command).
- (3) Target descent speed..... Set
- (4) Thrust levers..... Adjust
 - As required.
- (5) Target altitude Set
 - To the approach altitude or as instructed by ATC.

8. HOLDING

Prior to entering the hold, cruising speed should be reduced to holding speed ($V_{REF} + 30$ KIAS), by selecting the target speed using the speed bug knob on the FCP, and reducing thrust, as necessary. Altitude hold is selected to maintain the desired pressure altitude, and changes in heading are accomplished by going into the heading mode and selecting the appropriate headings, as required.

9. INSTRUMENT APPROACHES

Autopilot procedures for instrument approaches are essentially the same. The autopilot must not be used, however, at altitudes below 320 feet AGL, except when on an ILS approach, where the minimum allowable altitude for the operation of the autopilot is 80 feet AGL.

During approaches, descent is initiated by using the vertical speed mode, by selecting the VS switch which will automatically maintain the referenced vertical speed. When desired, the reference vertical speed can be changed by using the Speed/Pitch wheel adjacent to the VS switch.

The preselect altitude is set to the procedure turn altitude (if applicable), when tracking outbound from the fix. Maintaining headings and turns are accomplished in the heading mode by pressing the HDG switch, and rotating the heading select knob to the desired heading(s). If so desired, the half bank mode may be used in conjunction with the heading select mode.

When cleared for the approach on the inbound leg to the fix, or when on the localizer intercept heading for a straight-in approach, select the approach mode by pressing the APPR switch. The flight control computer then arms for glideslope capture, if on a front course approach, captures the glide slope (as annunciated by green GS on the PFDs), and maintains flight on the glide path. The half bank mode (if previously selected) will be automatically cleared in this instance.

At this point, the preselect altitude should be set to the missed approach altitude, to prepare for a possible overshoot.

In all instances, changes to the thrust setting must be made accordingly, in order to maintain the desired speed.

10. CIRCLING

If the circling altitude is lower than 320 feet AGL, the autopilot must not be used.

If the autopilot is used, altitude hold is accomplished by selecting ALT mode, which will maintain the pressure altitude existing at the time of selection. Turns are accomplished in the heading mode, and descents made with the vertical speed mode.

11. MISSED APPROACH

The autopilot may be engaged after a missed approach, at a safe altitude (not below 320 feet AGL). The missed approach altitude should already have been selected when tracking inbound towards the fix. Turns are accomplished in the heading mode, and climbs made with thrust adjustments to achieve the appropriate climb speed, enabling the autopilot to capture the preselected missed approach altitude.

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BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Auxiliary Power Unit	Vol. 1	06-04-1
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1. ELECTRICAL REQUIREMENTS

The APU 28 volt start motor normally gets its power directly from the APU battery, which is 24 volts DC. AC power from whatever source is not capable of augmenting a weak battery; therefore, the batteries must have adequate power to effect a successful normal start.

An alternate APU start utilizes DC power supplied from a ground power source, connected to the DC service receptacle located on the underside of the fuselage tail section.

The battery master switch must be ON for all types of start.

NOTE

APU BATTERY OFF caution message may appear momentarily during APU start.

2. ABNORMALITIES DURING APU START

A. APU Fails to Start

If the starter does not engage and the APU fails to start, stop the start sequence by pressing the APU START/STOP switch. The permitted APU start cycles are as follows:

- (1) Using airplane batteries on the ground or for normal in-flight starts:
 - Three start attempts, each of 30 seconds continuous cranking,
 - Followed by a 20 minute off-time,
 - Followed by two further attempts, each of 30 seconds continuous cranking.
- (2) Using ground power source:
 - Two start attempts, each of 15 seconds continuous cranking,
 - Followed by a 20 minute off-time,
 - Followed by two further attempts, each of 15 seconds continuous cranking.
- (3) If, in either case (1) or (2) above, a successful start is not obtained, a further start must not be attempted for a period of at least 35 minutes.

B. Hung Start

During start, if the APU rpm is not greater than 30%, within 60 seconds, stop the start sequence by pressing the APU START/STOP switch.

3. FAILURE OF APU TO SHUTDOWN

If the APU fails to shutdown when the START/STOP switch is pressed, push the APU FIRE PUSH switch/light on the glareshield to stop the APU. Keep in mind that the APU fire extinguishing SQUIBS will be armed for as long as the fire switch is pressed in. After the APU has stopped, reset the APU FIRE PUSH switch/light to the normal position.

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BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Flight Controls	Vol. 1	06-05-1
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1. CREW COORDINATION IN THE EVENT OF FLIGHT CONTROL JAM

In the event of a jam in one of the ailerons' cable runs, the control wheels can be disconnected using the ROLL DISC handle on the centre pedestal. When the ROLL DISC handle is pulled, the pilot has control over the left aileron and the copilot controls the right aileron.

In the event of a jam in one of the elevator's cable runs, the control columns can be disconnected using the PITCH DISC handle on the centre pedestal. When the PITCH DISC handle is pulled, the pilot has control over the left elevator and the copilot controls the right elevator.

In the event of a jam in one of the rudder's cable runs, an anti-jam breakout mechanism enables the operation of the unaffected side. A cross-coupling mechanism synchronizes the movements of the pilot's and copilot's breakout mechanisms. The rudder pedals should be overpowered in this case, and the anti-jam breakout and summing mechanisms in the system will compensate for this situation, enabling full travel and unimpeded movement of the rudder pedals.

The flight crew should keep these facts in mind while adhering to standard cockpit management procedures, and strive to be as coordinated as possible during all maneuvers. Trims should be adjusted, as required, and appropriate procedures performed.

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1. REFUELING/DEFUELING

A. Operation

The Fuel System Computer Unit (FSCU) Channel 1 facilitates automatic refueling operation. In the event of Channel 1 failure, Channel 2 will take over. If the fuselage refuel/defuel panel sends faulty transmission data, the FSCU will detect the fault, and send the information to the Refuel/Defuel panel.

The airplane may be refueled or defueled by pressure/suction or by gravity, using standard ground equipment. A single point adapter, located in the right hand wing leading edge filler, permits pressure fueling and defueling. All pressure fueling/defueling operations are controlled from a refuel/defuel panel.

NOTE

During the refueling/defueling operation, check for fuel spillage from the vent relief valves, water drain valves and the NACA vents.

Do not pull a main tank relief valve T-handle, if that main tank quantity is greater than 1,814 kg (4,000 lb). Do not pull the auxiliary tank relief valve T-handle, if the auxiliary tank quantity is greater than 2,041 kg (4,500 lb).

B. Refuel/Defuel Control Test

A test must be performed to verify the overfill protection of the refuel/defuel system. The following are checked during the test:

- High level sensor failure, and
- SOV failure.

NOTE

Fuel pressure is required to perform the refuel/defuel control test. Ensure that fuel tender pressure is applied before activating any switches.

(1) POWER switch On

- Check that the ON light comes on.

(2) Ensure that the fuel tender pressure is not more than 379 kPa (55 psi).

(3) SHUTOFF TEST TEST

- Check that the SOV OP lights come on, HIGH LEVEL SENSOR lights come on in sequence (left main, right main, tail and auxiliary tank), SOV OP lights go out, SOV CL lights come on.

1. REFUELING/DEFUELING (CONT'D)

C. Pressure Refueling

NOTE

Fuel pressure is required to perform the pressure refueling. Ensure that fuel tender pressure is applied before activating any switches.

- (1) Ensure that the airplane and the fuel tender are grounded.

NOTE

1. If the grounding wire has an alligator clip, attach the clamp to the grounding lug on the right side wall of the nose gear compartment.
2. If the grounding wire has a ground stud, put the stud in the nearest ground stud receptacle on the wing leading edge.

- (2) Fuel/Defuel panel Open

- (3) POWER FUEL switch..... POWER FUEL

The flight compartment refuel/defuel control panel, if installed, will override the fuselage refuel/defuel control panel.

- (4) Ensure that the fuel tender pressure is not more than 379 kPa (55 psi).

- (5) SOV switches Open (as required)

- To initiate pressure refueling.
- Check that the applicable SOV CL lights come on, when the preselected fuel quantity has been reached.

- (6) SOV switches CLOSED

- (7) POWER FUEL switch..... OFF

- (8) Refuel/Defuel panel Closed

- (9) Ensure that:

- Fuel nozzle is disconnected and the refuel/defuel adapter is capped.
- All grounding cables disconnected.
- Refuel/Defuel Adapter panel access door closed.

D. Gravity Refueling

Gravity refueling is enabled through overwing filler caps (3) on the upper wing surface. The main tanks have their respective filler caps on each side of the wings. The aux tank has its filler cap installed near the wing root on the right upper wing surface.

Caution must be observed not to overfill a tank. Gravity filler caps for the main tanks are located below the maximum fuel level, thus the tanks cannot be filled to the maximum. Do not open the gravity filler caps if the tanks are full, or if the fuel quantity is not known.

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1. REFUELING/DEFUELING (CONT'D)

E. Pressure Defueling

Automatic defueling is accomplished by selecting the POWER switch to DEFUEL, and SOV switches to ON. In the defuel mode, the SOVs are de-energized open with the SOV switches selected to ON. Suction is then applied through the refuel/defuel hose adapter, which is connected to the single-point adapter, and the tanks will then defuel.

F. Gravity Defueling

Gravity defueling may be accomplished through fuel tank drain valves (3), located on the underside of the wings. The main tanks' fuel drain valves are installed near the wing root on the underside of each wing. The center tank fuel drain valve is located near the wing root, underside of the left wing. A gravity defueler adapter can then be inserted into the valve, after it has been opened, to start gravity defueling.

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1. TAXIING WITH FLAT TIRE(S)

If the airplane experiences a tire blowout or a flat on one tire on the runway, clear the runway as soon as possible, so as not to obstruct other traffic. Taxi as slow as possible, using the least amount of braking necessary, to prevent heat build-up in the affected gear. Sharp turns should be avoided.

Taxiing completely clear of the runway is acceptable, provided only one tire per gear is flat. In cases where both tires of a gear are flat, taxiing is not recommended except in the interests of safety, because the airplane will then be rolling on the wheel rims.

NOTE

The duration of taxiing with a flat or blown-out tire should be kept to a minimum, since the remaining tire could also blow out.

2. TOWING WITH FLAT TIRE(S)

Normal towing procedures are acceptable when only one tire per gear is flat, and the airplane weight is not greater than the maximum landing weight. When both tires of a gear are flat, special maintenance handling/towing procedures will need to be employed, so as not to exceed the capacity of the nose gear for towing.

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1. FLIGHT MANAGEMENT SYSTEM

NOTE

FMS normal procedures are contained in the FMS-6000 Pilot's Guide.

On airplanes incorporating Service Bulletin 605-34-019:

FMS normal procedures are contained in the FMS-6200 Pilot's Guide.

A. Prior to FMS Use

- (1) FMS position information..... Initialize
- (2) CDU, FMS CONTROL page Select SYNC mode.

NOTE

1. The input parameters on the THRUST LIMIT, TAKEOFF REF and APPROACH REF CDU pages must be re-confirmed by the pilot following an FMS SYNC command, an FMS 3 reversion (if installed) or an FMS cold start.
2. The engine bleed selection used for the FMS thrust setting computations is based upon the sensed setting of the airplane engine bleed cockpit switches, unless manually overridden by the flight crew on the THRUST LIMIT CDU page.

The engine bleed selection used for the FMS V speed computations is based upon the default setting (as selected on the DEFAULTS 4/4 CDU page) or upon manual selection, by the flight crew, for the anticipated take-off or approach configuration.

The engine bleed setting on the TAKEOFF REF and APPROACH REF CDU pages represents the planned setting, and not the actual/sensed airplane setting.

3. In case of multiple obstacles, each obstacle clearance calculation is done separately. For each obstacle, obstacle height and obstacle distance from reference zero should be entered in the FMS. The FMS computes and displays the geometric gross level-off height and the required climb gradient for obstacle clearance. The highest gross level-off height and the greatest required climb gradient of all obstacles should be retained. Note that one obstacle may be the most limiting for the required climb gradient, while another is the most limiting for the gross level-off height.

The most limiting gross level-off height and required climb gradient should be entered in the FMS. The FMS computes and displays the take-off weight limited by obstacle clearance, which is used to calculate the maximum take-off weight, to ensure all obstacles will be cleared.



1. FLIGHT MANAGEMENT SYSTEM (CONT'D)

A. Prior to FMS Use (Cont'd)

On airplanes incorporating Service Bulletin 605-34-019:

- (1) FMS positionAuto initialize

NOTE

1. The engine bleed selection used for the FMS thrust setting computations is based upon the sensed setting of the airplane engine bleed cockpit switches, unless manually overridden by the flight crew on the THRUST LIMIT CDU page.

The engine bleed selection used for the FMS V speed computations is based upon the default setting (as selected on the DEFAULTS 5/5 CDU page) or upon manual selection, by the flight crew, for the anticipated take-off or approach configuration.

The engine bleed setting on the TAKEOFF REF and APPROACH REF CDU pages represents the planned setting and not the actual/sensed airplane setting.

2. In case of multiple obstacles, each obstacle clearance calculation is done separately. For each obstacle, obstacle height and obstacle distance from reference zero should be entered in the FMS. The FMS computes and displays the geometric gross level-off height and the required climb gradient for obstacle clearance. The highest gross level-off height and the greatest required climb gradient of all obstacles should be retained. Note that an obstacle may be the most limiting for the required climb gradient while another is the most limiting for the gross level-off height.

The most limiting gross level-off height and required climb gradient should be entered in the FMS. The FMS computes and displays the take-off weight limited by obstacle clearance, which is used to calculate the maximum take-off weight, to ensure all obstacles will be cleared.

B. After Periods of Dead Reckoning (DR) Navigation

- (1) FMS position information.....Check for accuracy.

C. Go-Around

On airplanes incorporating Service Bulletin 605-34-019:

NOTE

VNAV may be selected and used normally following the sequencing of the last waypoint on the missed approach procedure, or upon entering a new flight plan and activating a leg that is not part of the missed approach procedure.

2. INERTIAL REFERENCE SYSTEM

For the system description and operation of the inertial reference system, refer to the Flight Crew Operating Manual, Volume 2, Chapter 11; FLIGHT INSTRUMENTS.

A. Before Starting Engines

NOTE

The IRS automatically aligns using GPS position (normal IRS alignment time is approximately 7 minutes). If GPS is not available, it will align using the position entered to initialize the FMS.

On airplanes 5979 to 6049:

Following a successful initial alignment, do not manually re-enter the position in the FMS CDU either immediately before, or immediately after airplane motion. If this has occurred, a complete normal shutdown and restart of the airplane and IRSs must be completed. Failure to do so may lead to a total IRU loss during subsequent flight legs.

3. GLOBAL POSITIONING SYSTEM

A. Before Starting Engines

- (1) FMS..... Review and disposition messages.

NOTE

For any SBAS covered airspace (US RNAV routes, SIDs, and STARs), it is not required to run a pre-flight prediction of RAIM availability along the intended route if no satellites are NOTAM'd as out of service over the intended route.

For non-SBAS covered airspace, certain operations may require that an appropriate prediction program, such as Rockwell Collins part number 832-3443-008 or equivalent, be used prior to departure for RAIM availability along the intended route.

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TRAINING PURPOSES ONLY

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		REV 49, Nov 19/18	

1. QUICK TURN-AROUND STARTS

NOTE

If ITT is greater than 120°C prior to start, engine must be dry motored for a maximum of 90 seconds, with ignition off and affected thrust lever at SHUT OFF, in order to lower ITT below 120°C.

Refer to Chapter 2; LIMITATIONS – POWER PLANT for dry motoring cycle.

2. SINGLE ENGINE TAXI OPERATIONS

The purpose of this supplementary procedure is to provide operational guidance when conducting single engine taxi operations (both taxi out and taxi in) and should serve as a baseline reference if developing operator specific procedures.

Single engine taxi operations can provide fuel savings, emission reductions and conserve brake energy. However, there are many considerations that operators and pilots should review to ensure safe and efficient single engine taxi operations.

Consider all existing conditions when deciding whether or not to perform single engine taxi, including the following:

- Single engine taxi operations are prohibited if the OAT is 10°C (50°F) or below.
- Consider ramp size and congestion (increased jet blast associated with single engine taxi).
- Consider ramp and taxiway surface conditions (slipperiness, contamination, dust/dirt, FOD, etc.). Single engine taxi procedures may require higher than normal power settings for maneuvering and may result in reduced control on slippery surfaces. Heavy ramp weights should be of particular concern. Single engine taxi operations are not recommended on slippery or contaminated surfaces.
- Single engine breakaway thrust is approximately 39% N1 (ISA, MTOW, level surface).
- Consider brightness/lighting/visibility conditions (collision avoidance).
- Single engine taxi out operations (before take-off) are best suited for taxi time of 15 minutes or more.

A. Engine Start and Systems Management

It is strongly recommended to conduct single engine taxi with the right engine operating so that the hydraulic system 2 provides pressure to the outboard brakes.

If taxiing with the left engine only, the hydraulic 2B pump must be selected ON prior to aircraft movement.

The fuel boost pump for the inoperative engine should be deselected to prevent fuel imbalance during single engine operation.

For optimal cabin air management during single engine taxi operation, bleed air must be taken from the APU.

When performing single engine taxi with the engine as a bleed source, should both PACKs be needed, the 10th stage isolation valve must be selected OPEN (pressed in).

It is strongly recommended that the airplane be stationary when starting the second engine to maximize crew monitoring and minimize operational workload.

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2. SINGLE ENGINE TAXI OPERATIONS (CONT'D)

If using cross bleed to start the second engine while the airplane is moving, due to the higher thrust required by the cross bleed start procedure, additional consideration should be given to operational conditions, including taxiway contamination, width of taxiway, wind, brake temperature and distance to other aircraft or obstacles.

After the second engine is started, allow a minimum of 5 minutes for warm up before take-off.

If conducting single engine taxi in operations (after landing or return to ramp), wait at least 2 minutes at idle power before shutting down the engine.

B. Taxi

When taxiing with the left engine only, the hydraulic 2B must be selected ON prior to aircraft movement.

To prevent fuel imbalance during single engine operation, deselect the fuel boost pump for the inoperative engine.

During prolonged single engine taxi, fuel imbalance may occur. To prevent or correct fuel imbalance during single engine taxi operation, open the respective L to AUX or R to AUX XFLOW valve to balance the fuel tanks. Fuel imbalance must not exceed 400 lb during taxi.

Plan to start the second engine at least 5 minutes before take-off to allow for sufficient engine warm up time.

After starting the second engine, it is the responsibility of the crew to complete all checklist items that were deferred due to single engine taxi operations (i.e. after engine start and taxi checks).

The crew should plan to use more time during the first flight of the day due to the additional check requirements.

C. Take-Off

Before take-off is initiated, verify that all engines and airplane systems are within operational limits and are properly configured (e.g. BOOST PUMPS pressed in, L and R to AUX XFLOW pressed out).

3. ENGINE OIL REPLENISHMENT

If it is required to top-up the engine oil tank, the following outlines the procedures employed using the oil replenishment system.

A. Oil Level Control Panel Test

- (1) ON/OFF switch..... ON
- (2) ENG OIL, PRESS TO TEST
switch/light..... Press in and hold.

Check LH and/or RH FULL
lights come on for 2 to 4
seconds, then go out again.

NOTE

If either engine oil tank is full, the applicable FULL light would have been on and will remain on after the test. If the indications as stated above do not occur, the system is defective and must not be used.

B. Replenishment Procedures

- (1) Oil quantity indicator..... Verify and make a record of the oil level in the replenishment tank.
- (2) Manual selector valve..... Turn and hold to the R or L position, as applicable.
- (3) Oil quantity Monitor as the oil level in the replenishment tank decreases.
- (4) Manual selector valve..... Release as soon as the replenishing pump automatically stops (the LH or RH FULL light, as applicable, comes on), or until the oil quantity in the replenishment tank decreases by 2 U.S. quarts (1.9 liters), whichever comes first.

If the affected engine oil tank is not full after 2 U.S. quarts (1.9 liters) have been added:

- (5) Affected engine Dry motor for 30 seconds. The affected FULL light should then come on.

If the affected ENG OIL, FULL light still does not come on:

- (6) Steps (1) to (5) above..... Repeat within five (5) minutes. Confirm that the affected FULL light is now on.
- (7) Oil quantity indicator..... Verify and make a record of the oil quantity that was put into the engine.
- (8) Steps (1) to (7) above..... Repeat for the other engine, if required.



3. ENGINE OIL REPLENISHMENT (CONT'D)

B. Replenishment Procedures (Cont'd)

(9) ON/OFF switch.....OFF Check that all lights go out.

TRAINING PURPOSES ONLY

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NOTE

These supplementary procedures do not constitute approval to conduct Category II operations.

1. GENERAL

- I The CL–600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049) has been shown to meet the airworthiness requirements for Category II Operations contained in Appendix 1 of AC 120–29.

The following data must be used when conducting Category II operations.

These data complement or supersede data contained in the basic Flight Crew Operating Manual. This supplement must, therefore, be read in conjunction with the basic Flight Crew Operating Manual.

The effect of this Supplement on the basic Flight Crew Operating Manual is given in paragraphs (2) to (7).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

TRAINING PURPOSES ONLY

	CL–605 Flight Crew Operating Manual PSP 605–6	
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3. OPERATING LIMITATIONS

The operating limitations in Chapter 2 are applicable, except as modified by the following:

- An ILS approach to Category II minima must not be commenced or continued unless all required airborne equipment, specified in the Category II Required Equipment List below and their ground installations, are operating satisfactorily.

CATEGORY II REQUIRED EQUIPMENT LIST	
EQUIPMENT	PRIOR TO APPROACH
VHF NAV 1 and 2	Both must be operational and displayed onside.
PFD 1 and 2	One (1) PFD available and operational for each side.
STAB Channel 1 and 2	One (1) channel must be operational.
Radio Altimeter	One must be operational, with display on both sides.
IRS	Any two (2) must be operational.
ADC 1 and 2	Both must be operational.
FD 1 and 2	Both must be operational.
EFIS Comparator Monitors	Must be operational.
AFCs Pitch Trim	Must be operational.
Autopilot	Must be operational.
Hydraulics (3)	All systems must be on and operational.
Electrics	Two (2) generators on and sharing load (see NOTE).

NOTE

Approach to Category II minima may be continued with one generator, if an engine fails during final approach (Refer to ABNORMAL PROCEDURES).

- Operation of the autopilot is prohibited below 80 feet AGL.
- The approved flap configurations for approach are:
 - 45° for all engines operating, and
 - 20° for one engine inoperative.
- The minimum decision height for Category II operations is 100 feet AGL.
- Wind limits for autopilot approaches are:
 - 8 knots crosswind, and
 - 9 knots tailwind.
- The maximum ILS glidepath angle for Category II operations is 3.5 degrees.
- ATS must be disengaged.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

A. Autopilot Failure

Indication: Abnormal autopilot operation and/or **AFCS MSGS FAIL** warning message, “CAVALRY CHARGE” aural and/or flashing **AP** message on the primary flight displays.

Continue approach, if the following can be accomplished above 500 feet AGL:

- (1) Autopilot Disconnect
- (2) AFCS SEL 1(2) switch (centre pedestal)..... Select to operative AFCS channel.
Check **AFCS 1 (2) INOP** status message on.
- (3) Autopilot Engage and monitor operation.
- (4) Approach Continue to Category II minima.

If below 500 feet AGL, on a stabilized approach:

- (1) Autopilot Disconnect
- (2) Manual control Resume
- (3) Approach Continue to Category I minima.

If below 200 feet AGL and non-visual:

- (1) Go-around Initiate

— **END** —

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

A. Prior To Approach

Airplane must be properly configured prior to approach:

- | | | |
|---|--------|--|
| (1) Landing weight | Check | within limits (Refer to Chapter 6; PERFORMANCE – LANDING PERFORMANCE of the Airplane Flight Manual). |
| (2) NAV SOURCE | Set | to onside ILS (green) or armed (cyan) by FMS, for both PFDs. |
| (3) IRS and ADC source selectors | Set | to NORM. |
| (4) RA MINIMUMS | Set | on both PFDs. |
| (5) BARO MINIMUMS | Set | as a barometric DH, in case RA fails. |
| (6) MIN ALERT | Set | to RA on both PFDs. |
| (7) Runway Course | Set | on both PFDs, or armed (cyan) by FMS. |
| (8) EFIS | Check | that no comparator flags are displayed. |
| (9) CAS | Check | that the following messages are not displayed: <ul style="list-style-type: none"> • AFCS MSGS FAIL warning message, • AP PITCH TRIM, EFIS COMP INOP and EFIS MISCOMP caution messages, • FD 1 and 2 FAIL status message. |
| (10) Stabilizer and elevator position | Check | valid on EICAS or F/CTL synoptic page. |
| (11) Approach mode | Arm | |
| (12) Expanded lateral deviation indicator | Verify | displayed on both PFDs at 600 feet AGL. |

————— **END** —————

B. Before Landing

- | | | |
|---------------------|-----------|---|
| (1) Autopilot | Disengage | at an altitude not less than 80 feet AGL. |
|---------------------|-----------|---|

————— **END** —————

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

A. Single Engine Approach and Landing

- (1) Approach Conduct under Category I operations.

————— **END** —————

B. Engine Failure During Final Approach

Continue approach, if the following can be accomplished above 800 feet AGL:

- (1) Autopilot Disconnect
- (2) Operating engine Increase thrust, as required.
- (3) FLIGHT SPOILER lever Ensure retracted.
- (4) FLAPS Select to 20° for landing.
- (5) Airspeed Increase to $V_{REF} + 14$ KIAS.
- (6) Airplane Retrim
- (7) Autopilot Re-engage
- (8) Actual landing distance Increase by a factor of 1.55 (55%), for a landing without the use of reverse thrust.
- (9) TAWS WARNING, FLAPS OFF Press in to mute flap aural warning.

At glideslope intercept:

- (10) Airplane Retrim directionally as required.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

If above 800 feet AGL, and the above procedure cannot be accomplished:

- (11) Approach Conduct under Category I operations.

If below 800 feet AGL and non-visual:

- (12) Go-Around Initiate

After landing or during go-around, at a safe altitude:

- (13) Engine shutdown procedure Accomplish If applicable (Refer to Chapter 5; ABNORMAL PROCEDURES – POWERPLANT – In-Flight Engine Failure/Shutdown procedure).

————— **END** —————

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures

SYSTEM	INDICATION
VHF NAV 1 or 2	ILS receiver, on applicable side, is inoperative.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) NAV SOURCE Select X-SIDE. (2) Approach Continue to Category I minima, or initiate go-around. If below 500 feet AGL and non-visual: Failure of cross-side NAV: (1) Approach Continue to Category I minima, or initiate go-around. Failure of on-side NAV: (2) Go-around Initiate
SYSTEM	INDICATION
PFD 1 or 2	Primary flight display, on applicable side, goes blank.
	ACTION
	Failure during approach: (1) Display reversionary panel selector switch Select to MFD REV. Applicable MFD defaults to primary flight display.
SYSTEM	INDICATION
STAB CH 1 or 2	Horizontal stabilizer trim, on applicable side, is inoperative. STAB CH 1 (2) INOP status message displayed on EICAS.
	ACTION
	Single channel failure: (1) Approach Continue Dual channel failure (STAB TRIM caution message on; autopilot may disengage): (1) Go-around Initiate Dual channel failure during final approach (in stabilized trim condition): (2) Approach Continue to Category I minima.



6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
RADIO ALTIMETER	False radio altimeter height on both PFDs, and red or amber RA annunciation on both PFDs.
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate

SYSTEM	INDICATION
IRS	Erratic PFD data or ATT annunciation, and HDG/ROL/PIT annunciation on PFDs.
	ACTION
	First failure during the approach (AP will disconnect): If above 500 feet AGL, on a stabilized approach: (1) Manual controlResume (2) IRS source Select an operative IRS (If IRS 3 is installed, AP will be available). (3) ApproachContinue to Category I minima, if only one IRS available. If below 500 feet AGL and non-visual: (1) Manual controlResume (2) Go-aroundInitiate

SYSTEM	INDICATION
EFIS Comparator	EFIS COMP INOP (EFIS comparator failure) caution message or EFIS MISCOMP caution message displayed on EICAS and any of the ALT/HDG/IAS/PIT/ROL/LOC/GS annunciations on PFDs.
	ACTION
	Failure during approach: (1) Go-aroundInitiate (2) Applicable procedureAccomplish Refer to Chapter 5; ABNORMAL PROCEDURES – INSTRUMENTS SYSTEM.

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
ADC 1 or 2	Erratic PFD data or ADC FAIL annunciation, and ALT/IAS annunciation on PFDs.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) ADC Select 1 or 2. (2) ApproachContinue to Category I minima, or initiate go-around. If below 500 feet AGL and non-visual: Failure of cross-side ADC: (1) ApproachContinue to Category I minima, or initiate go-around. Failure of on-side ADC: (1) Go-aroundInitiate
SYSTEM	INDICATION
FD 1 or 2	Flight director, on applicable side is inoperative. FD 1 (2) FAIL status message displayed on EICAS.
	ACTION
	Continue approach, if the following is accomplished above 500 feet AGL: (1) Autopilot..... Disengage (2) AFCS SEL 1 (2) Select to operative AFCS channel. (3) FDs Engage (4) Autopilot..... Engage (5) ApproachContinue to Category II minima. If below 500 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage (2) Manual control Resume (3) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate



6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
AFCS PITCH TRIM	Autopilot pitch trim failure; AP PITCH TRIM caution message displayed on EICAS.
	ACTION
	During coupled approach: If above 200 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage Retrim, then re-engage. If caution message goes out: (2) Approach Continue If caution message persists: (2) Manual control Resume (3) Approach Continue to Category I minima. If below 200 feet AGL and non-visual: (1) Go-around Initiate
SYSTEM	INDICATION
AUTOPILOT DISCONNECT	Autopilot fails to disengage when intentionally disconnected.
	ACTION
	Autopilot fails to disconnect using AP/SP DISC switch on control wheel during approach to landing: (1) Autopilot..... Disconnect using AP switch on FCP or AP DISC switch on FCP or TOGA switch or AP/SP DISC switch on other control wheel. (2) Airplane Trim manually using stabilizer trim switch. If unable to disconnect autopilot, expect higher than normal control feel force loads during landing.



6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
ILS LOCALIZER or GLIDESLOPE	ILS deviation error is greater than Category II "window". Flashing localizer or glideslope scale and pointer.
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate
SYSTEM	INDICATION
HYDRAULICS	Any hydraulic system(s) failure(s) annunciated as caution messages displayed on EICAS (Refer to Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER).
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) Applicable procedureAccomplish Refer to Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER. (2) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate
SYSTEM	INDICATION
MAIN GENERATOR	Generator failure; GEN 1 (2) OFF caution message displayed on EICAS.
	ACTION
	NOTE AP will disconnect if AFCS 2 is active. YD 1 or YD 2 may disengage. (1) ApproachContinue to Category I minima, or initiate go-around.

————— **END** —————

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7. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual are applicable.

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NOTE

The data of this Supplement is to be used when JAA/EASA operators are conducting Category II operations under JAR-AWO at Amendment 3.

NOTE

These supplementary procedures do not constitute approval to conduct Category II operations. Such authorization must be obtained by the operator from the appropriate authorities.

1. GENERAL

The CL-600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049) has been shown to meet the airworthiness requirements for Category II Operations contained in Appendix 1 of AC 120-29 and Subpart 2 of JAR AWO, Amendment 3.

The following data must be used when conducting Category II operations.

These data complement or supersede data contained in the basic Flight Crew Operating Manual. This supplement must, therefore, be read in conjunction with the basic Flight Crew Operating Manual.

The effect of this Supplement on the basic Flight Crew Operating Manual is given in paragraphs (2) to (7).

2. INTRODUCTION

The general information in Chapter 1 is applicable.

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3. OPERATING LIMITATIONS

The operating limitations in Chapter 2 are applicable, except as modified by the following:

- An ILS approach to Category II minima must not be commenced or continued unless all required airborne equipment, specified in the Category II Required Equipment List below and their ground installations, are operating satisfactorily.

CATEGORY II REQUIRED EQUIPMENT LIST	
EQUIPMENT	PRIOR TO APPROACH
VHF NAV 1 and 2	Both must be operational and displayed onside.
PFD 1 and 2	One (1) PFD available and operational for each side.
STAB Channel 1 and 2	One (1) channel must be operational.
Radio Altimeter	One must be operational with display on both sides.
IRS	Any two (2) must be operational.
ADC 1 and 2	Both must be operational.
FD 1 and 2	Both must be operational.
EFIS Comparator Monitors	Must be operational.
AFCS Pitch Trim	Must be operational.
Autopilot	Must be operational.
Hydraulics (3)	All systems must be on and operational.
Electrics	Two (2) generators on and sharing load (see NOTE).

NOTE

Approach to Category II minima may be continued with one generator, if an engine fails during final approach (Refer to ABNORMAL PROCEDURES).

- Operation of the autopilot is prohibited below 80 feet AGL.
- The approved flap configurations for approach are:
 - 45° for all engines operating, and
 - 20° for one engine inoperative.
- The minimum decision height for Category II operations is 100 feet AGL.
- Wind limits for autopilot approaches are:
 - 8 knots crosswind, and
 - 9 knots tailwind.

NOTE

Demonstrated headwind is 16 knots.

- The maximum ILS glidepath angle for Category II operations is 3.5 degrees.
- ATS must be disengaged.
- Use of FMS computed approach performance is prohibited.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

A. Autopilot Failure

Indication: Abnormal autopilot operation and/or **AFCS MSGS FAIL** warning message, "CAVALRY CHARGE" aural and/or flashing **AP** message on the primary flight displays.

Continue approach, if the following can be accomplished above 500 feet AGL:

- (1) Autopilot Disconnect
- (2) AFCS SEL 1(2) switch (centre pedestal)..... Select to operative AFCS channel.
Check **AFCS 1 (2) INOP** status message on.
- (3) Autopilot Engage and monitor operation.
- (4) Approach Continue to Category II minima.

If below 500 feet AGL, on a stabilized approach:

- (1) Autopilot Disconnect
- (2) Manual control Resume
- (3) Approach Continue to Category I minima.

If below 200 feet AGL and non-visual:

- (1) Go-around Initiate

— END —

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

A. Prior To Approach

Airplane must be properly configured prior to approach:

- | | | |
|---|--------|--|
| (1) Landing weight | Check | within limits (Refer to Chapter 6; PERFORMANCE – LANDING PERFORMANCE of the Airplane Flight Manual). |
| (2) NAV SOURCE | Set | to onside ILS (green) or armed (cyan) by FMS, for both PFDs. |
| (3) IRS and ADC source selectors | Set | to NORM. |
| (4) RA MINIMUMS | Set | on both PFDs. |
| (5) BARO MINIMUMS | Set | as a barometric DH, in case RA fails. |
| (6) MIN ALERT | Set | to RA on both PFDs. |
| (7) Runway Course | Set | on both PFDs, or armed (cyan) by FMS. |
| (8) EFIS | Check | that no comparator flags are displayed. |
| (9) CAS | Check | that the following messages are not displayed: <ul style="list-style-type: none"> • AFCS MSGS FAIL warning message, • AP PITCH TRIM, EFIS COMP INOP and EFIS MISCOMP caution messages, • FD 1 and 2 FAIL status messages. |
| (10) Stabilizer and elevator position | Check | valid on EICAS or F/CTL synoptic page. |
| (11) Approach mode | Arm | |
| (12) Expanded lateral deviation indicator | Verify | displayed on both PFDs at 600 feet AGL. |

————— **END** —————

B. Before Landing

- | | | |
|---------------------|-----------|---|
| (1) Autopilot | Disengage | at an altitude not less than 80 feet AGL. |
|---------------------|-----------|---|

————— **END** —————

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

A. Single Engine Approach and Landing

- (1) Approach Conduct under Category I operations.

————— **END** —————

B. Engine Failure During Final Approach

Continue approach, if the following can be accomplished above 800 feet AGL:

- (1) Autopilot Disconnect
- (2) Operating engine Increase thrust, as required.
- (3) FLIGHT SPOILER lever Ensure retracted.
- (4) FLAPS Select to 20° for landing.
- (5) Airspeed Increase to $V_{REF} + 14$ KIAS.
- (6) Airplane Retrim
- (7) Autopilot Re-engage
- (8) Actual landing distance Increase by a factor of 1.55 (55%), for a landing without the use of reverse thrust.
- (9) TAWS WARNING, FLAPS OFF Press in to mute flap aural warning.

At glideslope intercept:

- (10) Airplane Retrim directionally as required.



With thrust reversers deployed, a nose-up pitching tendency will occur at high power settings, particularly at aft centre of gravity light weights. This tendency is controllable with elevator and may be minimized by ensuring that nose wheel touchdown is achieved and nose down elevator applied, before selecting reverse thrust.

If above 800 feet AGL, and the above procedure cannot be accomplished:

- (11) Approach Conduct under Category I operations.

If below 800 feet AGL and non-visual:

- (12) Go-Around Initiate

After landing or during go-around, at a safe altitude:

- (13) Engine shutdown procedure Accomplish If applicable (Refer to Chapter 5; ABNORMAL PROCEDURES – POWERPLANT – In-Flight Engine Failure/Shutdown procedure).

————— **END** —————

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures

SYSTEM	INDICATION
VHF NAV 1 or 2	ILS receiver, on applicable side, is inoperative.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) NAV SOURCE Select X-SIDE. (2) Approach Continue to Category I minima, or initiate go-around.
	If below 500 feet AGL and non-visual: Failure of cross-side NAV: (1) Approach Continue to Category I minima, or initiate go-around. Failure of on-side NAV: (2) Go-around Initiate
SYSTEM	INDICATION
PFD 1 or 2	Primary flight display, on applicable side, goes blank.
	ACTION
	Failure during approach: (1) Display reversionary panel selector switch Select to MFD REV. Applicable MFD defaults to primary flight display.
SYSTEM	INDICATION
STAB CH 1 or 2	Horizontal stabilizer trim, on applicable side, is inoperative. STAB CH 1 (2) INOP status message displayed on EICAS.
	ACTION
	Single channel failure: (1) Approach Continue Dual channel failure (STAB TRIM caution message on; autopilot may disengage): (1) Go-around Initiate Dual channel failure during final approach (in stabilized trim condition): (2) Approach Continue to Category I minima.



6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
RADIO ALTIMETER	False radio altimeter height on both PFDs, and red or amber RA annunciation on both PFDs.
	ACTION
	Failure (of 2nd radio altimeter) during approach: If above 200 feet AGL, on a stabilized approach: (1) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate
SYSTEM	INDICATION
IRS	Erratic PFD data or ATT annunciation, and HDG/ROL/PIT annunciation on PFDs.
	ACTION
	First failure during the approach (AP will disconnect): If above 500 feet AGL, on a stabilized approach: (1) Manual controlResume (2) IRS sourceSelect an operative IRS (If IRS 3 is installed, AP will be available). (3) ApproachContinue to Category I minima, if only one IRS available. If below 500 feet AGL and non-visual: (1) Manual controlResume (2) Go-aroundInitiate
SYSTEM	INDICATION
EFIS Comparator	EFIS COMP INOP (EFIS comparator failure) caution message or EFIS MISCOMP caution message displayed on EICAS and any of the ALT/HDG/IAS/PIT/ROL/LOC/GS annunciations on PFDs.
	ACTION
	Failure during approach: (1) Go-aroundInitiate (2) Applicable procedureAccomplish Refer to Chapter 5; ABNORMAL PROCEDURES – INSTRUMENTS SYSTEM.

6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
ADC 1 or 2	Erratic PFD data or ADC FAIL annunciation, and ALT/IAS annunciation on PFDs.
	ACTION
	First failure during approach (of the two required units): If above 500 feet AGL, on a stabilized approach: (1) ADC Select 1 or 2. (2) ApproachContinue to Category I minima, or initiate go-around. If below 500 feet AGL and non-visual: Failure of cross-side ADC: (1) ApproachContinue to Category I minima, or initiate go-around. Failure of on-side ADC: (1) Go-aroundInitiate
SYSTEM	INDICATION
FD 1 or 2	Flight director, on applicable side is inoperative. FD 1 (2) FAIL status message displayed on EICAS.
	ACTION
	Continue approach, if the following is accomplished above 500 feet AGL: (1) Autopilot..... Disengage (2) AFCS SEL 1 (2) Select to operative AFCS channel. (3) FDsEngage (4) Autopilot.....Engage (5) ApproachContinue to Category II minima. If below 500 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage (2) Manual controlResume (3) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate



6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
AFCS PITCH TRIM	Autopilot pitch trim failure; AP PITCH TRIM caution message displayed on EICAS.
	ACTION
	During coupled approach: If above 200 feet AGL, on a stabilized approach: (1) Autopilot..... Disengage Retrim, then re-engage. If caution message goes out: (2) Approach Continue If caution message persists: (2) Manual control Resume (3) Approach Continue to Category I minima. If below 200 feet AGL and non-visual: (1) Go-around Initiate
SYSTEM	INDICATION
AUTOPILOT DISCONNECT	Autopilot fails to disengage when intentionally disconnected.
	ACTION
	Autopilot fails to disconnect using AP/SP DISC switch on control wheel during approach to landing: (1) Autopilot..... Disconnect using AP switch on FCP or AP DISC switch on FCP or TOGA switch or AP/SP DISC switch on other control wheel. (2) Airplane Trim manually using stabilizer trim switch. If unable to disconnect autopilot, expect higher than normal control feel force loads during landing.



6. ABNORMAL PROCEDURES (CONT'D)

C. System Failures (Cont'd)

SYSTEM	INDICATION
ILS LOCALIZER or GLIDESLOPE	ILS deviation error is greater than Category II "window". Flashing localizer or glideslope scale and pointer.
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate
SYSTEM	INDICATION
HYDRAULICS	Any hydraulic system(s) failure(s) annunciated as caution messages displayed on EICAS (Refer to Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER).
	ACTION
	Failure during approach: If above 200 feet AGL, on a stabilized approach: (1) Applicable procedureAccomplish Refer to Chapter 5; ABNORMAL PROCEDURES – HYDRAULIC POWER. (2) ApproachContinue to Category I minima, or initiate go-around. If below 200 feet AGL and non-visual: (1) Go-aroundInitiate
SYSTEM	INDICATION
MAIN GENERATOR	Generator failure; GEN 1 (2) OFF caution message displayed on EICAS.
	ACTION
	NOTE AP will disconnect if AFCS 2 is active. YD 1 or YD 2 may disengage. (1) ApproachContinue to Category I minima, or initiate go-around.

————— **END** —————

7. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable, except as modified by Supplement 4A of the Airplane Flight Manual.

TRAINING PURPOSES ONLY

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TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Super-Cooled Large Droplet Icing	Vol. 1	06–11–1
		Oct 05/06	

1. ICING CONDITIONS

Icing conditions exist when the total air temperature is below 10°C (50°F), and visible moisture is present in any form. This includes cloud, fog, mist, rain, snow, sleet and ice crystals. Regardless of visible ambient moisture and temperature clues, icing conditions also exist when there are visible signs of ice accumulation on the airplane, or when indicated by the ice detection system.

2. CLOUD FORMS

In discussion of icing, cloud types can be categorized into two general classifications; stratiform (layer type clouds) or cumuliform (rising, thunderstorm) clouds. The certification requirements define icing envelopes conforming to these cloud types, corresponding to continuous (stratiform) icing and intermittent (cumulous) icing types.

3. ICING PROCESS

Icing results from super-cooled water droplets that remain in a liquid state, at temperatures below freezing. In general, leading edge structures, passing through such conditions, will cause a certain number of these droplets to impact the leading edge surface, and freeze. A relatively large or bluff body will generate a large pressure wave ahead of the leading edge, which forces the air and many of the smaller droplets around it. Only droplets with sufficient mass and inertia will impact the surface and freeze. Conversely, a narrow leading edge radius generates a smaller pressure wave, and so collects more of the lower mass inertia droplets. Ice will thus tend to accumulate at a greater rate on the (smaller, narrower) tail leading surfaces. Ice will also tend to accumulate in greater quantities, and cover a larger part of the leading edge, if the ambient liquid water droplets are relatively large.

4. ICE FORM

Three recognizable ice forms exist; rime ice (opaque), glaze ice (clear) and frost. It is also common to observe mixed form icing, comprised of mixed glaze and rime ice forms.

- Rime ice is rough and opaque in appearance, and generally forms a pointed or streamlined shape on the leading edge.
- Glaze ice is transparent, and often produces a wedge shape or concave ice shape with double horns. This is caused by partial run back of the impinging water droplets to positions aft of the stagnation point. Ice initially forms here as a thin layer of sandpaper ice, which then grows to form the glaze horns.
- Frost may form as a thin layer of crystalline ice on all exposed airplane surfaces. Frost is generally associated with ground operations.

5. SUPER-COOLED LARGE DROPLET ICING CONDITIONS

Super-cooled large droplet icing conditions are distinct from the icing described above, because of the propensity for the ambient liquid water to be contained in droplets of relatively large mass and inertia. This causes a larger proportion of the water to impact the leading edge surfaces. In addition, the droplets impacting the surface will do so further aft than smaller droplets. On the protected wing surfaces this may result in formation of ice ridges.

6. RECOGNITION OF SUPER-COOLED LARGE DROPLET ICING CONDITIONS

It is known that super-cooled large droplet (SLD) icing conditions may be prevalent in pristine atmospheres typical of coastal maritime environments, however, there are no defined means for prior indication of SLD icing conditions, or for differentiating SLD from other icing conditions.

The presence of SLD icing can only be determined by observation of the resulting ice accumulation on unprotected surfaces.

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6. RECOGNITION OF SUPER-COOLED LARGE DROPLET ICING CONDITIONS (CONT'D)

The indicator for differentiating SLD icing is observation of ice accumulation on the flight compartment (cockpit) side windows. Any ice accumulation on the side windows should be taken as the indication that SLD icing conditions are present.

7. PROCEDURES

Operation in SLD icing conditions is prohibited. Following recognition of SLD icing conditions by observation of side window icing, the engine cowl and wing anti-icing systems must be activated. Even with anti-icing systems being active, it is necessary to leave SLD icing conditions immediately.

After leaving SLD icing conditions, the wing leading edges should be observed for signs of ice formation aft of the heated leading edge area. If ice is observed aft of the leading edge, then the Ice Dispersal Procedure should be accomplished (Refer to Chapter 5; ABNORMAL PROCEDURES – ICE AND RAIN PROTECTION – Ice Dispersal Procedure).

TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Cold Weather Operations	Vol. 1	06-12-1
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1. GENERAL

The winter season presents additional problems to airplane operations, resulting from low temperatures, the potentially hazardous effects of precipitation contaminating the airplane and the aircraft movement area, and extreme turbulence. Removal of contaminants on runway surfaces, taxiways, aprons, holding bays and other areas, rests on the administration of the airports concerned, based on flight safety and schedule considerations. However, it is the ultimate responsibility of the pilot-in-command to make sure that the airplane is in a condition for safe flight prior to take-off. Use of the ATIS or other means to acquire accurate ambient temperature and other pertinent meteorological conditions can not be overemphasized. The indicated SAT on EFIS cannot be used before take-off, since the TAT probe gives inaccurate readings on the ground when the airplane is static or at low forward speed.

Adherence to the procedures in this section ensures an aerodynamically clean aircraft before take-off. When operating in such conditions, these procedures account for operational hazards associated with frozen contamination.

In all cases, it is assumed that the decision to operate is based on the general rules of good airmanship applicable in cold weather operations, and on the assurance that the operational and system limitations will not be exceeded (Refer to Chapter 2, LIMITATIONS). Under these provisions, the procedures given in the following section have been provided to supplement the normal operating procedures, with the goal of enhancing flight safety, and assisting in obtaining maximum performance from the airplane. In no circumstances, however, do they warrant operations in conditions imposing demands beyond the capabilities of the airplane or its flight crew.

2. DEFINITIONS

A. Cold Weather Operations

Cold weather operations refer to ground handling, take-offs and landings conducted on surface conditions where frozen moisture is present, or conditions are conducive to moisture freezing. These conditions are commonly encountered when the surface temperature is at or below 0°C (32°F), although frozen moisture may be present and persist for a significant time at higher temperatures. Examples of this latter condition are the penetration of heavy frozen precipitation to ground level when surface temperature is near freezing, and the formation of frozen condensation on airframe surfaces in contact with cold fuel.

Cold soaking is the effect cold fuel in the tanks may have on moisture present on the upper and lower wing surfaces. If fuel temperature is 0°C (32°F) or below, it is possible to have clear ice or frost on the wing, with the ambient air temperatures above freezing. The wing surfaces must be below freezing temperatures for frost to form, even though the ambient temperatures may be above freezing.

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2. DEFINITIONS (CONT'D)

B. Contaminants

(1) SLUSH

Slush is snow, saturated with water, which displaces with a splatter when stepped on firmly. It is encountered at temperatures up to 5°C (41°F).

(2) WET SNOW

Wet snow will easily stick together, and tends to form a snowball if compacted by hand.

(3) DRY SNOW

Dry snow is loose and can easily be blown. If compacted by hand, it will readily fall apart again.

(4) FROST

Frost forms from the slow deposition of ice crystals on cold surfaces directly from water vapor in the air. The frost-forming surface must be below freezing temperatures for frost to form, even though the ambient temperature may be above freezing. Frost appears as a white crystalline deposit that usually develops uniformly on exposed surfaces during below-freezing, calm and cloudless nights with a high ambient dewpoint. The deposit is thin enough for surface features underneath, such as paint lines, markings and lettering, to be distinguished.

(5) ICE

Two types of ice, rime and clear ice, commonly affect aircraft operations:

- Rime ice:

Although rime ice is more commonly found in flight, it may occur on the ground when conditions are favorable. Rime ice may occur on the ground in low temperatures, with a low concentration of small super-cooled water droplets and moderate winds. It appears as an opaque and rough ice surface that adheres to surfaces exposed to wind. It can easily be detected and is easily removed by application of deicing/anti-icing fluids.

- Clear ice:

Clear ice can occur in flight or on the ground. It forms at temperatures at or just below 0°C (32°F) with a high concentration of large super-cooled water droplets. Clear ice is hard, and appears as a smooth and glassy coating that can be very difficult to detect without a tactile inspection. Clear ice may not be seen during a walkaround, particularly if the wing is wet, or during night time operations. Clear ice adheres firmly to surfaces, and is difficult to remove, requiring special care during deicing/anti-icing.

(6) DEHYDRATED DEICING/ANTI-ICING FLUIDS

If deicing/anti-icing fluid is allowed to dry on airplane surfaces, this same fluid can become a contaminant. Deicing, and especially anti-icing, fluids are designed to adhere to airplane surfaces, and shear off at speeds approaching take-off speeds.

If left on airplane surfaces for long periods of time (overnight), they may dehydrate and form a gel or dried deposit that will not shear off, even at high speeds. This contaminant will severely affect airplane performance and lift.

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2. DEFINITIONS (CONT'D)

C. Critical Surfaces

Critical surfaces are defined to be wings, horizontal stabilizer, vertical stabilizer, control surfaces and engine inlets.

Although the upper fuselage is not defined as a critical surface, it must nonetheless be deiced to remove contamination, other than allowable frost, anytime the wing and tail surfaces require deicing. Contamination of the upper surface of the fuselage with frost (through which it is possible to distinguish surface features, markings and lines), is considered allowable.

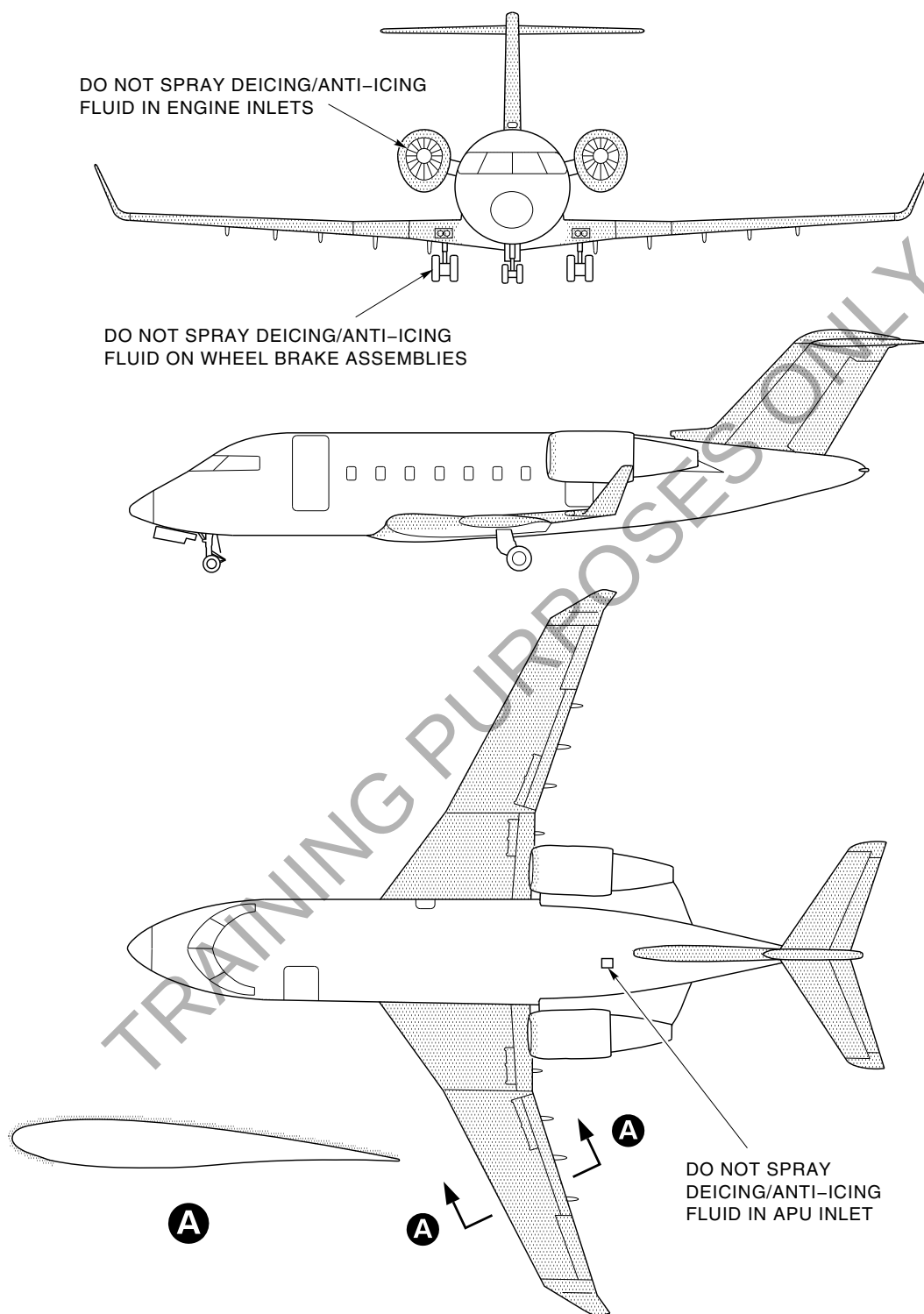
The aerodynamically critical surfaces of the horizontal stabilizer are the leading edge and the under surface. A visual inspection of these surfaces must be made to determine the extent of contamination before each flight.

The upper surface of the horizontal stabilizer may not be visible from the ground. A comparative analysis of the non-visible horizontal stabilizer upper surface may be used to validate the condition of the surface. The wing should be used as the comparative surface.

If the inspection of the wing dictates that there is a requirement to deice/anti-ice, then the horizontal stabilizer and elevator must also be deiced/anti-iced. Conversely, if the visual inspection of the wing dictates that there is not a requirement to deice/anti-ice then the horizontal stabilizer and elevator upper surfaces need not be deiced/anti-iced.

It is ultimately the responsibility of the pilot-in-command to see that the airplane is in a condition for safe flight prior to take-off. If the pilot-in-command has any doubt as to the cleanliness of the aircraft, then the aircraft must be deiced.

2. DEFINITIONS (CONT'D)

Critical Surfaces
Figure 06-12-1

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3. LIMITATIONS

Refer to Chapter 2; LIMITATIONS.

4. AIRFRAME CONTAMINATION

Cold weather operations present specific challenges in keeping an airplane free of contaminants.

A. Clean Aircraft Concept

The Clean Aircraft Concept (aerodynamically clean) prohibits take-off when frost, ice, snow, or other contaminants are present on the airplane's critical surfaces.

The performance data for this airplane are based on the clean aircraft concept. This means that all performance values are based on the airplane being aerodynamically clean prior to take-off. Failure to remove contaminants will result in adverse effects on airplane performance and flight characteristics. These adverse effects can include the following:

- Decreased thrust
- Decreased lift
- Increased drag
- Increased stall speeds
- Trim changes
- Altered stall characteristics
- Altered handling qualities.

The removal procedures for frost, ice and snow from the surfaces of the airplane prior to take-off, as described in this section, depend upon the deicing/anti-icing facilities, methods and types of fluid available at the airports involved. Deicing/anti-icing must be accomplished at the last possible time prior to take-off, to maximize the time that anti-icing will be able to provide protection (holdover time).

4. AIRFRAME CONTAMINATION (CONT'D)

B. General Precautions

The following general precautions must be observed in cold weather operations:

- (1) It must never be assumed that an apparently dry and loose form of frozen moisture, for example, dry snow, will be removed by the slipstream during the initial take-off roll. For instance, on an airplane removed from a warm hangar, a dry snowfall that remains free and uncompacted on the ground may melt, and later refreeze, to form ice that sticks to the surfaces of the airplane.
- (2) Before each flight, a thorough inspection of critical surfaces must be made to determine the extent of contamination on them. This inspection must be made by the pilot-in-command (PIC) or by other trained and approved personnel qualified to report its results directly to the PIC. Deicing and anti-icing are part of flight operations, and remain under the authority of the PIC.
- (3) After deicing, another inspection, subject to the same qualifications mentioned in paragraph (2) above, must be made to confirm that all contamination is removed.
- (4) If, during the period between the completion of deicing and take-off, there is the possibility that the airplane may again be contaminated, anti-icing protection, usually in the form of deicing/anti-icing fluid, must be provided. The period of effective anti-icing, known as holdover time, must be longer than the period between deicing and take-off. Holdover times start at the beginning of the final deicing procedure.
- (5) If, during the conditions described in paragraph (4) above, take-off cannot be started prior to the expiration of the holdover time, the airplane must again be inspected and deiced, if necessary, before attempting take-off.

A one-step deicing/anti-icing procedure is only possible on a clean aircraft. If ice, snow or frost is found on the airplane, or when visible precipitation continues to fall, extended anti-ice protection is required. In these conditions, the application of deicing/anti-icing fluids must be accomplished in two separate steps, which will provide adequate protection in most environmental conditions. The characteristics of the various deicing/anti-icing fluids used in carrying out these operations are described later in this section.

C. Wet Aircraft and Temperatures Greater Than 0°C (32°F) but Less Than 5°C (41°F)

Consideration of the following should be made as to whether the aircraft should be deiced/anti-iced:

- (1) Conditions such as wind and forecast temperature. If temperatures are dropping, or are forecasted to drop, treatment with Type I fluid should be considered.
- (2) When an aircraft is wet due to light rain or mist, and the AFM icing definition of "visible moisture" and "less than 5°C (41°F)" is satisfied. In such a situation, the AFM limitations require the use of wing anti-ice for take-off.
- (3) If the aircraft is wet because it has been cleaned with hot water, but there is no visible moisture in the air, then the wing is at the same risk of being contaminated as if the aircraft was taxiing in slush or pooled water on taxiways/runways. The use of wing anti-ice is required for such conditions.

D. Clear Ice Due to Cold Fuel

Pilots must be aware of the effect that cold fuel in the tanks may have on moisture present on the wing upper and lower surfaces. If fuel temperature is 0°C (32°F) or below, it is possible to have clear ice on the wing with the temperature above freezing.

4. AIRFRAME CONTAMINATION (CONT'D)

Pilots must ensure that the wing upper surface is free of clear ice by means of a tactile (touch) check. Clear ice must be removed.

E. Frost Due to Cold Soaked Fuel

Wing frost, caused by cold soaked fuel, can form on the upper and lower surfaces of the wing even at temperatures significantly above freezing. Frost on the upper surface of the wing must be removed.

Take-off with the following accumulation of frost, due to cold soaked fuel, on the underside of the wing fuel tank area is permissible:

- Maximum 3 mm (1/8 inch) layer of frost.

F. Frost on the Upper Surface of the Fuselage

Frost on the upper fuselage surface is not required to be removed, if it is possible to distinguish surface features (markings and lines). Frost in excess of this must be removed from the fuselage.

5. PRE-FLIGHT PREPARATION**A. External Safety Inspection**

The removal of contaminants from the airplane is a maintenance function: however, the flight crew should be diligent during the pre-flight preparation to inspect areas where adherence and accumulation of frost, ice, and snow could seriously affect normal systems operations.

- | | | |
|-----------------------------------|---------------|---|
| (1) All protective covers | Removed | <ul style="list-style-type: none">• Probe covers (pitot and static), TAT, ice detector, AOA vane, and fuel NACA vent covers.• Wheel covers (nose and main landing gear)• Intake and exhaust covers (engines, APU, ram air scoop, and air-conditioning packs). |
| (2) Pitot and static probes | Clear | and not obstructed. |
| (3) AOA vanes | Free movement | |
| (4) Windshield | Clear | of ice and snow. |



5. PRE-FLIGHT PREPARATION (CONT'D)

A. External Safety Inspection (Cont'd)

- (5) Airplane surfaces..... Clear of frost, ice, snow and dehydrated deicing/anti-icing fluids.

NOTE

1. During snowfall, freezing rain and drifting snow, it is possible for snow and melting ice to penetrate into hinges, operating linkages, drainage openings and vents, and then refreeze. The above mentioned areas should be checked with diligence.
2. Take-off is permitted with frost on the upper surface of the fuselage, through which it is possible to distinguish surface features (markings and lines).
3. Take-off is permitted with frost adhering to the underside of the wing that is caused by cold soaked fuel. Maximum 3 mm (1/8 inch) layer of frost.

- (6) Nosewheel and main landing gear area Clear of frost, ice and snow.

Check the following:

- Latching and operating mechanisms are free and clear of any accumulation (uplocks/downlocks).
- Electrical components (connectors, cables and micro-switches) for evidence of water ingress.
- Gear doors are free from accumulations.
- Tire pressure is acceptable and wheels are not frozen to the ground.

- (7) APU and air-conditioning intake and exhaust areas Clear of frost, ice, and snow.

- (8) Engine inlet and cowlings Clear of frost, ice, and snow.

Check the following:

- Fan rotation, as applicable.
- Drain lines are clear.

- (9) Fuel tanks and hydraulic components Check for evidence of leaks.

- (10) Water system and drain masts Check for evidence of freezing.



5. PRE-FLIGHT PREPARATION (CONT'D)

A. External Safety Inspection (Cont'd)

- (11) APU and main airplane batteryCheck installed and charged.
- Precautionary measures for extremely low temperatures may have entailed overnight removal of the airplane batteries.

NOTE

If the batteries were removed, the fuel used indication on the FMS will remain from the previous flight. When the batteries are reinstalled, the fuel used should be manually reset to zero.

————— **END** —————

B. Cockpit and Cabin Preparation

- (1) BATT MASTER switch ON
- (2) APU Start
- (3) AIR-CONDITIONING, L and R
PACK switch/lights Press in Check L and R PACK OFF lights out.
- (4) AIR-CONDITIONING, CABIN and
CKPT mode selectors Set to NORM.
- (5) AIR-CONDITIONING, CABIN and
CKPT TEMP control switches As required

If airplane is cold soaked at temperatures below –20°C (–4°F):

- (6) AIR-CONDITIONING, CABIN
TEMP control switch..... HOT to warm up the cabin to a comfortable level (approximately 10°C (50° F) or higher) before the passengers board aircraft.
- (7) Main door and exitsCheck operation Check that the main passenger door can be opened properly.
- Check that there are no messages on the door synoptic page.

————— **END** —————

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION

A. Deicing/Anti-Icing Fluids

The application of deicing/anti-icing fluid is the most common means of effecting ground deicing and anti-ice protection. These fluids are water/glycol solutions, commonly referred to as Type I (AMS 1424), Type II (AMS 1428), Type III (AMS 1428) and Type IV (AMS 1428). Any deicing/anti-icing fluids, approved to these specifications, can be used to deice/anti-ice the airplane, provided the manufacturer's recommendations and appropriate holdover times are respected.

(1) TYPE I FLUIDS

In concentrated form, these fluids contain glycols to a minimum concentration of 80%, but with no thickening agents. Their resulting low viscosity and very short holdover time provide very limited anti-icing protection.

It is apparent that except in the case of frost or freezing fog, the duration of anti-icing protection provided by Type I fluid is inadequate, unless take-off can be made almost immediately after deicing. Therefore, if conditions require effective ground anti-icing, it is imperative that Type II, III or IV fluid be available for use.

(2) TYPE II AND TYPE IV FLUIDS

These fluids contain glycols to a minimum concentration of 50%, as well as thickening agents. Their relatively high viscosity permits the application of a layer of fluid that is effective in deicing, and persists for a significant holdover time to provide anti-icing. During take-off, the slipstream imparts a shear stress to the fluid layer, causing it to flow off the surface to which it was applied.

(3) TYPE III FLUIDS

Type III is a thickened fluid which has properties that lie between Types I and II. Therefore, it provides a longer holdover time than Type I but less than Type II.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

B. Holdover and Allowance Times

Having flaps extended prior to deicing/anti-icing, accelerates anti-icing fluid degradation off aircraft wings and decreases the holdover and allowance to 76% of the flap retracted times. Holdover and allowance time tables are provided for both retracted and deployed flaps. The holdover and allowance time tables for flaps retracted prior to deicing/anti-icing, can be used if flaps are deployed as close to departure as safety allows.

(1) HOLDOVER TIME

Holdover time is the estimated time that an application of an approved deicing/anti-icing fluid is effective in preventing frost, ice, or snow from adhering to treated surfaces. Holdover time is calculated as beginning at the start of the final application of an approved deicing/anti-icing fluid, and as expiring when the fluid is no longer effective.

The fluid is no longer effective when its ability to absorb more precipitation has been exceeded. This produces a visible surface build-up of contamination.

Anti-icing effectiveness, however, is subject to many more variables than are usually present in deicing. Of fundamental concern to the aircrew is the calculation of the anti-icing holdover time available after deicing, given prevailing conditions and use of a particular fluid. To provide some assistance, tables are provided to show holdover times for Type I, Type II, Type III, and Type IV fluids, as influenced by the kind of freezing precipitation present.

These tables do not account for all the factors that influence holdover time. Diverse and individually variable factors, such as fluid temperature, relative humidity, wind direction and speed, can significantly shorten the holdover times shown in these tables.

NOTE

Guidelines for holdover times and deicing fluid application are revised every year, according to the Transport Canada TP 14052 and associated holdover time guidelines document. The current holdover time documents are available at the following web-sites:

www.tc.gc.ca/eng/civilaviation/standards/commerce-holdovertime-menu-1877.htm

www.faa.gov/other_visit/aviation_industry/airline_operators/airline_safety/deicing

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

(2) SNOWFALL INTENSITY

Use the following table to determine the snowfall intensity (including snow grains and snow pellets).

SNOWFALL INTENSITY AS A FUNCTION OF VISIBILITY						
Lighting	Temperature Range		Visibility in Snow in Statute Miles (Meters)			
	°C	°F	Heavy	Moderate	Light	Very Light
Darkness	-1 and above	30 and above	≤ 1 (≤ 1600)	> 1 to 2 1/2 (> 1600 to 4000)	> 2 1/2 to 4 (> 4000 to 6400)	> 4 (> 6400)
	below -1	below 30	≤ 3/4 (≤ 1200)	> 3/4 to 1 1/2 (> 1200 to 2400)	> 1 1/2 to 3 (> 2400 to 4800)	> 3 (> 4800)
Daylight	-1 and above	30 and above	≤ 1/2 (≤ 800)	> 1/2 to 1 1/2 (> 800 to 2400)	> 1 1/2 to 3 (> 2400 to 4800)	> 3 (> 4800)
	below -1	below 30	≤ 3/8 (≤ 600)	> 3/8 to 7/8 (> 600 to 1400)	> 7/8 to 2 (> 1400 to 3200)	> 2 (> 3200)

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Snowfall Intensity as a Function of Visibility
Figure 06-12-2

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

(3) HOLDOVER TABLES



1. Takeoff after the longest applicable holdover time has been exceeded is not permitted for Type I fluids.
2. The only acceptable decision-making criteria for takeoff without a pre-takeoff contamination inspection, is the shortest time within the applicable holdover timetable cell.
3. For all types of deicing fluid, the time of protection will be shortened in heavy weather conditions, heavy precipitation rates or high moisture content. High wind velocity or jet blast may reduce holdover time below the lowest time stated in the range. Holdover time may also be reduced when airplane skin temperature is lower than OAT.
4. Fluids used during ground deicing/anti-icing do not provide in-flight icing protection.

TRAINING PURPOSES ONLY

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

SAE TYPE I FLUIDS – FLAPS RETRACTED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS									
OAT [1], [2]		Holdover Times (minutes)							
°C	°F	Freezing Fog, Freezing Mist ^[7] or Ice Crystals	Snow, Snow Grains or Snow Pellets ^[8]			Freezing Drizzle ^[4]	Light Freezing Rain	Rain on Cold Soaked Wing ^[5]	Other ^[6]
			Very Light ^[3]	Light ^[3]	Moderate				
-3 and above	27 and above	11 – 17	18 – 22	11 – 18	6 – 11	9 – 13	4 – 6	2 – 5	No holdover time guidelines exist
below -3 to -6	below 27 to 21	8 – 13	14 – 17	8 – 14	5 – 8	5 – 9	4 – 6		
below -6 to -10	below 21 to 14	6 – 10	11 – 13	6 – 11	4 – 6	4 – 7	2 – 5		
below -10	below 14	5 – 9	7 – 8	4 – 7	2 – 4				
[1] Type I fluid/water mixture must be selected so that the freezing point of the mixture is at least 10°C (18°F) below outside air temperature. [2] Ensure that the lowest operational use temperature (LOUT) is respected. [3] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle. [4] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible. [5] No holdover time guidelines exist for this condition for 0°C (32°F) and below. [6] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail. [7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0° C (32° F) and below. [8] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.									

GF0701_040C

Type I Fluid Holdover Times – Flaps Retracted Prior to Deicing/Anti-Icing
Figure 06-12-3

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

SAE TYPE I FLUIDS – FLAPS DEPLOYED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS									
OAT [1], [2]		Holdover Times (minutes)							
°C	°F	Freezing Fog, Freezing Mist ^[7] or Ice Crystals	Snow, Snow Grains or Snow Pellets ^[8]			Freezing Drizzle ^[4]	Light Freezing Rain	Rain on Cold Soaked Wing ^[5]	Other ^[6]
			Very Light ^[3]	Light ^[3]	Moderate				
−3 and above	27 and above	8 – 13	14–17	8 – 14	5 – 8	7 – 10	3 – 5	2 – 4	No holdover time guidelines exist
below −3 to −6	below 27 to 21	6 – 10	11 – 13	6 – 11	4 – 6	4 – 7	3 – 5		
below −6 to −10	below 21 to 14	5 – 8	8 – 10	5 – 8	3 – 5	3 – 5	2 – 4		
below −10	below 14	4 – 7	5 – 6	3 – 5	2 – 3				
[1] Type I fluid/water mixture must be selected so that the freezing point of the mixture is at least 10°C (18°F) below outside air temperature. [2] Ensure that the lowest operational use temperature (LOUT) is respected. [3] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle. [4] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible. [5] No holdover time guidelines exist for this condition for 0°C (32°F) and below. [6] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail. [7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0°C (32°F) and below. [8] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.									

GF0701_037D

Type I Fluid Holdover Times – Flaps Deployed Prior to Deicing/Anti-Icing
Figure 06-12-4

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

SAE TYPE II FLUIDS – FLAPS RETRACTED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS									
OAT [1]		Type II Fluid Concentration	Holdover Times (minutes)						
°C	°F	Fluid/Water (% by Volume)	Freezing Fog, Freezing Mist ^[7] or Ice Crystals	Snow, Snow Grains or Snow Pellets ^{[2][9]}	Freezing Drizzle ^[3]	Light Freezing Rain	Rain on Cold Soaked Wing ^[4]	Other ^[5]	
-3 and above	27 and above	100 / 0	55 – 110	25 – 50	30 – 60	20 – 35	7 – 45	No holdover time guidelines exist	
		75 / 25	25 – 55	15 – 25	15 – 40	10 – 20	4 – 25		
		50 / 50	15 – 25	5 – 10	8 – 15	6 – 9			
below -3 to -8	below 27 to 18	100 / 0	30 – 45	20 – 35	20 – 45	15 – 20			
		75 / 25	25 – 50	10 – 20	15 – 25	8 – 15			
below -8 to -14	below 18 to 7	100 / 0	30 – 45	15 – 30	20 – 45 ^[8]	15 – 20 ^[8]			
		75 / 25	25 – 50	8 – 20	15 – 25 ^[8]	8 – 15 ^[8]			
below -14 to -18	below 7 to 0	100 / 0	15 – 20	2 – 7	No holdover time guidelines exist				
below -18 to -25	below 0 to -13	100 / 0	15 – 20 ^[6]	1 – 3 ^[6]					
below -25 to LOUT	below -13 to LOUT	100 / 0	15 – 20 ^[6]	0 – 1 ^[6]					

[1] Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I when Type II fluid cannot be used.

[2] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle.

[3] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.

[4] No holdover time guidelines exist for this condition for 0° C (32° F) and below.

[5] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

[6] If the LOUT is unknown, no holdover time guidelines exist below -25° C (-13° F).

[7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0° C (32° F) and below.

[8] No holdover time guidelines exist for this condition below -10° C (14° F).

[9] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.

GF0701_027E

Type II Fluid Holdover Times – Flaps Retracted Prior to Deicing/Anti-Icing
Figure 06-12-5

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

SAE TYPE II FLUIDS – FLAPS DEPLOYED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS									
OAT [1]		Type II Fluid Concentration	Holdover Times (minutes)						
°C	°F	Fluid/Water (% by Volume)	Freezing Fog, Freezing Mist ^[7] or Ice Crystals	Snow, Snow Grains or Snow Pellets ^{[2][9]}	Freezing Drizzle ^[3]	Light Freezing Rain	Rain on Cold Soaked Wing ^[4]	Other ^[5]	
−3 and above	27 and above	100 / 0	42 – 84	19 – 38	23 – 46	15 – 27	5 – 34		
		75 / 25	19 – 42	11 – 19	11 – 30	8 – 15	3 – 19		
		50 / 50	11 – 19	4 – 8	6 – 11	5 – 7			
below −3 to −8	below 27 to 18	100 / 0	23 – 34	15 – 27	15 – 34	11 – 15			
		75 / 25	19 – 38	8 – 15	11 – 19	6 – 11			
below −8 to −14	below 18 to 7	100 / 0	23 – 34	11 – 23	15 – 34 ^[8]	11 – 15 ^[8]			
		75 / 25	19 – 38	6 – 15	11 – 19 ^[8]	6 – 11 ^[8]			
below −14 to −18	below 7 to 0	100 / 0	11 – 15	2 – 5	No holdover time guidelines exist				
below −18 to −25	below 0 to −13	100 / 0	11 – 15 ^[6]	1 – 2 ^[6]					
below −25 to LOUT	below −13 to LOUT	100 / 0	11 – 15 ^[6]	0 – 1 ^[6]					

[1] Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I when Type II fluid cannot be used.

[2] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle.

[3] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.

[4] No holdover time guidelines exist for this condition for 0° C (32° F) and below.

[5] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

[6] If the LOUT is unknown, no holdover time guidelines exist below −25° C (−13° F).

[7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0° C (32° F) and below.

[8] No holdover time guidelines exist for this condition below -10° C (14° F).

[9] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.

GF0701_028E

Type II Fluid Holdover Times – Flaps Deployed Prior to Deicing/Anti-Icing
Figure 06-12-6

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

ALLCLEAR AEROCLEAR MAX HIGH SPEED TYPE III FLUID – APPLIED UNHEATED FLAPS RETRACTED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS										
OAT ^[2]		Type III Fluid Concentration	Holdover Times (minutes) ^[1]							
°C	°F	Fluid/Water (% by Volume)	Freezing Fog, Freezing Mist ^[7] or Ice Crystals	Snow, Snow Grains or Snow Pellets ^[8]			Freezing Drizzle ^[4]	Light Freezing Rain	Rain on Cold Soaked Wing ^[5]	Other ^[6]
				Very Light ^[3]	Light ^[3]	Moderate				
−3 and above	27 and above	100 / 0	45 – 115	80 – 105	40 – 80	18 – 40	25 – 50	14 – 25	5 – 40	No holdover time guidelines exist
below −3 to −10	below 27 to 14	100 / 0	50 – 100	80 – 105	40 – 80	18 – 40	25 – 45	15 – 25		
below −10 to −25	below 14 to −13	100 / 0	40 – 105	80 – 105	40 – 80	18 – 40				
below −25 to −35	below −13 to −31	100 / 0	25 – 60	45 – 60	20 – 45	10 – 20				
[1] Fluid must be applied unheated. No holdover time guidelines exist for this fluid applied heated. [2] Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I when Type III fluid cannot be used. [3] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle. [4] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible. [5] No holdover time guidelines exist for this condition for 0° C (32° F) and below. [6] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail. [7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0° C (32° F) and below. [8] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.										

GF0701_031D

AllClear Unheated Type III Fluid Holdover Times – Flaps Retracted Prior to Deicing/Anti-Icing
Figure 06-12-7

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

ALLCLEAR AEROCLEAR MAX HIGH SPEED TYPE III FLUID – APPLIED UNHEATED FLAPS DEPLOYED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS										
OAT [2]		Type III Fluid Concentration	Holdover Times (minutes) [1]							
°C	°F	Fluid/Water (% by Volume)	Freezing Fog, Freezing Mist [7] or Ice Crystals	Snow, Snow Grains or Snow Pellets [8]			Freezing Drizzle [4]	Light Freezing Rain	Rain on Cold Soaked Wing [5]	Other [6]
				Very Light [3]	Light [3]	Moderate				
-3 and above	27 and above	100 / 0	34 – 87	61 – 80	30 – 61	14 – 30	19 – 38	11 – 19	4 – 30	
below -3 to -10	below 27 to 14	100 / 0	38 – 76	61 – 80	30 – 61	14 – 30	19 – 34	11 – 19		
below -10 to -25	below 14 to -13	100 / 0	30 – 80	61 – 80	30 – 61	14 – 30	No holdover time guidelines exist			
below -25 to -35	below -13 to -31	100 / 0	19 – 46	34 – 46	15 – 34	8 – 15				

[1] Fluid must be applied unheated. No holdover time guidelines exist for this fluid applied heated.
 [2] Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I when Type III fluid cannot be used.
 [3] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle.
 [4] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
 [5] No holdover time guidelines exist for this condition for 0°C (32°F) and below.
 [6] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
 [7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0°C (32°F) and below.
 [8] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.

GF0701_032D

AllClear Unheated Type III Fluid Holdover Times – Flaps Deployed Prior to Deicing/Anti-Icing
Figure 06-12-8

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

SAE TYPE IV FLUIDS – FLAPS RETRACTED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS											
OAT [1]		Type IV Fluid Concentration	Holdover Times (minutes)								
°C	°F	Fluid/Water (% by Volume)	Freezing Fog, Freezing Mist ^[7] or Ice Crystals	Snow, Snow Grains or Snow Pellets ^[9]			Freezing Drizzle ^[3]	Light Freezing Rain	Rain on Cold Soaked Wing ^[4]	Other ^[5]	
				Very Light ^[2]	Light ^[2]	Moderate					
-3 and above	27 and above	100 / 0	75 – 160	115 – 120	60 – 115	30 – 60	40 – 70	20 – 35	8 – 65		
		75 / 25	85 – 160	120	75 – 120	40 – 75	50 – 80	30 – 45	9 – 75		
		50 / 50	30 – 55	60 – 70	25 – 60	10 – 25	15 – 40	9 – 20			
below -3 to -8	below 27 to 18	100 / 0	20 – 95	105 – 120	55 – 105	25 – 55	25 – 70	20 – 25			
		75 / 25	30 – 80	110 – 120	60 – 110	30 – 60	20 – 65	15 – 25			
below -8 to -14	below 18 to 7	100 / 0	20 – 95	80 – 100	45 – 80	25 – 45	25 – 70 ^[8]	20 – 25 ^[8]			
		75 / 25	30 – 80	100 – 120	45 – 100	20 – 45	20 – 65 ^[8]	15 – 25 ^[8]			
below -14 to -18	below 7 to 0	100 / 0	20 – 35	30 – 45	9 – 30	2 – 9	No holdover time guidelines exist				
below -18 to -25	below 0 to -13	100 / 0	20 – 35 ^[6]	10 – 20 ^[6]	3 – 10 ^[6]	1 – 3 ^[6]					
below -25 to LOUT	below -13 to LOUT	100 / 0	20 – 35 ^[6]	7 – 10 ^[6]	2 – 7 ^[6]	0 – 2 ^[6]					

[1] Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I when Type IV fluid cannot be used.

[2] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle.

[3] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.

[4] No holdover time guidelines exist for this condition for 0° C (32° F) and below.

[5] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

[6] If the LOUT is unknown, no holdover time guidelines exist below -23.5° C (-10° F)

[7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0° C (32° F) and below.

[8] No holdover time guidelines exist for this condition below -10° C (14° F).

[9] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.

GF0701_041D

Type IV Fluid Holdover Times – Flaps Retracted Prior to Deicing/Anti-Icing
Figure 06-12-9

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

SAE TYPE IV FLUIDS – FLAPS DEPLOYED PRIOR TO DEICING/ANTI-ICING APPROXIMATE HOLDOVER TIMES ANTICIPATED UNDER VARIOUS WEATHER CONDITIONS											
OAT [1]		Type IV Fluid Concentration	Holdover Times (minutes)								
°C	°F	Fluid/Water (% by Volume)	Freezing Fog, Freezing Mist ^[7] or Ice Crystals	Snow, Snow Grains or Snow Pellets ^[9]			Freezing Drizzle ^[3]	Light Freezing Rain	Rain on Cold Soaked Wing ^[4]	Other ^[5]	
				Very Light ^[2]	Light ^[2]	Moderate					
-3 and above	27 and above	100 / 0	57 – 122	87 – 106	46 – 87	23 – 46	30 – 53	15 – 27	6 – 49		
		75 / 25	65 – 122	95 – 110	57 – 95	30 – 57	38 – 61	23 – 34	7 – 57		
		50 / 50	23 – 42	46 – 53	19 – 46	8 – 19	11 – 30	7 – 15			
below -3 to -8	below 27 to 18	100 / 0	15 – 72	80 – 95	42 – 80	19 – 42	19 – 53	15 – 19			
		75 / 25	23 – 61	84 – 99	46 – 84	23 – 46	15 – 49	11 – 19			
below -8 to -14	below 18 to 7	100 / 0	15 – 72	61 – 76	34 – 61	19 – 34	19 – 53 ^[8]	15 – 19 ^[8]			
		75 / 25	23 – 61	76 – 91	34 – 76	15 – 34	15 – 49 ^[8]	11 – 19 ^[8]			
below -14 to -18	below 7 to 0	100 / 0	15 – 27	23 – 34	7 – 23	2 – 7	No holdover time guidelines exist				
below -18 to -25	below 0 to -13	100 / 0	15 – 27 ^[6]	8 – 15 ^[6]	2 – 8 ^[6]	1 – 2 ^[6]					
below -25 to LOUT	below -13 to LOUT	100 / 0	15 – 27 ^[6]	5 – 8 ^[6]	2 – 5 ^[6]	0 – 2 ^[6]					

[1] Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I when Type IV fluid cannot be used.

[2] Use light freezing rain holdover times in conditions of very light or light snow mixed with light rain or drizzle.

[3] Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.

[4] No holdover time guidelines exist for this condition for 0°C (32°F) and below.

[5] Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

[6] If the LOUT is unknown, no holdover time guidelines exist below -23.5°C (-10°F)

[7] Freezing Mist is best confirmed by observation. It is never reported by METAR however it can occur when mist is present at 0°C (32°F) and below.

[8] No holdover time guidelines exist for this condition below -10°C (14°F).

[9] To determine snowfall intensity, the Snowfall Intensities as a Function of Visibility table is required.

GF0701_033E

Type IV Fluid Holdover Times – Flaps Deployed Prior to Deicing/Anti-Icing
Figure 06-12-10

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

ACTIVE FROST – TYPE I FLUID FLAPS RETRACTED			ACTIVE FROST – TYPE II, III AND IV FLUIDS FLAPS RETRACTED					
OAT ^{[1], [2], [3]}		Approximate Holdover Times (hours:minutes)	OAT ^{[2], [3]}		Fluid Concentration	Approximate Holdover Times (hours:minutes)		
°C	°F	Type I	°C	°F	Fluid/Water (% by Volume)	Type II	Type III ^[4]	Type IV
–1 and above	30 and above	0:45	–1 and above	30 and above	100 / 0	8:00	2:00	12:00
					75 / 25	5:00	1:00	5:00
					50 / 50	2:00	0:30	3:00
below –1 to –3	below 30 to 27		100 / 0	8:00	2:00	12:00		
			75 / 25	5:00	1:00	5:00		
			50 / 50	1:30	0:30	3:00		
below –3 to –10	below 27 to 14		100 / 0	8:00	2:00	10:00		
			75 / 25	4:00	1:00	5:00		
below –10 to –14	below 14 to 7		100 / 0	6:00	2:00	6:00		
			75 / 25	1:00	1:00	1:00		
below –14 to –21	below 7 to –6		below –14 to –21	below 7 to –6	100 / 0	3:00	2:00	6:00
below –21 to –25	below –6 to –13		below –21 to –25	below –6 to –13	100 / 0	2:00	2:00	4:00
below –25 to LOUT	below –13 to LOUT		below –25 to LOUT	below –13 to LOUT	100 / 0	No holdover time guidelines exist		
[1] Type I fluid/water mixture must be selected so that the freezing point of the mixture is at least 10°C (18°F) below outside air temperature. [2] Ensure that the lowest operational use temperature (LOUT) is respected. [3] Changes in outside air temperature (OAT) over the course of longer frost events can be significant; the appropriate holdover time to use is the one provided for the coldest OAT that has occurred in the time between the deicing/anti-icing fluid application and takeoff. [4] To use Type III fluid frost holdover times, the fluid brand being used must be known. AllClear AeroClear MAX must be applied unheated.								

GF0701_038A

Active Frost Holdover Times – Flaps Retracted Prior to Deicing/Anti-Icing
Figure 06-12-11

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

ACTIVE FROST – TYPE I FLUID FLAPS DEPLOYED			ACTIVE FROST – TYPE II, III AND IV FLUIDS FLAPS DEPLOYED					
OAT ^{[1], [2], [3]}		Approximate Holdover Times (hours:minutes)	OAT ^{[2], [3]}		Fluid Concentration	Approximate Holdover Times (hours:minutes)		
°C	°F	Type I	°C	°F	Fluid/Water (% by Volume)	Type II	Type III ^[4]	Type IV
–1 and above	30 and above	0:34	–1 and above	30 and above	100 / 0	6:05	1:31	9:07
					75 / 25	3:48	0:46	3:48
					50 / 50	1:31	0:23	2:17
below –1 to –3	below 30 to 27		100 / 0	6:05	1:31	9:07		
			75 / 25	3:48	0:46	3:48		
			50 / 50	1:08	0:23	2:17		
below –3 to –10	below 27 to 14		below –3 to –10	below 27 to 14	100 / 0	6:05	1:31	7:36
					75 / 25	3:02	0:46	3:48
below –10 to –14	below 14 to 7		below –10 to –14	below 14 to 7	100 / 0	4:34	1:31	4:34
					75 / 25	0:46	0:46	0:46
below –14 to –21	below 7 to –6		below –14 to –21	below 7 to –6	100 / 0	2:17	1:31	4:34
below –21 to –25	below –6 to –13		below –21 to –25	below –6 to –13	100 / 0	1:31	1:31	3:02
below –25 to LOUT	below –13 to LOUT		below –25	below –13	100 / 0	No holdover time guidelines exist		
[1] Type I fluid/water mixture must be selected so that the freezing point of the mixture is at least 10°C (18°F) below outside air temperature. [2] Ensure that the lowest operational use temperature (LOUT) is respected. [3] Changes in outside air temperature (OAT) over the course of longer frost events can be significant; the appropriate holdover time to use is the one provided for the coldest OAT that has occurred in the time between the deicing/anti-icing fluid application and takeoff. [4] To use Type III fluid frost holdover times, the fluid brand being used must be known. AllClear AeroClear MAX must be applied unheated.								

GF0701_039B

Active Frost Holdover Times – Flaps Deployed Prior to Deicing/Anti-Icing
Figure 06-12-12

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)**(4) ICE PELLET AND SMALL HAIL ALLOWANCE**

Operators with a deicing program updated to include the allowance time information will be allowed, in the specified ice pellet and small hail conditions, up to the specific allowance time, to commence the takeoff with the following restrictions:

- (a) The aircraft critical surfaces must be free of contaminants before applying anti-icing fluid. If not, the aircraft must be properly deiced and checked to be free of contaminants before the application of anti-icing fluid.
- (b) The allowance time is valid only if the aircraft is anti-iced with undiluted Type III or Type IV fluid.
- (c) The Type III allowance times are only applicable for unheated anti-icing fluid applications.
- (d) Due to the shearing qualities of Type III and Type IV fluids with imbedded ice pellets, allowance times are limited to aircraft with rotation speeds as indicated in the allowance times tables.
- (e) If takeoff is not accomplished within the applicable allowance time, the aircraft must be completely deiced, and if precipitation is still present, anti-iced again prior to a subsequent takeoff. If the precipitation stops at or before the time limits of the applicable allowance time and does not restart, the aircraft may takeoff up to 90 minutes after the start of the application of the Type III or Type IV anti-icing fluid. These allowance times are from the start of the anti-icing fluid application. Additionally, if the ice pellet condition stops, and the allowance time has not been exceeded, the operator is permitted to consider the anti-icing fluid effective without any further action up to 90 minutes after the start of the application time of the anti-icing fluid. To use this guidance in the following conditions, the outside air temperature (OAT) must remain constant or increase during the 90-minute period:
 - light ice pellets mixed with freezing drizzle;
 - light ice pellets mixed with freezing rain and
 - light ice pellets mixed with rain.
- (f) A pre-takeoff contamination inspection is not required. The allowance time cannot be extended by an internal or external inspection of the aircraft critical surfaces.
- (g) If ice pellet precipitation becomes heavier than moderate or if the light ice pellets mixed with other forms of allowable precipitation exceeds the listed intensities or temperature range, the allowance time cannot be used.
- (h) If the temperature decreases below the temperature on which the allowance time was based, and the new lower temperature has an associated allowance time for the precipitation condition and the present time is within the new allowance time, then that new time must be used as the allowance time limit.
- (i) If the temperature decreases below the temperature on which the allowance time was based, and the allowance time has expired (within the 90 minute post anti-icing window if the precipitation has stopped within the allowance time), the aircraft may not takeoff and must be completely deiced and, if applicable, anti-iced before a subsequent takeoff.
- (j) If an intensity is reported with small hail, the ice pellet condition with the equivalent intensity can be used, i.e. if light small hail is reported, the light ice pellets allowance times can be used. This also applies in mixed conditions, i.e. if light small hail mixed with snow is reported, use the light ice pellets mixed with snow allowance times.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)



1. Allowance time can not be extended by an inspection of the aircraft critical surfaces.
2. Take off is allowed up to 90 minutes after start of the fluid application if the precipitation stops at or before the allowance time expires and does not restart. The OAT must not decrease during the 90 minutes to use this guidance in conditions of light ice pellets mixed with either freezing drizzle, freezing rain or rain.

TRAINING PURPOSES ONLY

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

Precipitation Types or Combinations	Applicable METAR Codes	Outside Air Temperature		
		-5 °C and above (23 °F and above)	Below -5 to -10 °C (Below 23 to 14 °F)	Below -10 °C ^[1] (Below 14 °F) ^[1]
Light Ice Pellets	-PL	10 minutes	10 minutes	Caution: No allowance times currently exist
Light Ice Pellets Mixed with Light Snow	-PLSN, -SNPL	10 minutes	10 minutes	
Light Ice Pellets Mixed with Light Freezing Drizzle or Moderate Freezing Drizzle	-PLFZDZ, -FZDZPL, FZDZPL	7 minutes	5 minutes	
Light Ice Pellets Mixed with Light Freezing Rain	-PLFZRA, -FZRAPL	7 minutes	5 minutes	
Light Ice Pellets Mixed with Light Rain	-PLRA, -RAPL	7 minutes ^[2]		
Moderate Ice Pellets (or Small Hail ^[3])	PL, GS	5 minutes	5 minutes	

GF0701_047

[1] Ensure that the lowest operational use temperature (LOUT) is respected.

[2] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below; consider use of light ice pellets mixed with light freezing rain.

[3] In Canada (and outside the US) the code GS is used when hail is less than 5 mm and GR when it is 5mm or greater. In the US, small hail is included with regular hail and the remarks section is used saying "GR LESS THAN ¼". If no intensity is reported with small hail, use the "moderate ice pellets or small hail" allowance times. If an intensity is reported with small hail, the ice pellet condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

Type III Ice Pellet and Small Hail Allowance Times – Slats/Flaps Retracted Prior to
Deicing/Anti-Icing
Figure 06-12-13

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

Precipitation Types or Combinations	Applicable METAR Codes	Outside Air Temperature		
		-5 °C and above (23 °F and above)	Below -5 to -10 °C (Below 23 to 14 °F)	Below -10 °C ^[1] (Below 14 °F) ^[1]
Light Ice Pellets	-PL	8 minutes	8 minutes	Caution: No allowance times currently exist
Light Ice Pellets Mixed with Light Snow	-PLSN, -SNPL	8 minutes	8 minutes	
Light Ice Pellets Mixed with Light Freezing Drizzle or Moderate Freezing Drizzle	-PLFZDZ, -FZDZPL, FZDZPL	5 minutes	4 minutes	
Light Ice Pellets Mixed with Light Freezing Rain	-PLFZRA, -FZRAPL	5 minutes	4 minutes	
Light Ice Pellets Mixed with Light Rain	-PLRA, -RAPL	5 minutes ^[2]		
Moderate Ice Pellets (or Small Hail ^[3])	PL, GS	4 minutes	4 minutes	

GF0701_048

[1] Ensure that the lowest operational use temperature (LOUT) is respected.

[2] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below; consider use of light ice pellets mixed with light freezing rain.

[3] In Canada (and outside the US) the code GS is used when hail is less than 5 mm and GR when it is 5mm or greater. In the US, small hail is included with regular hail and the remarks section is used saying "GR LESS THAN ¼". If no intensity is reported with small hail, use the "moderate ice pellets or small hail" allowance times. If an intensity is reported with small hail, the ice pellet condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

Type III Ice Pellet and Small Hail Allowance Times – Slats/Flaps Deployed Prior to Deicing/Anti-Icing

Figure 06-12-14

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

Precipitation Types or Combinations	Applicable METAR Codes	Outside Air Temperature			
		-5 °C and above (23 °F and above)	Below -5 to -10 °C (Below 23 to 14 °F)	Below -10 to -16 °C (Below 14 to 3.2 °F)	Below -16 to -22 °C ^[1] (Below 3 to -7.6 °F) ^[1]
Light Ice Pellets	-PL	70 minutes	50 minutes	30 minutes	30 minutes
Light Ice Pellets Mixed with Light Snow	-PLSN, -SNPL	50 minutes	30 minutes	15 minutes	Caution: No allowance times currently exist
Light Ice Pellets Mixed with Light Freezing Drizzle or Moderate Freezing Drizzle	-PLFZDZ, -FZDZPL, FZDZPL	40 minutes	30 minutes		
Light Ice Pellets Mixed with Light Freezing Rain	-PLFZRA, -FZRAPL	40 minutes	30 minutes		
Light Ice Pellets Mixed with Light Rain	-PLRA, -RAPL	40 minutes ^[2]			
Moderate Ice Pellets (or Small Hail ^[3])	PL, GS	35 minutes	25 minutes	10 minutes	10 minutes
Moderate Ice Pellets (or Small Hail ^[3]) Mixed with Moderate Freezing Drizzle	PLFZDZ, GSFZDZ,	20 minutes	10 minutes	Caution: No allowance times currently exist	
Moderate Ice Pellets (or Small Hail ^[3]) Mixed with Moderate Rain	PLRA, GSRA, RAPL, RAGS	15 minutes ^[4]			

GF0701_049

[1] Ensure that the lowest operational use temperature (LOUT) is respected.

[2] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below; consider use of light ice pellets mixed with light freezing rain.

[3] In Canada (and outside the US) the code GS is used when hail is less than 5 mm and GR when it is 5mm or greater. In the US, small hail is included with regular hail and the remarks section is used saying "GR LESS THAN ¼". If no intensity is reported with small hail, use the "moderate ice pellets or small hail" allowance times. If an intensity is reported with small hail, the ice pellet condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

[4] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below.

[5] If the glycol type is unknown, the allowance times for SAE Type IV Propylene Glycol (PG) fluids should be used.

Type IV Fluids Ethylene Glycol (EG) Fluids Ice Pellet and Small Hail Allowance Times – Slats/Flaps Retracted Prior to Deicing/Anti-Icing
Figure 06-12-15

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

Precipitation Types or Combinations	Applicable METAR Codes	Outside Air Temperature			
		-5 °C and above (23 °F and above)	Below -5 to -10 °C (Below 23 to 14 °F)	Below -10 to -16 °C (Below 14 to 3.2 °F)	Below -16 to -22 °C ^[1] (Below 3 to -7.6 °F) ^[1]
Light Ice Pellets	-PL	53 minutes	38 minutes	23 minutes	23 minutes
Light Ice Pellets Mixed with Light Snow	-PLSN, -SNPL	38 minutes	23 minutes	11 minutes	Caution: No allowance times currently exist
Light Ice Pellets Mixed with Light Freezing Drizzle or Moderate Freezing Drizzle	-PLFZDZ, -FZDZPL, FZDZPL	30 minutes	23 minutes		
Light Ice Pellets Mixed with Light Freezing Rain	-PLFZRA, -FZRAPL	30 minutes	23 minutes		
Light Ice Pellets Mixed with Light Rain	-PLRA, -RAPL	30 minutes ^[2]			
Moderate Ice Pellets (or Small Hail) ^[3]	PL, GS	27 minutes	19 minutes	8 minutes	8 minutes
Moderate Ice Pellets (or Small Hail) ^[3] Mixed with Moderate Freezing Drizzle	PLFZDZ, GSFZDZ,	15 minutes	8 minutes	Caution: No allowance times currently exist	
Moderate Ice Pellets (or Small Hail) ^[3] Mixed with Moderate Rain	PLRA, GSRA, RAPL, RAGS	11 minutes ^[4]			

GF0701_050

[1] Ensure that the lowest operational use temperature (LOUT) is respected.

[2] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below; consider use of light ice pellets mixed with light freezing rain.

[3] In Canada (and outside the US) the code GS is used when hail is less than 5 mm and GR when it is 5mm or greater. In the US, small hail is included with regular hail and the remarks section is used saying "GR LESS THAN ¼". If no intensity is reported with small hail, use the "moderate ice pellets or small hail" allowance times. If an intensity is reported with small hail, the ice pellet condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

[4] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below.

[5] If the glycol type is unknown, the allowance times for SAE Type IV Propylene Glycol (PG) fluids should be used.

Type IV Fluids Ethylene Glycol (EG) Fluids Ice Pellet and Small Hail Allowance Times – Slats/Flaps Deployed Prior to Deicing/Anti-Icing
Figure 06-12-16

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

Precipitation Types or Combinations	Applicable METAR Codes	Outside Air Temperature			
		-5 °C and above (23 °F and above)	Below -5 to -10 °C (Below 23 to 14 °F)	Below -10 to -16 °C (Below 14 to 3.2 °F)	Below -16 to -22 °C ^[1] (Below 3 to -7.6 °F) ^[1]
Light Ice Pellets	-PL	50 minutes	30 minutes	30 minutes ^[2]	30 minutes ^[2]
Light Ice Pellets Mixed with Light Snow	-PLSN, -SNPL	40 minutes	15 minutes	15 minutes ^[2]	Caution: No allowance times currently exist
Light Ice Pellets Mixed with Light Freezing Drizzle or Moderate Freezing Drizzle	-PLFZDZ, -FZDZPL, FZDZPL	25 minutes	10 minutes		
Light Ice Pellets Mixed with Light Freezing Rain	-PLFZRA, -FZRAPL	25 minutes	10 minutes		
Light Ice Pellets Mixed with Light Rain	-PLRA, -RAPL	25 minutes ^[3]			
Moderate Ice Pellets (or Small Hail ^[4])	PL, GS	15 minutes	10 minutes	10 minutes ^[2]	Caution: No allowance times currently exist
Moderate Ice Pellets (or Small Hail ^[4]) Mixed with Moderate Freezing Drizzle	PLFZDZ, GSFZDZ	10 minutes	7 minutes		
Moderate Ice Pellets (or Small Hail ^[4]) Mixed with Moderate Rain	PLRA, GSRA, RAPL, RAGS	10 minutes ^[5]			

GF0701_051

[1] Ensure that the lowest operational use temperature (LOUT) is respected.

[2] No allowance times exist for propylene glycol (PG) fluids when used on aircraft with rotation speeds less than 115 knots.

[3] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below; consider use of light ice pellets mixed with light freezing rain.

[4] In Canada (and outside the US) the code GS is used when hail is less than 5 mm and GR when it is 5mm or greater. In the US, small hail is included with regular hail and the remarks section is used saying "GR LESS THAN ¼". If no intensity is reported with small hail, use the "moderate ice pellets or small hail" allowance times. If an intensity is reported with small hail, the ice pellet condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

[5] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below.

[6] If the glycol type is unknown, the allowance times for SAE Type IV Propylene Glycol (PG) fluids should be used.

Type IV Fluids Propylene Glycol (PG) Fluids Ice Pellet and Small Hail Allowance Times – Slats/Flaps Retracted Prior to Deicing/Anti-Icing
Figure 06-12-17

	CL-605 Flight Crew Operating Manual PSP 605-6	
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6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

Precipitation Types or Combinations	Applicable METAR Codes	Outside Air Temperature			
		-5 °C and above (23 °F and above)	Below -5 to -10 °C (Below 23 to 14 °F)	Below -10 to -16 °C (Below 14 to 3.2 °F)	Below -16 to -22 °C ^[1] (Below 3 to -7.6 °F) ^[1]
Light Ice Pellets	-PL	38 minutes	23 minutes	23 minutes ^[2]	23 minutes ^[2]
Light Ice Pellets Mixed with Light Snow	-PLSN, -SNPL	30 minutes	11 minutes	11 minutes ^[2]	Caution: No allowance times currently exist
Light Ice Pellets Mixed with Light Freezing Drizzle or Moderate Freezing Drizzle	-PLFZDZ, -FZDZPL, FZDZPL	19 minutes	8 minutes		
Light Ice Pellets Mixed with Light Freezing Rain	-PLFZRA, -FZRAPL	19 minutes	8 minutes		
Light Ice Pellets Mixed with Light Rain	-PLRA, -RAPL	19 minutes ^[3]			
Moderate Ice Pellets (or Small Hail ^[4])	PL, GS	14 minutes	8 minutes	8 minutes ^[2]	Caution: No allowance times currently exist
Moderate Ice Pellets (or Small Hail ^[4]) Mixed with Moderate Freezing Drizzle	PLFZDZ, GSFZDZ	8 minutes	5 minutes		
Moderate Ice Pellets (or Small Hail ^[4]) Mixed with Moderate Rain	PLRA, GSRA, RAPL, RAGS	8 minutes ^[5]			

GF0701_052

[1] Ensure that the lowest operational use temperature (LOUT) is respected.

[2] No allowance times exist for propylene glycol (PG) fluids when used on aircraft with rotation speeds less than 115 knots.

[3] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below; consider use of light ice pellets mixed with light freezing rain.

[4] In Canada (and outside the US) the code GS is used when hail is less than 5 mm and GR when it is 5mm or greater. In the US, small hail is included with regular hail and the remarks section is used saying "GR LESS THAN ¼". If no intensity is reported with small hail, use the "moderate ice pellets or small hail" allowance times. If an intensity is reported with small hail, the ice pellet condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

[5] No allowance times exist in this condition for temperatures of 0 °C (32 °F) and below.

[6] If the glycol type is unknown, the allowance times for SAE Type IV Propylene Glycol (PG) fluids should be used.

Type IV Fluids Propylene Glycol (PG) Fluids Ice Pellet and Small Hail Allowance Times – Slats/Flaps Deployed Prior to Deicing/Anti-Icing
Figure 06-12-18

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

C. Deicing/Anti-Icing Procedures

Deicing is the removal of snow, ice or frost from airplane surfaces, using mechanical means, hot water or a heated mixture of water and deicing/anti-icing fluid, or other means which satisfy the requirement of the clean aircraft concept.

Anti-icing is the application of deicing/anti-icing fluid with a useful holdover time, to prevent the accumulation of snow, ice or frost on airplane surfaces after deicing.

Current practice prescribes the following general methods for effecting deicing/anti-icing:

(1) MECHANICAL REMOVAL OF LOOSE CONTAMINATION

If a significant amount of loose snow is on the airplane, the expenditure of a relatively large amount of deicing fluid can be avoided if the snow is removed mechanically. Subject to the results of an inspection as outlined below (Removal of Loose Contamination), this may achieve complete deicing of the airplane.

(2) ONE-STEP DEICING ONLY OR ANTI-ICING ONLY

To deice, fluid is applied in one step to remove frozen contamination with no minimum fluid quantity required. Use as much fluid as necessary to ensure the aircraft is free of frozen contamination.

One step anti-icing is only possible on a clean aircraft. In this process, the residual fluid film, regardless of the type of fluid used, will provide only a very limited duration of anti-icing protection.

If deicing and anti-icing are required, a two-step procedure should be followed.

(3) TWO-STEP DEICING/ANTI-ICING

Two fluid applications are made: the first to deice, using hot water or a water/fluid mixture; the second to anti-ice, using undiluted (100%) fluid, or a water/fluid mixture. This method ensures that the full anti-icing holdover time available from the fluid will be obtained.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)**(4) INFRARED DEICING**

Deicing using infrared energy is an alternative procedure for removing frozen precipitation. It is accomplished through heat that breaks the bond of adhering frozen contamination. The application of infrared energy may be continued to melt and evaporate frozen contaminants.

WARNING

When using infrared energy to deice, wet surfaces require an application of heated deicing fluids to prevent refreezing after removal of infrared energy source.

When required, for operation other than frost or leading edge ice removal, and when OAT is at or below 0°C (32°F), an additional treatment with hot deicing fluid must be done within the infrared deicing facility to prevent refreezing of water, which may remain in hidden areas.

CAUTION

If the aircraft requires deicing again and deicing/anti-icing fluids had been applied before flight, conventional deicing/anti-icing with fluids must be done.

When using infrared energy to deice, refer also to the following SAE industry standard practices and FAA Advisory Circulars for procedures and precautions:

- SAE Aerospace Recommended Practices ARP 4737
- FAA Advisory Circular No. 150/5300-14 Appendix A
- FAA Advisory Circular No. 120-89

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

D. Fluid Application Guidelines

TYPE I FLUID APPLICATION GUIDELINES				
Outside Air Temperature (OAT) [1]	One-Step Procedures		Two-Step Procedure	
	Deicing Only [2]	Anti-icing Only [3]	First Step Deicing [2]	Second Step Anti-icing [4]
0°C (32°F) and above	Heated Type I fluid/water mixture with a freezing point of at least 10°C (18°F) below OAT.	Fluid/water mixture heated to at least 60°C (140°F) at the nozzle.	Heated water or a heated fluid/water mixture	Fluid/water mixture heated to at least 60°C (140°F) at the nozzle with a freezing point of at least 10°C (18°F) below OAT.
Below -0°C (32°F) to LOUT			Heated fluid/water mixture with a freezing point at OAT or below	
[1] Fluids must not be used at temperature below their lowest operational use temperature (LOUT). [2] When deicing, there is no minimum fluid quantity required, use as much fluid as necessary to ensure the aircraft is free of frozen contamination. A fluid temperature of at least 60°C (140°F) at the nozzle is desirable. Deicing is complete after all contamination is removed. There is no holdover time for a deicing only operation. [3] One-step anti-icing only procedure is only possible on a clean aircraft. If deicing and anti-icing are required, a two-step procedure should be followed. [4] To be applied before first fluid freezes, typically within 3 minutes. (This time may be higher than 3 minutes in some conditions, but potentially lower in heavy precipitation, colder temperatures, or for critical surfaces constructed of composite materials. If necessary, the second step shall be applied area by area.)				

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)



The TYPE I FLUID APPLICATION GUIDELINES table is applicable for the use of Type I holdover time guidelines in all conditions, including active frost.

To use Type I holdover time guidelines in all conditions including active frost, an additional minimum of 1 litre/m² (2 gal/100 ft²) of heated Type I fluid mixture must be applied to the surfaces after all frozen contamination is removed. This application is necessary to heat the surfaces, as heat contributes significantly to the Type I fluid holdover times. If deicing and anti-icing are required, use a two-step procedure.

If holdover times are not required, a temperature of 60°C (140°F) at the nozzle is desirable. If holdover times are required, the temperature of water or fluid/water mixtures shall be at least 60°C (140°F) at the nozzle. Upper temperature limit shall not exceed fluid and aircraft manufacturers recommendations.

The lowest operational use temperature (LOUT) for a given Type I fluid is the higher (warmer) of:

- The lowest temperature at which the fluid meets the aerodynamic acceptance test for a given aircraft type; or
- The actual freezing point of the fluid plus its freezing point buffer of 10°C (18°F).

Wing skin temperatures may differ and in some cases may be lower than the OAT. A stronger mix (more glycol) may be needed under these conditions.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

TYPE II AND TYPE IV FLUID APPLICATION GUIDELINES (CONCENTRATION IN % BY VOLUME)				
Outside Air Temperature (OAT) [1]	One-Step Procedure		Two-Step Procedure	
	Deicing [2]	Anti-icing [3]	First Step Deicing [2]	Second Step Anti-icing [4]
0°C (32°F) and above	Heated Type II or IV fluid/water mixture	100/0, 75/25 or 50/50 Heated or unheated Type II or IV fluid/water mixture	Heated water or a heated fluid/water mixture	100/0, 75/25 or 50/50 Heated or unheated Type II or IV fluid/water mixture
Below 0°C (32°F) to -3°C (27°F)		100/0, 75/25 or 50/50 Heated or unheated Type II or IV fluid/water mixture	Heated fluid/water mixture with a freezing point at OAT or below	100/0, 75/25 or 50/50 Heated or unheated Type II or IV fluid/water mixture
Below -3°C (27°F) to -14°C (7°F)		100/0 or 75/25 Heated or unheated Type II or IV fluid/water mixture		100/0 or 75/25 Heated or unheated Type II or IV fluid/water mixture
Below -14°C (7°F) to LOUT		100/0 Heated or unheated Type II or IV fluid/water mixture		100/0 Heated or unheated Type II or IV fluid/water mixture
[1] One step or second step fluids must not be used at temperatures below their lowest operational use temperature (LOUT). First step fluids must not be used below their freezing points. [2] When deicing, there is no minimum fluid quantity required, use as much fluid as necessary to ensure the aircraft is free of frozen contamination. A fluid temperature of at least 60°C (140°F) at the nozzle is desirable. Deicing is complete after all contamination is removed. There is no holdover time for a deicing only operation. [3] One-step anti-icing only procedure is only possible on a clean aircraft. If deicing and anti-icing are required, a two-step procedure should be followed. [4] To be applied before first fluid freezes, typically within 3 minutes. (This time may be higher than 3 minutes in some conditions, but potentially lower in heavy precipitation, colder temperatures, or for critical surfaces constructed of composite materials. If necessary, the second step shall be applied area by area.)				

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)



Consider the use of Type I or III fluid when Type II or IV fluid cannot be used due to LOUT limitations. The LOUT for a given Type II or IV fluid is the higher (warmer) of:

- The lowest temperature at which the fluid meets the aerodynamic acceptance test for a given airplane type.
- The actual freezing point of the fluid plus its freezing point buffer of 7°C (13°F).
- For diluted Type II or IV fluids, the coldest temperature for which holdover times are published.

For heated fluids, a fluid temperature not less than 60°C (140°F) at the nozzle is desirable.

Upper temperature limit shall not exceed fluid and aircraft manufacturers recommendations.

Wing skin temperatures may differ and in some cases may be lower than the OAT. Consult the appropriate HOT tables for the contaminant in question.

Whenever frost or ice occurs on the lower surface of the wing in the area of the fuel tank, indicating a cold soaked wing, the 50/50 dilutions of Type II or IV shall not be used for the anti-icing step because fluid freezing may occur.

An insufficient amount of anti-icing fluid may cause a substantial loss of holdover time. This is particularly true when using a Type I fluid mixture for the first step in a two-step procedure.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

TYPE III FLUID APPLICATION GUIDELINES (CONCENTRATION IN % BY VOLUME)				
Outside Air Temperature (OAT) [1]	One-Step Procedure		Two-Step Procedure	
	Deicing [2]	Anti-icing [3]	First Step Deicing [2]	Second Step Anti-icing [4]
0°C (32°F) and above	Heated Type III fluid/water mixture	100/0, 75/25 or 50/50 Unheated Type III fluid/water mixture	Heated water or a heated fluid/water mixture	100/0, 75/25 or 50/50 Unheated Type III fluid/water mixture
Below 0°C (32°F) to -3°C (27°F)		100/0, 75/25 or 50/50 Unheated Type III fluid/water mixture	Heated fluid/water mixture with a freezing point at OAT or below	100/0, 75/25 or 50/50 Unheated Type III fluid/water mixture
Below -3°C (27°F) to -10°C (14°F)		100/0 or 75/25 Unheated Type III fluid/water mixture		100/0 or 75/25 Unheated Type III fluid/water mixture
Below -10°C (14°F) to LOUT		100/0 Unheated Type III fluid/water mixture		100/0 Unheated Type III fluid/water mixture
[1] Fluids used for anti-icing must not be used at temperatures below their lowest operational use temperature (LOUT). First step fluids must not be used below their freezing points. [2] When deicing, there is no minimum fluid quantity required, use as much fluid as necessary to ensure the aircraft is free of frozen contamination. A fluid temperature of at least 60°C (140°F) at the nozzle is desirable. Deicing is complete after all contamination is removed. There is no holdover time for a deicing only operation. [3] One-step anti-icing only procedure is only possible on a clean aircraft. If deicing and anti-icing are required, a two-step procedure should be followed. [4] To be applied before first fluid freezes, typically within 3 minutes. (This time may be higher than 3 minutes in some conditions, but potentially lower in heavy precipitation, colder temperatures, or for critical surfaces constructed of composite materials. If necessary, the second step shall be applied area by area.)				

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)



Consideration should be given to the use of Type I when Type III fluid cannot be used. The LOUT for a given Type III fluid is the higher (warmer) of:

- The lowest temperature at which the fluid meets the aerodynamic acceptance test for a given airplane type.
- The actual freezing point of the fluid plus its freezing point buffer of 7°C (13°F).
- For diluted Type III fluids, the coldest temperature for which holdover times are published.

Upper temperature limit shall not exceed fluid and aircraft manufacturers recommendations.

Wing skin temperatures may differ and in some cases may be lower than the OAT. Consult the appropriate HOT tables for the contaminant in question.

Whenever frost or ice occurs on the lower surface of the wing in the area of the fuel tank, indicating a cold soaked wing, the 50/50 dilutions of Type III shall not be used for the anti-icing step because fluid freezing may occur.

An insufficient amount of anti-icing fluid may cause a substantial loss of holdover time. This is particularly true when using a Type I fluid mixture for the first step in a two-step procedure.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

E. Removal of Loose Contamination

If significant amounts of loose snow are on the airplane:

- | | |
|--|---|
| <p>(1) Flaps..... As required</p> <p>(2) Snow Remove</p> | <p>Flaps 45° recommended.</p> <p>from the following areas, and inspect for presence of adhering ice, frost or snow:</p> <ul style="list-style-type: none"> • Wings – leading edges, upper and lower surfaces; • Upper fuselage; • Vertical and horizontal stabilizers – leading edges, upper/lower surfaces and side panels; • Flaps, flap tracks and flap drives; • Ailerons, elevators, rudder and spoilers; • Air data probes/sensors, AOA vanes; • Antennas; • Fuel drains and NACA vent scoops; • Engine and APU intakes; APU exhaust; • Landing gear and landing gear bays; • Windshield, windows; door sills and surrounds. |
|--|---|



Excessive temperatures in the windshield may result in permanent damage.

NOTE

A light accumulation of snow may be cleared from the windshield, by setting the WSHLD/WIND heat switches on the overhead panel to LOW.

- (3) Tactile check Accomplish

If deicing/anti-icing is not planned:

- (4) Limitations Review Refer to Chapter 2; LIMITATIONS.



6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)**E. Removal of Loose Contamination (Cont'd)**

If frozen contamination is found adhering to critical surfaces:

- (5) Deicing/anti-icing Accomplish

F. Preparation for Deicing/Anti-Icing

Before fluid is applied:

NOTE

It is recommended that the application of deicing/anti-icing fluid be carried out with the engines and APU off. If this is not possible, steps (2) to (7), following, must be observed.

If engines/APU are off:

- (1) Proceed to step (8).

If engines/APU are operating:

- (2) Thrust levers IDLE for the duration of the operation.

NOTE

If the APU is running, ensure that personnel carrying out the fluid application are aware of the location of the APU air intake, and have been instructed to avoid fluid spray that can be ingested by the APU.

- (3) AIR-CONDITIONING, L and R
PACK switch/lights Press out Check L and R PACK OFF lights on.
- (4) BLEED AIR, 10TH STAGE L and
R switch/lights Press out Check L and R 10TH STAGE CLOSED lights on.
- (5) ANTI-ICE, WING switch OFF
- (6) ANTI-ICE, COWL L and R
switch/lights Press out Check L and R COWL ON lights out.
- (7) BLEED AIR, 14TH STAGE L and
R switch/lights Press out Check L and R 14TH STAGE CLOSED lights on.
- (8) Stabilizer trim As required In most cases, it is advisable to set the stabilizer trim to nose up, in order for the deicing fluid and contaminants to run off more easily. Set stabilizer trim for take-off after completion of the deicing procedure.



6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

F. Preparation for Deicing/Anti-Icing (Cont'd)

- (9) Flaps..... As required If the flaps require deicing, extend to (or leave at) 45° during the deicing procedure. Upon completion, retract to 0° or set for take-off, as applicable. Otherwise, leave flaps retracted or in their present position, as applicable.
- (10) Deicing operator Advise to avoid direct spraying of the anti-icing fluid jet, to prevent damage on:
- Windshields, side windows and cabin windows;
 - Pitot/static and TAT probes;
 - AOA vanes;
 - Static pressure ports;
 - Engine air intake (to prevent ingestion)/engine exhaust, APU inlet, and APU exhaust;
 - Antennas;
 - Vents and drains;
 - Winglets;
 - Wheels and brakes (to prevent thermal damage).



Application of deicing/anti-icing fluid on wheel brake assemblies will seriously degrade braking performance.

NOTE

A spray trajectory of 3 meters (10 feet) is recommended to ensure that direct spray does not damage airplane surfaces.

- (11) Airplane Head into wind if possible.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)**G. Deicing/Anti-Icing****CAUTION**

1. Under no circumstances can an airplane that has been anti-iced receive another coat of Type II or Type IV fluid on top of the existing film. If the holdover time is exceeded, surfaces must first be deiced with a mixture of hot water and deicing fluid, before another application of Type II or Type IV fluid is made.
2. Type II, III and IV fluid must never be applied to the windshields and side windows.
3. Application of deicing/anti-icing fluid on wheel brake assemblies will seriously degrade braking performance.
4. With the APU operating, ingestion of deicing fluid will contaminate the air-conditioning system, and cause objectionable fumes (causing throat irritation) and odours to enter the airplane. This may also cause erratic operation and possible damage to the APU.
5. Under no circumstances should spray be directed at the trailing edges of control surfaces. Such spray may force partially melted contamination into hinge mechanisms, and under control shrouds, with risk of later re-freezing.

NOTE

1. If heated pure water is used for the first step, the second step must be completed before re-freezing occurs; as a general rule, within 3 minutes of the beginning of the deicing step. This short period makes it necessary to deice/anti-ice relatively small areas of the airplane successively.
 2. Considering the nature of the precipitation present and the likely duration of the delay between the completion of the anti-icing step and take-off, use the applicable holdover table to determine the fluid/water mixture that will provide the most effective anti-ice protection.
 3. A spray trajectory of 3 meters (10 feet) is recommended to ensure that direct spray does not damage airplane surfaces.
 4. On flight control surfaces, application should always be from leading edge to trailing edge and from outer panels to inner panels.
- (1) Fluid.....Apply in the following sequence:
- Horizontal stabilizer
 - Vertical stabilizer
 - Top of fuselage
 - Sides of fuselage
 - Wings.



6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

After Deicing/Anti-icing Spraying:

- (2) Deicing

Inspect

areas listed below, to confirm complete deicing:

 - Wings – leading edges, upper and lower surfaces;
 - Upper fuselage;
 - Vertical and horizontal stabilizers – leading edges, upper/lower surfaces and side panels;
 - Flaps, flap tracks and flap drives;
 - Ailerons, elevators, rudder and spoilers;
 - Air data probes/sensors, AOA vanes;
 - Antennas;
 - Fuel drains and NACA vent scoops;
 - Engine and APU intakes; APU exhaust;
 - Landing gear and landing gear bays;
 - Windshield, windows; door sills and surrounds.

If engines/APU are operating:

- (3) Thrust

Increase to 60% N₁

for 20 seconds, to clear residual fluid.
- (4) BLEED AIR, 14TH STAGE L and R switch/lights

Press in

Check L and R 14TH STAGE CLOSED lights out.
- (5) ANTI-ICE, WING and COWL

As required

NOTE

Wait 2 minutes before opening the 10th stage bleed air valves and turning on the packs, to avoid contaminating the air-conditioning system with deicing/anti-icing fluid. Contamination will cause objectionable fumes (causing throat irritation) and odours to enter the airplane.

- (6) BLEED AIR, 10TH STAGE L and R switch/lights

Press in

Check L and R 10TH STAGE CLOSED lights out.

6. AIRFRAME DEICING, ANTI-ICING, AND INSPECTION (CONT'D)

- (7) AIR CONDITIONING, L and R
PACK switch/lights Press in Check L and R PACK OFF
lights out.
- (8) Take-off configuration..... Recheck Check that **T/O CONFIG OK**
advisory message is on.

7. AIRPLANE PROCEDURES DURING GANTRY DEICING**WARNING**

For personal safety, and to avoid occurrence of incidents caused by engine suction or blast, engines must be run at idle.

CAUTION

The gantry should be programmed to avoid direct impingement on engine intakes, windshields and cabin windows, AOA vanes, pitot heads, static vents, wheels and brakes. If the airplane is deiced using manual direct spray jets, the above areas should be avoided.

NOTE

Direct spraying is defined as anti-ice fluid jet impact due to spray velocity, which may cause mechanical damage. If direct spraying occurs to the engine(s), the pilot should request a wash of both engines on the airplane's return to base.

For all deicing/anti-icing procedures, refer to AIRFRAME DEICING, ANTI-ICING AND INSPECTION earlier in this Supplement.

8. PHASE OF FLIGHT PROCEDURES

NOTE

For all deicing/anti-icing procedures, refer to AIRFRAME DEICING, ANTI-ICING AND INSPECTION earlier in this Supplement.

A. Push Back

When the parking position is covered with ice or snow, the tow truck may not be capable of developing normal power to effect a push back due to reduced friction. It is recommended, therefore, to delay engine start until push back or towing is completed. The flight crew must be aware that in extreme conditions, the application of the parking brake may not be sufficient to restrain the forward motion of the airplane, with the engines at idle thrust.

TRAINING PURPOSES ONLY

8. PHASE OF FLIGHT PROCEDURES (CONT'D)**B. Engine Start**

- (1) Normal engine start procedures Accomplish
- Refer to Chapter 4; NORMAL PROCEDURES – ENGINE STARTING.
 - If engines have not run for 6 hours or more and have been in ambient conditions conducive to moisture freezing, it may be necessary to verify normal operation of the starter valve prior to engine start by carrying out a dry motor.
 - It may be necessary to use ground heating to warm the starter valve, fuel control unit and ignition system, if the engine fails to start normally.
 - If ground heating was used to remove ice build-up from the fan, start the engine immediately to prevent refreezing.
- (2) Engine Instruments Observe for normal operation
- During cold weather starts, initial oil pressure response may be slow, and is not unusual. The oil pressure indication may then indicate higher than the normal range, and should progressively reduce to normal readings as the engine achieves normal operating temperatures (130 psid at idle, 10 minutes maximum).

NOTE

No oil pressure indication by the time idle RPM is achieved requires an immediate shutdown.

8. PHASE OF FLIGHT PROCEDURES (CONT'D)

C. After Engine Start

- (1) Electrical power Transfer to generators Expect generators to be slow in producing steady power due to cold oil in the IDGs. Five minutes may be required for the IDGs to stabilize sufficiently.
- (2) ANTI-ICE, WING and COWL As required

WARNING

When Type II, Type III or Type IV anti-icing fluids have been applied, use of wing anti-ice heat prior to take-off can cook the fluid onto the leading edges of the wing, creating contamination.

NOTE

1. If snow, ice or frost is detected in any amount on the wings and tail surfaces of the airplane, then the airplane must be treated with deicing fluids prior to take-off.
2. The following procedure should not be used if the airplane has been treated with deicing fluids:

Prior to the first flight of the day, whenever the OAT is 5°C (41°F) or below, and it is suspected that the overnight conditions have been conducive to frost formation, select the wing anti-ice system on and advance the thrust levers, as required, until the **L WING A/ICE** and **R WING A/ICE** caution messages are extinguished, to remove any potential uncertainty about the state of the leading edges.

Whenever the aircraft has been held over for a considerable time on the ground in conditions conducive to frost formation, just prior to take-off, select the wing anti-ice system on and advance the thrust levers, as required, until the **L WING A/ICE** and **R WING A/ICE** caution messages are extinguished, to remove any potential uncertainty about the state of the leading edges.

WARNING

Ground use of the wing anti-ice system is intended to complement, **and not replace**, ground deicing/anti-icing and inspection procedures. Close inspection is still required to ensure that no frost, snow or ice is adhering to critical surfaces.



8. PHASE OF FLIGHT PROCEDURES (CONT'D)

C. After Engine Start (Cont'd)

- (3) Engine instruments.....Monitor
- Do not operate engines above idle power until engine indications have returned to normal. Make thrust lever movements cautiously.
 - If a sudden unexplainable rise in ITT occurs, shut down engine immediately.
 - When moderate to severe icing conditions are present during prolonged ground operation, periodic engine run-ups to as high a thrust setting as practical are recommended. This run-up should be done for a minimum of 15 seconds each at 10 minute intervals. This action is effective in dissipating fan blade icing.
 - Do not increase to take-off thrust until normal engine operation has been achieved and indications stabilized.

CAUTION

At temperatures below -20°C (-4°F), the condensation and freezing of moisture in thrust reverser components may degrade the performance of the thrust reversers. The risk of this occurrence is highest when airplanes hangared in relatively warm ambient conditions are readied for flight in freezing conditions.

NOTE

To clear the thrust reversers of frozen moisture during the above described conditions, it is recommended that the thrust reversers be actuated until the deploy and stow cycles are less than 5 seconds.

8. PHASE OF FLIGHT PROCEDURES (CONT'D)**D. Taxi-Out**

Considerations:

- Avoid large nosewheel steering inputs.
- Always taxi at a manageable speed depending on surface condition and contamination.
- Maintain a greater distance than normal between airplanes, especially when slush, standing water, ice or snow is present on surfaces of the movement area.
- Application of brakes should be kept to a minimum during turns.
- Be aware of increased residual thrust at cold temperatures.
- Be aware of increased stopping distances, engine exhaust of the airplane ahead, obscured runway, taxiway or ramp markings, snowbanks and crosswind conditions.
- DO NOT use thrust reversers if movement area surfaces are covered with slush, ice, standing water or snow, except in the interest of safety.
- If taxiways are contaminated, delay flap extension and taxi checklist until prior to take-off.

- (1) Brakes Warm Perform the following number of firm brake applications from 25 knots to 10 knots. Perform the prescribed number of brake applications during the last mile/kilometer of taxi prior to and not including the final stop or snub before take-off. Do not drag the brakes.

AIRPLANE WEIGHT	NUMBER OF BRAKE APPLICATIONS
Greater than 42,000 lb	5
36,000 to 42,000 lb	6
Less than 36,000 lb	7

NOTE

1. Use of symmetric braking is recommended, to ensure uniform brake heating.
 2. Warming of the brakes will preclude the chance of water saturated brakes freezing at altitude and being locked for landing touchdown.
- (2) Nose wheel steering Check Nose wheel steering should be exercised in both directions during taxi.



8. PHASE OF FLIGHT PROCEDURES (CONT'D)**D. Taxi-Out (Cont'd)**

- (3) Flaps.....Check Closely monitor the associated indicator and annunciators for positive movement, when operating the flaps during low temperatures. In the event of stoppage in a detented position, immediately place the flaps selector in the same position as indicated.

Prior to positioning for take-off on the active runway:

- (4) Taxi Check Complete
- (5) Take-off configuration.....Check Check that **T/O CONFIG OK** advisory message is on.

E. Take-Off

If take-off cannot be started prior to the expiration of the holdover time, the airplane must again be inspected, and deiced if necessary, before attempting take-off.

The maximum depths of runway contaminants covering an appreciable part of the runway are:

- Standing water is more than 6.4 millimeters (0.25 inch).
- Slush is more than 6.4 millimeters (0.25 inch).
- Wet snow is more than 12.7 millimeters (0.50 inch).
- Dry snow is more than 30.5 millimeters (1.20 inch).

Take-off is not recommended:

- During extreme weather conditions (i.e. freezing rain).
- When braking action is reported to be poor.
- When crosswind component exceeds 15 knots and the runway is slippery.



8. PHASE OF FLIGHT PROCEDURES (CONT'D)

E. Take-Off (Cont'd)

WARNING

When Type II, Type III or Type IV anti-icing fluids have been applied, use of wing anti-ice heat prior to take-off can cook the fluid onto the leading edges of the wing, creating contamination.

CAUTION

1. Operating on ramps or taxiways which are contaminated with surface snow, slush or standing water when the OAT is 5°C (41°F) or below, can cause the wing leading edge to become contaminated with ice, e.g., from nose wheel splashing or jet blast spray. Just prior to take-off, select the wing anti-ice system on and advance the thrust levers, as required, until the **L WING A/ICE** and **R WING A/ICE** caution messages are extinguished, to remove any leading edge ice contamination that may have accumulated during taxi.
2. The same procedure should be performed whenever the PIC has any doubt of the cleanliness of the wing leading edge prior to take-off.

Considerations:

- Ensure all engine bleed and runway condition penalties have been considered in take-off performance calculations.
- Power application should be done as symmetrically as possible, to avoid yawing moments during engine acceleration.
- Ensure that the cleared runway width available is sufficient.
- If the airplane starts to creep or slide on the ice or snow during thrust application, release the brakes and begin the take-off roll. Anticipate lag in nosewheel steering response and nosewheel skidding, and apply corrections as necessary.
- Do not exceed 3 degrees/second rate of rotation. Anticipate and be prepared to accept a higher than normal initial climb speed. This increased initial climb speed will not adversely affect the climb profile.
- At departure, unless weather conditions or performance requirements prohibit, delay retraction of the landing gear until excess water, snow or slush is thrown off by wheel rotation and/or slip stream force.
- If the airplane tends to pitch-up or roll-off once airborne, immediately reduce the pitch to reduce the angle of attack and simultaneously apply maximum thrust. Be prepared to accept altitude loss to recover the aircraft. Use ailerons as required to level the wings.
- If the decision is made to reject the take-off, normal rejected take-off procedures are applicable. In addition, the following should be considered:
 - Anticipate the possibility of skidding on contaminated runways, and be prepared to make the necessary corrections.
 - If a loss of directional control occurs, reduce reverse thrust to idle reverse and, if necessary, return the engines to idle forward thrust. Regain the centerline with nosewheel steering, rudder and/or differential braking.

8. PHASE OF FLIGHT PROCEDURES (CONT'D)**F. Descent – Approach**

Considerations:

- Anticipate wing anti-icing requirements during descent. This may require increased thrust settings and associated lower rate of descent. Descent planning should be adjusted accordingly.
- Flaps should not be extended in icing conditions except as required. However, if flaps are deployed in icing conditions for extended periods or in severe icing, light to moderate buffet may be encountered. No handling difficulties will result and normal landing flaps can be used. If the buffet is alleviated by reducing the flap setting, a landing in this configuration may be made at the discretion of the pilot. Refer to Chapter 5; ABNORMAL PROCEDURES – FLIGHT CONTROLS, Flaps Failure, for factors to be applied to approach speed and landing distance.
- If holding for extended periods in severe icing and handling difficulties result, i.e. excessive tailplane angle required to trim, landing may be made at a reduced flap setting at the discretion of the pilot. Refer to Chapter 5; ABNORMAL PROCEDURES – FLIGHT CONTROLS, Flaps Failure, for factors to be applied to approach speed and landing distance.
- Ensure all engine bleed and runway condition penalties have been considered in landing/go-around performance calculations.
- Avoid holding in icing conditions for longer than is necessary.
- Ensure that the cleared runway width available is sufficient.
- Take note of Pilot Reports (PIREPs) on braking conditions.

The maximum depths of runway contaminants covering an appreciable part of the runway are:

- Standing water is more than 12.7 millimeters (0.50 inch).
- Slush is more than 12.7 millimeters (0.50 inch).
- Wet snow is more than 25.4 millimeters (1.00 inch).
- Dry snow is more than 61.0 millimeters (2.40 inch).

A diversion should be considered:

- During extreme weather conditions (i.e. freezing rain).
- When braking action is reported to be poor.
- When crosswind component exceeds 15 knots, and the runway is slippery.

8. PHASE OF FLIGHT PROCEDURES (CONT'D)

G. Landing

Considerations:

- Carry out a positive touchdown to ensure initial wheel spin up and breakout of frozen brakes, if icing accumulation on the wheels and brakes is suspected.
- Lower the nose wheel immediately.
- Anticipate skidding and hydroplaning to occur, and be prepared to make the necessary corrections.
- Deploy spoilers and use maximum reverse thrust as soon as possible after touchdown. Thrust reversers are most effective at high speed. Maximum reverse thrust may be used to a complete stop in case of an emergency situation.
- If a loss of directional control occurs, reduce reverse thrust to idle reverse and, if necessary, return the engines to forward idle thrust to return to the centerline. Regain the centerline with nosewheel steering, rudder and/or differential braking.
- DO NOT pump the brakes, as this will only diminish braking effectiveness. Apply brakes normally with steadily increasing pressure, allowing the anti-skid system to modulate brake pressures to obtain maximum braking.

H. Taxi-In and Parking

Considerations:

- DO NOT retract the flaps to less than 20° if the landing occurred on a contaminated runway, to avoid possible damage to the structure and mechanism by frozen slush and/or snow.
- A ground crew check should be performed, and flap retraction completed, when the ground crew has confirmed that there is no significant amount of ice, slush or snow in the flap mechanism.
- Anticipate the movement areas may be slippery, and the use of reverse thrust may be necessary to stop the airplane.
- If able, park on a clear or sanded spot.
- Park into wind, if possible.

8. PHASE OF FLIGHT PROCEDURES (CONT'D)**I. Securing the Airplane**

Considerations

- In addition to the normal procedures for securing the airplane, the flight crew must ensure that ground personnel provide for special servicing of the airplane, especially for prolonged or overnight stops.
- The APU may be operated, and air-conditioning packs used, to provide for heating of the airplane interior and essential compartments, to preclude freezing of the battery or water containers.

At airports where normal support is not available, the flight crew should make certain that the following procedures are performed:

- | | | |
|---|----------------|--|
| (1) Wheel chocks | Check in place | |
| (2) Parking brake | Off | Wheel chocks must be in place, and parking brake off, to eliminate the possibility of the brakes freezing. |
| (3) All protective covers | Installed | |
| (4) Water, lavatory and waste systems | As required | If adequate airplane interior heating cannot be provided, the water, lavatory and waste systems must be drained. |

NOTE

For water and lavatory system draining, refer to the WATER SYSTEM PURGING PROCEDURE placard in the galley.

- | | | |
|-------------------------|---------|--------------|
| (5) Battery (ies) | Removed | if required. |
|-------------------------|---------|--------------|

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1. GENERAL

Experience with operations conducted from wet or contaminated runways has shown that standing water, slush, snow or ice cause a deteriorating effect on take-off and landing performance. Braking effectiveness is reduced due to low tire-to-runway friction and is further reduced if tire hydroplaning occurs.

During take-off on a contaminated runway, airplane acceleration is reduced. Energy normally available for acceleration is dissipated in compression or displacement of the contaminant, and the resulting impingement of the contaminant onto the airplane can aggravate this energy loss.

Landing on a contaminated runway, on the other hand, increases the stopping distance of the airplane, and may present directional control difficulties.

The following supplement is provided to serve as a guideline when operating under such conditions.

2. DEFINITIONS

1. DAMP

- A runway is said to be damp when it is not perfectly dry, and when the water on it does not give it a shiny appearance.

NOTE

There should be no performance penalties on a damp runway.

2. WET

- A runway is said to be wet when it has a shiny appearance due to a thin layer of water on it, not leading to hydroplaning. There should be no standing water covering large areas of the runway.

3. STANDING WATER

- Standing water is usually caused by heavy rainfall and/or insufficient runway drainage, with a depth of more than 3 millimeters (1/8 inch), and covering more than 25% of the runway surface.

4. SLUSH

- Slush is snow saturated with water, which displaces with a splatter when stepped firmly on. It is encountered at temperatures around 5°C (41°F), and has a density of approximately 800 kg per cubic meter (50 pounds per cubic foot), and a specific gravity of 0.85.

5. WET SNOW

- Wet snow will easily stick together and tend to form a snowball if compacted by hand. It has a density of approximately 400 kg per cubic meter (25 pounds per cubic foot).

6. DRY SNOW

- Dry light snow is loose and can easily be blown. If compacted by hand, it will readily fall apart again. It has a density of approximately 192 kg per cubic meter (12 pounds per cubic foot). The coverage is more than 25% of the runway surface.

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2. DEFINITIONS (CONT'D)

7. DRY ICE

- Ice covered runway having a dull weathered appearance. The temperature range is from -40°C to -5°C (-40°F to 23°F).

8. WET ICE

- Ice covered runway having a shiny wet appearance. The temperature range is from -5°C to 4°C (23°F to 40°F).

3. LIMITATIONS AND RECOMMENDATIONS

A. Limitations

The limitations established in Chapter 2 concerning the operation of the airplane on contaminated runways and icing conditions, and the operation of the thrust reversers, must be complied with.

The maximum depths of runway contaminants covering an appreciable part of the runway are:

CONTAMINANT	TAKE-OFF	LANDING
SLUSH/STANDING WATER	6.4 mm (0.25 in)	12.7 mm (0.5 in)
WET SNOW	12.7 mm (0.5 in)	25.4 mm (1.0 in)
DRY SNOW	30.5 mm (1.2 in)	61.0 mm (2.4 in)

B. Recommendations

Compliance with the following take-off and landing practices is recommended.

TAKE-OFF

- Take-off is not recommended on runways which have an appreciable area covered with ice.
- Take-off is not recommended unless APR and both thrust reversers are operational and armed.

LANDING

- Landing is not recommended unless both thrust reversers are operational.
- Landing is not recommended following a system failure (e.g. anti-skid, spoilers, etc.) that would adversely affect the landing distance (Refer to Chapter 5; ABNORMAL PROCEDURES).

4. HYDROPLANING

Hydroplaning can occur on runways contaminated with standing water or slush, and results when hydrodynamic lift forces generated between the tires and the contaminant are sufficient to lift the tires and airplane clear of the runway surface. In this condition, the tires are no longer capable of providing directional control or effective braking.

Hydroplaning does not normally occur in water depths of less than 5 millimeters (1/5 inch), although under some conditions the minimum depth may be as low as 2.5 millimeters (1/10 inch). Once hydroplaning has commenced, it can be sustained over areas where the water depth is less than that required to initiate hydroplaning, and at speeds lower than that required for initiation.

Initiation of hydroplaning is calculated to be at the following ground speeds (KTS):

CONTAMINANT	TIRE PRESSURE (PSI)	HYDROPLANING SPEED (KTS)
Standing Water ($\delta = 1.00$)	MAIN: 184	122
	NOSE: 155	112
Slush ($\delta = 0.85$)	MAIN: 184	132
	NOSE: 155	122
Wet Snow ($\delta = 0.50$)	MAIN: 184	173
	NOSE: 155	158
Dry Snow ($\delta = 0.20$)	MAIN: 184	273
	NOSE: 155	251

NOTE

- Tire pressures are determined with the airplane on the ground.
- δ = specific gravity of contaminant.
- Hydroplaning speed = $9 (\text{tire pressure}/\delta)^{1/2}$

5. TAKE-OFF

Runway contamination, in the form of standing water, slush, snow and ice, has a negative impact on the accelerate-stop distance and the ground run portions of the take-off run and take-off distance values. Under these conditions, corrections have to be applied to the take-off speeds and distances, which would consequently affect take-off performance values.

Take-off field length is the greater of the distance required to accelerate to the critical engine failure recognition speed (V_1) and then come to a full stop, or the distance required to accelerate to V_1 and then continue acceleration with an engine failed to a height of 35 feet above the runway surface. Braking coefficients used during the braking phase of a rejected take-off are based on analysis. APR must be operational and armed for take-off.

The take-off procedures mentioned in the Cold Weather Operation supplement in this chapter may be used, as applicable.

For the appropriate take-off data and take-off performance, refer to Chapter 6; PERFORMANCE of the Airplane Flight Manual.

6. LANDING

Landing on an icy runway is not recommended. If landing is unavoidable under such conditions, it is recommended that reverse thrust be used as soon as possible after touchdown. For dry runways, the most effective means of stopping the airplane are the brakes. However, for icing conditions and contaminated runway operations, the thrust reversers were found to be the most effective deceleration device.

DO NOT try to offset a poor runway braking condition by landing short. It is equally important not to land long. However, landing short can have far more serious consequences than overrunning the far end of the runway at low speed. The desired touchdown point is always about 1,000 feet from the approach end of the runway.

Maintain close control over approach speeds, and maintain the recommended speed for the existing condition. The recommended wind additives (half steady state crosswind plus all gust (regardless of direction), to a maximum of 20 KIAS) should provide adequate safety margins for both the approach and the landing roll. Control the glide slope path to accomplish a touchdown on the desired touchdown point. Fly the airplane firmly toward the runway, keeping the aim point even if the approach speed will be overshoot. If an unsatisfactory approach will result in a touchdown far down the runway, go around and make another approach.



Once the airplane has been landed and the deceleration effort commenced, attempting a go-around is not recommended.

If the wing anti-ice system is inoperative, and large ice formations remain on the wing leading edge, 30 knots must be added to the reference speed to maintain normal handling characteristics (Refer to Chapter 5; ABNORMAL PROCEDURES – ICE AND RAIN PROTECTION).

A. Crosswind Landings

A slippery runway and a crosswind are obviously a bad combination. When landing on a wet runway, a 25 knot crosswind is generally considered maximum. If the runway is known to be slippery (snow or ice covered), a crosswind component of 15 knots should be the maximum.

In crosswind conditions, the crosswind crab angle should be maintained for as long as possible, until prior to touchdown. Aim for the centerline or slightly on the upwind side, and avoid touching down on the downwind side of the runway, because of the possibility of the airplane weathervaning towards the wind after touchdown, and drifting toward the downwind side of the runway.

B. Touchdown

After touchdown, the early employment of all means of decelerating the airplane (particularly the thrust reversers) cannot be overemphasized. The use of rudder pedals for steering at high speeds is recommended, and use the nosewheel steering tiller with great care. Apply a slight forward pressure on the control column, to increase main gear loading and improve directional control.

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6. LANDING (CONT'D)

If the airplane starts to skid or drift gradually, it may be necessary to move the thrust levers out of reverse thrust, and to go to forward idle thrust to recover the centerline. It may also be necessary to reduce the brake pressure, to regain control and re-establish alignment with the centerline. For additional procedures and techniques during the landing roll, refer to the applicable procedures mentioned in the Cold Weather Operations supplement in this chapter.

NOTE

Pilots should seat themselves so as to ensure that they can achieve maximum braking with full rudder pedal deflection in either direction.

C. Use of Reverse Thrust

The intensity and duration of reverse thrust used at low speeds should be minimized. Using reverse thrust at low speeds, on snow and ice covered runways, can cause foreign object damage (FOD) to engines, increase the possibility of loss of forward visibility due to “whiteout”, as well as increase the possibility of ice build-up on the wing and empennage sections, which could cause considerable delays specially during quick turn-around schedules.

It should be realized, however, that maximum reverse thrust may be used up to a full stop during emergency situations, or if the safety of the airplane will be jeopardized.

The use of reverse thrust during crosswind conditions may aggravate possible directional control problems encountered during such conditions. If the airplane is allowed to weathervane into the wind, the reverse thrust side force component will add to the crosswind component, drifting the airplane towards the downwind side of the runway at a faster rate than normal. To correct the situation, it will be necessary to reduce the reverse thrust to reverse idle and release the brakes. In extreme conditions, it may even be necessary to move the thrust levers out of reverse thrust and to go to forward idle thrust. Use rudder, steering and differential braking, as required, to prevent overcorrecting past the runway centerline. When re-established on the runway centreline, reapply steady brakes and reverse thrust, as required, to stop the airplane.

Do not attempt to turn off from a slippery runway until the speed is reduced to a safe level, to prevent skidding. Anticipate low friction when approaching the touchdown zone at the far end of the runway. The touchdown zone may be very slippery when wet, due to heavy rubber and oil deposits.

For the corrected landing distance and performance data, refer to Chapter 6; PERFORMANCE – LANDING PERFORMANCE of the Airplane Flight Manual.

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BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Operation on Contaminated Runways	Vol. 1	06-13-6
		Oct 05/06	

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1. GENERAL

Extremely high temperature conditions present problems to airplane operations of a different nature than those associated with cold weather operations. The main concerns focus primarily on passenger and crew comfort, and the significant decrease in airplane performance which high ground temperatures can effect. The following recommended procedures have been provided to supplement the normal operating procedures, and should be observed, as applicable.

2. PRE-FLIGHT PREPARATION

NOTE

During airplane operations when the flight compartment and cabin temperatures are above 30°C (86°F), the air-conditioning packs or the low pressure ground conditioned air supply must be operating in order to maintain display temperatures within a range that prevents a display shutdown.

The importance of keeping the interior of the airplane as cool as possible cannot be overemphasized. All entrances and access doors to the airplane should be kept closed as much as possible, and the cargo bay door should not be left open any longer than is necessary. Every effort to reduce the heat being generated in the flight compartment while the airplane is on the ground should be made.

The flight crew should see to it that the following procedures are performed:

- (1) APU bleed air pressure Check
 - (2) BLEED AIR, APU LCV switch/light Press in to open APU load control valve.
Check APU LCV OPEN light on.
- Then:**
- (3) BLEED AIR, 10TH STAGE ISOL switch/light Press in to open ISOL valve.
Check 10TH ISOL OPEN light on.
 - (4) BLEED AIR, 10TH STAGE, L and R switch/lights Press out to close bleed air valves.
Check L and R 10TH STAGE CLOSED lights on.
 - (5) AIR-CONDITIONING, L and R PACK switch/lights Press in to select both packs on.
Check L and R PACK OFF lights off.

NOTE

If possible both air-conditioning packs should be used for maximum cooling. If using left pack only, close 10TH STAGE ISOL valve.



2. PRE-FLIGHT PREPARATION (CONT'D)

(6) AIR-CONDITIONING, CKPT
TEMP and CABIN TEMP
selectors COLD

(7) ANTI-ICE, WSHLD/WIND L and R
switches OFF

NOTE

Normally, the L and R WSHLD/WIND switches are selected to HI or LOW during the Take-Off Check. However, when ambient temperatures are above 30°C (86°F), it is recommended to delay selection of the WSHLD/WIND switches until the After Take-Off Check. The **L (R) WSHLD HEAT** and/or **L (R) WINDOW HEAT** caution messages may still come on momentarily.

(8) Weather radar selectors OFF

(9) Non-essential avionics equipment Off

(10) All air outlets and gaspers Open

(11) Cabin gaspers Open

(12) Window shades Closed On the sun-exposed side of the passenger cabin.

3. TAXI-OUT AND TAKE-OFF

A. Taxi-Out

Operation in areas of high ambient temperatures may cause brake temperature limits to be exceeded. This condition would lead to the fusible plug(s) melting, resulting in the deflation of the tire(s) involved. When operating on runways and taxiways exposed to high temperatures, considerations for brake cooling should always be taken into account. These areas usually maintain temperatures well above the ambient, therefore, excessive use and riding of brakes should be avoided. Braking should be done to a minimum so as to allow the airplane to accelerate, then brake to a very slow taxi speed, and release the brakes completely.

Reverse thrust may be used, at the pilot's discretion, to control the taxiing speed of the airplane without the use of braking. The use of single (or both) reverse thrust should be considered when operating under the following conditions:

- High ambient temperatures
- Heavy braking after landing
- Downslope taxi
- Tailwinds
- Light gross weight
- Any combination of the above.

The application of reverse thrust should be done with caution, taking into account the surface condition of the movement area. The use of thrust reversers during taxi on airports with contaminated runways and taxiways (i.e. dirt, dust, loose debris, etc.) is not recommended.

3. TAXI-OUT AND TAKE-OFF (CONT'D)

B. Take-Off

High ambient temperatures in combination with short runways or high elevation airports could cause substantial penalties on airplane performance. These penalties should be taken into account early in the pre-flight preparation and the limiting effects to the payload which can be carried, seriously considered. The use of alternate take-off procedures (e.g. Zero-Bleeds Take-off, etc.) may be necessary under such conditions. Review the various airplane, powerplant and fuel temperature limitations in the applicable sections of the Airplane Flight Manual. Use the longest runway available, if possible.

4. LANDING

A. Landing

The landing phase of flight during high ambient temperature conditions is not as demanding as with cold weather operations. It should be noted, however, that runway conditions, particularly at the touchdown area, could be slick due to the heavy deposits of rubber and oil which have melted. The occurrence of hydroplaning is not remote, and the procedures mentioned in the 'Landing' portion of the "Operation on Contaminated Runways" supplement in this chapter can be used.

This condition also holds true at the opposite approach area at the end of the runway in use, therefore, slow down to a manageable speed before turning off from the runway, to preclude skidding and departing from the intended path.

Always opt for the longest runway available for landing, if possible.

B. Brake Cooling

For quick turnaround operations, bear in mind that the energy absorbed by the brakes following each landing is accumulative. This could prove detrimental and may cause considerable delays at stops, if it results in overheating of the brakes and the possible melting of the wheel fusible plug(s) resulting in the deflation of the tire(s) involved.

The brake cooling times established in the Airplane Flight Manual, Chapter 6; PERFORMANCE – TURN-AROUND TIME – Cooling of Wheels, Brakes and Tires, should be adhered to.

Every effort to maintain cool brakes should be attempted. Early extension of the landing gear prior to the approach is recommended. Additional in-flight cooling for every segment of the route should be sufficient enough to cool the brakes and the tires before landing.

The recommended deceleration technique after landing should be adhered to. Excessive braking should be avoided, and the thrust reversers should be used to their full advantage.

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BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Flight in Turbulence	Vol. 1	06–15–1 Oct 05/06
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1. GENERAL

Flight through known turbulence should be avoided as much as possible. During cruise, areas with known severe turbulence should be overflown or circumnavigated, if possible. In the departure or approach area, when severe turbulence or thunderstorms have been reported or observed, this may entail a delay of the take-off or the approach. If flight through turbulence is unavoidable, the following procedures are recommended and should be observed, as applicable.

2. TURBULENCE PENETRATION

Before entering an area of known turbulence, secure all loose equipment in the flight compartment. Shoulder harnesses should be worn by both flight crew, and checked for security.

The recommended procedures for flight in turbulence are as follows:

1. AUTOPILOT

- The autopilot may be used when flying through turbulence, and in most cases can be regarded as the primary means of aircraft control in turbulence. To engage the autopilot in the turbulence mode, select the turbulence (TURB) switch on the flight control panel (FCP). In the turbulence mode, the autopilot gains are reduced to compensate for, and provide a smoother flight during, turbulent conditions. When the autopilot is used, guard the AP/SP DISC button on the control wheel to permit immediate disengagement should the need arise.

2. AIRSPEED

- The best airspeed and flight configuration to use in severe turbulence is that which affords the best overall protection from inadvertent stall and high speed buffet while retaining structural integrity. Most turbulence encounters at high altitude are not severe and do not require an airspeed reduction. DO NOT fly less than the minimum maneuvering speed for the existing flight condition. The maximum recommended airspeed for turbulence penetration is 280 KIAS or 0.75M, whichever is less. (For maneuvering speeds, refer to the Airplane Flight Manual).
- Severe gusts or drafts will cause large and often rapid variations in IAS. Do not “chase” the airspeed in order to maintain the recommended speed.

3. ALTITUDE

- Large altitude variations are possible in severe gusts or drafts, but do not “chase” altitude. Allow it to vary (provided there is adequate terrain clearance), that is, sacrifice altitude in order to maintain the desired attitude.

4. ATTITUDE

- If the airplane is flown manually, hold wings level and desired pitch attitude using the flight director as the primary instrument. In extreme drafts, large attitude changes may occur. Do not make sudden large control inputs when making corrections; instead, use moderate inputs to resist changes in attitude. After establishing the trim setting for penetration, do not change stabilizer trim.

5. ENGINE

- Adjust thrust to maintain the recommended turbulence penetration airspeed as required for the phase of flight. Once power is set to maintain the desired airspeed, avoid further power changes. Change the thrust setting only in case of extreme airspeed variation. Remember that a transient increase is always more advisable than a loss in speed, which decreases buffet margins and is difficult to recover.

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2. TURBULENCE PENETRATION (CONT'D)

- Select CONT IGNITION to ON. This action will reduce the possibility of flameout due to engine instability. Once out of the area of turbulence, deselect CONT IGNITION unless required further.

6. FLAPS/SPOILERS

- Flaps extension in an area of known turbulence should be delayed for as long as possible, because the airplane can withstand higher gust loads in the clean configuration. If severe turbulence persists in the area, consider diverting to the alternate station.
- Whenever the flight spoilers are employed, keep hand on the flight spoilers lever unless it is necessary to remove hand in order to perform some other specific function.

7. PASSENGER/CABIN CREW CONSIDERATIONS

- Whenever possible, advise the passenger cabin before or shortly after take-off of anticipated enroute turbulence, so that they can plan their activities accordingly.
- When mild turbulence is encountered, turn on the 'Fasten Seat Belt' sign and advise the passengers over the PA system of light turbulence and request that they fasten their seatbelts.
- If other than mild turbulence is encountered, turn on the 'Fasten Seat Belt' and 'No Smoking' signs. Make an announcement over the PA system and advise the passengers of the expected rough air and request that they fasten their seatbelts snugly. After passing through the area of turbulence, turn off the 'No Smoking' sign and make a PA announcement that the flight is clear of the area of turbulence. The 'Fasten Seat Belt' sign may then be turned off.

1. GENERAL

Flight operations in areas of known volcanic activity should be avoided. This consideration is most important during hours of darkness or daytime instrument meteorological conditions, when volcanic ash/dust clouds may not be visible.

Flight planning considerations should include the review of pertinent NOTAMs, PIREPs and other directives concerning the status of volcanic activity, when a flight is planned into areas of possible volcanic activity. When volcanic activity is currently reported, remain well clear of the area, or, if possible, stay on the upwind side of the volcanic ash.

2. DETECTION

The airplane's weather radar is not capable of detecting volcanic ash/dust clouds and is therefore not reliable under these circumstances. The presence of volcanic ash/dust may be indicated by:

- Smoke or dust appearing in the flight compartment.
- An acrid odor similar to electrical smoke.
- Multiple engine malfunctions such as power loss, fluctuating RPM, stalls, increasing ITTs, flameouts, etc.
- At night, static discharges (also known as Saint Elmo's fire or Saint Elmo's light) can be observed around the windshield and/or windows, accompanied by a bright orange glow in the engine inlets.

3. EFFECTS

Flight into volcanic ash/dust clouds can result in the degradation of airplane and engine performance. The adverse effects of volcanic ash/dust encounters are as follows:

- Rapid erosion and damage to the internal components of the engines.
- Ash/dust build-up and blockage of the guide vanes and cooling holes, which can cause surge, loss of thrust and/or high ITT.
- Ash/dust blockage of the pitot system, which can result in erratic airspeed indications.
- The abrasive properties of volcanic material can cause serious damage to the engines, wing and tail leading edge surfaces, windshields, landing lights, etc.
- Volcanic ash/dust can also cause the windshield to become translucent, effectively reducing visibility.

4. CORRECTIVE ACTIONS

The best course of action to take is still avoidance. However, volcanic ash/dust clouds may sometimes extend for hundreds of miles and an encounter could be unavoidable. As previously stated, volcanic ash/dust can cause engine malfunctions, and the need to exit the area as quickly as possible cannot be overemphasized.

If an inadvertent encounter is experienced, the following procedures are recommended:

If altitude permits:

- (1) Thrust levers..... IDLE

NOTE

Moving thrust levers to idle will maximize the engine stall margin, lower the ITT and reduce the build-up of volcanic material on the turbine vanes.

- (2) ANTI-ICE, L and R COWL switch/lights Press in to select cowl anti-ice ON.
- (3) ANTI-ICE, WING switch Select to NORM.

NOTE

Using anti-ice will increase bleed air extraction from the engines and further improve the engine stall margin.

- (4) ITT Monitor to ensure that limits are not exceeded.

If ITT continues to increase:

Affected engine:

- (5) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5; ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown.

NOTE

- If it becomes necessary to shutdown an engine to prevent exceeding ITT limits, restart engine once it has cooled down.
- If the engine fails to start, repeated attempts within starter limits should be made immediately.
- A successful start may not be possible until the airplane is clear of the volcanic ash/dust, and the airspeed and altitude is within the airstart envelope. The engines can be very slow to accelerate to idle at high altitude, and this could be interpreted as a failure to start, or as an engine malfunction.

After exiting area of volcanic ash/dust cloud and engine(s) restarted:

- (6) Affected systems Restore
- (7) ATC Notify

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Windshear	Vol. 1	06–17–1 REV 40, Aug 30/16
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1. GENERAL

Windshear is a weather phenomenon of sudden windspeed and/or direction changes over a short distance. The “downburst” is the most dangerous type of windshear, and recent studies have confirmed the existence of a more hazardous yet smaller-scale form of downburst, known as a “microburst”.

Microbursts are characterized by sudden intense downdrafts, which spread outwards from the downflow centre upon reaching the surface, causing both vertical and horizontal wind shear activity. It varies in size, sometimes extending to about 1 mile (1.6 km) in diameter at 2,000 feet AGL, and ranges typically from 1 to 2 1/2 miles (1.6 to 4 km) horizontally. The vertical winds could be as high as 6,000 feet per minute, with horizontal winds running up to approximately 45 knots at the surface.

The duration of microbursts, from the initial downburst to dissipation, seldom exceeds 15 minutes, with the maximum intensity winds lasting for about 2 to 4 minutes. Multiple microbursts have been known to occur in the same general area, and tend to take a line structure such that the downburst activity could last for an hour, or even longer. Once microburst activity starts, be prepared for further windshear encounters, because they can occur in groups.

2. DETECTION

There is no established method in predicting or forecasting when and where windshear can occur. There are certain conditions, however, that the flight crew can look for, which would indicate the possible existence of windshear along the flight path, such as:

1. Thunderstorms accompanied by heavy rain (where the air is very humid).
2. The presence of ‘virga’ (rain that evaporates before reaching the ground, usually in drier air).
3. Frontal activity.
4. Low level jetstream.
5. A combination of the following conditions:
 - Extreme variations in wind speed and/or direction in a relatively short time span.
 - Evidence of a gust front, such as blowing dust over the airport surface.
 - Surface temperatures higher than 30°C (86°F).
 - Dew point spread of 4°C (7.2°F) or more.

Strong microbursts can sometimes occur without precipitation and even during sunny weather conditions. Windshear can also be caused by strong surface winds in the vicinity of small hills or large buildings, and by sea breeze fronts.

Additionally, the following procedures are recommended to allow for the detection of possible windshear activity:

1. Be aware of windshear PIREPs.
2. Determine the presence of thunderstorm cells in the approach or take-off areas of the airport, using the weather radar.
3. Use the Low Level Windshear Alerting System (LLWAS), if available.
4. Exercise frequent flight instruments scanning.

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3. PRECAUTIONARY ACTIONS

Avoidance is still the best course of action to take, in the event that the occurrence of windshear is known or suspected. Stay clear of thunderstorm cells, heavy precipitation, virga and areas of known windshear.

Flight crew awareness, preparation and coordination cannot be overemphasized. A thorough knowledge of, and familiarity with, the phenomenon and the various meteorological conditions indicative of windshear activity is necessary, if one is to be prepared for the possibility of an inadvertent encounter. Pilots should be trained adequately to enable them to readily recognize the indications and/or occurrence of windshear/downburst and take proper action.

When windshear activity is known or suspected at departure:

1. Delay the take-off until conditions are more favorable.
2. Assess the situation during the delay and ensure that a safe take-off is possible.
3. Select the longest runway length available.
4. During the take-off roll, carefully monitor the flight instruments, particularly for airspeed fluctuations, to detect at the earliest time the possible occurrence of windshear.
5. Be familiar with the normal (all-engine) initial climb pitch attitude, and rotate the airplane at the normal rate to this attitude. Reduction of the pitch attitude should be delayed until terrain and/or obstruction clearance is assured, or when the stick shaker is activated.
6. A thorough knowledge of the normal take-off indications of the flight instruments, in particular those pertaining to the vertical flight path, is a must. Closely monitor the flight instruments. Any deviation from the normal values of airspeed, airspeed build-up, attitude, vertical speed, or altitude trend should be called out by the pilot-not-flying.
7. If windshear is encountered near V_R and the airspeed suddenly decreases, there may be not enough runway length left to accelerate back to the normal V_R . If the runway remaining is not sufficient to reject the take-off, rotate the airplane at a normal rate at least 2,000 feet before the end of the runway, even if the speed is below V_R .
8. Follow the windshear recovery procedures.
9. Remain prepared for further windshear encounters.

When windshear activity is known or suspected at arrival:

1. Delay the approach until conditions are more favorable, or divert to another airport.
2. Use the longest and most favorable runway available.
3. Add the appropriate crosswind and gust corrections to the final approach speed.
4. Fly an "on the slot" approach, and consider the use of the autopilot during the approach, to allow for more monitoring and recognition time.
5. Be aware of the normal values of the flight instruments during the approach, and constantly cross-check flight director commands with the vertical flight path instruments.
6. The pilot-not-flying should closely and constantly monitor the flight instruments, paying particular attention to the vertical flight path instruments (i.e. vertical speed, altitude, glideslope and airspeed), and call out any deviation from the normal.
7. Establish crew coordination and awareness, especially at night or in marginal weather conditions, and be alert for the possibility of an inadvertent encounter.
8. DO NOT make large thrust reductions or trim changes to correct fluctuations in airspeed, for they could vary from an increase to a decrease in a very short span of time.

3. PRECAUTIONARY ACTIONS (CONT'D)

9. If windshear is encountered during the approach, and a significant reduction and/or increased rate of descent is experienced, immediately execute a go-around and follow the windshear recovery procedures.
10. Trade the airspeed for altitude. Maintain the go-around pitch attitude until terrain and/or obstacle clearance is assured, unless the stick shaker is activated.
11. Anticipate the encounter of subsequent windshear activity along the flight path.

If windshear activity was encountered during take-off or approach, promptly inform ATC of the encounter. The PIREP should include:

- Location where windshear was encountered,
- Altitude where windshear was encountered,
- Airspeed fluctuations experienced (speed gained or lost), and
- Type of aircraft.

4. RECOVERY PROCEDURES

A. Windshear Warning

Windshear warnings are indicated as follows:

- (Siren) “WINDSHEAR, WINDSHEAR, WINDSHEAR” aural.
- WINDSHEAR warning message on PFD.

NOTE

The autopilot (if engaged) will automatically disconnect, and the flight director bars will be removed within two (2) seconds after a windshear warning.

During take-off, approach or landing:

- | | | | |
|---|--------------------------------|---------|--|
| I | (1) ATS DISC switch..... | Depress | |
| I | (2) Thrust levers..... | Advance | to maximum take-off thrust. |
| I | (3) FLIGHT SPOILER lever | Select | to RETRACT. |
| | (4) Airplane | Rotate | smoothly to go-around/take-off pitch attitude, and allow airspeed to decrease, if necessary. |

NOTE

Stick shaker must be respected. Reduce pitch attitude sufficient to stop shaker.

If airplane is descending:

- | | | |
|-------------------------|----------|---|
| (5) Pitch attitude..... | Increase | smoothly and in small increments, reducing airspeed as necessary to stop descent. |
|-------------------------|----------|---|



4. RECOVERY PROCEDURES (CONT'D)

A. Windshear Warning (Cont'd)

- (6) Airplane configuration..... Maintain until positively out of windshear condition:
- Do not change flap, gear or trim position until WINDSHEAR warning message is removed and above 1,500 feet AGL.

WARNING

If ground or obstacle contact is imminent, apply maximum forward throttle travel.

B. Windshear Caution

Windshear cautions are indicated as follows:

- WINDSHEAR caution message on PFD.

During take-off, approach or landing:

- (1) Thrust levers/TOGA switch Advance/press
- Manually advance thrust levers to maximum take-off thrust, while simultaneously pressing either take-off or go-around (TOGA) switch and ATS DISC switch.
- (2) Normal go-around Initiate

1. TERRAIN AWARENESS AND WARNING SYSTEM

A. TAWS Warning During Flight

TAWS Warnings are indicate as follows:

- “TERRAIN, TERRAIN, PULL UP”, “OBSTACLE, OBSTACLE, PULL UP” and “PULL UP” aural.
 - **PULL UP** warning message on the PFDs.
- (1) ATS DISC switch Depress
 - (2) Autopilot Disconnect
 - (3) Thrust levers.....Advance to maximum take-off thrust.
 - (4) FLIGHT SPOILER leverSelect to RETRACT.
 - (5) Immediately initiate a smooth pull up.
 - (6) Landing gear Retract
 - (7) Maintain climb until warning ceases and it is confirmed that a safe ground clearance exists.

B. TAWS Caution During Flight

TAWS cautions are indicated as follows:

- “CAUTION TERRAIN”, “CAUTION OBSTACLE”, “TERRAIN”, “TOO LOW TERRAIN”, “TOO LOW GEAR”, “TOO LOW FLAPS”, “SINK RATE”, “DON’T SINK” OR “GLIDESLOPE” aural.
 - GND PROX message on the PFDs.
- (1) Airplane flight path or configuration..... Correct

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2. TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM

A. TCAS Resolution Advisory During Flight:

Resolution advisories are indicated as follows:

- Vertical maneuver commands (red/green arc) on vertical speed indicator,
- Threat level symbology (red square symbol) on multi-function display, including closure rates and relative altitude, and
- Vertical maneuver voice aural.



Compliance with a TCAS resolution advisory is necessary unless the pilot considers the maneuver unsafe to accomplish, or unless the pilot has accurate information about the cause of the resolution advisory and can maintain safe separation from a nearby airplane.

NOTE

The TCAS RA algorithms are based on the pilot initiating the initial maneuver within approximately 5 seconds, and if an additional corrective RA, for example, increase or reversal, is issued, the maneuver must be initiated within approximately 2 1/2 seconds.

The consequences of not following an RA may result in additional RAs in which aural alert and visual commands may not agree with each other.

During flight:

- (1) Autopilot Disconnect
 - (2) Applicable maneuver Accomplish immediately, adjusting engine thrust as required.
- If an additional maneuver is indicated (e.g. a reversal), it also must be accomplished immediately.



Once an RA maneuver has been initiated, the maneuver must be continued until a "CLEAR OF CONFLICT" aural comes on.

If stabilized in approach:

- (1) Autopilot Disconnect
- (2) Go-around Initiate



2. TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (CONT'D)**A. TCAS Resolution Advisory During Flight: (Cont'd)**

If RA is accompanied with a TAWS or STALL warning:

- (1) Applicable RA maneuver Discontinue
- (2) Applicable recovery procedure
(GPWS or STALL) Accomplish

When TCAS “CLEAR OF CONFLICT” aural comes on:

- (3) Airplane Re-position to assigned altitude.

B. TCAS Traffic Advisory During Flight:

Traffic advisories are indicated as follows:

- Threat level symbology on multi-function display, including closure rate and relative altitude symbols and
- “TRAFFIC – TRAFFIC” voice aural.

NOTE

1. The pilot should not initiate evasive maneuvers using information from the traffic display only, or on a traffic advisory (TA), without visually sighting the traffic.
2. These displays and advisories are intended only for assistance in visually locating the traffic, and lack the flight path trends necessary for use in evasive maneuvering. However, while climbing or descending, modest changes in vertical speed, based on traffic display information, is not considered evasive maneuvering.

- (1) Display control panel, TFC switch Press to select TCAS traffic display on MFD.

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TRAINING PURPOSES ONLY

1. INTRODUCTION

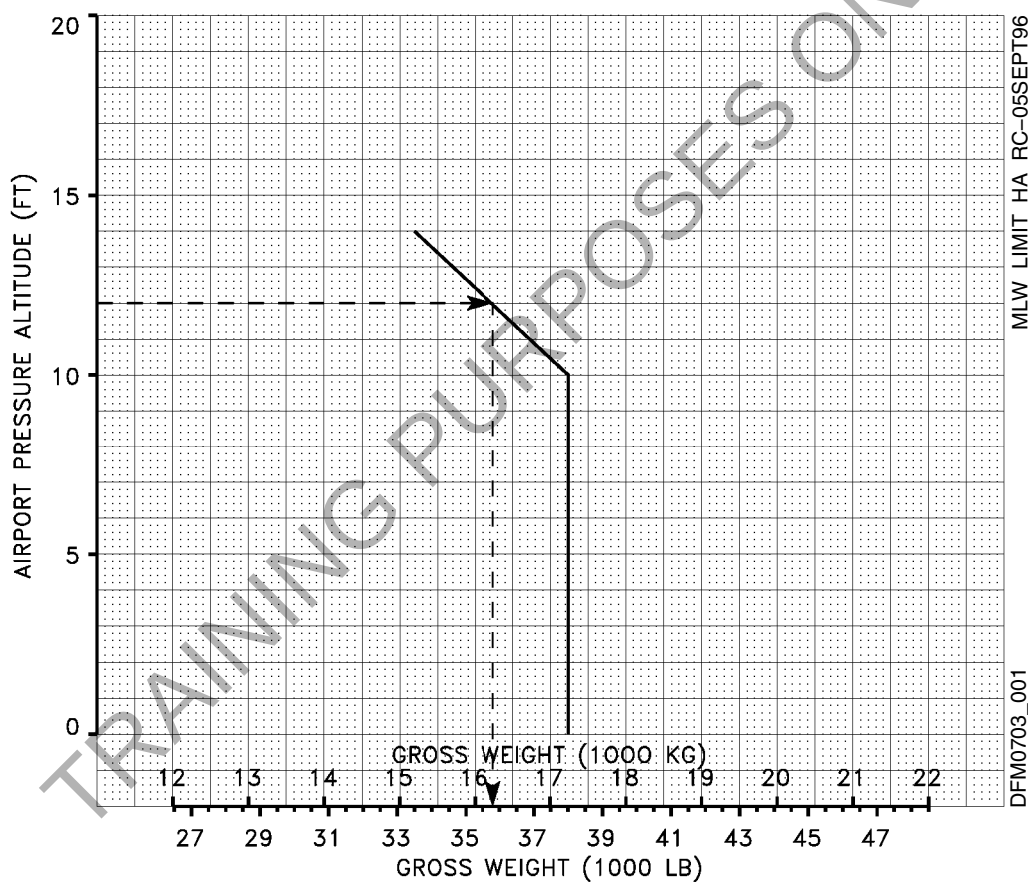
These data must be used on the CL 600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049) when operating from airport pressure altitudes above 10,000 feet up to 14,000 feet. These data complement or supersede some data contained in the basic Flight Crew Operating Manual. This Supplement must, therefore, be read in conjunction with the basic Flight Crew Operating Manual.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below:

2. OPERATING LIMITATIONS

The operating limitations in Chapter 2 are applicable, except as modified by the following:

The maximum landing weight limit is shown in Figure 06-19-1.

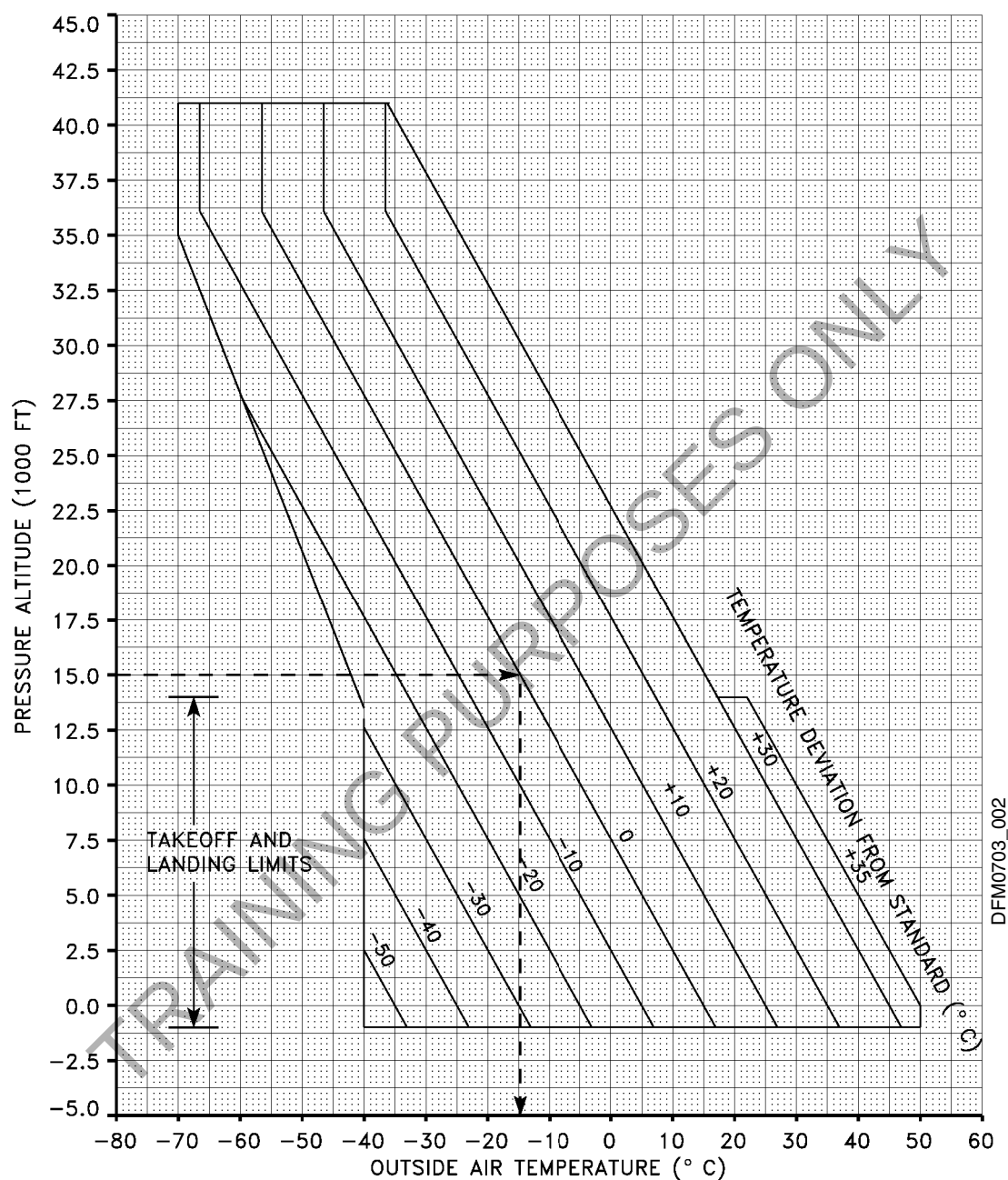


Maximum Landing Weight Limit
Figure 06-19-1

2. OPERATING LIMITATIONS (CONT'D)

The maximum altitude and temperature operating limits are shown in Figure 06-19-2.

- Maximum airport pressure altitude for take-off and landing is 14,000 feet.



Altitude and Temperature Operating Limits
Figure 06-19-2

3. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

4. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

A. Engine Start

The engine start procedures in the basic FCOM are applicable, in addition to the following:

NOTE

If difficulty is experienced in starting a cold engine using the APU, due to slow acceleration, start the other engine and perform a crossbleed start on the affected engine. If required, repeat sequence to obtain a successful start and perform a crossbleed start for the opposite engine.

B. Climb/Descent Rate

With air-conditioning packs selected on and cabin pressurization rate set at PIP position, the minimum time required to climb to 25,000 feet or to descend from 25,000 feet without causing discomfort to passengers is as follows:

AIRFIELD ELEVATION	MINIMUM TIME
14,000 feet	10.0 minutes
13,000 feet	8.4 minutes
12,000 feet	6.7 minutes
11,000 feet	5.0 minutes
10,000 feet	3.4 minutes

NOTE

1. During take-off, the minimum time may be reduced by selecting a higher cabin pressurization rate.
2. During climb, do not exceed 25,000 feet until cabin pressure altitude has decreased to 8,000 feet or less.

C. Take-Off

Before take-off:

- (1) CABIN PRESSURIZATION, LDG

ALT selector Set the following:

- Destination airfield altitude,
- Desired cabin pressure rate of change, and
- Barometric pressure.



4. NORMAL PROCEDURES (CONT'D)

C. Take-Off (Cont'd)

- (2) Oxygen masks..... Don Crew will don oxygen masks and establish communications.

NOTE

1. The use of crew oxygen is required in accordance with the applicable operating regulations.
2. Carry out a high crosswind take-off procedure if there is an audible engine noise accompanied by engine indication instabilities, when setting take-off power.

When cabin altitude is below 10,000 feet:

- (3) Oxygen masks..... Remove

D. Descent

- (1) CABIN PRESSURIZATION, LDG
ALT selector Check selected to the following:
- Destination airfield altitude.
 - Destination airfield barometric pressure.
 - Desired cabin pressure descent rate.
- Check FAULT lamp out.

E. Landing

NOTE

The cabin altitude indicator will be in the red area when cabin altitude is above 10,000 feet.

This indication is normal.

When cabin altitude is above 10,000 feet:

- (1) Oxygen masks..... Don Crew will don oxygen masks and establish communications.

NOTE

The use of crew oxygen is required in accordance with the applicable operating regulations.

Prior to landing:

- (2) Cabin altitude Check

Prior to opening passenger door:

- (3) Cabin differential pressure Ensure at zero.

5. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

A. Flaps Failure

Operation at high airport elevations increases the probability of exceeding the maximum tire speed. The landing weight resulting in a touchdown speed equal to the tire speed of 182 knots ground speed, for varying conditions of airport pressure altitude, temperature and wind is shown for the following flap configurations. Landings exceeding the weights derived will result in a tire speed greater than 182 knots ground speed at touchdown.

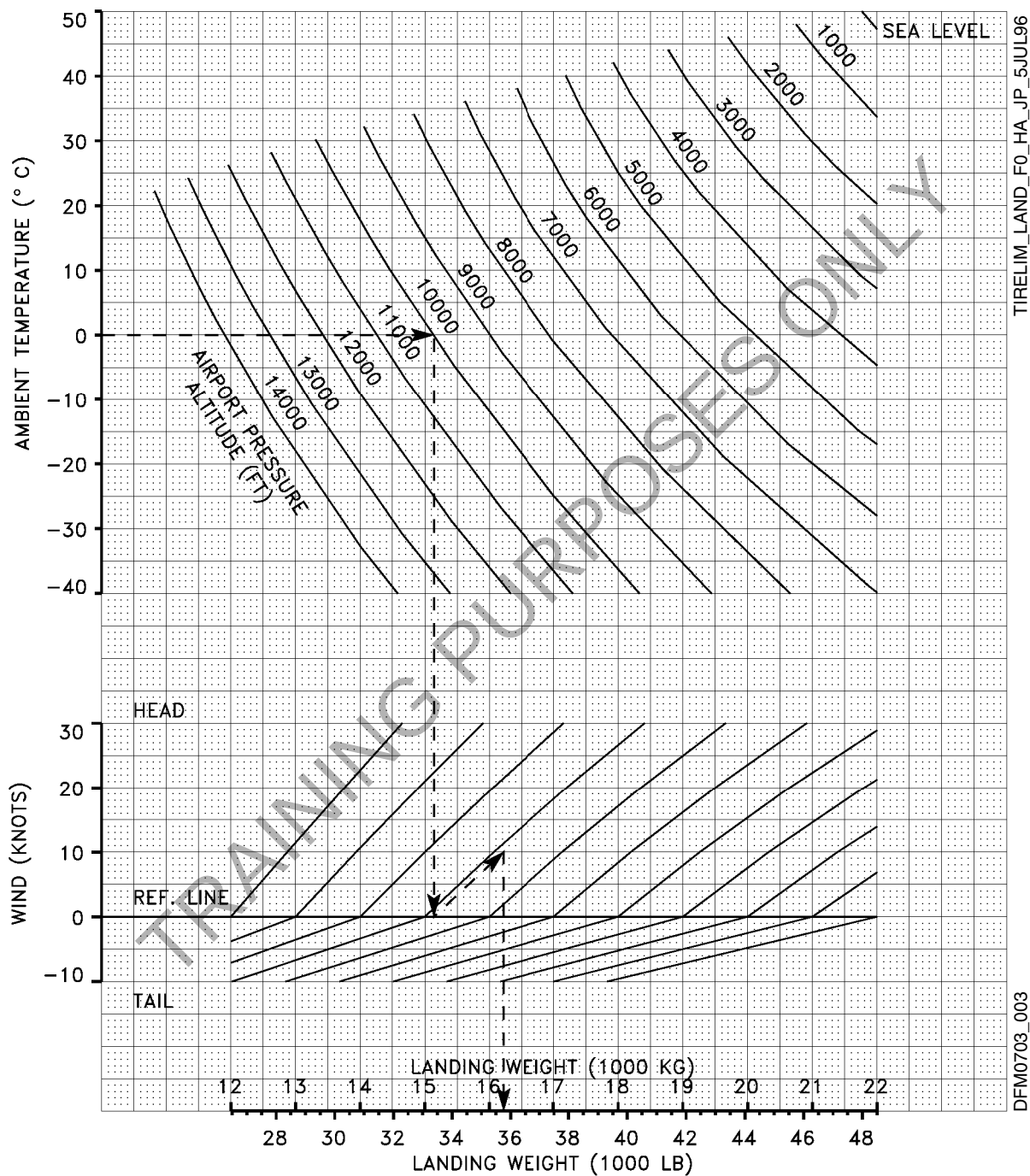
FLAPS 0°

Associated Conditions:

Temperature	= 0°C
Airport pressure altitude	= 10,000 ft
Wind	= 10 knots, headwind

The example in Figure 06–19–3 shows that for the above associated conditions, the landing weight limited by tire speed of 182 knots ground speed with flaps 0° is 16,200 kg (35,715 lb).

5. ABNORMAL PROCEDURES (CONT'D)

APPROACH SPEED = $V_{REF} + 30$ KIASLanding Weight Limited by Maximum Tire Speed, Flaps 0°
Figure 06-19-3

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Operation at High Airport Elevations	Vol. 1	06-19-7
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5. ABNORMAL PROCEDURES (CONT'D)

FLAPS 20°

Associated Conditions:

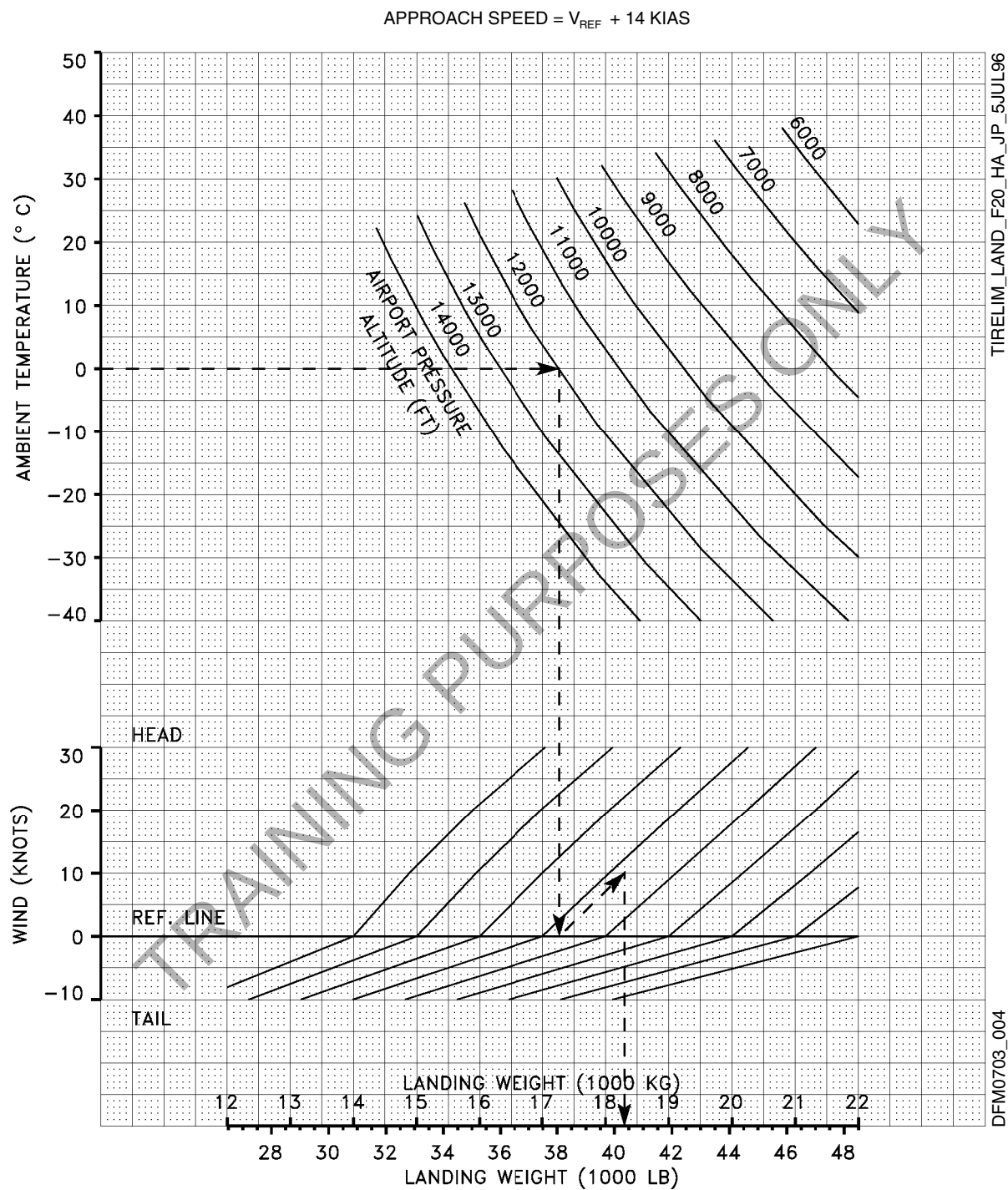
Temperature	= 0°C
Airport pressure altitude	= 12,000 ft
Wind	= 10 knots, headwind

The example in Figure 06-19-4 shows that for the above associated conditions, the landing weight limited by tire speed of 182 knots ground speed with flaps 20° is 18,300 kg (40,345 lb).

TRAINING PURPOSES ONLY

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5. ABNORMAL PROCEDURES (CONT'D)



Landing Weight Limited by Maximum Tire Speed, Flaps 20°
Figure 06-19-4

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5. ABNORMAL PROCEDURES (CONT'D)

FLAPS 30°

Associated Conditions:

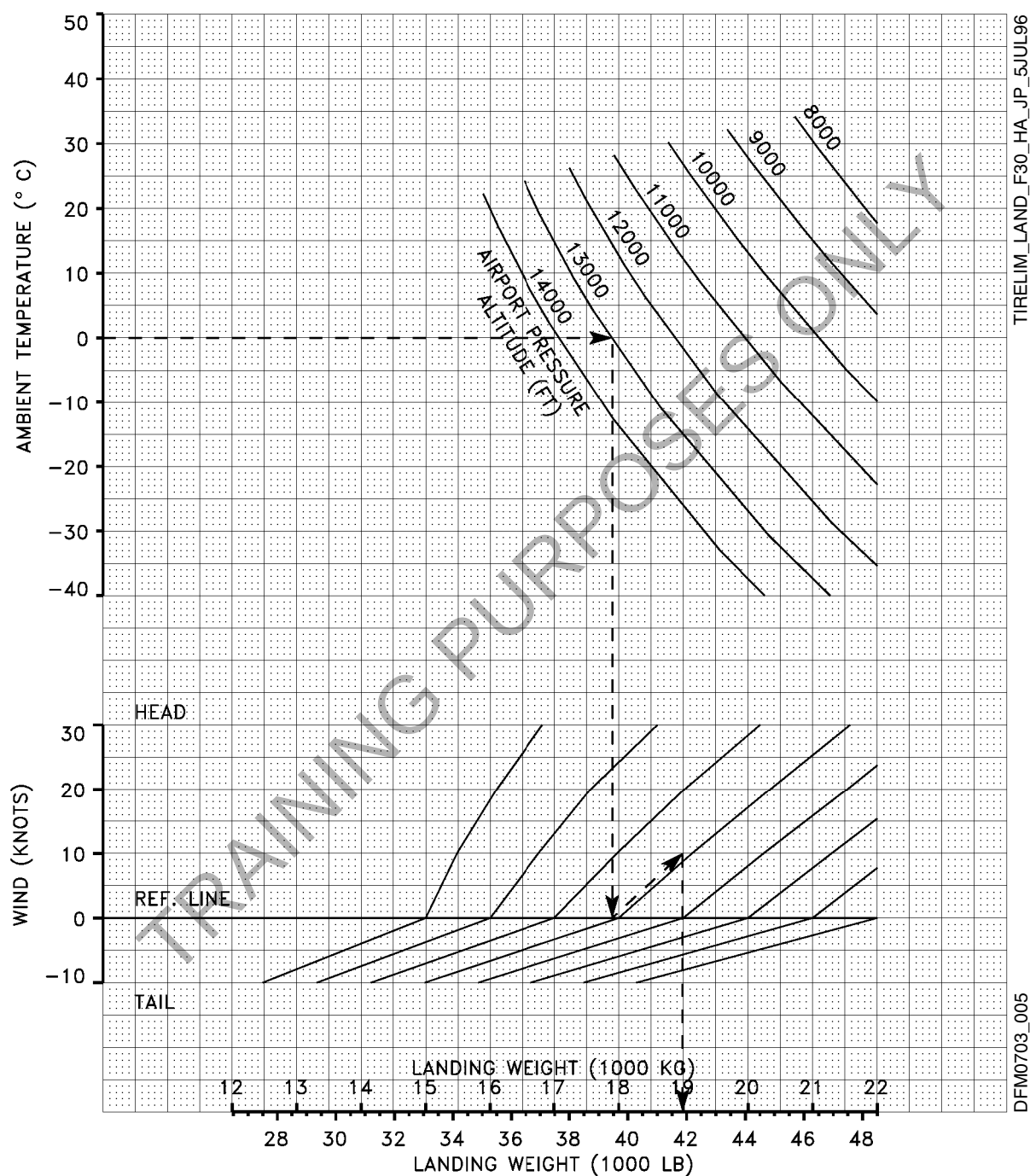
Temperature	= 0°C
Airport pressure altitude	= 13,000 ft
Wind	= 10 knots, headwind

The example in Figure 06-19-5 shows that for the above associated conditions, the landing weight limited by tire speed of 182 knots ground speed with flaps 30° is 19,000 kg (41,888 lb).

TRAINING PURPOSES ONLY

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5. ABNORMAL PROCEDURES (CONT'D)

APPROACH SPEED = $V_{REF} + 7$ KIASLanding Weight Limited by Maximum Tire Speed, Flaps 30°
Figure 06-19-5

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6. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual are applicable.

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TRAINING PURPOSES ONLY

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NOTE

These supplementary procedures are NOT applicable to airplanes incorporating Service Bulletin 605–34–019.

1. INTRODUCTION

These supplementary procedures must be used on the CL 600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049) when operating the airplane in Polar Regions.

These data complement or supersede data contained in the basic Flight Crew Operating Manual. These supplementary procedures must, therefore, be used in conjunction with the basic Flight Crew Operating Manual.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below.

2. GENERAL

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

A. Navigation Procedures

Navigation procedures identified below are NOT supported by the FMS 6000 in Polar Regions:

- Search and Rescue Patterns (SAR)
- Holding Patterns
- Terminal Functions (such as approaches and procedure turns)
- Radial and Abeam Based Fixes
- Place–Bearing/Distance
- Place–Bearing/Place–Bearing
- Along Track Waypoints with Reference Fix.

NOTE

Attempting to insert these procedures will result in a “N/A IN POLAR REGION” message on the CDU. For the purpose of these limitations, Polar Region is defined as latitude greater than 89° north or south.

B. Minimum Navigation Equipment

The following minimum navigation equipment is required for operations within Polar Regions:

- Dual FMS
- Dual IRS
- Dual GPS

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

A. MAG/TRUE Transition

The FMS automatically transitions from MAG to TRUE when operating in the region bounded by latitudes greater than 82 N, or when the latitude is greater than 73 N with a longitude between 80 W and 130 W, otherwise known as the northern latitude cutout region (keyhole). The FMS also automatically transitions from MAG to TRUE when operating in the region bounded by latitudes greater than 82 S, or when the latitude is greater than 60 S with a longitude between 120 E and 160 E, otherwise known as the southern latitude cutout region (keyhole). Whenever the aircraft is operating within the keyhole region, selection of MAG is not possible. MAG must be selected manually by the operator when the aircraft is outside of the keyhole region, before the FMS will transition back to MAG. Transition will NOT occur automatically.

NOTE

The pilot must follow the proper operational requirements for selection of MAG/TRUE modes. FMS automatic switching between MAG and TRUE modes may not correspond with domestic airspace requirements.

B. Polar Region

Select FMS as NAV source when entering the Polar Region.

NOTE

The EFIS compass cards on the PFDs may begin to deviate when approaching and transitioning the Pole and an **EFIS MISCOMP** caution message may be displayed and a boxed **HDG** may be posted on both PFD's.

The divergence and convergence of the HDG numbers displayed results from each display using its inside IRS sensor as its heading source. The split in IRS positions does not adversely affect FMS steering, flight directors, or cross track deviation as displayed on the HSI.

The compass information will normalize after transitioning the pole, and no pilot action is required.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

7. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable.

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1. GENERAL

These supplementary procedures must be used on the CL 600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049) when operating with fixed landing gear down.

These data complement or supersede data contained in the basic Flight Crew Operating Manual. These supplementary procedures must therefore be used in conjunction with the basic Flight Crew Operating Manual.

These supplementary procedures are applicable only when used in conjunction with a Minimum Equipment List approved by the appropriate authority.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. OPERATING LIMITATIONS

The operating limitations in Chapter 2 are applicable, except as modified by the following:

- Airspeed must not exceed V_{LE} (250 KIAS).
- Flight altitude above 20,000 ft is prohibited.
- Flight in known or anticipated icing conditions is not permitted.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

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7. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual are applicable, except as modified by the following:

A. Take-Off Performance

The take-off speeds (V_1 , V_R , and V_2) must be increased by 2 knots. The field length limited take-off weight must be reduced by 2%. For a given take-off weight, the take-off field length must be increased by 4%. The take-off weight limited by tire speed must be reduced by 4%. The take-off weight limited by maximum brake energy must be reduced by 2%.

B. Weight Altitude Temperature Limits

The following penalty must be applied to the performance data as a weight decrement:

FLIGHT SEGMENT	WEIGHT DECREMENT (%)
Take-off – Flaps 20°	16%
Landing – Approach Flaps 20° – Landing Flaps 45°	13%

The take-off weight is limited by the second segment climb gradient requirement (2.4% gross gradient – one engine inoperative). In landing, the limiting requirement is the approach climb gradient (2.1% gross gradient, one engine inoperative).

C. Climb Gradients and Climb Speeds

The following gradient penalty must be subtracted from the climb gradient data:

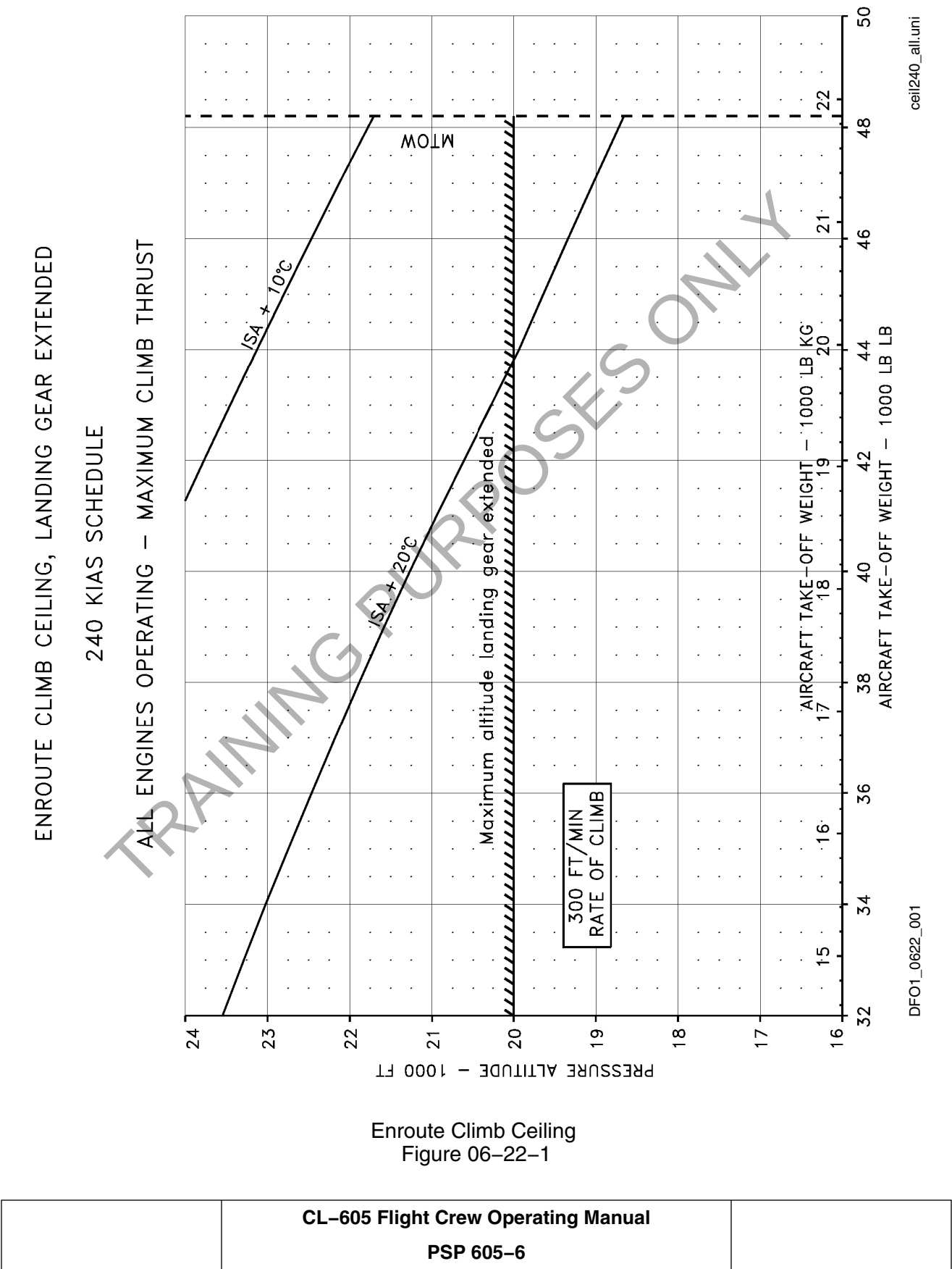
CLIMB GRADIENT	GRADIENT PENALTY (%)
Obstacle Clearance Reference Climb Gradient (Flaps 20°)	2.4%
Final Segment/Enroute (Flaps 0°)	3.3%
Approach (Flaps 20°)	2.4%
Landing (Flaps 45°)	Not applicable

The approach climb speed in the flaps 20° configuration must be increased by 2 knots.

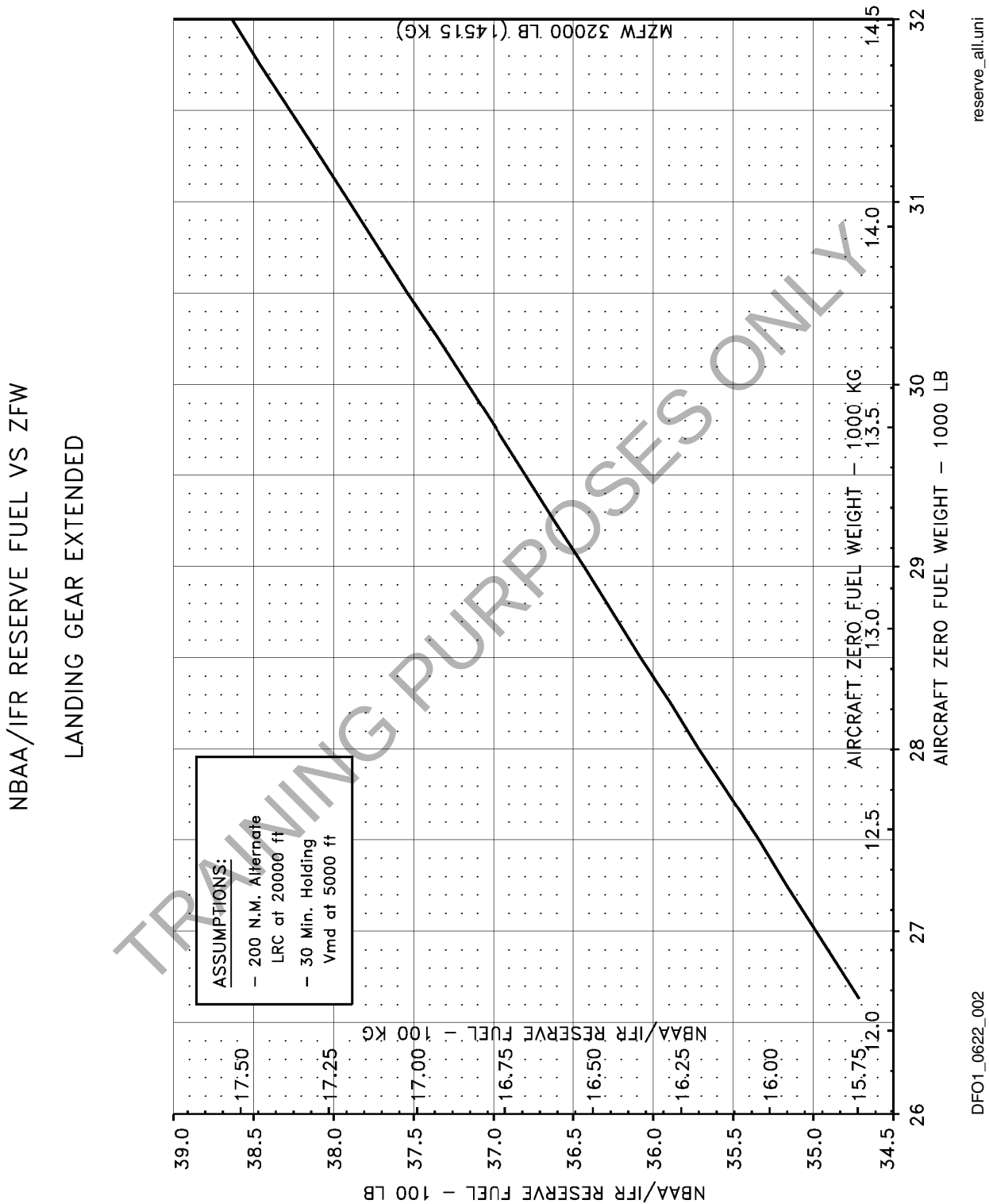
D. Operational Performance

The effect of landing gear down on operational performance is presented as follows. Enroute climb ceiling, fuel reserve, block fuel and block time charts are included as well as a table for long range cruise speed.

7. PERFORMANCE (CONT'D)

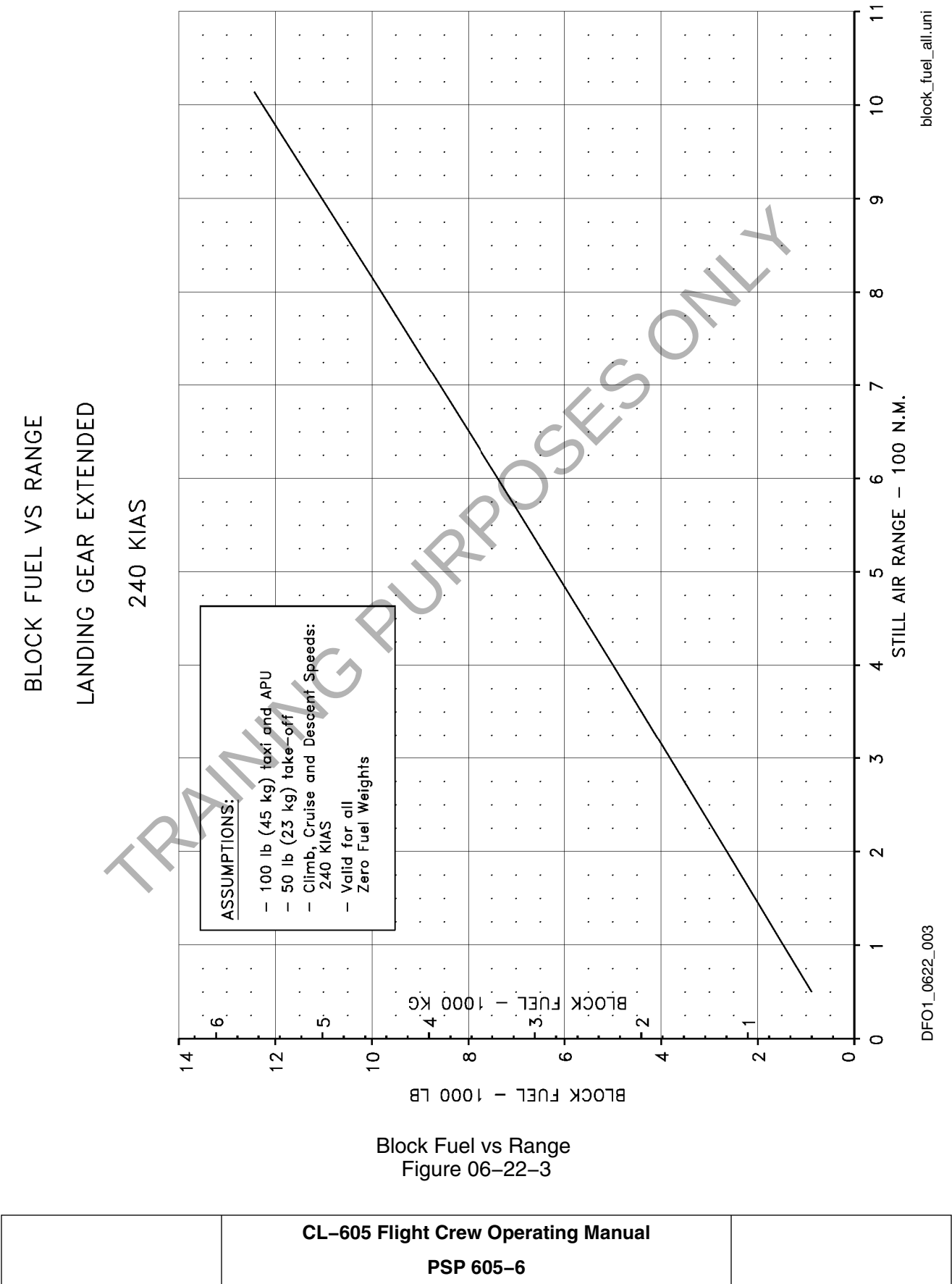


7. PERFORMANCE (CONT'D)

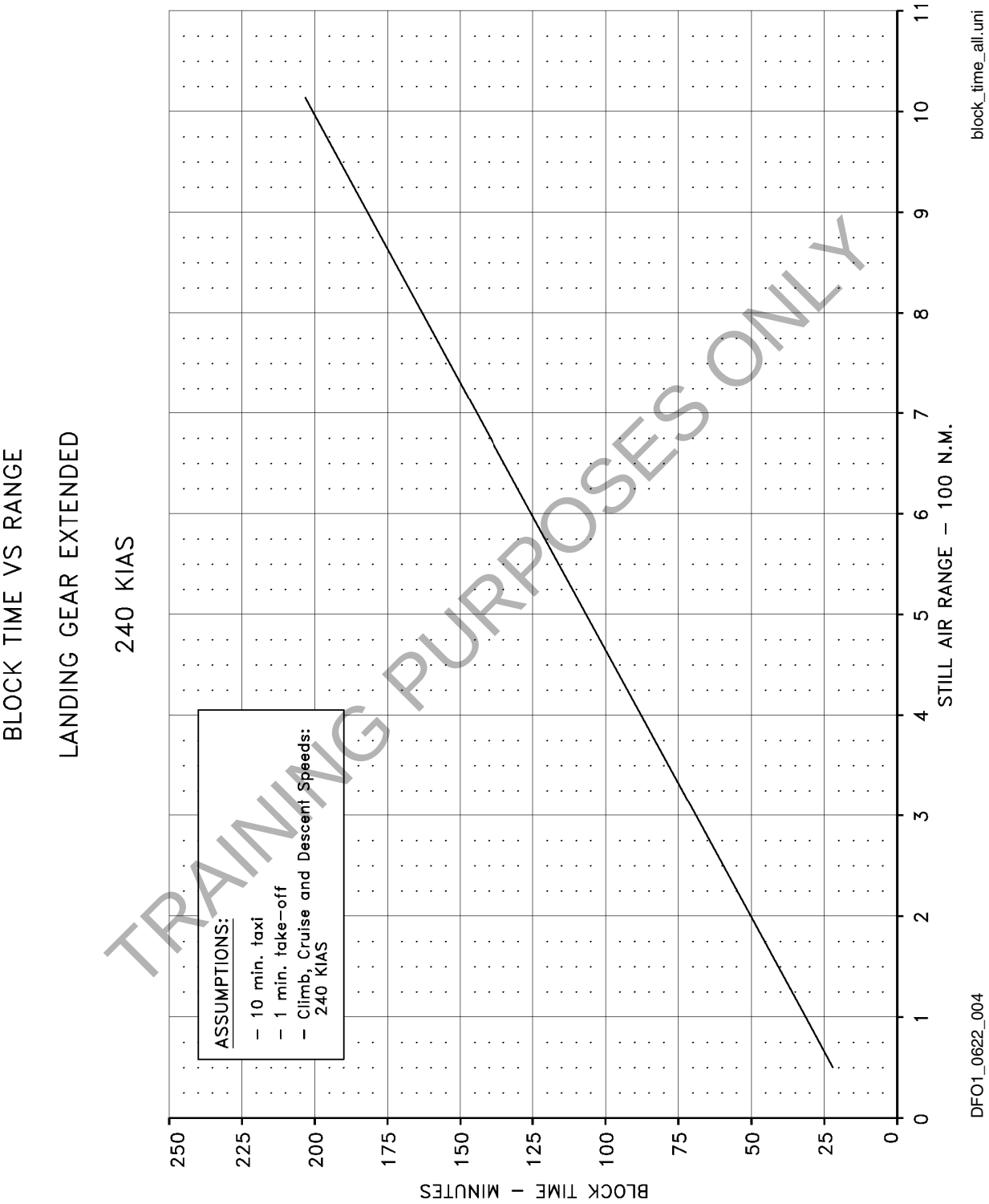


NBAA/IFR Reserve Fuel vs ZFW
 Figure 06-22-2

7. PERFORMANCE (CONT'D)



7. PERFORMANCE (CONT'D)



Block Time vs Range
 Figure 06-22-4

7. PERFORMANCE (CONT'D)

Long Range Cruise Speed (KIAS)

CANADAIR CHALLENGER CL-604 – G.E. CF34-3B ENGINES – LANDING GEAR EXTENDED																
Altitude ft	Weight – lb															
	26000	28000	30000	32000	34000	36000	38000	40000	42000	44000	46000	48000	50000			
0	179	181	184	186	189	193	198	201	204	207	210	213	216			
1500	176	178	182	184	188	192	196	199	203	205	209	212	215			
5000	170	174	178	181	185	188	192	196	199	203	206	214	221			
10000	167	170	173	177	181	185	188	195	202	206	211	220	226			
15000	161	166	170	174	185	189	193	198	204	209	214	221	225			
16000	161	166	170	175	182	188	193	200	204	210	215	220	225			
17000	161	165	171	180	183	188	193	201	204	210	214	221	226			
18000	160	166	172	179	183	188	195	199	204	210	216	222	225			
19000	160	166	175	179	184	190	194	198	205	211	217	221	224			
20000	160	167	175	180	186	187	191	201	206	213	216	221	227			

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Long Range Cruise Speed (KIAS)
Figure 06-22-5

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BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Operation with Systems Inoperative	Vol. 1	06-24-1 REV 36, Nov 27/15
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1. GENERAL

These supplementary procedures must be used on the CL 600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049) when conducting operations with some specific airplane systems inoperative.

These data complement or supersede data contained in the basic Flight Crew Operating Manual. This supplement must, therefore, be used in conjunction with the basic Flight Crew Operating Manual.

These supplementary procedures are applicable only when used in conjunction with a Minimum Equipment List approved by the appropriate authority.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. OPERATING LIMITATIONS

The operating limitations in Chapter 2 are applicable, except per MMEL or as modified by the following:

A. Hydraulic AC Motor Pump Inoperative

APR must be selected off, when dispatching with one hydraulic AC motor pump (1B or 2B) inoperative.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

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7. SUPPLEMENTARY PROCEDURES

The supplementary procedures in Chapter 6 are applicable, except as modified by the following:

A. Category II Operations (JAA/EASA Operators)

The maximum landing weight limited by climb requirements when dispatching with one hydraulic AC motor pump inoperative is determined from Figure 06-24-1.

Example: ANTI-ICE ON

Associated Conditions:

Temperature	= -8°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= On
10th stage bleeds	= Closed

As shown in the example (Figure 06-24-1, Anti-ice On), the maximum landing weight limited by climb requirements is 21,100 kg (46,518 lb).

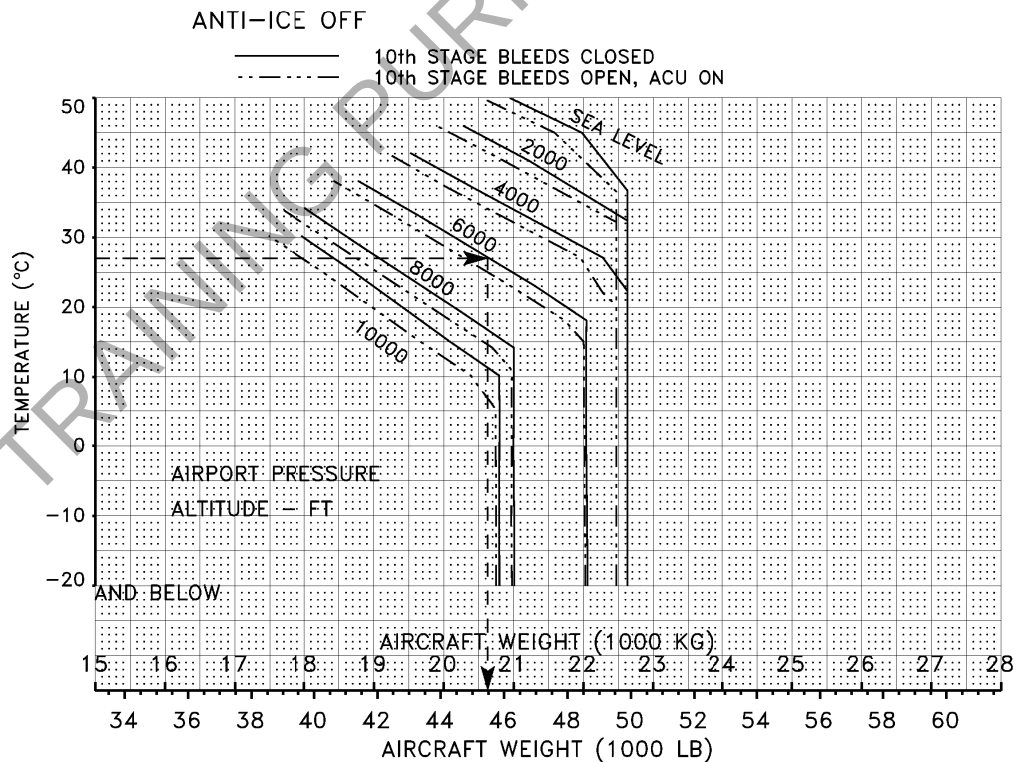
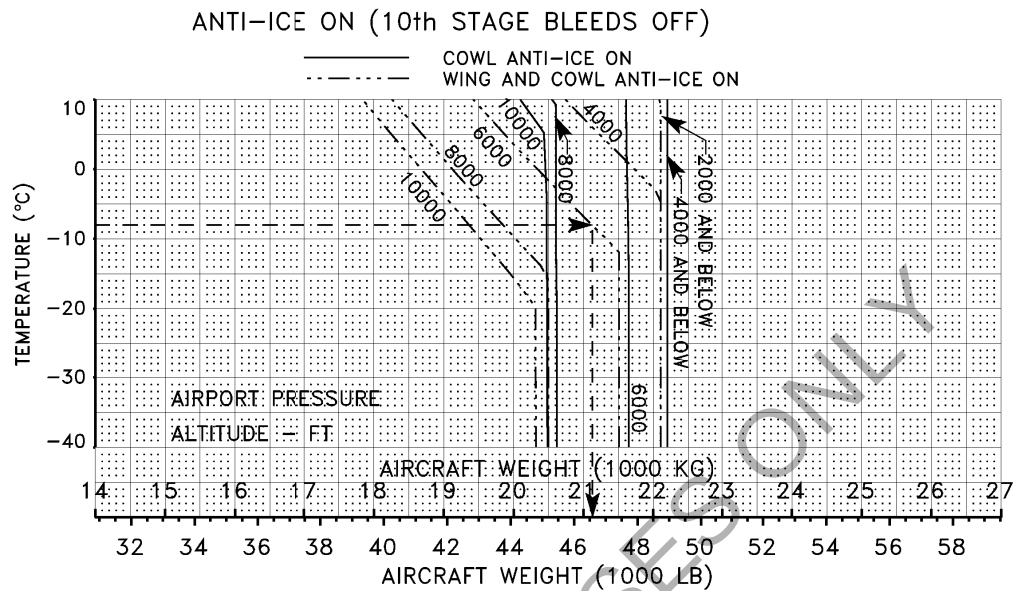
Example: ANTI-ICE OFF

Associated Conditions:

Temperature	= 27°C
Airport pressure altitude	= 6,000 ft
Wing and cowl anti-ice	= Off
10th stage bleeds	= Closed

As shown in the example (Figure 06-24-1, Anti-ice Off), the maximum landing weight limited by climb requirements is 20,600 kg (45,415 lb).

7. SUPPLEMENTARY PROCEDURES (CONT'D)



DFM0708_014

Landing Weight Limited by Climb Requirements – Category II Operations
Figure 06-24-1

7. SUPPLEMENTARY PROCEDURES (CONT'D)**B. Flight with Landing Gear Down****(1) WEIGHT ALTITUDE TEMPERATURE LIMITS**

When dispatching with one hydraulic AC motor pump inoperative, the following penalty must be applied to the performance data as a weight decrement:

FLIGHT SEGMENT	WEIGHT DECREMENT (%)
Take-off – Flaps 20°	16%
Landing – Approach Flaps 20° – Landing Flaps 45°	13%

(2) CLIMB GRADIENTS AND CLIMB SPEEDS

When dispatching with one hydraulic AC motor pump inoperative, the following gradient penalty must be subtracted from the climb gradient data:

CLIMB GRADIENT	GRADIENT PENALTY (%)
Obstacle Clearance Reference Climb Gradient (Flaps 20°)	2.9%
Final Segment / Enroute (Flaps 0°)	3.3%
Approach (Flaps 20°)	2.6%
Landing (Flaps 45°)	Not applicable

The approach climb speed in the flaps 20° configuration must be increased by 2 knots.

8. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable, except as modified by Supplement 8 of the Airplane Flight Manual.

9. SPECIAL OPERATIONS

The special operations in Chapter 8 are applicable, except as modified by the following:

A. Operation on Wet and Contaminated Runways**(1) LIMITATIONS**

Operation on contaminated runways with hydraulic AC motor pump 2B inoperative is prohibited.

(2) TAKE-OFF DISTANCE

When dispatching with hydraulic AC motor pump 1B inoperative:

Take-off distance	Multiply by a factor of 1.3 (30%)
-------------------	-----------------------------------

When dispatching with hydraulic AC motor pump 2B inoperative:

Take-off distance	Multiply by a factor of 2.0 (100%)
-------------------	------------------------------------

9. SPECIAL OPERATIONS (CONT'D)

(3) TAKE-OFF SPEEDS

When dispatching with one hydraulic AC motor pump inoperative, and operating at or below 4,000 feet pressure altitude, apply V_R and V_2 corrections as shown on Figure 06-24-2 and Figure 06-24-3 respectively.

For V_1 , when dispatching with hydraulic AC motor pump 2B inoperative, use the highest V_{1MCG} or multiply the V_R from Figure 06-24-2 by a factor of 0.86 (-14%).

Altitude	Temperature	Weight	Correction
Sea level	ISA + 15 and below	below 37,000 lb	0
		37,000 and above	-1.5
	ISA + 20	below 37,000 lb	0.0
		37,000 and above	-1.0
2,000 ft	ISA + 25 and above	all weights	0
	ISA + 15 and below	below 37,000 lb	0
		37,000 and above	-1.0
	ISA + 20 and above	below 37,000 lb	0.0
		37,000 and above	-0.5
4,000 ft	all temperatures	all weights	0

DFM0708_008

V_R Adjustment
Figure 06-24-2

Altitude	Temperature	Weight	Correction
Sea level	ISA + 15 and below	24,000 lb	-3.0
		30,000 lb	-2.5
		36,000 lb	-1.5
		37,000 and above	0
	ISA + 20	24,000 lb	-1.5
		30,000 lb	-1.0
		36,000 lb	-0.5
		37,000 and above	0
	ISA + 25 and above	all weights	0
2,000 ft	ISA + 15 and below	24,000 lb	-2.0
		30,000 lb	-1.5
		36,000 lb	-1.0
		37,000 and above	0
	ISA + 20 and above	all weights	0
4,000 ft	all temperatures	all weights	0

DFM0708_007A

V_2 Adjustment
Figure 06-24-3

9. SPECIAL OPERATIONS (CONT'D)

B. Operation on Wet Grooved Runways

When dispatching with anti-skid inoperative, use of the landing performance on a wet grooved runway is prohibited. Use the landing distance and landing field length factor of Supplement 8 of the Airplane Flight Manual.

TRAINING PURPOSES ONLY

1. GENERAL

These supplementary procedures must be used when conducting approaches above 3.5 degrees, but below 4.5 degrees.

These data complement or supersede data contained in the basic Operating Manual. These supplementary procedures must, therefore, be used in conjunction with the basic Operating Manual.

The effect of these supplementary procedures on the basic Operating Manual is given below.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. OPERATING LIMITATIONS

The operating limitations in Chapter 2 are applicable, except as modified by the following:

A. Operating Limitations

(1) APPROACHES

On airplanes NOT incorporating Service Bulletin 605-34-008:

Operations with a glidepath angle exceeding the maximum shown in area A of Figure 06-25-1 are prohibited.

On airplanes incorporating Service Bulletin 605-34-008:

Operations with a glidepath angle exceeding the maximum shown in area B of Figure 06-25-1 are prohibited.

B. Systems Limitations

(1) AUTOMATIC FLIGHT CONTROL SYSTEM

The minimum autopilot use height for precision approaches (ILS) with a glidepath angle that exceeds 3.5 degrees is 160 feet AGL.

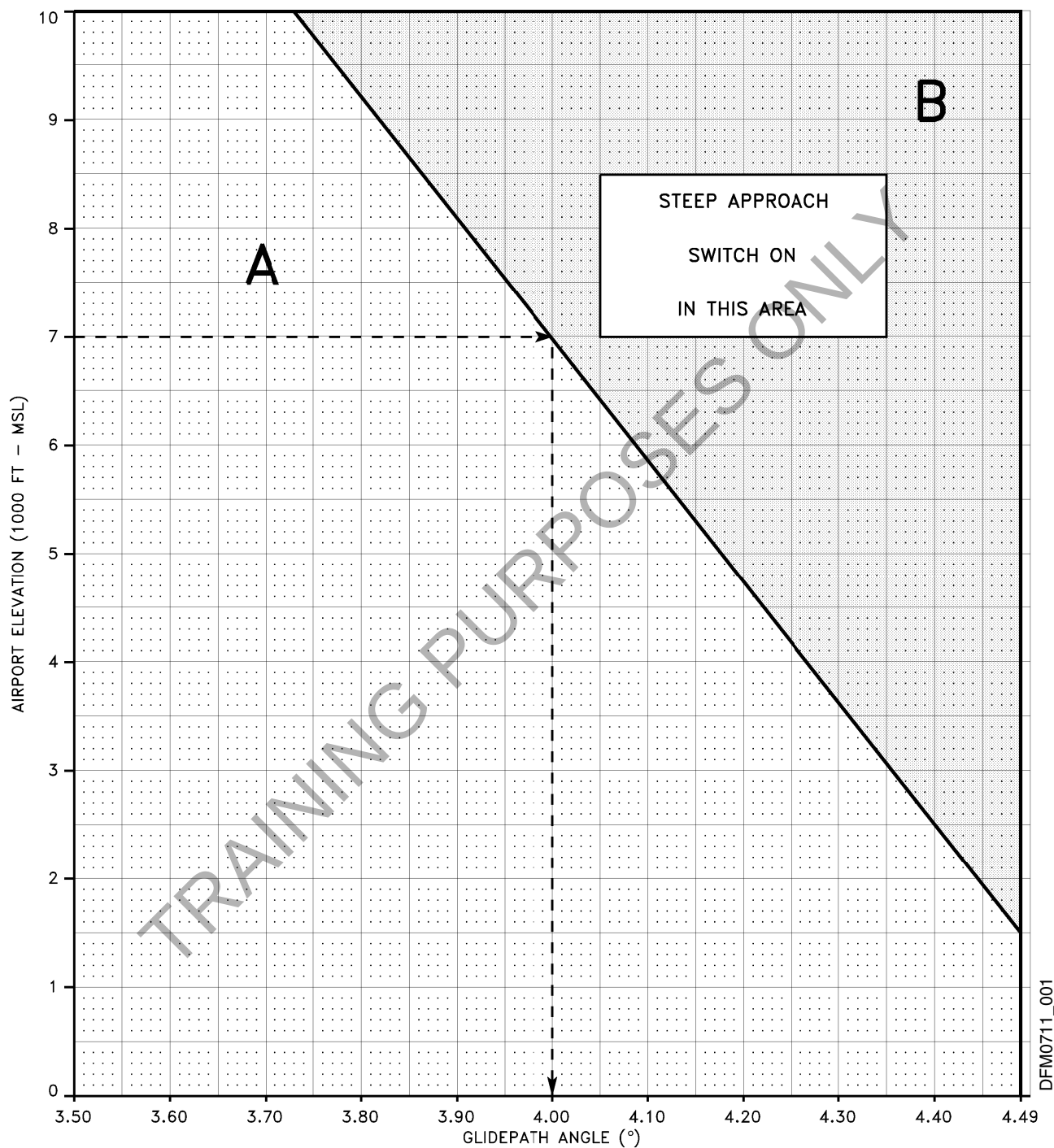
C. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

On airplanes incorporating Service Bulletin 605-34-019:

FMS approaches with LPV guidance are prohibited.

3. OPERATING LIMITATIONS (CONT'D)



Approach Glidepath Angle Operating Limit
Figure 06-25-1

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following;

A. Before Start Check

On airplanes incorporating Service Bulletin 605-34-008:

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

B. Precision (ILS/PAR) Approach

Precision approaches utilizing the autopilot and/or the flight director are essentially the same.

Approaching the fix (or beacon), when performing a full approach (with procedure turn), or coming from a hold:

NOTE

When performing an ILS autopilot-coupled approach, ensure that both flight directors are displayed.

- (1) Airspeed Flaps 20° speed
+ 20 KIAS

Fix outbound:

- (2) Flaps.....20 degrees

For straight-in approach (radar vectored):

- (1) Flaps.....20 degrees
(2) Airspeed Flaps 20° speed
+ 20 KIAS

NOTE

When performing a straight-in approach, assume the initial approach configuration (as in 1 and 2 above), at a distance of 12 miles away from the airport.

Cleared for the approach; Fix inbound (for a full approach), or when on the localizer intercept heading (for a straight-in approach):

- (3) FCP mode APPR
(4) Mode annunciationsVerify
- LOC 1 (2) and GS armed – white indications on both PFDs.
 - LOC 1 (2) – green indications on both PFDs.

For operations in area B of Figure 06-25-1:

- (5) TAWS WARNING, STEEP APPR
switch/light.....Press in Check ON light on.



5. NORMAL PROCEDURES (CONT'D)

B. Precision (ILS/PAR) Approach (Cont'd)

When 1.5 dots below the glideslope:

- (6) LDG GEAR lever Set to DN.
- (7) Flaps.....30 degrees

When 1 dot below the glideslope:

- (8) Flaps.....45 degrees
- (9) Airspeed V_{REF}

NOTE

- The airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept.
- Operations in icing conditions may result in the N_2 being displayed in the cautionary (amber) range (below 78% N_2). In this condition, the wing and cowl anti-ice systems should be monitored. If an abnormal condition occurs, it may not be possible to complete the approach when the associated procedure is followed.

- (10) Before landing check..... Complete
- (11) Preselect ALT Set to missed approach altitude.

At 1,500 feet AGL:

- (12) Airspeed $V_{REF} + WIND$ Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction). Maximum correction is + 20 KIAS.

At the final approach fix (FAF):

- (13) Fix name, altitude and flags Recheck correct fix name, at correct altitude and no flags in view.

At decision height (DH) or decision altitude (DA) and landing is feasible:

- (14) Autopilot (if used) Disconnect at no less than 160 feet AGL.

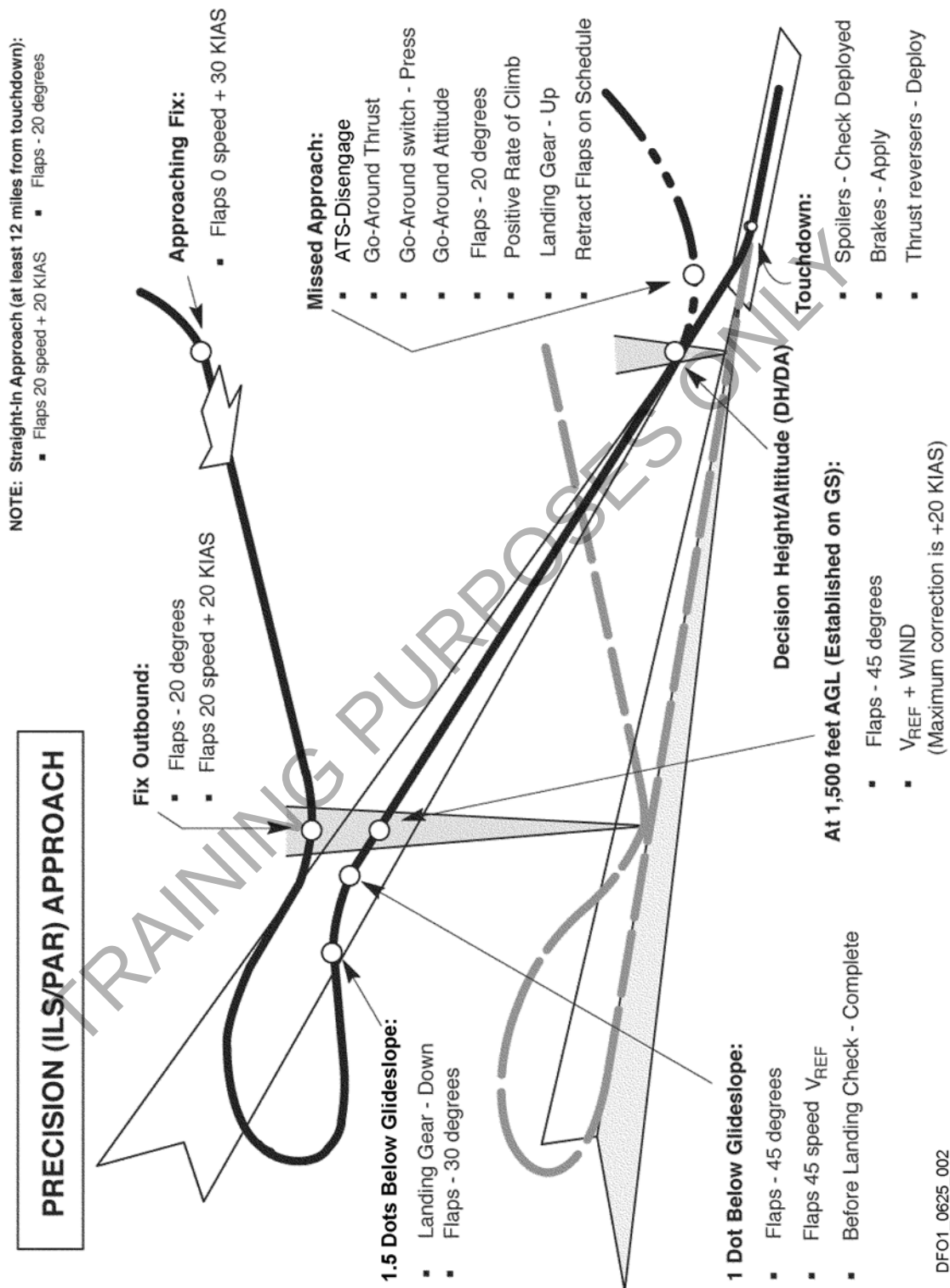


PFD negative barometric altitude is indicated by a yellow NEG flag within the barometric altitude window.



5. NORMAL PROCEDURES (CONT'D)

B. Precision (ILS/PAR) Approach (Cont'd)

Precision (ILS/PAR) Approaches below 4.5 Degrees
Figure 06-25-2

5. NORMAL PROCEDURES (CONT'D)

C. In Range Check

For operations in area B of Figure 06-25-1:

- (1) TAWS WARNING, STEEP APPR
switch/light..... Press in Check ON light on.

D. Before Landing Check

NOTE

- For approaches at angles above 4.0 degrees, the airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction).
- Intermittent operation at idle when wing anti-ice and/or cowl anti-ice is ON is acceptable during final approach provided no associated EICAS caution messages are annunciated.

E. Go-Around

For operations in area B of Figure 06-25-1:

- (1) TAWS WARNING, STEEP APPR
switch/light..... Press out Check ON light out.

F. After Landing Check

On airplanes incorporating Service Bulletin 605-34-008:

- (1) TAWS WARNING, STEEP APPR
switch/light..... Press out Check ON light out.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

A. Flaps Failure

The following applies, in addition to the existing flap failure procedure:

If flaps fail at less than 20 degrees, divert to a suitable landing field with a glidepath angle of 3.5 degrees or less.

————— **END** —————

7. SUPPLEMENTARY PROCEDURES

The supplementary procedures in Chapter 6 are applicable.

8. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual are applicable.

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Steep Approaches up to 5.5 Degrees	Vol. 1	06-26-1 REV 45, Dec 04/17
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NOTE

These supplementary procedures are only applicable to airplanes incorporating Service Bulletin 605-34-008.

These supplementary procedures are NOT applicable to FAA registered airplanes.

These supplementary procedures do not constitute approval to conduct Steep Approach operations.

1. GENERAL

These supplementary procedures must be used when conducting steep approaches from 4.5 degrees to 5.5 degrees, inclusive.

These data complement or supersede data contained in the basic Flight Crew Operating Manual. These supplementary procedures must, therefore, be used in conjunction with the basic Flight Crew Operating Manual.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

A. Structural Weight

Maximum landing weight: 15,875 kg (35,000 lb)

B. Operating Limitations

(1) ALTITUDE AND TEMPERATURE OPERATING LIMIT

Maximum landing altitude is 5,000 feet.

(2) OPERATION IN ICING CONDITIONS

Steep approaches in known icing conditions are prohibited.

(3) WIND LIMITS

No tailwind component is permitted for steep approach landings.

The maximum crosswind component for steep approach landings is 20 knots.

Increasing landing reference speed by more than 10 KIAS for crosswind and/or gusts is prohibited. If wind conditions require a speed adder of more than 10 KIAS, divert to a suitable landing field that does not require a steep approach procedure.

(4) APPROACHES

Operations with a glidepath angle that exceeds 5.5 degrees is prohibited.

Deliberate one engine inoperative steep approaches are prohibited.

Partial flap steep approaches and landings are prohibited.

Steep approach landing on contaminated runways is prohibited.

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3. LIMITATIONS (CONT'D)

(5) LANDING DECISION POINT

The minimum height for go-around following an engine failure (Landing Decision Point) is 200 feet AGL.

C. Power Plant

(1) AUTOTHROTTLE (ATS)

Use of the ATS for steep approaches or landing is prohibited.

D. Systems Limitations

(1) AUTOMATIC FLIGHT CONTROL SYSTEM

Use of autopilot for steep approaches is prohibited.

(2) ANTI-ICE SYSTEM

Approach and landing with wing and/or cowl anti-ice selected ON is prohibited.

(3) SYSTEMS INOPERATIVE

Landings at a steep approach airfield are prohibited if any of the following systems are inoperative:

- Flaps
- Ground Spoilers
- Anti-Skid
- Nose Wheel Steering
- Radio Altimeter
- Thrust Reverser

E. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

Use of the FMS approach performance function for steep approach is prohibited.

On airplanes incorporating Service Bulletin 605-34-019:

FMS approaches with LPV guidance are prohibited.

(2) TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

The TAWS steep approach switch/light must be selected ON as directed in the normal procedures of this supplement.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

- I** Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:
- an approach speed greater than V_{REF} ,
 - a landing with flaps other than 45 degrees,
 - an increase in actual landing distance by a factor greater than 1.50 (50%) or
 - a landing with gear up or unsafe.
 - a landing with any of the systems inoperative as listed in the system limitations of this supplement.

TRAINING PURPOSES ONLY

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

NOTE

Use of ILS or FMS VNAV guidance with or without the flight director is authorized.

A. Before Start Check

- (1) TAWS WARNING, STEEP APPR
switch/light Press out Check ON light out.

B. Precision (ILS/PAR) Approach

Approaching the fix (or beacon), when performing a full approach (with procedure turn), or coming from a hold:

- (1) Airspeed Flaps 20° speed
+ 20 KIAS

Fix outbound:

- (2) Flaps 20 degrees

For straight-in approach (radar vectored):

- (1) Flaps 20 degrees
(2) Airspeed Flaps 20° speed
+ 20 KIAS

NOTE

When performing a straight-in approach, assume the initial approach configuration (as in 1 and 2 above), at a distance of 12 miles away from the airport.

Cleared for the approach; Fix inbound (for a full approach), or when on the localizer intercept heading (for a straight-in approach):

- (3) FCP mode APPR
(4) Mode annunciations Verify
- LOC 1 (2) and GS armed – white indications on both PFDs.
 - LOC 1 (2) – green indications on both PFDs.
- (5) TAWS WARNING, STEEP APPR
switch/light Press in Check ON light on.

When 1.5 dots below the glideslope:

- (6) LDG GEAR lever Set to DN.
(7) Flaps 30 degrees

When 1 dot below the glideslope:

- (8) Flaps 45 degrees



5. NORMAL PROCEDURES (CONT'D)

B. Precision (ILS/PAR) Approach (Cont'd)

(9) Airspeed $V_{REF} + WIND$

NOTE

The airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction). Maximum correction is + 10 KIAS.

(10) Before landing check Complete

(11) Preselect ALT Set to missed approach altitude.

At 1,500 feet AGL:

(12) Airspeed $V_{REF} + WIND$ Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction). Maximum correction is + 10 KIAS.

At the final approach fix (FAF):

(13) Fix name, altitude and flags Recheck correct fix name, at correct altitude and no flags in view.

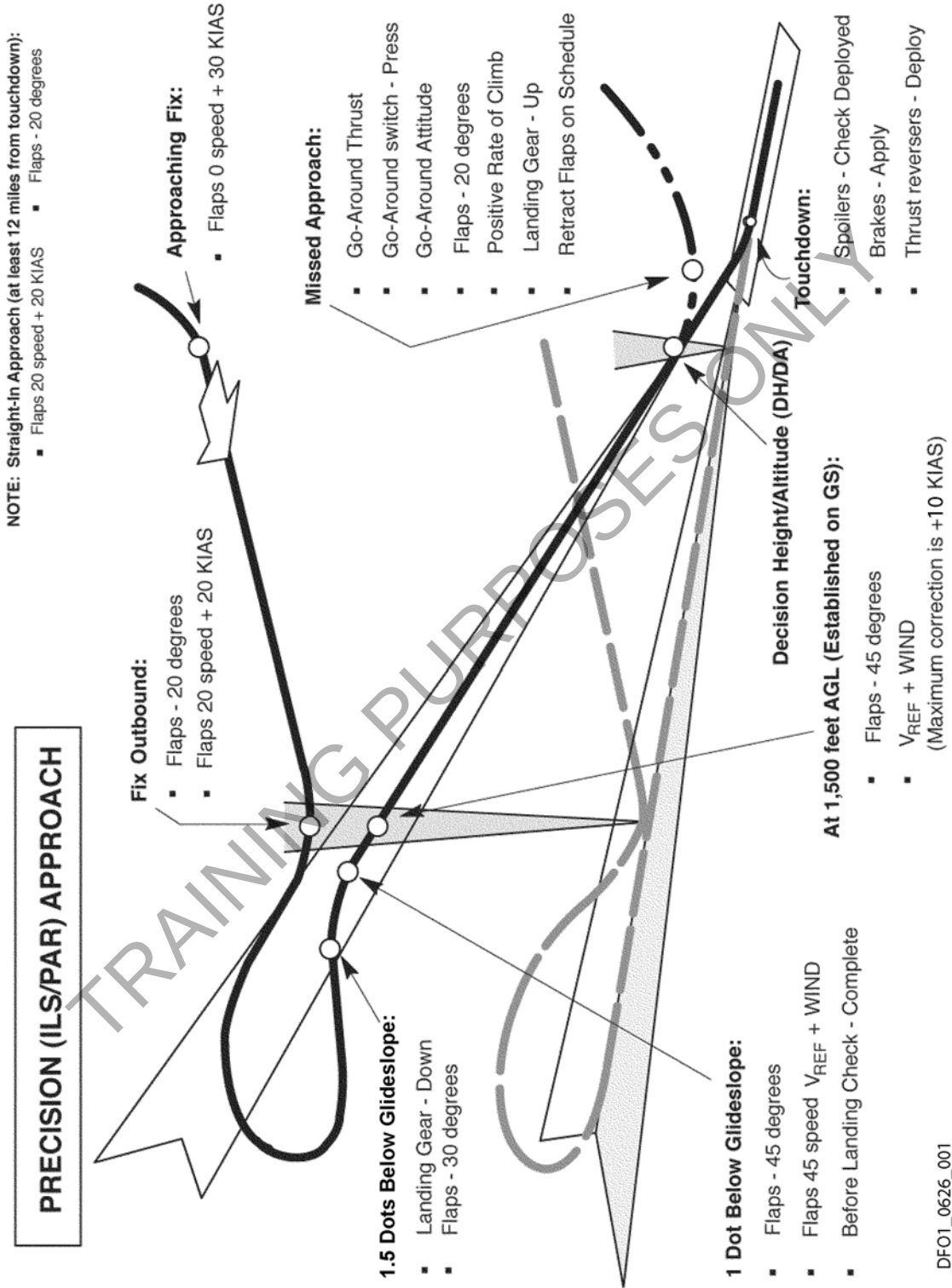


PFD negative barometric altitude is indicated by a yellow NEG flag within the barometric altitude window.



5. NORMAL PROCEDURES (CONT'D)

B. Precision (ILS/PAR) Approach (Cont'd)



Precision (ILS/PAR) Approaches up to 5.5 Degrees
Figure 06-26-1

5. NORMAL PROCEDURES (CONT'D)

C. In Range Check

- (1) TAWS WARNING, STEEP APPR
switch/light..... Press in Check ON light on.

D. Before Landing Check

NOTE

The airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction) up to a maximum of $V_{REF} + 10$ KIAS.

E. Go-Around

- (1) TAWS WARNING, STEEP APPR
switch/light..... Press out Check ON light out.

F. After Landing Check

- (1) TAWS WARNING, STEEP APPR
switch/light..... Press out Check ON light out.

TRAINING PURPOSES ONLY

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

I Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:

- an approach speed greater than V_{REF} ,
- a landing with flaps other than 45 degrees or
- an increase in actual landing distance by a factor greater than 1.50 (50%).
- a landing with any of the systems inoperative as listed in the system limitations of this supplement.

A. Engine Failure During Approach

- I**
- | | | |
|---|----------|--|
| (1) Operating engine | Increase | thrust as required. |
| (2) FLIGHT SPOILER lever
(If extended) | Select | to RETRACT.
Check EICAS indicates
spoilers retracted. |
| (3) FLAPS | Select | to 20°. |
| (4) Airspeed | Increase | to V_{REF} (Flaps 45°) + 14 KIAS. |
| (5) Airplane | Retrim | and go-around (Refer to
Single Engine Go-Around in
this section).

If required, autopilot may be
engaged, if above 800 feet
AGL.

Divert to a suitable landing
field that does not require a
steep approach procedure. |

NOTE

The height loss during a single engine go-around is no more than 200 feet.

————— **END** —————

6. ABNORMAL PROCEDURES (CONT'D)

B. Single Engine Go-Around

NOTE

1. The minimum fuel for go-around is 230 kg (500 lb) per wing, with a maximum airplane climb attitude of 5° nose up.
2. Single engine go-around capability is not assured with flaps greater than 20°.

- | | | |
|--|---------|--|
| (1) Single Engine Climb Thrust (APR rating)..... | Set | immediately (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS). |
| (2) TOGA push switch | Depress | |
| (3) FLIGHT SPOILER lever (If extended) | Select | to RETRACT.

Check EICAS indicates spoilers retracted. |
| (4) Airplane | Rotate | smoothly to initial pitch attitude of 10°, and adjust pitch as required to achieve the approach climb speed. (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – APPROACH AND LANDING CLIMB). |
| (5) FLAPS | Select | to 20°. |
- When a positive rate of climb is achieved:**
- | | | |
|--------------------------|--------|--------|
| (6) LDG GEAR lever | Select | to UP. |
|--------------------------|--------|--------|
- At a safe altitude (not below 400 feet AGL):**
- | | | |
|---|--------|---|
| (7) FLAPS | Select | to 0° at a speed of not less than V _{FTO} – 5 KIAS. |
| (8) THRUST REVERSER, L and R switches | Select | to OFF.

Check L and R REV ARMED advisory messages out. |

If go-around was due to engine failure during approach:

- | | | |
|---|------------|---|
| (9) In-Flight Engine Failure/Shutdown | Accomplish | Refer to Chapter 5 – ABNORMAL PROCEDURES – POWER PLANT – In-Flight Engine Failure/Shutdown. |
|---|------------|---|

————— **END** —————

7. SUPPLEMENTARY PROCEDURES

The supplementary procedures in Chapter 6 are applicable except as modified by the following:

A. FAA Steep Approach into London City Airport

Use of the supplementary procedures for steep approaches into London City Airport is prohibited.

8. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable, except as modified by Supplement 12 of the Airplane Flight Manual.

TRAINING PURPOSES ONLY

9. SPECIAL OPERATIONS

The special operations in Chapter 8 are applicable, except as modified by the following:

A. Operation on Wet Grooved Runways**(1) RUNWAY CONDITIONS**

The data is applicable to concrete runway surfaces with transverse grooves constructed and maintained to FAA standard AC 150/5320-12C, or equivalent, and on which there are no areas of significant rubber deposit in sections where the high-speed portion of braking is expected.

(a) Wet Runway

A grooved runway is considered wet when there is sufficient moisture on the runway surface to cause it to appear partially reflective, but without significant areas of standing water (runway grooves not overflowing).

(2) LIMITATIONS

The limitations in Chapter 2 are applicable, with the addition of the following:

- The data may be used only for wet grooved runways.
- Non-performance related limitations and procedures that are applicable to a wet runway, are also applicable to a wet grooved runway.
- The use of this Special Operation for landing performance requires both thrust reversers to be operational.
- The use of this Special Operation for landing performance requires the anti-skid system to be fully operational.
- The use of this Special Operation requires that :
 - the runway is maintained to FAA standard AC 150/5320-12C, or equivalent,
 - there are no areas of significant rubber deposit in sections where the high-speed portion of braking is to occur, and
 - there is no indication, via NOTAMs, PIREPs or braking action reports, that degraded braking exists on the runway.
- This Special Operation is applicable only to grooved concrete surfaces.
- The use of a wet grooved runway corrected landing distance which is shorter than the dry runway landing field length is prohibited.

On EASA registered airplanes:

This Special Operation is approved for use on concrete runways having a grooved surface that has been constructed and maintained to the FAA standard of AC 150/5320-12C or equivalent. Operational approval must be obtained in order to use the performance data presented in this Special Operation.

The wet grooved landing distance determined during flight planning must account for the conditions expected at the time of arrival, including reported wind, runway slope and V_{REF} speed increment.

9. SPECIAL OPERATIONS (CONT'D)

(3) EMERGENCY PROCEDURES

The emergency procedures in the supplementary procedures for Steep Approaches up to 5.5 Degrees are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

(4) ABNORMAL PROCEDURES

The abnormal procedures in the supplementary procedures for Steep Approaches up to 5.5 Degrees are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

(5) PERFORMANCE

The performance for operation on wet grooved runways is provided in Supplement 12 of the Airplane Flight Manual.

TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Operational Capabilities	Vol. 1	06-27-1
		REV 36, Nov 27/15	

1. GENERAL

These supplementary procedures provides data for operational capabilities and the applicable standards of certain airplane systems.

NOTE

Compliance with the standards noted below does not constitute an operational approval.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below:

A. Introduction

The general information in Chapter 1 is applicable.

B. Limitations

The limitations in Chapter 2 are applicable.

C. Emergency Procedures

The emergency procedures in Chapter 3 are applicable.

D. Normal Procedures

The normal procedures in Chapter 4 are applicable.

E. Abnormal Procedures

The abnormal procedures in Chapter 5 are applicable.

F. Supplementary Procedures

The supplementary procedures in Chapter 6 are applicable.

G. Performance

The performance data in Chapter 6 of the Airplane Flight Manual are applicable.

	CL-605 Flight Crew Operating Manual PSP 605-6	
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2. NAVIGATION

A. RVSM

The airplane is certified capable of RVSM operations in accordance with FAA "Interim guidance material on the approval of operations/aircraft for RVSM operations" 91-RVSM, dated March 14, 1994, or later applicable revision.

RVSM operations must not be commenced or continued unless all of the required equipment specified below is operational:

RVSM REQUIRED EQUIPMENT LIST	
EQUIPMENT	REQUIREMENTS FOR RVSM
AUTOPILOT (FCC, FCP, Elevator servo)	MUST BE OPERATIONAL.
ALTITUDE ALERTING SYSTEM	TWO (2) AFDs MUST BE OPERATIONAL DISPLAYING 2 PRIMARY ALTIMETERS
ALTITUDE REPORTING TRANSPONDER	ONE (1) MUST BE OPERATIONAL.
AIR DATA COMPUTER	TWO (2) MUST BE OPERATIONAL.

B. Traffic Alert and Collision Avoidance System (TCAS)

The TCAS installation is in accordance with JAA TGL No. 8.

Pilots are authorized to deviate from their Air Traffic Control (ATC) clearance in order to comply with a TCAS resolution advisory (RA) command.

AR Certified Airplanes

Pilots are authorized to deviate from their Air Traffic Control (ATC) clearance in order to comply with a TCAS resolution advisory (RA) command. The pilots shall inform ATC of the deviation from the ATC clearance following the response to the RA.

Maneuvers must not be based solely on information presented on the traffic display.

2. NAVIGATION (CONT'D)

I C. Flight Management System (FMS) – FMS-6000

The FMS has been demonstrated capable of, and has been shown to meet the requirements for, the following operation:

(1) OCEANIC AND REMOTE

- Use of the FMS with the GPS is approved for supplemental means of navigation source for oceanic and remote operations.
- Use of the FMS with GPS has been found to comply with the requirements of AC 20-138A, Appendix 1 for GPS primary means of navigation in oceanic and remote airspace, with two operational GPS receivers and with two operational FMS systems.

(2) NORTH ATLANTIC (NAT) MINIMUM NAVIGATIONAL PERFORMANCE SPECIFICATION (MNPS) AIRSPACE

- Provided two FMS installations are operating with each receiving information from at least two inertial reference systems (IRS), the FMS has been demonstrated capable of flight into North Atlantic (NAT) minimum navigational performance specification (MNPS) airspace, and has been shown to meet the accuracy specification in accordance with AC 120-33 or AC 91-49.

TRAINING PURPOSES ONLY

2. NAVIGATION (CONT'D)

(3) ENROUTE, TERMINAL AREA AND NON-PRECISION APPROACH OPERATIONS

- Use of the FMS with the GPS is approved for supplemental means of navigation source for enroute, terminal area and non-precision approach operations.
- The FMS installation meets the performance/accuracy criteria of AC 20-130A, Airworthiness Approval of Navigation or Flight Management Systems Integrating Multiple Navigation Sensors, for enroute, terminal area and non-precision approach operations.

GPS TSO C129a/AC 20-138A

- The installed FMS and GPS comply with AC 20-138A for IFR navigation using GPS for enroute, terminal area and non-precision approach operations (including "GPS", "or GPS", and "RNAV" approaches).

NOTE

1. The system does not meet the requirements for LPV approaches.
2. The system does not meet the requirements for procedures or routes with RF legs.

On airplanes 5863 to 6049 and airplanes incorporating Service Bulletin 605-34-011:

GPS/WAAS TSO C145a/AC 20-138A

- The installed FMS and GPS comply with AC 20-138A for navigation using GPS and WAAS (within the coverage of a Space-Based Augmentation System (SBAS) complying with ICAO Annex 10) for enroute, terminal area and non-precision approach operations (including "GPS", "or GPS", and "RNAV" approaches).

NOTE

1. The system does not meet the requirements for LPV approaches.
2. The system does not meet the requirements for procedures or routes with RF legs.

2. NAVIGATION (CONT'D)

(4) RNAV–1/PRNAV

- The FMS installation meets the airworthiness certification requirements of JAA Temporary Guidance Leaflet No. 10, Airworthiness and Operational Approval for Precision RNAV Operations in Designated European Airspace.
- Precision RNAV operations must not be conducted unless all of the required equipment specified below is operational.

PRNAV REQUIRED EQUIPMENT LIST	
EQUIPMENT	REQUIREMENTS FOR PRNAV
FLIGHT MANAGEMENT COMPUTER	ONE (1) MUST BE OPERATIONAL.
FMS CONTROL DISPLAY UNIT	ONE (1) MUST BE OPERATIONAL.
VHF NAV, DME, GPS	ONE (1) VHF NAV and ONE (1) DME MUST BE OPERATIONAL OR ONE (1) GPS MUST BE OPERATIONAL.
IRS	IRS ONLY message must not be displayed.

Compliance with AC 90–100A, US Terminal and Enroute Area Navigation (RNAV) Operations

- With an operating FMS that is receiving valid signals from either:
 - GPS, or
 - DME and IRS

the aircraft meets the functional and accuracy requirements of AC 90–100A, US Terminal and Enroute Area Navigation (RNAV) Operations for RNAV–1 and RNAV–2 routes, provided that:

1) None of the following messages are displayed:

- FMS DR (PFD, MFD or CDU)
- IRS ONLY (PFD, MFD or CDU)
- VOR/DME ONLY (CDU or PFD)
- VOR/DME DIST > 40.0 NM (CDU)
- CHK POS (PFD or CDU)

and,

2) For procedures that specifically require GPS, or when GPS is the only available sensor, none of the following messages are posted on the CDU:

- NO GPS RAIM
- GPS NOT AVAILABLE
- GPS-FMS DISAGREE

and,

3) The crew has entered NOTAMed nav aids on the CDU VOR/DME CONTROL page.

2. NAVIGATION (CONT'D)

(5) RNAV-5/BRNAV

- The FMS installation meets the requirements of AC90-96, Approval of US Operators and Aircraft to Operate under Instrument Flight Rules (IFR) in European Airspace Designated for Basic Area Navigation and JAA Temporary Guidance Leaflet No. 2, rev. 1, AMJ 20X2, JAA Guidance Material on Airworthiness Approval and Operational Criteria for the use of Navigation Systems in European Airspace Designated for Basic RNAV Operations.

(6) RNP-10

- The FMS installation with the IRS has been demonstrated to meet the criteria of FAA Order 8400.12A "Required Navigation Performance 10 (RNP-10) Operational Approval" as a primary means of navigation without time limitation and without updating, based on compliance with the IRS accuracy requirements of FAR 121, Appendix G.
- The FMS with the GPS with RAIM has been demonstrated to meet the criteria of FAA Order 8400.12A "Required Navigation Performance 10 (RNP-10) Operational Approval" as a means of navigation for flights without time limitations.

(7) VNAV

- The FMS installation meets the performance/accuracy criteria for enroute, terminal and approach VNAV operation as per AC 20-129, titled Airworthiness Approval of Vertical Navigation (VNAV) Systems for use in the US National Airspace System (NAS) and Alaska.

2. NAVIGATION (CONT'D)

D. Flight Management System (FMS) – FMS-6200

On airplanes incorporating Service Bulletin 605-34-019:

Approval Status

Provided the FMS is receiving adequate usable signals, the FMS has been demonstrated capable of and has been shown to meet the performance requirements for the following operations:

Lateral Navigation

- 1) VFR/IFR en route oceanic and remote, en route domestic, terminal operation, and instrument approach using WGS84 (or NAD83) coordinate reference datum in accordance with the performance/accuracy requirements of AC 20-138D.
- 2) Global Navigation Satellite System (GNSS) TSO C145a B3/AC 20-138D and Flight Management System (FMS) TSO-C115b and TSO-C146c D4/AC 20-138D: The installed equipment complies with AC 20-138D, AMC 20-27A and AMC 20-28; for navigation using GNSS and GNSS/SBAS (within the coverage of a satellite-based augmentation system complying with ICAO Annex 10) for enroute, terminal area, non-precision approach, and approach procedures with vertical guidance operations. Non-precision approach operations include those based on conventional navigation aids with or without GNSS in the title and those with GNSS and RNAV (GNSS) in the title to LNAV minimums. Approach procedures with vertical guidance include RNAV (GNSS) and LNAV/VNAV minimums as well as LPV approaches.

The installed system is compatible with the WAAS (North America), EGNOS (Europe), GAGAN (India) and MSAS (Japan) SBAS Service Providers, complying with ICAO Annex 10 and developed per RTCA DO-229().

RNP Operations

RNP 10 (RNAV 10) – In accordance with AC 90-105A Appendix G, ICAO Doc 9613 PBN Volume II, Part B, Chapter 1, and AMC 20-12:

- the FMS installation with the IRS has been demonstrated to meet the criteria as a primary means of navigation for flight up to 6.2 hours in duration without updating. The determination of flight duration starts when the aircraft first begins to move after the system is placed in navigation mode.
- the FMS installation with the GNSS with RAIM has been demonstrated to meet the criteria as a means of navigation for flights without time limitations.

On airplanes incorporating Service Bulletin 605-34-021:

RNP-AR – In accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 6; AMC 20-26; and AC 20-138C, Appendix 2, the FMS has been demonstrated capable of primary navigation for RNP AR approach operations.

2. NAVIGATION (CONT'D)

RNP-2 – In accordance with ICAO PBN document 9613 Edition 4, Volume II, Part C, Chapter 2, (Implementing RNP-2):

- the FMS has been demonstrated capable of primary navigation in RNP-2 for oceanic and remote airspace operations provided that two FMSs are receiving information from at least two GNSS and operating in GNSS navigation mode prior to initiating an RNP procedure.
- the FMS has been demonstrated capable of primary navigation in RNP-2 for continental operations provided that one FMS is receiving information from at least one GNSS and operating in GNSS navigation mode prior to initiating an RNP procedure, provided that an alternative means of navigation is available if required for the class of operation.

RNP-4 – In accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 1 and AC 90-105A Appendix F, the FMS has been demonstrated capable of primary navigation for RNP-4 Oceanic and remote area operations provided that two FMS systems are available and receiving information from two independent GNSS sensors and operating in GNSS navigation mode.

Basic RNP-1 – In accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 3 and AC 20-138D Chap 9 for FAA RNP 1 (Terminal) operations, the FMS has been demonstrated capable of primary navigation for RNP-1 operations provided that the FMS is receiving information from a GNSS sensor and operating in GNSS navigation mode.

RNP APCH – The FMS has been demonstrated capable of primary navigation for RNP APCH operations in accordance with ICAO Doc 9613 PBN Volume II, Part C, Chapter 5, AC 20-138D for FAA RNP instrument approach with Baro-VNAV and AMC 20-27A for RNP APCH with Baro-VNAV Provided FMS is receiving information from a GNSS sensor and operating in:

- GNSS navigation mode; and
- RNP APPR annunciation is displayed at the Final Approach Fix.

LPV Approaches – The FMS and GNSS have been demonstrated capable of primary navigation for LPV operations in accordance with ICAO Doc 9613 PBN Volume II, Part B, Chapter 5, AC 20-138D for FAA LPV and AMC 20-28 for EASA LPV.

NOTE

LP approaches cannot be performed.

Baro-VNAV – Provided at least a single FMS, a single GNSS and single ADC are installed and operational and the NO APPR (on PFD) messages are NOT posted, this FMS installation has been demonstrated to meet the requirements of AC 20-138D for performing Baro-VNAV.

Vertical Navigation (VNAV) – The FMS has been demonstrated capable of and has been shown to meet the accuracy requirements of: VFR/IFR en route, terminal and approach VNAV operation within the contiguous United States and Alaska in accordance with the criteria of AC 20-138D. VNAV Temperature Compensation meets the requirements of TCCA AC 500.020 FMS Barometric VNAV Temperature Compensation and RTCA DO-236B MASPS RNP for Area Navigation, Appendix H.

2. NAVIGATION (CONT'D)

Primary Means Oceanic and Remote – Use of the FMS with GNSS has been found to comply with the requirements of AC 20-138D, Appendix 1 for GNSS primary means of navigation in oceanic and remote airspace, with two operational GNSS receivers and with two operational FMS systems.

NAT MNPS – Provided dual FMS and any two independent long range navigation systems, consisting of any combination of GNSS or IRS are installed and operating, the FMS with GNSS inputs has been demonstrated capable of flight into North Atlantic (NAT) minimum navigation performance specification (MNPS) and has been shown to meet the accuracy specification in accordance with AC 120.33 or AC 91-49.

RNAV-1 / RNAV-2 – In accordance with AC90-100A, the FMS has been demonstrated capable of primary navigation in RNAV-1 and RNAV-2 airspace provided that the FMS is operating in GNSS or DME/DME navigation mode.

RNAV-5 / BRNAV – This FMS installation has been demonstrated to meet the BRNAV requirements of AC 90-96A and JAA AMJ 20X2 Leaflet 2, Revision 1.

PRNAV – The FMS installation has been demonstrated to meet the PRNAV requirements of JAA TGL 10.

Compliance with AC 90-100A – When equipped with an operating FMS and GNSS, the aircraft meets the functional and the ± 1 nm accuracy requirements of AC 90-100A, U.S. Terminal and En Route Area Navigation (RNAV) Operations

E. FM Immunity

The VHF NAV radios installed comply with FM immunity requirements per ICAO Annex 10, and:

- ED-22B / DO-196 for VOR receivers
- ED-46B / DO-195 for LOC receivers

3. SURVEILLANCE

A. Mode S Elementary Surveillance

The Mode S transponder has been certified to meet the requirements of Elementary Mode S Surveillance as defined by JAA TGL 13.

B. Enhanced Mode S Surveillance

The installed Mode S system satisfies the data requirements of ICAO Doc 7030/4, Regional Supplementary Procedures for SSR Mode S Enhanced Surveillance in designated airspace. The capability to transmit data parameters is shown in the following table:

PARAMETER	AVAILABLE/NOT AVAILABLE
MAGNETIC HEADING	AVAILABLE
INDICATED AIRSPEED	AVAILABLE
MACH NUMBER	AVAILABLE
VERTICAL RATE	AVAILABLE (SEE NOTE 1)
ROLL ANGLE	AVAILABLE
TRACK ANGLE RATE	AVAILABLE
TRUE TRACK ANGLE	AVAILABLE (SEE NOTE 2)
GROUND SPEED	AVAILABLE (SEE NOTE 2)
SELECTED ALTITUDE	AVAILABLE
BAROMETRIC PRESSURE SETTING	AVAILABLE

NOTE

1. Barometric rate of climb/descent as well as Inertial Rate of climb/descent are available as the aircraft is equipped with IRS.
2. Aircraft must be equipped with at least one functioning Flight Management Computer and FMS Control Display Unit.

3. SURVEILLANCE (CONT'D)

C. Automatic Dependent Surveillance Broadcast (ADS-B)

On airplanes incorporating Service Bulletin 605-34-013:

The installed ADS-B Out system meets Section 8 of EASA AMC 20-24, dated 2 May 2008.

ADS-B operations in Non-Radar Areas (ADS-B NRA) cannot be commenced or continued unless the required equipment specified below is operational:

ADS-B OUT REQUIRED EQUIPMENT LIST	
EQUIPMENT	REQUIREMENTS FOR ADS-B OUT
ATC/MODE S TRANSPONDER	ONE (1) MUST BE OPERATIONAL.
AIR DATA COMPUTER	ONE (1) MUST BE OPERATIONAL.
GLOBAL POSITIONING SYSTEM RECEIVER or GNSS SBAS RECEIVER	ONE (1) MUST BE OPERATIONAL.

On airplanes incorporating Service Bulletin 605-34-030:

The installed ADS-B Out system has been certified in accordance with the guidance provided in the FAA AC 20-165A, the EASA CS.ACNS.D.ADS and the Australian AC 21-45, and meets the equipment requirements of FAA 14 CFR 91.227 and those of CASA CAO 20.18.

ADS-B operations cannot be commenced or continued unless the required equipment specified below is operational:

ADS-B OUT REQUIRED EQUIPMENT LIST	
EQUIPMENT	REQUIREMENTS FOR ADS-B OUT
ATC/MODE S TRANSPONDER	ONE (1) MUST BE OPERATIONAL.
AIR DATA COMPUTER	ONE (1) MUST BE OPERATIONAL.
GLOBAL POSITIONING SYSTEM RECEIVER or GNSS SBAS RECEIVER	ONE (1) MUST BE OPERATIONAL.

4. COMMUNICATIONS

A. FM Immunity

The VHF COM radios installed comply with FM immunity requirements per ICAO Annex 10, and ED-23B / DO-186 for VHF communication receivers.

B. Datalink

(1) ATN B1 CPDLC (CONTROLLER TO PILOT DATALINK COMMUNICATION)

On airplanes incorporating Service Bulletin 605-23-007:

The aircraft Datalink system has been demonstrated to comply with the applicable safety, performance and interoperability requirements for continental ATN B1 Datalink using VDL Mode-2.

The aircraft ATC Datalink system does support multi-frequency operation as defined in ARINC Spec. 631-5.

(2) FANS 1/A+

On airplanes incorporating Service Bulletin 605-23-008:

The aircraft FANS 1/A+ system enables CPDLC and ADS-C capabilities in support of reduced aircraft separations in remote and oceanic areas, by using Satellite Communications and ACARS ground networks. The system complies with the interoperability requirements of RTCA DO-258A and the guidelines of ICAO GOLD.

The system adopts FAA AC 20-140A as an acceptable means of compliance for the following datalink capabilities:

Interoperability Designators:

- FANS 1/A+ with automation (including "push to load")
- ATN B1
- ACARS ATS

ATS Subnetworks:

- VDL Mode-A
- VDL Mode-2
- SATCOM Inmarsat Classic Aero
- SATCOM Iridium (Short Burst Data (SBD))

Aircraft-Allocated Performance:

- CPDLC: RCP 240
- ADS-C: RSP 180

FANS 1/A+ system includes message latency monitoring as indicated by the + symbol as per DO258A 4.6.6.9 .

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Operation with Cracked Windshield Faceply	Vol. 1	06-28-1
		REV 21, Jan 18/12	

NOTE

These supplementary procedures are only applicable to airplanes on the FAA registry.

1. GENERAL

These supplementary procedures provide data for operation with a cracked windshield faceply.

These data complement or supersede data contained in the basic Operating Manual. These supplementary procedures must therefore be used in conjunction with the basic Operating Manual.

The effect of these supplementary procedures on the basic Operating Manual is given below.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable with the addition of the following:

A. Systems Limitations

(1) WINDSHIELD AND SIDE WINDOWS

The faceply of one windshield may be cracked provided:

- Cracked faceply is repaired in accordance with Structural Repair Manual, Chapter 51,
- There is no FOD damage to the airframe, nacelles or engines, on initial discovery or if faceply pieces are missing,
- Vision is not impaired through remaining windows,
- Tape and cracked faceply is checked for general integrity prior to each flight,
- Visibility through the affected windshield is acceptable to the flight crew,
- Affected windshield heater is considered inoperative,
- Remaining windshield heater and both side window heaters are operative,
- Flight into known or forecast icing conditions, precipitation, thunderstorms or other conditions which could cause fogging or misting is prohibited and
- Airspeed is limited to 280 KIAS or less below 8,000 feet MSL.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

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7. SUPPLEMENTARY PROCEDURES

The supplementary procedures in Chapter 6 are applicable.

8. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable with the addition of the following:

A. General

One windshield faceply may be cracked provided the mission fuel requirements are increased by:

Fuel consumption	0.30%
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TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES Runway Awareness and Advisory System	Vol. 1	06–29–1 REV 36, Nov 27/15
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NOTE

These supplementary procedures are only applicable to airplanes incorporating Service Bulletin 605–34–022.

1. GENERAL

These supplementary procedures provide data for operation of the Runway Awareness and Advisory System (RAAS).

These data complement or supersede data contained in the basic Flight Crew Operating Manual. These supplementary procedures must therefore be used in conjunction with the basic Flight Crew Operating Manual.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below.

Functional Overview

The Runway Awareness Advisory System (RAAS) is a software enhancement hosted in the Terrain Awareness Warning System (TAWS) unit, which provides aural and visual advisories to supplement flight crew awareness of position during ground and flight operations. RAAS uses existing TAWS inputs to generate aural advisories in the flight deck. RAAS advisories are annunciated over the same audio system as the TAWS. Unique voices are used for RAAS advisories to allow adequate differentiation from the current TAWS Mode 1 – 7 and Terrain / Obstacle Awareness voice call outs. RAAS advisories are succinct by design. RAAS is designed to be fully compatible with normal aircraft operations. There is no requirement for pilot input to the system.

RAAS is configurable (during installation/activation) for voice gender, units (feet or meters) and most advisories may be configured ON or OFF. This flight manual supplement is arranged so that each configurable advisory is maintained in its own sub-section. This allows an operator to utilize this supplement by choosing each section that applies to them.

RAAS Implementation and Configuration

RAAS may be configured for a variety of advisory callout options via the RAAS configuration database.

The only customer selectable options are:

- Voice gender: Female/Male
- Units of measurement: Feet/Meter
- Altimeter Setting callout: ON/OFF.

A RAAS inhibit switch allows the crew to inhibit all RAAS visual indications and aural alerts. The white OFF switch/light illuminates when selected.

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1. GENERAL (CONT'D)

The table below provides the settings of alert options certified on the aircraft. These options and configurations are set during installation and activation of the RAAS on the aircraft, and are NOT pilot selectable.

GROUP	ADVISORY/ ALERT	ENABLED	SETTINGS	AUDIO	VISUAL
	GPS Antenna Location		15 feet	–	–
	Advisory Suppress Window	YES	450 – 550 feet	–	–
RAAS	Approaching Runway – In Air/On Ground	YES		“Approaching XX” “Approaching Runways” (if more than one runway meets the qualifying condition)	APP XX APP RWYS (if more than one runway meets the qualifying condition)
	On Runway	YES		“On Runway XX” “On Runways” (if more than one runway meets the qualifying condition)	ON XX ON RWYS (if more than one runway meets the qualifying condition)
	Distance Remaining (Landing & Rollout)	YES	2,000 feet max	“ZZ Remaining”	NONE
	Runway End (< 100 feet)	YES		“One Hundred Remaining” (in feet) “Thirty Remaining” (in meters)	NONE
	Insufficient Runway Length (In Air)	YES	5,000 feet	“Approaching Runway XX, ZZ available”	APP XX ZZ
	Short Runway Landing	NO			
	Insufficient Runway Length (On Ground)	YES	6,000 feet	“On Runway XX, ZZ remaining”	ON XX ZZ
	Short Runway Takeoff	NO			
	Extended Hold On Runway Initial/Repeats	YES	120 sec/ 120 sec	“On Runway XX – On Runway XX”	ON XX
	Taxiway Takeoff	YES		“Caution, On Taxiway! On Taxiway”	ON TAXIWAY GND PROX alert on PFD also illuminated
	Distance Remaining (Rejected Takeoff)	YES	50% of Runway	“ZZ Remaining”	NONE
	Taxiway Landing	YES		“Caution, Taxiway! Caution, Taxiway!”	TAXIWAY GND PROX alert on PFD also illuminated
	Takeoff Flaps	YES	Min: 16°; Max: 24°	“On Runway XX, Flaps – Flaps”	FLAPS GND PROX alert on PFD also illuminated
Stability Approach Monitor	Landing Flaps (Not Aligned With Runway)	NO			
	Landing Flaps (Aligned With Runway)	YES	700 feet	“Flaps – Flaps”	FLAPS
	Too High (During Approach)	YES		“Too High – Too High”	TOO HIGH
	Too Fast (During Approach)	YES		“Too Fast – Too Fast”	TOO FAST
	Unstable (During Approach)	YES		“Unstable – Unstable”	UNSTABLE GND PROX alert on PFD also illuminated

1. GENERAL (CONT'D)

GROUP	ADVISORY/ ALERT	ENABLED	SETTINGS	AUDIO	VISUAL
Long Landing Monitor	Long Landing Distance	YES	3,000 ft past approach end	"Long Landing – Long Landing"	LONG LANDING GND PROX alert on PFD also illuminated
	Long Landing Percentage	YES	67% remaining	"Long Landing – Long Landing"	LONG LANDING GND PROX alert on PFD also illuminated
	Long Landing Distance Remaining	YES	3,500 ft past approach end	"ZZ Remaining"	
	Long Landing Distance Remaining Percentage	YES	60% remaining	"ZZ Remaining"	
Corrected Altitude Monitor	Altimeter Setting (Below Transition Altitude)	YES		"Altimeter Setting"	ALTM SETTING
	Altimeter Setting (Above Transition Altitude)	YES		"Altimeter Setting"	ALTM SETTING

XX = Runway Identifier

ZZ = Available runway distance in feet or meters

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable with the addition of the following:

Refer to the Honeywell MK V and MK VII Enhanced Ground Proximity Warning System (EGPWS) and Flight Safety Functions TSO C151b Class A TAWS Pilot Guide, document 060-4241-000 revision G or later appropriate revision for descriptive information.

A. Circling Approach

- (1) TAWS WARNING,
RAAS OFF switch/light..... Press in Check OFF light on.

————— **END** —————

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

7. SUPPLEMENTARY PROCEDURES

The supplementary procedures in Chapter 6 are applicable.

8. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable.

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES RNP Authorization Required Approach Operations	Vol. 1	06–30–1 REV 36, Nov 27/15
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NOTE

These supplementary procedures are only applicable to airplanes incorporating Service Bulletin 605–34–021.

1. GENERAL

These supplementary procedures contain information and procedures for Required Navigation Performance (RNP) Authorization Required (AR) – approach operations.

NOTE

Compliance with these standards noted below does not constitute approval to conduct RNP AR approach operations.

These data complement or supersede data contained in the basic Flight Crew Operating Manual. These supplementary procedures must therefore, be read in conjunction with the basic Flight Crew Operating Manual.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

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3. LIMITATIONS

The limitations in Chapter 2 are applicable, with the addition of the following:

A. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

An RNP AR approach must not be commenced or continued unless the required navigation equipment, as specified in the RNP AR Approach Required Equipment List below, are operating satisfactorily.

RNP AR APPROACH – REQUIRED EQUIPMENT LIST	
EQUIPMENT	PRIOR TO APPROACH
YD 1 and 2	Two (2) yaw dampers must be available to commence the approach, one (1) yaw damper is required to continue the approach
PFD 1 and 2	One (1) PFD available and operational for each side
ADC 1 and 2	Two (2) ADCs must be available to commence and continue the approach
GNSS 1 and 2	Two (2) GNSS Sensors must be available to commence the approach, one (1) GNSS Sensor is required to continue the approach
FMS 1, 2 and 3	Two (2) FMSs must be available and operational for each side
IRS 1, 2 and 3 (ATT and HDG)	Two (2) IRSs must be available to commence the approach, one (1) IRS is required to continue the approach
TAWS	TAWS must be operational and displayed
FD 1 and 2	Two (2) must be operational and independently displayed

RNP AR approach operations beyond the IAF are prohibited unless RNP AR APPR is annunciated on the PFD.

RNP AR approach with RF Legs operations are approved for RNP greater than or equal to RNP 0.3 in the approach segment and RNP AR 1.0 in the missed approach segment.

RNP AR approach operations are approved with autopilot or flight director engaged only.

RNP AR approach operations are approved for autothrottle engaged or disengaged.

NOTE

It is recommended to have A/T OFF for FD only RNP AR approaches.

RNP AR approach operations are approved for any flap configuration up to the IF (Intermediate Fix) for all engines operating, or for flaps 0° and 20° with one engine inoperative, and with flaps 45° from IF to MAP (Missed Approach Point) for all engines operating, or for flaps 20° from IF to MAP with one engine inoperative.

RNP AR approach operations must be accomplished in accordance with instrument approach procedures that are retrieved from the FMS navigation database.

RNP AR approach operations are approved following failure of one engine.

RNP AR approach operations must be accomplished at or below published maximum speed limits for the approach segment depicted on the Instrument Approach Procedure. In the event of a missed approach the aircraft speed must not exceed the maximum speed for the segment of the approach.

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, with the addition of the following:

A. Approach

- (1) FMS navigation source..... Set independent source on each PFD.
- (2) IRS and ADC Set independent source on each PFD.
- (3) Flight director mode..... APPR LNV
- (4) RNP AR APPR annunciations..... Verify annunciated at IAF

————— **END** —————

TRAINING PURPOSES ONLY

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, with the addition of the following:

A. Loss of GNSS

Indication: **NO APPR** caution message on PFD.

- (1) RNP AR approach Discontinue

NOTE

Conduct a go-around continuing to follow the RNP AR approach lateral track and speed constraints until established at or above the minimum safe altitude.

————— **END** —————

B. Excessive Lateral or Vertical Deviation

Indication: Amber flashing lateral or vertical deviation indication.

- (1) RNP AR approach Discontinue

NOTE

Conduct a go-around continuing to follow the RNP AR approach lateral track and speed constraints until established at or above the minimum safe altitude.

————— **END** —————

C. EFIS Parameter Mismatch

Indication: **EFIS MISMATCH** caution message on and **ALT** annunciation on PFD.

- (1) RNP AR approach Discontinue

NOTE

Conduct a go-around continuing to follow the RNP AR approach lateral track and speed constraints until established at or above the minimum safe altitude.

————— **END** —————

7. SUPPLEMENTARY PROCEDURES

The supplementary procedures in Chapter 6 are applicable.

8. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable.

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES FAA Steep Approach into London City Airport	Vol. 1	06–31–1 REV 45, Dec 04/17
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NOTE

These supplementary procedures are only applicable to airplanes incorporating Service Bulletin 605–34–008.

These supplementary procedures are **ONLY** applicable to FAA registered airplanes operating into London City Airport (LCY).

These supplementary procedures do not constitute approval to conduct Steep Approach operations.

1. GENERAL

These supplementary procedures must be used when conducting steep approaches into London City Airport (LCY).

These data complement or supersede data contained in the basic Flight Crew Operating Manual. These supplementary procedures must, therefore, be used in conjunction with the basic Flight Crew Operating Manual.

The effect of these supplementary procedures on the basic Flight Crew Operating Manual is given below.

2. INTRODUCTION

The general information in Chapter 1 is applicable.

3. LIMITATIONS

The limitations in Chapter 2 are applicable, except as modified by the following:

A. Structural Weight

Maximum landing weight: 14,742 kg (32,500 lb)

B. Operating Limitations

(1) OPERATION IN ICING CONDITIONS

Steep approaches in known icing conditions are prohibited.

(2) WIND LIMITS

No tailwind component is permitted for steep approach landings.

The maximum crosswind component for steep approach landings is 20 knots.

Increasing landing reference speed by more than 10 KIAS for crosswind and/or gusts is prohibited. If wind conditions require a speed adder of more than 10 KIAS, divert to a suitable landing field that does not require a steep approach procedure.

(3) APPROACHES

Operations at airports other than London City Airport (LCY) are prohibited.

Deliberate one engine inoperative steep approaches are prohibited.

Partial flap steep approaches and landings are prohibited.

Steep approach landing on contaminated runways is prohibited.

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3. LIMITATIONS (CONT'D)

C. Power Plant

(1) AUTOTHROTTLE (ATS)

Use of the ATS for steep approaches or landing is prohibited.

D. Systems Limitations

(1) AUTOMATIC FLIGHT CONTROL SYSTEM

Use of autopilot for steep approaches is prohibited.

(2) ANTI-ICE SYSTEM

Approach and landing with wing and/or cowl anti-ice selected ON is prohibited.

(3) SYSTEMS INOPERATIVE

Landings at a steep approach airfield are prohibited if any of the following systems are inoperative:

- Flaps
- Ground Spoilers
- Anti-Skid
- Nose Wheel Steering
- Radio Altimeter
- Thrust Reverser

E. Navigation Systems Limitations

(1) FLIGHT MANAGEMENT SYSTEM

Use of the FMS approach performance function for steep approach is prohibited.

On airplanes incorporating Service Bulletin 605-34-019:

FMS approaches with LPV guidance are prohibited.

(2) TERRAIN AWARENESS AND WARNING SYSTEM (TAWS)

The TAWS steep approach switch/light must be selected ON as directed in the normal procedures of these supplementary procedures.

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES FAA Steep Approach into London City Airport	Vol. 1	06-31-3
		REV 43, Mar 13/17	

4. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:

- an approach speed greater than V_{REF} ,
- a landing with flaps other than 45 degrees,
- an increase in actual landing distance by a factor greater than 1.50 (50%) or
- a landing with gear up or unsafe.
- a landing with any of the systems inoperative as listed in the system limitations of these supplementary procedures.

TRAINING PURPOSES ONLY

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5. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable, except as modified by the following:

NOTE

Use of ILS or FMS VNAV guidance with or without the flight director is authorized.

A. Before Start Check

- (1) TAWS WARNING, STEEP APPR
switch/light Press out Check ON light out.

B. Precision (ILS/PAR) Approach

Approaching the fix (or beacon), when performing a full approach (with procedure turn), or coming from a hold:

- (1) Airspeed Flaps 20° speed
+ 20 KIAS

Fix outbound:

- (2) Flaps 20 degrees

For straight-in approach (radar vectored):

- (1) Flaps 20 degrees
(2) Airspeed Flaps 20° speed
+ 20 KIAS

NOTE

When performing a straight-in approach, assume the initial approach configuration (as in 1 and 2 above), at a distance of 12 miles away from the airport.

Cleared for the approach; Fix inbound (for a full approach), or when on the localizer intercept heading (for a straight-in approach):

- (3) FCP mode APPR
(4) Mode annunciations Verify
- LOC 1 (2) and GS armed – white indications on both PFDs.
 - LOC 1 (2) – green indications on both PFDs.
- (5) TAWS WARNING, STEEP APPR
switch/light Press in Check ON light on.

When 1.5 dots below the glideslope:

- (6) LDG GEAR lever Set to DN.
(7) Flaps 30 degrees



5. NORMAL PROCEDURES (CONT'D)

B. Precision (ILS/PAR) Approach (Cont'd)

When 1 dot below the glideslope:

- (8) Flaps.....45 degrees
- (9) Airspeed $V_{REF} + WIND$

NOTE

The airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction). Maximum correction is + 10 KIAS.

- (10) Before landing check..... Complete
- (11) Preselect ALT Set to missed approach altitude.

At 1,500 feet AGL:

- (12) Airspeed $V_{REF} + WIND$ Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction). Maximum correction is + 10 KIAS.

At the final approach fix (FAF):

- (13) Fix name, altitude and flags Recheck correct fix name, at correct altitude and no flags in view.

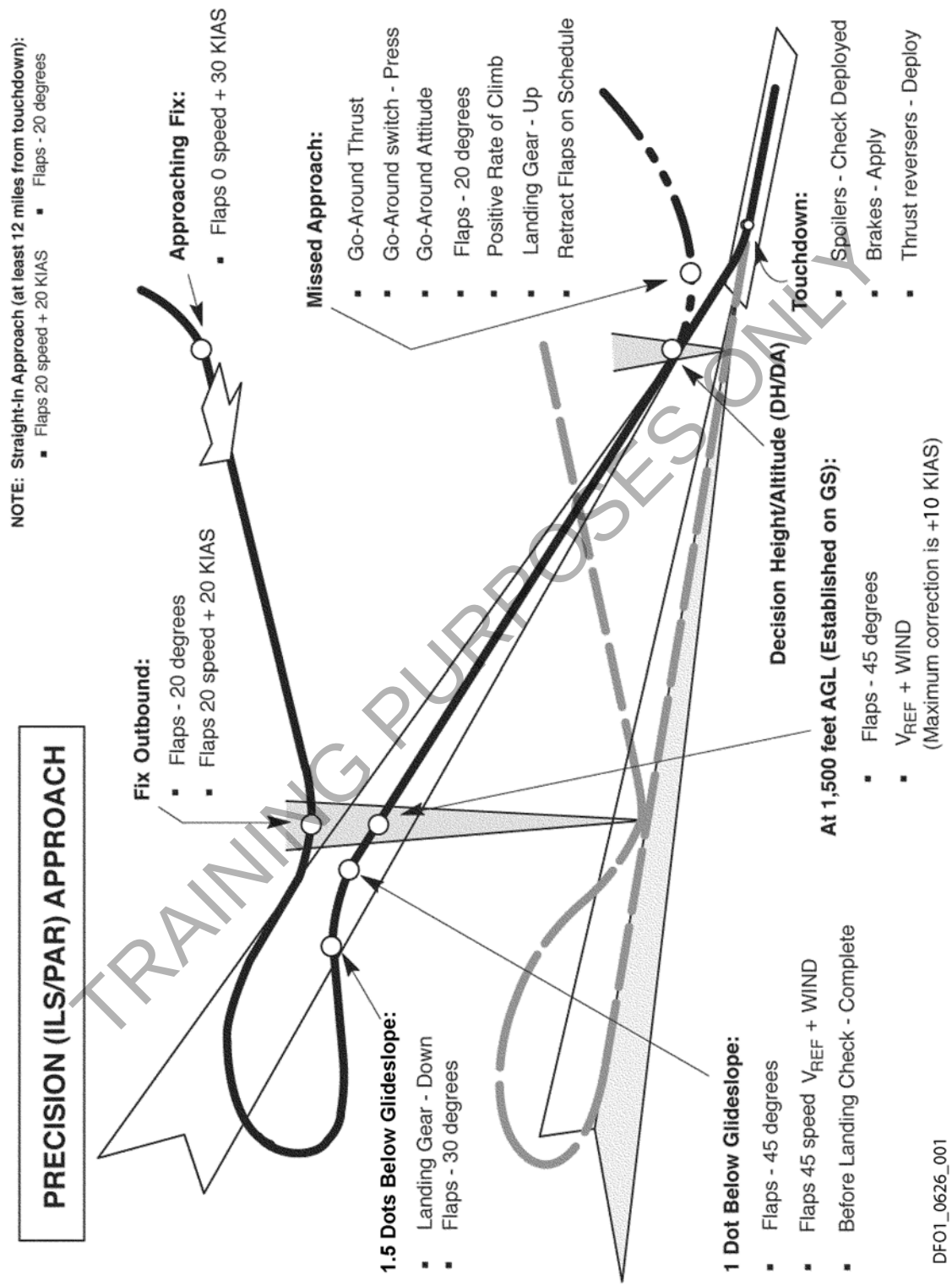


PFD negative barometric altitude is indicated by a yellow NEG flag within the barometric altitude window.



5. NORMAL PROCEDURES (CONT'D)

B. Precision (ILS/PAR) Approach (Cont'd)



Precision (ILS/PAR) Approach
Figure 06-31-1

5. NORMAL PROCEDURES (CONT'D)

C. In Range Check

- (1) TAWS WARNING, STEEP APPR
switch/light..... Press in Check ON light on.

D. Before Landing Check

NOTE

The airplane should be fully configured for landing and slowed to $V_{REF} + WIND$ prior to glideslope intercept. Wind correction is half steady state crosswind component, plus all of the gust (regardless of direction) up to a maximum of $V_{REF} + 10$ KIAS.

E. Go-Around

- (1) Thrust levers/TOGA switch Advance/Press
- (2) FLIGHT SPOILER lever
(if extended) Select to RETRACT.
Check EICAS indicates
spoilers retracted.
- (3) Pitch attitude..... Increase rapidly but smoothly to initial
pitch attitude of 10° at a speed
of not less than V_{REF} , then
adjust pitch attitude to
maintain approach climb
speed.

NOTE

- Contact with the ground may be unavoidable if go-around is initiated from a low altitude or high sink rate. If the aircraft contacts the ground during the go-around, maintain the normal landing attitude or slightly higher. Aircraft configuration should not be changed until the aircraft is climbing if ground contact is made or not.
- A momentary **GND SPLRS** caution message may be displayed if ground contact occurs.

- (4) FLAPS Select to 20° .

When a positive rate of climb is achieved:

- (5) LDG GEAR lever Select to UP.

At a safe altitude (not below 400 feet AGL):

- (6) FLAPS Select to 0° at a speed of not less
than $V_{FTO} - 5$ KIAS.
- (7) THRUST REVERSER, L and R
switches..... Select to OFF.
Check **L** and **R REV ARMED**
advisory messages out.
- (8) TAWS WARNING, STEEP APPR
switch/light..... Press out Check ON light out.

5. NORMAL PROCEDURES (CONT'D)

F. After Landing Check

- (1) TAWS WARNING, STEEP APPR
switch/light.....Press out Check ON light out.

6. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure necessitates:

- an approach speed greater than V_{REF} ,
- a landing with flaps other than 45 degrees or
- an increase in actual landing distance by a factor greater than 1.50 (50%).
- a landing with any of the systems inoperative as listed in the system limitations of these supplementary procedures.

A. Engine Failure During Approach

- (1) Operating engine Increase thrust as required.
- (2) FLIGHT SPOILER lever
(If extended) Select to RETRACT.
Check EICAS indicates
spoilers retracted.
- (3) FLAPS Select to 20°.
- (4) Airspeed Increase to V_{REF} (Flaps 45°) + 14 KIAS.
- (5) Airplane Retrim and go-around (Refer to
Single Engine Go-Around in
this section).

If required, autopilot may be
engaged, if above 800 feet
AGL.

Divert to a suitable landing
field that does not require a
steep approach procedure.

NOTE

1. The height loss during a single engine go-around is no more than 200 feet.
2. After an engine failure at or below 200 feet, single engine approach to landing is authorized with flaps at 45 degrees.

————— **END** —————

6. ABNORMAL PROCEDURES (CONT'D)

B. Single Engine Go-Around

NOTE

- The minimum fuel for go-around is 230 kg (500 lb) per wing, with a maximum airplane climb attitude of 5° nose up.
- Single engine go-around capability is not assured with flaps greater than 20°.

- | | | |
|---|--------|--|
| (1) Normal Take-off Thrust | Set | immediately (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – THRUST SETTINGS). |
| (2) TOGA switch | Press | |
| (3) FLIGHT SPOILER lever
(If extended) | Select | to RETRACT.

Check EICAS indicates spoilers retracted. |
| (4) Airplane | Rotate | rapidly but smoothly to initial pitch attitude of 10°, and adjust pitch as required to achieve the approach climb speed. (Refer to the Airplane Flight Manual, Chapter 6; PERFORMANCE – APPROACH AND LANDING CLIMB). |

NOTE

- Contact with the ground may be unavoidable if go-around is initiated from a low altitude or high sink rate. If the aircraft contacts the ground during the go-around, maintain the normal landing attitude or slightly higher. Aircraft configuration should not be changed until the aircraft is climbing if ground contact is made or not.
- A momentary **GND SPLRS** caution message may be displayed if ground contact occurs.

- (5) FLAPS Select to 20°.

When a positive rate of climb is achieved:

- (6) LDG GEAR lever Select to UP.

At a safe altitude (not below 400 feet AGL):

- (7) FLAPS Select to 0° at a speed of not less than $V_{FTO} - 5$ KIAS.

- (8) THRUST REVERSER, L and R switches Select to OFF.

Check **L** and **R REV ARMED** advisory messages out.



6. ABNORMAL PROCEDURES (CONT'D)

B. Single Engine Go-Around (Cont'd)

If go-around was due to engine failure during approach:

- (9) In-Flight Engine Failure/Shutdown Accomplish Refer to Chapter 5 –
 ABNORMAL PROCEDURES
 – POWER PLANT – In-Flight
 Engine Failure/Shutdown.

————— **END** —————

7. SUPPLEMENTARY PROCEDURES

The supplementary procedures in Chapter 6 are applicable, except as modified by the following:

A. Steep Approaches up to 5.5 Degrees

Use of the supplementary procedures for steep approaches up to 5.5 degrees is prohibited.

8. PERFORMANCE

The performance data in Chapter 6 of the Airplane Flight Manual is applicable, except as modified by Supplement 20 of the Airplane Flight Manual.

TRAINING PURPOSES ONLY

9. SPECIAL OPERATIONS

The special operations in Chapter 8 are applicable, except as modified by the following:

A. Operation on Wet Grooved Runways

(1) RUNWAY CONDITIONS

The data is applicable to concrete runway surfaces with transverse grooves constructed and maintained to FAA standard AC 150/5320-12C, or equivalent, and on which there are no areas of significant rubber deposit in sections where the high-speed portion of braking is expected.

(a) Wet Runway

A grooved runway is considered wet when there is sufficient moisture on the runway surface to cause it to appear partially reflective, but without significant areas of standing water (runway grooves not overflowing).

(2) LIMITATIONS

The limitations in Chapter 2 are applicable, with the addition of the following:

- The data may be used only for wet grooved runways.
- Non-performance related limitations and procedures that are applicable to a wet runway, are also applicable to a wet grooved runway.
- The use of these supplementary procedures for landing performance requires both thrust reversers to be operational.
- The use of these supplementary procedures for landing performance requires the anti-skid system to be fully operational.
- The use of these supplementary procedures requires that :
 - the runway is maintained to FAA standard AC 150/5320-12C, or equivalent,
 - there are no areas of significant rubber deposit in sections where the high-speed portion of braking is to occur, and
 - there is no indication, via NOTAMs, PIREPs or braking action reports, that degraded braking exists on the runway.
- These supplementary procedures are applicable only to grooved concrete surfaces.
- The use of a wet grooved runway corrected landing distance which is shorter than the dry runway landing field length is prohibited.

(3) EMERGENCY PROCEDURES

The emergency procedures in these supplementary procedures are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

(4) ABNORMAL PROCEDURES

The abnormal procedures in these supplementary procedures are applicable, except as modified by the following:

Discontinue steep approach and divert to the nearest suitable airport if any procedure requires an increase in the actual landing distance.

BOMBARDIER CHALLENGER 605	SUPPLEMENTARY PROCEDURES FAA Steep Approach into London City Airport	Vol. 1	06-31-12
		REV 49, Nov 19/18	

I 9. SPECIAL OPERATIONS (CONT'D)

I (5) PERFORMANCE

I The performance in these supplementary procedures is applicable, except as modified by
I Supplement 20 of the Airplane Flight Manual.

TRAINING PURPOSES ONLY

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CHAPTER 7 - PERFORMANCE

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TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	PERFORMANCE Introduction	Vol. 1	07-01-1
		Oct 05/06	

1. GENERAL

The Challenger 605 airplane performance data and information is found in the Airplane Flight Manual.

TRAINING PURPOSES ONLY

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TRAINING PURPOSES ONLY

BOMBARDIER CHALLENGER 605	PERFORMANCE Operational Landing Distance	Vol. 1	07-02-1
		REV 58, Dec 08/20	

1. GENERAL

The Challenger 605 QRH allows for the calculation of Operational Landing Distances (OLD) for normal landing configurations. The OLD are calculated based on guidelines of FAA AC 25-32 Landing Performance Data for Time of Arrival Landing Performance Assessments and provide a better assessment of landing distances required than Actual Landing Distances (ALD).

The QRH FOLD tables account for:

- Actual OAT (covers ISA and non-ISA conditions)
- Runway Slope
- Altitude
- Winds
- V_{REF} increment
- Use of thrust reversers
- Runway Condition (dry/Wet/contaminated)
- MMEL items.

A minimum 15% operational distance factor is included in the FOLD data.

Data includes weights that are greater than the MLW of 17,237 kg (38,000 lb).

NOTE

The FOLD calculation using the QRH FOLD tables may be different than the FMS ALD with FOLD factor applied due to conservative QRH corrections. The lesser distance may be used.

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2. RUNWAY SURFACE CONDITION - PILOT-REPORTED BRAKING ACTION - WHEEL BRAKING CORRELATION MATRIX

RUNWAY CONDITION CODE (RCC)	RUNWAY SURFACE CONDITION DESCRIPTION	PILOT-REPORTED BRAKING ACTION
6	Dry	–
5	- Frost - Wet (includes damp and 1/8in (3mm) depth or less of water) 1/8in (3mm) depth or less of: - Slush - Dry snow - Wet snow	Good
4	–15°C and colder OAT: Compacted snow	Good to Medium [1]
3	- Wet (“Slippery when wet” runway) - Dry snow or wet snow (any depth) over compacted snow - Greater than 1/8in (3mm) depth of: - Dry snow - Wet snow - Warmer than –15°C OAT: - Compacted snow	Medium [1]
2	Greater than 1/8in (3mm) depth of: - Water - Slush	Medium [1] to Poor
1	Ice	Poor
0	- Wet ice - Water on top of compacted snow - Dry snow or wet snow over ice	Nil

[1] The braking action term “FAIR” is in the process of being changed to “MEDIUM” throughout the FAA. Until an official change is published, the term “FAIR” may be used.

Runway Surface Condition – Pilot-reported braking Action – Wheel Braking Correlation Matrix
Table 07-02-1

NOTE

- Do not operate on a runway surface condition corresponding to Runway Condition Code 0.
- Initial selection of the Runway Condition Code must be based on the RCC reported and if not available then the runway surface condition description. The Equivalent Pilot-reported braking action must only be considered if it downgrades the runway condition (i.e. reduces the Runway Condition Code).

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3. PERFORMANCES

OLD was designed to be used in flight to assess landing conditions at time of arrival. The more precise OLD model provides greater accuracy in estimating landing distances. The OLD landing distances can be calculated manually (Example 1 to 4) or with the FMS (Example 5) using the associated QRH tables, in a more conservative way.

Example 1: OLD LANDING DATA - QRH TABLE CALCULATIONS (IMPERIAL)

Associated Conditions:

Airplane weight	36,000 lb
Pressure altitude	4,000 feet
Ambient Temperature	15°C (ISA + 8°C)
Runway Condition	ATIS reports, "... runway covered with slush less than 1/8"...", RCC5,
V _{REF} increment	5 knots
Reported wind	10 knots (headwind)
Runway slope	0%
Thrust Reversers Operative	1

1. Determine the Runway Condition Code, refer to Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), Runway Surface Condition – Pilot – Reported Braking Action – Wheel Braking Correction Matrix.

- If RCC is not reported, in the column Runway Surface Condition Description, locate the condition described in the problem, "... runway covered with slush less than 1/8"...": RCC5.

2. Determine the OLD (uncorrected), Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – FT) – RCC 5. using the pressure altitude, airplane weight from the example:

- Associated OLD: 5,233 feet

3. Apply the applicable corrections to determine the OLD, Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – FT) – RCC 5 .

V _{REF}	5 x 1.2% = 6.0%
Temperature correction	8 x 0.4% = 3.2%
Headwind correction	10 x -0.5% = -5.0%
Runway slope	0%
Total correction	4.2%
FOLD	5,233 x 1.042 = 5,453 feet

4. Multiply, the thrust reverser correction:

One thrust reverser inoperative	x 1.12
FOLD	5,453 x 1.12 = 6,107 feet

5. The final Factored Operational Landing Distance (FOLD) is **6,107 feet** .

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3. PERFORMANCES (CONT'D)

VREF
TWO THRUST REVERSERS
RUNWAY CONDITION CODE 5

ZERO WIND
ZERO SLOPE
ISA

FACTORED OPERATIONAL LANDING DISTANCE (FOLD - FT) - RCC 5								
WEIGHT (LB)	PRESSURE ALTITUDE (FT)							
	0	2000	4000	6000	8000	10000	12000	14000
28000	3862	4053	4260	4544	4874	5240	5645	6068
30000	3950	4199	4490	4811	5164	5556	5990	6443
32000	4161	4427	4737	5079	5455	5873	6337	6821
34000	4373	4655	4984	5348	5748	6193	6686	7201
36000	4585	4884	5233	5618	6042	6514	7037	7583
38000	4797	5113	5482	5889	6338	6837	7389	7967
40000	5045	5380	5771	6203	6681	7211	7797	8412
42000	5260	5613	6024	6480	6983	7540	8157	8807
44000	5476	5846	6279	6758	7285	7872	8521	9205
46000	5675	6062	6515	7015	7567	8180	8860	9577

<CL604_FOLD_RCC5_IMP_06JUL20.ps>

Add the following corrections together based on the conditions and apply the total correction on the uncorrected FOLD value obtained from the table above:

- + 1.2 % / kt above V_{REF}
- + 0.4 % / °C above ISA, - 0.3 % / °C below ISA
- + 2.4 % / kt tailwind, - 0.5 % / kt headwind
- + 2.4 % / % downhill slope, - 1.3 % / % uphill slope

If required, multiply the corrections below to the result above:

Thrust Reversers	
One thrust reverser inoperative	X 1.12
Two thrust reversers inoperative	X 1.34
MMEL	
Ground spoilers inoperative	X 1.14
Anti-skid inoperative	X 1.95

NOTE: The Factored Operational Landing Distance includes a 15% operational distance factor.

VREF – Two Thrust Reversers – Runway Condition Code 5 – RCC 5 – Imperial
Figure 07-02-1

BOMBARDIER CHALLENGER 605	PERFORMANCE Operational Landing Distance	Vol. 1	07-02-5 REV 57, Sep 11/20
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3. PERFORMANCES (CONT'D)

Example 2: OLD LANDING DATA - QRH TABLE CALCULATIONS (IMPERIAL)

Associated Conditions:

Airplane weight	34,000 lb
Pressure altitude	1,417 feet (Airport Elevation) @29.82 = 1,317 feet
Ambient Temperature	8°C (ISA – 4°C) Engine Cowls required.
Runway Condition	ATIS reports, "... wet, slippery when wet...", RCC3
V _{REF} increment	5 knots
Reported wind	10 knots (headwind)
Runway slope	0%
Thrust Reversers Operative	2

1. Determine the Runway Condition Code, refer to Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), Runway Surface Condition – Pilot – Reported Braking Action – Wheel Braking Correction Matrix.
 - If RCC is not reported, in the column Runway Surface Condition Description, locate the condition described in the problem, "... wet, slippery when wet...": RCC3.

2. Determine the OLD (uncorrected), Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – FT) – RCC 3. using the pressure altitude, airplane weight from the example:

For an airplane weight of 34,000 lb:

- OLD at 0 feet (Altitude Pressure): 4,835 feet.
- OLD at 2,000 feet (Altitude Pressure): 5,124 feet.
- $\Delta\text{OLD} : 5,124 - 4,835 = 289$ feet.
- Associated OLD at 1,317 feet: $4,835 + (289/2\ 000) \times 1,317$.
- Associated OLD at 1,317 feet: 5,025 feet.

3. Apply the applicable corrections to determine the OLD, Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – FT) – RCC 3 .

V _{REF}	5 x 1.0% = 5.0%
Temperature correction	4 x –0.2% = –0.8%
Headwind correction	10 x –0.4% = –4.0%
Runway slope	0%
Total correction	0.2%
FOLD	5,025 feet x 1.002 = 5,035 feet

4. No correction to apply for Thrust reversers or MMEL
5. The final Factored Operational Landing Distance (FOLD) is **5,035 feet** .

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3. PERFORMANCES (CONT'D)

VREF
TWO THRUST REVERSERS
RUNWAY CONDITION CODE 3

ZERO WIND
ZERO SLOPE
ISA

FACTORED OPERATIONAL LANDING DISTANCE (FOLD - FT) - RCC 3								
WEIGHT (LB)	PRESSURE ALTITUDE (FT)							
	0	2000	4000	6000	8000	10000	12000	14000
28000	4294	4499	4717	5010	5346	5715	6119	6538
30000	4398	4657	4955	5282	5639	6030	6460	6905
32000	4617	4891	5207	5553	5930	6345	6799	7270
34000	4835	5124	5458	5822	6220	6658	7137	7633
36000	5051	5356	5707	6091	6509	6969	7473	7995
38000	5267	5587	5955	6358	6797	7280	7808	8357
40000	5515	5852	6240	6664	7126	7635	8192	8770
42000	5731	6083	6488	6931	7414	7946	8528	9133
44000	5945	6313	6735	7197	7702	8256	8864	9496
46000	6143	6524	6963	7443	7967	8543	9175	9831

<CL604_FOLD_RCC3_IMP_06JUL20.ps>

Add the following corrections together based on the conditions and apply the total correction on the uncorrected FOLD value obtained from the table above:

- + 1.0 % / kt above V_{REF}
- + 0.4 % / °C above ISA, - 0.2 % / °C below ISA
- + 2.3 % / kt tailwind, - 0.4 % / kt headwind
- + 2.8 % / % downhill slope, - 1.8 % / % uphill slope

If required, multiply the corrections below to the result above:

Thrust Reversers

- One thrust reverser inoperative X 1.13
- Two thrust reversers inoperative X 1.36

MMEL

- Ground spoilers inoperative X 1.14
- Anti-skid inoperative Prohibited

NOTE: The Factored Operational Landing Distance includes a 15% operational distance factor.

VREF – Two Thrust Reversers – Runway Condition Code 3 – RCC 3 – Imperial
Figure 07-02-2

BOMBARDIER CHALLENGER 605	PERFORMANCE Operational Landing Distance	Vol. 1	07-02-7 REV 57, Sep 11/20
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3. PERFORMANCES (CONT'D)

Example 3: OLD LANDING DATA - QRH TABLE CALCULATIONS (METRIC)

Associated Conditions:

Airplane weight	16,000 kg
Pressure altitude	4,000 feet
Ambient Temperature	15°C (ISA + 8°C)
Runway Condition	ATIS reports, "... runway covered with slush less than 1/8"... , RCC5
V _{REF} increment	5 knots
Reported wind	10 knots (headwind)
Runway slope	0%
Thrust Reversers Operative	1

1. Determine the Runway Condition Code, refer to Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), Runway Surface Condition – Pilot – Reported Braking Action – Wheel Braking Correction Matrix.

- If RCC is not reported, in the column Runway Surface Condition Description, locate the condition described in the problem, "... runway covered with slush less than 1/8"... : RCC5

2. Determine the OLD (uncorrected), Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – M) – RCC 5. using the pressure altitude, airplane weight from the example:

- Associated OLD: 1,567 m.

3. Apply the applicable corrections to determine the OLD, Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – M) – RCC 5 .

V _{REF}	5 x 1.2% = 6.0%
Temperature correction	8 x 0.4% = 3.2%
Headwind correction	10 x -0.5% = -5.0%
Runway slope	0%
Total correction	4.2%
FOLD	1,567 x 1.042 = 1,633 m

4. Multiply, the thrust reverser correction:

One thrust reverser inoperative	x 1.12
FOLD	1,633 x 1.12 = 1,829 m

5. The final Factored Operational Landing Distance (FOLD) is **1,829 m**.

3. PERFORMANCES (CONT'D)

VREF
TWO THRUST REVERSERS
RUNWAY CONDITION CODE 5

ZERO WIND
ZERO SLOPE
ISA

FACTORED OPERATIONAL LANDING DISTANCE (FOLD – M) – RCC 5								
WEIGHT (KG)	PRESSURE ALTITUDE (FT)							
	0	2000	4000	6000	8000	10000	12000	14000
12700	1177	1235	1298	1385	1486	1597	1720	1849
13000	1180	1239	1318	1412	1515	1629	1755	1887
14000	1232	1310	1401	1502	1612	1735	1871	2014
15000	1303	1386	1484	1592	1710	1842	1988	2141
16000	1374	1463	1567	1682	1809	1950	2106	2269
17000	1445	1540	1651	1773	1908	2058	2224	2398
18000	1527	1629	1747	1877	2022	2182	2359	2545
19000	1599	1707	1832	1970	2123	2292	2480	2677
20000	1672	1785	1917	2064	2225	2404	2602	2811
21000	1738	1857	1996	2149	2318	2507	2715	2935

<CL604_FOLD_RCC5_MET_08JUL20.ps>

Add the following corrections together based on the conditions and apply the total correction on the uncorrected FOLD value obtained from the table above:

- + 1.2 % / kt above V_{REF}
- + 0.4 % / °C above ISA, – 0.3 % / °C below ISA
- + 2.4 % / kt tailwind, – 0.5 % / kt headwind
- + 2.4 % / % downhill slope, – 1.3 % / % uphill slope

If required, multiply the corrections below to the result above:

Thrust Reversers

- One thrust reverser inoperative X 1.12
- Two thrust reversers inoperative X 1.34

MMEL

- Ground spoilers inoperative X 1.14
- Anti-skid inoperative X 1.95

NOTE: The Factored Operational Landing Distance includes a 15% operational distance factor.

VREF – Two Thrust Reversers – Runway Condition Code 5 – RCC 5 – Metric
Figure 07-02-3

BOMBARDIER CHALLENGER 605	PERFORMANCE Operational Landing Distance	Vol. 1	07-02-9 REV 57, Sep 11/20
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3. PERFORMANCES (CONT'D)

Example 4: OLD LANDING DATA - QRH TABLE CALCULATIONS (METRIC)

Associated Conditions:

Airplane weight	15,000 kg
Pressure altitude	1,417 feet (Airport Elevation) @29.82 = 1,317 feet
Ambient Temperature	8°C (ISA – 4°C) Engine Cowls required.
Runway Condition	ATIS reports, "... wet, slippery when wet...", RCC3
V _{REF} increment	5 knots
Reported wind	10 knots (headwind)
Runway slope	0%
Thrust Reversers Operative	2

- Determine the Runway Condition Code, refer to Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), Runway Surface Condition – Pilot – Reported Braking Action – Wheel Braking Correction Matrix.
 - If RCC is not reported, in the column Runway Surface Condition Description, locate the condition described in the problem, "... wet, slippery when wet...": RCC3
- Determine the OLD (uncorrected), Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – M) – RCC 3. using the pressure altitude, airplane weight from the example:

For an airplane weight of 15,000 kg:

- OLD at 0 feet (Altitude Pressure): 1,443 m.
- OLD at 2,000 feet (Altitude Pressure): 1,529 m.
- $\Delta\text{OLD} : 1,529 - 1,443 = 86 \text{ m}$
- Associated OLD at 1,317 feet: $1,443 + (86/2,000) * 1,317$.
- Associated OLD at 1,317 feet: 1,500 m.

- Apply the applicable corrections to determine the OLD, Quick Reference Handbook Volume 1 – OPERATIONAL LANDING DISTANCE (OLD), FACTORED OPERATIONAL LANDING DISTANCE (FOLD – M) – RCC 3 .

V _{REF}	5 x 1.0% = 5.0%
Temperature correction	4 x -0.2% = -0.8%
Headwind correction	10 x -0.4% = -4.0%
Runway slope	0%
Total correction	0.2%
FOLD	1,500 m x 1.002 = 1,503 m

- No correction to apply for Thrust reversers or MMEL
- The final Factored Operational Landing Distance (FOLD) is **1,503 m**.

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3. PERFORMANCES (CONT'D)

VREF
TWO THRUST REVERSERS
RUNWAY CONDITION CODE 3

ZERO WIND
ZERO SLOPE
ISA

FACTORED OPERATIONAL LANDING DISTANCE (FOLD – M) – RCC 3								
WEIGHT (KG)	PRESSURE ALTITUDE (FT)							
	0	2000	4000	6000	8000	10000	12000	14000
12700	1309	1371	1438	1527	1629	1742	1865	1993
13000	1314	1377	1459	1554	1659	1774	1899	2030
14000	1369	1450	1543	1646	1757	1879	2014	2153
15000	1443	1529	1628	1736	1855	1985	2127	2275
16000	1516	1607	1712	1827	1952	2090	2240	2397
17000	1588	1685	1795	1917	2049	2194	2353	2518
18000	1670	1772	1890	2018	2158	2312	2480	2655
19000	1743	1850	1973	2108	2255	2416	2593	2777
20000	1815	1927	2056	2197	2351	2521	2706	2899
21000	1881	1998	2132	2279	2440	2616	2810	3011

<CL604_FOLD_RCC3_MET_08JUL20.ps>

Add the following corrections together based on the conditions and apply the total correction on the uncorrected FOLD value obtained from the table above:

- + 1.0 % / kt above V_{REF}
- + 0.4 % / °C above ISA, – 0.2 % / °C below ISA
- + 2.3 % / kt tailwind, – 0.4 % / kt headwind
- + 2.8 % / % downhill slope, – 1.8 % / % uphill slope

If required, multiply the corrections below to the result above:

Thrust Reversers

- One thrust reverser inoperative X 1.13
- Two thrust reversers inoperative X 1.36

MMEL

- Ground spoilers inoperative X 1.14
- Anti-skid inoperative Prohibited

NOTE: The Factored Operational Landing Distance includes a 15% operational distance factor.

VREF – Two Thrust Reversers – Runway Condition Code 3 – RCC 3 – Metric
Figure 07-02-4

BOMBARDIER CHALLENGER 605	PERFORMANCE Operational Landing Distance	Vol. 1	07-02-11
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3. PERFORMANCES (CONT'D)

Example 5: OLD LANDING DATA - FMS CALCULATION METHOD

Airplane Weight: 34,000 lb
 Pressure Altitude: 1,506 feet (1010 HPA equivalent to 29.82)
 Ambient
 Temperature: 8 deg C (ISA – 4 deg C) Engine Cowls required
 Reported Wind: 10 knots (headwind)
 Runway Condition
 Code RCC5
 V_{REF} increment 5 knots
 Thrust Reversers
 Operative 2

1. Enter all applicable landing configuration and condition inputs into the FMS Arrival Performance page.
2. Select **DRY** runway in the FMS.
FMS ALD (1.00 LAND FACT): 2,516 feet
3. Determine the factor using the tables presented in this section based on the runway condition code:

FACTOR 2.13 @ V_{REF}+5 KTS (interpolation between 2.03 at V_{REF}
 and 2.23 at V_{REF}+10 KTS)

4. **On airplanes with Proline 21 FMS-6000 (NOT incorporating SB 605-34-019):**

Manually calculate the corrected OLD using alternate means (calculator, etc...)

2,516 x 2.13 = 5,360 feet

- On airplanes with Proline 21 Advanced FMS-6200 (Post SB 605-34-019):**

Enter 2.13 into the FACTOR field of the FMS, as shown in Figure 07-02-6 .

FMS ALD (2.13 LAND FACT): 5,360 feet.

3. PERFORMANCES (CONT'D)

TWO THRUST REVERSERS

FACTORS TO CONVERT FMS DRY ALD TO FOLD						
RCC	ISA + 15°C AND BELOW			ISA + 35°C		
	V _{REF}	V _{REF} + 10 KTS	V _{REF} + 20 KTS	V _{REF}	V _{REF} + 10 KTS	V _{REF} + 20 KTS
6	1.48	1.67	1.88	1.56	1.76	1.98
5	2.03	2.23	2.43	2.19	2.41	2.63
4	2.09	2.29	2.48	2.24	2.46	2.67
3	2.15	2.34	2.53	2.29	2.50	2.71
2	2.56	2.75	2.96	2.77	2.99	3.22
1	2.78	2.95	3.13	2.97	3.16	3.38

<CH604_FOLD_FACTORS_03JUL20_2TR>

FACTORS TO CONVERT FMS DRY ALD TO FOLD – TWO THRUST REVERSERS
Figure 07-02-5

3. PERFORMANCES (CONT'D)

On airplanes incorporating Service Bulletin 605-34-019:



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1. GENERAL

The operating procedures outlined in this chapter are particular to the CL-605 airplane. These special operations are not routinely performed on every flight, and are done only on an “as required” basis, in order to cope with minor system irregularities and abnormal flight conditions, to ensure the safe continued operation of the airplane.

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1. INTRODUCTION

A. General

This Special Operation contains information and procedures for operation of the CL 600, Model 2B16 Challenger airplane (Serial No. 5701 to 6049) on wet runways, and on runways contaminated by standing water, slush, loose snow, compacted snow or ice.

This information has been prepared by the manufacturer and approved as guidance material, to assist operators in developing suitable guidance, recommendations or instructions for use by their flight crews, when operating on wet and contaminated runway surface conditions.

The data have been prepared using reasonable estimates of the effects of wet and contaminated runway surface conditions on the accelerating ground roll and the braking ground roll. The effects of actual conditions may differ from those used to establish the data.

The level of safety is decreased when operating on contaminated runways, therefore, every effort should be made to ensure that the runway surface is cleared adequately of any significant precipitation.

B. Runway Conditions

(1) DRY RUNWAY

- A runway is considered to be dry when the runway surface is not “wet” or “contaminated”.

(2) WET RUNWAY

- A runway is considered to be wet when there is sufficient moisture on the runway surface to cause it to appear reflective, but without significant areas of standing water.

NOTE

A runway with standing water would be a contaminated runway.

(3) RUNWAY CONTAMINATED BY STANDING WATER, SLUSH OR LOOSE SNOW

- A runway is considered to be contaminated when more than 25% of the runway surface area (whether in isolated areas or not), within the required length and width being used, is covered by more than 3 millimeters (1/8 inch) of standing water or slush, or by loose snow, equivalent to more than 3 millimeters (1/8 inch) of water.

(4) RUNWAY CONTAMINATED BY COMPACTED SNOW

- A runway is considered to be contaminated by compacted snow when covered by snow which has been compacted into a solid mass, which resists further compression and will hold together or break into lumps if picked up.

(5) RUNWAY CONTAMINATED BY WET ICE

- A runway surface condition where braking action is expected to be very low, due to the presence of wet ice.

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2. LIMITATIONS

The limitations established in Chapter 2 are applicable, with the addition of the following:

- (a) The maximum depths of runway contaminants covering an appreciable part of the runway are:

CONTAMINANT	TAKE-OFF	LANDING
SLUSH/STANDING WATER	6.4 mm (0.25 in)	12.7 mm (0.5 in)
WET SNOW	12.7 mm (0.5 in)	25.4 mm (1.0 in)
DRY SNOW	30.5 mm (1.2 in)	61.0 mm (2.4 in)

- (b) The use of a take-off distance on a wet or contaminated runway that is shorter than the take-off distance on a dry runway is prohibited.
- (c) The use of actual landing distance on contaminated runways that is shorter than the dry runway landing field length is prohibited.
- (d) During taxi, do not use thrust reversers if movement area surfaces are covered with slush, ice, standing water or snow, except in the interest of safety.

3. EMERGENCY PROCEDURES

The emergency procedures in Chapter 3 are applicable.

4. NORMAL PROCEDURES

The normal procedures in Chapter 4 are applicable.

5. ABNORMAL PROCEDURES

The abnormal procedures in Chapter 5 are applicable.

6. PERFORMANCE

The performance data given in Chapter 6 of the Airplane Flight Manual is applicable, except as modified by Supplement 2 of the Airplane Flight Manual.

CHAPTER 9 - IN-FLIGHT CHECKS

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TRAINING PURPOSES ONLY

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1. GENERAL

In-flight checks may include operating procedures not routinely performed and may need to be accomplished when certain airplane systems are to be observed/verified in flight.

Other than EASA Authorities, there are no special requirements for flight crews to perform the ADG in-flight check; flight crew must be type-rated and current on the airplane per the applicable Authority. No other maintenance check flights are required by Bombardier for this airplane to be returned to service, unless specifically required by Bombardier engineering in special cases.

2. CONDUCT OF IN-FLIGHT CHECKS

Prior to conducting any in-flight checks, flight crew should thoroughly review and understand the applicable procedure(s). This should be done pre-flight. There should be a complete assessment which includes but may not be limited to the following:

- Expected systems indications and behaviors.
- Roles and responsibilities of each crew member during the task, including those of a jump seat crew member if one is present.
- Coordination that may be required with Air Traffic Control unit(s).
- Meteorological conditions and their impact on the flight (VMC may be required).
- Any weight and balance changes.
- FMS settings – defaults may have changed, and settings or sensor selections may be non-standard.
- Switches and circuit breakers may have been moved out of normal position.
- Any contingencies that may be required if systems do not function as expected.
- Any other items or issues that could impact safe conduct of the flight.

Determining airplane status, must not be underestimated. This following questions should be addressed from the onset:

- Has the airplane been in storage?
- Has maintenance been performed?
- Has the aircraft been modified (e.g. Service Bulletins, STC)?
- What is the status of airplane systems?

Flight crew should thoroughly review what system(s) may be impacted:

- Note the affected system(s) / sub-system(s),
- Evaluate and brief the impact of affected items as a crew,
- Review the associated AFM Supplements if the aircraft configuration has changed (e.g. Service Bulletins, STC),
- Pay close attention to those items during walk around and pre-flight checks, and
- Evaluate mitigations for potential in-flight issues.

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2. CONDUCT OF IN-FLIGHT CHECKS (CONT'D)

Any discrepancies should be noted and addressed with maintenance personnel prior to the flight. The procedures outlined in the following pages are designed to be completed as presented. Complete all applicable procedures. Do not change sequences within procedures.

NOTE

For flights following maintenance on flight control systems, close attention should be paid to flight control checks. Full travel and proper sense of movement should be determined, including spoiler behavior where applicable. If flight control surfaces cannot be directly observed from the flight deck, an external observer with audio communication to the flight deck should be used for movement verification.

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1. GENERAL

As EASA regulations stipulate, the operational in-flight check of the Air-Driven Generator (ADG) falls under the requirements of a Maintenance Check Flight (MCF). Under EASA regulations, MCF can only be conducted by flight crew who have received training and are approved to conduct MCF. Exceptions have been granted for crew who have relatively recent previous experience performing a similar task (EASA Authority approval required).

Other Authorities do not have special requirements for flight crews to perform the ADG in-flight check; flight crew must be type-rated and current on the airplane per the Authority under which the airplane is operated.



The presence of foreign objects in the ADG bay could cause damage to the ADG blades during deployment. Before flight, the ADG bay must be inspected to ensure that foreign objects are not present.

NOTE

Once deployed, the ADG cannot be stowed during flight. Therefore, this check should be performed at the end of a test flight. Normal electrical power must be restored before final landing approach is commenced.

If icing conditions are encountered while the ADG is the only source of electrical power, the WING anti-ice system must be operated only in the standby (STBY) mode.

For flight planning, a fuel burn correction of 5% should be used when the ADG is deployed.

2. IN-FLIGHT ADG CHECK

Prior to commencing the ADG operational check, the crew should be familiar with the In-Flight Checks introduction section of this chapter (09-01-01).

Flight crews should review the complete procedure prior to its commencement to understand the sequence and review expected system indications/behaviors.

The following is the operational in-flight check of the ADG. Only perform this check in visual meteorological conditions (VMC). Ensure that this check is not performed during icing conditions.

Complete all sequences as presented.

3. MANUAL DEPLOYMENT OPERATIONAL CHECK

A. Before Flight

- (1) ADG bay Check
 - Clear of foreign objects.
- (2) Fwd avionics bay Check
 - Verify that ADG MAN and AUTO circuit breakers (6B4 and 6B5) are closed.



3. MANUAL DEPLOYMENT OPERATIONAL CHECK (CONT'D)

A. Before Flight (Cont'd)

- (3) STAB TRIM, CH 2 and MACH TRIM switch/lights.....Press in
- Check **STAB CH 1 INOP** status message on.
 - Check for independent CH 2 operation from both stabilizer trim switches, in both directions, until trim clacker is activated.
- (4) Pilot's or copilot's STAB TRIM DISC switch Press
- Check **STAB TRIM** caution message on.
 - Check **MACH TRIM** caution message on.
 - Check stabilizer trim is inoperative.
- (5) STAB TRIM, CH 1 and CH 2 switch/lights.....Press in
- Check **STAB CH 1** and **CH 2 INOP** status messages out.
 - Check **STAB TRIM** caution message out.
- (6) MACH TRIM switch/light.....Press in
- Check **MACH TRIM** caution message out.

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3. MANUAL DEPLOYMENT OPERATIONAL CHECK (CONT'D)

B. In-Flight Procedures

- (1) BATT MASTER..... Check ON
- (2) APU Start
- (3) APU GEN..... OFF
- (4) VHF Comm, VHF Nav, ADF and transponder Set to #1

NOTE

The No. 2 VHF Comm, VHF Nav, ADF and transponder become inoperative when the main generators (GEN 1 and GEN 2) are turned off in the steps to follow.

- (5) HYDRAULIC, 1 and 2 switches ON
- (6) Airspeed..... Maintain between 180 and 197 KIAS
- (7) Flaps Set to 20°
- (8) Altitude Maintain 10,000 feet or below
- (9) ADG DEPLOY CONT – AUTO circuit breaker (2N6)..... Open
- (10) LDG GEAR lever Select to DN
- (11) HYDRAULIC, 3A and 3B switches OFF
 - Bleed off No. 3 system pressure and check that **HYD 3 LO PRESS** caution message comes on.
- (12) AC ELECTRICAL synoptic page Select
- (13) ADG manual deploy handle..... Pull and leave extended.
- (14) AC ELECTRICAL page..... Check
 - Verify that within six (6) seconds, the following indications occur:
 - ADG voltage 115 ±5 VAC
 - Frequency..... 400 ±40 Hz
 - AC ESS BUS Powered
 - AC essential bus temporarily not powered (**AC ESS BUS** caution message on EICAS comes on briefly, then out).
 - ADG output..... Green

WARNING

If at anytime the **STAB CH 2 INOP** status message comes on or the ADG fails to deliver the required electrical power, discontinue the ADG test. Stow the ADG manual deploy handle and restore normal supply to the essential busses by pressing the PWR TXFR OVERRIDE switch.



3. MANUAL DEPLOYMENT OPERATIONAL CHECK (CONT'D)

B. In-Flight Procedures (Cont'd)

- (15) DC ELECTRICAL synoptic page Check
- Verify the following indications:
 - ESS TRU 1, DC ESS BUS
and BATT BUS Powered

- (16) HYDRAULIC synoptic page Check
- That system pressure shows 3,000 ±200 psi.

NOTE

ATS is inoperative with the loss of normal AC power.

- (17) GEN 1 OFF
- Check that **GEN 1 OFF** caution message comes on.
- (18) GEN 2 OFF
- Check that **GEN 2 OFF** caution message comes on.
- (19) Primary display Check
- That **EMER PWR ONLY** warning message comes on, and
 - All other indications are normal, except VIB = 0.0 .
- (20) Pilot's and copilot's intercoms Check
- (21) STAB TRIM, CH 2 switch/light Check engaged
- (22) YAW DAMPER, YD 1 switch/light Check engaged
- YD 1 INOP** status message out.
- (23) Pitch trim Operate
- To check that stabilizer trim operates normally.



3. MANUAL DEPLOYMENT OPERATIONAL CHECK (CONT'D)

B. In-Flight Procedures (Cont'd)

(24) AC and DC essential busses Check

- Verify that power is available on each bus. It is sufficient to confirm the normal operation of one of the services powered by the bus. The following is a list of the essential busses and the services appropriate for this check:

A. DC Essential Bus:

- VHF Nav 1 operational.
- ADF 1 operational.
- Transponder 1 operational.
- Pilot's MFD and PFD with complete displays.
- Wing anti-ice, STBY (Check for ITT variation).

B. AC Essential Bus:

- **ICE DETECTOR 1 FAIL** status message out.
- ANTI-ICE, TEST switchSelect to DET

NOTE

ICE caution message will come on during test.

- Verify that **L AOA HEAT**, **L PITOT BASE HEAT**, **L PITOT TUBE HEAT**, **STBY PITOT HEAT** and **L STATIC HEAT** caution messages out.

NOTE

L AOA HEAT, **L PITOT BASE HEAT**, **L PITOT TUBE HEAT**, and **STBY PITOT HEAT** caution messages may come on during ADG deployment. If this occurs, reset circuit breaker 4A11.

If **L WINDOW HEAT** caution message comes on:

- ANTI-ICE, WSHLD/WIND
L switchSelect to RESET
then LO or HI, as required.

C. Battery Bus:

- RAM AIR SOV operational.
- Map light operational.
- INTEG instruments panel lights operational.

(25) Hydraulic system No. 3 pressure Stabilized at 3,000 ±200 psi

(26) LDG GEAR lever Cycle; UP then DN

- Check that hydraulic No. 3 system pressure does not go below 2,150 psi.



3. MANUAL DEPLOYMENT OPERATIONAL CHECK (CONT'D)**B. In-Flight Procedures (Cont'd)**

- (27) Airspeed..... Reduce
- Slow down to 130 KIAS or slowest practical safe speed and confirm the ADG output:
 - ADG voltage 115 ±5 VAC
 - Frequency.....340 Hz minimum
- (28) Airspeed..... Resume between 180 and 197 KIAS
- (29) GEN 1 and GEN 2 ON
- Check that **GEN 1** and **2 OFF** caution messages go out;
 - Check that **EMER PWR ONLY** warning message goes out.
- (30) ADG manual deploy handle..... Stow
- (31) ADG, PWR TXFR OVERRIDE switch Press
- To restore normal power supply.
 - Check that all indications return to normal.
- (32) EFIS altimeters (2) Cross-check
- (33) ADG DEPLOY CONT – AUTO circuit breaker (2N6)..... Close
- The hydraulic 3A switch is normally set to ON and 3B switch is normally set to AUTO.
- If required:**
- (34) Landing gear lever Select to UP
- (35) Flaps As required
- (36) Airspeed..... As required
- Resume normal airspeed for phase of flight, in accordance with the operating limitations as established in Chapter 2.

CHAPTER 10 - ROPAT

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1. INTRODUCTION

A. General

The purpose of this ROPAT (Recommended Operational Procedures And Techniques) section is to provide additional guidance for operations of the Challenger 605 airplane. This chapter has been created for future use and only contains the Nosewheel Steering (NWS) Inoperative (INOP) guidance at this time.

2. NON-NORMAL OPERATIONS

A. Nosewheel Steering (NWS) Inoperative (INOP)

(1) DISPATCH WITH NWS INOP/STEERING INOP

Dispatch with NWS INOP, if authorized, must be in accordance with the approved MEL.

NOTE

1. Takeoff with nosewheel steering (NWS) inoperative (INOP) must be performed in accordance with DDG/MMEL specific variant guidance. Included with the guidance is AFM Supplement 8 (Operation with Airplane Systems Inoperative) performance guidance.
2. For FAA-registered Challenger 605 airplanes, takeoff with NWS INOP is not authorized.
3. From a stopped position on the centerline of the runway, the Rolling Takeoff technique is recommended as described in the Flight Crew Operating Manual (FCOM) Chapter 4 (Normal Procedures) with the takeoff distance impact per Airplane Flight Manual (AFM) Chapter 6 (Takeoff Performance).

When takeoff with NWS INOP is planned, the runway should be chosen to minimize crosswinds if possible. Maximum crosswind component for takeoff and landing is 20 knots.

Takeoff distance available should be greater than the calculated takeoff distance. As per the AFM Supplement 8 and MMEL/DDG guidance, the Dry and Wet Takeoff Field Length obtained from the applicable AFM data must be increased by 2.0% (1.02 factor) when dispatching with NWS INOP. In addition to this MMEL correction, the recommended rolling takeoff technique adds an extra 200 feet to the Takeoff Distance. Both corrections are cumulative.

The nosewheel steering system must be selected OFF. The nose wheel will be free casting.

Dispatch with NWS INOP is prohibited for the following conditions:

- Contaminated Runways
- **Operations with Steep Approaches up to 5.5 degrees (AFM Supplement 12)**

(2) NWS INOP TAXI TECHNIQUE

When nosewheel steering (NWS) is inoperative (INOP), the nosewheel is free casting and taxi can be accomplished using differential braking. Application of differential braking will turn the airplane while it is moving. Differential thrust can be used in moderation to facilitate turns and to prevent slowing down excessively as a result of braking.

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2. NON-NORMAL OPERATIONS (CONT'D)

Anticipate turns and turn reversals or termination of turns because smooth opposite direction differential braking must be anticipated to facilitate tracking the desired route and to avoid jerky braking and associated discomfort for passengers. Above all, brake temperature management can be largely controlled by anticipating long in advance taxi needs through rudder and differential thrust. Sharp turns should be avoided as they require more braking action either to enter, exit or both.

Minimize prolonged use of sustained brake applications to not cause excessive increase of brake temperatures. This is especially important for NWS INOP taxi following a landing or RTO with aggressive braking.

Care must be exercised when making tight turns. Avoid castering the nosewheel to high angles approaching 90 degrees. Larger angles beyond 55 degrees up to 90 should be avoided. At extreme nosewheel angles, forward progress may no longer be possible and excessive lateral forces could be placed on nosewheel tires, potentially causing tire damage, tire coming off the wheel, deflation or damage to the nose gear. A tow bar may be required to straighten the nosewheel. If there is any doubt, the airplane should be stopped until a tow can be arranged.

(3) NWS INOP TAKEOFF TECHNIQUE

NOTE

NWS failure could occur during the takeoff roll. The NWS CAS message is Takeoff Inhibited above 80 KIAS, thus the TO below 80 KIAS should be aborted and care needs to be taken. After the decision to reject the takeoff, use the NWS Landing Technique to safely stop on the runway. The nosewheel will be free castering after the NWS failure. Once at safe taxi speed, use the NWS INOP taxi technique. After 80 KIAS you should not notice the steering inoperative and once airborne the CAS message should appear which you will need for landing.

Minimize sharp turns entering the runway. Confirm that the airplane is lined up with the runway, ensure the nosewheel is also tracking in line with the runway before application of takeoff thrust.

As the airplane accelerates, maximize use of rudder via pedal input to maintain centerline control. If additional directional control is required at slower speed, modulate differential brake input (enough input to prevent or correct an undesired drift, but not enough to grossly impact normal acceleration) until aerodynamic forces on the rudder provide control. Supplement 8 provides for differential braking during acceleration.

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2. NON-NORMAL OPERATIONS (CONT'D)

Additional takeoff distance must be considered when using this technique.

NOTE

1. Higher crosswinds will adversely impact centerline control.
2. Do not ride the brakes.
3. When runway length available is approaching runway length required, operators should consider not dispatching with NWS INOP.
4. In the event of rejected takeoff, the crew should apply maximum braking as much as practicable and if required, slightly reduce pedal pressure on applicable brake to maintain directional control.
i.e. reduce the brake on the pedal opposite of the desired direction.

(4) LANDING WITH NOSEWHEEL STEERING INOP

When a landing with NWS INOP is planned, the following apply:

- Choose the longest, widest available runway with minimum crosswind and turbulence to provide maximum deviation space and reduce pilot workload.
- Landing distances may be greater than for a normal landing because of the requirement to use differential or asymmetric braking for directional control as aerodynamic rudder control diminishes. The calculated landing distances are based upon a maximum symmetric brake application to a stop. Deviations from this technique including asymmetric braking and less than maximum brake application will increase the required stopping distance. Crews must properly account for this increase when selecting a landing runway.
- Select NOSE STEER to the OFF position.

If NWS fails during landing rollout, despite possible associated CAS message(s)/aural tone, crew awareness may be delayed until the airplane path deviates directionally even with significant pedal input to counter drift. Once aware, the pilot requires prompt transition to appropriate technique. Do not attempt to troubleshoot during the rollout.

(5) NWS INOP AFTER LANDING TECHNIQUE

The nosewheel will be free casting. As speed decreases, directional control will gradually transition from primarily aerodynamic control via rudder input, to differential braking. This will become apparent with ever increasing rudder input to deflection limit. Approaching pedal travel limit, start applying appropriate differential brake input to control centerline.

During aggressive braking (large brake pedal application), precise differential braking may be more difficult to modulate. With large rudder pedal deflections, precise differential brake application on either pedal may require more conscious effort. In all cases, strict attention should be paid to prevent inadvertent brake application on the aft-moving brake pedal. Elevated crosswinds will adversely impact centerline control, making the task more difficult and may result in high asymmetric brake temperature.

At taxi speed, use differential braking and differential thrust to clear the runway. Use of high-speed turnoffs should be considered only when directional control is well in hand and conditions permit.

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2. NON-NORMAL OPERATIONS (CONT'D)

If directional control becomes difficult, the airplane should be stopped until a tow can be arranged.

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CHAPTER 11 - SERVICING

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1. INTRODUCTION

This chapter contains guidance to accomplish certain procedures or elementary maintenance tasks to facilitate continued operation of the airplane.

Some operators may require that flight crew undergo specific training in accordance with authority guidance to be allowed to complete some or all of these procedures.

The description of the procedures contained herein does not constitute authority to carry out these procedures.

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1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES

Before disconnecting the batteries, ensure the following:

- (a) SECURING CHECK is complete.
- (b) AC and DC External power are **NOT** connected to the aircraft.
- (c) No one is performing maintenance or servicing the aircraft.

WARNING

DO NOT CONNECT EXTERNAL POWER WHEN THE BATTERIES ARE DISCONNECTED. IF EXTERNAL POWER IS NEEDED, RECONNECT BATTERIES BEFORE APPLYING EXTERNAL POWER.

NOTE

It is recommended that the passenger door is closed before the main battery is disconnected.

WARNING

Be careful when you touch or do maintenance on or near the battery. Remove all metal objects and jewelry from your person. If you accidentally touch the battery with a metal object, you can cause personal injury and damage to equipment. Also, due to the elevated position, you may also scratch or damage the skin of the aircraft.

A. Main Battery Disconnect Procedure

CAUTION

A ladder is required to gain access to the Main Battery electrical power connector. Always pay attention to the environment when positioning the ladder for access (ramp contamination, wind, slippery surfaces, etc.). Failure to observe this caution could cause personal injury and/or damage to equipment. For the same reason, stow the ladder when not required. Never leave the ladder unattended. If you must leave the ladder unattended, leave it lying flat on the ground.



1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

A. Main Battery Disconnect Procedure (Cont'd)

- (1) Open the left forward upper equipment bay door by lifting the aft and forward latches. Refer to Figure 11-02-1 .



Upper Equipment Bay Door Latches
Figure 11-02-1



1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

A. Main Battery Disconnect Procedure (Cont'd)

- (2) Secure the left forward upper equipment bay door open via the holding rod. Refer to Figure 11-02-2



Upper Equipment Bay Holding Rod
Figure 11-02-2



1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

A. Main Battery Disconnect Procedure (Cont'd)

- (3) Disconnect the main battery electrical power connector by rotating the knob counterclockwise. Refer to Figure 11-02-3



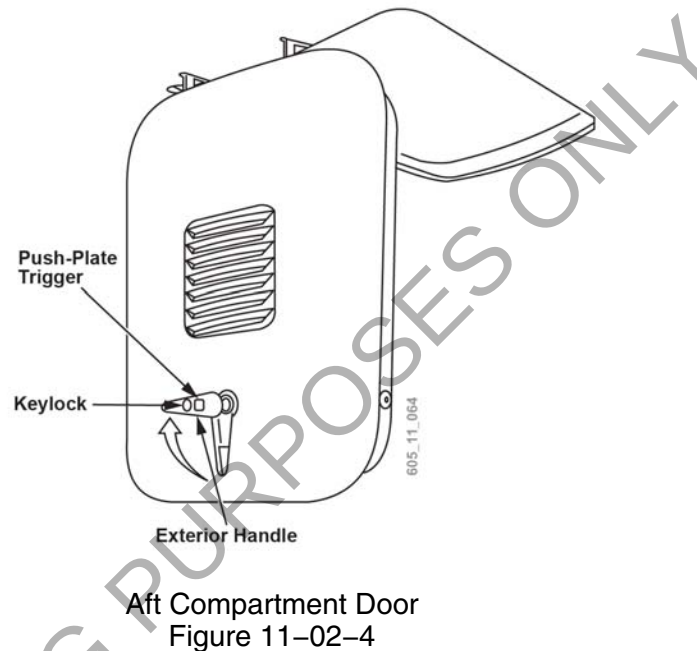
Main Battery Disconnected
Figure 11-02-3

- (4) Position the main battery electrical power connector away from the battery terminals and adjacent components. Refer to Figure 11-02-3.
- (5) Close the left forward upper equipment bay door by engaging the aft and forward latches. Make sure they are flush to the aircraft skin. Refer to Figure 11-02-1

1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Disconnect Procedure

- (1) Open the aft equipment compartment access door by pressing the PUSH on the trigger plate and let the handle come out of its receptacle. While holding the door, rotate the handle clockwise to retract the plunger. Open the door slowly. (Refer to Figure 11-02-4). Hold the aft compartment bay door open by engaging the door support rods. The aft compartment door is equipped with integrated steps to ease access. Refer to Figure 11-02-5



1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Disconnect Procedure (Cont'd)



Aft Compartment Door Support Rods and Steps
Figure 11-02-5

1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Disconnect Procedure (Cont'd)

- (2) Disconnect the Auxiliary Power Unit (APU) battery electrical power connector by rotating the knob counterclockwise. Refer to Figure 11-02-6



APU Battery Disconnected
Figure 11-02-6

- (3) Position the APU battery electrical connector away from the battery terminals and adjacent components.



1. PROCEDURES TO DISCONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Disconnect Procedure (Cont'd)

- (4) Close the aft equipment compartment access light, release the door support rods and close the door. While holding the door closed, turn the handle counterclockwise to engage the plunger (refer to Figure 11-02-7) then push the handle upward to stow and latch the door in the trigger lock. Make sure the latch handle is fully engaged and flush with the aircraft skin. Refer to Figure 11-02-8



Aft Equipment Compartment Latch
Figure 11-02-7



Compartment Latch Locked
Figure 11-02-8

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2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES

Before connecting the batteries, ensure the following:

- (a) SECURING CHECK is complete.
- (b) AC and DC External power are **NOT** connected to the aircraft.
- (c) No one is performing maintenance or servicing the aircraft.

WARNING

DO NOT CONNECT EXTERNAL POWER WHEN THE BATTERIES ARE DISCONNECTED. IF EXTERNAL POWER IS NEEDED, RECONNECT BATTERIES BEFORE APPLYING EXTERNAL POWER.

TRAINING PURPOSES ONLY

2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

WARNING

Be careful when you touch or do maintenance on or near the battery. Remove all metal objects and jewelry from your person. If you accidentally touch the battery with a metal object, you can cause personal injury and damage to equipment. Also, due to the elevated position, you may also scratch or damage the skin of the aircraft.

A. Main Battery Connect Procedure

CAUTION

A ladder is required to gain access to the Main Battery electrical power connector. Always pay attention to the environment when positioning the ladder for access (ramp contamination, wind, slippery surfaces, etc.). Failure to observe this caution could cause personal injury and/or damage to equipment. For the same reason, stow the ladder when not required. Never leave the ladder unattended. If you must leave the ladder unattended, leave it lying flat on the ground.

- (1) Open the left forward upper equipment bay door by lifting the aft and forward latches. Refer to Figure 11-02-9.



Upper Equipment Bay Door Latches
Figure 11-02-9



2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

A. Main Battery Connect Procedure (Cont'd)

- (2) Secure the left forward upper equipment bay door open via the holding rod. Refer to Figure 11-02-10



Upper Equipment Bay Holding Rod
Figure 11-02-10

- (3) Connect the electrical power connector to the main battery by engaging the connector guide pin in the battery receptacle and then rotating the knob clockwise. Refer to Figure 11-02-11

NOTE

Make sure that the mating surfaces are clean with no corrosion, paint, grease or preservation fluids.



2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

A. Main Battery Connect Procedure (Cont'd)



Battery
Connector

Main Battery Connected
Figure 11-02-11

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2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

A. Main Battery Connect Procedure (Cont'd)

- (4) Close the left forward upper equipment bay door by engaging the aft and forward latches. Make sure they are flush to the aircraft skin. Refer to Figure 11-02-9.



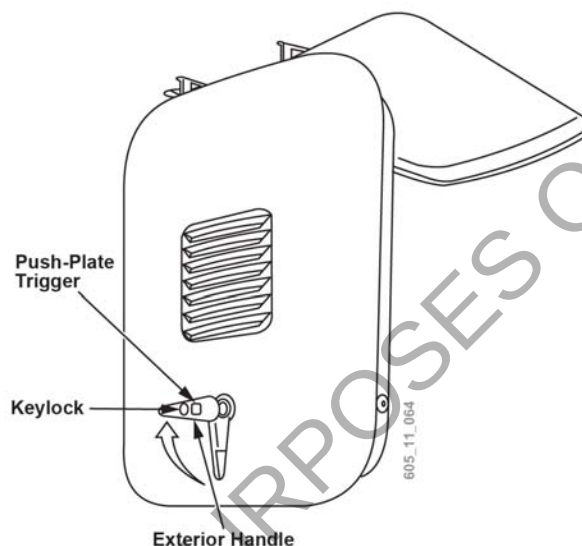
Make sure both latches are properly secured and flush with the surrounding airframe. Failure to properly secure the latches could result in disrupted airflow over the System 1 pitot static probe and possibly cause erroneous airspeed/altitude indications.

TRAINING PURPOSES ONLY

2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Connect Procedure

- (1) Open the aft equipment compartment access door by pressing the PUSH on the trigger plate and let the handle come out of its receptacle. While holding the door, rotate the handle clockwise to retract the plunger. Open the door slowly. (Refer to Figure 11-02-12). Hold the aft compartment bay door open by engaging the door support rods. The aft compartment door is equipped with integrated steps to ease access. Refer to Figure 11-02-13



Aft Compartment Door
Figure 11-02-12



2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Connect Procedure (Cont'd)



Aft Compartment Door Support Rods and Steps
Figure 11-02-13

2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Connect Procedure (Cont'd)

- (2) Connect the electrical power connector on the auxiliary power unit (APU) battery by engaging the connector guide pin in the battery receptacle and then rotating the knob clockwise. Figure 11-02-14

NOTE

Make sure that the mating surfaces are clean with no corrosion, paint, grease or preservation fluids.



APU Battery Connected
Figure 11-02-14



2. PROCEDURES TO CONNECT MAIN & AUXILIARY POWER UNIT (APU) BATTERIES (CONT'D)

B. APU Battery Connect Procedure (Cont'd)

- (3) Close the aft equipment compartment access light, release the door support rods and close the door. While holding the door closed, turn the handle counterclockwise to engage the plunger (refer to Figure 11-02-15) then push the handle upward to stow and latch the door in the trigger lock. Make sure the latch handle is fully engaged and flush with the aircraft skin. Refer to Figure 11-02-16



Aft Equipment Compartment Latch
Figure 11-02-15



Compartment Latch Locked
Figure 11-02-16

3. MAIN & AUXILIARY POWER UNIT (APU) BATTERIES CONNECTION CHECK

NOTE

If approaching the aircraft with the intent of departure, proceed to the AFM Normal Power-Up Check. If not, perform steps A.(1) to A.(5) below.

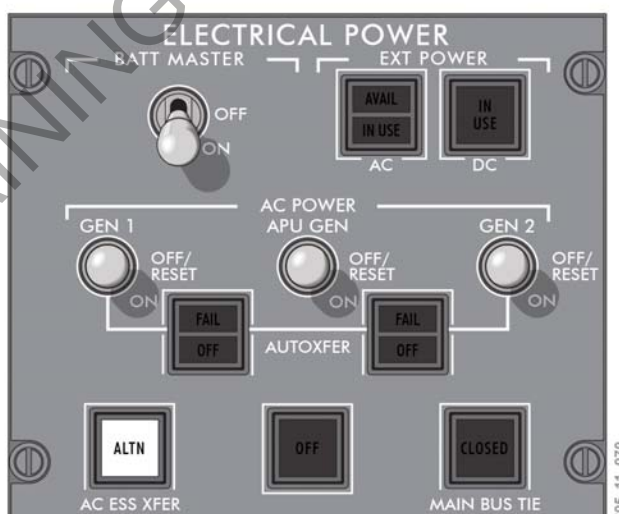
A. Main & Auxiliary Power Unit (APU) Batteries Connection Check

- (1) On the ELECTRICAL POWER control panel, set the BATT MASTER toggle switch to ON. Refer to Figure 11-02-17.
- (2) On the Cursor Control Panel (CCP), select the DC ELEC key. Refer to Figure 11-02-18.
- (3) On the DC ELECTRICAL synoptic page, confirm that both the MAIN and the APU Batteries display green outlines and valid voltages. Make sure that BATT BUS, APU BATT DIR BUS, MAIN BATT DIR BUS and EMER BUS are powered (green outlines) with associated green flow tubes. (Refer to Figure 11-02-19.)



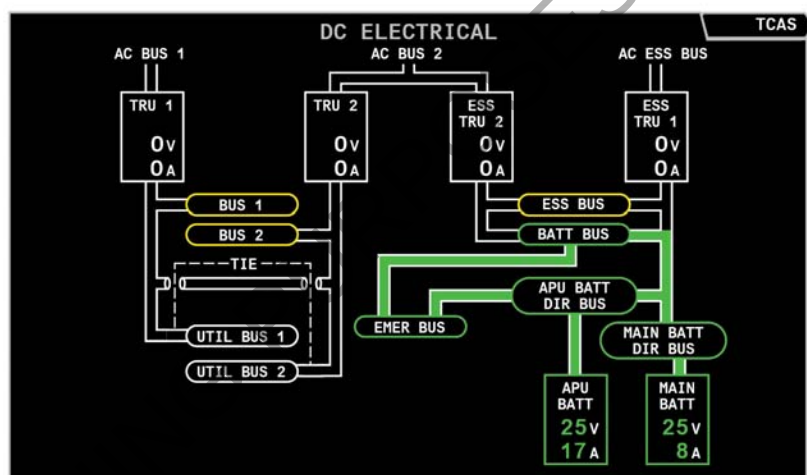
Wait for the FMS to be available before removing aircraft power (40 seconds minimum). Failure to do this may result in FMC lockup which will require FMC card removal and reset by Collins (ref AW600-34-2509).

- (4) On the ELECTRICAL POWER control panel, set the BATT MASTER toggle switch to OFF. Refer to Figure 11-02-17.
- (5) Make sure the aircraft is completely powered off and confirm Boarding, stairs and entrance lights are OFF (Refer to Figure 11-02-20)



ELECTRICAL POWER Control Panel
Figure 11-02-17

3. MAIN & AUXILIARY POWER UNIT (APU) BATTERIES CONNECTION CHECK (CONT'D)

Cursor Control Panel (CCP)
Figure 11-02-18DC ELECTRICAL Synoptic Page
Figure 11-02-19

BOARD Switch
Controls the door
boarding light

Misc Lighting Panel
Figure 11-02-20

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